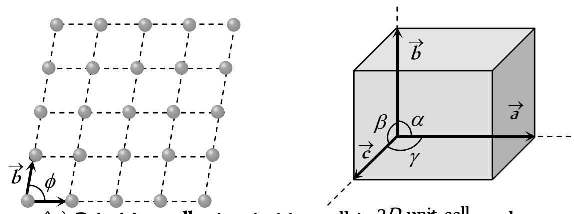
(3) **Unit cell :** Is defined as that volume of the solid from which the entire crystal structure can be constructed by the translational repetition in three dimensions. The length of three sides of a unit cell (3*D*) are called primitives or lattice constant they are denoted by *a*, *b*, *c*



2*D* unit cell 3*D* unit cell

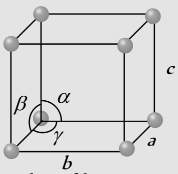
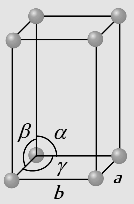
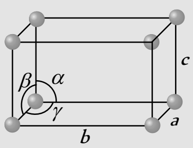
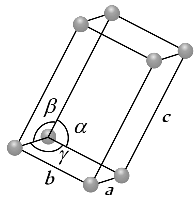
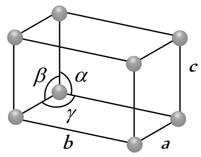
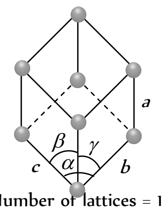
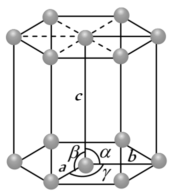
Fig. 27.1

(4) **Primitive cell :** A primitive cell is a minimum volume unit cell or the simple unit cell with particles only at the corners is a primitive unit cell and other types of unit cells are called non-primitive unit cells. There is only one lattice point per primitive cell.

(5) **Crystallographic axis :** The lines drawn parallel to the lines of intersection of the faces of the unit cell are called crystallographic axis.

All the crystals on the basis of the shape of their unit cells, have been divided into seven crystal systems as shown in the following table.

Table 27.1 : Different crystal systems

System	Lattice constants	Angle between lattice constants	Examples
Cubic  Number of lattices = 3	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	Diamond, *NaCl*, *Li*, *Ag*, *Cu*, NH_4Cl, *Pb etc.*
Tetragonal  Number of lattices = 2	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	White tin, $NiSO_4$ *etc.*
Orthorhombic  Number of lattices = 4	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	$HgCl_2$, KNO_3, gallium *etc.*
Monoclinic  Number of lattices = 2	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ$ and $\beta \neq 90^\circ$	$KclO_3$, $FeSO_4$ *etc.*
Triclinic  Number of lattices = 1	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	$K_2Cr_2O_7$, $CuSO_4$ *etc.*
Rhombo-hedral or Trigonal  Number of lattices = 1	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	Calcite, *As*, *Sb*, *Bi etc.*
Hexagonal  Number of lattices = 1	$a = b \neq c$	$\alpha = \beta = 90^\circ$ and $\gamma = 120^\circ$	*Zn*, *Cd*, *Ni etc.*

Different Types of Symmetry in Cubic Lattices

(1) **Centre of symmetry :** An imaginary point within the crystal such that any line drawn through it intersects the surface of the crystal at equal distances in both directions.

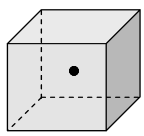
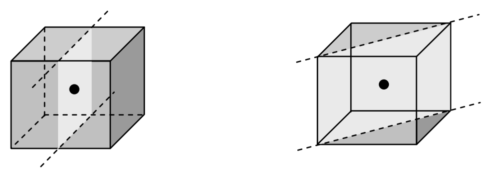


Fig. 27.2

(2) **Plane of symmetry :** It is an imaginary plane which passes through the centre of a crystal and divides it into two equal portions such that one part is exactly the mirror image of the other.



(A) Rectangular plane of symmetry (B) Diagonal plane of symmetry

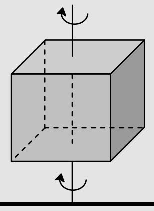
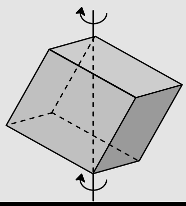
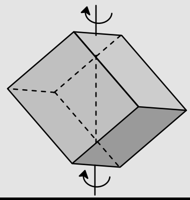
Fig. 27.3

A cubical crystal possesses six diagonal plane of symmetry and three rectangular plane of symmetry.

(3) **Axis of symmetry** : It is an imaginary straight line about which, if the crystal is rotated, it will present the same appearance more than once during the complete revolution.

In general, if the same appearance of a crystal is repeated on rotating through an angle $\frac{360^o}{n}$, around an imaginary axis, the axis is called an *n*-fold axis.

Table 27.2 : A cubical crystal possesses in all 13 axis of symmetry

Axis of four-fold symmetry = 3 (Because of six faces)	Axis of three-fold symmetry = 4 (Because of eight corners)	Axis of two-fold symmetry = 6 (Because of twelve edges)
		

(4) **Elements of symmetry :** The total number of planes, axes and centre of symmetry possessed by a crystal are termed as elements of symmetry. A cubic crystal possesses a total of 23 elements of symmetry.

Planes of symmetry $= (3+6) = 9$,

Axes of symmetry $= (3+4+6) = 13$,

Centre of symmetry = 1.

Total number of symmetry elements = 23

More About Cubic Crystals

(1) **Different lattice in cubic crystals :** There are three lattice in the cubic system.

(i) The simple cubic (*sc*) lattice.

(ii) The body-centered cubic (*bcc*).

(iii) The face-centered cubic (*fcc*).

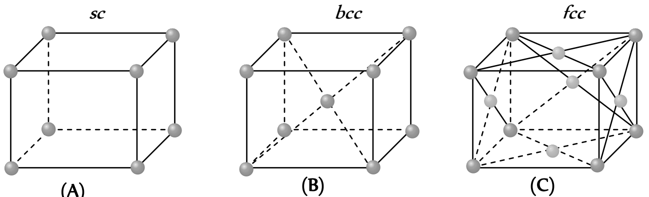
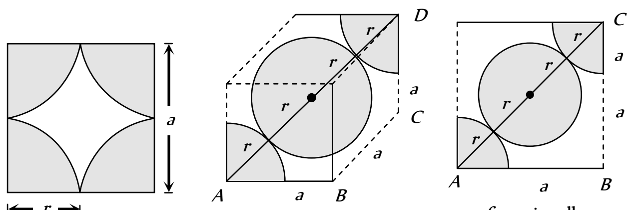


Fig. 27.4

(2) **Atomic radius :** The half of the distance between two atoms in contact is defined as atomic radius.



sc unit cell
$r = a/2$

bcc unit cell
$r = \sqrt{3}\, a/4$

fcc unit cell
$r = a/2\sqrt{2}$

Fig. 27.5

(3) **Atoms per unit cell :** An atom located at the corner of a unit cell of a lattice is shared equally by eight other unit cells in the three dimensional lattice. Therefore, each unit cell has 1/8 share of an atom at its each corner. Similarly, a face of the unit cell is common to the two unit cells in the lattice. Therefore, each unit cell has 1/2 share of an atom at its each face. The atom located at the centre of the unit cell belongs completely to the unit cell.

Let N_c, N_b and N_f be the number of atoms at the corners, centre and face of the unit cell respectively. Therefore the number of atoms per unit cell is given by $N = N_b + \frac{N_f}{2} + \frac{N_c}{8}$

(i) In *sc* lattice : $N_b = 0,\ N_f = 0,\ N_c = 8$ so $N = 1$

(ii) In *bcc* lattice : $N_b = 1,\ N_f = 0,\ N_c = 8$ so $N = 2$

(iii) In *fcc* lattice : $N_b = 0,\ N_f = 6,\ N_c = 8$ so $N = 4$

(4) **Co-ordination number :** It is defined as the number of nearest neighbours that an atom has in a unit cell. It depends upon structure.

(i) Simple cubic structure : Each atom has two neighbours along *X*-axis, two along *Y*-axis and two along *Z*-axis so co-ordination number = 6.

(ii) Face-centred cubic structure: Every corner atom has four neighbours in each of the three planes *XY*, *YZ*, and *ZX* so co-ordination number = 12

(iii) Body-centred cubic structure: The atom of the body of the cell has eight neighbours at eight corner of the unit cell so co-ordination number = 8.

(5) **Atomic packing fraction (or packing factor or relative packing density)**

The atomic packing fraction indicates how close the atoms are packed together in the given crystal structure or the ratio of the volume occupied by atoms in a unit cell in a crystal and the volume of unit cell is defined as APF.

(i) **For *sc* crystal :** Volume occupied by the atom in the unit cell $= \frac{4}{3}\pi r^3 = \frac{\pi a^3}{6}$. Volume of the unit cell $= a^3$

Thus P.F. $= \frac{\pi a^3/6}{a^3} = \frac{\pi}{6} = 0.52 = 52\%$

(ii) **For *bcc* :** P.F. $= \frac{\sqrt{3}\pi}{8} = 68\%$

(iii) **For *fcc* :** P.F. $= \frac{\pi}{3\sqrt{2}} = 74\%$

(6) **Density of unit cell :** Density of unit cell $= \frac{\text{Mass of the unit cell}}{\text{Volume of the unit cell}} = \frac{nA}{NV} = \frac{nA}{Na^3}$

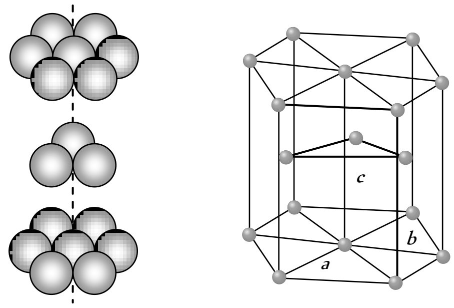
where *n* = Number of atoms in unit cell (For *sc* lattice *n* = 1, for *bcc* lattice *n* = 2, for *fcc* lattice *n* = 4), *A* = atomic weight, *N* = Avogadro's number, *V* = Volume of the unit cell.

(7) **Bond length :** The distance between two nearest atoms in a unit cell of a crystal is defined as bond length.

(i) In a *sc* lattice : Bond length = *a* (ii) In a *bcc* lattice : Bond length $= \frac{\sqrt{3}a}{2}$ (iii) In a *fcc* lattice : Bond length $= \frac{a}{\sqrt{2}}$

Hexagonal Close Packed (HCP) Structure

The HCP structure also maximizes the packing fraction



(A) (B)

Fig. 27.6

(1) $a = b \neq c$

(2) Number of atoms per unit cell = 6

(3) The volume of the hexagonal cell = $3\sqrt{2}\,a^3$

(4) The packing fraction $= \dfrac{\pi\sqrt{2}}{6}$

(5) Coordination number = 12

(6) Magnesium is a special example of HCP lattice structure.

Bonding Forces in Crystals

The properties of a solid are mainly determined by the type of bonding that exists between the atoms. According to bonding in crystals they are classified into following types.

(1) **Ionic crystal :** This type of bonding is formed due to transfer of electrons between atoms and consequent attraction between them.

(i) In *NaCl* crystal, the electron of *Na* atom is transferred to chlorine atom. In this way *Na* atom changes in to *Na* ion and *Cl* atom changes into *Cl* ion.

(ii) Cause of binding is electrostatic force between positive and negative ion.

(iii) These crystal are usually hard, brittle and possesses high melting and boiling point.

(iv) These are bad conductor of electricity.

(v) Common example are *NaCl*, *CsCl*, *LiF etc.*

(2) **Covalent crystal :** Covalent bonding is formed by sharing of electrons of opposite spins between two atoms

(i) The conductivity of these solids rise with rise in temperature.

(ii) These crystal posses high melting point.

(iii) Bonding between H_2, Cl_2 molecules *Ge*, *Si*, Quartz, diamond *etc*. are common example of covalent bonding

(3) **Metallic bonds :** This type of bonding is formed due to attraction of valence (free) electrons with the positive ion cores

(i) Their conductivity decreases with rise of temperature.

(ii) When visible light falls on a metallic crystal, the electrons of atom absorb visible light, so they are opaque to visible light. However some orbital electrons absorb energy and reach in excited state. They then return to their normal states, remitting light of same frequency.

Common examples are *Na*, *Li*, *K*, *Cs*, *Au*, *Hg etc*.

(4) **Vander waal's crystal :** These crystal consists of neutral atoms or molecules bonded together in solid phase by weak, short range attractive forces called vander Waal's forces.

(i) This bonding is weakest and occurs in solid CO_2, methane, paraffin, ice, etc.

(ii) They are normally insulator, they are soft, easily compressible and posses low melting point.

(5) **Hydrogen bonding :** Hydrogen bonding is due to permanent dipole interaction.

(i) This bond is stronger than vander Waal's bond but much weaker than ionic and covalent bond.

(ii) They possesses low melting point.

(iii) Common examples are H_2O, *HF etc*.

Single, Poly and Liquid Crystals

(1) **Single crystal :** The crystals in which the periodicity of the pattern extends throughout the piece of the crystal are known as single crystals. Single crystals have anisotropic behaviour *i.e.* their physical properties (like mechanical strength, refractive index, thermal and electrical conductivity) are different along different directions. The small sized single crystals are called mono-crystals.

(2) **Poly-crystals :** A poly-crystal is the aggregate of the monocrystals whose well developed faces are joined together so that it has isotropic properties. Ceramics are the important illustrations of the poly-crystalline solids.

(3) **Liquid crystals :** The organic crystalline solid which on heating, to a certain temperature range becomes fluid like but its molecules remain oriented in a particular directions, showing that they retain their anisotropic properties, is called liquid crystal. These crystals are used in a liquid crystal displays (L.C.D.) which are commonly used in electronic watches, clocks and micro-calculators *etc*.

Energy Bands

This theory is based on the Pauli exclusion principle.

In isolated atom the valence electrons can exist only in one of the allowed orbitals each of a sharply defined energy called energy levels. But when two atoms are brought nearer to each other, there are alterations in energy levels and they spread in the form of bands.

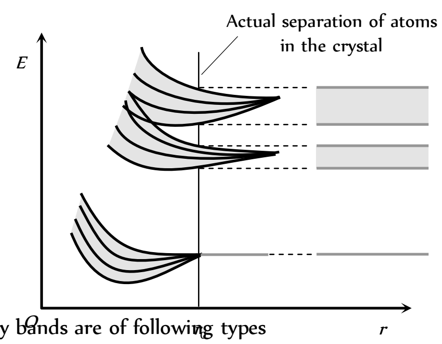


Fig. 27.7 Origin of energy bands in a solid

Energy bands are of following types

(1) **Valence band :** The energy band formed by a series of energy levels containing valence electrons is known as valence band. At 0 *K*, the electrons fills the energy levels in valence band starting from lowest one.

(i) This band is always filled with electrons.

(ii) This is the band of maximum energy.

(iii) Electrons are not capable of gaining energy from external electric field.

(iv) No flow of current due to electrons present in this band.

(v) The highest energy level which can be occupied by an electron in valence band at 0 *K* is called fermi level.

(2) **Conduction band :** The higher energy level band is called the conduction band.

(i) It is also called empty band of minimum energy.

(ii) This band is partially filled by the electrons.

(iii) In this band the electrons can gain energy from external electric field.

(iv) The electrons in the conduction band are called the free electrons. They are able to move any where within the volume of the solid.

(v) Current flows due to such electrons.

(3) **Forbidden energy gap (ΔE) :** Energy gap between conduction band and valence band $\Delta E_g = (C.B.)_{\min} - (V.B.)_{\max}$

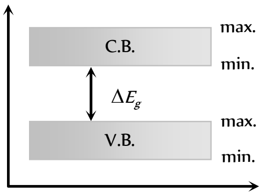


Fig. 27.8

(i) No free electron is present in forbidden energy gap.

(ii) Width of forbidden energy gap depends upon the nature of substance.

(iii) As temperature increases (↑), forbidden energy gap decreases (↓) very slightly.

Table 27.3 : Types of solid

Properties	**Conductors**	**Insulators**	**Semiconductors**
Electrical conductivity	10^2 to 10^8 ℧/m	10^{-8} ℧/m	10^{-5} to 10^0 ℧/m
Resistivity	10^{-2} to 10^{-8} Ω-m (negligible)	10^8 Ω-m	10^5 to 10^0 Ω-m
Band structure	C.B. V.B.	C.B. ΔE_g (large) V.B.	C.B. ΔE_g (small) V.B.
Energy gap (E_g)	Zero or very small	Very large; for diamond it is 6 *eV*	*Ge* → 0.7 *eV* *Si* → 1.1 *eV* *GaAs* →1.3 *eV* GaF_2 → 2.8 *eV*
Current carriers	Free electrons	––	Free electrons and holes
Condition of V.B. and C.B. at ordinary temperature	V.B. and C.B. are completely filled or C.B. is some what empty	V.B. – completely filled C.B. – completely unfilled	V.B. – somewhat empty C.B. – somewhat filled
Temperature co-efficient of resistance	Positive	Zero	Negative
Effect of temperature on conductivity	Decreases	—	Increases
Effect of temperature on resistance	Increases	—	Decreases
Examples	*Cu, Ag, Au, Na, Pt, Hg* etc.	Wood, plastic, mica, diamond, glass etc.	*Ge, Si, Ga, As* etc.
Electron density	$10^{29}/m^3$	—	*Ge* ~ 10^{19} $/m^3$ *Si* ~ 10^{16} $/m^3$

Holes in Semiconductors

(1) When an electron is removed from a covalent bond, it leaves a vacancy behind. An electron from a neighbouring atom can move into this vacancy, leaving the neighbour with a vacancy. In this way the vacancy formed is called hole (or cotter), and can travel through the material and serve as an additional current carriers.

(2) A hole is considered as a seat of positive charge, having magnitude of charge equal to that of an electron.

(3) Holes acts as virtual charge, although there is no physical charge on it.

(4) Effective mass of hole is more than electron.

(5) Mobility of hole is less than electron.

Intrinsic Semiconductors

(1) A pure semiconductor is called intrinsic semiconductor. It has thermally generated current carriers

(2) They have four electrons in the outermost orbit of atom and atoms are held together by covalent bond

(3) Free electrons and holes both are charge carriers and n_e (in C.B.) $= n_h$ (in V.B.)

(4) The drift velocity of electrons (v_e) is greater than that of holes (v_h)

(5) For them fermi energy level lies at the centre of the C.B. and V.B.

(6) In pure semiconductor, impurity must be less than 1 in 10^8 parts of semiconductor.

(7) In intrinsic semiconductor

$n_e^{(o)} = n_h^{(o)} = n_i$; where $n_e^{(o)} =$ Electron density in conduction band, $n_h^{(o)} =$ Hole density in V.B., $n_i =$ Density of intrinsic carriers.

(8) The fraction of electrons of valance band present in conduction band is given by $f \propto e^{-E_g/kT}$; where E_f = Fermi energy or k = Boltzmann's constant and T = Absolute temperature

(9) Because of less number of charge carriers at room temperature, intrinsic semiconductors have low conductivity so they have no practical use.

(10) Number of electrons reaching from valence band to conduction band $n = AT^{3/2}e^{-E_g/2kT}$

Extrinsic Semiconductor

(1) An impure semiconductor is called extrinsic semiconductor

(2) When pure semiconductor material is mixed with small amounts of certain specific impurities with valency different from that of the parent material, the number of mobile electrons/holes drastically changes. The process of addition of impurity is called doping.

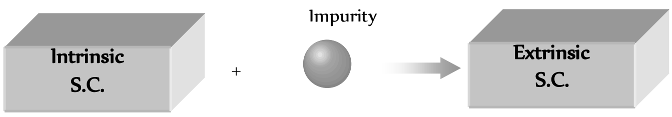


Fig. 27.9

(3) **Pentavalent impurities :** The elements whose atom has five valance electrons are called pentavalent impurities e.g. *As, P, Sb etc.* These impurities are also called donor impurities because they donate extra free electron.

(4) **Trivalent impurities :** The elements whose each atom has three valance electrons are called trivalent impurities e.g. *In, Ga, Al, B, etc.* These impurities are also called acceptor impurities as they accept electron.

(5) The compounds of trivalent and pentavalent elements also behaves like semiconductors *e.g. GaAs, InSb, In P, GaP etc.*

(6) The number of atoms of impurity element is about 1 in 10^8 atoms of the semiconductor.

(7) In extrinsic semiconductors $n_e \neq n_h$

(8) In extrinsic semiconductors fermi level shifts towards valence or conduction energy bands.

(9) Their conductivity is high and they are used for practical purposes.

(10) In a doped extrinsic semiconductor, the number density of e^- of the conduction band (n_e) and the number density of holes in the valence band (n_h) differs from that in a pure semiconductor. If n_i is the number density of electron in conduction band or the number density of holes in valence band in a pure semiconductor then $\boldsymbol{n_e n_h = n_i^2}$ (mass action law)

(11) Extrinsic semiconductors are of two types

(i) *N*-type semiconductor (ii) *P*-type semiconductor

N-Type Semiconductor

These are obtained by adding a small amount of pentavalent impurity to a pure sample of semiconductor (*Ge*).

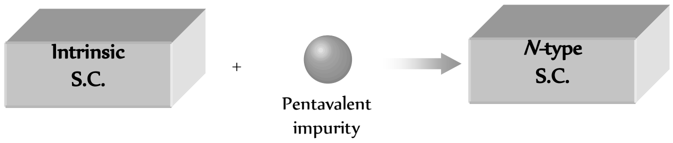


Fig. 27.10

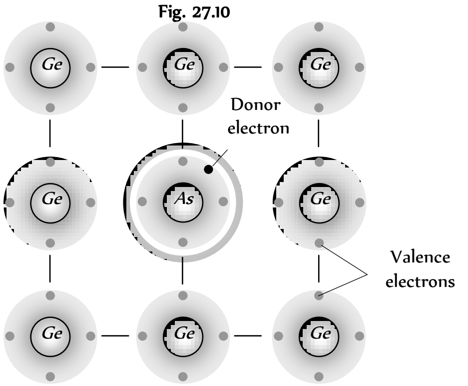


Fig. 27.11

(1) Majority charge carriers – electrons

Minority charge carriers – holes

(2) $n_e >> n_h$; $i_e >> i_h$

(3) Conductivity $\sigma \approx n_e \mu_e e$

(4) *N*-type semiconductor is electrically neutral (not negatively charged)

(5) Impurity is called Donar impurity because one impurity atom generate one electron.

(6) Donor energy level lies just below the conduction band.

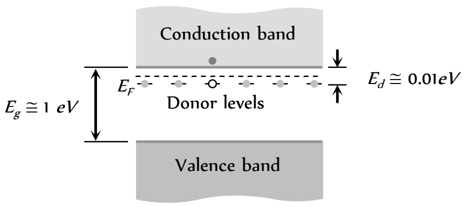
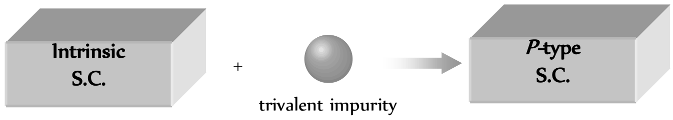


Fig. 27.12

P-Type Semiconductor

These are obtained by adding a small amount of trivalent impurity to a pure sample of semiconductor (*Ge*).



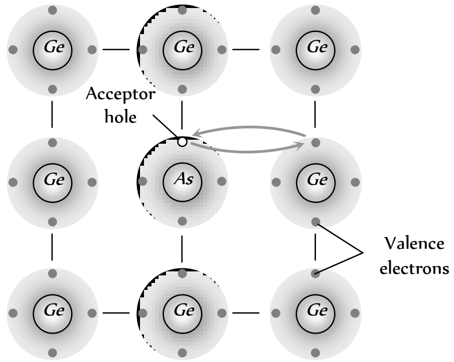


Fig. 27.13

(1) Majority charge carriers – holes

Minority charge carriers – electrons

(2) $n_h >> n_e$; $i_h >> i_e$

(3) Conductivity $\sigma \approx n_h \mu_h e$

(4) *P*-type semiconductor is also electrically neutral (not positively charged)

(5) Impurity is called Acceptor impurity.

(6) Acceptor energy level lies just above the valence band.

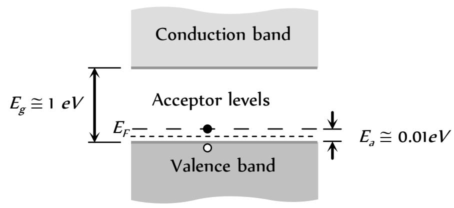


Fig. 27.14

Density of Charge Carriers

Due to thermal collisions, an electron can take up or release energy. Thus, occasionally a valence electron takes up energy and the bond is broken. The electron goes to the conduction band and a hole is created. And occasionally, an electron from the conduction band loses some energy, comes to the valence band and fills up a hole. Thus, new electron-hole pairs are formed as well as old electron-hole disappear. A steady-state situation is reached and the number of electron-hole pairs takes a nearly constant value. For silicon at room temperature (300 *K*), the number of these pairs is about 7×10^{15} *m*. For germanium, this number is about 6×10^{19} /*m*.

Table 27. 4 : Densities of charge carriers

Material	Type	Density of conduction electrons (m^{-3})	Density of holes (m^{-3})
Copper	Conductor	9×10^{28}	0
Silicon	Intrinsic semiconductor	7×10^{15}	7×10^{15}
Silicon doped with phosphorus (1 part in 10^6)	*N*-type semiconductor	5×10^{22}	$1 \quad \times 10^9$

Silicon doped with aluminium (1 part in 10^6	*P*-type semiconductor	1×10^9	5×10^{22}

Conductivity of Semiconductor

(1) In intrinsic semiconductors $n_e = n_h$. Both electron and holes contributes in current conduction.

(2) When some potential difference is applied across a piece of intrinsic semiconductor current flows in it due to both electron and holes *i.e.* $i = i_e + i_h \Rightarrow i = eA[n_e v_e + n_h v_h]$

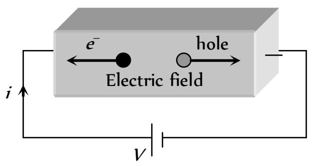


Fig. 27.15

(3) As we know $\sigma = \dfrac{J}{E} = \dfrac{i}{AE}$. Hence conductivity of semiconductor

$\sigma = e[n_e \mu_e + n_h \mu_h]$; where v_e = drift velocity of electron, v_h = drift velocity of holes, E = Applied electric field $\mu_e = \dfrac{v_e}{E}$ = mobility of electron and $\mu_h = \dfrac{v_h}{E}$ = mobility of holes

(4) Motion of electrons in the conduction band and of holes the valence band under the action of electric field is shown below

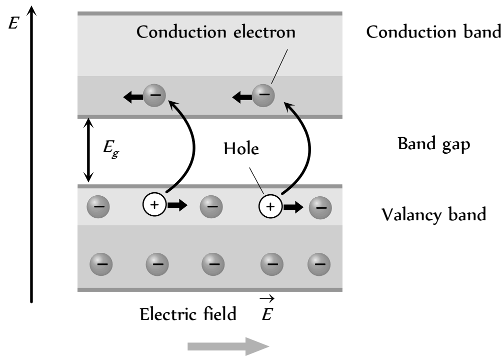


Fig. 27.16

(5) At absolute zero temperature (0 *K*) conduction band of semiconductor is completely empty *i.e.* $\sigma = 0$. Hence the semiconductor behaves as an insulator.

P-N Junction Diode

When a *P*-type semiconductor is suitably joined to an *N*-type semiconductor, then resulting arrangement is called *P-N* junction or *P-N* junction diode

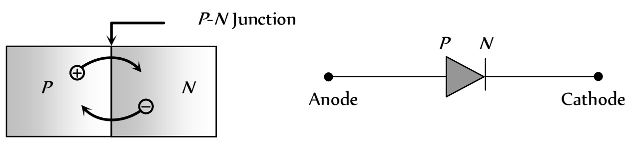


Fig. 27.17

(1) **Depletion region :** On account of difference in concentration of charge carrier in the two sections of *P-N* junction, the electrons from *N*-region diffuse through the junction into *P*-region and the hole from *P* region diffuse into *N*-region.

Due to diffusion, neutrality of both *N* and *P*-type semiconductor is disturbed, a layer of negative charged ions appear near the junction in the *P*-crystal and a layer of positive ions appears near the junction in *N*-crystal. This layer is called depletion layer

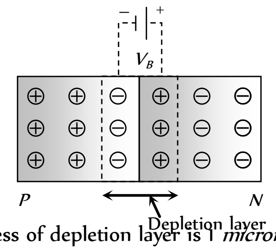


Fig. 27.18

(i) The thickness of depletion layer is 1 *micron* = 10^{-6} *m*.

(ii) Width of depletion layer $\propto \dfrac{1}{\text{Dopping}}$

(iii) Depletion is directly proportional to temperature.

(iv) The *P-N* junction diode is equivalent to capacitor in which the depletion layer acts as a dielectric.

(2) **Potential barrier :** The potential difference created across the *P-N* junction due to the diffusion of electron and holes is called potential barrier.

For *Ge* $V_B = 0.3V$ and for silicon $V_B = 0.7V$

On the average the potential barrier in *P-N* junction is ~ 0.5 *V* and the width of depletion region ~ 10^{-6} *m*.

So the barrier electric field $E = \dfrac{V}{d} = \dfrac{0.5}{10^{-6}} = 5 \times 10^5\ V/m$

(3) **Some important graphs**

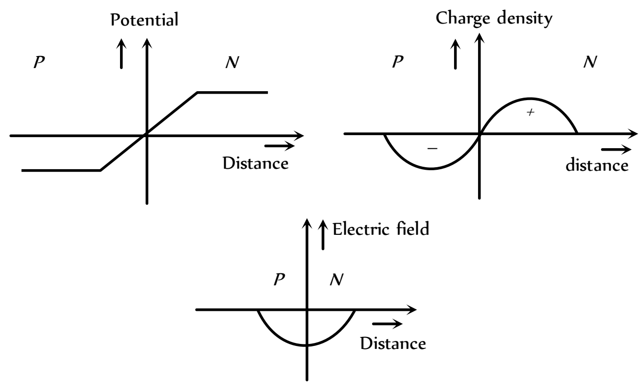


Fig. 27.19

(4) **Diffusion and drift current :** Because of concentration difference holes/electron try to diffuse from their side to other side. Only those holes/electrons crosses the junction, which have high kinetic energy. This diffusion results in an electric current from the *P*-side to the *N*-side known as diffusion current (i_{df})

As electron hole pair (because of thermal collisions) are continuously created in the depletion region. There is a regular flow of electrons towards the *N*-side and of holes towards the *P*-side. This makes a current from the *N*-side to the *P*-side. This current is called the drift current (i_{dr}).

Biasing

It means the way of connecting emf source to *P-N* junction diode. It is of following two types

(1) **Forward biasing :** Positive terminal of the battery is connected to the *P*-crystal and negative terminal of the battery is connected to *N*-crystal

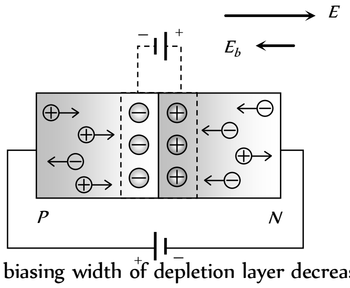


Fig. 27.20

(i) In forward biasing width of depletion layer decreases

(ii) In forward biasing resistance offered $R_{forward} \approx 10\Omega$ - 25Ω

(iii) Forward bias opposes the potential barrier and for $V > V_{\cdot}$ a forward current is set up across the junction.

(iv) The current is given by $i = i_s(e^{eV/kT} - 1)$; where

$i_s =$ Saturation current, In the exponent $e = 1.6 \times 10^{-}$ *C*,

k = Boltzmann's constant

(v) Cut-in (Knee) voltage : The voltage at which the current starts to increase rapidily. For *Ge* it is 0.3 *V* and for *Si* it is 0.7 *V*.

(vi) *df* – diffusion

dr – drift

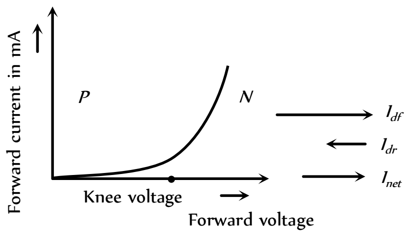


Fig. 27.21

(2) **Reverse biasing :** Positive terminal of the battery is connected to the *N*-crystal and negative terminal of the battery is connected to *P*-crystal

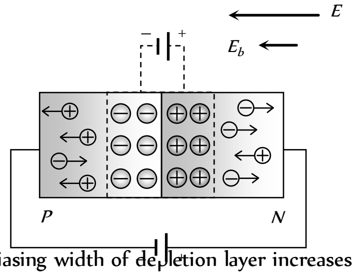


Fig. 27.22

(i) In reverse biasing width of depletion layer increases

(ii) In reverse biasing resistance offered $R_{Reverse} \approx 10\Omega$

(iii) Reverse bias supports the potential barrier and no current flows across the junction due to the diffusion of the majority carriers.

(A very small reverse currents may exist in the circuit due to the drifting of minority carriers across the junction)

(iv) Break down voltage : Reverse voltage at which break down of semiconductor occurs. For *Ge* it is 25 *V* and for *Si* it is 35 *V*.

(v)

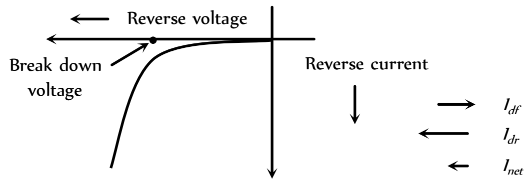


Fig. 27.23

Reverse Breakdown

If the reverse biased voltage is too high, then breakdown of *P-N* junction diode occurs. It is of following two types

(1) **Zener breakdown :** When reverse bias is increased the electric field across the junction also increases. At some stage the electric field becomes so high that it breaks the covalent bonds creating electron, hole pairs. Thus a large number of carriers are generated. This causes a large current to flow. This mechanism is known as **Zener breakdown.**

(2) **Avalanche breakdown :** At high reverse voltage, due to high electric field, the minority charge carriers, while crossing the junction acquires very high velocities. These by collision breaks down the covalent bonds, generating more carriers. A chain reaction is established, giving rise to high current. This mechanism is called **avalanche breakdown.**

Special Purpose Diodes

(1) **Zener diode :** It is a highly doped *p-n* junction which is not damaged by high reverse current. It can operate continuously, without being damaged in the region of reverse background voltage. In the forward bias, the zener diode acts as ordinary diode. It can be used as voltage regulator

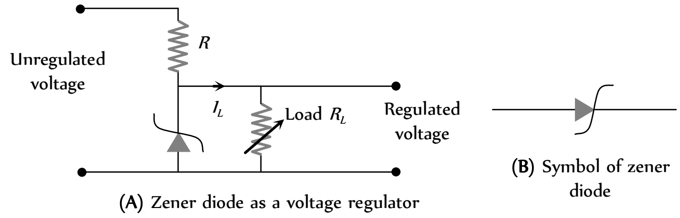


Fig. 27.24

(2) **Light emitting diode (LED :** Specially designed diodes, which give out light radiations when forward biases. LED'S are made of *GaAsp*, *Gap etc*.

These are forward biased *P-N*-junctions which emits spontaneous radiation.

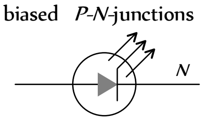


Fig. 27.25

(3) **Photo diode:** Photodiode is a special type of photo-detector. Suppose an optical photons of frequency ν is incident on a semiconductor, such that its energy is greater than the band gap of the semiconductor (*i.e.* $h\nu > E_{\cdot}$) This photon will excite an electron from the valence band to the conduction band leaving a vacancy or hole in the valence band.

Which obviously increase the conductivity of the semiconductor. Therefore, by measuring the change in the conductance (or resistance) of the semiconductor, one can measure the intensity of the optical signal.

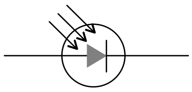


Fig. 27.26

(4) **Solar cells :** It is based on the photovoltic effect. One of the semiconductor region is made so thin that the light incident on it reaches the *P-N*-junction and gets absorbed. It converts solar energy into electrical energy.

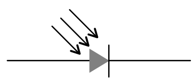


Fig. 27.27

P-N Junction Diode as a Rectifier

Rectifier is a circuit which converts *ac* to unidirectional pulsating output. In other words it converts *ac* to *dc*. It is of following two types

(1) **Half wave rectifier :** When the *P-N* junction diode rectifies half of the *ac* wave, it is called half wave rectifier

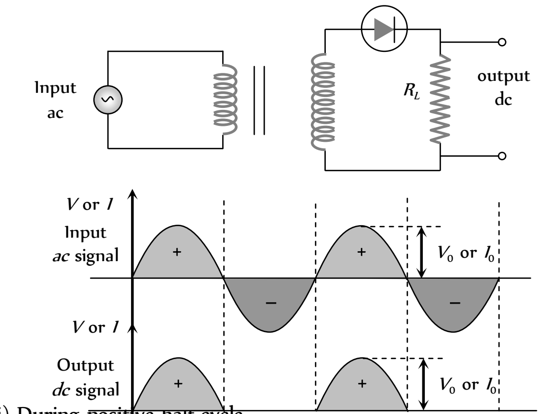


Fig. 27.28

(i) During positive half cycle

Diode $\longrightarrow$ forward biased

Output signal $\longrightarrow$ obtained

(ii) During negative half cycle

Diode $\longrightarrow$ reverse biased

Output signal $\longrightarrow$ not obtained

(iii) Output voltage is obtained across the load resistance R_L. It is not constant but pulsating (mixture of *ac* and *dc*) in nature .

(iv) Average output in one cycle

$$I_{dc} = \frac{I_0}{\pi} \text{ and } V_{dc} = \frac{V_0}{\pi};\ I_0 = \frac{V_0}{r_f + R_L}$$

(r_f = forward biased resistance)

(v) r.m.s. output : $I_{rms} = \frac{I_0}{2}, V_{rms} = \frac{V_0}{2}$

(vi) The ratio of the effective alternating component of the output voltage or current to the *dc* component is known as ripple factor.

$$r = \frac{I_{ac}}{I_{dc}} = \left[\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1\right]^{1/2} = 1.21$$

(vii) Peak inverse voltage (PIV) : The maximum reverse biased voltage that can be applied before commoncement of Zener region is called the PIV. When diode is not conducting PIV across it = V_0

(viii) Efficiency : It is given by $\%\,\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{40.6}{1 + \frac{r_f}{R_L}}$

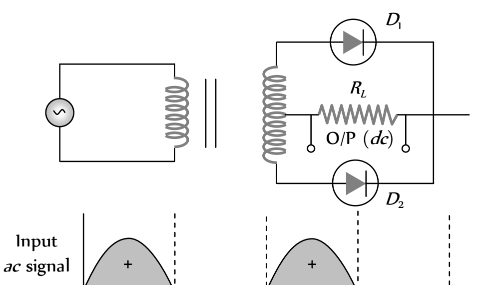
If $R_L >> r_f$ then η = 40.6%

If $R_L = r_f$ then η = 20.3%

(ix) Form factor = $\frac{I_{rms}}{I_{dc}} = \frac{\pi}{2} = 1.57$

(x) The ripple frequency (ω) for half wave rectifier is same as that of *ac*.

(2) **Full wave rectifier :** It rectifies both halves of *ac* input signal.



(i) During positive half cycle

Diode : D_1 forward biased

$D_2 \longrightarrow$ reverse biased

Output signal $\longrightarrow$ obtained due to D_1 only

(ii) During negative half cycle

Diode : $D_1 \longrightarrow$ reverse biased

$D_2 \longrightarrow$ forward biased

Output signal $\longrightarrow$ obtained due to D_2 only

(iii) Fluctuating *dc* $\longrightarrow$ Filter $\longrightarrow$ constant *dc*.

(iv) Output voltage is obtained across the load resistance R_L. It is not constant but pulsating in nature.

(v) Average output : $V_{av} = \frac{2V_0}{\pi}, I_{av} = \frac{2I_0}{\pi}$

(vi) *r.m.s.* output : $V_{rms} = \frac{V_0}{\sqrt{2}}, I_{rms} = \frac{I_0}{\sqrt{2}}$

(vii) Ripple factor : r = 0.48 = 48%

(viii) Ripple frequency : The ripple frequency of full wave rectifier = 2 × (Frequency of input *ac*)

(ix) Peak inverse voltage (PIV) : It's value is $2V_0$

(x) Efficiency : $\eta_{\%} = \frac{81.2}{1 + \frac{r_f}{R_L}}$ for $r_f << R_L$, $\eta = 81.2\%$

(3) **Full wave bridge rectifier :** Four diodes D_1, D_2, D_3 and D_4 are used in the circuit.

During positive half cycle D_1 and D_3 are forward biased and D_2 and D_4 are reverse biased

During negative half cycle D_2 and D_4 are forward biased and D_1 and D_3 are reverse biased

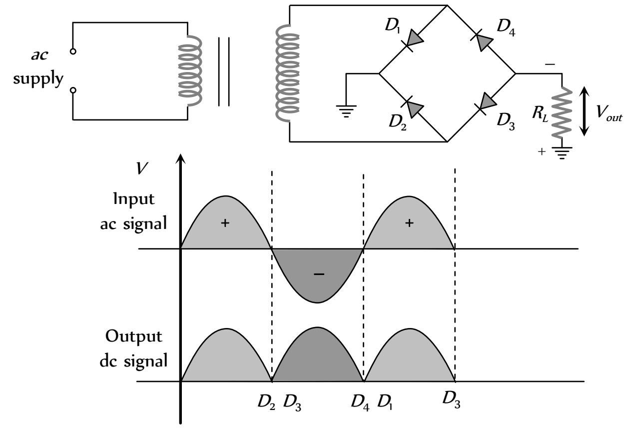


Fig. 27.30

Transistor

(1) The name of this electronic device is derived from it's fundamental action transfer resistor.

(2) Transistor does not need any heater or hot filament, transistor is small in size and light in weight.

(3) Transistor in general is known as bipolar junction transistor.

(4) Transistor is a current operated device.

(5) It consists of three main regions

(i) **Emitter (*E*) :** It provides majority charge carriers by which current flows in the transistor. Therefore the emitter semiconductor is heavily doped.

(ii) **Base (*B*) :** The based region is lightly doped and thin.

(iii) **Collector (*C*) :** The size of collector region is larger than the two other regions.

(6) Junction transistor are of two types :

(i) *NPN* transistor : It is formed by sandwiching a thin layer of *P*-type semiconductor between two *N*-type semiconductors

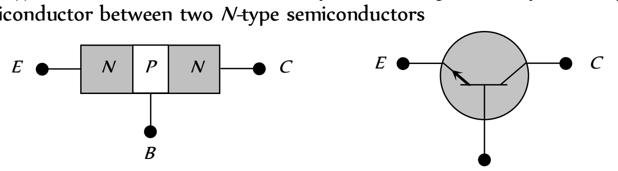


Fig. 27.31

In *NPN* transistor electrons are majority charge carriers and flow from emitter to base.

(ii) *PNP* transistor : It is formed by sandwiching a thin layer of *N*-type semiconductor between two *P*-type semiconductor

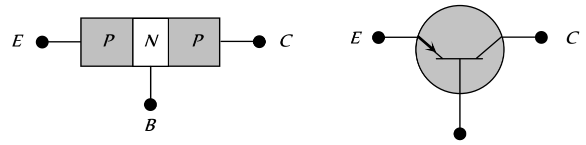


Fig. 27.32

In *PNP* transistor holes are majority charge carriers and flow from emitter to base.

In the symbols of both *NPN* and *PNP* transistor, arrow indicates the direction of conventional current.

Working of Transistor

(1) There are four possible ways of biasing the two *P-N* junctions (emitter junction and collector junction) of transistor.

(i) Active mode : Also known as linear mode operation.

(ii) Saturation mode : Maximum collector current flows and transistor acts as a closed switch from collector to emitter terminals.

(iii) Cut-off mode : Denotes operation like an open switch where only leakage current flows.

(iv) Inverse mode : The emitter and collector are inter changed.

Table 27.5 : Different modes of operation of a transistor

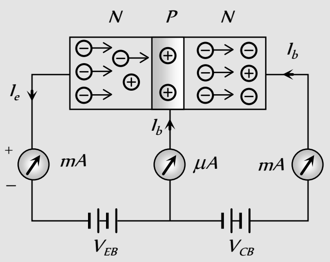
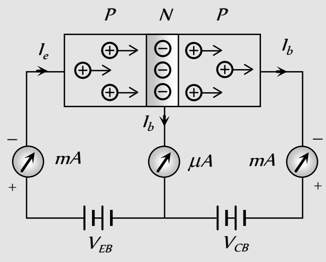
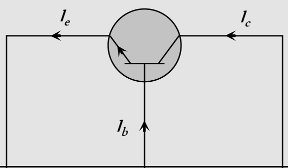
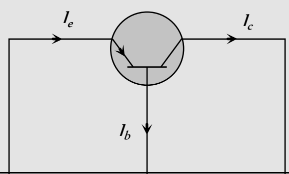
Operating mode	Emitter base bias	Collector base bias
Active	Forward	Reverse
Saturation	forward	Forward
Cut off	Reverse	Reverse
Inverse	Reverse	Forward

(2) A transistor is mostly used in the active region of operation *i.e.* emitter base junction is forward biased and collector base junction is reverse biased.

(3) From the operation of junction transistor it is found that when the current in emitter circuit changes. There is corresponding change in collector current.

(4) In each state of the transistor there is an input port and an output port. In general each electrical quantity (V or I) obtained at the output is controlled by the input.

Table 27.6 : Circuit diagram of *PNP*/*NPN* transistor

NPN – transistor	*PNP* – transistor
	
	
5% emitter electron combine with the holes in the base region resulting in small base current. Remaining 95% electrons enter the collector region.	5% emitter holes combine with the electrons in the base region resulting in small base current. Remaining 95% holes enter the collector region.
$I_e > I_c$, and $I_e = I_b + I_c$	$I_e > I_c$, and $I_e = I_b + I_c$

Transistor Configurations

A transistor can be connected in a circuit in the following three different configurations.

Common base (CB), Common emitter (CE) and Common collector (CC) configuration.

(1) **CB configurations :** Base is common to both emitter and collector .

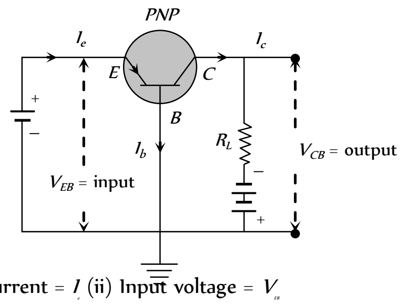


Fig. 27.33

(i) Input current = I_e (ii) Input voltage = V_{eb}

(iii) Output voltage = V_{cb} (iv) Output current = I_c

With small increase in emitter-base voltage V_{eb}, the emitter current I_e increases rapidly due to small input resistance.

(v) **Input characteristics :** If V_{cb} = constant, curve between I_e and V_{eb} is known as input characteristics. It is also known as emitter characteristics

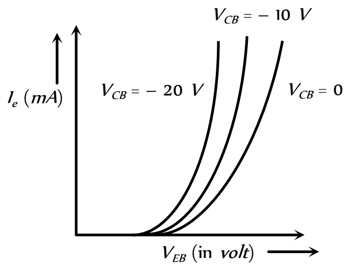


Fig. 27.34

Input characteristics of *NPN* transistor are also similar to the above figure but I_e and V_{eb} both are negative and V_{cb} is positive.

Dynamic input resistance of a transistor is given by

$$R_i = \left(\frac{\Delta V_{EB}}{\Delta I_e}\right)_{V_{CB}=\text{constant}} \qquad \{R_i \text{ is of the order of } 100\ \Omega\}$$

(vi) **Output characteristics :** Taking the emitter current i_e constant, the curve drawn between I_c and V_{cb} are known as output characteristics of *CB* configuration.

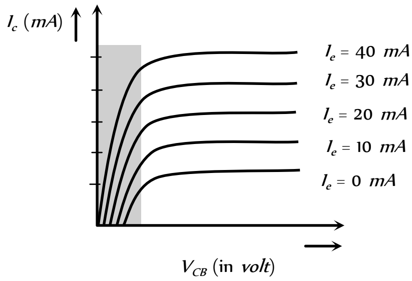


Fig. 27.35

Dynamic output resistance $R_o = \left(\frac{\Delta V_{CB}}{\Delta i_C}\right)_{i_e=\text{constant}}$

(2) **CE configurations :** Emitter is common to both base and collector.

The graphs between voltages and currents when emitter of a transistor is common to input and output circuits are known as CE characteristics of a transistor.

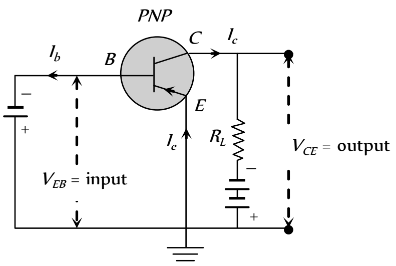


Fig. 27.36

Input characteristics : Input characteristic curve is drawn between base current I_b and emitter base voltage V_{eb}, at constant collector emitter voltage V_{ce}.

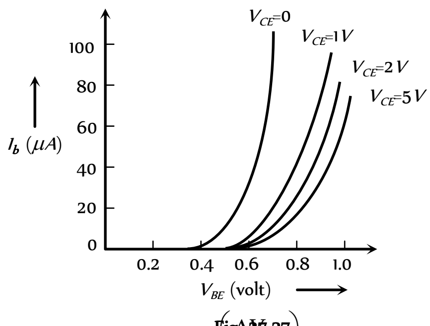


Fig. 27.37

Dynamic input resistance $R_i = \left(\frac{\Delta V_{BE}}{\Delta I_B}\right)_{V_{CE}\to\text{constant}}$

Output characteristics : Variation of collector current I_c with V_{ce} can be noticed for V_{ce} between 0 to 1 *V* only. The value of V_{ce} up to which the I_c changes with V_{ce} is called knee voltage. The transistor are operated in the region above knee voltage.

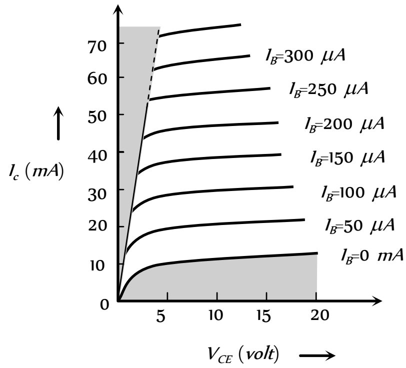


Fig. 27.38

Dynamic output resistance $R_0 = \left(\frac{\Delta V_{CE}}{\Delta I_C}\right)_{I_B\to\text{constant}}$

Field-Effect Transistor

The low input impedance of the junction transistor is a handicap in certain applications. In addition, it is difficult to incorporate large numbers of them in an integrated circuit and they consume relatively large amounts of power. The field-effect transistor (FET) lacks these disadvantages and is widely used today although slower in operation than junction transistors.

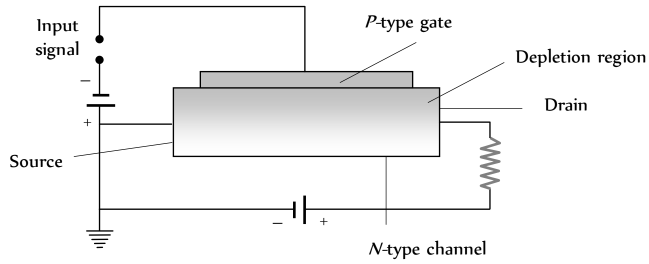


Fig. 27.39

An *n*-channel FET consists of a block of *N*-type material with contacts at each end together with a strip of *P*-type material on one side that is called the gate. When connected as shown, electrons move from the source terminal to the drain terminal through the *N*-type channel. the *PN* junction is given a reverse bias, and as a result both the *N* and *P* materials near the junction are depleted on charge carriers. The higher the reverse potential on the gate, the larger the depleted region in the channel and the fewer the electrons available to carry the current. Thus the gate voltage controls the channel current. Very little current passes through the gate circuit owing to the reverse bias, and the result is an extremely high input impedance. FET is uni-polar.

Transistor as an Amplifier

A device which increases the amplitude of the input signal is called amplifier.

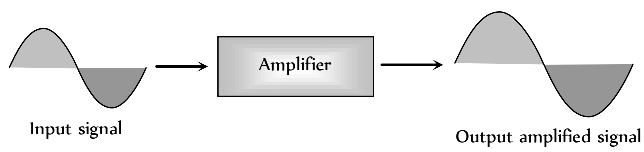


Fig. 27.40

The transistor can be used as an amplifier in the following three configuration

(i) CB amplifier (ii) CE amplifier (iii) CC amplifier

(1) ***NPN* transistor as CB amplifier**

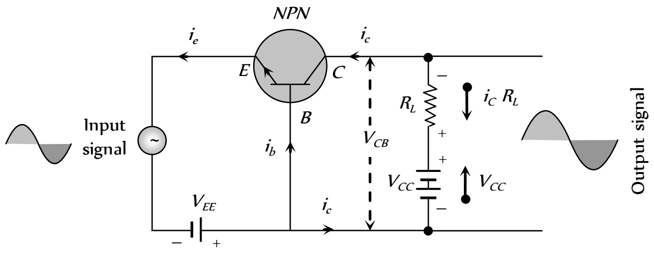


Fig. 27.41

(i) $i_e = i_b + i_C$; i_b = 5% of i_e and i_C = 95% of i_e

(ii) $V_{EE} < V_{CC}$

(iii) Net collector voltage $V_{CB} = V_{CC} - i_C R_L$

When the input signal (signal to be amplified) is fed to the emitter base circuit, it will change the emitter voltage and hence emitter current. This in turn will change the collector current (i_c). This will vary the collector voltage V_{CB}. This variation of V_{CB} will appear as an amplified output.

(iv) Input and output signals are in same phase

(2) ***NPN* transistor as *CE* amplifier**

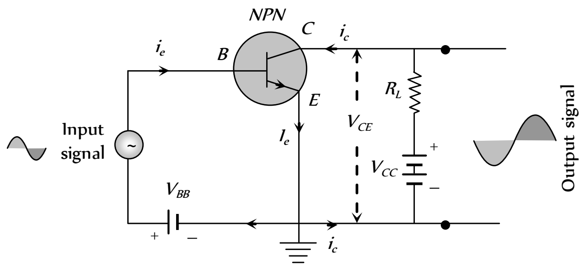


Fig. 27.42

(i) $i_e = i_b + i_C$; i_b = 5% of i_e and i_C = 95% of i_e

(ii) $V_{CC} > V_{BB}$

(iii) Net collector voltage $V_{CE} = V_{CC} - i_C R_L$

(iv) Input and output signals are 180° out of phase.

Different Gains in CE/CB Amplifiers

(1) **Transistor as CB amplifier**

(i) *ac* current gain $\alpha_{ac} = \dfrac{\text{Small change in collector current } (\Delta i_c)}{\text{Small change in collector current } (\Delta i_e)}$

V_{B} (constant)

(ii) *dc* current gain $\alpha_{dc}(\text{or}\,\alpha) = \dfrac{\text{Collector current}(i_c)}{\text{Emitter current }(i_e)}$

valve of α_{dc} lies between 0.95 to 0.99

(iii) Voltage gain $A_v = \dfrac{\text{Change in output voltage}(\Delta V_o)}{\text{Change in input voltage}(\Delta V_i)}$

$\Rightarrow$ $A_v = \alpha_{ac} \times$ Resistance gain

(iv) $\text{Power gain} = \dfrac{\text{Change in output power}(\Delta P_o)}{\text{Change in input power}(\Delta P_c)}$

$\Rightarrow \text{Power gain} = \alpha_{ac}^2 \times \text{Resistance gain}$

(2) **Transistor as CE amplifier**

(i) *ac* current gain $\beta_{ac} = \left(\dfrac{\Delta i_c}{\Delta i_b}\right)$ V_{CE} = constant

(ii) *dc* current gain $\beta_{dc} = \dfrac{i_c}{i_b}$

(iii) Voltage gain : $A_v = \dfrac{\Delta V_o}{\Delta V_i} = \beta_{ac} \times \text{Resistance gain}$

(iv) $\text{Power gain} = \dfrac{\Delta P_o}{\Delta P_i} = \beta_{ac}^2 \times \text{Resistance gain}$

(v) Trans conductance (g_m) : The ratio of the change in collector current to the change in emitter base voltage is called trans conductance. *i.e.* $g_m = \dfrac{\Delta i_c}{\Delta V_{EB}}$. Also $g_m = \dfrac{A_V}{R_L}$; R_L = Load resistance

(3) **Relation between α and β** : $\beta = \dfrac{\alpha}{1-\alpha}$ or $\alpha = \dfrac{\beta}{1+\beta}$

Transistor as an Oscillator

(1) It is defined as a circuit which generates an *ac* output signal without any externally applied input signal.

Audio frequency oscillators generates signals of frequencies ranging from a few *Hz* to 20 *kHz* and radio frequency oscillators have a range from few *kHz* to *MHz*.

(2) In an oscillator the frequency, waveform, and magnitude of *ac* power generated is controlled by circuit itself.

(3) An oscillator may be considered as amplifier which provides it's own input signal.

(4) The essential of a transistor oscillator are

(i) **Tank circuit :** Parallel combination of *L* and *C*. This network resonates at a frequency $\nu_0 = \dfrac{1}{2\pi}\sqrt{\dfrac{1}{LC}}$.

(ii) **Amplifier :** It receives *dc* power from the battery and converts into *ac* power.

The amplifier increases the strength of oscillations.

(iii) **Feed back circuit :** This circuit supplies a part of the collector energy to the tank circuit.

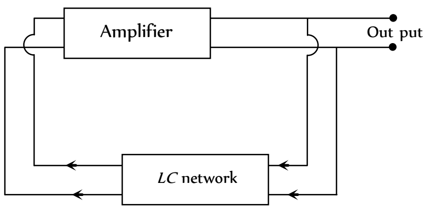


Fig. 27.43

(5) A basic common-emitter *NPN* oscillator is shown in the figure.

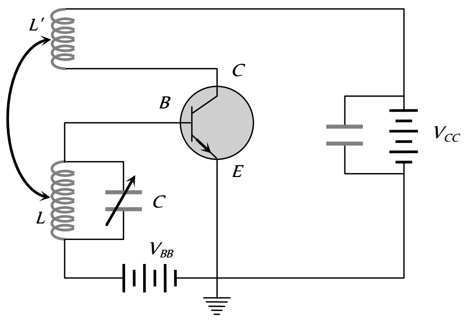


Fig. 27.44

A tank circuit (L-C circuit) is connected in the base-emitter circuit, in which the capacitance C is kept variable. By changing C oscillations of a desired frequency can be obtained. An inductance coil L' connected in the collector-emitter circuit is coupled to coil L.

On completion of the circuit electrical oscillations are developed in the tank circuit. The circuit amplifies these oscillations. A part of the amplifies signal in the collector circuit is fed back in the base circuit by the coupling between L and L'. Due to this feed back amplitude of oscillation builds up till power dissipation in the oscillatory circuit becomes equal to power fed-back. In this state the amplitude of oscillations becomes constant.

The oscillations can be transferred to an external circuit by mutual induction in a coil connected in that circuit.

(6) **Need for positive feedback :** The oscillations are damped due to the presence of some inherent electrical resistance in the circuit. Consequently, the amplitude of oscillations decreases rapidly and the oscillations ultimately stop. Such oscillations are of little practical importance. In order to obtain oscillations of constant amplitude, we make an arrangement for regenerative or positive feedback from the output circuit to the input circuit so that the losses in the circuit can be compensated.

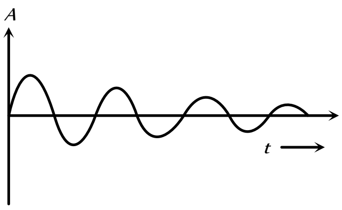


Fig. 27.45

Table 27.7: Comparison between CB, CE and CC amplifier

Characteristic	Amplifier		
	CB	CE	CC
Input resistance (R_i)	≈ 50 to 200 Ω low	≈ 1 to 2 $k\Omega$ medium	≈ 150 – 800 $k\Omega$ high
Output resistance (R_o)	≈ 1 – 2 $k\Omega$ high	≈ 50 $k\Omega$ medium	≈ $k\Omega$ low
Current gain	0.8 – 0.9 low	20 – 200 high	20 – 200 high
Voltage gain	Medium	High	Low
Power gain	Medium	High	Low
Phase difference between input and output voltages	Zero	180°	Zero
Used as amplifier for	current	Power	Voltage

Digital Electronics

Decimal and Binary Number System

(1) **Decimal number system :** In a decimal number system, we have ten digits *i.e.* 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

A decimal number system has a base of ten (10)

e.g. 1971 = 1000 + 900 + 70 + 1

$= 1 \times 10^3 + 9 \times 10^2 + 7 \times 10^1 + 1 \times 10^0$

MSD LSD

LSD = Least significant digit

MSD = Most significant digit

(2) **Binary number system :** A number system which has only two digits *i.e.* 0 (Low) and 1 (High) is known as binary system. The base of binary number system is 2.

(i) Each digit in binary system is known as a bit and a group of bits is known as a byte.

(ii) The electrical circuit which operates only in these two state *i.e.* 1 (On or High) and 0 (*i.e.* Off or Low) are known as digital circuits.

Table 27. 8 : Different names for the digital signals

State Code	1	0
Name for the State	On	Off
	Up	Down
	Close	Open
	Excited	Unexcited
	True	False
	Pulse	No pulse
	High	Low
	Yes	No

(3) **Decimal to binary conversion**

(i) Divide the given decimal number by 2 and the successive quotients by 2 till the quotient becomes zero.

(ii) The sequence of remainders obtained during divisions gives the binary equivalent of decimal number.

(iii) the most significant digit (or bit) of the binary number so obtained is the last remainder and the least significant digit (or bit) is the first remainder obtained during the division.

For Example **:** Binary equivalence of 61

2	61	Remainder
2	30	1 LSD
2	15	0

2	7	1
2	3	1
2	1	1
	0	1 MSD

$\Rightarrow (61)_{10} = (111101)_2$

(4) **Binary to decimal conversion :** The least significant digit in the binary number is the coefficient of 2 with power zero. As we move towards the left side of LSD, the power of 2 goes on increasing.

For Example **:** $(11111100101)_2 = 1 \times 2^{10} + 1 \times 2^9 + 1 \times 2^8 + 1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 2021$

Voltage Signal

(1) **Analogue voltage signal :** The signal which represents the continuous variation of voltage with time is known as analogue voltage signal

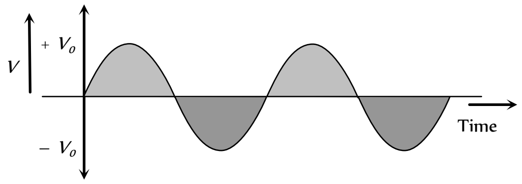


Fig. 27.46

(2) **Digital voltage signal :** The signal which has only two values. *i.e.* either a constant high value of voltage or zero value is called digital voltage signal

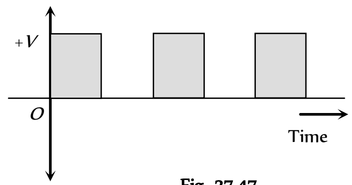


Fig. 27.47

Boolean Algebra

(1) In Boolean algebra only two states of variables (0 and 1) are allowed.

(2) The variables (A, B, C ....) of Boolean Algebra are subjected to three operations.

(i) **OR Operation :** Represented by (+) sign

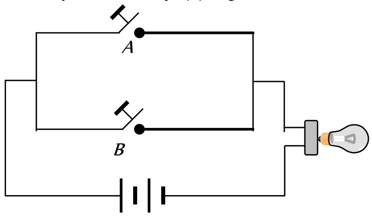


Fig. 27.48

Boolean expression $Y = A + B$

When switch A or B is closed – Bulb glows

(ii) AND **Operation :** Represented by (·) sign

Boolean expression $Y = A \cdot B$

When switches A and B both are closed – Bulb glows

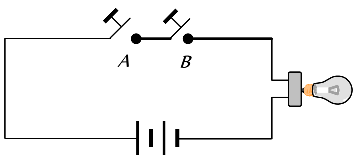
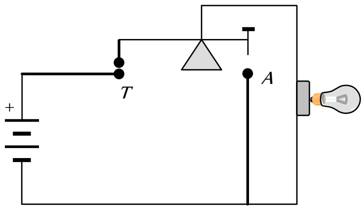


Fig. 27.49

(iii) **NOT Operation :** Represented by bar over the variables

Boolean expression $Y = \overline{A}$



A OFF → Lamp ON

A ON → Contact at T is broken

→ Lamp OFF

Fig. 27.50

(3) **Basic Boolean postulates and laws**

(i) Boolean Postulates : $0 + A = A$, $1 \cdot A = A$,

$1 + A = 1$, $0 \cdot A = 0$,

$A + \overline{A} = 1$

(ii) Identity law : $A + A = A$, $A \cdot A = A$

(iii) Negation law : $\overline{\overline{A}} = A$

(iv) Commutative law : $A + B = B + A$, $A \cdot B = B \cdot A$

(v) Associative law : $(A+B) + C = A + (B+C)$,

$(A \cdot B) \cdot C = A \cdot (B \cdot C)$

(vi) Distributive law : $A \cdot (B+C) = A \cdot B + A \cdot C$

$(A + B) \cdot (A + C) = A + BC$

(vii) Absorption laws : $A + A \cdot B = A$, $A \cdot (A + B) = A$

$\overline{A} \cdot (A + B) = \overline{A} \cdot B$

(viii) Boolean identities : $A + \overline{A}B = A + B$, $A(\overline{A} + B) = AB$,

$A + BC = (A + B)(A + C)$, $(\overline{A} + B) \cdot (A + C) = \overline{A}C + AB$

(ix) **De Morgan's theorem** : It states that the complement of the whole sum is equal to the product of individual complements and vice versa *i.e.* $\overline{A + B} = \overline{A} \cdot \overline{B}$ and $\overline{A \cdot B} = \overline{A} + \overline{B}$

Logic Gates and Truth Table

(1) **Logic gate :** The digital circuit that can be analysed with the help of Boolean algebra is called logic gate or logic circuit. A logic gate has two or more inputs but only one output.

There are primarily three logic gates namely the OR gate, the AND gate and the NOT gate.

(2) **Truth table :** The operation of a logic gate or circuit can be represented in a table which contains all possible inputs and their corresponding outputs is called the truth table. To write the truth table we use binary digits 1 and 0.

The 'OR' Gate

(1) It has two inputs (A and B) and only one output (Y)

(2) Boolean expression is $Y = A + B$ and is read as "Y equals A **OR** B"

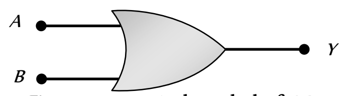
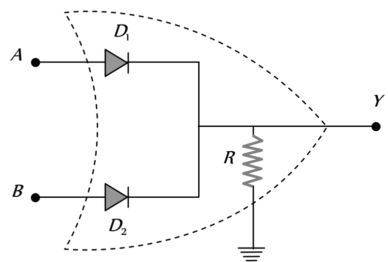


Fig. 27.51 : Logical symbol of OR gate

(3) **Realization of OR gate**



(i) $A = 0$, $B = 0$

Both diodes D_1 and D_2 do not conduct and hence $Y = 0$

(ii) $A = 0$, $B = 1$

D_1 = Does not conducts, D_2 = Conducts, hence $Y = 1$

(iii) $A = 1$, $B = 0$

D_1 = Conducts, D_2 = Does not conduct, hence $Y = 1$

(iv) $A = 1$, $B = 1$

Both D_1 and D_2 conducts, hence $Y = 1$

(4) **Truth table for 'OR' gate**

A	*B*	*Y = A + B*
0	0	0
0	1	1
1	0	1
1	1	1

The 'AND' Gate

(1) It has two inputs (A and B) and only one output (Y)

(2) Boolean expression is $Y = A \cdot B$ is read as " Y equals A AND B"

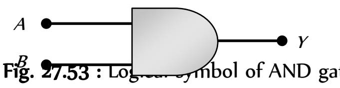


Fig. 27.53 : Logical symbol of AND gate

(3) **Realization of AND gate**

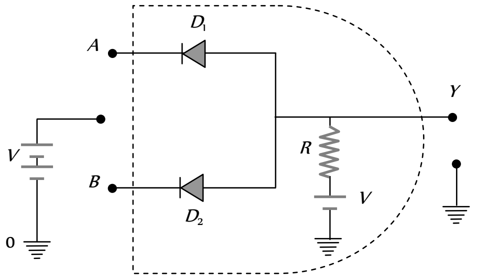


Fig. 27.54

(i) $A = 0$, $B = 0$

The voltage supply through R is forward biasing diodes D_1 and D_2 (offers low resistance) the voltage V would drop across R

The output voltage at Y = the voltage across diode = 0

(ii) $A = 0$, $B = 1$

D_1 = conducts, D_2 = Not Conducts

the out voltage at Y= The voltage across the diode (D_1) =0

(iii) $A = 1$, $B = 0$

D_1 = Conducts, D_2 = Not conducts

the out voltage at Y= The voltage across the diode (D_2) =0

iv) $A = 1$, $B = 1$

None of the diode conducts

the out voltage at Y= Battery voltage =1

(4) **Truth table for 'AND' gate**

A	*B*	*Y = A . B*
0	0	0
0	1	0
1	0	0
1	1	1

The 'NOT' Gate

(1) It has only one input and only one output.

(2) Boolean expression is $Y = \overline{A}$ and is read as "y equals not A"

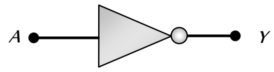


Fig 27.55 : Logical symbol of NOT gate

(3) **Realization of NOT gate :** The transistor is so biased that the collector voltage $V_{cc} = V$ (Voltage corresponding to 1 state)

The resistors R and R_1 are so chosen that if the input is low *i.e.* *O*, the transistor is in the cut off and hence the voltage appearing at the output will be the same as applied V. Hence $Y = V$ (or state 1)

If the input is high, the transistor current is in saturation and the net voltage at the output Y is 0 (in state 0)

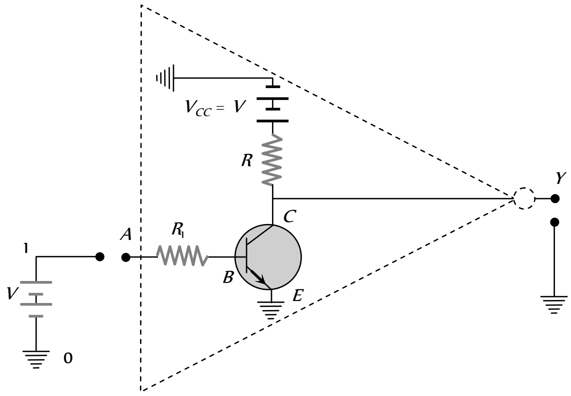


Fig. 27.56

(4) **Truth table for NOT gate :**

A	$Y = \overline{A}$
0	1
1	0

Combination of Logic Gates

(1) **The 'NAND' gate :** From 'AND' and 'NOT' gate

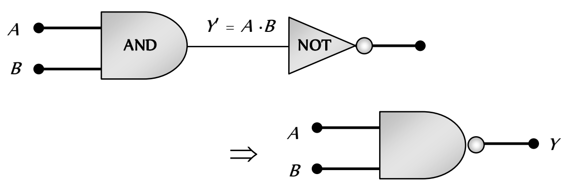


Fig. 27.57

Boolean expression and truth table : $Y = \overline{A \cdot B}$

A	B	$Y' = A \cdot B$	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

(2) **The 'NOR' gate :** From 'OR' and 'NOT' gate

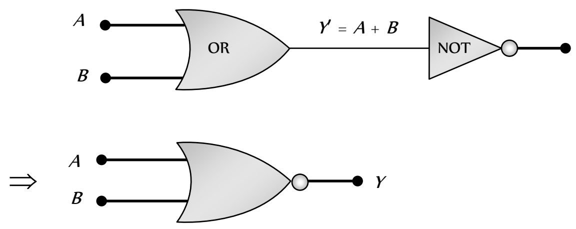


Fig. 27.58

Boolean expression and truth table : $Y = \overline{A + B}$

A	B	$Y' = A + B$	Y
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

(3) **The 'XOR' gate :** From 'NOT', 'AND' and 'OR' gate. Known as exclusive OR gate.

or

The logic gate which gives high output (*i.e.*, 1) if either input A or input B but not both are high (*i.e.* 1) is called exclusive OR gate or the XOR gate.

It may be noted that if both the inputs of the XOR gate are high, then the output is low (*i.e.*, 0).

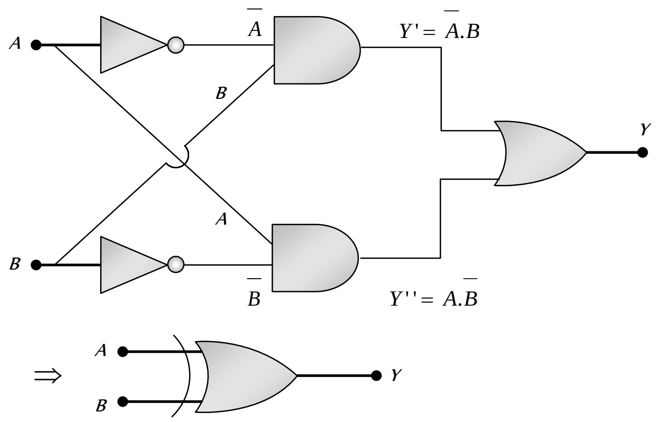


Fig. 27.59

Boolean expression and truth table : $Y = A \oplus B = \overline{A}B + A\overline{B}$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(4) **The exclusive nor (XNOR) gate**

XOR + NOT $\longrightarrow$ XNOR

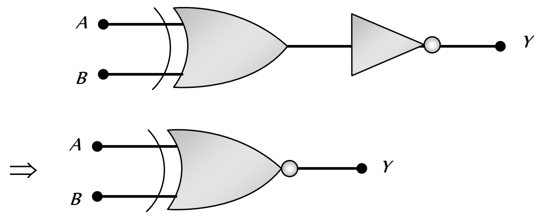


Fig. 27.60

Boolean expression : $Y = A \odot B = \overline{A}\,\overline{B} + AB$

Logic Gates Using 'NAND' Gate

The NAND gate is the building block of the digital electronics. All the logic gates like the OR, the AND and the NOT can be constructed from the NAND gates.

(1) **Construction of the 'NOT' gate from the 'NAND' gate**

(i) When both the inputs (A and B) of the NAND gate are joined together then it works as the NOT gate.

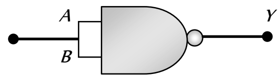


Fig. 27.61

(ii) Truth table and logic symbol

Input	Output
$A = B$	Y
0	1
1	0

(2) **Construction of the 'AND' gate from the 'NAND' gate**

(i) When the output of the NAND gate is given to the input of the NOT gate (made from the NAND gate), then the resultant logic gate works as the AND gate

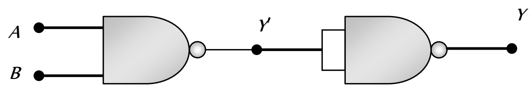


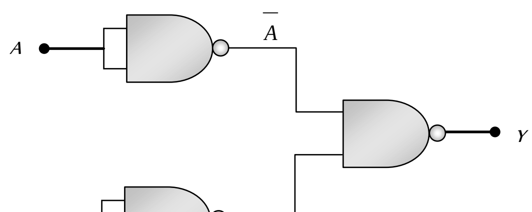
Fig. 27.62

(ii) Truth table and logic symbol

A	B	Y'	Y
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

(3) **Construction of the 'OR' gate by the 'NAND' gate**

(i) When the outputs of two NOT gates (obtained from the NAND gate) is given to the inputs of the NAND gate, the resultant logic gate works as the OR gate



(ii) Truth table and logic symbol

A	B	$\overline{A}$	$\overline{B}$	Y
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

Valve Electronics

Electron Emission from Metal

(1) Free electron in metal experiences a barrier on surface due to attractive Coulombian force.

(2) When kinetic energy of electron becomes greater than barrier potential energy (or binding energy E_b) then electron can come out of the surface of metal.

(3) **Fermi energy (E_f) :** Is the maximum possible energy possessed by free electron in metal at $0K$ temperature

(i) In this energy level, probability of finding electron is 50%.

(ii) This is a reference level and it is different for different metals.

(4) **Threshold energy (or work function W_0) :** Is the minimum energy required to take out an electron from the surface of metal. Also $W_0 = E_b - E_f$

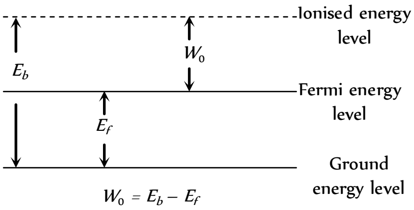


Fig. 27.64

Work function for different materials

$(W)_{\text{Pure tungsten}}$ = 4.5 eV

$(W)_{\text{Thoriated tungsten}}$ = 2.6 eV

$(W)_{\text{Oxide coated tungsten}}$ = 1 eV

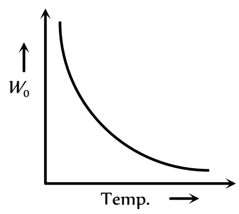


Fig. 27.65

(5) Four processes of electron emission from a metal are

(i) Thermionic emission

(ii) Photoelectric emission

(iii) Field emission

(iv) Secondary emission

Thermionic Emission

(1) The phenomenon of ejection of electrons from a metal surface by the application of heat is called thermionic emission and emitted electrons are called thermions and current flowing is called thermion current.

(2) Thermions have different velocities.

(3) This was discovered by Edison

(4) Richardson – Dushman equation for current density (*i.e.* electric current emitted per unit area of metal surface) is given as

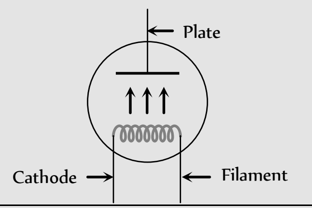
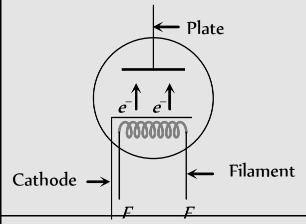
$$J = AT^2 e^{-W_0/kT} = AT^2 e^{-\frac{qV}{kT}} = AT^2 e^{-\frac{11600\,V}{T}}$$

where A = emission constant = 12×10^4 *amp/ m-K*, k = Boltzmann's constant, T = Absolute temp and W_0 = work function.

(5) The number of thermions emitted per second per unit area (J) depends upon following :

(i) $J \propto T^2$ (ii) $J \propto e^{-W_0}$

Table 27.9: Types of thermionic emitters

Directly heated emitter	Indirectly heated emitter
	
Cathode is directly heated by passing current.	Cathode is indirectly heated.
Thermionic current is less.	Thermionic current is more.
Energy consumption and life is small.	Energy consumption and life is more.

Vacuum Tubes and Thermionic Valves

(1) Those tubes in which electrons flows in vacuum are called vacuum tubes.

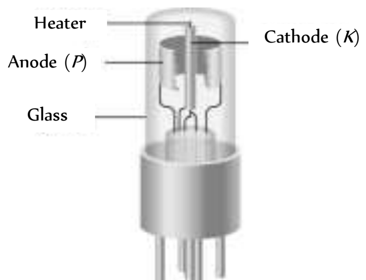
(2) These are also called valves because current flow in them is unidirectional.

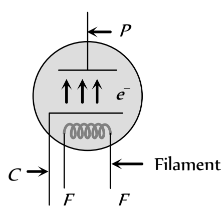
(3) Vacuum in vacuum tubes prevents the emission of secondary electrons and burning of heated filament (which will happen if we use air in place of vacuum)

(4) Every vacuum tube necessarily contains two electrodes out of which one is always electron emitter (cathode) and another one is electron collector (anode or plate).

(5) Depending upon the number of electrodes used the vacuum tubes are named as diode, triode, tetrode, pentode.... respectively, if the number of electrodes used are 2, 3, 4, 5..... respectively.

Diode Valve





(1) **Inventor :** Fleming

(2) **Principle :** Thermionic emission

(3) **Number of electrodes :** Two

(4) **Working :** When plate potential (V_p) is positive, plate current (i_p) flows in the circuit (because some emitted electrons reaches to plate). If $+V_p$ increases i_p also increases and finally becomes maximum (saturation).

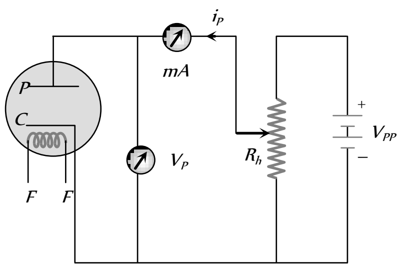


Fig. 27.67

(5) **Space charge :** If V_p is zero or negative, then electrons collect around the plate as a cloud which is called space charge. space charge decreases the emission of electrons from the cathode.

Characteristic Curves of a Diode

A graph represents the variation of i_p with V_p at a given filament current (i_f) is known as characteristic curve.

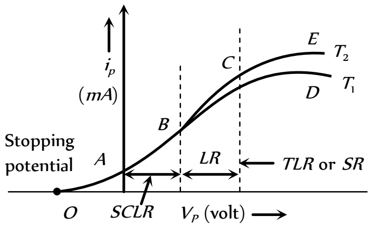


Fig. 27.68

The curve is not linear hence diode valve is a non-ohmic device.

(1) **Space charge limited region (SCLR)** : In this region current is space charge limited current.

Also $i_p \propto V_p^{3/2} \Rightarrow i_p = kV_p^{3/2}$; where k is a constant depending on metal as well as on the shape and area of the cathode. This is called child's law.

(2) **Linear region (LR) :** In this region $i_p \propto V_p$

(3) **Saturated region (SR) or temperature limited region (TLR) :** In this part, the current is independent of potential difference applied between the cathode and anode.

$i_p \neq f(V_p)$, $i_p = f$ (Temperature)

The saturation current follows Richardson Dushman equation *i.e.* $i = AST^2 e^{-\phi_0/kT}$; Here

A = Emission constant = $\dfrac{4\pi\, mek^2}{h^3}\, amp/m^2 - k^2$

S = Area of emitter in m^2 ; T = Absolute temperature in K

ϕ_0 =Work function of metal in Joule; k =Boltzmann constant

The small increase in i_p after saturation stage due to field emission is known as Shottkey effect.

(4) **Diode resistance**

(i) Static plate resistance or dc plate resistance : $R_p = \dfrac{V_p}{i_p}$.

(ii) Dynamic or ac plate resistance : If at constant filament current, a small change ΔV_p in the plate potential produces a small change Δi_p in the plate current, then the ratio $\Delta V_p / \Delta i_p$ is called the dynamic resistance, or the 'plate resistance' of the diode $r_p = \dfrac{\Delta V_p}{\Delta i_p}$.

(iii) In SCLR : $r_p < R_p$, (iv) In TLR : $R_p < r_p$ and $r_p = \infty$.

(5) **Uses of diode valve**

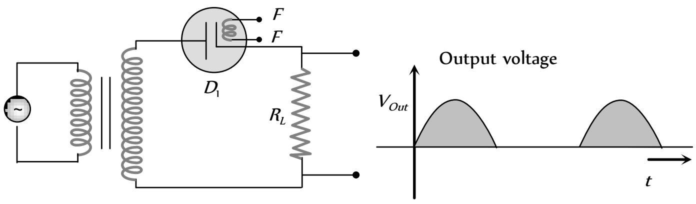
(i) As a rectifier (ii) As a detector

(iii) As a transmitter (iv) As a modulator

Diode Valve as a Rectifier

Rectifier is a device which converts ac into dc

(1) **Half wave rectifier :** The circuit of half wave rectifier is shown below. In the first half cycle of *ac* input the diode conducts and in the second half cycle it does not conducts. Thus half of the input cycle appear as output.



(A) Half wave rectifier (B) Output signal

Fig. 27.69

(i) Output voltage is not constant but pulsating in nature.

(ii) It is a mixture of *ac* and *dc.*

(iii) The *dc* values of the half wave output are given by

$$V_{d.c.} = \frac{V_0}{\pi} \text{ and } i_{d.c.} = \frac{i_0}{\pi}$$

(iv) The *r.m.s.* values of the half wave output are given by

$$V_{rms} = \frac{V_0}{2} \text{ and } i_{rms} = \frac{i_0}{2}$$

(v) The ratio of the effective alternating component to the direct component of the output voltage or current is called ripple factor $r = \dfrac{i_{a.c.}}{i_{d.c.}} = \sqrt{\left(\dfrac{i_{rms}}{i_{d.c.}}\right)^2 - 1} = \sqrt{\left(\dfrac{\pi}{2}\right)^2 - 1} = 1.21 = 121\%$.

(vi) Efficiency of half wave rectifier is given by

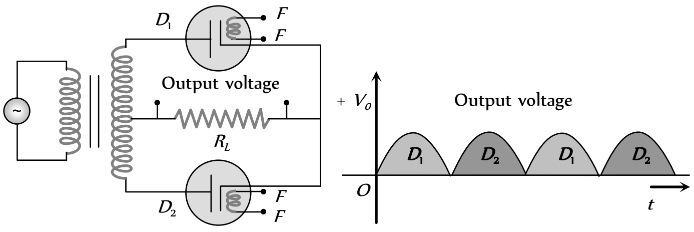
$$\eta = \frac{P_{d.c.}}{P_{a.c.}} \times 100\% = \frac{40.6}{1+\frac{r_p}{R_L}}\%$$

The maximum efficiency (for $R_L >> r_p$) = 40.6%

(vii) Form factor $= \frac{i_{rms}}{i_{d.c.}} = \frac{V_{rms}}{V_{d.c.}} = \frac{\pi}{2} = 1.57$

(viii) Ripple frequency = Frequency of input ac = ω

(2) **Full wave rectifier :** It consist of two diodes D_1 and D_2. They conducts alternately during positive and negative half cycle of input *ac* and a unidirectional (or *dc*) current flows in output



(A) Full wave rectifier (B) Output signal

Fig. 27.70

(i) The average or *dc* output values are

$$V_{d.c.} = \frac{2V_0}{\pi} \text{ and } i_{d.c.} = \frac{2i_0}{\pi}$$

(ii) It is a mixture of *ac* and *dc*

(iii) The *r.m.s.* values of the half wave output are given by

$$V_{rms} = \frac{V_0}{\sqrt{2}} \text{ and } i_{rms} = \frac{i_0}{\sqrt{2}}$$

(iv) Ripple factor $r = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} = 0.48 = 48\%$.

(v) Efficiency of half wave rectifier is given by

$$\eta = \frac{P_{d.c.}}{P_{a.c.}} \times 100\% = \frac{81.2}{1+\frac{r_p}{R_L}}\%$$

The maximum efficiency (for $R_L >> r_p$) = 81.2%

(vii) Form factor $= \frac{i_{rms}}{i_{d.c.}} = \frac{V_{rms}}{V_{d.c.}} = \frac{\pi}{2\sqrt{2}} = 1.11$

(viii) Ripple frequency = Double of frequency of input ac = 2ω

Filter Circuit

Filter circuits smooth out the fluctuations in amplitude of ac ripple of the output voltage obtained from a rectifier.

(i) Filter circuit consists of capacitors or/ and choke coils.

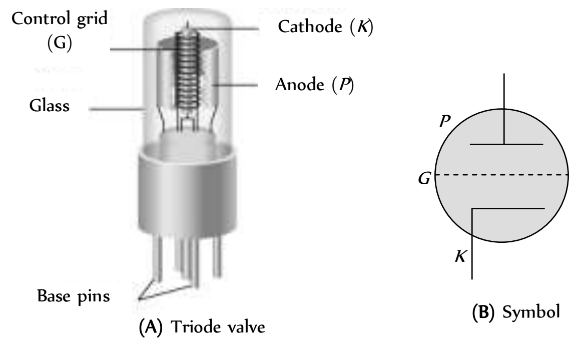
(ii) A capacitor offers a high resistance to low frequency ac ripple (infinite resistance to dc) and a low resistance to high frequency ac ripple. Therefore, it is always used as a shunt to the load.

(iii) A choke coil offers high resistance to high frequency ac, and almost zero resistance to dc. It is used in series.

(iv) π - Filter is best for ripple control.

(v) For voltage regulation choke input filter (*L*-filter) is best.

Triode Valve



(A) Triode valve (B) Symbol

Fig. 27.71

(1) **Inventor :** Dr. Lee De Forest

(2) **Principle :** Thermionic emission

(3) **Number of electrodes :** It consists of three electrodes.

(i) Filament (*F*) : It emits electron on heating.

(ii) Plate or anode (*P*) : It collect the electrons.

(iii) Control grid : It is a third electrode, also known as control grid, which controls the electrons going from cathode to plate. As a result grid controls the plate current. It is kept near the cathode with low negative potential.

When grid is given positive potential then plate current increases but in this case triode cannot be used for amplifier and therefore grid is normally not given positive potential.

When grid is given negative potential then plate current decreases but in this case grid controls plate current most effectively.

(4) **Working :** Plate of triode valve is always kept at positive potential *w.r.t.* cathode. The potential of plate is more than that of grid.

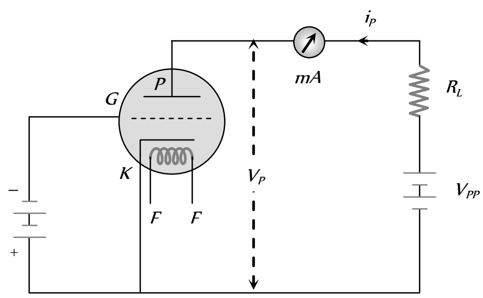


Fig. 27.72

The variation of plate potential affects the plate current as follows $i_p = k\left(V_G + \frac{V_p}{\mu}\right)^{3/2}$; where μ = Amplification factor of triode valve, k = Constant of triode valve.

The valve of V_G for which the plate current becomes zero is known as the cut off voltage. For a given V_p, it is given by $V_G = -\frac{V_p}{\mu}$.

Characteristics of Triode

The triode characteristics can be obtained under two sets of condition as

Static characteristics and dynamic characteristics

(1) **Static characteristics :** Graphical representation of V_p or V_G and i_p without any load

(i) **Static plate characteristic curve :** Graphical representation of i_p and V_p at constant V_g.

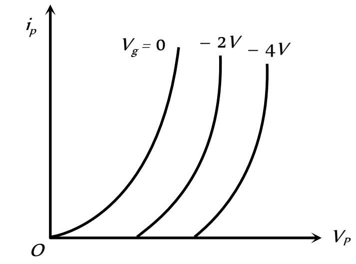


Fig. 27.73

(ii) **Static mutual characteristics curve :** Graphical representation of i_p and V_g when V_p is kept constant

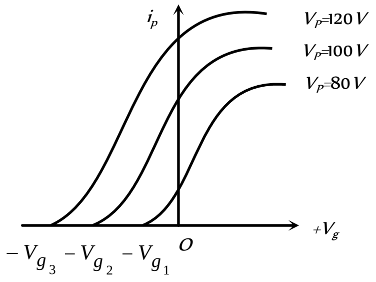


Fig. 27.74

(iii) **Constant current characteristic curve :** Graphical representation between V_p and V_g when i_p is constant.

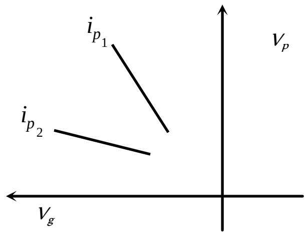


Fig. 27.75

(2) **Dynamic characteristics :** The curve plotted between i_p, V_g and V_p when the triode contains load in the plate circuit are called dynamics characteristics of diode.

(i) **Load line :** Voltage drop i_pR_L across load R_L which decreases the plate potential will be less then the supply voltage.

Plate voltage $V_p = V_{pp} - i_pR_L \Rightarrow i_p = -\dfrac{1}{R_L}V_p + \dfrac{V_{pp}}{R_L}$

This equation represents a straight line on the static plate characteristics, joining the points (V_{pp}, 0) on plate voltage axis and $(0, V_{pp}/R_L)$ on plate current axis. This line known as load line.

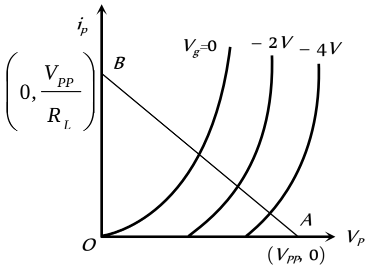


Fig. 27.76

(a) Points at which load line cuts the plate characteristic curves are called operating points.

(b) The slope of load line $AB = \dfrac{di_p}{dV_p} = -\dfrac{1}{R_L}$

(c) In graph, $OA = V_{pp}$ = intercept of load line on V_p axis and $OB = V_{pp}/R_L$ = intercept of load line on i_p axis.

(d) Static plate characteristic + load line

→ Dynamic plate characteristic

Static mutual characteristic + load line

→ Dynamic mutual characteristic

Constants of Triode Valve

(1) **Plate or dynamic resistance (r_p)**

(i) The slope of plate characteristic curve is equal to $\dfrac{1}{\text{plate resistance}}$

or It is the ratio of small change in plate voltage to the change in plate current produced by it, the grid voltage remaining constant. That is, $r_p = \dfrac{\Delta V_p}{\Delta i_p}, V_G = \text{constant}$.

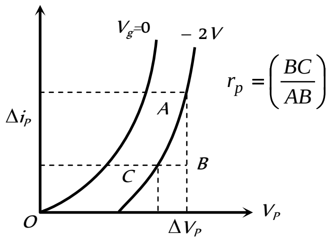


Fig. 27.77

(ii) It is expressed in kilo ohms ($k\Omega$). Typically, it ranges from $8k\Omega$ to $40k\Omega$. The r_p can be determined from plate characteristics. It represents the reciprocal of the slope of the plate characteristic curve.

(iii) If the distance between plate and cathode is increased the r_p increases. The value of r_p is infinity in the state of cut off bias or saturation state.

(2) **Mutual conductance (or *trans* conductance) (g_m)**

(i) It is defined as the ratio of small change in plate current (Δi_p) to the corresponding small change in grid potential (ΔV_g) when plate potential V_p is kept constant *i.e.* $g_m = \left(\dfrac{\Delta i_p}{\Delta V_g}\right)_{V_p \text{ is constant}}$

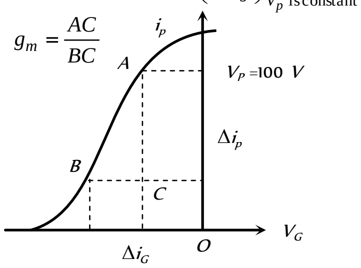


Fig. 27.78

(ii) The value of g_m is equal to the slope of mutual characteristics of triode.

(iii) The value of g_m depends upon the separation between grid and cathode. The smaller is this separation, the larger is the value of g_m and vice versa.

(iv) In the saturation state, the value of $\Delta i_p = 0$, $g_m = 0$

(3) **Amplification factor (μ) :** It is defined as the ratio of change in plate potential (ΔV_p) to produce certain change in plate current (Δi_p) to the change in grid potential (ΔV_g) for the same change in plate current (Δi_p) *i.e.* $\mu = -\left(\frac{\Delta V_p}{\Delta V_g}\right)_{\Delta I_p = \text{a constant}}$; negative sign indicates that V_p and V_g are in opposite phase.

(i) Amplification factor depends upon the distance between plate and cathode (d_{pk}), plate and grid (d_{pg}) and grid and cathode(d_{gk}).

i.e. $\mu \propto d_{pg} \propto d_{pk} \propto \frac{1}{d_{gk}}$

(ii) The value of μ is greater than one.

(iii) Amplification factor is unitless and dimensionless.

(4) **Relation between triode constants :** The triode constants are not independent of each other. They are related by the relation $\mu = r_p \times g_m$

The r_p and g_m depends on i_p in the following manner

$r_p \propto i_p^{-1/3}$, $g_m \propto i_p^{1/3}$, μ does not depend on i_p.

Above three constants may be determined from any one set of characteristic curves.

$$r_p = \frac{V_{P1} - V_{P2}}{I_{PA} - I_{PB}},$$

$$g_m = \frac{I_{PA} - I_{PB}}{V_{G1} - V_{G2}},$$

$$\mu = -\frac{V_{P1} - V_{P2}}{V_{G2} - V_{G1}}$$

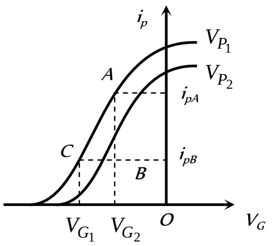


Fig. 27.79

Triode as an Amplifiers

Amplifier is a device by which the amplitude of variation of *ac* signal voltage / current/ power can be increased

(1) The signal to be amplified (V_i) is applied in the grid circuit and amplified output is obtained from the plate circuit

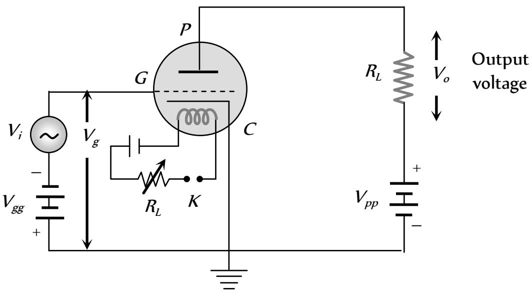


Fig. 27.80

(2) The voltage at grid is the sum of signal V_i and grid bias V_{gg}.

$V_g = V_{gg} + V_i$

(3) Small change in grid voltage results in a large change in plate current so results in a large change in voltage across R_L $(V_0 = i_p R_L \Rightarrow \Delta V_0 = \Delta i_p R_L)$

(4) The linear portion of the mutual characteristic with maximum slope is chosen for amplification without distortion.

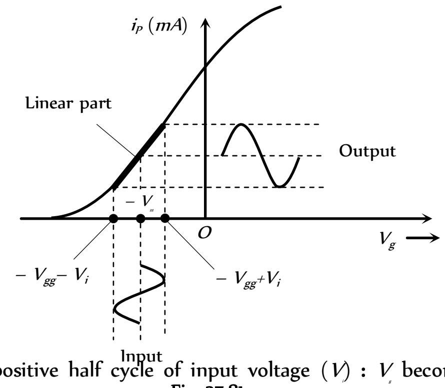


Fig. 27.81

(i) For the positive half cycle of input voltage (V_i) : V_g becomes less negative, so i_p increases

(ii) For the negative half cycle of input voltage (V_i) : V_g becomes more negative, so i_p decreases

(iii) The phase difference between the output signal and input signal is 180° (or π)

(5) **Voltage amplification**

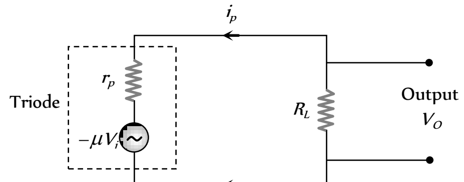


Fig. 27. 82 : Equivalent circuit of triode amplifier

Current through the load resistance is given by $i_p = -\frac{\mu V_i}{r_p + R_L}$

$$\Rightarrow V_0 = i_p R_L = \frac{-\mu V_i R_L}{r_p + R_L} \;\Rightarrow\; \text{Voltage gain} = \frac{V_0}{V_i} = -\frac{\mu R_L}{r_p + R_L}$$

Numerically $A = \frac{\mu R_L}{r_p + R_L} = \frac{\mu}{1 + \frac{r_p}{R_L}}$

(i) If $R_L = \infty \Rightarrow A$ will be maximum and $A_{max} = \mu$

(Practically $A < \mu$)

(ii) If $r_p = R_L \Rightarrow A = \frac{\mu}{2}$

(iii) Power at load resistance $P = i_p V_0 = i_p^2 R_L$

Condition for maximum power $R_L = r_p$

$$\therefore\; P_{max} = \left(\frac{\mu V_i}{R_L + R_L}\right)^2 \times R_L = \frac{\mu^2 V_i^2}{4 R_L}$$

Tips & Tricks

✍ The most efficient packing of atoms in cubic lattice structure occurs for *fcc*.

- The lattice for *NaCl* crystal is *fcc*.

- The space lattice of diamond is *fcc*. (The diamond structure may be viewed as two *fcc* structures displaced from each other by one quarter of a body diagonal).

- Carbon, silicon, germanium, tin can crystallize in the diamond structure.

- At room temperature $\sigma_{Ge} > \sigma_{Si}$

- $(n_i)_{Ge} \simeq 2.4 \times 10^{19} / m^3$ and $(n_i)_{Si} \simeq 1.5 \times 10^{16} / m^3$

- In a transistor circuit the reverse bias is high as compared to the forward bias. So that it may exert a large attractive force on the charge carriers to enter the collector region.

- *Ge* is more sensitive to heat since it's forbidden energy gap is smaller than that of silicon. Electrons from the valence band of *Ge* requires less energy to move from the valence band to conduction band.

- Both *N*-type as well as *P*-type semiconductor are neutral.

- Semiconductor devices are current control devices.

- The semiconductor devices are temperature sensitive devices.

- The electric field setup across the potential barrier is of the order of 3×10^5 *V/m* for *Ge* and 7×10^5 *V/m* for *Si*.

- An ideal junction diode when forward biased offers zero resistance. Voltage drop across such a junction diode is zero. In reverse biased diode offers infinite resistance and voltage drop across it is equal to voltage applied.

- A *P-N* junction diode can be considered to be equivalent to a capacitor with *P* and *N* regions acting as the plates of the capacitors and depletion layer as the dielectric medium.

- The mobility of electron is two-three times the mobility of holes. Therefore NPN devices are fast and hence preferred.

- If $E_g \simeq 0\, eV$, the material is good conductor or metal and if $E_g \simeq 1\, eV$, the material is a semiconductor. If $E_g \simeq 6\, eV$ then the material is an insulator.

- A *P-N* junction or diode acts like a valve or voltage controlled switch. When forward biased, it acts like ON switch. When reverse biased, it acts like an OFF switch.

- The current due to minority carriers in the junction diode is independent of the applied voltage. It only depends upon the temperature of the diode.

- Voltage obtained from a diode rectifier is a mixture of alternating and direct voltage.

- Cross sectional area of base is very large as compared to emitter. Cross sectional area of collector is less than base but greater than emitter.

- C.C (common collector) amplifier is called power amplifier or current booster or emitter follower.

- Devices like tunnel diode, tetrode and thyrisisters have negative resistance.

- Transistor provides good power amplification when they are use in CE configuration.

- **MOSFETS :** In a MOSFET, a type of three-terminal transistor, a potential applied to the gate terminal *G* controls the internal flow of electrons from the source terminal *S* to the drain terminal *D*. Commonly, a MOSFET is operated only in its ON (conducting) or OFF (not conducting condition. Installed by the thousands and millions on silicon wafers (chips) to form integrated circuits, MOSFETs form the basis for computer hardware.

- When a *PN* junction is forward biased, it can emit light, hence can serve as a light-emitting diode (LED). The wavelength of the emitted light is $\lambda = \dfrac{c}{f} = \dfrac{hc}{E_g}$

- The fermi energy of a given material is the energy of a quantum state that has the probability 0.5 of being occupied by an electron.

- Number of conduction electrons per unit volume

$$= \frac{\text{(Material's density)}}{(\text{Molar mass } M)/N_A}$$

(N_A = Avogadro's number = 6.02×10^{23} */mol*)

- The occupancy probability *P*(*E*) : Electrical conduction of a metal depends on the probability that if an energy level is available at energy *E*, is it actually occupied by an electron.

the expression for occupancy probability *P*(*E*) is given by

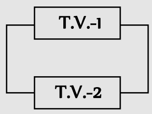
Fermi-Dirac statistics $P(E) = \dfrac{1}{\exp\left(\dfrac{E - E_F}{kT}\right) + 1}$; E_F=Fermi energy

- A good emitter should have low work function, high melting point, high working temperature, high electrical and mechanical strength.

- When triode amplifier are in series, total voltage gain

$A = A_1 A_2 A_3$ ....

- When two triode valve are in parallel



Total plate resistance $\dfrac{1}{r_p} = \dfrac{1}{r_{p_1}} + \dfrac{1}{r_{p_2}}$

Total mutual conductance $G_m = g_{m_1} + g_{m_2}$

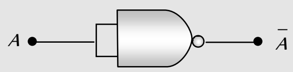
Total amplification factor $\mu = G_m R_p$

Voltage amplification $A = \dfrac{\mu R_L}{r_p + R_L}$

- **NOR** gate is a universal gate because it can be used to perform the basic logic function, AND, OR and NOT.

- Output in Ex-OR gate is '1' only when inputs are different.

- If both inputs of NAND gate are shorted then it will become 'NOT gate



NOT gate

Ordinary Thinking

Objective Questions

Solids and Crystals

1. The nature of binding for a crystal with alternate and evenly spaced positive and negative ions is [CBSE PMT 2000]
   (a) Covalent (b) Metallic
   (c) Dipolar (d) Ionic
2. For a crystal system, $a = b = c$, $\alpha = \beta = \gamma \neq 90^\circ$, the system is [BHU 2000]
   (a) Tetragonal system (b) Cubic system
   (c) Orthorhombic system (d) Rhombohedral system
3. Biaxial crystal among the following is [Pb. CET 1998]
   (a) Calcite (b) Quartz
   (c) Selenite (d) Tourmaline
4. The temperature coefficient of resistance of a conductor is [AFMC 1998]
   (a) Positive always (b) Negative always
   (c) Zero (d) Infinite
5. Potassium has a *bcc* structure with nearest neighbour distance 4.525 Å. Its molecular weight is 39. Its density in kg/m is [DCE 1997]
   (a) 900 (b) 494
   (c) 602 (d) 802
6. The expected energy of the electrons at absolute zero is called [RPET 1996]
   (a) Fermi energy (b) Emission energy
   (c) Work function (d) Potential energy
7. In a triclinic crystal system [EAMCET (Med.) 1995]
   (a) $a \neq b \neq c$, $\alpha \neq \beta \neq \gamma$ (b) $a = b = c$, $\alpha \neq \beta \neq \gamma$
   (c) $a \neq b \neq c$, $\alpha \neq \beta = \gamma$ (d) $a = b \neq c$, $\alpha = \beta = \gamma$
8. Metallic solids are always opaque because [AFMC 1994]
   (a) Solids effect the incident light
   (b) Incident light is readily absorbed by the free electron in a metal
   (c) Incident light is scattered by solid molecules
   (d) Energy band traps the incident light
9. In which of the following ionic bond is present [EAMCET (Med.) 1994]
   (a) $NaCl$ (b) Ar
   (c) Si (d) Ge
10. Which of the following materials is non crystalline [CBSE PMT 1993]
    (a) Copper (b) Sodium chloride
    (c) Wood (d) Diamond

11. The coordination number of Cu is [AMU 1992]

(a) 1 (b) 6

(c) 8 (d) 12

12. Which one of the following is the weakest kind of bonding in solids[CBSE PMT 1992; KCET 1992]

(a) Ionic (b) Metallic

(c) Vander Waals (d) Covalent

13. In a crystal, the atoms are located at the position of [AMU 1985]

(a) Maximum potential energy

(b) Minimum potential energy

(c) Zero potential energy

(d) Infinite potential energy

14. Crystal structure of $NaCl$ is [NCERT 1982]

(a) Fcc (b) Bcc

(c) Both of the above (d) None of the above

15. What is the coordination number of sodium ions in the case of sodium chloride structure [CBSE PMT 1988]

(a) 6 (b) 8

(c) 4 (d) 12

16. The distance between the body centred atom and a corner atom in sodium (a = 4.225 $\mathring{A}$) is [CBSE PMT 1995]

(a) 3.66 $\mathring{A}$ (b) 3.17 $\mathring{A}$

(c) 2.99 $\mathring{A}$ (d) 2.54 $\mathring{A}$

17. A solid that transmits light in visible region and has a very low melting point possesses [J & K CET 2001]

(a) Metallic bonding (b) Ionic bonding

(c) Covalent bonding (d) Vander Waal's bonding

18. Atomic radius of *fcc* is [J & K CET 2001]

(a) $\frac{a}{2}$ (b) $\frac{a}{2\sqrt{2}}$

(c) $\frac{\sqrt{3}}{4}a$ (d) $\frac{\sqrt{3}}{2}a$

19. A solid reflects incident light and it's electrical conductivity decreases with temperature. The binding in this solids

(a) Ionic (b) Covalent

(c) Metallic (d) Molecular

20. The laptop PC's modern electronic watches and calculators use the following for display

(a) Single crystal (b) Poly crystal

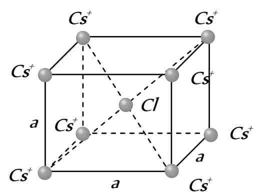
(c) Liquid crystal (d) Semiconductors

21. The nearest distance between two atoms in case of a bcc lattice is equal to [J & K CET 2004]

(a) $a\frac{\sqrt{2}}{3}$ (b) $a\frac{\sqrt{3}}{2}$

(c) $q\sqrt{3}$ (d) $\frac{a}{\sqrt{2}}$

22. What is the net force on a *Cl* placed at the centre of the *bcc* structure of *CsCl* [DCE 2003; AIIMS 2004]



(a) Zero (b) ke^2/a^2

(c) ke^2a^2 (d) Data is incomplete

23. Sodium has body centred packing. If the distance between two nearest atoms is 3.7 Å, then its lattice parameter is [Pb. PET 2002]

(a) 4.8 Å (b) 4.3 Å

(c) 3.9 Å (d) 3.3 Å

24. Which of the following is an amorphous solid [AIIMS 2005; J & K CET 2004]

(a) Glass (b) Diamond

(c) Salt (d) Sugar

25. Copper has face centered cubic (*fcc*) lattice with interatomic spacing equal to 2.54Å. The value of the lattice constant for this lattice is

(a) 1.27 Å (b) 5.08 Å

(c) 2.54 Å (d) 3.59 Å

26. In good conductors of electricity, the type of bonding that exists is

(a) Ionic (b) Vander Waals

(c) Covalent (d) Metallic

27. Bonding in a germanium crystal (semi- conductor) is [CPMT 1986; KCET 1992; EAMCET (Med.) 1995; MP PET/PMT 2004]

(a) Metallic (b) Ionic

(c) Vander Waal's type (d) Covalent

28. The ionic bond is absent in [J & K CET 2005]

(a) $NaCl$ (b) $CsCl$

(c) LiF (d) H_2O

Semiconductors

1. The majority charge carriers in *P*-type semiconductor are [MP PMT 1999; CBSE PMT 1999; MP PET 1991; MP PET/PMT 1998; MH CET 2003]

(a) Electrons (b) Protons

(c) Holes (d) Neutrons

2. A *P*-type semiconductor can be obtained by adding [NCERT 1979; BIT 1988; MP PMT 1987; 90]

(a) Arsenic to pure silicon

(b) Gallium to pure silicon

(c) Antimony to pure germanium

(d) Phosphorous to pure germanium

3. The valence of an impurity added to germanium crystal in order to convert it into a *P*-type semi conductor is [MP PMT 1989; CPMT 1987]

(a) 6 (b) 5

(c) 4 (d) 3

4. In a semiconductor, the concentration of electrons is $8\times10^{14}/cm^3$ and that of the holes is $5\times10^{12}cm^3$. The semiconductor is [MP PMT 1997; RPET 1999; Kerala PET 2002]
(a) *P*-type (b) *N*-type
(c) Intrinsic (d) *PNP*-type

5. In *P*-type semiconductor, there is [MP PMT 1989]
(a) An excess of one electron
(b) Absence of one electron
(c) A missing atom
(d) A donar level

6. The valence of the impurity atom that is to be added to germanium crystal so as to make it a *N*-type semiconductor, is
[MNR 1993; MP PET 1994; CBSE PMT 1999; AIIMS 2000]
(a) 6 (b) 5
(c) 4 (d) 3

7. Silicon is a semiconductor. If a small amount of *As* is added to it, then its electrical conductivity [MP PMT 1996]
(a) Decreases (b) Increases
(c) Remains unchanged (d) Becomes zero

8. When the electrical conductivity of a semi- conductor is due to the breaking of its covalent bonds, then the semiconductor is said to be
[AIIMS 1997; KCET (Engg.) 2002]
(a) Donar (b) Acceptor
(c) Intrinsic (d) Extrinsic

9. A piece of copper and the other of germanium are cooled from the room temperature to 80 K, then which of the following would be a correct statement
[IIT-JEE 1988; Bihar CEE 1992; CBSE PMT 1993; MP PET 1997; RPET 1999; AIEEE 2004]
(a) Resistance of each increases
(b) Resistance of each decreases
(c) Resistance of copper increases while that of germanium decreases
(d) Resistance of copper decreases while that of germanium increases

10. To obtain *P*-type Si semiconductor, we need to dope pure Si with [IIT-JEE 1988; MP PET 1997, 93; Pb. PMT 2001, 02; UPSEAT 2004]
(a) Aluminium (b) Phosphorous
(c) Oxygen (d) Germanium

11. Electrical conductivity of a semiconductor
[MP PMT 1993, 2000; RPET 1996]
(a) Decreases with the rise in its temperature
(b) Increases with the rise in its temperature
(c) Does not change with the rise in its temperature
(d) First increases and then decreases with the rise in its temperature

12. Three semi-conductors are arranged in the increasing order of their energy gap as follows. The correct arrangement is
[MP PMT 1993]
(a) Tellurium, germanium, silicon
(b) Tellurium, silicon, germanium
(c) Silicon, germanium, tellurium
(d) Silicon, tellurium, germanium

13. When a semiconductor is heated, its resistance
[KCET 1992; MP PMT 1994; MP PET 1992, 2002; RPMT 2001; DCE 2001]
(a) Decreases (b) Increases
(c) Remains unchanged (d) Nothing is definite

14. In an insulator, the forbidden energy gap between the valence band and conduction band is of the order of
[DPMT 1988; EAMCET (Engg.) 1995; MP PET 1996]
(a) $1\,MeV$ (b) $0.1\,MeV$
(c) $1\,eV$ (d) $5\,eV$

15. A *N*-type semiconductor is [AFMC 1988; RPMT 1999]
(a) Negatively charged (b) Positively charged
(c) Neutral (d) None of these

16. The energy band gap of Si is
[MP PET 1994, 2002; BHU 1995; RPMT 2000]
(a) $0.70\,eV$
(b) 1.1 *eV*
(c) Between $0.70\,eV$ to $1.1\,eV$
(d) 5 *eV*

17. The forbidden energy band gap in conductors, semiconductors and insulators are EG_1, EG_2 and EG_3 respectively. The relation among them is
[MP PMT 1994; RPMT 1997]
(a) $EG_1 = EG_2 = EG_3$ (b) $EG_1 < EG_2 < EG_3$
(c) $EG_1 > EG_2 > EG_3$ (d) $EG_1 < EG_2 > EG_3$

18. Which statement is correct [MP PMT 1994]
(a) *N*-type germanium is negatively charged and *P*-type germanium is positively charged
(b) Both *N*-type and *P*-type germanium are neutral
(c) *N*-type germanium is positively charged and *P*-type germanium is negatively charged
(d) Both *N*-type and *P*-type germanium are negatively charged

19. When *Ge* crystals are doped with phosphorus atom, then it becomes [AFMC 1995; Orissa PMT 2004]
(a) Insulator (b) *P*-type
(c) *N*-type (d) Superconductor

20. Let n_P and n_e be the number of holes and conduction electrons respectively in a semiconductor. Then
[MP PET 1995]
(a) $n_P > n_e$ in an intrinsic semiconductor
(b) $n_P = n_e$ in an extrinsic semiconductor
(c) $n_P = n_e$ in an intrinsic semiconductor
(d) $n_e > n_P$ in an intrinsic semiconductor

21. Wires P and Q have the same resistance at ordinary (room) temperature. When heated, resistance of P increases and that of Q decreases. We conclude that

[MP PMT 1995; MP PET 2001]

(a) P and Q are conductors of different materials

(b) P is N-type semiconductor and Q is P-type semiconductor

(c) P is semiconductor and Q is conductor

(d) P is conductor and Q is semiconductor

22. The impurity atoms which are mixed with pure silicon to make a P-type semiconductor are those of [MP PMT 1995]

(a) Phosphorus (b) Boron

(c) Antimony (d) Copper

23. Holes are charge carriers in [IIT-JEE 1996]

(a) Intrinsic semiconductors (b) Ionic solids

(c) P-type semiconductors (d) Metals

24. In extrinsic P and N-type, semiconductor materials, the ratio of the impurity atoms to the pure semiconductor atoms is about

(a) 1 (b) 10^{-1}

(c) 10^{-4} (d) 10^{-7}

25. A hole in a P-type semiconductor is [MP PET 1996]

(a) An excess electron (b) A missing electron

(c) A missing atom (d) A donor level

26. The forbidden gap in the energy bands of germanium at room temperature is about [MP PMT/PET 1998]

(a) $1.1 eV$ (b) $0.1 eV$

(c) $0.67 eV$ (d) $6.7 eV$

27. In P-type semiconductor the majority and minority charge carriers are respectively

[EAMCET 1994; MP PMT/PET 1998; MH CET 2000]

(a) Protons and electrons (b) Electrons and protons

(c) Electrons and holes (d) Holes and electrons

28. At zero Kelvin a piece of germanium [MP PET 1999]

(a) Becomes semiconductor

(b) Becomes good conductor

(c) Becomes bad conductor

(d) Has maximum conductivity

29. Electronic configuration of germanium is 2, 8, 18 and 4. To make it extrinsic semiconductor small quantity of antimony is added

(a) The material obtained will be N-type germanium in which electrons and holes are equal in number

(b) The material obtained will be P-type germanium

(c) The material obtained will be N-type germanium which has more electrons than holes at room temperature

(d) The material obtained will be N-type germanium which has less electrons than holes at room temperature

30. A semiconductor is cooled from $T_1 K$ to $T_2 K$. Its resistance

[MP PET 1999]

(a) Will decrease

(b) Will increase

(c) Will first decrease and then increase

(d) Will not change

31. If N_P and N_e be the numbers of holes and conduction electrons in an extrinsic semiconductor, then

[MP PMT 1999; AMU 2001]

(a) $N_P > N_e$

(b) $N_P = N_e$

(c) $N_P < N_e$

(d) $N_P > N_e$ or $N_P < N_e$ depending on the nature of impurity

32. In intrinsic semiconductor at room temperature, number of electrons and holes are

[EAMCET (Engg.) 1995; JIPMER 2001, 02]

(a) Equal (b) Zero

(c) Unequal (d) Infinite

33. (USS 133) Indium impurity in germanium makes

[EAMCET (Engg.) 1995]

[MP PET 2003]

(a) N-type (b) P-type

(c) Insulator (d) Intrinsic

34. Fermi level of energy of an intrinsic semiconductor lies

[EAMCET (Med.) 1995]

(a) In the middle of forbidden gap

(b) Below the middle of forbidden gap

(c) Above the middle of forbidden gap

(d) Outside the forbidden gap

35. In a semiconductor the separation between conduction band and valence band is of the order of

[EAMCET (Med.) 1995; AIIMS 2000]

(a) $100 eV$ (b) $10 eV$

(c) $1 eV$ (d) $0\ eV$

36. The intrinsic semiconductor becomes an insulator at

[EAMCET (Med.) 1995; KCET (Engg./Med.) 1999; MP PET 2000; CBSE PMT 2001]

(a) $0°C$ (b) $-100°C$

(c) $300\ K$ (d) $0\ K$

37. The addition of antimony atoms to a sample of intrinsic germanium transforms it to a material which is

[AMU 1995]

[MP PET 1999]

(a) Superconductor (b) An insulator

(c) N-type semiconductor (d) P-type semiconductor

38. Resistance of semiconductor at $0°K$ is [RPET 1997]

(a) Zero (b) Infinite

(c) Large (d) Small

39. In a good conductor the energy gap between the conduction band and the valence band is

[KCET 1993; EMCET (Med.) 1994]

(a) Infinite (b) Wide

(c) Narrow (d) Zero

40. The impurity atom added to germanium to make it N-type semiconductor is [KCET 1993; KCET (Engg./Med.) 2000]

(a) Arsenic (b) Iridium
(c) Aluminium (d) Iodine

41. When *N*-type of semiconductor is heated [CBSE PMT 1993; DPMT 2000]

(a) Number of electrons increases while that of holes decreases
(b) Number of holes increases while that of electrons decreases
(c) Number of electrons and holes remains same
(d) Number of electrons and holes increases equally

42. To obtain a *P*-type germanium semiconductor, it must be doped with [CBSE PMT 1997; Pb. PET 2000]

(a) Arsenic (b) Antimony
(c) Indium (d) Phosphorus

43. The temperature coefficient of resistance of a semiconductor [AFMC 1998, MNR 1998]

(a) Is always positive
(b) Is always negative
(c) Is zero
(d) May be positive or negative or zero

44. *P*-type semiconductor is formed when [RPET 1999]

A. *As* impurity is mixed in *Si*
B. *Al* impurity is mixed in *Si*
C. *B* impurity is mixed in *Ge*
D. *P* impurity is mixed in *Ge*

(a) A and C (b) A and D
(c) B and C (d) B and D

45. In case of a semiconductor, which of the following statement is wrong [Pb. PMT 1999]

(a) Doping increases conductivity
(b) Temperature coefficient of resistance is negative
(c) Resisitivity is in between that of a conductor and insulator
(d) At absolute zero temperature, it behaves like a conductor

46. Energy bands in solids are a consequence of [DCE 1999, 2000; AIEEE 2004]

(a) Ohm's Law
(b) Pauli's exclusion principle
(c) Bohr's theory
(d) Heisenberg's uncertainty principle

47. In a *P*-type semiconductor [AIIMS 1997; Orissa JEE 2002; MP PET 2003]

(a) Current is mainly carried by holes
(b) Current is mainly carried by electrons
(c) The material is always positively charged
(d) Doping is done by pentavalent material

48. At ordinary temperatures, the electrical conductivity of semi conductors in $mho/meter$ is in the range [MP PET 2003]

(a) 10^{-3} to 10^{-4} (b) 10^{6} to 10^{9}
(c) 10^{-6} to 10^{-10} (d) 10^{-10} to 10^{-16}

49. When the temperature of silicon sample is increased from $27°C$ to $100°C$, the conductivity of silicon will be [RPMT 1999]

(a) Increased (b) Decreased
(c) Remain same (d) Zero

50. In a *P*-type semiconductor, germanium is doped with [AFMC 1999]

(a) Boron (b) Gallium
(c) Aluminium (d) All of these

51. In *N*-type semiconductors, majority charge carriers are [AIIMS 1999]

(a) Holes (b) Protons
(c) Neutrons (d) Electrons

52. Semiconductor is damaged by the strong current due to [MH CET 2000]

(a) Lack of free electron (b) Excess of electrons
(c) Excess of proton (d) None of these

53. GaAs is [RPMT 2000]

(a) Element semiconductor
(b) Alloy semiconductor
(c) Bad conductor
(d) Metallic semiconductor

54. If n_e and n_h are the number of electrons and holes in a semiconductor heavily doped with phosphorus, then [MP PMT 2000]

(a) $n_e >> n_h$ (b) $n_e << n_h$
(c) $n_e \leq n_h$ (d) $n_e = n_h$

55. An *N*-type and *P*-type silicon can be obtained by doping pure silicon with [EAMCET (Med.) 2000]

(a) Arsenic and Phosphorous (b) Indium and Aluminium
(c) Phosphorous and Indium (d) Aluminium and Boron

56. *N*-type semiconductors will be obtained, when germanium is doped with [AIIMS 2000]

(a) Phosphorus (b) Aluminium
(c) Arsenic (d) Both (a) or (c)

57. The state of the energy gained by valance electrons when the temperature is raised or when electric field is applied is called as

(a) Valance band (b) Conduction band
(c) Forbidden band (d) None of these

58. To obtain electrons as majority charge carriers in a semiconductor, the impurity mixed is [MP PET 2000]

(a) Monovalent (b) Divalent
(c) Trivalent (d) Pentavalent

59. For germanium crystal, the forbidden energy gap in joules is [MP PET 2000]

(a) 1.12×10^{-19} (b) 1.76×10^{-19}
(c) 1.6×10^{-19} (d) Zero

60. A pure semiconductor behaves slightly as a conductor at [MH CET (Med.) 2001; BHU 2000; AFMC 2001]

(a) Room temperature (b) Low temperature

(c) High temperature (d) Both (b) and (c)

61. Which is the correct relation for forbidden energy gap in conductor, semi conductor and insulator [RPMT 2001; AIEEE 2002]

(a) $\Delta Eg_{c} > \Delta Eg_{sc} > \Delta Eg_{\text{insulator}}$

(b) $\Delta Eg_{\text{insulator}} > \Delta Eg_{sc} > \Delta Eg_{\text{conductor}}$

(c) $\Delta Eg_{\text{conductor}} > \Delta Eg_{\text{insulator}} > \Delta Eg_{sc}$

(d) $\Delta Eg_{sc} > \Delta Eg_{\text{conductor}} > \Delta Eg_{\text{insulator}}$

62. The band gap in Germanium and silicon in *eV* respectively is

(a) 0.7, 1.1 (b) 1.1, 0.7

(c) 1.1, 0 (d) 0, 1.1

63. *P*-type semiconductors are made by adding impurity element

(a) *As* (b) *P*

(c) *B* (d) *Bi*

64. At room temperature, a *P*-type semiconductor has [Kerala PMT 2002]

(a) Large number of holes and few electrons

(b) Large number of free electrons and few holes

(c) Equal number of free electrons and holes

(d) No electrons or holes

65. In intrinsic semiconductor at room temperature, number of electrons and holes are [JIPMER 2001, 02; MP PMT 2002]

(a) Unequal (b) Equal

(c) Infinite (d) Zero

66. The valence band and conduction band of a solid overlap at low temperature, the solid may be [Orissa JEE 2002; BCECE 2004]

(a) A metal (b) A semiconductor

(c) An insulator (d) None of these

67. Which impurity is doped in *Si* to form *N*-type semi-conductor? [CBSE PMT 1996; AIEEE 2002]

(a) *Al* (b) *B*

(c) *As* (d) None of these

68. In a semiconductor [AIEEE 2002; AIIMS 2002]

(a) There are no free electrons at any temperature

(b) The number of free electrons is more than that in a conductor

(c) There are no free electrons at 0 *K*

(d) None of these

69. The energy band gap is maximum in [AIEEE 2002]

(a) Metals (b) Superconductors

(c) Insulators (d) Semiconductors

70. The process of adding impurities to the pure semiconductor is called [MH CET 2002]

(a) Drouping (b) Drooping

(c) Doping (d) None of these

71. When phosphorus and antimony are mixed in zermaniun, then

(a) *P*-type semiconductor is formed

(b) *N*-type semiconductor is formed

(c) Both (a) and (b)

(d) None of these

72. To a germanium sample, traces of gallium are added as an impurity. The resultant sample would behave like [AIIMS 2003]

(a) A conductor

(b) A *P*-type semiconductor

(c) An *N*-type semiconductor

(d) An insulator [MP PMT 2001]

73. For non-conductors, the energy gap is [EAMCET (Engg.) 1995; MP PET 1996; RPET 2003]

(a) 6 *eV* [MP PMT 2001] (b) 1.1 *eV*

(c) 0.8 *eV* (d) 0.3 *eV*

74. Donor type impurity is found in [RPET 2003]

(a) Trivalent elements (b) Pentavalent elements

(c) In both the above (d) None of these

75. The difference in the variation of resistance with temperature in a metal and a semiconductor arises essentially due to the difference in the [AIEEE 2003]

(a) Variation of scattering mechanism with temperature

(b) Crystal structure

(c) Variation of the number of charge carriers with temperature

(d) Type of bon

76. The charge on a hole is equal to the charge of [MP PMT 2004]

(a) Zero (b) Proton

(c) Neutron (d) Electron

77. When germanium is doped with phosphorus, the doped material has

(a) Excess positive charge

(b) Excess negative charge

(c) More negative current carriers

(d) More positive current carriers

78. A *Ge* specimen is doped with *Al*. The concentration of acceptor atoms is $\sim 10^{21}$ *atoms*/m^3. Given that the intrinsic concentration of electron hole pairs is $\sim 10^{19}/m^3$, the concentration of electrons in the specimen is [AIIMS 2004]

(a) $10^{17}/m^3$ (b) $10^{15}/m^3$

(c) $10^{4}/m^3$ (d) $10^{2}/m^3$

79. Which of the following has negative temperature coefficient of resistance [AFMC 2004]

(a) Copper (b) Aluminium

(c) Iron (d) Germanium

80. In semiconductors at a room temperature [CPMT 2003] [CBSE PMT 2004]

(a) The valence band is partially empty and the conduction band is partially filled

(b) The valence band is completely filled and the conduction band is partially filled

(c) The valence band is completely filled

(d) The conduction band is completely empty

81. Regarding a semiconductor which one of the following is wrong

(a) There are no free electrons at room temperature

(b) There are no free electrons at 0 K

(c) The number of free electrons increases with rise of temperature

(d) The charge carriers are electrons and holes

82. Which of the following statements is true for an N-type semi-conductor **[CPMT 2004]**

(a) The donor level lies closely below the bottom of the conduction band

(b) The donor level lies closely above the top of the valence band

(c) The donor level lies at the halfway mark of the forbidden energy gap

(d) None of above

83. Choose the correct statement **[DCE 2004]**

(a) When we heat a semiconductor its resistance increases

(b) When we heat a semiconductor its resistance decreases

(c) When we cool a semiconductor to 0 K then it becomes super conductor

(d) Resistance of a semiconductor is independent of temperature

84. In a P-type semi-conductor, germanium is dopped with **[MH CET 2003]**

(a) Gallium (b) Boron

(c) Aluminium (d) All of these

85. A piece of semiconductor is connected in series in an electric circuit. On increasing the temperature, the current in the circuit will

(a) Decrease (b) Remain unchanged

(c) Increase (d) Stop flowing

86. Intrinsic semiconductor is electrically neutral. Extrinsic semiconductor having large number of current carriers would be

(a) Positively charged

(b) Negatively charged

(c) Positively charged or negatively charged depending upon the type of impurity that has been added

(d) Electrically neutral

87. If n_e and v_d be the number of electrons and drift velocity in a semiconductor. When the temperature is increased **[Pb. CET 2000]**

(a) n_e increases and v_d decreases

(b) n_e decreases and v_d increases

(c) Both n_e and v_d increases

(d) Both n_e and v_d decreases

88. In extrinsic semiconductors **[EAMCET (Engg.) 1999]**

(a) The conduction band and valence band overlap

(b) The gap between conduction band and valence band is more than 16 eV

(c) The gap between conduction band and valence band is near about 1 eV

(d) The gap between conduction band and valence band will be 100 eV and more **[CPMT 2004]**

89. Resistivity of a semiconductor depends on **[MP PMT 1999]**

(a) Shape of semiconductor

(b) Atomic nature of semiconductor

(c) Length of semiconductor

(d) Shape and atomic nature of semiconductor

90. Electric current is due to drift of electrons in **[CPMT 1996]**

(a) Metallic conductors

(b) Semi-conductors

(c) Both (a) and (b)

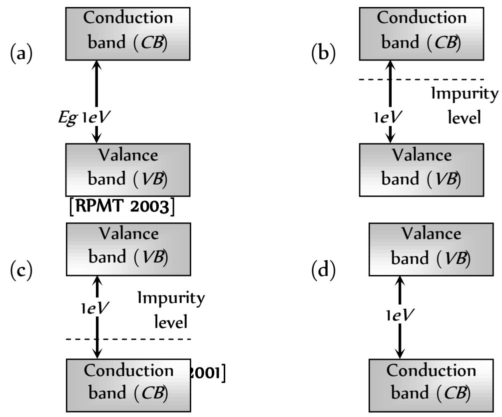
(d) None of these

91. The energy gap of silicon is 1.14 eV. The maximum wavelength at which silicon will begin absorbing energy is **[MP PMT 1993]**

(a) 10888 $Å$ (b) 1088.8 $Å$

(c) 108.88 $Å$ (d) 10.888 $Å$

92. Which of the following energy band diagram shows the N-type semiconductor **[RPET 1986]**



93. The mobility of free electron is greater than that of free holes because

(a) The carry negative charge

(b) They are light

(c) They mutually collide less

(d) They require low energy to continue their motion

94. The relation between the number of free electrons in semiconductors (n) and its temperature (T) is

(a) $n \propto T^2$ (b) $n \propto T$

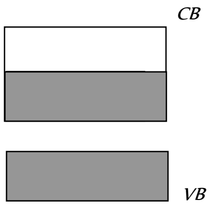
(c) $n \propto \sqrt{T}$ (d) $n \propto T^{3/2}$

95. The electron mobility in *N*-type germanium is 3900 $cm^2/v\text{-}s$ and its conductivity is 6.24 *mho/cm*, then impurity concentration will be if the effect of cotters is negligible

(a) $10^{15}\ cm^3$ (b) $10^{13}\ /cm^3$

(c) $10^{12}\ /cm^3$ (d) $10^{16}\ /cm^3$

96. Which of the energy band diagrams shown in the figure corresponds to that of a semiconductor [Orissa JEE 2003]

(a)


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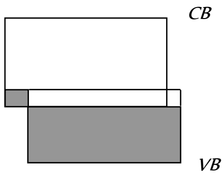
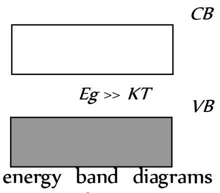
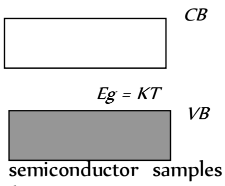
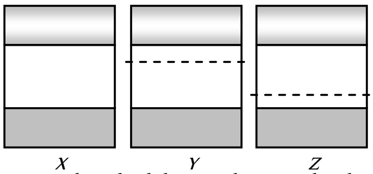


(c)


(d)


97. The energy band diagrams for three semiconductor samples of silicon are as shown. We can then assert that [Haryana CEE 1996]



(a) Sample *X* is undoped while samples *Y* and *Z* have been doped with a third group and a fifth group impurity respectively

(b) Sample *X* is undoped while both samples *Y* and *Z* have been doped with a fifth group impurity

(c) Sample *X* has been doped with equal amounts of third and fifth group impurities while samples *Y* and *Z* are undoped

(d) Sample *X* is undoped while samples *Y* and *Z* have been doped with a fifth group and a third group impurity respectively

98. Carbon, silicon and Germanium atoms have four valence electrons each. Their valence and conduction band are separated by energy band gaps represented by $(E_g)_C$. $(E_g)_{Si}$ and $(E_g)_{Ge}$ respectively. Which one of the following relationship is true in their case

(a) $(E_g)_C > (E_g)_{Si}$ (b) $(E_g)_C = (E_g)_{Si}$

(c) $(E_g)_C < (E_g)_{Ge}$ (d) $(E_g)_C < (E_g)_{Si}$

99. A semiconductor dopped with a donor impurity is [AFMC 2005]

(a) *P*-type (b) *N*-type

(c) *NPN* type (d) *PNP* type

100. In a semiconducting material the mobilities of electrons and holes are μ_e and μ_h respectively. Which of the following is true

(a) $\mu_e > \mu_h$ (b) $\mu_e < \mu_h$

(c) $\mu_e = \mu_h$ (d) $\mu_e < 0; \mu_h > 0$

101. Doping of intrinsic semiconductor is done [Orissa JEE 2005]

(a) To neutralize charge carriers

(b) To increase the concentration of majority charge carriers

(c) To make it neutral before disposal

(d) To carry out further purification

Semiconductor Diode

1. In the forward bias arrangement of a *PN*-junction diode [MP PMT 1994, 96, 99]

(a) The *N*-end is connected to the positive terminal of the battery

(b) The *P*-end is connected to the positive terminal of the battery

(c) The direction of current is from *N*-end to *P*-end in the diode

(d) The *P*-end is connected to the negative terminal of battery

2. In a *PN*-junction diode [MP PET 1993]

(a) The current in the reverse biased condition is generally very small

(b) The current in the reverse biased condition is small but the forward biased current is independent of the bias voltage

(c) The reverse biased current is strongly dependent on the applied bias voltage

(d) The forward biased current is very small in comparison to reverse biased current

3. The cut-in voltage for silicon diode is approximately

(a) 0.2 V (b) 0.6 V

(c) 1.1 V (d) 1.4 V

4. The electrical circuit used to get smooth dc output from a rectifier circuit is called [KCET 2003]

(a) Oscillator (b) Filter

(c) Amplifier (d) Logic gates

5. *PN*-junction diode works as a insulator, if connected [CPMT 1987]

(a) To A.C. (b) In forward bias

(c) In reverse bias (d) None of these

6. The reverse biasing in a *PN* junction diode [MP PMT 1991; EAMCET 1994; CBSE PMT 2003]

(a) Decreases the potential barrier

(b) Increases the potential barrier

(c) Increases the number of minority charge carriers

[CBSE PMT 2005]

(d) Increases the number of majority charge carriers

7. The electrical resistance of depletion layer is large because

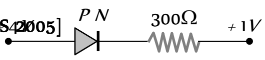
(a) It has no charge carriers

(b) It has a large number of charge carriers

(c) It contains electrons as charge carriers

(d) It has holes as charge carriers

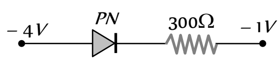
8. In the circuit given below, the value of the current is [AIIMS 2005]



(a) 0 *amp* (b) 10^{-2} *amp*

(c) 10^2 *amp* (d) 10^{-3} *amp*

9. What is the current in the circuit shown below [AFMC 2000; RPMT 2001]



(a) 0 *amp* (b) 10^{-2} *amp*

(c) 1 *amp* (d) 0.10 *amp*

10. If the forward voltage in a semiconductor diode is doubled, the width of the depletion layer will [MP PMT 1996]

(a) Become half (b) Become one-fourth

(c) Remain unchanged (d) Become double

11. The *PN* junction diode is used as [CPMT 1972; AFMC 1997; CBSE PMT 1999; AIIMS 1999; RPMT 2000; MP PMT 04]

(a) An amplifier (b) A rectifier

(c) An oscillator (d) A modulator

12. When a *PN* junction diode is reverse biased

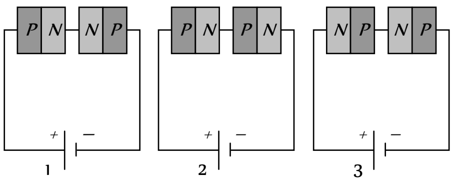
(a) Electrons and holes are attracted towards each other and move towards the depletion region

(b) Electrons and holes move away from the junction depletion region

(c) Height of the potential barrier decreases

(d) No change in the current takes place

13. Two *PN*-junctions can be connected in series by three different methods as shown in the figure. If the potential difference in the junctions is the same, then the correct connections will be



(a) In the circuit (1) and (2) (b) In the circuit (2) and (3)

(c) In the circuit (1) and (3) (d) Only in the circuit (1)

14. A *PN*- junction has a thickness of the order of [BIT 1990]

(a) $1\,cm$ (b) $1\,mm$

(c) $10^{-6} m$ (d) $10^{-12}\,cm$

15. In the depletion region of an unbiased *P-N* junction diode there are [KCET 1999; CBSE PMT 1999; RPMT 2001; MP PMT 1994, 2003]

(a) Only electrons

(b) Only holes

(c) Both electrons and holes

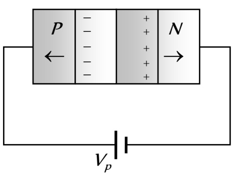
(d) Only fixed ions

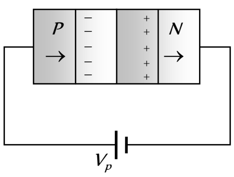
16. On increasing the reverse bias to a large value in a *PN*-junction diode, current [MP PMT 1994; BHU 2002]

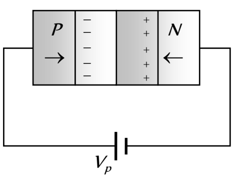
(a) Increases slowly (b) Remains fixed

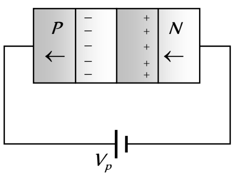
(c) Suddenly increases (d) Decreases slowly

17. In the case of forward biasing of *PN*-junction, which one of the following figures correctly depicts the direction of flow of carriers

(a) 

(b) 

(c) 

(d) 

18. Which of the following statements concerning the depletion zone of an unbiased *PN* junction is (are) true [IIT-JEE 1995]

(a) The width of the zone is independent of the densities of the dopants (impurities)

(b) The width of the zone is dependent on the densities of the dopants

(c) The electric field in the zone is produced by the ionized dopant atoms

(d) The electric field in the zone is provided by the electrons in the conduction band and the holes in the valence band

19. A semiconductor device is connected in a series circuit with a battery and a resistance. A current is found to pass through the circuit. If the polarity of the battery is reversed, the current drops almost to zero. The device may be [MP PET 1995; CBSE PMT 1998]

(a) A *P*-type semiconductor (b) An *N*-type semiconductor

(c) A *PN*-junction[IIT-JEE 1989] (d) An intrinsic semiconductor

20. The approximate ratio of resistances in the forward and reverse bias of the *PN*-junction diode is [MP PET 2000; MP PMT 1999, 2002, 03; Pb. PMT 2003]

(a) $10^2 : 1$ (b) $10^{-2} : 1$

(c) $1 : 10^{-4}$ (d) $1 : 10^4$

21. In a junction diode, the holes are due to [CBSE PMT 1999; Pb. PMT 2003]

(a) Protons (b) Neutrons

(c) Extra electrons (d) Missing of electrons

22. In forward bias, the width of potential barrier in a *P-N* junction diode [EAMCET (Engg.) 1995; CBSE PMT 1999 RPMT 1997, 2002, 03]

(a) Increases

(b) Decreases

(c) Remains constant

(d) First increases then decreases

23. The cause of the potential barrier in a *P-N* diode is [CBSE PMT 1998; RPMT 2001]

(a) Depletion of positive charges near the junction

(b) Concentration of positive charges near the junction

(c) Depletion of negative charges near the junction

(d) Concentration of positive and negative charges near the junction

24. In a *PN*[CBSE PMT 1998] junction diode not connected to any circuit [IIT-JEE 1998]

(a) The potential is the same everywhere

(b) The *P*-type is a higher potential than the *N*-type side

(c) There is an electric field at the junction directed from the *N*-type side to the *P*- type side

(d) There is an electric field at the junction directed from the *P*-type side to the *N*-type side

25. Which of the following statements is not true

[IIT-JEE 1997 Re-Exam]

(a) The resistance of intrinsic semiconductors decrease with increase of temperature

(b) Doping pure Si with trivalent impurities give *P*-type semiconductors

(c) The majority carriers in *N*-type semiconductors are holes

(d) A *PN*-junction can act as a semiconductor diode

26. The dominant mechanisms for motion of charge carriers in forward and reverse biased silicon *P-N* junctions are

[IIT-JEE 1997 Cancelled; RPMT 2000; AIIMS 2000]

(a) Drift in forward bias, diffusion in reverse bias

(b) Diffusion in forward bias, drift in reverse bias

(c) Diffusion in both forward and reverse bias

(d) Drift in both forward and reverse bias

27. In *P-N* junction, avalanche current flows in circuit when biasing is

(a) Forward (b) Reverse

(c) Zero (d) Excess

28. The depletion layer in the *P-N* junction region is caused by

[CBSE PMT 1994]

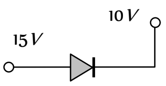
(a) Drift of holes

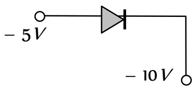
(b) Diffusion of charge carriers

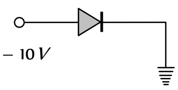
(c) Migration of impurity ions

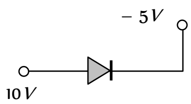
(d) Drift of electrons

29. Which one is reverse-biased [DCE 1999]

(a) 

(b) 

(c) 

(d) 

30. In a *P-N* junction diode if *P* region is heavily doped than *n* region then the depletion layer is [RPMT 1999]

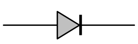
(a) Greater in *P* region

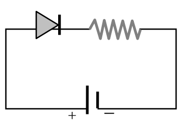
(b) Greater in *N* region

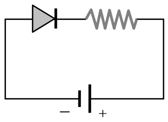
(c) Equal in both region

(d) No depletion layer is formed in this case

31. Which one is in forward bias [RPMT 2000]

(a) 

(b) 

(c) 

(d) None of these

32. The reason of current flow in *P-N* junction in forward bias is

[RPMT 2000]

(a) Drifting of charge carriers

(b) Minority charge carriers

(c) Diffusion of charge carriers

(d) All of these

33. The resistance of a reverse biased *P-N* junction diode is about

(a) 1 *ohm* (b) 10^2 *ohm*

(c) 10^3 *ohm* (d) 10^6 *ohm*

34. Consider the following statements *A* and *B* and identify the correct choice of the given answers

A: The width of the depletion layer in a *P-N* junction diode increases in forwards bias

B: In an intrinsic semiconductor the fermi energy level is exactly in the middle of the forbidden gap

[EAMCET (Engg.) 2000]

(a) *A* is true and *B* is false (b) Both *A* and *B* are false

(c) *A* is false and *B* is true (d) Both *A* and *B* are true

[RPET 1997]

35. In comparison to a half wave rectifier, the full wave rectifier gives lower [AFMC 2001]

(a) Efficiency (b) Average *dc*

(c) Average output voltage (d) None of these

36. Avalanche breakdown is due to [RPMT 2001]

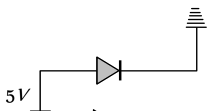
(a) Collision of minority charge carrier

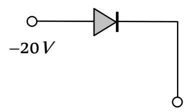
(b) Increase in depletion layer thickness

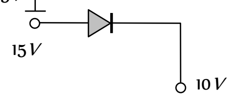
(c) Decrease in depletion layer thickness

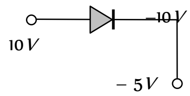
(d) None of these

37. Which is reverse biased diode [DCE 2001]

(a) 

(b) 

(c) 

(d) 

38. Zener breakdown in a semi-conductor diode occurs when

[UPSEAT 2002]

(a) Forward current exceeds certain value

(b) Reverse bias exceeds certain value

(c) Forward bias exceeds certain value

(d) Potential barrier is reduced to zero

39. When forward bias is applied to a *P-N* junction, then what happens to the potential barrier V_B, and the width of charge depleted region x [UPSEAT 2002, 03; Roorkee 1999; RPET 2003; AIEEE 2004]

(a) V_B increases, x decreases

(b) V_B decreases, x increases

(c) V_B increases, x increases

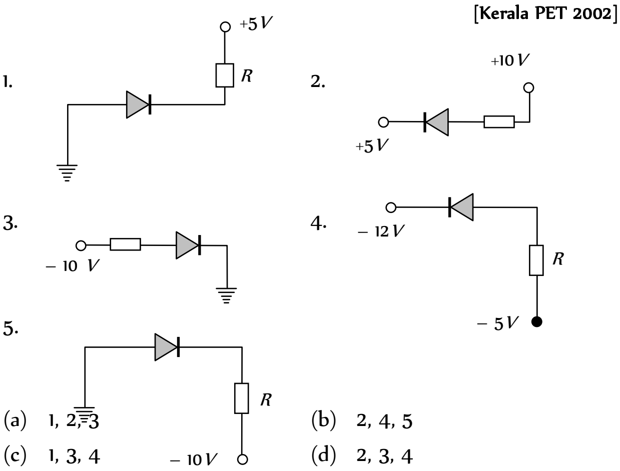
(d) V_B decreases, x decreases

40. The potential barrier, in the depletion layer, is due to [EAMCET (Engg.) 1998; Pb. PMT 1999; Pb. PET 2001; AIIMS 2002]

(a) Ions (b) Holes

(c) Electrons (d) Both (b) and (c)

41. In the given figure, which of the diodes are forward biased ? [Kerala PET 2002]



(a) 1, 2, 3 (b) 2, 4, 5

(c) 1, 3, 4 (d) 2, 3, 4

42. Function of rectifier is [AFMC 2002, 04]

(a) To convert *ac* into *dc* (b) To convert *dc* into *ac*

(c) Both (a) and (b) (d) None of these

43. When the *P* end of *P-N* junction is connected to the negative terminal of the battery and the *N* end to the positive terminal of the battery, then the *P-N* junction behaves like

(a) A conductor (b) An insulator

(c) A super-conductor (d) A semi-conductor

44. If the two ends *P* and *N* of a *P-N* diode junction are joined by a wire [MP PMT 2002]

(a) There will not be a steady current in the circuit

(b) There will be a steady current from *N* side to *P* side

(c) There will be a steady current from *P* side to *N* side

(d) There may not be a current depending upon the resistance of the connecting wire

45. A potential barrier of 0.50 *V* exists across a *P-N* junction. If the depletion region is 5.0×10^{-7} m wide, the intensity of the electric field in this region is [UPSEAT 2002]

(a) $1.0\times10^{6}\ V/m$ (b) $1.0\times10^{5}\ V/m$

(c) $2.0\times10^{5}\ V/m$ (d) $2.0\times10^{6}\ V/m$

46. If no external voltage is applied across *P-N* junction, there would be

(a) No electric field across the junction

(b) An electric field pointing from *N*-type to *P*-type side across the junction

(c) An electric field pointing from *P*-type to *N*-type side across the junction

(d) A temporary electric field during formation of *P-N* junction that would subsequently disappear

47. In a *PN*-junction [CBSE PMT 2002]

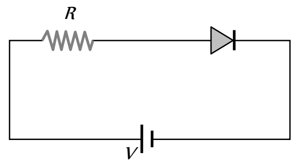
(a) *P* and *N* both are at same potential

(b) High potential at *N* side and low potential at *P* side

(c) High potential at *P* side and low potential at *N* side

(d) Low potential at *N* side and zero potential at *P* side

48. For the given circuit of *PN*-junction diode, which of the following statement is correct [CBSE PMT 2002]



(a) In forward biasing the voltage across *R* is *V*

(b) In forward biasing the voltage across *R* is 2*V*

(c) In reverse biasing the voltage across *R* is *V*

(d) In reverse biasing the voltage across *R* is 2*V*

49. On adjusting the *P-N* junction diode in forward biased [RPET 2003]

(a) Depletion layer increases (b) Resistance increases

(c) Both decreases (d) None of these

50. In the middle of the depletion layer of a reverse-biased *PN* junction, the [AIEEE 2003]

(a) Potential is zero (b) Electric field is zero

(c) Potential is maximum (d) Electric field is maximum

51. Barrier potential of a *P-N* junction diode does not depend on

(a) Temperature (b) Forward bias

(c) Doping density [MP PET 2002] (d) Diode design

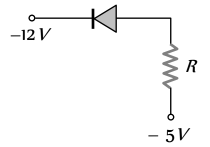
52. A crystal diode is a [MP PET 2004]

(a) Non-linear device (b) Amplifying device

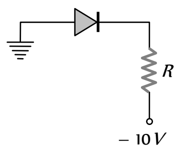
(c) Linear device (d) Fluctuating device

53. Of the diodes shown in the following diagrams, which one is reverse biased [CBSE PMT 2004]

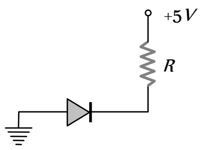
(a)



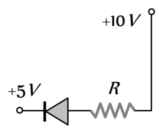
(b)



(c)



(d)



54. In a *PN* junction photo cell [Orissa JEE 2002], the value of photo-electromotive force produced by monochromatic light is proportional to [CBSE PMT 2004]

(a) The voltage applied at the *PN* junction

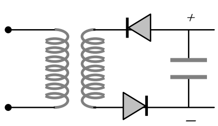
(b) The barrier voltage at the *PN* junction

(c) The intensity of the light falling on the cell

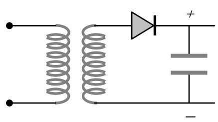
(d) The frequency of the light falling on the cell

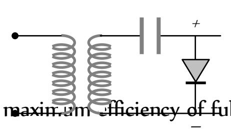
55. Which is the correct diagram of a half-wave rectifier [Orissa PMT 2004]

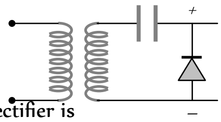
(a)



(b)



(c) 

(d) 

56. The maximum efficiency of full wave rectifier is [J & K CET 2004]

(a) 100% (b) 25.20%

(c) 40.2% (d) 81.2%

57. Serious draw back of the semiconductor device is [Pb. PMT 2004]

(a) They cannot be used with high voltage

(b) They pollute the environment

(c) They are costly

(d) They do not last for long time

58. Select the correct statement [RPMT 2003]

(a) In a full wave rectifier, two diodes work alternately

(b) In a full wave rectifier, two diodes work simultaneously

(c) The efficiency of full wave and half wave rectifiers is same

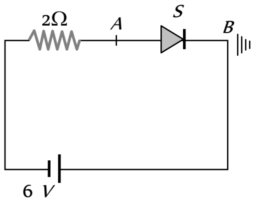
(d) The full wave rectifier is bi-directional.

59. In order to forward bias a *PN* junction, the negative terminal of battery is connected to [RPMT 2003]

(a) *P*–side (b) Either *P*–side or *N*–side

(c) *N*–side (d) None of these

60. The diode shown in the circuit is a silicon diode. The potential difference between the points *A* and *B* will be [RPMT 2002]



(a) 6 *V*

(b) 0.6 *V*

(c) 0.7 *V*

(d) 0 *V*

61. Zener breakdown takes place if [RPMT 2000]

(a) Doped impurity is low (b) Doped impurity is high

(c) Less impurity in *N*-part (d) Less impurity in *P*-type

62. Consider the following statements *A* and *B* and identify the correct choice of the given answers

(A) A zener diode is always connected in reverse bias

(B) The potential barrier of a *PN* junction lies between 0.1 to 0.3 *V* approximately [EAMCET 2000]

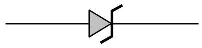
(a) *A* and *B* are correct

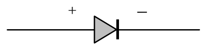
(b) *A* and *B* are wrong

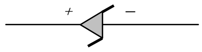
(c) *A* is correct but *B* is wrong

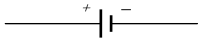
(d) *A* is wrong but *B* is correct

63. The correct symbol for zener diode is [RPMT 2000]

(a) 

(b) 

(c) 

(d) 

64. Which one of the following statements is not correct [SCRA 2000]

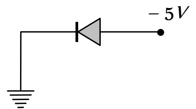
(a) A diode does not obey Ohm's law

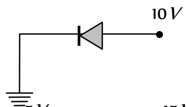
(b) A *PN* junction diode symbol shows an arrow identifying the direction of current (forward) flow

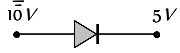
(c) An ideal diode is an open switch

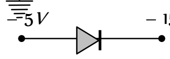
(d) An ideal diode is an ideal one way conductor

65. Which of the following semi-conductor diodes is reverse biased

(a) 

(b) 

(c) 

(d) 

66. No bias is applied to a *P-N* junction, then the current [RPMT 1999]

(a) Is zero because the number of charge carriers flowing on both sides is same

(b) Is zero because the charge carriers do not move

(c) Is non-zero

(d) None of these

67. Zener diode is used as [CBSE PMT 1999]

(a) Half wave rectifier (b) Full wave rectifier

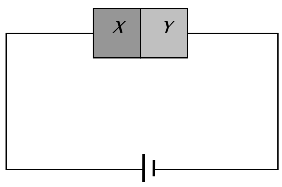
(c) ac voltage stabilizer (d) dc voltage stabilizer

68. The width of forbidden gap in silicon crystal is 1.1 *eV*. When the crystal is converted in to a *N*-type semiconductor the distance of Fermi level from conduction band is [EAMCET (Med.) 1999]

(a) Greater than 0.55 *eV* (b) Equal to 0.55 *eV*

(c) Lesser than 0.55 *eV* (d) Equal to 1.1 *eV*

69. A semiconductor *X* is made by doping a germanium crystal with arsenic (Z = 33). A second semiconductor *Y* is made by doping germanium with indium (Z = 49). The two are joined end to end and connected to a battery as shown. Which of the following statements is correct [Orissa JEE 1998]



(a) *X* is *P*-type, *Y* is *N*-type and the junction is forward biased

(b) *X* is *N*-type, *Y* is *P*-type and the junction is forward biased

(c) *X* is *P*-type, *Y* is *N*-type and the junction is reverse biased

(d) *X* is *N*-type, *Y* is *P*-type and the junction is reverse biased

70. In *P-N* junction, the barrier potential offers resistance to [AMU 1995, 96]

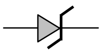
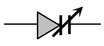
(a) Free electrons in *N* region and holes in *P* region

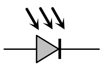
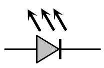
(b) Free electrons in *P* region and holes in *N* region

(c) Only free electrons in *N* region

(d) Only holes in *P* region

71. Symbolic representation of photodiode is [RPMT 1995]

(a)  (b) 

(c)  (d) 

72. To make a *PN* junction conducting [IIT-JEE 1994]

(a) The value of forward bias should be more than the barrier potential

(b) The value of forward bias should be less than the barrier potential

(c) The value of reverse bias should be more than the barrier potential

(d) The value of reverse bias should be less than the barrier potential

73. Which is the wrong statement in following sentences? A device in which *P* and *N*-type semiconductors are used is more useful then a vacuum type because [MP PET 1992]

(a) Power is not necessary to heat the filament

(b) It is more stable

(c) Very less heat is produced in it

(d) Its efficiency is high due to a high voltage across the junction

74. The depletion layer in silicon diode is 1 μm wide and the knee potential is 0.6 *V*, then the electric field in the depletion layer will be

(a) Zero

(b) 0.6 *Vm*

(c) 6×10 *V/m*

(d) 6×10 *V/m*

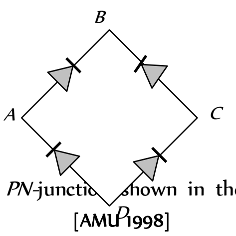
75. In the diagram, the input is across the terminals *A* and *C* and the output is across the terminals *B* and *D*, then the output is

(a) Zero

(b) Same as input

(c) Full wave rectifier

(d) Half wave rectifier



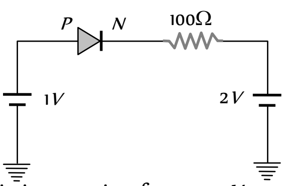
76. The current through an ideal *PN*-junction shown in the following circuit diagram will be [AMU 1998]

(a) Zero

(b) 1 *mA*

(c) 10 *mA*

(d) 30 *mA*



77. If a full wave rectifier circuit is operating from 50 *Hz* mains, the fundamental frequency in the ripple will be

[UPSEAT 2000; CBSE PMT 2003; AIEEE 2005]

(a) 50 *Hz* (b) 70.7 *Hz*

(c) 100 *Hz* (d) 25 *Hz*

78. In a full wave rectifiers, input *ac* current has a frequency 'ν'. The output frequency of current is [BHU 2005]

(a) $\nu/2$ (b) ν

(c) 2ν (d) None of these

79. A diode having potential difference 0.5 *V* across its junction which does not depend on current, is connected in series with resistance of 20 Ω across source. If 0.1 *A* passes through resistance then what is the voltage of the source

[DCE 2005]

(a) 1.5 *V* (b) 2.0 *V*

(c) 2.5 *V* (d) 5 *V*

Junction Transistor

1. When *NPN* transistor is used as an amplifier

[AIEEE 2004]

(a) Electrons move from base to collector

(b) Holes move from emitter to base

(c) Electrons move from collector to base

(d) Holes move from base to emitter

2. The phase difference between input and output voltages of a CE circuit is [MP PET 2004]

(a) 0 (b) 90

(c) 180 (d) 270

3. An oscillator is nothing but an amplifier with

[MP PET 2004]

(a) Positive feed back (b) Large gain

(c) No feedback (d) Negative feedback

4. The emitter-base junction of a transistor is ...... biased while the collector-base junction is .... biased

[CBSE PMT 1994] [KCET 2004]

(a) Reverse, forward (b) Reverse, reverse

(c) Forward, forward (d) Forward, reverse

5. In an *NPN* transistor the collector current is 24 *mA*. If 80% of electrons reach collector its base current in *mA* is

[Kerala PMT 2004]

(a) 36 (b) 26

(c) 16 (d) 6

6. A *NPN* transistor conducts when [CPMT 2003]

(a) Both collector and emitter are positive with respect to the base

(b) Collector is positive and emitter is negative with respect to the base

(c) Collector is positive and emitter is at same potential as the base

(d) Both collector and emitter are negative with respect to the base

7. In the case of constants α and β of a transistor [CET 2003]

(a) $\alpha = \beta$ (b) $\beta < 1$ $\alpha > 1$

(c) $\alpha\beta = 1$ (d) $\beta > 1$ $\alpha < 1$

8. Which of the following is true [DPMT 2002]

(a) Common base transistor is commonly used because current gain is maximum

(b) Common emitter is commonly used because current gain is maximum

(c) Common collector is commonly used because current gain is maximum

(d) Common emitter is the least used transistor

9. If $\alpha = 0.98$ and current through emitter $i_e = 20$ mA, the value of β is [DPMT 2002]

(a) 4.9 (b) 49

(c) 96 (d) 9.6

10. For a common base configuration of PNP transistor $\frac{I_C}{I_E} = 0.98$ then maximum current gain in common emitter configuration will be [CBSE PMT 2002]

(a) 12 (b) 24

(c) 6 (d) 5

11. In a PNP transistor working as a common-base amplifier, current gain is 0.96 and emitter current is 7.2 mA. The base current is [AFMC 2002; Pb. PET 2002]

(a) 0.4 mA (b) 0.2 mA

(c) 0.29 mA (d) 0.35 mA

12. If l_1, l_2, l_3 are the lengths of the emitter, base and collector of a transistor then [KCET 2002]

(a) $l_1 = l_2 = l_3$ (b) $l_3 < l_2 > l_1$

(c) $l_3 < l_1 < l_2$ (d) $l_3 > l_1 > l_2$

13. In an NPN transistor circuit, the collector current is 10 mA. If 90% of the electrons emitted reach the collector, the emitter current (i_e) and base current (i_b) are given by [KCET 2001]

(a) $i_e = -1$ mA, $i_b = 9$ mA

(b) $i_e = 9$ mA, $i_b = -1$ mA

(c) $i_e = 1$ mA, $i_b = 11$ mA

(d) $i_e = 11$ mA, $i_b = 1$ mA

14. In a common emitter transistor, the current gain is 80. What is the change in collector current, when the change in base current is 250 μA [CBSE PMT 2000]

(a) 80×250 μA (b) $(250 - 80)$ μA

(c) $(250 + 80)$ μA (d) 250/80 μA

15. Least doped region in a transistor [KCET 2000]

(a) Either emitter or collector

(b) Base

(c) Emitter

(d) Collector

16. The transistors provide good power amplification when they are used in [AMU 1999]

(a) Common collector configuration

(b) Common emitter configuration

(c) Common base configuration

(d) None of these

17. The transfer ratio of a transistor is 50. The input resistance of the transistor when used in the common-emitter configuration is 1 $K\Omega$. The peak value for an A.C input voltage of 0.01 V peak is

(a) 100 μA (b) 0.01 mA

(c) 0.25 mA (d) 500 μA

18. For a transistor the parameter $\beta = 99$. The value of the parameter α is [Pb CET 1998]

(a) 0.9 (b) 0.99

(c) 1 (d) 9

19. A transistor is used in common emitter mode as an amplifier. Then

(a) The base-emitter junction is forward biased

(b) The base-emitter junction is reverse biased

(c) The input signal is connected in series with the voltage applied to the base-emitter junction

(d) The input signal is connected in series with the voltage applied to bias the base collector junction

20. In a PNP transistor the base is the N-region. Its width relative to the P-region is [DCE 1997]

(a) Smaller (b) Larger

(c) Same (d) Not related

21. A common emitter amplifier is designed with NPN transistor ($\alpha = 0.99$). The input impedance is 1 $K\Omega$ and load is 10 $K\Omega$. The voltage gain will be [CPMT 1996]

(a) 9.9 (b) 99

(c) 990 (d) 9900

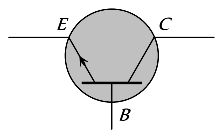
22. The symbol given in figure represents [AMU 1995, 96]

(a) NPN transistor

(b) PNP transistor

(c) Forward biased PN junction diode

(d) Reverse biased NP junction diode

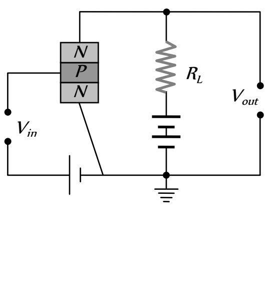


23. The most commonly used material for making transistor is [MNR 1995]

(a) Copper (b) Silicon

(c) Ebonite (d) Silver

24. An NPN-transistor circuit is arranged as shown in figure. It is [BHU 1994]



(a) A common base amplifier circuit

(b) A common emitter amplifier circuit

(c) A common collector amplifier circuit

(d) Neither of the above

25. The part of a transistor which is heavily doped to produce a large number of majority carriers, is [CBSE PMT 1993]

(a) Base (b) Emitter

(c) Collector (d) None of these

26. For a transistor, the current amplification factor is 0.8. The transistor is connected in common emitter configuration. The change in the collector current when the base current changes by 6 *mA* is [Haryana CET 1991]

(a) 6 *mA* (b) 4.8 *mA*

(c) 24 *mA* (d) 8 *mA*

27. In a common base amplifier circuit, calculate the change in base current if that in the emitter current is 2 *mA* and $\alpha = 0.98$

[BHU 1995]

(a) 0.04 *mA* (b) 1.96 *mA*

(c) 0.98 *mA* (d) 2 *mA*

28. In case of *NPN*-transistors the collector current is always less than the emitter current because [AIIMS 1983]

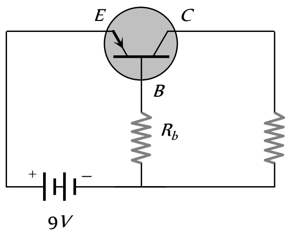
(a) Collector side is reverse biased and emitter side is forward biased

(b) After electrons are lost in the base and only remaining ones reach the collector

(c) Collector side is forward biased and emitter side is reverse biased

(d) Collector being reverse biased attracts less electrons

29. In a transistor circuit shown here the base current is 35 μA. The value of the resistor R is



(a) 123.5 $k\Omega$

(b) 257 $k\Omega$

(c) 380.05 $k\Omega$

(d) None of these

30. In a transistor, a change of 8.0 *mA* in the emitter current produces a change of 7.8 *mA* in the collector current. What change in the base current is necessary to produce the same change in the collector current

(a) 50 μA (b) 100 μA

(c) 150 μA (d) 200 μA

31. In a transistor configuration β-parameter is

[Orissa PMT 2004]

(a) $\frac{l_b}{l_c}$ (b) $\frac{l_c}{l_b}$

(c) $\frac{l_c}{l_a}$ (d) $\frac{l_a}{l_c}$

32. Which of these is unipolar transistor [Pb PMT 2004]

(a) Point contact transistor (b) Field effect transistor

(c) *PNP* transistor (d) None of these

33. For a transistor, in a common emitter arrangement, the alternating current gain β is given by [DPMT 2004]

(a) $\beta = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_C}$ (b) $\beta = \left(\frac{\Delta I_B}{\Delta I_C}\right)_{V_C}$

(c) $\beta = \left(\frac{\Delta I_C}{\Delta I_E}\right)_{V_C}$ (d) $\beta = \left(\frac{\Delta I_E}{\Delta I_C}\right)_{V_C}$

34. The relation between α and β parameters of current gains for a transistors is given by [Pb. PET 2000]

(a) $\alpha = \frac{\beta}{1-\beta}$ (b) $\alpha = \frac{\beta}{1+\beta}$

(c) $\alpha = \frac{1-\beta}{\beta}$ (d) $\alpha = \frac{1+\beta}{\beta}$

35. When *NPN* transistor is used as an amplifier [DCE 2002]

(a) Electrons move from base to emitter

(b) Electrons move from emitter to base

(c) Electrons moves from base to emitter

(d) Holes moves from base to emitter

36. In the *CB* mode of a transistor, when the collector voltage is changed by 0.5 *volt*. The collector current changes by 0.05 *mA*. The output resistance will be [Pb. PMT 2003]

(a) 10 $k\Omega$ (b) 20 $k\Omega$

(c) 5 $k\Omega$ (d) 2.5 $k\Omega$

37. Which of the following is used to produce radio waves of constant amplitude [DCE 2004]

(a) Oscillator (b) FET

(c) Rectifier (d) Amplifier

38. While a collector to emitter voltage is constant in a transistor, the collector current changes by 8.2 *mA* when the emitter current changes by 8.3 *mA*. The value of forward current ratio h is

(a) 82 (b) 83

(c) 8.2 (d) 8.3

39. Consider an *NPN* transistor amplifier in common-emitter configuration. The current gain of the transistor is 100. If the collector current changes by 1 *mA*, what will be the change in emitter current [AIIMS 2005]

(a) 1.1 *mA* (b) 1.01 *mA*

(c) 0.01 *mA* (d) 10 *mA*

40. In a common base amplifier the phase difference between the input signal voltage and the output voltage is

[CBSE PMT 1990; AIEEE 2005]

(a) 0 (b) $\pi / 4$

(c) $\pi / 2$ (d) π

41. In *NPN* transistor the collector current is 10 *mA*. If 90% of electrons emitted reach the collector, then [Kerala PMT 2005]

(a) Emitter current will be 9 *mA*
(b) Emitter current will be 11.1 *mA*
(c) Base current will be 0.1 *mA*
(d) Base current will be 0.01 *mA*

42. *NPN* transistor are preferred to *PNP* transistor because they have [J & K CET 2005]

(a) Low cost
(b) Low dissipation energy
(c) Capability of handing large power
(d) Electrons having high mobility than holes

43. In a transistor in CE configuration, the ratio of power gain to voltage gain is [J & K CET 2005]

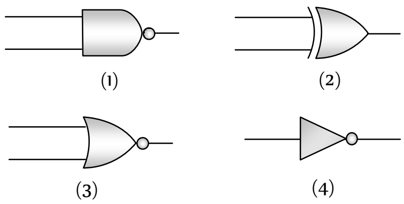
(a) α (b) β / α
(c) $\beta\alpha$ (d) β

44. In the study of transistor as an amplifier, if $\alpha = I_c / I_e$ and $\beta = I_c / I_b$, where I_c, I_b and I_e are the collector, base and emitter currents, then [CBSE PMT 2000; KCET 2000; Orissa JEE 2005]

(a) $\beta = \frac{1-\alpha}{\alpha}$ (b) $\beta = \frac{\alpha}{1-\alpha}$
(c) $\beta = \frac{\alpha}{1+\alpha}$ (d) $\beta = \frac{1+\alpha}{\alpha}$

Digital Electronics

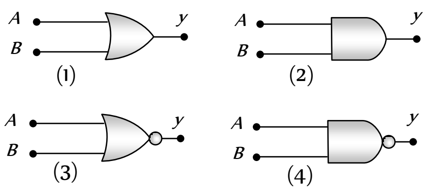
1. Given below are symbols for some logic gates



The XOR gate and NOR gate respectively are [AFMC 1994]

(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1 and 4

2. Given below are four logic gate symbol (figure). Those for OR, NOR and NAND are respectively [NSEP 1994]



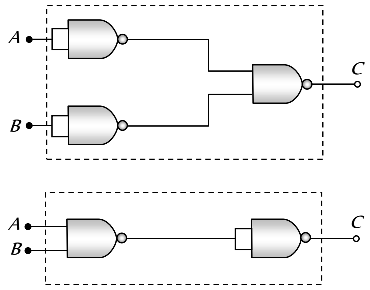
(a) 1, 4, 3 (b) 4, 1, 2
(c) 1, 3, 4 (d) 4, 2, 1

3. The following truth table corresponds to the logic gate [BHU 1994; CPMT 2000; J & K CET 2004]

A	0	0	1	1
B	0	1	0	1
X	0	1	1	1

(a) NAND (b) OR
(c) AND (d) XOR

4. The combination of 'NAND' gates shown here under (figure) are equivalent to [Haryana CEET 1996]



(a) An OR gate and an AND gate respectively
(b) An AND gate and a NOT gate respectively
(c) An AND gate and an OR gate respectively
(d) An OR gate and a NOT gate respectively.

5. A truth table is given below. Which of the following has this type of truth table [CBSE PMT 1996; UPSEAT 2002]

A	0	1	0	1
B	0	0	1	1
y	1	0	0	0

(a) XOR gate (b) NOR gate
(c) AND gate (d) OR gate

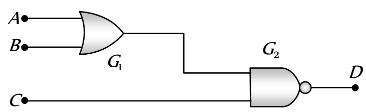
6. The truth table shown in figure is for [Pb. CET 1998]

A	0	0	1	1
B	0	1	0	1
Y	1	0	0	1

(a) XOR (b) AND
(c) XNOR (d) OR

7. For the given combination of gates, if the logic states of inputs *A*, *B*, *C* are as follows *A* = *B* = *C* = 0 and *A* = *B* = 1, *C* = 0 then the logic states of output *D* are

(a) 0, 0
(b) 0, 1
(c) 1, 0
(d) 1, 1



8. Boolean algebra is essentially based on [AIIMS 1999]

(a) Truth (b) Logic
(c) Symbol (d) Numbers

9. The logic behind 'NOR' gate is that it gives [CPMT 1999, AFMC 1999]

(a) High output when both the inputs are low
(b) Low output when both the inputs are low
(c) High output when both the inputs are high

(d) None of these

10. A logic gate is an electronic circuit which **[BHU 2000]**

(a) Makes logic decisions

(b) Allows electrons flow only in one direction

(c) Works binary algebra

(d) Alternates between 0 and 1 values

11. A gate has the following truth table **[CBSE PMT 2000]**

P	1	1	0	0
Q	1	0	1	0
R	1	0	0	0

The gate is

(a) NOR (b) OR

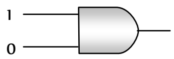
(c) NAND (d) AND

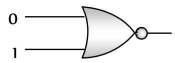
12. How many NAND gates are used to form an AND gate **[MP PET 2004]**

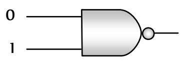
(a) 1 (b) 2

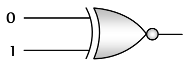
(c) 3 (d) 4

13. Which of the following gates will have an output of 1 **[CBSE PMT 1998]**

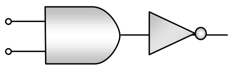
(a) 

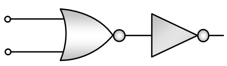
(b) 

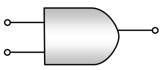
(c) 

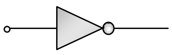
(d) 

14. Which represents NAND gate **[DCE 2002]**

(a) 

(b) 

(c) 

(d) 

15. The given truth table is of **[AMU 1998; J & K CET 2002]**

A	X
0	1
1	0

(a) OR gate (b) AND gate

(c) NOT gate (d) None of above

16. What will be the input of A and B for the Boolean expression $\overline{(A+B)}\cdot\overline{(A\cdot B)}=1$ **[TNPCEE 2002]**

(a) 0, 0 (b) 0, 1

(c) 1, 0 (d) 1, 1

17. If A and B are two inputs in AND gate, then AND gate has an output of 1 when the values of A and B are **[TNPCEE 2002]**

(a) $A = 0,\ B = 0$ (b) $A = 1,\ B = 1$

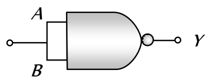
(c) $A = 1,\ B = 0$ (d) $A = 0,\ B = 1$

18. The Boolean equation of NOR gate is **[Haryana CET 2002]**

(a) $C = A + B$ (b) $C = \overline{A+B}$

(c) $C = A\cdot B$ (d) $C = \overline{A\cdot B}$

19. This symbol represents **[CBSE PMT 1996]**



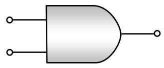
(a) NOT gate

(b) OR gate

(c) AND gate

(d) NOR gate

20. Which logic gate is represented by following diagram **[DCE 2001]**

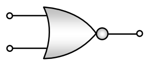


(a) AND

(b) OR

(c) NOR

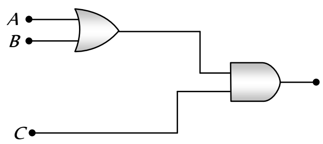
(d) XOR

21. Symbol  represents **[Kerala PMT 2001]**

(a) NAND gate (b) NOR gate

(c) NOT gate (d) XNOR gate

22. To get an output 1 from the circuit shown in the figure, the input must be **[UPSEAT 2002]**



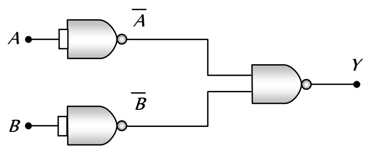
(a) $A=0,\ B=1,\ C=0$

(b) $A=1,\ B=0,\ C=0$

(c) $A=1,\ B=0,\ C=1$

(d) $A=1,\ B=1,\ C=0$

23. The combination of the gates shown in the figure below produces



(a) NOR gate

(b) OR gate

(c) AND gate

(d) XOR gate

24. The output of a NAND gate is 0 **[UPSEAT 2004]**

(a) If both inputs are 0

(b) If one input is 0 and the other input is 1

(c) If both inputs are 1

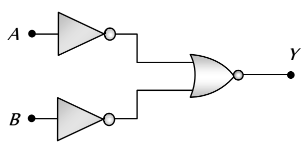
(d) Either if both inputs are 1 or if one of the inputs is 1 and the other 0

25. A gate in which all the inputs must be low to get a high output is called **[UPSEAT 2004]**

(a) A NAND gate (b) An inverter

(c) A NOR gate (d) An AND gate

26. Which logic gate is represented by the following combination of logic gates **[AIIMS 2004]**



(a) OR (b) NAND

(c) AND (d) NOR

27. The output of OR gate is 1 [CBSE PMT 2004]

(a) If both inputs are zero

(b) If either or both inputs are 1

(c) Only if both input are 1

(d) If either input is zero

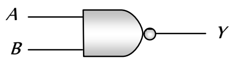
28. Which gates is represented by this figure [DCE 2003]

(a) NAND gate

(b) AND gate

(c) NOT gate

(d) OR gate



29. Sum of the two binary numbers $(1000010)_2$ and $(11011)_2$ is

(a) $(111101)_2$ (b) $(111111)_2$

(c) $(101111)_2$ (d) $(111001)_2$

30. The truth-table given below is for which gate

[CBSE PMT 1994, 98 2002; DPMT 2002; BCECE 2005]

A	0	0	1	1
B	0	1	0	1
C	1	1	1	0

(a) XOR (b) OR

(c) AND (d) NAND

31. Which of the following logic gate is an universal gate

[AIIMS 2005]

(a) OR (b) NOT

(c) AND (d) NOR

Valve Electronics (Diode and Triode)

1. Thermionic emission from a heated filament varies with its temperature *T* as

[CBSE PMT 1990; RPMT 2000; CPMT 2002]

(a) T^{-1} (b) T

(c) T^2 (d) $T^{3/2}$

2. Number of secondary electrons emitted per number of primary electrons depends on [RPET 2000]

(a) Material of target

(b) Frequency of primary electrons

(c) Intensity

(d) None of the above

3. Due to S.C.R in vacuum tube [RPET 2000]

(a) $I_p \to$ Decrease (b) $I_p -$ Increase

(c) $V_p =$ Increase (d) $V_g =$ Increase

4. In diode, when there is saturation current, the plate resistance (r_p) is [AIIMS 1997; Haryana PMT 2000]

(a) Zero (b) Infinite

(c) Some finite quantity (d) Data is insufficient

5. The grid voltage of any triode valve is changed from –1 volt to – 3 *volt* and the mutual conductance is 3×10^{-4} *mho*. The change in plate circuit current will be [MNR 1999]

(a) 0.8 *mA* (b) 0.6 *mA*

(c) 0.4 *mA* (d) 1 *mA*

6. In a triode, $g_m = 2 \times 10^{-3} ohm^{-1}; \mu = 42$, resistance load, $R = 50$ kilo ohm. The voltage amplification obtained from this triode will be [MNR 1999]

(a) 30.42 (b) 29.57

(c) 28.18 (d) 27.15

7. In an amplifier the load resistance R_L is equal to the plate resistance (r_p). The voltage amplification is equal to [DCE 2004]

[CPMT 1995]

(a) μ (b) 2μ

(c) $\mu / 2$ (d) $\mu / 4$

8. For a given plate-voltage, the plate current in a triode is maximum when the potential of

[IIT-JEE 1985; CPMT 1995; AFMC 1999]

(a) The grid is positive and plate is negative

(b) The grid is positive and plate is positive

(c) The grid is zero and plate is positive

(d) The grid is negative and plate is positive

9. If $R_p = 7K\Omega, g_m = 2.5$ *millimho*, then on increasing plate voltage by $50V$, how much the grid voltage is changed so that plate current remains the same [RPET 1996]

(a) – 2.86 *V* (b) – 4 *V*

(c) + 4 *V* (d) + 2 *V*

10. The amplification factor of a triode is 20 and trans-conductance is 3 *milli mho* and load resistance $3 \times 10^4 \Omega$, then the voltage gain is

(a) 16.36 (b) 28

(c) 78 (d) 108

11. In a triode amplifier, $\mu = 25, r_p = 40$ kilo ohm and load resistance $R_L = 10$ kilo ohm. If the input signal voltage is 0.5 volt, then output signal voltage will be [RPMT 1995]

(a) 1.25 *volt* (b) 5 *volt*

(c) 2.5 *volt* (d) 10 *volt*

12. The amplification factor of a triode is 20. If the grid potential is reduced by 0.2 *volt* then to keep the plate current constant its plate voltage is to be increased by

[RPMT 1993, 95]

(a) 10 *volt* (b) 4 *volt*

(c) 40 *volt* (d) 100 *volt*

13. For a triode $r_p = 10$ *kilo ohm* and $g_m = 3$ *milli mho*. If the load resistance is double of plate resistance, then the value of voltage gain will be [RPMT 1994]

(a) 10 (b) 20

(c) 15 (d) 30

14. The amplification produced by a triode is due to the action of

(a) Filament (b) Cathode

(c) Grid (d) Plate

15. In an experiment, the saturation in the plate current in a diode is observed at 240 *V*. But a student still wants to increase the plate current. It can be done, if [MNR 1994]

(a) The plate voltage is increased further

(b) The plate voltage is decreased

(c) The filament current is decreased

(d) The filament current is increased

16. In a triode amplifier, the value of maximum gain is equal to

[MP PMT 1992]

(a) Half the amplification factor

(b) Amplification factor

(c) Twice the amplification factor

(d) Infinity

17. For a given triode $\mu = 20$. The load resistance is 1.5 times the anode resistance. The maximum gain will be

[CPMT 1992]

(a) 16 (b) 12

(c) 10 (d) None of the above

18. The voltage gain of a triode depends upon [CPMT 1992]

(a) Filament voltage (b) Plate voltage

(c) Plate resistance (d) Plate current

19. In a triode valve [MP PET 1992]

(a) If the grid voltage is zero then plate current will be zero

(b) If the temperature of filament is doubled, then the thermionic current will also be doubled

(c) If the temperature of filament is doubled, then the thermionic current will nearly be four times [RPMT 1990]

(d) At a definite grid voltage the plate current varies with plate voltage according to Ohm's law

20. The amplification factor of a triode valve is 15. If the grid voltage is changed by 0.3 *volt* the change in plate voltage in order to keep the plate current constant (in volt) is

[CPMT 1990]

(a) 0.02 (b) 0.002

(c) 4.5 (d) 5.0

21. The slope of plate characteristic of a vacuum tube diode for certain operating point on the curve is $10^{-3} \frac{mA}{V}$. The plate resistance of the diode and its nature respectively

[MP PMT 1990]

(a) 100 kilo-ohms static (b) 1000 *kilo-ohms* static

(c) 1000 *kilo-ohms* dynamic (d) 100 *kilo-ohms* dynamic

22. A triode has a mutual conductance of $2 \times 10^{-3} mho$ and an amplification factor of 50. The anode is connected through a resistance of 25×10^3 *ohms* to a 250 *volts* supply. The voltage gain of this amplifier is [MP PMT 1989]

(a) 50 (b) 25

(c) 100 [AFMC 1994] (d) 12.5

23. 14×10^{15} electrons reach the anode per second. If the power consumed is 448 *milliwatts*, then the plate (anode) voltage is

(a) 150 *V* (b) 200 *V*

(c) $14 \times 448V$ (d) 448/14 *V*

24. In the circuit of a triode valve, there is no change in the plate current, when the plate potential is increased from 200 *volt* to 220 *volt* and the grid potential is decreased from – 0.5 *volt* to –1.3 *volt*. The amplification factor of this valve is

[MP PMT 1989]

(a) 15 (b) 20

(c) 25 (d) 35

25. If the amplification factor of a triode (μ) is 22 and its plate resistance is 6600 *ohm*, then the mutual conductance of this valve is mho is [MP PMT 1989]

(a) $\frac{1}{300}$ (b) 25×10^{-2}

(c) 2.5×10^{-2} (d) 0.25×10^{-2}

26. For a triode, at $V_g = -1$ *volt*, the following observations were taken $V_p = 75V, I_p = 2mA$, $V_p = 100V, I_p = 4mA$. The value of plate resistance will be [MP PMT 1989]

(a) 25 $k\Omega$ (b) 20.8 $k\Omega$

(c) 12.5 $k\Omega$ (d) 100 $k\Omega$

27. The triode constant is out of the following **[RPMT 1989]**

(a) Plate resistance (b) Amplification factor

(c) Mutual conductance (d) All the above

28. The unit of mutual conductance of a triode valve is **[MP PMT 1988]**

(a) Siemen (b) *Ohm*

(c) *Ohm metre* (d) *Joule Coulomb*

29. With a change of load resistance of a triode, used as an amplifier, from 50 *kilo ohms* to 100 *kilo ohms*, its voltage amplification changes from 25 to 30. Plate resistance of the triode is

(a) 25 $k\Omega$ (b) 75 $k\Omega$

(c) 7.5 $k\Omega$ (d) 2.5 $k\Omega$

30. Select the correct statements from the following **[IIT-JEE 1984]**

(a) A diode can be used as a rectifier

(b) A triode cannot be used as a rectifier

(c) The current in a diode is always proportional to the applied voltage

(d) The linear portion of the I–V characteristic of a triode is used for amplification without distortion

31. The introduction of a grid in a triode valve affects plate current by

(a) Making the thermionic emission easier at low temperature

(b) Releasing more electrons from the plate

(c) By increasing plate voltage

(d) By neutralising space charge

32. Before the saturation state of a diode at the plate voltages of 400 V and 200 V respectively the currents are i_1 and i_2 respectively. The ratio i_1/i_2 will be

(a) $\sqrt{2}/4$ (b) $2\sqrt{2}$

(c) 2 (d) 1/2

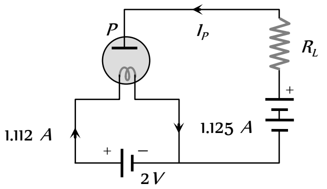
33. The value of plate current in the given circuit diagram will be

(a) 3 mA

(b) 8 mA

(c) 13 mA

(d) 18 mA



34. Coating of strontium oxide on Tungsten cathode in a valve is good for thermionic emission because **[RPMT 1998]**

(a) Work function decreases

(b) Work function increases

(c) Conductivity of cathode increases

(d) Cathode can be heated to high temperature

35. Correct relation for triode is **[RPMT 2000]**

(a) $\mu = g_m \times r_p$ (b) $\mu = \dfrac{g_m}{r_p}$

(c) $\mu = 2g_m \times r_p$ (d) None of these

36. Following is the relation between current and charge $I = AT^2 e^{qt/V_L}$ then value of V_L will be **[RPMT 2000]**

(a) $\dfrac{V}{kT}$ (b) $\dfrac{kV}{T}$

(c) $\dfrac{kT}{V}$ (d) $\dfrac{VT}{k}$

37. Which one is correct relation for thermionic emission **[RPMT 2000]**

(a) $J = AT^{1/2} e^{-\phi/kT}$ (b) $J = AT^2 e^{-\phi/kT}$

(c) $J = AT^{3/2} e^{-\phi/kT}$ (d) $J = AT^2 e^{-\phi/2kT}$

[MP PET 1986]

38. When plate voltage in diode valve is increased from 100 *volt* to 150 *volt* then plate current increases from 7.5 *mA* to 12 *mA*. The dynamic plastic resistance will be **[RPMT 2000]**

(a) 10 $k\Omega$ (b) 11 $k\Omega$

(c) 15 $k\Omega$ (d) 11.1 $k\Omega$

39. In a diode valve, the state of saturation can be obtained easily by

(a) High plate voltage and high filament current

(b) Low filament current and high plate voltage

(c) Low plate voltage and high cathode temperature

(d) High filament current and high plate voltage

40. Plate resistance of two triode valves is 2 $k\Omega$ and 4 $k\Omega$, amplification factor of each of the valves is 40. The ratio of voltage amplification, when used with 4 $k\Omega$ load resistance, will be **[CPMT 1975, 90]**

(a) 10 (b) $\dfrac{4}{3}$

(c) $\dfrac{3}{4}$ (d) $\dfrac{16}{3}$

41. Diode is used as a/an **[AIIMS 1999]**

(a) Oscillator (b) Amplifier

(c) Rectifier (d) Modulator

42. The electrical circuits used to get smooth d.c. output from a rectifier circuit is called **[KCET 2000]**

(a) Filter (b) Amplifier

(c) Full wave rectifier (d) Oscillator

43. Which of the following does not vary with plate or grid voltages

(a) g_m (b) R_p

(c) μ (d) Each of them varies

44. The grid in a triode valve is used **[UPSEAT 2000]**

(a) To increases the thermionic emission

(b) To control the plate to cathode current

(c) To reduce the inter-electrode capacity

(d) To keep cathode at constant potential

45. In a triode valve the amplification factor is 20 and mutual conductance is 10^{-3} *mho*. The plate resistance is **[UPSEAT 2000]**

(a) $2 \times 10^3\ \Omega$ (b) $4 \times 10^3\ \Omega$

(c) 2 × 10 Ω (d) 2 × 10 Ω

46. The thermionic emission of electron is due to

[UPSEAT 2000]

(a) Electromagnetic field (b) Electrostatic field

(c) High temperature (d) Photoelectric effect

47. The amplification factor of a triode is 50. If the grid potential is decreased by 0.20 *V*, what increase in plate potential will keep the plate current unchanged [RPMT 1999]

(a) 5 *V* (b) 10 *V*

(c) 0.2 *V* (d) 50 *V*

48. The slope of plate characteristic of a vacuum diode is $2\times10^{-2}\, mA / V$. The plate resistance of diode will be

[RPMT 1999]

(a) 50 Ω (b) 50 *k*Ω

(c) 500 *k*Ω (d) 500 *k*Ω

49. The transconductance of a triode amplifier is 2.5 *mili mho* having plate resistance of 20 *k*Ω, amplification 10. Find the load resistance

(a) 5 *k*Ω (b) 25 *k*Ω

(c) 20 *k*Ω (d) 50 *k*Ω

50. The amplification factor of a triode is 18 and its plate resistance is 8 × 10Ω. A load resistance of 10Ω is connected in the plate circuit. The voltage gain will be

[RPMT 2002]

(a) 30 (b) 20

(c) 10 (d) 1

51. The ripple factor in a half wave rectifier is [RPMT 2002]

(a) 1.21 (b) 0.48

(c) 0.6 (d) None of these

52. The correct relation for a triode is [RPET 2000, 02]

(a) $g_m = \left.\frac{\Delta I_p}{\Delta V_p}\right|_{V_g = constt.}$ (b) $g_m = \left.\frac{\Delta I_p}{\Delta V_g}\right|_{V_p = constt.}$

(c) Both (d) None of these

53. In a diode valve the cathode temperature must be (ϕ = work function) [RPET 2002]

(a) High and ϕ should be high

(b) High and ϕ should be low

(c) Low and ϕ should be high

(d) Low and ϕ should be high

54. The plate resistance of a triode is 2.5 × 10 Ω and mutual conductance is 2 × 10 *mho*. What will be the value of amplification factor [RPET 2002]

(a) 50 (b) 1.25 × 10

(c) 75 (d) 2.25 × 10

55. Plate voltage of a triode is increased from 200 *V* to 225 *V*. To maintain the plate current, change in grid voltage from 5 *V* to 5.75 *V* is needed. The amplification factor is

[RPET 2002]

(a) 40 (b) 45

(c) 33.3 (d) 25

56. The current in a triode at anode potential 100 *V* and grid potential – 1.2 *V* is 7.5 *mA*. If grid potential is changed to – 2.2 *V*, the current becomes 5.5 *mA*. the value of trans conductance (g) will be

(a) 2 *mili mho* (b) 3 *mili mho*

(c) 4 *mili mho* (d) 0.2 *mili mho*

57. Select the correct statement [RPMT 2003]

(a) In a full wave rectifier, two diodes work alternately

(b) In a full wave rectifier, two diodes work simultaneously

(c) The efficiency of full wave and half wave rectifiers is same

(d) The full wave rectifier is bi-directional

58. The amplification factor of a triode is 20. Its plate resistance is 10 kilo ohms. Mutual conductance is

[MNR 1992; Orissa JEE 2005]

(a) 2×10^{5} *mho* (b) 2×10^{4} *mho*

(c) 500 *mho* (d) 2×10^{-3} *mho*

[RPMT 2001]

Critical Thinking

Objective Questions

1. A silicon speciman is made into a *P*-type semi-conductor by dopping, on an average, one Indium atom per 5×10^{7} silicon atoms. If the number density of atoms in the silicon specimen is 5×10^{28} atoms $/m^3$ then the number of acceptor atoms in silicon per cubic centimetre will be [MP PMT 1993, 2003]

(a) $2.5\times10^{30}\,atoms/cm^3$ (b) $1.0\times10^{13}\,atoms/cm^3$

(c) $1.0\times10^{15}\,atoms/cm^3$ (d) $2.5\times10^{36}\,atoms/cm^3$

2. The probability of electrons to be found in the conduction band of an intrinsic semiconductor at a finite temperature [IIT-JEE 1995; DPMT 2004]

(a) Decreases exponentially with increasing band gap

(b) Increases exponentially with increasing band gap

(c) Decreases with increasing temperature

(d) Is independent of the temperature and the band gap

3. The typical ionisation energy of a donor in silicon is [IIT-JEE 1992]

(a) $10.0\,eV$ (b) $1.0\,eV$

(c) $0.1\,eV$ (d) $0.001\,eV$

4. In *PN*-junction diode the reverse saturation current is 10^{-5} *amp* at $27°C$. The forward current for a voltage of $0.2 volt$ is

(a) $2037.6\times10^{-3}amp$ (b) $203.76\times10^{-3}amp$

(c) $20.376\times10^{-3}amp$ (d) $2.0376\times10^{3}amp$

$[\exp(7.62)=2038.6, K=1.4\times10^{-23}\,J/K]$

5. When a potential difference is applied across, the current passing through [IIT-JEE 1999]

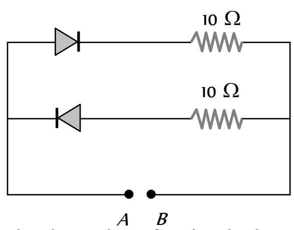
(a) An insulator at $0K$ is zero

(b) A semiconductor at $0K$ is zero

(c) A metal at $0K$ is finite

(d) A *P-N* diode at $300K$ is finite, if it is reverse biased

6. A 2 *V* battery is connected across the points *A* and *B* as shown in the figure given below. Assuming that the resistance of each diode is zero in forward bias and infinity in reverse bias, the current supplied by the battery when its positive terminal is connected to *A* is [UPSEAT 2002]



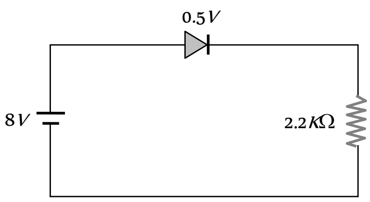
(a) 0.2 *A*

(b) 0.4 *A*

(c) Zero

(d) 0.1 *A*

7. In the circuit, if the forward voltage drop for the diode is 0.5 *V*, the current will be [UPSEAT 2003]



(a) 3.4 *mA*

(b) 2 *mA*

(c) 2.5 *mA*

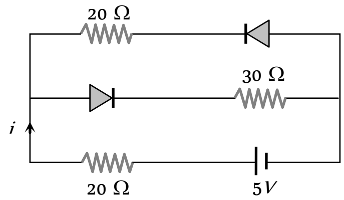
(d) 3 *mA*

8. A *P*-type semiconductor has acceptor levels 57 *meV* above the valence band. The maximum wavelength of light required to create a hole is (Planck's constant h = 6.6×10^{-34} *J-s*)

(a) 57 Å (b) 57×10^{-3} Å

(c) 217100 Å (d) 11.61×10^{-33} Å

9. Current in the circuit will be [CBSE PMT 2001]



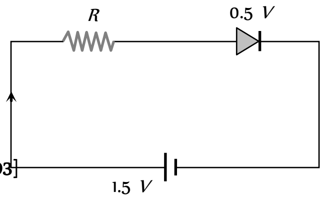
(a) $\dfrac{5}{40}A$

(b) $\dfrac{5}{50}A$

(c) $\dfrac{5}{10}A$

(d) $\dfrac{5}{20}A$

10. The diode used in the circuit shown in the figure has a constant voltage drop of 0.5 *V* at all currents and a maximum power rating of 100 *milliwatts*. What should be the value of the resistor *R*, connected in series with the diode for obtaining maximum current [CBSE PMT



(a) 1.5 Ω

(b) 5 Ω

(c) 6.67 Ω

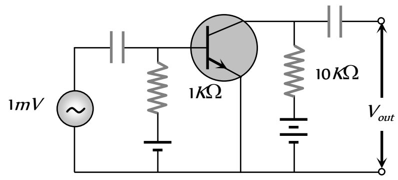
(d) 200 Ω [MP PMT 1993]

11. For a transistor amplifier in common emitter configuration for load impedance of 1 *k*Ω (h_{fe} = 50 and h_{oe} = 25 *μA/V*) the current gain is

(a) – 5.2 (b) – 15.7

(c) – 24.8 (d) – 48.78

12. In the following common emitter configuration an *NPN* transistor with current gain β = 100 is used. The output voltage of the amplifier will be [AIIMS 2003]



(a) 10 *mV*

(b) 0.1 *V*

(c) 1.0 *V*

(d) 10 *V*

13. In semiconductor the concentrations of electrons and holes are $8\times10^{18}/m$ and $5\times10^{18}/m$ respectively. If the mobilities of electrons and hole are 2.3 *m/volt-sec* and 0.01 *m/volt-sec* respectively, then semiconductor is

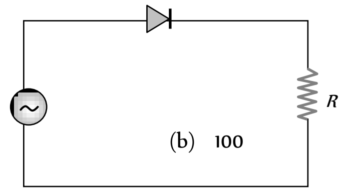
(a) *N*-type and its resistivity is 0.34 *ohm-metre*

(b) *P*-type and its resistivity is 0.034 *ohm-metre*

(c) *N*-type and its resistivity is 0.034 *ohm-metre*

(d) *P*-type and its resistivity is 3.40 *ohm-metre*

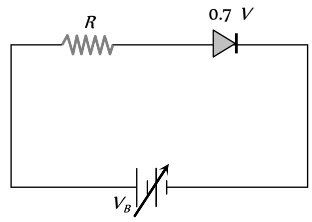
14. A sinusoidal voltage of peak value 200 *volt* is connected to a diode and resistor R in the circuit shown so that half wave rectification occurs. If the forward resistance of the diode is negligible compared to R the *rms* voltage (in *volt*) across R is approximately



(a) 200 (b) 100

(c) $\frac{200}{\sqrt{2}}$ (d) 280

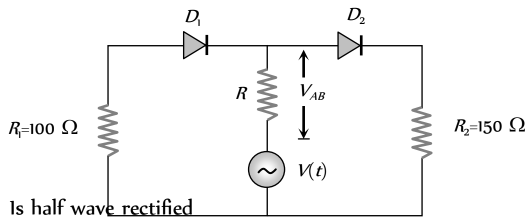
15. The junction diode in the following circuit requires a minimum current of 1 *mA* to be above the knee point (0.7 *V*) of its I-V characteristic curve. The voltage across the diode is independent of current above the knee point. If V_B = 5 *V*, then the maximum value of R so that the voltage is above the knee point, will be



(a) 4.3 $k\Omega$ (b) 860 $k\Omega$

(c) 4.3 Ω (d) 860 Ω

16. In the circuit given below, $V(t)$ is the sinusoidal voltage source, voltage drop $V_0(t)$ across the resistance R is [IIT 1993]



(a) Is half wave rectified

(b) Is full wave rectified

(c) Has the same peak value in the positive and negative half cycles

(d) Has different peak values during positive and negative half cycle

17. The peak voltage in the output of a half-wave diode rectifier fed with a sinusoidal signal without filter is 10 *V*. The dc component of the output voltage is [CBSE PMT 2004]

(a) $10/\sqrt{2}\ V$ (b) $10/\pi\ V$

(c) 10 *V* (d) $20/\pi\ V$

18. A transistor is used as an amplifier in *CB* mode with a load resistance of 5 $k\,\Omega$ the current gain of amplifier is 0.98 and the input resistance is 70 Ω, the voltage gain and power gain respectively are [Pb. PET 2003]

(a) 70, 68.6 (b) 80, 75.6

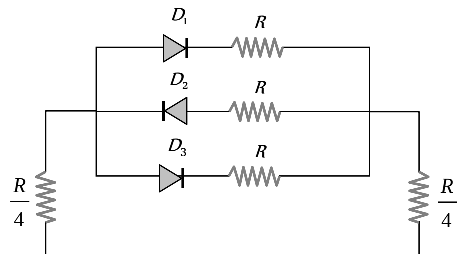
(c) 60, 66.6 (d) 90, 96.6

19. The Bohr radius of the fifth electron of phosphorus (atomic number = 15) acting as dopant in silicon (relative dielectric constant = 12) is

(a) 10.6 Å (b) 0.53 Å

(c) 21.2 Å (d) None of these

20. In the following circuits *PN*-junction diodes D_1, D_2 and D_3 are ideal for the following potential of A and B, the correct increasing order of resistance between A and B will be



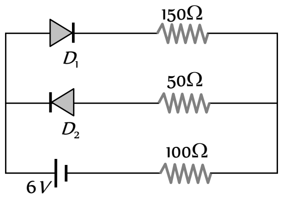
(i) – 10 *V*, – 5 *V* (ii) – 5 *V*, – 10 *V*

(iii) – 4 *V*, – 12 *V*

(a) (i) < (ii) < (iii) (b) (iii) < (ii) < (i)

(c) (ii) = (iii) < (i) (d) (i) = (iii) < (ii)

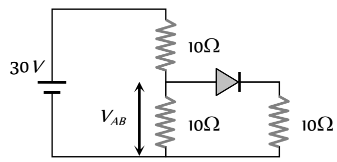
21. The circuit shown in following figure contains two diode D_1 and D_2 each with a forward resistance of 50 *ohms* and with infinite backward resistance. If the battery voltage is 6 *V*, the current through the 100 *ohm* resistance (in *amperes*) is [IIT-JEE 1997]



(a) Zero (b) 0.02

(c) 0.03 (d) 0.036

22. Find V_{AB} [RPMT 2000]



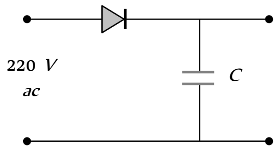
(a) 10 *V*

(b) 20 *V*

(c) 30 *V*

(d) None of these

23. A diode is connected to 220 *V* (*rms*) *ac* in series with a capacitor as shown in figure. The voltage across the capacitor is



(a) 220 *V*

(b) 110 *V*

(c) 311.1 *V*

(d) $\frac{220}{\sqrt{2}}V$

24. A potential difference of 2 *V* is applied between the opposite faces of a *Ge* crystal plate of area 1 *cm* and thickness 0.5 *mm*. If the concentration of electrons in *Ge* is $2 \times 10^{-}/m$ and mobilities of electrons and holes are $0.36\frac{m^2}{volt-\sec}$ and $0.14\frac{m^2}{volt-\sec}$ respectively, then the current flowing through the plate will be

(a) 0.25 A (b) 0.45 A

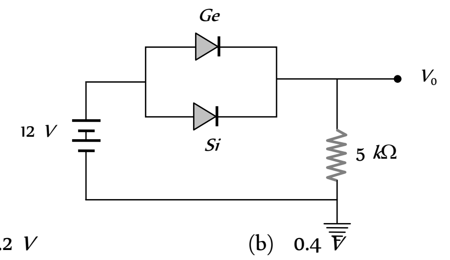
(c) 0.56 A (d) 0.64 A

25. The contribution in the total current flowing through a semiconductor due to electrons and holes are $\frac{3}{4}$ and $\frac{1}{4}$ respectively. If the drift velocity of electrons is $\frac{5}{2}$ times that of holes at this temperature, then the ratio of concentration of electrons and holes is

(a) 6 : 5 (b) 5 : 6

(c) 3 : 2 (d) 2 : 3

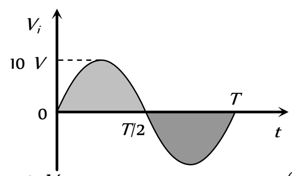
26. Ge and Si diodes conduct at 0.3 V and 0.7 V respectively. In the following figure if Ge diode connection are reversed, the valve of V_0 changes by **[Based on Roorkee 2000]**

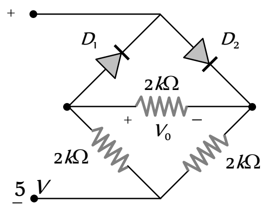


(a) 0.2 V (b) 0.4 V

(c) 0.6 V (d) 0.8 V

27. In the circuit shown in figure the maximum output voltage V_0 is

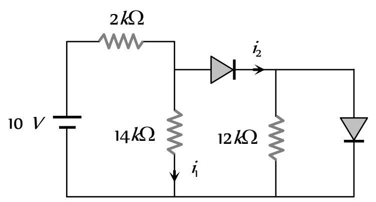




(a) 0 V (b) 5 V

(c) 10 V (d) $\frac{5}{\sqrt{2}}V$

28. In the following circuit find I_1 and I_2



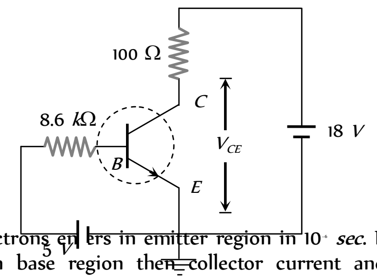
(a) 0, 0

(b) 5 mA, 5 mA

(c) 5 mA, 0

(d) 0, 5 mA

29. For the transistor circuit shown below, if β = 100, voltage drop between emitter and base is 0.7 V then value of V_{CE} will be



(a) 10 V

(b) 5 V

(c) 13 V

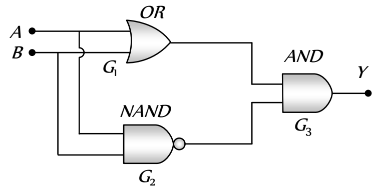
(d) 0 V

30. In NPN transistor, 10^{10} electrons enters in emitter region in 10^{-6} sec. If 2% electrons are lost in base region then collector current and current amplification factor (β) respectively are

(a) 1.57 mA, 49 (b) 1.92 mA, 70

(c) 2 mA, 25 (d) 2.25 mA, 100

31. The following configuration of gate is equivalent to **[AMU 1999]**



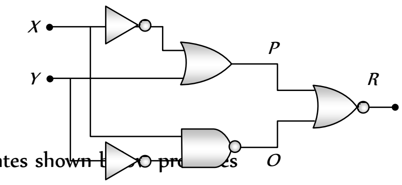
(a) NAND

(b) XOR

(c) OR

(d) None of these

32. Figure gives a system of logic gates. From the study of truth table it can be found that to produce a high output (1) at R, we must have



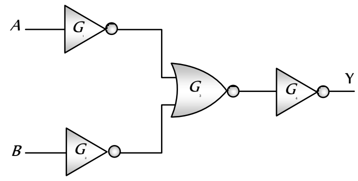
(a) X = 0, Y = 1

(b) X = 1, Y = 1

(c) X = 1, Y = 0

(d) X = 0, Y = 0

33. The combination of gates shown below produces



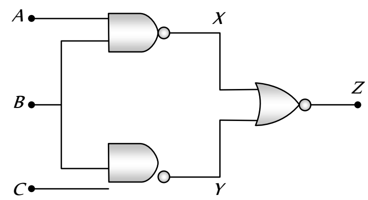
(a) AND gate

(b) XOR gate

(c) NOR gate

(d) NAND gate

34. The shows two NAND gates followed by a NOR gate. The system is equivalent to the following logic gate



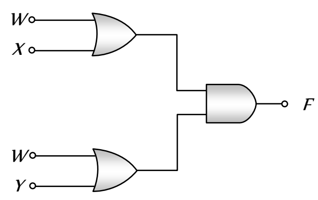
(a) OR

(b) AND

(c) NAND

(d) None of these

35. The diagram of a logic circuit is given below. The output F of the circuit is represented by



(a) $W.(X+Y)$

(b) $W\cdot(X\cdot Y)$

(c) $W+(X\cdot Y)$

(d) $W+(X+Y)$

36. The plate current i_p in a triode valve is given $i_p = K(V_p + \mu V_g)^{3/2}$ where i_p is in milliampere and V_p and V_g are in $volt$. If r_p = 10^4 ohm, and $g_m = 5\times10^{-3}\,mho$, then for $i_p = 8\,mA$ and $V_p = 300\,volt$, what is the value of K and grid cut off voltage **[Roorkee 1992]**

(a) – 6 V, $(30)^{3/2}$ (b) $-6V, (1/30)^{3/2}$

(c) + 6 V, $(30)^{3/2}$ (d) + 6 V, $(1/30)^{3/2}$

37. The linear portions of the characteristic curves of a triode valve give the following readings **[Roorkee 1985]**

V_g ($volt$)	0	– 2	– 4	– 6

$I_p(mA)$ for $V_p = 150$ *volts*	15	12.5	10	7.5
$I_p(mA)$ for $V_p = 120$ *volts*	10	7.5	5	2.5

The plate resistance is

(a) 2000 *ohms* (b) 4000 *ohms*

(c) 8000 *ohms* (d) 6000 *ohms*

38. The relation between dynamic plate resistance (r) of a vacuum diode and plate current in the space charge limited region, is

(a) $r_p \propto I_p$ (b) $r_p \propto I_p^{3/2}$

(c) $r_p \propto \frac{1}{I_p}$ (d) $r_p \propto \frac{1}{(I_p)^{1/3}}$

39. The relation between I_p and V_p for a triode is

$$I_p = (0.125V_p - 7.5)mA$$

Keeping the grid potential constant at 1*V*, the value of r_p will be

(a) 8 $k\Omega$ (b) 4 $k\Omega$

(c) 2 $k\Omega$ (d) 8 $k\Omega$

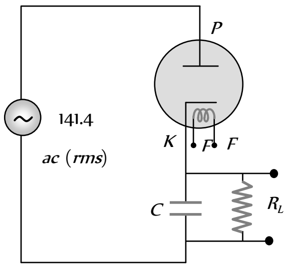
40. An alternating voltage of 141.4 *V* (*rms*) is applied to a vacuum diode as shown in the figure. The maximum potential difference across the condenser will be

(a) 100 *V*

(b) 200 *V*

(c) $100\sqrt{2}V$

(d) $200\sqrt{2}V$



41. A metallic surface with work function of 2 *eV*, on heating to a temperature of 800 *K* gives an emission current of 1 *mA*. If another metallic surface having the same surface area, same emission constant but work function 4 *eV* is heated to a temperature of 1600 *K*, then the emission current will be

(a) 1 *mA* (b) 2 *mA*

(c) 4 *mA* (d) None of these

42. A change of 0.8 *mA* in the anode current of a triode occurs when the anode potential is changed by 10 *V*. If μ = 8 for the triode, then what change in the grid voltage would be required to produce a change of 4 *mA* in the anode current

(a) 6.25 *V* (b) 0.16 *V*

(c) 15.2 *V* (d) None of these

43. The plate current in a triode is given by

$$I_p = 0.004\,(V_p + 10V_g)^{3/2} mA$$

where I_p, V_p and V_g are the values of plate current, plate voltage and grid voltage, respectively. What are the triode parameters μ, r_p and g_m for the operating point at $V_p = 120\,volt$ and $V_g = -2\,volt$?

(a) 10, 16.7 $k\Omega$, 0.6 *m mho* (b) 15, 16.7 $k\Omega$, 0.06 *m mho*

(c) 20, 6 $k\Omega$, 16.7 *m mho* (d) None of these

44. A triode whose mutual conductance is 2.5 *m A/volt* and anode resistance is 20 *kilo ohm*, is used as an amplifier whose amplification is 10. The resistance connected in plate circuit will be **[MP PET 1989; RPMT 1998]**

(a) 1 $k\Omega$ (b) 5 $k\Omega$

(c) 10 $k\Omega$ (d) 20 $k\Omega$

45. In the grid circuit of a triode a signal $E = 2\sqrt{2}\cos\omega t$ is applied. If μ = 14 and r_p =10 $k\Omega$ then root mean square current flowing through $R_L = 12\,k\Omega$ will be

(a) 1.27 *mA* (b) 10 *mA*

(c) 1.5 *mA* (d) 12.4 *mA*

46. For a triode μ = 64 and g_m =1600 μ *mho*. It is used as an amplifier and an input signal of 1 *V* (*rms*) is applied. The signal power in the load of 40 $k\Omega$ will be

(a) 23.5 *mW* (b) 48.7 *mW*

(c) 25.6 *mW* (d) None of these

47. Amplification factor of a triode is 10. When the plate potential is 200 *volt* and grid potential is – 4 *volt*, then the plate current of 4*mA* is observed. If plate potential is changed to 160 *volt* and grid potential is kept at – 7 *volt*, then the plate current will be

(a) 1.69 *mA* (b) 3.95 *mA*

(c) 2.87 (d) 7.02 *mA*

48. On applying a potential of – 1 *volt* at the grid of a triode, the following relation between plate voltage V_p (*volt*) and plate current $I_p(\text{in}\, mA)$ is found

$$I_p = 0.125\,V_p - 7.5$$

If on applying – 3 *volt* potential at grid and 300 *V* potential at plate, the plate current is found to be 5*mA*, then amplification factor of the triode is

(a) 100 (b) 50

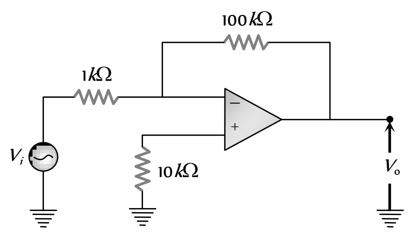
(c) 30 (d) 20

49. The slopes of anode and mutual characteristics of a triode are 0.02 *mA V* and 1 *mA V* respectively. What is the amplification factor of the valve **[MP PMT 1990]**

(a) 5 (b) 50

(c) 500 (d) 0.5

50. The voltage gain of the following amplifier is **[AIIMS 2005]**



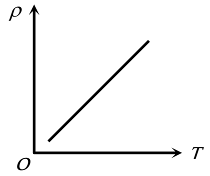
(a) 10 (b) 100

(c) 1000 (d) 9.9

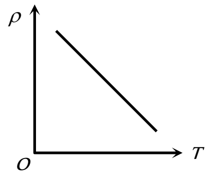
Graphical Questions

1. The temperature (T) dependence of resistivity (ρ) of a semiconductor is represented by **[AIIMS 2004]**

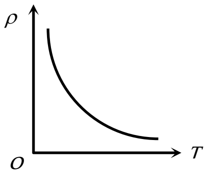
(a)



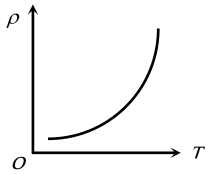
(b)



(c)

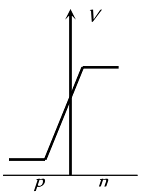


(d)

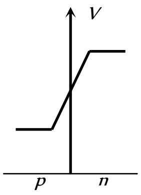


2. In a forward biased *PN*-junction diode, the potential barrier in the depletion region is of the form ... **[KCET 2004]**

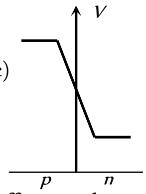
(a)



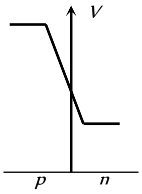
(b)



(c)



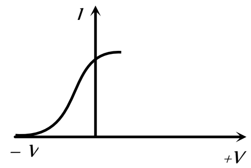
(d)



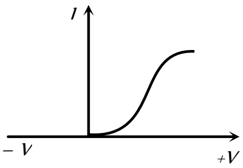
3. Different voltages are applied across a *P-N* junction and the currents are measured for each value. Which of the following graphs is obtained between voltage and current

[MP PET 1996; UPSEAT 2002]

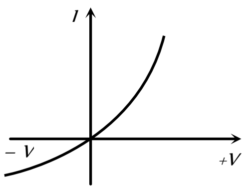
(a)



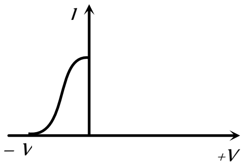
(b)



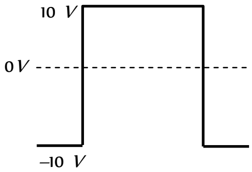
(c)

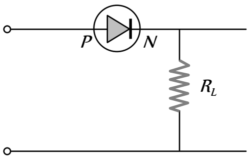


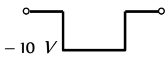
(d)

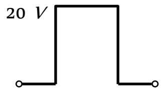


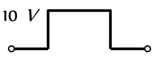
4. If the following input signal is sent through a *PN*-junction diode, then the output signal across R_L will be

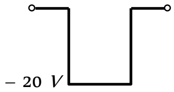




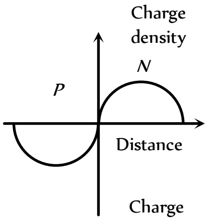
(a) 

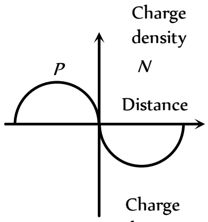
(b) 

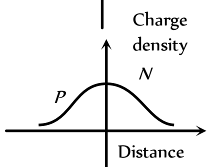
(c) 

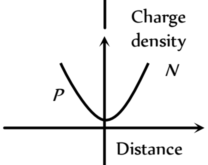
(d) 

5. The curve between charge density and distance near *P-N* junction will be

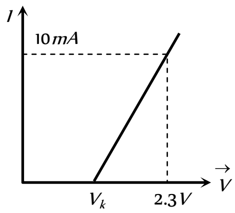
(a) 

(b) 

(c) 

(d) 

6. The resistance of a germanium junction diode whose $V-I$ is shown in figure is $(V_k = 0.3V)$



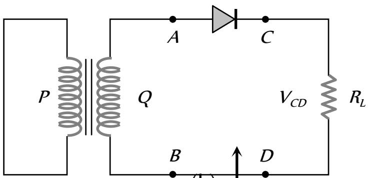
(a) 5 $k\Omega$

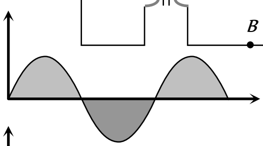
(b) 0.2 $k\Omega$

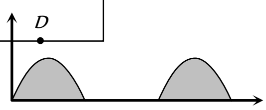
(c) 2.3 $k\Omega$

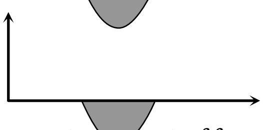
(d) $\left(\frac{10}{2.3}\right)k\Omega$

7. In the half-wave rectifier circuit shown. Which one of the following wave forms is true for V_{CD}, the output across *C* and *D*?



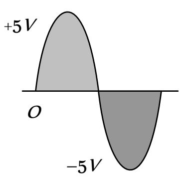
(a) 

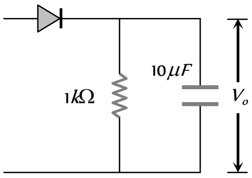
(b) 

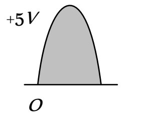
(c) 

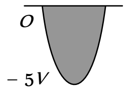
(d) 

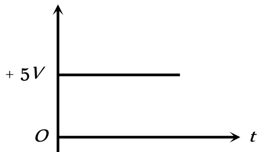
8. The output in the circuit of figure is taken across a capacitor. It is as shown in figure

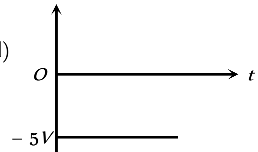




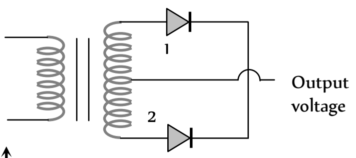
(a) 

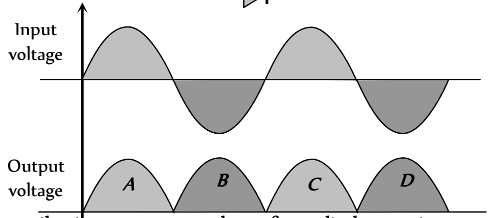
(b) 

(c) 

(d) 

9. A full wave rectifier circuit along with the input and output voltages is shown in the figure





The contribution to output voltage from diode – 2 is

[MP PMT 2001]

(a) *A*, *C*
(b) *B*, *D*
(c) *B*, *C*
(d) *A*, *D*

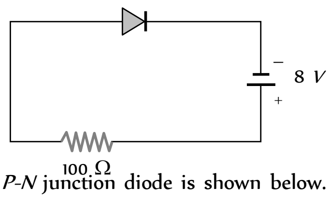
10. A source voltage of 8 *V* drives the diode in fig. through a current-limiting resistor of 100 *ohm*. Then the magnitude of the slope load line on the *V-I* characteristics of the diode is

(a) 0.01

(b) 100

(c) 0.08

(d) 12.5



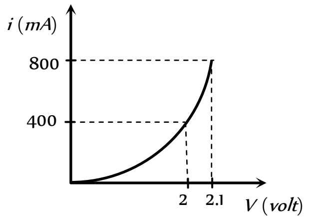
11. The *i-V* characteristic of a *P-N* junction diode is shown below. The approximate dynamic resistance of the *P-N* junction when a forward bias of 2 *volt* is applied

(a) 1 Ω

(b) 0.25 Ω

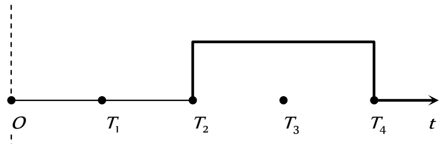
(c) 0.5 Ω

(d) 5 Ω

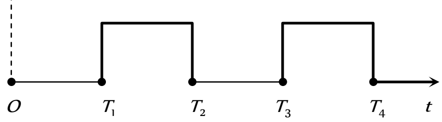


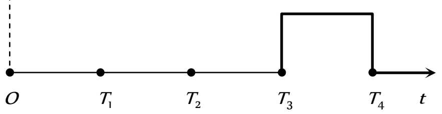
12. The given figure shows the wave forms for two inputs *A* and *B* and that for the output *Y* of a logic circuit. The logic circuit is

(A)



(B)



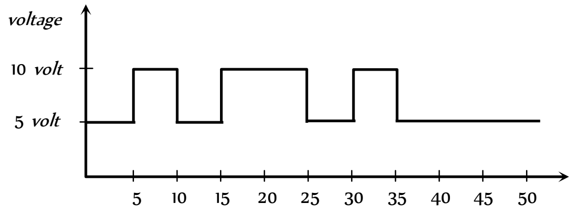


(Y)

(a) An AND gate (b) An OR gate

(c) A NAND gate (d) An NOT gate

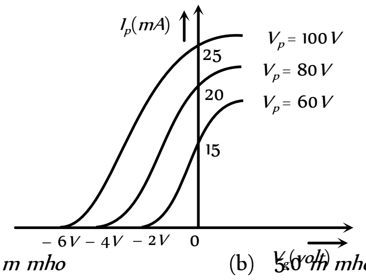
13. In a negative logic the following wave form corresponds to the



(a) 0000000000 000

(c) 111111111 (d) 1010010111

14. The variation of anode current in a triode corresponding to a change in grid potential at three different values of the plate potential is shown in the diagram. The mutual conductance of the triode is

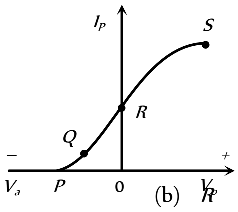


[CPMT 1986, 88]

(a) 2.5 $m\ mho$ (b) 5.0 $m\ mho$

(c) 7.5 $m\ mho$ (d) 10.0 $m\ mho$

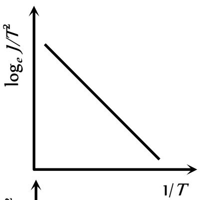
15. The point representing the cut off grid voltage on the mutual characteristic of triode is

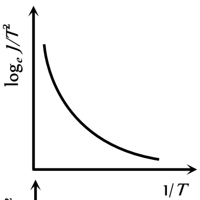


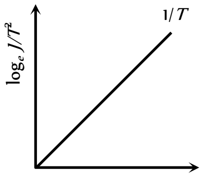
(a) S (b) R

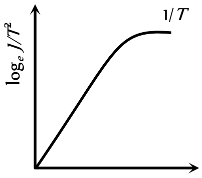
(c) O (d) P

16. For a thermionic emitter (metallic) if J represents the current density and T is its absolute temperature then the correct curve between $\log_e \frac{J}{T^2}$ and $\frac{1}{T}$ is

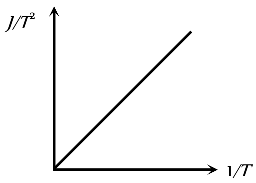
(a) 

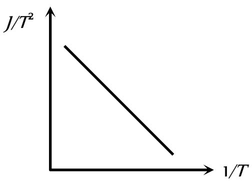
(b) 

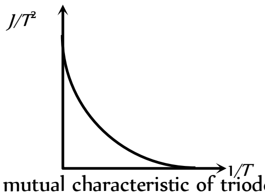
(c) 

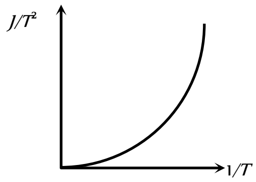
(d) 

17. If the thermionic current density is J and emitter temperature is T then the curve between $\frac{J}{T^2}$ and $\frac{1}{T}$ will be

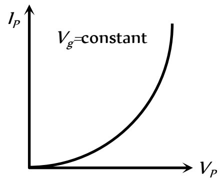
(a) 

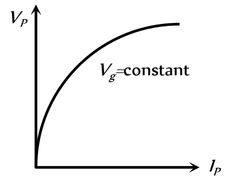
(b) 

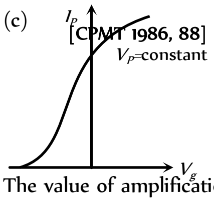
(c) 

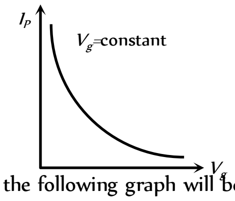
(d) 

18. The mutual characteristic of triode is

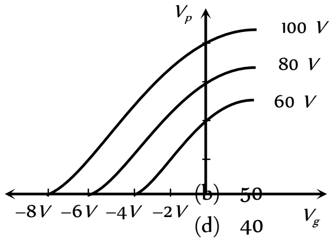
(a) 

(b) 

(c) 

(d) 

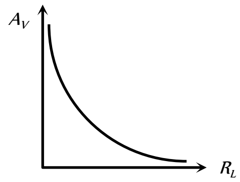
19. The value of amplification factor from the following graph will be

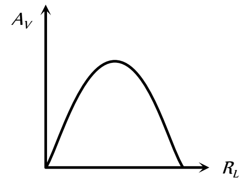


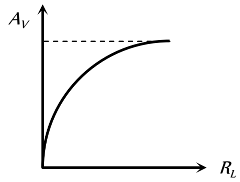
(a) 10 (b) 50

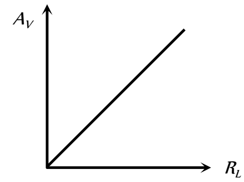
(c) 25 (d) 40

20. The correct curve between voltage gain (A_v) and load resistance (R_L) is

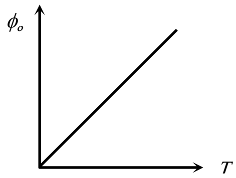
(a) 

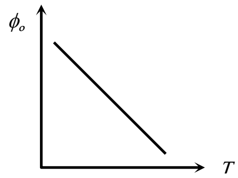
(b) 

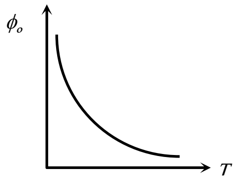
(c) 

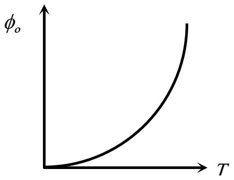
(d) 

21. The curve between the work function of a metal (ϕ_o) and its temperature (T) will be

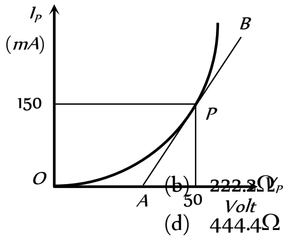
(a) 

(b) 

(c) 

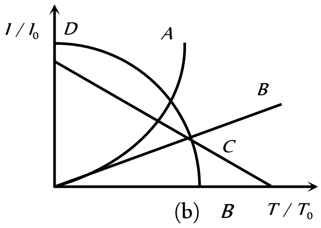
(d) 

22. The plate characteristic curve of a diode in space charge limited region is as shown in the figure. The slope of curve at point P is 5.0 mA/V. The static plate resistance of diode will be



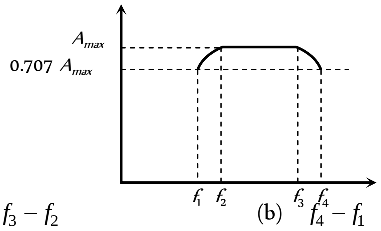
(a) 111.1Ω (b) 222.2Ω
(c) 333.3Ω (d) 444.4Ω

23. The ratio of thermionic currents (I/I_0) for a metal when the temperature is slowly increased T_0 to T as shown in figure. (I and I_0 are currents at T and respectively). Then which one is correct?



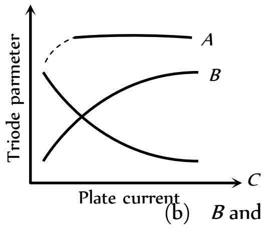
(a) A (b) B
(c) C (d) D

24. The frequency response curve of RC coupled amplifier is shown in figure. The band width of the amplifier will be



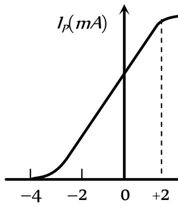
(a) $f_3 - f_2$ (b) $f_4 - f_1$
(c) $\frac{f_4 - f_2}{2}$ (d) $f_3 - f_1$

25. The figure represents variation of triode parameter (μ or r_p or g_m) with the plate current. The correct variation of μ and r_p are given, respectively by the curves



(a) A and B (b) B and C
(c) A and C (d) None of the above

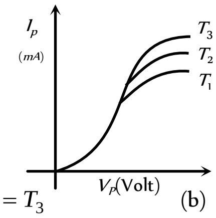
26. The mutual characteristic curves of a triode are as shown in figure. The cut off voltage for the triode is



(a) 0 V (b) 2 V
(c) – 4 V (d) 6 V

27. For the diode, the characteristic curves are given at different temperature. The relation between the temperatures is

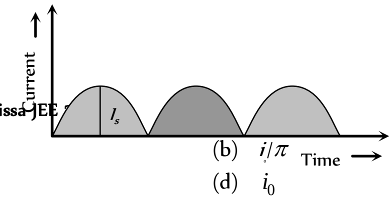
[RPET 1990]



(a) $T_1 = T_2 = T_3$ (b) $T_1 < T_2 < T_3$
(c) $T_1 > T_2 > T_3$ (d) None of the above

28. The output current versus time curve of a rectifier is shown in the figure. The average value of the output current in this case is

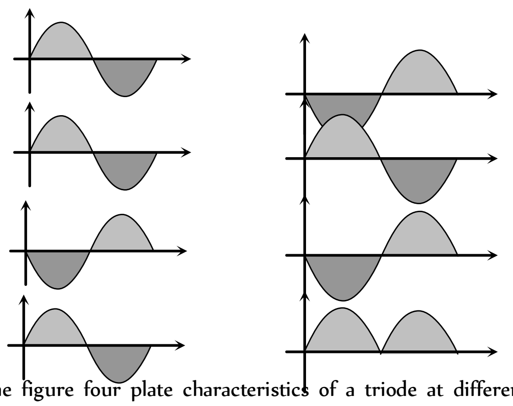
[Orissa JEE



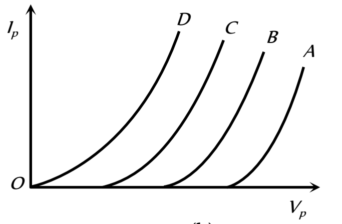
(a) 0 (b) i_0/π
(c) $2i_0/\pi$ (d) i_0

29. Which of the following figures correctly shows the phase relation between the input signal and the output signal of triode amplifier

(a)
(b)
(c)
(d)



30. In the figure four plate characteristics of a triode at different grid voltage are shown. The difference between successive grid voltage is 1 V. Which curve will have maximum grid voltage and what is its value?



(a) A, V_g = + 4 V (b) B, V_g = + 4 V
(c) A, V_g = 0 (d) D, V_g = 0

AR Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) *If both assertion and reason are true and the reason is the correct explanation of the assertion.*
(b) *If both assertion and reason are true but reason is not the correct explanation of the assertion.*
(c) *If assertion is true but reason is false.*
(d) *If the assertion and reason both are false.*
(e) *If assertion is false but reason is true.*

1. Assertion : The logic gate NOT can be built using diode.

Reason : The output voltage and the input voltage of the diode have 180° phase difference.

[AIIMS 2005]

2. Assertion : The number of electrons in a *P*-type silicon semiconductor is less than the number of electrons in a pure silicon semiconductor at room temperature.

Reason : It is due to law of mass action. [AIIMS 2005]

3. Assertion : In a common emitter transistor amplifier the input current is much less than the output current.

Reason : The common emitter transistor amplifier has very high input impedance. [AIIMS 2005]

4. Assertion : A transistor amplifier in common emitter configuration has a low input impedence.

Reason : The base to emitter region is forward biased.

[AIIMS 2004]

5. Assertion : The resistivity of a semiconductor increases with temperature.

Reason : The atoms of a semiconductor vibrate with larger amplitude at higher temperature there by increasing it's resistivity. [AIIMS 2003]

6. Assertion : If the temperature of a semiconductor is increased then it's resistance decreases.

Reason : The energy gap between conduction band and valence band is very small [AIIMS 1997]

7. Assertion : The temperature coefficient of resistance is positive for metals and negative for *P-type* semiconductor.

Reason : The effective charge carriers in metals are negatively charged whereas in *P*-type semiconductor they are positively charged.

[AIIMS 1996]

8. Assertion : Electron has higher mobility than hole in a semiconductor.

Reason : Mass of electron is less than the mass of hole.

9. Assertion : An *N*-type semiconductor has a large number of electrons but still it is electrically neutral.

Reason : An *N*-type semiconductor is obtained by doping an intrinsic semiconductor with a pentavalent impurity.

10. Assertion : The crystalline solids have a sharp melting point.

Reason : All the bonds between the atoms or molecules of a crystalline solids are equally strong, that they get broken at the same temperature.

11. Assertion : Silicon is preferred over germanium for making semiconductor devices.

Reason : The energy gap for germanium is more than the energy gap of silicon.

12. Assertion : We can measure the potential barrier of a *PN* junction by putting a sensitive voltmeter across its terminals.

Reason : The current through the *PN* junction is not same in forward and reversed bias.

13. Assertion : Semiconductors do not Obey's Ohm's law.

Reason : Current is determined by the rate of flow of charge carriers.

14. Assertion : Two *P-N* junction diodes placed back to back, will work as a *NPN* transistor.

Reason : The *P*-region of two *PN* junction diodes back to back will form the base of *NPN* transistor.

15. Assertion : In transistor common emitter mode as an amplifier is preferred over common base mode.

Reason : In common emitter mode the input signal is connected in series with the voltage applied to the base emitter function.

16. Assertion : The dominant mechanism for motion of charge carriers in forward and reverse biased silicon *P-N* junction are drift in both forward and reverse bias.

Reason : In reverse biasing, no current flow through the junction.

17. Assertion : A transistor is a voltage-operating device.

Reason : Base current is greater than the collector current.

18. Assertion : NAND or NOR gates are called digital building blocks.

Reason : The repeated use of NAND (or NOR) gates can produce all the basic or complicated gates.

19. Assertion : At 0 *K* Germanium is a superconductor.

Reason : At 0 *K* Germanium offers zero resistance.

20. Assertion : Base in a transistor is made very thin as compared to collector and emitter regions.

Reason : Due to thin base power gain and voltage gain is obtained by a transistor.

21. Assertion : The current gain in common base circuit is always less than one.

Reason : At constant collector voltage the change in collector current is more than the change in emitter current.

22. Assertion : *V-i* characteristic of *P-N* junction diode is same as that of any other conductor.

Reason : *P-N* junction diode behave as conductor at room temperature.

23. Assertion : Zener diode works on a principle of breakdown voltage.

Reason : Current increases suddenly after breakdown voltage.

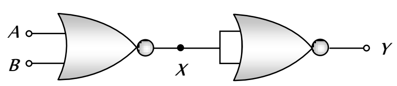
24. Assertion : NOT gate is also called inverter circuit.

Reason : NOT gate inverts the input order.

25. Assertion : In vacuum tubes (valves), vacuum is necessary for the movement of electrons between electrodes otherwise electrons collide with air particle and loses their energy.

Reason : In semiconductors devices, external heating or vacuum is not required.

26. Assertion : The following circuit represents 'OR' gate



Reason : For the above circuit $Y = \overline{X} = \overline{\overline{A+B}} = A + B$

27. Assertion : A *P-N* photodiode is made from a semiconductor for which E_g = 2.8 *eV*. This photo diode will not detect the wavelength of 6000 *nm*.

Reason : A *PN* photodiode detect wavelength λ if $\frac{hc}{\lambda} > E_g$.

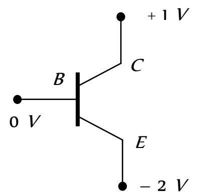
28. Assertion : 29 is the equivalent decimal number of binary number 11101.

Reason : $(11101)_2 = (1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0)_{10}$

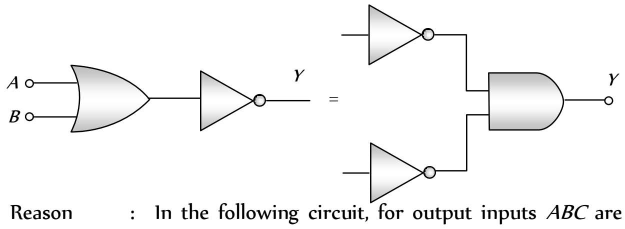
$= (16 + 8 + 4 + 0 + 1)_{10} = (29)_{10}$

29. Assertion : When *PN*-junction is forward biased then motion of charge carriers at junction is due to diffusion. In reverse biasing. The cause of motion of charge is drifting.

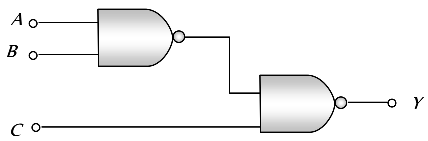
Reason : In the following circuit emitter is reverse biased and collector is forward biased.



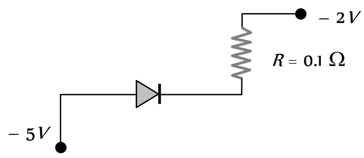
30. Assertion : De-morgan's theorem $\overline{A+B} = \overline{A} \cdot \overline{B}$ may be explained by the following circuit



Reason : In the following circuit, for output inputs *ABC* are 101



31. Assertion : In the following circuit the potential drop across the resistance is zero.



Reason : The given resistance has low value.

Answers

Solids and Crystals

1	d	2	d	3	d	4	a	5	a
6	a	7	a	8	b	9	a	10	c
11	d	12	c	13	b	14	a	15	a
16	a	17	d	18	b	19	c	20	c
21	b	22	a	23	b	24	a	25	d
26	d	27	d	28	d				

Semiconductors

1	c	2	b	3	d	4	b	5	b
6	b	7	b	8	c	9	d	10	a
11	b	12	a	13	a	14	d	15	c
16	b	17	b	18	b	19	c	20	c
21	d	22	b	23	ac	24	d	25	b
26	c	27	d	28	c	29	c	30	b
31	d	32	a	33	b	34	a	35	c
36	d	37	c	38	b	39	d	40	a
41	d	42	c	43	b	44	c	45	d
46	b	47	a	48	b	49	a	50	d
51	d	52	b	53	b	54	a	55	c
56	d	57	b	58	d	59	a	60	a
61	b	62	a	63	c	64	a	65	b
66	a	67	c	68	c	69	c	70	c
71	b	72	b	73	a	74	b	75	c
76	b	77	c	78	a	79	d	80	a
81	a	82	a	83	b	84	d	85	c
86	d	87	a	88	c	89	b	90	c
91	a	92	b	93	d	94	d	95	d
96	d	97	d	98	a	99	b	100	a
101	b								

Semiconductor Diode

1	b	2	a	3	b	4	b	5	c
6	b	7	a	8	b	9	a	10	a
11	b	12	b	13	b	14	c	15	d
16	c	17	c	18	bc	19	c	20	d
21	d	22	b	23	d	24	c	25	c
26	b	27	b	28	b	29	c	30	b
31	b	32	c	33	d	34	c	35	d
36	a	37	b	38	b	39	d	40	a
41	b	42	a	43	b	44	a	45	a
46	b	47	b	48	a	49	c	50	d
51	d	52	a	53	c	54	c	55	b
56	d	57	a	58	a	59	c	60	a
61	b	62	c	63	a	64	c	65	a
66	b	67	c	68	c	69	d	70	a
71	c	72	a	73	d	74	d	75	c
76	a	77	c	78	c	79	c		

Junction Transistor

1	a	2	c	3	a	4	d	5	d
6	b	7	d	8	b	9	b	10	b
11	c	12	d	13	d	14	a	15	b
16	b	17	d	18	b	19	ac	20	a
21	c	22	a	23	b	24	b	25	b
26	c	27	a	28	b	29	b	30	d
31	b	32	b	33	a	34	b	35	b
36	a	37	a	38	a	39	b	40	a
41	b	42	d	43	d	44	b		

Digital Electronics

1	b	2	c	3	b	4	a	5	b
6	c	7	d	8	b	9	a	10	a
11	d	12	b	13	c	14	a	15	c
16	a	17	b	18	b	19	a	20	a
21	b	22	c	23	b	24	c	25	b
26	c	27	b	28	a	29	a	30	d
31	d								

Valve Electronics

1	c	2	c	3	a	4	b	5	b
6	b	7	c	8	b	9	a	10	a
11	c	12	b	13	b	14	c	15	d
16	b	17	b	18	c	19	c	20	c
21	b	22	b	23	b	24	c	25	a
26	c	27	d	28	a	29	a	30	ad
31	d	32	c	33	c	34	a	35	a
36	c	37	b	38	d	39	b	40	c
41	c	42	b	43	d	44	b	45	c
46	c	47	b	48	b	49	a	50	c
51	a	52	b	53	b	54	a	55	c
56	a	57	a	58	d				

Critical Thinking Questions

1	c	2	a	3	c	4	c	5	abd
6	a	7	a	8	c	9	b	10	b
11	d	12	c	13	a	14	b	15	a
16	d	17	b	18	a	19	a	20	c
21	b	22	a	23	d	24	d	25	a
26	b	27	b	28	d	29	c	30	a
31	b	32	c	33	d	34	b	35	c
36	b	37	d	38	d	39	d	40	b
41	c	42	a	43	a	44	b	45	a
46	c	47	a	48	a	49	b	50	b

Graphical Questions

1	c	2	b	3	c	4	c	5	a

6	b	7	b	8	c	9	b	10	a
11	b	12	a	13	d	14	a	15	d
16	a	17	c	18	c	19	a	20	c
21	c	22	c	23	a	24	b	25	c
26	c	27	b	28	c	29	a	30	d

Assertion and Reason

1	d	2	a	3	c	4	a	5	d
6	a	7	b	8	a	9	b	10	a
11	c	12	e	13	e	14	d	15	b
16	d	17	d	18	a	19	d	20	a
21	c	22	d	23	a	24	a	25	b
26	a	27	a	28	a	29	b	30	c
31	b								

AS Answers and Solutions

Solids and Crystals

1. (d) Ionic bonds cone into being when atoms that have low ionization energies, and hence lose electrons rapidly, interact with other atoms that and to acquire excess electrons. The former atoms give up electrons to the latter and they there upon become positive and negative ions respectively.

2. (d) For tetragonal, cubic and orthorhombic system $\alpha = \beta = \gamma = 90°$.

3. (d) Tourmaline crystal is biaxial.

4. (a) The temperature co-efficient of resistance of conductor is positive.

5. (a) Density $\rho = \dfrac{nA}{N(a)^3}$

where $n = 2$ for bcc structure , $A = 39 \times 10^{-3}$ kg,

$N = 6.02 \times 10^{23}$, $a = \dfrac{2}{\sqrt{3}} d = \dfrac{2}{\sqrt{3}} \times (4.525 \times 10^{-10}) m$

(d = nearest neighbour distance = distance between centres of two neighbouring atoms $= \dfrac{a}{\sqrt{2}}$)

On putting the values we get $\rho = 907$

6. (a) The highest energy level which an electron can occupy in the valence band at 0 K, is called Fermi energy level.

7. (a) In a triclinic crystal $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90°$

8. (b) Metallic solids are opaque because incident light is absorbed by the free electrons in a metal.

9. (a) In ionic bonding electrons are transferred from one type of atoms to the other type creating positive and negative ions. For example in $NaCl$, Na loses one electrons and Cl gains one so that Na and Cl ions have a stable shell structure.

10. (c) Wood is non-crystalline.

11. (d) Cu has fcc structure, for fcc structure co-ordination number = 12

12. (c) Vander Waal force is weak dipole-dipole interaction.

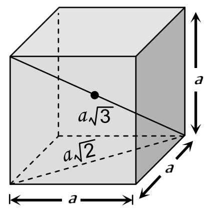
13. (b)

14. (a) The sodium chloride crystal structure has a fcc lattice with one chloride ion at each lattice point and one sodium ion half a cube length above it.

15. (a) In $NaCl$ crystal Na ion is surrounded by $6\,Cl^-$ ion, therefore coordination number of Na is 6.

16. (a) Sodium has bcc structure. The distance between body centre and a corner $= \dfrac{\sqrt{3}\,a}{2}$

$= \dfrac{\sqrt{3} \times 4.225}{2} = 3.66\,Å$

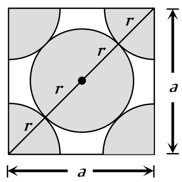


17. (d)

18. (b) For the fcc structure

$4r = (a^2 + a^2)^{1/2} = a\sqrt{2}$

$\Rightarrow r = \dfrac{a\sqrt{2}}{4} = \dfrac{a}{2\sqrt{2}}$



19. (c) Metals reflects incident light by the vibrations of free electrons under the influence of electric field of incident wave. The conductivity of metals decreases with increase of temperature due to increase in random motion of free electrons. The bonding is therefore metallic.

20. (c)

21. (b) The nearest distance between two atoms in a bcc lattice = 2 (atomic radius) $= 2 \times \left(\dfrac{\sqrt{3}\,a}{4}\right) = \dfrac{\sqrt{3}a}{2}$

22. (a) The net force on electron placed at the centre of bcc structure is zero. (By the principle of superposition of couloumb forces)

23. (b) For bcc packing, distance between two nearest atoms $d = 2r = 2\left(\dfrac{\sqrt{3}\,a}{4}\right)$

$\Rightarrow$ Lattice constant $a = \dfrac{2d}{\sqrt{3}} = \dfrac{2 \times 3.7}{\sqrt{3}} = 4.3Å$

24. (a)

25. (d) $\sqrt{2}\,a = 4r \Rightarrow a = \dfrac{4r}{\sqrt{2}} = \sqrt{2}(2r) = \sqrt{2} \times 2.54 = 3.59\,Å$

26. (d)

27. (d) Covalent bonding exists in semi-conductor.

28. (d) In H_2O covalent bonding is present.

Semiconductors

1. (c) In P-type semiconductors, holes are the majority charge carriers

2. (b) Ga has a valancy of 3.

3. (d) $\boxed{Ge} + \boxed{\text{Trivalent impurity}} \rightarrow \boxed{P\text{-type semiconductor}}$

4. (b) Since $n_e > n_h$; the semiconductor is N-type.

5. (b) Absence of one electron, creates the positive charge of magnitude equal to that of electronic charge.

6. (b) $\boxed{Ge}$ + $\boxed{\text{Pantavalent impurity}}$ $\boxed{N\text{ - type semiconductor}}$

7. (b) Impurity increases the conductivity.

8. (c) $\boxed{\text{Intrinsic semiconductor}}$ → Conductivity is due to the breaking of covalent bond

$\boxed{\text{Extrisnsic semiconductor}}$ → Conductivity is due to the breaking of covalent bond and excess of charge carriers due to impurity.

9. (d) Resistance of conductors (Cu) decreases with decrease in temperature while that of semi-conductors (Ge) increases with decrease in temperature.

10. (a) Aluminum is trivalent impurity.

11. (b) With temperature rise conductivity of semiconductors increases.

12. (a)

13. (a) Similar to Q. 11

14. (d) In insulators, the forbidden energy gap is largest and it is of the order of 6 eV.

15. (c) N-type semiconductors are neutral because neutral atoms are added during doping.

16. (b)

17. (b) In insulators, the forbidden energy gap is very large, in case of semiconductor it is moderate and in conductors the energy gap is zero.

18. (b) Similar to Q. 15.

19. (c) Phosphorus is pentavalent.

20. (c) In intrinsic semiconductors, the creation or liberation of one free electron by the thermal energy has created one hole. Thus in intrinsic semiconductors $n_e = n_h$

21. (d) Conductor has positive temperature coefficient of resistance but semiconductor has negative temperature coefficient of resistance.

22. (b) Boron is trivalent.

23. (a, c) In intrinsic semiconductors, electrons and holes both are charge carriers. In P-type semiconductors (Extrinsic semiconductors) holes are majority charge carriers.

24. (d)

25. (b)

26. (c) $\Delta E_{g(Germanium)} = 0.67\ eV$

27. (d) In P-type semiconductors, holes are majority charge carrier and electrons are minority charge carriers.

28. (c) At zero Kelvin, there is no thermal agitation and therefore no electrons from valence band are able to shift to conduction band.

29. (c) Antimony is a fifth group impurity and is therefore a donor of electrons.

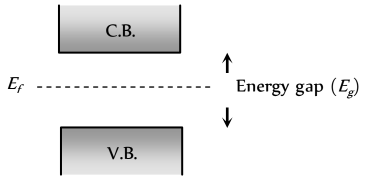
30. (b) Resistance of semiconductor $\propto \dfrac{1}{\text{Temperature}}$

31. (d) $\boxed{\text{Extrinsic S.C.}}$ → $P - \text{Type}\,(n_p >> n_e)$
→ $N - \text{Type}\,(n_e >> n_p)$

32. (a) At room temperature the number of electrons and holes are equal in the intrinsic semiconductor.

33. (b) Indium is trivalent, hence on doping with it, the intrinsic semiconductor becomes P-type semiconductor.

34. (a)



35. (c) In semiconductors, Forbidden energy gap is of the order of 1 eV.

36. (d) At $0K$ temperature semiconductor behaves as an insulator, because at very low temperature electrons cannot jump from the valence band to conduction band.

37. (c) Antimony is pentavalent.

38. (b) At $0K$ semiconductor behaves as insulator so it's resistance is infinite.

39. (d) The conduction and valence bands in the conductors merge into each other.

40. (a) For N-type semiconductor, the impurity should be pentavalent.

41. (d) When a free electron is produced, simultaneously a hole is also produced.

42. (c) For P – type semiconductor the doping impurity should be trivalent.

43. (b) The temperature co-efficient of resistance of a semiconductor is always negative.

44. (c) The resistance of semiconductor decreases with the increase in temperature.

45. (d) At absolute zero temperature, semiconductor.

46. (b) Formation of energy bands in solids are due to Pauli's exclusion principle.

47. (a) In P-type semiconductors, holes are majority charge carriers.

48. (b)

49. (a) Conductivity of semiconductors increases with rise in temperature.

50. (d) All are trivalent in nature.

51. (d) In N-type semiconductors, electrons are majority charge corners.

52. (b) When a strong current passes through the semiconductor it heats up the crystal and covalent bond are broken. Hence because of excess number of free electrons it behaves like a conductor.

53. (b)

54. (a) Phosphorus is a pentavalent impurity so $n_e > n_h$.

55. (c) Phosphorus is pentavalent while Indium is trivalent.

56. (d) Phosphorus and Arsenic both are pentavalent.

57. (b)

58. (d)

59. (a) For Ge, $E_g = 0.7\,eV = 0.7 \times 1.6 \times 10^{-19}\,J = 1.12 \times 10^{-19}\,J$

60. (a) At room temperature some covalent bond breaks and semiconductor behaves slightly as a conductor.

61. (b)

62. (a)

63. (c) Because boron is a trivalent impurity.
64. (a) In P-type semi conductor, holes are majority charge carriers.
65. (b) In intrinsic semiconductors, at room temperature $n_e = n_h$.
66. (a) In conductors valence band and conduction band overlaps.
67. (c) Because As is pentavalent impurity.
68. (c) At 0 K semiconductor behaves as an insulator.
69. (c)
70. (c)
71. (b) Antimony and phosphorous both are pentavalent.
72. (b) Gallium is trivalent impurity.
73. (a)
74. (b) One atom of pentavalent impurity, donates one electron.
75. (c)
76. (b) The charge on hole is positive.
77. (c) Phosphorus is pentavalent impurity.
78. (a) $n_i^2 = n_h n_e \Rightarrow (10^{19})^2 = 10^{21} \times n_e \Rightarrow n_e = 10^{17}/m^3$.
79. (d) Temperature co-efficient of semiconductor is negative.
80. (a) Copper, Aluminum, Iron are conductors, while Ge is semiconductor.
81. (a) At room temperature, few bonds breaks and electron hole pair generates inside the semiconductor.
82. (a)
83. (b) With rise in temperature, conductivity of semiconductor increases while resistance decreases.
84. (d) Gallium, boron and aluminum are trivalent.
85. (c) Because with rise in temperature, resistance of semiconductor decreases, hence overall resistance of the circuit increases, which in turn increases the current in the circuit.
86. (d) Extrinsic semiconductor (N-type or P-type) are neutral.
87. (a) Because $v_d = \dfrac{i}{(n_e)eA}$
88. (c)
89. (b) Resistivity is the intrinsic property, it doesn't depend upon length and shape of the semiconductors.
90. (c)
91. (a) $\lambda_{\max} = \dfrac{hc}{E} = \dfrac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.14 \times 1.6 \times 10^{-19}} = 10888 Å$
92. (b) In N-type semiconductor impurity energy level lies just below the conduction band.
93. (d)
94. (d)
95. (d) $\sigma = en_e\mu_e$

$$\Rightarrow n_e = \frac{\sigma}{e\mu_e} = \frac{6.24}{1.6 \times 10^{-19 \times 3900}} = 10^{16}/cm^3$$

96. (d) In semiconductors, the forbidden energy gap between the valence band and conduction band is very small, almost equal to kT. Moreover, valence band is completely filled where as conduction band is empty.
97. (d) In sample x no impurity level seen, so it is undoped. In sample y impurity energy level lies below the conduction bond so it is doped with fifth group impurity.

    In sample z, impurity energy level lies above the valence band so it is doped with third group impurity.
98. (a) Forbidden energy gap for carbon is greater than that of silicon.
99. (b)
100. (a) Because electrons needed less energy to move.
101. (b)

Semiconductor Diode

1. (b)

2. (a) In forward biased *PN*-junction, external voltage decreases the potential barrier, so current is maximum. While in reversed biased *PN*-junction, external voltage increases the potential barrier, so the current is very small.

3. (b)

4. (b) Filter circuits are used to get smooth *dc* π -filter is the best filter.

5. (c) In reverse bias no current flows.

6. (b) In reverse biasing, width of depletion layer increases.

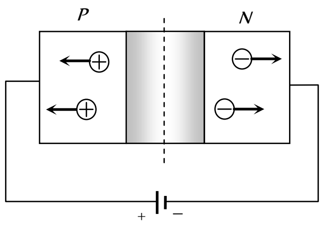
7. (a) Depletion layer consist of mainly stationary ions.

8. (b) Current flow is possible and $i=\frac{V}{R}=\frac{(4-1)}{300}=10^{-2}A$

9. (a) The potential of *P*-side is more negative that of *N*-side, hence diode is in reverse biasing. In reverse biasing it acts as open circuit, hence no current flows.

10. (a)

11. (b) It is used to convert *ac* into *dc* (rectifier)



12. (b)

13. (b) Because in case (1) N is connected with N. This is not a series combination of transistor.

14. (c)

15. (d)

16. (c) After a large reverse voltage is *PN*-junction diode, a huge current flows in the reverse direction suddenly. This is called Breakdown of *PN*-junction diode.

17. (c) In forward biasing both positive and negative charge carriers move towards the junction.

18. (b,c)

19. (c) When polarity of the battery is reversed, the *P-N* junction becomes reverse biased so no current flows.

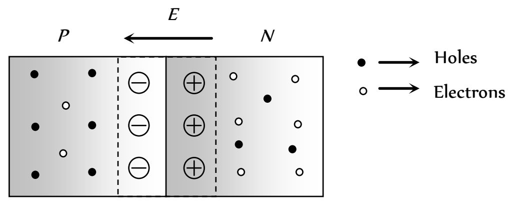
20. (d) Resistance in forward biasing $R_{fr}\approx 10\,\Omega$ and resistance in reverse biasing $R_{Rw}\approx 10^5\Omega \Rightarrow \frac{R_{fr}}{R_{Rw}}=\frac{1}{10^4}$

21. (d)

22. (b) In forward biasing width of depletion layer decreases.

23. (d)

24. (c) At junction a potential barrier/depletion layer is formed, with *N*-side at higher potential and *P*-side at lower potential. Therefore there is an electric field at the junction directed from the *N*-side to *P*-side



25. (c) In *N*-type semiconductor majority charge carriers are electrons.

26. (b) In forward biasing the diffusion current increases and drift current remains constant so not current is due to the diffusion.

In reverse biasing diffusion becomes more difficult so net current (very small) is due to the drift.

27. (b) At a particular reverse voltage in *PN*-junction, a huge current flows in reverse direction known as avalanche current.

28. (b) Due to the large concentration of electrons in *N*-side and holes in *P*-side, they diffuses from their own side to other side. Hence depletion region produces.

29. (c) Only in option (c), *P*-side is more negative as compared to *N*-side.

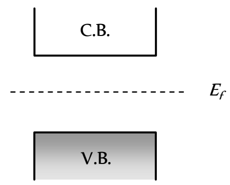
30. (b) Depletion layer is more in less doped side.

31. (b) In forward biasing *P*-side is connected with positive terminal and *N*-side with negative terminal of the battery

32. (c) In forward biasing of *PN*-junction diode, current mainly flows due to the diffusion of majority charge carriers.

33. (d)

34. (c) In forward biasing of *PN* junction diode width of depletion layer decreases. In intrinsic semiconductor fermi energy level is exactly in the middle of the forbidden gap



35. (d)

36. (a) At high reverse voltage, the minority charge carriers, acquires very high velocities. These by collision break down the covalent bonds, generating more carriers. This mechanism is called Avalanche breakdown.

37. (b) Because *P*-side is more negative as compared to *N*-side.

38. (b) When reverse bias is increased, the electric field at the junction also increases. At some stage the electric field breaks the covalent bond, thus the large number of charge carriers are generated. This is called Zener breakdown.

39. (d) In forward biasing both V and x decreases.

40. (a)

41. (b) In figure 2,4 and 5. *P*-crystals are more positive as compared to *N*-crystals.

42. (a) *ac* → $\boxed{\text{Rectifier}}$ → *dc*

43. (b) In this condition $P-N$ junction is reverse biased.

44. (a)

45. (a) $E=\frac{V}{d}=\frac{0.5}{5\times 10^{-7}}=10^6\ V/m$.

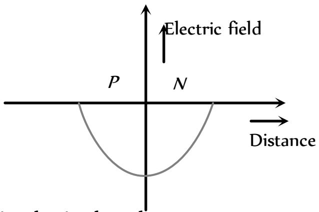
46. (b) Across the $P-N$ junction, a barrier potential is developed whose direction is from N region to P region.

47. (b)

48. (a) In forward biasing, resistance of PN junction diode is zero, so whole voltage appears across the resistance.

49. (c)

50. (d) The electric field strength versus distance curve across the *P-N* junction is as follows



51. (d)

52. (a) It doesn't Obey's ohms law.

53. (c) Because *N*-side is more positive as compared to *P*-side.

54. (c) When a light (wavelength sufficient to break the covalent bond) falls on the junction, new hole electron pairs are created. No. of produced electron hole pair deponed upon no. of photons. So photo emf or current proportional to intensity of light.

55. (b)

56. (d) For full wave rectifier $\eta = \dfrac{81.2}{1+\dfrac{r_f}{R_L}}$

$\Rightarrow n_{\max} = 81.2\%$ $(r_f << R_L)$

57. (a)

58. (a)

59. (c) In reverse biasing negative terminal of the battery is connected to *N*-side.

60. (a) In the given condition diode is in reverse biasing so it acts as open circuit. Hence potential difference between *A* and *B* is 6 *V*

61. (b) Zener breakdown can occur in heavily doped diodes. In lightly doped diodes the necessary voltage is higher, and avalanche multiplication is then the chief process involved.

62. (c)

63. (a)

64. (c) Diode acts as open switch only when it is reverse biased

65. (a) Because *P*-side is more negative than *N*-side.

66. (b) In unbiased condition of *PN*-junction, depletion region is generated which stops the movement of charge carriers.

67. (c) For a wide range of values of load resistance, the current in the zener diode may change but the voltage across it remains unaffected. Thus the output voltage across the zener diode is a regulated voltage.

68. (c)

69. (d) Arsenic has five valence electrons, so it a donor impurity. Hence *X* becomes *N*-type semiconductor. Indium has only three outer electrons, so it is an acceptor impurity. Hence *Y* becomes *P*-type semiconductor. Also *N* (*i.e. X*) is connected to positive terminal of battery and *P*(*i.e. Y*) is connected to negative terminal of battery so *PN*-junction is reverse biased.

70. (a)

71. (c) In photodiode, it is illuminated by light radiations, which in turn produces electric current.

72. (a)

73. (d)

74. (d) By using $E = \dfrac{V}{d} = \dfrac{0.6}{10^{-6}} = 6\times10^{5}\ V/m$

75. (c) The given circuit is full wave rectifier.

76. (a) The diode is in reverse biasing so current through it is zero.

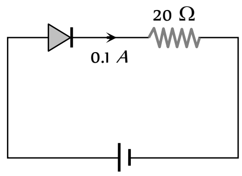
77. (c) In full wave rectifier, the fundamental frequency in ripple is twice that of input frequency.

78. (c)

79. (c) $V' = V + IR$

= 0.5 + 0.1 × 20

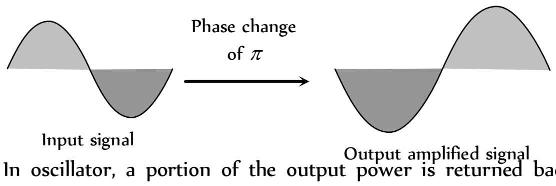
= 2.5 *V*



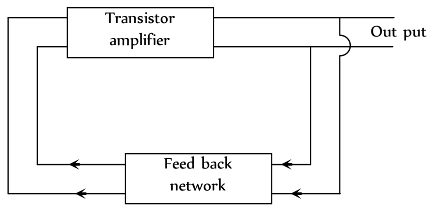
Junction Transistor

1. (a) When *NPN* transistor is used as an amplifier, majority charge carrier electrons of *N*-type emitter move from emitter to base and than base to collector.

2. (c)



3. (a) In oscillator, a portion of the output power is returned back (feed back) to the input in phase with the starting power. This process is termed as positive feedback.

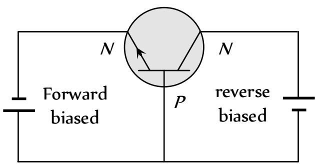


4. (d) The emitter base junction is forward biased while collector base junction is reversed biased.

5. (d) Given $i_c = \dfrac{80}{100}\times i_e \Rightarrow 24 = \dfrac{80}{100}\times i_e \Rightarrow i_e = 30\,mA$

By using $i_e = i_b + i_c \Rightarrow i_b$ = 30 – 24 = 6 *mA*.

6. (b)



7. (d) α is the ratio of collector current and emitter current while β is the ratio of collector current and base current.

8. (b)

9. (b) $\beta = \dfrac{\alpha}{1-\alpha} = \dfrac{0.98}{1-0.98}$ =49.

10. (b) $\beta = \dfrac{\alpha}{1-\alpha} = \dfrac{0.96}{1-0.96}$ =24.

11. (c) $\alpha = \frac{i_c}{i_e}$ =0.96 and i_e = 7.2 mA

$\Rightarrow i_c = 0.96 \times i_e$ = 0.96 × 7.2 = 6.91 mA

$\therefore i_e = i_c + i_b \Rightarrow$ 7.2 = 6.91 + $i_b \Rightarrow i_b$ = 0.29 mA.

12. (d)

13. (d) $i_C = \frac{90}{100} \times i_E \Rightarrow$ 10 = 0.9 × i_e = 11mA

Also $i_E = i_B + i_C \Rightarrow i_B = 11 - 10 = 1mA$.

14. (a) Current gain $\beta = \frac{\Delta i_c}{\Delta i_b} \Rightarrow \Delta i_c = \beta \times \Delta i_b$ = 80 × 250 μA.

15. (b) In transistor, base is least doped.

16. (b)

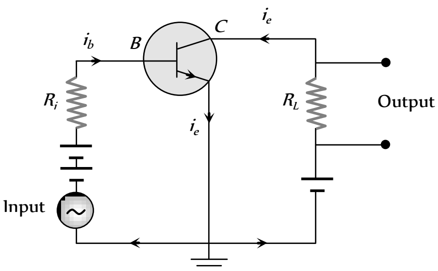
17. (d) β = 50, R_i = 1000 Ω, V_i = 0.01V

$$\beta = \frac{i_c}{i_b} \text{ and } i_b = \frac{V_i}{R_i} = \frac{0.01}{10^3} = 10^{-5} A$$

Hence $i_c = 50 \times 10^{-5} A = 500\,\mu A$.

18. (b) $\alpha = \frac{\beta}{1+\beta} = \frac{99}{1+99} = 0.99$.

19. (a,c) The circuit of a CE amplifier is as shown below.



This has been shown a NPN transistor. Therefore base emitter are forward, biased and input signal is connected between base and emitter.

20. (a) The base is always thin

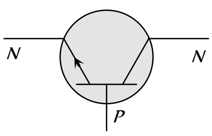
21. (c) Voltage gain = β × Resistance gain

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.99}{(1-0.99)} = 99$$

Resistance gain $= \frac{10 \times 10^3}{10^3} = 10$

$\Rightarrow$ Voltage gain = 99 × 10 = 990.

22. (a) The arrow head in the transistor symbol always shows the direction of hole flow in the emitter region.



23. (b)

24. (b) Because emitter (N) is common to both, base (P) and collector (N).

25. (b) Emitter is heavily doped.

26. (c) α = 0.8 $\Rightarrow \beta = \frac{0.8}{(1-0.8)}$ =4

Also $\beta = \frac{\Delta i_c}{\Delta i_b} \Rightarrow \Delta i_c = \beta \times \Delta i_b$ = 4 × 6 = 24mA.

27. (a) $\Delta i_c = \alpha\, \Delta i_e = 0.98 \times 2 = 1.96\, mA$

$\therefore \quad \Delta i_b = \Delta i_e - \Delta i_c = 2 - 1.96 = 0.04\, mA$.

28. (b) $i_e = i_b + i_c \Rightarrow i_c = i_e - i_b$

29. (b) $V_b = i_b R_b \Rightarrow R_b = \frac{9}{35 \times 10^{-6}}$ =257 $k\Omega$.

30. (d) $\Delta i_e = \Delta i_c + \Delta i_b$

$\Rightarrow$ 8 = 7.8 + $\Delta i_b \Rightarrow \Delta i_b$ =0.2mA = 200 μA.

31. (b) $\beta = \frac{i_c}{i_b}$

32. (b) FET is unipolar.

33. (a)

34. (b) $i_e = i_b + i_c \Rightarrow \frac{i_e}{i_c} = \frac{i_b}{i_c} + 1 \Rightarrow \frac{1}{\alpha} = \frac{1}{\beta} + 1 \Rightarrow \alpha = \frac{\beta}{(1+\beta)}$.

35. (b) In NPN transistor when emitter-base is forward biased, electrons move from emitter to base.

36. (a) Here $\Delta V_c = 0.5\, V$, Δi_c =0.05mA = 0.05 × $10^{-3}A$

Output resistance is given by

$$R_{out} = \frac{\Delta V_c}{\Delta i_c} = \frac{0.5}{0.05 \times 10^{-3}} = 10^4 \Omega = 10k\Omega.$$

37. (a) Oscillator can produce radio waves of constant amplitude.

38. (a) $h_{fe} = \left(\frac{\Delta i_c}{\Delta i_b}\right)_{V_{ce}} = \frac{8.2}{8.3 - 8.2} = 82$

39. (b) Current gain $\beta = \frac{\Delta i_c}{\Delta i_b} \Rightarrow \Delta i_b = \frac{1 \times 10^{-3}}{100} = 10^{-5} A$ =0.01mA.

By using $\Delta i_e = \Delta i_b + \Delta i_c \Rightarrow \Delta i_e$ = 1.01 + 1 = 1.01mA.

40. (a) In CB amplifier Input and output voltage signal are in same phase.

41. (b)

42. (d)

43. (d) For CE configuration voltage gain $= \beta \times R_L / R_i$

Power gain $= \beta^2 \times R_L / R_i \Rightarrow \frac{\text{Power gain}}{\text{Voltage gain}} = \beta$

44. (b) As we know $i_E = i_C + i_B$

$$\Rightarrow \frac{i_e}{i_c} = 1 + \frac{i_b}{i_c} \Rightarrow \frac{1}{\alpha} = 1 + \frac{1}{\beta} \Rightarrow \beta = \frac{\alpha}{1-\alpha}.$$

Digital Electronics

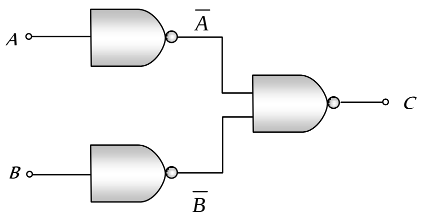
1. (b)

2. (c)

3. (b) For 'OR' gate $X = A + B$

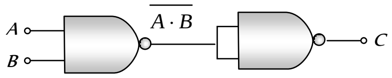
i.e. $0+0=0$, $0+1=1$, $1+0=1$, $1+1=1$

4. (a)



$C=\overline{\overline{A}.\overline{B}}=\overline{\overline{A}}+\overline{\overline{B}}=A+B$ (De morgan's theorem)

Hence output C is equivalent to OR gate.



$C=\overline{\overline{AB}.\overline{AB}}=\overline{\overline{AB}}+\overline{\overline{AB}}=AB+AB=AB$

In this case output C is equivalent to AND gate.

5. (b) In 'NOR' gate $Y=\overline{A+B}$

i.e. $\overline{0+0}=\overline{0}=1$, $\overline{1+0}=\overline{1}=0$

$\overline{0+1}=\overline{1}=0$, $\overline{1+1}=\overline{1}=0$

6. (c) For 'XNOR' gate $Y=\overline{A}\,\overline{B}+AB$

i.e. $\overline{0}.\overline{0}+0.0=1.1+0.0=1+0=1$

$\overline{0}.\overline{1}+0.1=1.0+0.1=0+0=0$

$\overline{1}.\overline{0}+1.0=0.1+1.0=0+0=0$

$\overline{1}.\overline{1}+1.1=0.0+1.1=0+1=1$

7. (d) The output D for the given combination

$D=\overline{(A+B).C}=\overline{(A+B)}+\overline{C}$

If $A=B=C=0$ then $D=\overline{(0+0)}+\overline{0}=\overline{0}+\overline{0}=1+1=1$

If $A=B=1, C=0$ then $D=\overline{(1+1)}+\overline{0}=\overline{1}+\overline{0}=0+1=1$

8. (b)

9. (a) The Boolean expression for 'NOR' gate is $Y=\overline{A+B}$

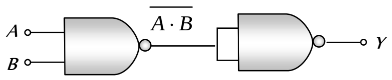
i.e. if $A=B=0$ (Low), $Y=\overline{0+0}=\overline{0}=1$ (High)

10. (a)

11. (d) The Boolean expression for 'AND' gate is $R=P.Q$

$\Rightarrow 1.1=1,\ 1.0=0,\ 0.1=0,\ 0.0=0$

12. (b) Two 'NAND' gates are required as follows



$Y=\overline{\overline{AB}.\overline{AB}}=AB$

13. (c) For 'NAND' gate (option c), output $=\overline{0.1}=\overline{0}=1$

14. (a) AND + NOT $\rightarrow$ NAND

15. (c) For 'NOT' gate $X=\overline{A}$

16. (a) The given Boolean expression can be written as

$Y=\overline{(A+B)}.\overline{(A.B)}=(\overline{A}.\overline{B}).(\overline{A}+\overline{B})=(\overline{A}\,\overline{A}).\overline{B}+\overline{A}(\overline{B}.\overline{B})$

$=\overline{A}.\overline{B}+\overline{A}\,\overline{B}=\overline{A}\,\overline{B}$

A	B	Y
0	0	1
1	0	0
0	1	0
1	1	0

17. (b) For 'AND' gate, if output is 1 then both inputs must be 1.

18. (b)

19. (a)

20. (a) The given symbol is of 'AND' gate.

21. (b) It is the symbol of 'NOR' gate.

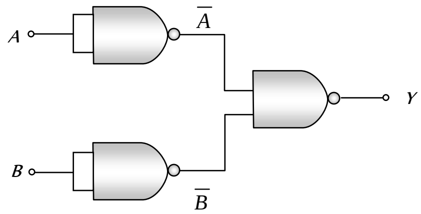
22. (c) The Boolean expression for the given combination is output $Y=(A+B).C$

The truth table is

A	B	C	Y =(A+B).C
0	0	0	0
1	0	0	0
0	1	0	0
0	0	1	0
1	1	0	0
0	1	1	1
1	0	1	1
1	1	1	1

Hence $A=1$, $B=0$, $C=1$

23. (b)



$Y=\overline{\overline{A}.\overline{B}}=\overline{\overline{A}}+\overline{\overline{B}}=A+B$

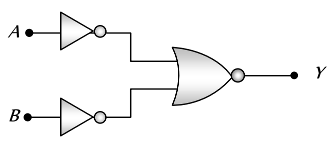
This output equation is equivalent to OR gate.

24. (c) If inputs are A and B then output for NAND gate is $Y=\overline{AB}$

$\Rightarrow$ If $A=B=1$, $Y=\overline{1.1}=\overline{1}=0$

25. (b)

26. (c)



$Y=\overline{\overline{A}+\overline{B}}$

According to De morgan's theorem

$Y=\overline{\overline{A}+\overline{B}}=\overline{\overline{A}}.\overline{\overline{B}}=A.B$

This is the output equation of 'AND' gate.

27. (b) The output of OR gate is $Y = A + B$.

28. (a) The given symbol is of NAND gate.

29. (a) $(100010)_2 = 2^5 \times 1 + 2^4 \times 0 + 2^3 \times 0 + 2^2 \times 0 +$

$$2^1 \times 1 + 2^0 \times 0 = 32 + 0 + 0 + 0 + 2 + 0 = (34)_{10}$$

and $(11011)_2 = 2^4 \times 1 + 2^3 \times 1 + 2^2 \times 0 + 2^1 \times 1 + 2^0 \times 1$

$$= 16 + 8 + 0 + 2 + 1 = (27)_{10}$$

$\therefore$ Sum $(100010)_2 + (11011)_2 = (34)_{10} + (27)_{10} = (61)_{10}$

Now

2	61	Remainder
2	30	1 LSD
2	15	0
2	7	1
2	3	1
2	1	1
	0	1 MSD

$\therefore$ Required sum (in binary system)

$(100010)_2 + (11011)_2 = (111101)_2$

30. (d) For 'NAND' gate $C = \overline{A.B}$

i.e. $\overline{0.0} = \overline{0} = 1$, $\overline{0.1} = \overline{0} = 1$

$\overline{1.0} = \overline{0} = 1$, $\overline{1.1} = \overline{1} = 0$

31. (d) 'NOR' gates are considered as universal gates, because all the gates like AND, OR, NOT can be obtained by using only NOR gates.

Valve Electronics (Diode and Triode)

1. (c) According to Richardson-Dushman equation, number of thermions emitted per sec per unit area $J = AT^2 e^{-W_0/kT}$ $\Rightarrow$ $J \propto T^2$

2. (c) Intensity $\propto$ Number of electrons

3. (a) In SCR (Space charge region) electrons collect around the plate, this cloud decreases the emission of electrons from the cathode, hence plate current decreases.

4. (b)

5. (b) By using $g_m = \dfrac{\Delta i_p}{\Delta v_g} \Rightarrow 3 \times 10^{-4} = \dfrac{\Delta i_p}{-1-(-3)}$

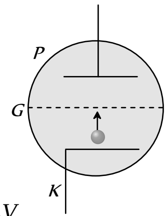
$$\Rightarrow \Delta i_p = 6 \times 10^{-4} A = 0.6\, mA$$

6. (b) Voltage gain $A_v = \dfrac{\mu}{1 + \dfrac{r_p}{R_L}}$ and $\mu = r_p \times g_m$

$$\Rightarrow r_p = \frac{42}{2 \times 10^{-3}} = 21000\,\Omega \Rightarrow A_v = \frac{42}{1 + \dfrac{21000}{50 \times 10^3}} = 29.57$$

7. (c) Voltage gain $A_v = \dfrac{\mu}{1 + \dfrac{r_p}{R_L}}$, for $r_p = R_L \Rightarrow A_v = \dfrac{\mu}{2}$

8. (b) When grid is given positive potential more electrons will cross the grid to reach the positive plate *P*. Hence current increases.



9. (a) By using $\mu = -\dfrac{\Delta V_p}{\Delta V_g} = r_p \times g_m$

$$\Rightarrow 7 \times 10^3 \times 2.5 \times 10^{-3} = -\frac{50}{\Delta V_g} \Rightarrow \Delta V_g = -2.86\,V.$$

10. (a) Using voltage gain $A_v = \dfrac{\mu}{1 + \dfrac{r_p}{R_L}}$ also $\mu = r_p \times g_m$

$$\Rightarrow r_p = \frac{\mu}{g_m} = \frac{20}{3 \times 10^{-3}}$$

$$\therefore A_v = \frac{20}{1 + \dfrac{20}{3 \times 10^{-3} \times 3 \times 10^4}} = \frac{180}{11} = 16.36.$$

11. (c) Voltage gain $= \dfrac{V_{out}}{V_{in}} = \dfrac{\mu}{1 + \dfrac{r_p}{R_L}} \Rightarrow \dfrac{V_{out}}{0.5} = \dfrac{25}{1 + \dfrac{40 \times 10^3}{10 \times 10^3}}$

$$\Rightarrow V_{out} = 2.5\,V.$$

12. (b) $\mu = -\dfrac{\Delta V_p}{\Delta V_G} \Rightarrow \Delta V_p = -\mu \Delta V_G = -20 \times (-0.2) = 4\,V.$

13. (b) Voltage gain $A_V = \dfrac{\mu}{1 + \dfrac{r_p}{R_L}}$ and $\mu = r_p \times g_m$

$$\Rightarrow \mu = 10 \times 10^3 \times 3 \times 10^{-3} = 30$$

$$\therefore A_v = \frac{\mu}{1 + \dfrac{r_p}{2r_p}} = \frac{2}{3}\mu = \frac{2}{3} \times 30 = 20.$$

14. (c)

15. (d) After saturation plate current can be increased by increasing the temperature of filament. It can be done by increasing the filament current.

16. (b) The maximum voltage gain $(A)_{max} = \mu$

(Which is obtained when $R_L = \infty$).

17. (b) Voltage gain $A_v = \dfrac{\mu}{1 + \dfrac{r_p}{R_L}}$

$$\because R_L = 1.5\, r_p \Rightarrow A_v = \frac{\mu}{1 + \dfrac{r_p}{1.5\, r_p}} = \frac{3}{5}\mu = \frac{3}{5} \times 20 = 12.$$

18. (c)

19. (c)

20. (c) $\mu = \dfrac{\Delta V_p}{\Delta V_g} \Rightarrow \Delta V_p = \mu \Delta V_g = 15 \times 0.3 = 4.5$ *volt*.

21. (b) Plate resistance $=\frac{1}{\text{slope}}=\frac{1}{10^{-3}\times 10^{-3}}=10^{6}\Omega$

$= 1000\ k\Omega$ (static).

22. (b) Using $A_v=\frac{\mu}{1+\frac{r_p}{R_L}}$ and $\mu = r_p \times g_m$

$$\Rightarrow r_p=\frac{\mu}{g_m}=\frac{50}{2\times 10^{-3}}=25\times 10^{3}\Omega$$

$$\therefore A_v=\frac{50}{1+\frac{25\times 10^3}{25\times 10^3}}=25.$$

23. (b) $P = Vi \Rightarrow V=\frac{P}{i}=\frac{448\times 10^{-3}}{14\times 10^{15}\times 1.6\times 10^{-19}}=200V$

24. (c) $\mu=\frac{(V_{p_1}-V_{p_2})}{(V_{G_1}-V_{G_2})}=\frac{(200-220)}{(0.5-1.3)}=25.$

25. (a) $\mu=r_p\times g_m \Rightarrow g_m=\frac{\mu}{r_p}=\frac{22}{6600}=\frac{1}{300}.$

26. (c) $r_p=\frac{V_{p_1}-V_{p_2}}{I_{p_1}-I_{P_2}}=\frac{75-100}{(2-4)\times 10^{-3}}=12.5\times 10^{3}\Omega=12.5k\Omega.$

27. (d)

28. (a)

29. (a) Voltage amplification $A_v=\frac{\mu}{1+\frac{r_p}{R_L}}$

$$\Rightarrow \quad 25=\frac{\mu}{1+\frac{r_p}{50\times 10^3}} \qquad \text{......(i)}$$

and $30=\frac{\mu}{1+\frac{r_p}{100\times 10^3}}$ ......(ii)

an solving equation (i) and (ii), $r_p = 25k\Omega$.

30. (a, d)

31. (d)

32. (c) Before saturation region, linear region comes. In linear region $i_p \propto V_p$

$$\Rightarrow \frac{i_1}{i_2}=\frac{V_{p_1}}{V_{p_2}}=\frac{400}{200}=\frac{2}{1}.$$

33. (c) $i_p = 1.125 - 1.112 = 0.013A = 13\ mA$.

34. (a)

35. (a)

36. (c) Comparing the given equation with standard equation

$$i=AT^2e^{qV/kT} \Rightarrow V_L=\frac{kT}{V}.$$

37. (b)

38. (d) $r_p=\frac{\Delta V_p}{\Delta i_p}=\frac{150-100}{(12-7.5)\times 10^{-3}}=\frac{50}{4.5}\times 10^{3}=11.1k\Omega.$

39. (b)

40. (c) Voltage amplification $A_v=\frac{\mu}{1+\frac{r_p}{R_L}}=\frac{\mu R_L}{R_L+r_p}$

$$\Rightarrow \frac{A_1}{A_2}=\frac{2+4}{4+4}=\frac{3}{4}.$$

41. (c) A diode is used as a rectifier to convert *ac* in to *dc*.

42. (b) Fluctuating *dc* → Filter circuit → smooth *dc*.

43. (d)

44. (b)

45. (c) $\mu=r_p\times g_m \Rightarrow r_p=\frac{20}{10^{-3}}=2\times 10^{4}\Omega.$

46. (c)

47. (b) $\mu=-\frac{\Delta V_p}{\Delta V_g}$

$\Rightarrow \Delta V_p=-\mu\times\Delta V_g=-50(-0.20)=10\ V.$

48. (b) $r_p=\frac{1}{\text{slope}}=\frac{1}{2\times 10^{-2}\times 10^{-3}}=50k\Omega.$

49. (a) Voltage amplification $A_v=\frac{\mu}{1+\frac{r_p}{R_L}}=\frac{r_p\times g_m\times R_L}{R_L+r_p}$

$$\Rightarrow 10=\frac{20\times 10^3\times 2.5\times 10^{-3}\times R_L}{(R_L+20\times 10^3)} \Rightarrow R_L=5k\Omega.$$

50. (c) Voltage gain $A_v=\frac{\mu}{1+\frac{r_p}{R_L}}=\frac{18}{1+\frac{8\times 10^3}{10^4}}=10.$

51. (a) Ripple factor $r=\sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2-1}=\sqrt{\frac{(I_0/2)^2}{(I_0/\pi)^2}-1}=1.21.$

52. (b)

53. (b)

54. (a) $\mu=r_p\times g_m=2.5\times 10^{4}\times 2\times 10^{-3}=50.$

55. (c) $\mu=\left(\frac{\Delta V_p}{\Delta V_g}\right)_{i_p=\text{constant}}=\frac{(225-200)}{(5.75-5)}=33.3$

56. (a) $g_m=\left(\frac{\Delta I_p}{\Delta V_g}\right)_{V_p=\text{constant}}=\frac{(7.5-5.5)}{-1.2-(-2.2)}=2m\ mho$

57. (a)

58. (d) Using $\mu=r_p\times g_m \Rightarrow g_m=\frac{20}{10\times 10^3}=2\times 10^{-3}.$

Critical Thinking Questions

1. (c) Number density of atoms in silicon specimen $= 5\times 10^{28}$ *atom/m*^{3} $= 5\times 10^{22}$ *atom/cm*^{3}

Since one atom of indium is doped in 5×10^{7} *Si* atom. So number of indium atoms doped per *cm*^{3} of silicon.

$$n=\frac{5\times 10^{22}}{5\times 10^{7}}=1\times 10^{15}\ atom/cm^3.$$

2. (a) The probability of electrons to be found in the conduction band of an intrinsic semiconductor

$$P(E)=\frac{1}{1+e^{\frac{(E-E_F)}{kT}}}$$; where k = Boltzmann's constant

Hence, at a finite temperature, the probability decreases exponentially with increasing band gap.

3. (c) When donor impurity (+5 valence) added to a pure silicon (+4 valence), the +5 valence donor atom sits in the place of + 4 valence silicon atom. So it has a net additional + 1 electronic charge. The four valence electron form covalent bond and get fixed in the lattice. The fifth electron (with net – 1 electronic charge) can be approximated to revolve around + 1 additional charge. The situation is like the hydrogen atom for which energy is given by $E=-\frac{13.6}{n^2}eV$. For the case of hydrogen, the permittivity was taken as ε_0. However, if the medium has a permittivity ε_r relative to ε_0 then $E=-\frac{13.6}{\varepsilon_r^2 n^2}eV$

For Si, ε_r = 12 and for n = 1, $E \simeq 0.1\,eV$

4. (c) The forward current

$$i=i_s(e^{eV/kT}-1)=10^{-5}\left[e^{\frac{1.6\times10^{-19}\times0.2}{1.4\times10^{-23}\times300}}-1\right]$$

$$=10^{-5}[2038.6-1]=20.376\times10^{-3}A$$

5. (a,b,d) At 0 K, a semiconductor becomes a perfect insulator. Therefore at 0 K, if some potential difference is applied across an insulator or a semiconductor, current is zero. But a conductor will become a superconductor at 0 K. Therefore, current will be infinite. In reverse biasing at 300 K through a P-N junction diode, a small finite current flows due to minority charge carriers.

6. (a) Since diode in upper branch is forward biased and in lower branch is reversed biased. So current through circuit $i=\frac{V}{R+r_d}$; here r_d = diode resistance in forward biasing = 0

$$\Rightarrow\ i=\frac{V}{R}=\frac{2}{10}=0.2A.$$

7. (a) The voltage drop across resistance = 8 – 0.5 = 7.5 V

$\therefore$ Current $i=\frac{7.5}{2.2\times10^3}=3.4\,mA$

8. (c) $E=\frac{hc}{\lambda}\Rightarrow\lambda=\frac{hc}{E}=\frac{6.6\times10^{-34}\times3\times10^8}{57\times10^{-3}\times1.6\times10^{-19}}$ =217100Å.

9. (b) The diode in lower branch is forward biased and diode in upper branch is reverse biased

$$\therefore\quad i=\frac{5}{20+30}=\frac{5}{50}A.$$

10. (b) The current through circuit $i=\frac{P}{V}=\frac{100\times10^{-3}}{0.5}=0.2A$

$\therefore$ voltage drop across resistance = 1.5 – 0.5 = 1 V

$$\Rightarrow R=\frac{1}{0.2}=5\,\Omega.$$

11. (d) In common emitter configuration current gain

$$A_i=\frac{-h_{fe}}{1+h_{oe}R_L}=\frac{-50}{1+25\times10^{-6}\times10^3}$$ = – 48.78.

12. (c) Voltage gain $=\frac{\text{Output voltage}}{\text{Input voltage}}$

$\Rightarrow V_{out}=V_{in}\times$ Voltage gain

$\Rightarrow V_{out}=V_{in}\times$ Current gain $\times$ Resistance gain

$$=V_{in}\times\beta\times\frac{R_L}{R_{BE}}=10^{-3}\times100\times\frac{10}{1}=1V.$$

13. (a) $n_e=8\times10^{18}/m^3,\ n_h=5\times10^{18}/m^3$

$$\mu_e=2.3\frac{m^2}{volt-\sec},\ \mu_h=0.01\frac{m^2}{volt-\sec}$$

$\because n_e>n_h$ so semiconductor is N-type

Also conductivity $\sigma=\frac{1}{\text{Resistivity}(\rho)}=e(n_e\mu_e+n_h\mu_h)$

$$\Rightarrow\frac{1}{\rho}=1.6\times10^{-19}[8\times10^{18}\times2.3+5\times10^{18}\times0.01]$$

$\Rightarrow \rho$ = 0.34 Ω-m.

14. (b) $V_{rms}=\frac{V_0}{2}=\frac{200}{2}=100V$

15. (a) At knee point voltage across the diode is 0.7 V.

Hence voltage across resistance R is 5 – 0.7 = 4.3 V.

$\Rightarrow$ using $V=iR\Rightarrow 4.3=1\times10^{-3}\times R\Rightarrow R=4.3\ k\Omega$.

16. (d) In positive half cycle one diode is in forward biasing and other is in reverse biasing while in negative half cycle their polarity reverses, and direction of current is opposite through R for positive and negative half cycles so out put is not rectified.

Since R_1 and R_2 are different hence the peaks during positive half and negative half of the input signal will be different.

17. (b) In half wave rectifier $V_{dc}=\frac{V_0}{\pi}=\frac{10}{\pi}$

18. (a) In common base mode α = 0.98, R = 5 $k\Omega$, R_{in} = 70Ω

$\therefore$ voltage gain $A_v=\alpha\times\frac{R}{R_{in}}=0.98\times\frac{5\times10^3}{70}=70$

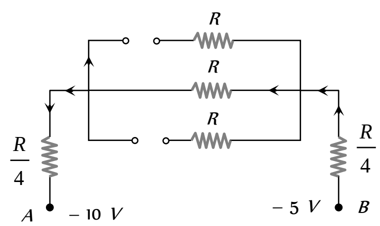
Power gain = Current gain $\times$ Voltage gain

= 0.98 $\times$ 70 = 68.6

19. (a) $r_n=\varepsilon_r\left(\frac{n^2}{Z}\right)a_o=12\times\frac{(5^2)}{15}\times0.53$ = 10.6 Å.

20. (c) (i) V_A = –10 V and V_B = –5 V

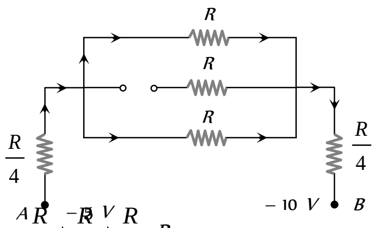
Diodes D_1 and D_3 are reveres biased and D_2 is forward biased.



$$\Rightarrow R_{AB} = R + \frac{R}{4} + \frac{R}{4} = \frac{3}{2}R.$$

(ii) When $V = -5\,V$ and $V = -10\,V$

Diodes D is reverse biased D and D are forward biased



$$\Rightarrow R_{AB} = \frac{R}{4} + \frac{R}{2} + \frac{R}{4} = R.$$

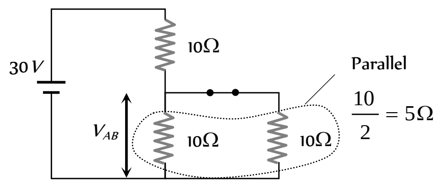
(iii) In this case equivalent resistance between A and B is also R.

Hence (ii) = (iii) < (i).

21. (b) According to the given polarity, diode D is forward biased while D is reverse biased. Hence current will pass through D only.

So current $i = \dfrac{6}{(150+50+100)} = 0.02\,A$

22. (a) Diode is in forwards biasing hence the circuit can be redrawn as follows



$$V_{AB} = \frac{30}{(10+5)} \times 5 = 10\,V$$

23. (d) The diode D will conduct for positive half cycle of *a.c.* supply because this is forward biased. For negative half cycle of *a.c.* supply, this is reverse biased and does not conduct. So out put would be half wave rectified and for half wave rectified out put

$$V_{rms} = \frac{V_0}{2} = \frac{200\sqrt{2}}{2} = \frac{200}{\sqrt{2}}$$

24. (d) $\sigma = ne(\mu_e + \mu_h) = 2\times10^{19} \times 1.6\times10^{-19}(0.36+0.14)$

$= 1.6\,(\Omega\text{-}m)^{-1}$

$$R = \rho\frac{l}{A} = \frac{l}{\sigma A} = \frac{0.5\times10^{-3}}{1.6\times10^{-4}} = \frac{25}{8}\Omega$$

$$\therefore\ i = \frac{V}{R} = \frac{2}{25/8} = \frac{16}{25}A = 0.64\,A$$

25. (a) As we know current density $J = nqv$

$\Rightarrow J_e = n_e q v_e$ and $J_h = n_h q v_h$

$$\Rightarrow \frac{J_e}{J_h} = \frac{n_e}{n_h} \times \frac{v_e}{v_h} \Rightarrow \frac{3/4}{1/4} = \frac{n_e}{n_h} \times \frac{5}{20} \Rightarrow \frac{n_e}{n_h} = \frac{6}{5}$$

26. (b) Consider the case when *Ge* and *Si* diodes are connected as show in the given figure.

Equivalent voltage drop across the combination *Ge* and *Si* diode = 0.3 V

$$\Rightarrow \text{Current } i = \frac{12-0.3}{5\,k\Omega} = 2.34\,mA$$

$\therefore$ Out put voltage $V = Ri = 5\ k\Omega \times 2.34\ mA = 11.7\ V$

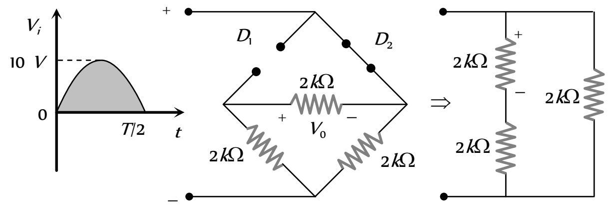
Now consider the case when diode connection are reversed. In this case voltage drop across the diode's combination = 0.7 V

$$\Rightarrow \text{Current } i = \frac{12-0.7}{5\,k\Omega} = 2.26\,mA$$

$$\therefore\ V_0 = iR = 2.26\,mA \times 5\,k\Omega = 11.3\,V$$

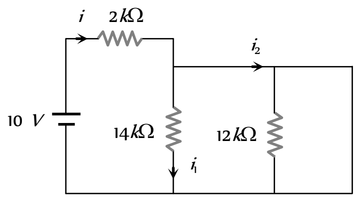
Hence charge in the value of V = 11.7 – 11.3 = 0.4 V

27. (b) For the positive half cycle of input the resulting network is shown below



$$\Rightarrow (V)_- = \frac{1}{2}(V_i)_{\max} = \frac{1}{2} \times 10 = 5\,V.$$

28. (d) The equivalent circuit can be redrawn as follows



From figure it is clear that current drawn from the battery

$$i = i_2 = \frac{10}{2} = 5mA \text{ and } i_1 = 0.$$

29. (c) $i_b = \dfrac{5-0.7}{8.6} = 0.5\,mA \Rightarrow I_c = \beta I_b = 100 \times 0.5\,mA$

By using $V_{CE} = V_{CC} - I_c R_L = 18 - 50\times10^{-3}\times100 = 13V$

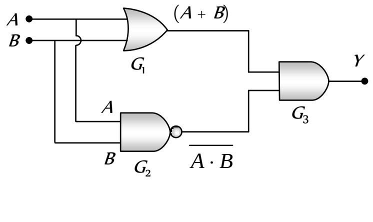
30. (a) $I_e = 10^{10} \times 1.6\times10^{-19} \times \dfrac{1}{10^{-6}} = 1.6\,mA \quad \left(\because I = \dfrac{Q}{t}\right)$

Since 2% electrons are absorbed by base, hence 98% electrons reaches the collector *i.e.* $\alpha = 0.98$

$$\Rightarrow I_c = \alpha I_e = 0.98 \times 1.6 = 1.568\,mA \approx 1.57\,mA$$

Also current amplification factor $\beta = \dfrac{\alpha}{1-\alpha} = \dfrac{0.98}{0.02} = 49$

31. (b)



$Y = (A+B).\overline{AB}$

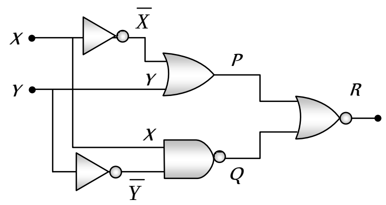
The given output equation can also be written as

$Y = (A+B).(\overline{A}+\overline{B})$ (De morgan's theorem)

$= A\overline{A} + A\overline{B} + B\overline{A} + B\overline{B} = 0 + A\overline{B} + \overline{A}B + 0 = \overline{A}B + A\overline{B}$

This is the expression for XOR gate.

32. (c)

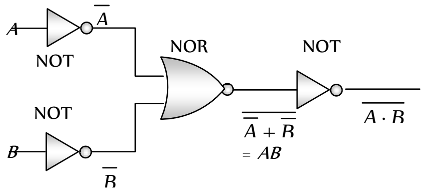


The truth table can be written as

X	Y	$\overline{X}$	$\overline{Y}$	$P = \overline{X}+Y$	$Q = \overline{X.\overline{Y}}$	$R = \overline{P+Q}$
0	1	1	0	1	1	0
1	1	0	0	1	1	0
1	0	0	1	0	0	1
0	0	1	1	1	1	0

Hence X = 1, Y = 0 gives output R =1

33. (d)



Hence option (d) is correct.

34. (b) The truth table of the circuit is given

A	B	C	$X = \overline{AB}$	$Y = \overline{BC}$	$Z = \overline{X+Y}$
0	0	0	1	1	0
1	0	0	1	1	0
0	0	1	1	1	0
1	0	1	1	1	0
0	1	0	1	1	0
1	1	0	0	1	0
0	1	1	1	0	0
1	1	1	0	0	1

Output Z of single three input gate is that of AND gate.

35. (c) Output of upper OR gate = $W + X$

Output of lower OR gate = $W + Y$

Net output $F = (W + X)\,(W + Y)$

$= WW + WY + XW + XY$ (Since $WW = W$)

$= W(1 + Y) + XW + XY$ (Since $1 + Y = 1$)

$= W + XW + XY = W(1 + X) + XY = W + XY$

36. (b) $\mu = r_p\, g_m = 50$

From $i_p = KV_p^{3/2} \Rightarrow \frac{\Delta V_p}{\Delta i_p} = r_p = \frac{2i_p^{-1/3}}{3K^{2/3}}$

$\Rightarrow g_m = \frac{\mu}{r_p} = \frac{3\mu K^{2/3} i_p^{1/3}}{2} = \frac{3}{2}\mu K^{2/3}\left[K^{1/3}(V_p+\mu V_g)^{1/2}\right]$

$= \frac{3}{2}\mu K(V_p+\mu V_g)^{1/2}$ = 75 $K\,(i/K)$

Because i_p was in mA, g_m is substituted as 5 $m℧$

$\Rightarrow 5 = 75K^{2/3}i_p^{1/3} = 75\,K^{2/3}(8)^{1/3} \Rightarrow K = \left(\frac{1}{30}\right)^{3/2}$

Cut off grid voltage $V_G = -\frac{V_p}{\mu} = -\frac{300}{50} = -6V$

37. (d) $g_m = \left(\frac{\Delta i_p}{\Delta V_g}\right)_{V_p = \text{constant}} = \frac{(15-10)\times 10^{-3}}{0-(-4)} = 1.25\times 10^{-3}\Omega$

$\mu = \left(\frac{\Delta V_p}{\Delta V_g}\right)_{I_p = \text{constant}} = \frac{150-120}{0-(-4)} = 7.5$

$\therefore\ r_p = \frac{\mu}{g_m} = \frac{7.5}{1.25\times 10^{-3}} = 6000\ ohms$

38. (d) The dynamic plate resistance is $r_p = \frac{\Delta V_p}{\Delta i_p}$

Now for a vacuum diode $i_p = KV_p^{3/2} \Rightarrow V_p = \left(\frac{i_p}{K}\right)^{2/3}$

$\Rightarrow \frac{\Delta V_p}{\Delta i_p} = \frac{2}{3\,K^{2/3}}\, i_p^{\left(\frac{2}{3}-1\right)}$

$\Rightarrow r_p = (\text{constant})I_p^{-1/3} \Rightarrow r_p \propto \frac{1}{I_p^{1/3}}$

39. (d) $i_p = [0.125\,V_p - 7.5]\times 10^{-3}\ amp$

Differentiating this equation $w.r.t.\ V_p$

$\frac{\Delta i_p}{\Delta V_p} = 0.125\times 10^{-3}$ or $\frac{1}{r_p} = 0.125\times 10^{-3} \Rightarrow r_p = 8\,k\Omega$

40. (b) $V_{peak} = \sqrt{2}\ V_{rms} = \sqrt{2}\times 141.4 = 200\ V$

41. (c) The emission current $i = AT^2Se^{-\phi/kT}$

For the two surfaces $A_1 = A_2$, $S_1 = S_2$, $T_1 = 800\ K$, $T_2 = 1600\,K, \phi_1/T_1 = \phi_2/T_2$

Therefore, $\frac{i_2}{i_1} = \left(\frac{T_2}{T_1}\right)^2 = (2)^2 = 4 \Rightarrow i_2 = 4i_1 = 4\ mA.$

42. (a) The first data gives value of plate resistance $r_p = \frac{\Delta V_p}{\Delta i_p} = \frac{10}{0.8\times 10^{-3}} = \frac{10^5}{8}\Omega$

Also $g_m = \frac{\Delta i_p}{\Delta V_g}$ and $g_m = \frac{\mu}{r_p}$

$$\Rightarrow \Delta V_g = \frac{\Delta i_p \times r_p}{\mu} = \frac{4\times10^{-3}\times10^5/8}{8} = 6.25\ V$$

43. (a) $I_p = 0.004\,(V_p + 10V_g)^{3/2}$

$$\Rightarrow \frac{\Delta I_p}{\Delta V_g} = 0.004\left[\frac{3}{2}(V_p+10V_g)^{1/2}\times 10\right]$$

$$\Rightarrow g_m = 0.004\times\frac{3}{2}(120+10\times -2)^{1/2}\times 10$$

$$\Rightarrow g_m = 6\times10^{-4}\,mho = 0.6\,m\,mho$$

Comparing the given equation of I_p with standard equation $I_p = K(V_p+\mu V_g)^{3/2}$ we get $\mu = 10$

Also from $\mu = r_p \times g_m \Rightarrow r_p = \dfrac{\mu}{g_m} = \dfrac{10}{0.6\times10^{-3}}$

$$\Rightarrow r_p = 16.67\times10^3\,\Omega = 16.67\,k\Omega.$$

44. (b) $\mu = r_P \times g_m = 20\times 2.5 = 50$

From $A = \dfrac{\mu R_L}{r_P + R_L} \Rightarrow r_P + R_L = \dfrac{\mu R_L}{A} = \dfrac{50R_L}{10} = 5R_L$

$$\Rightarrow 4R_L = r_p \Rightarrow R_L = \frac{r_p}{4} = \frac{20}{4} = 5k\Omega$$

45. (a) $A = \dfrac{\mu R_L}{r_p + R_L} = \dfrac{14\times12}{10+12} = \dfrac{84}{11}$. Peak value of output signal

$$V_0 = \frac{84}{11}\times 2\sqrt{2}V \Rightarrow V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{84\times 2}{11}V$$

$\Rightarrow$ *r.m.s.* value of current through the load

$$= \frac{84\times 2}{11\times12\times10^3}A = 1.27\,mA$$

46. (c) $r_p = \dfrac{\mu}{g_m} = \dfrac{64}{1600\times10^{-6}} = 4\times10^4\,\Omega$

Voltage gain $A_v = \dfrac{\mu}{1+\dfrac{r_p}{R_L}} = \dfrac{64}{1+\dfrac{4\times10^4}{40\times10^3}} = 32$

$\therefore$ Output signal voltage

$$V_0 = A_v \times V_i = 32\times1 = 32\,V(r.m.s.)$$

Signal power in load $= \dfrac{V_0^2}{R_L} = \dfrac{(32)^2}{40\times10^3} = 25.6\,mW$

47. (a) $i_p = k(V_p+\mu V_g)^{3/2}mA$

$\Rightarrow 4 = k(200 - 10\times 4)^{3/2} = k\times(160)^{3/2}$ ....(i)

and $i_p = k(160 - 10\times 7)^{3/2} = k\times(90)^{3/2}$ ....(ii)

From equation (i) and (ii) we get

$$i_p = 4\times\left(\frac{90}{160}\right)^{3/2} = 4\times\left(\frac{3}{4}\right)^3 = 1.69\,mA$$

48. (a) At $V_g = -3V, V_p = 300\ V$ and $I_p = 5mA$

At $V_g = -1V$, for constant plate current *i.e.* $I_p = 5mA$

From $I_p = 0.125\,V_p - 7.5$

$$\Rightarrow 5 = 0.125V_p - 7.5 \Rightarrow V_p = 100V$$

$\therefore$ change in plate voltage $\Delta V_p = 300 - 100 = 200V$

Change in grid voltage $\Delta V_g = -1-(-3) = 2V$

So, $\mu = \dfrac{\Delta V_p}{\Delta V_g} = \dfrac{200}{2} = 100$

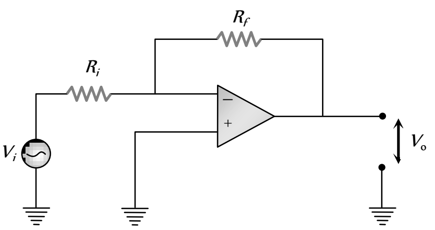
49. (b) The slope of anode characteristic curve $= \dfrac{1}{r_p}$

$$\Rightarrow r_p = \frac{1}{0.02\,mA/V} = 50\frac{V}{mA} = 50\times10^3\frac{V}{A}$$

The slope of mutual characteristic curve $= g_m$ $= 1\times10^{-3}$ *A/V*.

$$\therefore \mu = r_p\times g_m = 50\times10^3\times10^{-3} = 50.$$

50. (b) Voltage gain $A = \dfrac{V_o}{V_i} = \dfrac{R_f}{R_i} = \dfrac{100\,k\Omega}{1\,k\Omega} = 100$.



Graphical Questions

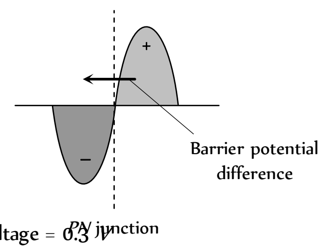
1. (c) With rise in temperature, resistivity of semiconductors decreases exponentially.

2. (b) Potential across the *PN* junction varies symmetrically linear, having *P* side negative and *N* side positive.

3. (c) *PN* junction has low resistance in one direction of potential difference $+V$, so a large current flows (forward biasing). It has a high resistance in the opposite potential difference direction $-V$, so a very small current flows (Reverse biasing).

4. (c) When input voltage is -10 *V*, the diode is reverse biased and no output is obtained. On the other hand, when input is $+10$ *V*, the diode is forward biased and output is obtained which is $+10$ *V*. Therefore the output is of the form as show in the following figure.

5. (a) In the depletion layer of *PN* junction, stationary, positive ions exists in the *N*-side and stationary negative ions exists in the *P* side.



6. (b) V_k = knee voltage = 0.3 *V*

$$\therefore \text{Resistance} = \frac{\Delta V}{\Delta i} = \frac{(2.3-0.3)}{(10-0)\times10^{-3}} = 200\Omega = 0.2k\Omega$$

7. (b) Half wave rectifier, rectifies only the half cycle of input ac signal and it blocks the other half.

8. (c) As RC time constant of the capacitor is quite large ($\tau = RC = 10\times10^3\times10\times10^{-6} = 0.1\,sec$), if will not discharge appreciably. Hence voltage remains nearly constant.

9. (b) In the positive half cycle of input *ac* signal diode D_1 is forward biased and D_2 is reverse biased so in the output voltage signal, A and C are due to D_1. In negative half cycle of Input *ac* signal D_2 conducts, hence output signals B and D are due to D_2.

10. (a) If i is the current in the diode and V is voltage drop across it, then for given figure voltage equation is

$$i\times100 + V = 8 \Rightarrow i = -\frac{1}{100}V + \frac{8}{100} \Rightarrow i = -(0.01)V + 0.08$$

Thus the slope of i-V graph $= \frac{1}{R_L} = 0.01$

11. (b) The current at 2 V is 400 mA and at 2.1 V it is 800 mA. The dynamic resistance in this region

$$R = \frac{\Delta V}{\Delta i} = \frac{(2.1-2)}{(800-400)\times10^{-3}} = \frac{1}{4} = 0.25\Omega$$

12. (a) From the given waveforms, the following truth table can be made

Time interval	Inputs		Output
	A	B	Y
$0 \to T_1$	0	0	0
$T_1 \to T_2$	0	1	0
$T_2 \to T_3$	1	0	0
$T_3 \to T_4$	1	1	1

This truth table is equivalent to 'AND' gate.

13. (d) 5 *volt* is low signal (0) and 10 *volt* is high signal (1) and taking 5 μ-*sec* as 1 unit. In a negative logic, low signal (0) gives high output (1) and high signal (1) gives low output (0). The output is therefore 1010010111.

14. (a) $g_m = \frac{\Delta i_p}{\Delta V_g} = \frac{(20-15)\times10^{-3}}{(4-2)} = 2.5\,millimho$

15. (d) The cut off grid voltage is that negative grid bias corresponding to which the plate current becomes zero. At point P, $i_p = 0$

16. (a) According to Richardson-Dushman equation $J = AT^2e^{-b/T}$

Taking log of this equation $\log_e\frac{J}{T^2} = \log_e A - \frac{b}{T}$

i.e. graph between $\log_e\frac{J}{T^2}$ and $\frac{1}{T}$ will be a straight line having negative slope and positive intercept ($\log_e A$) on $\log_e\frac{J}{T^2}$ axis.

17. (c) $J = AT^2e^{-b/T} \Rightarrow \frac{J}{T^2} \propto e^{-b/T}$

i.e. $\frac{J}{T^2}$ will vary exponentially with $\frac{1}{T}$, having negative slope.

18. (c) This is the graph between i_p and V_g and i_p becomes zero at certain negative potential.

19. (a) $\mu = -\left(\frac{\Delta V_p}{\Delta V_g}\right)_{\Delta i_p = \text{const.}} = \frac{-(80-60)}{[-6-(-4)]} = \frac{20}{2} = 10$

20. (c) According to $|A_v| = \dfrac{\mu}{1+\dfrac{r_p}{R_L}}$

as R_L increases A_v also increases. When R_L becomes too high then A_v = maximum = μ

Hence only option (c) is correct.

21. (c) With rise in temperature, work function decreases (non-linearly).

22. (c) $R_p = \frac{V_p}{i_p} = \frac{50}{150\times10^{-3}} = 333.3\,\Omega$

23. (a) $i \propto T^2 \Rightarrow \frac{i}{i_0} = \left(\frac{T}{T_0}\right)^2$

This is the equation of a parabola.

24. (b) The band width is defined as the frequency band in which the amplifier gain remains above $\frac{1}{\sqrt{2}} = 0.707$ of the mid frequency gain (A_{max}). The low frequency f_1 at which the gain falls to $\frac{1}{\sqrt{2}}$ *i.e.* 0-.707 times it's mid frequency value is called lower cut off frequency and the high frequency f_4 at which the gain falls to $\frac{1}{\sqrt{2}}$ *i.e.* 0.707 times of it's mid frequency is known as higher cut off frequency so band width = $f_4 - f_1$.

25. (c) r_p varies with i_p according to relation $r_p \propto i_p^{-1/3}$ *i.e.* when i_p increases, r_p decreases, hence graph C represents the variation of r_p.

μ doesn't depends upon i_p, hence graph A is correct.

26. (c) From the graph it is clear that of for $V_g = -4V, i_p = 0$, so cut off voltage is – 4 *volt*.

27. (b) As temperature increases saturation current also increases.

28. (c)

29. (a) Output signal voltage has phase difference of 180° with respect to input.

30. (d) Grid is maintained between 0 *volt* to certain negative voltage.

Assertion and Reason

1. (d) In diode the output is in same phase with the input therefore it cannot be used to built NOT gate.

2. (a) According to law of mass action, $n_i^2 = n_e n_h$. In intrinsic semiconductors $n_i = n_e = n_h$ and for P-type semiconductor n_e would be less than n_i, since n_h is necessarily more than n_i.

3. (c) In common emitter transistor amplifier current gain $\beta > 1$, so output current > Input current, hence assertion is correct.

Also, input circuit has low resistance due to forward biasing to emitter base junction, hence reason is false.

4. (a) Input impedance of common emitter configuration

$$= \left.\frac{\Delta V_{BE}}{\Delta i_B}\right|_{V_{CE} = \text{constant}}$$

where ΔV_{BE} = voltage across base and emitter (base emitter region is forward biased)

Δi_B = base current which is order of few microampere.

Thus input impedance of common emitter is low.

5. (d) Resistivity of semiconductors decreases with temperature. The atoms of a semiconductor vibrate with larger amplitudes at higher temperatures there by increasing it's conductivity not resistivity.

6. (a) In semiconductors the energy gap between conduction band and valence band is small ($\approx$ 1 *eV*). Due to temperature rise, electron in the valence band gained thermal energy and may jump across the small energy gap, goes in to the conduction band. Thus conductivity increases and hence resistance decreases.

7. (b)

8. (a) The ratio of the velocity to the applied field is called the mobility. Since electron is lighter than holes, they move faster in applied field than holes.

9. (b)

Intrinsic semiconductor (Neutral) + Pentalvalent impurity (Neutral) → *N*-type semiconductor (Neutral)

10. (a) At a particular temperature all the bonds of crystalline solids breaks and show sharp melting point.

11. (c) The energy gap for germanium is less $(0.72 eV)$ than the energy gap of silicon $(1.1 eV)$. Therefore, silicon is preferred over germanium for making semiconductor devices.

12. (e) We cannot measure the potential barrier of a *PN*-junction by connecting a sensitive voltmeter across its terminals because in the depletion region, there are no free electrons and holes and in the absence of forward biasing, *PN*- junction offers infinite resistance.

13. (e) The assertion is not true. In fact, semiconductor Obeys Ohm's law for low values of electric field (~ 10^6 *V/m*). Above this, the current becomes almost independent of electric field.

14. (d) Two *PN*-junctions placed back to back cannot work as *NPN* transistor because in transistor the width and concentration of doping of *P*-semiconductor is less as compared to width doping of *N*-type semiconductor type.

15. (b) Common emitter is prepared over common base because all the current, voltage and power gain of common emitter amplifier is much more than the gains of common base amplifier.

16. (d) In *PN*-junction, the diffusion of majority carriers takes place when junction is forward biased and drifting of minority carriers takes place across the function, when reverse biased. The reverse bias opposes the majority carriers but makes the minority carriers to cross the *PN*-junction. Thus the small current in μA flows during reverse bias.

17. (d) A transistor is a current operating device because the action of transistor is controlled by the charge carriers (electrons or holes). Base current is very much lesser than the collector current.

18. (a) These gates are called digital building blocks because using these gates only (either NAND or NOR) we can compile all other gates also (like OR, AND, NOT, XOR).

19. (d) At 0*K*, Germanium offers infinite resistance, and it behaves as an insulator.

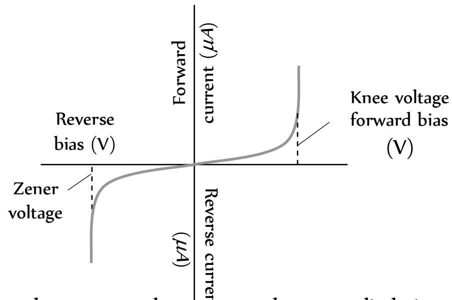
20. (a) In a transistor, the base is made extremely thin to reduce the combinations of holes and electrons. Under this condition, most of the holes (or electrons) arriving from the emitter diffuses across the base and reach the collector. Hence, the collector current, is almost equal to the emitter current, the base current being comparatively much smaller. This is the main reason that power gain and voltage gain are obtained by a transistor. If the base region was made quite thick, then majority of carriers from emitter will combine with the carriers in the base and only small number of carriers will reach the collector, so there would be little collector current and the purpose of transistor would be defeated.

21. (c) The current gain in common base circuit $\alpha = \left(\frac{\Delta I_C}{\Delta I_E}\right)_{V_C}$

The change in collector current is always less than the change in emitter current.

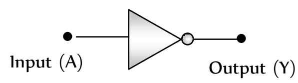
$\Delta I_C < \Delta I_E$. Therefore, $\alpha < 1$.

22. (d) The *V-i* characteristic of *PN*- diode depends whether the junction is forward biased or reverse biased. This can be showed by graph between voltage and current.



23. (a) When the reverse voltage across the zener diode is equal to or more than the breakdown voltage, the reverse current increases sharply.

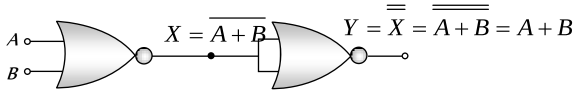
24. (a)



If $A = 0$, $Y = 1$ and $A = 1$, $Y = 0$.

25. (b) In vacuum tubes, vacuum is necessary and the working of semiconductor devices is independent of heating or vacuum.

26. (a)



This is the Boolean expression for 'OR' gate.

27. (a) For detection of a particular wavelength (λ) by a *PN* photo diode, energy of incident light > E_g $\Rightarrow \frac{hc}{E_g} > \lambda$

For $E_g = 2.8\, eV$, $\frac{hc}{E_g} = \frac{6.6\times10^{-34}\times3\times10^8}{2.8\times1.6\times10^{-19}} = 441.9\, nm$

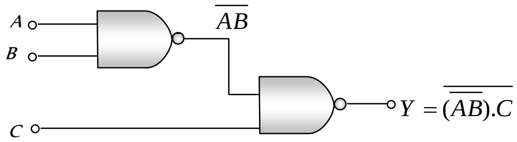
i.e. $\frac{hc}{E_g} < 6000\, nm$, so diode will not detect the wavelength of 6000Å.

28. (a)

29. (b) In forward biasing of *PN* junction current flows due to diffusion of majority charge carriers. While in reverse biasing current flows due to drifting of minority charge carriers.

The circuit given in the reason is a *PNP* transistor having emitter is more negative *w.r.t.* base so it is reverse biased and collector is more positive *w.r.t.* base so it is forward biased.

30. (c) Assertion is true but reason is false



If $A = 1$, $B = 0$, $C = 1$ then $Y = 0$

31. (b) Both assertion and reason are true but potential difference across the resistance is zero, because diode is in reverse biasing hence no current flows.

Electronics

SET Self Evaluation Test - 27

1. In a pure silicon ($n_i = 10^{16}/m^3$) crystal at 300 K, 10^{21} atoms of phosphorus are added per cubic meter. The new hole concentration will be

   (a) 10^{21} per m^3 (b) 10^{19} per m^3

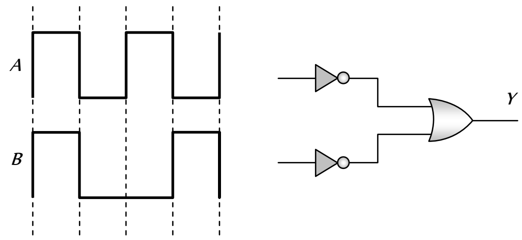
   (c) 10^{11} per m^3 (d) 10^{5} per m^3

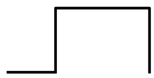
2. In the Boolean algebra $(\overline{\overline{A}\cdot\overline{B}})\cdot A$ equals to

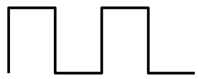
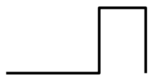
   (a) $\overline{A+B}$ (b) A

   (c) $\overline{A\cdot B}$ (d) $A + B$

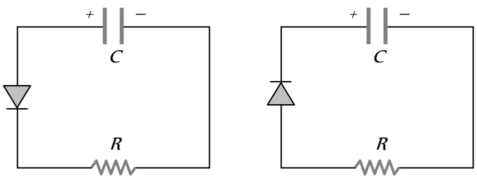
3. In a given circuit as shown the two input waveform A and B are applied simultaneously. The resultant waveform Y is



   (a)  (b) 

   (c)  (d) 

4. Two identical capacitors A and B are charged to the same potential V and are connected in two circuits at $t = 0$, as shown in figure. The charge on the capacitors at time $t = CR$ are respectively



(i) (ii)

   (a) VC, VC (b) $\frac{VC}{e}$, VC

   (c) VC, $\frac{VC}{e}$ (d) $\frac{VC}{e}$, $\frac{VC}{e}$

5. In transistor, forward bias is always smaller than the reverse bias. The correct reason is

   (a) To avoid excessive heating of transistor

   (b) To maintain a constant base current

   (c) To produce large voltage gain

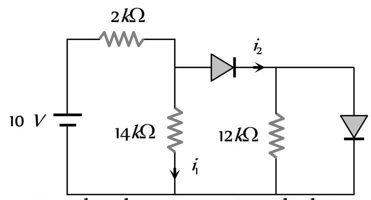
   (d) None of these

6. In NPN transistor, if doping in base region is increased then collector current

   (a) Increases (b) Decreases

   (c) Remain same (d) None of these

7. In the following circuit I_1 and I_2 are respectively



   (a) 0, 0

   (b) 5 mA, 5 mA

   (c) 5 mA, 0

   (d) 0, 5 mA

8. In space charge limited region, the plate current in a diode is 10 mA for plate voltage 150 V. If the plate voltage is increased to 600 V, then the plate current will be

   (a) 10 mA (b) 40 mA

   (c) 80 mA (d) 160 mA

9. A triode has a plate resistance of 10 $k\Omega$ and amplification factor 24. If the input signal voltage is 0.4 V (*r.m.s.*), and the load resistance is 10 k *ohm*, then, the output voltage (*r.m.s.*) is

   (a) 4.8 V (b) 9.6 V

   (c) 12.0 V (d) None of these

10. Pure sodium (Na) is a good conductor of electricity because the $3s$ and $3p$ atomic bands overlap to form a partially filled conduction band. By contrast the ionic sodium chloride ($NaCl$) crystal is

    (a) Insulator (b) Conductor

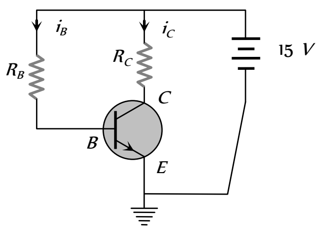
    (c) Semiconductor (d) None of these

11. Would there be any advantage to adding n-type or p-type impurities to copper

    (a) Yes (b) No

    (c) May be (d) Information is insufficient

12. In the following common emitter circuit if $\beta = 100$, $V_{CE} = 7V$, V_{BE} = Negligible $R_C = 2\ k\Omega$ then $I_B = ?$



    (a) 0.01 mA

    (b) 0.04 mA

    (c) 0.02 mA

    (d) 0.03 mA

13. When a battery is connected to a P-type semiconductor with a metallic wire, the current in the semiconductor (predominantly), inside the metallic wire and that inside the battery respectively due to

    (a) Holes, electrons, ions (b) Holes, ions, electrons

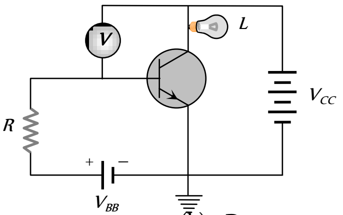
    (c) Electrons, ions, holes (d) Ions, electrons, holes

14. Is the ionisation energy of an isolated free atom different from the ionisation energy E_g for the atoms in a crystalline lattice

    (a) Yes (b) No

    (c) May be (d) None of these

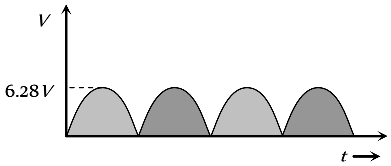
15. In the following circuit, a voltmeter V is connected across a lamp L. What change would occur in voltmeter reading if the resistance R is reduced in value



(a) Increases (b) Decreases

(c) Remains same (d) None of these

16. For given electric voltage signal dc value is



(a) 6.28 V (b) 3.14 V

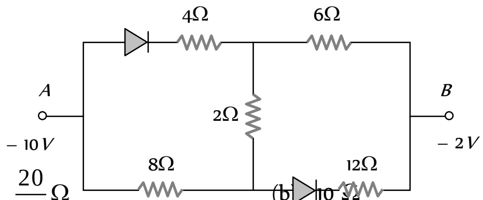
(c) 4 V (d) 0 V

17. When a silicon PN junction is in forward biased condition with series resistance, it has knee voltage of 0.6 V. Current flow in it is 5 mA, when PN junction is connected with 2.6 V battery, the value of series resistance is

(a) 100 Ω (b) 200 Ω

(c) 400 Ω (d) 500 Ω

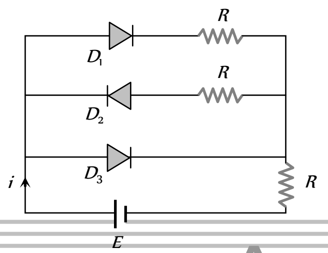
18. In the following circuit the equivalent resistance between A and B is



(a) $\frac{20}{3}\Omega$ (b) 10 Ω

(c) 16 Ω (d) 20 Ω

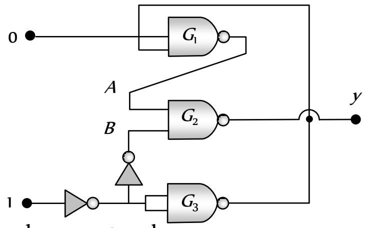
19. In the following circuit of PN junction diodes D_1, D_2 and D_3 are ideal then i is



(a) E/R (b) $E/2R$

(c) $2E/3R$ (d) Zero

20. In circuit in following fig. the value of Y is



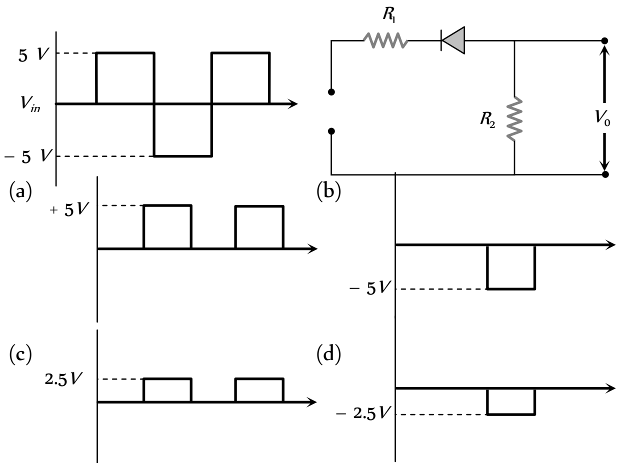
(a) 0

(b) 1

(c) Fluctuates between 0 and 1

(d) Indeterminate as the circuit can't be realised

21. A waveform shown when applied to the following circuit will produce which of the following output waveform. Assuming ideal diode configuration and $R_1 = R_2$



22. In a triode, cathode, grid and plate are at 0, – 2 and 80 V respectively. The electrons is emitted from the cathode with energy 3 eV. The energy of the electron reaching the plate is

(a) 77 eV (b) 85 eV

(c) 81 eV (d) 83 eV

23. The energy gap of silicon is 1.5 eV. At what wavelength the silicon will stop to absorb the photon

(a) 8250 Å (b) 7250 Å

(c) 6875.5 Å (d) 5000 Å

Answers and Solutions

(SET -27)

1. (c) By using mass action law $n_i^2 = n_e n_h$

$$\Rightarrow n_h = \frac{n_i^2}{n_e} = \frac{(10^{16})^2}{10^{21}} = 10^{11}\ per\ m^3$$

2. (b) $(\overline{\overline{A} \cdot \overline{B}}) \cdot A = (\overline{\overline{A}} + \overline{\overline{B}}) \cdot A = (A + B) \cdot A$

$= A \cdot A + AB = A + AB = A(1 + B) = A$

3. (a) (1 = high, 0 = low)

Input to A is in the sequence, 1,0,1,0.

Input to B is in the sequence, 1, 0, 0, 1.

Sequence is inverted by NOT gate.

Thus inputs to OR gate becomes 0, 1, 0, 1 and output of OR gate becomes 0, 1, 1, 1

Since for OR gate 0 + 1 = 1. Hence choice (a) is correct.

4. (b) Time $t = CR$ is known as time constant. It is time in which charge on the capacitor decreases to $\frac{1}{e}$ times of it's initial charge (steady state charge).

In figure (i) PN junction diode is in forward bias, so current will flow the circuit *i.e.*, charge on the capacitor decrease and in time t it becomes $Q = \frac{1}{e}(Q_o)$; where $Q_o = CV$

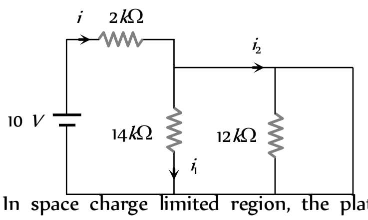
$\Rightarrow Q = \frac{CV}{e}$

In figure (ii) P-N junction diode is in reverse bias, so no current will flow through the circuit hence change on capacitor will not decay and it remains same *i.e.* CV after time t.

5. (a) If forward bias is made large, the majority charge carriers would move from the emitter to the collector through the base with high velocity. This would give rise to excessive heat causing damage to transistor.

6. (b) Number of holes in base region increases hence recombination of electron and hole are also increases in this region. As result base current increases which in turn decreases the collector current.

7. (d) Equivalent circuit can be redrawn as follows



$i = \frac{10}{2} = 5\,mA = i_2$

$i_1 = 0$

8. (c) In space charge limited region, the plate current is given by Child's law $i_p = KV_p^{3/2}$

Thus, $\frac{i_{p_2}}{i_{p_1}} = \left(\frac{V_{p_2}}{V_{p_1}}\right)^{3/2} = \left(\frac{600}{150}\right)^{3/2} = (4)^{3/2} = 8$

or $i_{p_2} = i_{p_1} \times 8 = 10 \times 8\,mA = 80\,mA$.

9. (a) Use $V_0 = AV_s$

Now $A = \frac{24 \times 10k}{10k + 10k} = \frac{24 \times 10}{20} = 12$

Therefore, $V_0 = 12 \times 0.4 = 4.8\,volt(r.m.s.)$

10. (a) In sodium chloride the Na^+ and Cl^- ions both have noble gas electron configuration corresponding to completely filled bands. Since the bands do not overlap, there must be a gap between the filled bands and the empty bands above them, so $NaCl$ is an insulator.

11. (b) Pure Cu is already an excellent conductor, since it has a partially filled conduction band, furthermore, Cu forms a metallic crystal as opposed to the covalent crystals of silicon or germanium, so the scheme of using an impurity to donate or accept an electron does not work for copper. In fact adding impurities to copper decreases the conductivity because an impurity tends to scatter electrons, impeding the flow of current.

12. (b) $V = V_{CE} + I_C R_L$

$\Rightarrow 15 = 7 + I_c \times 2 \times 10^3 \Rightarrow i_c = 4\,mA$

$\because \beta = \frac{i_C}{i_B} \Rightarrow i_B = \frac{4}{100} = 0.04\,mA$

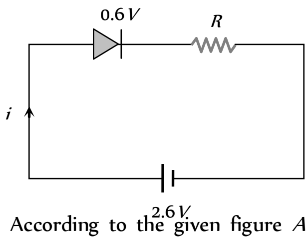
13. (a) Charge carriers inside the P-type semiconductor are holes (mainly). Inside the conductor charge carriers are electrons and for cell ions are the charge carriers.

14. (a) The ionisation energy of an isolated atom is different from it's value in crystalline lattice, because in the latter case each bound electron is influenced by many atoms in the periodic crystalline lattice.

15. (a) Here the emitter base junction of N-P-N transistor is forward biased with battery $V_{}$ through resistance R. When the value of R is reduced, then the emitter current $i_{}$ will increase. As a result the collector current will also increase. ($i_{} = i_{} - i_{}$). Due to increase in $i_{}$, the potential difference across L increases and hence the reading of voltmeter will increases.

16. (c) $V_{dc} = V_{ac} = \frac{2V_0}{\pi} = \frac{2 \times 6.28}{3.14} = 4\,V$.

17. (c)

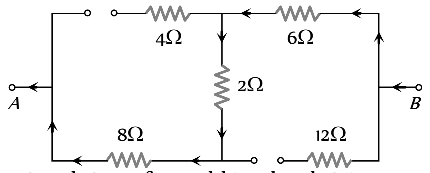


$R = \frac{(2.6 - 0.6)}{5 \times 10^{-3}} = 400\ \Omega$.

18. (c) According to the given figure A is at lower potential *w.r.t.* B. Hence both diodes are in reverse biasing, so equivalent, circuit can be redrawn as follows.

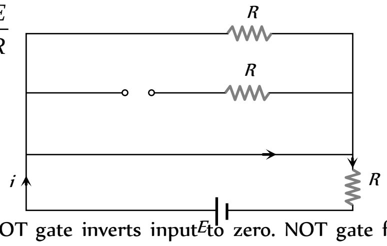
$\Rightarrow$ Equivalent resistance between A and B

$R = 8 + 2 + 6 = 16\ \Omega$.



19. (a) Diodes $D_{}$ and $D_{}$ are forward biased and $D_{}$ is reverse biased so the circuit can be redrawn as follows.

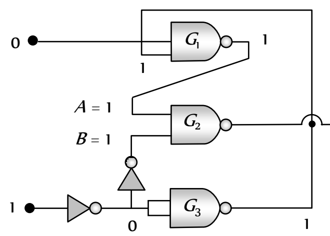
$\Rightarrow i = \frac{E}{R}$



20. (a) Lower NOT gate inverts input to zero. NOT gate from NAND gate inverts this output to 1 upper NAND gate converts this input 1 and input 0 to 1.

Thus $A = 1$ and $B = 1$ become inputs of NAND gate giving final output as zero. Choice A is correct.

***

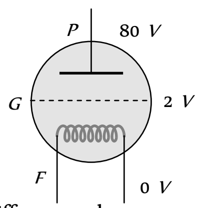


$\overline{A \cdot B} = \overline{1 \cdot 1} = 0$

21. (d) The *P-N* junction will conduct only when it is forward biased *i.e.* when $-5V$ is fed to it, so it will conduct only for 3rd quarter part of signal shown and when it conducts potential drop 5 *volt* will be across both the resistors, so output voltage across R_2 is 2.5 *V*.

$\therefore\ V_0 = -2.5\ V$

22. (d) There is a loss of kinetic energy of 2 *eV* from filament to grid. The energy of the electron after passing through the grid will be 3 – 2 = 1 *eV*



The potential difference between plate and grid is $80-(-2)=82V$. The electron will gain energy 82 *eV* from the grid to the plate. The energy of electron reaching the plate = 1 + 82 = 83 *eV*

23. (a) $\lambda = \dfrac{hc}{E} = \dfrac{6.6\times10^{-34}\times3\times10^{8}}{1.5\times1.6\times10^{-19}} = 8.25\times10^{-7} m = 8250Å$

The photon having wavelength equal to 8250Å or more than this will not able to overcome the energy gap of silicon.

Chapter
28
Communication

The term communication refers to the transmitting, receiving and processing of information by electronic means.

Basic Communication System

A basic communication system consists of an information source, a transmitter, a link and a receiver.

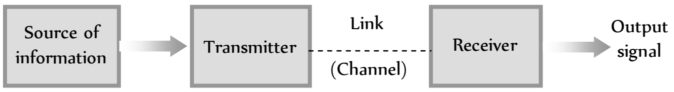


Fig. 28.1

(1) **Information :** The idea/message that is to be conveyed is information. The message may be individual one or a set of messages. The message may be a symbol, code, group of words or any pre decided unit.

(2) **Transmitter :** In radio transmission, the transmitter consists of a transducer, modulator, amplifier and transmitting anteena.

Transducer :	Converts sound signals into electric signal.
Modulator :	Mixing of audio electric signal with high frequency radio wave.
Amplifier :	Boosting the power of modulated signal.
Anteena :	Signal is radiated in the space with the aid of an anteena.

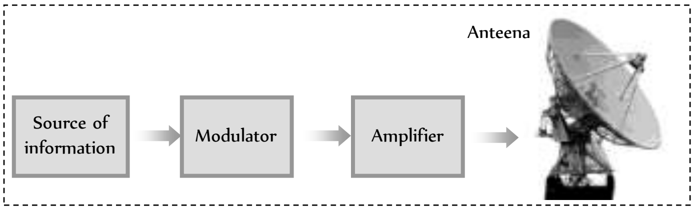


Fig. 28.2 : Transmitter

(3) **Communication channel :** The function of communication channel is to carry the modulated signal from transmitter to receiver. The communication channel is also called transmission medium or link.

The term channel refers to the frequency range allocated to a particular service or transmission.

Table 28.1 : Different channels

Type of communication	Channels or links
Radio communication	Free space
Telephony and Telegraphy communication	Transmission line
Optical communication	Optical fibre

(4) **Receiver :** The receiver consists of

Pickup anteena :	To pick the signal
Demodulator :	To separate out the audio signal from the modulated signal
Amplifier :	To boost up the weak audio signal
Transducer :	To convert back audio signal in the form of electrical pulses into sound waves.

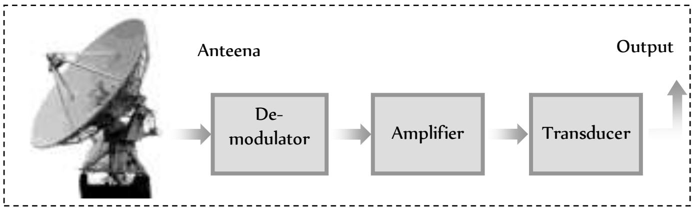


Fig. 28.3 : Receiver

Types of Communication System

Communication systems can be classified according to the nature of information or mode of transmission or types of transmission channel.

(1) **According to the nature of information source**

(i) Speech transmission

(ii) Picture transmission

(iii) **Facsimile transmission (FAX) :** This involves exact reproduction of a document or picture which are static.

(2) **According to the mode of transmission**

(i) **Analog communication :** The communication system, which make use of analog signals are called analog communication system.

Table 28.2 : Few analog communication system

System	Specification
Telegraphy	Message in the form of codes are sent.
Television broadcast	Both sound as well as pictures are sent.
Telephony	It sends voice signal from one place to another by means of wire.
Radar	It means radio detection and ranging. It is used for determining the distance and direction of objects using microwave.
Teleprinting	Message can be typed and telegraphed to distant receivers

(ii) **Digital communication :** In this system digital signals are used.

Table 28.3 : Few digital communication system

System	Specification
Facsimile transmission (FAX)	This involves exact reproduction of a document or picture which are static
Mobile phone	Such telephones are also called cellular phones, because they operate within a network of radio cells.
E-mail	the message sent via a computer network are called e-mail
Tele conferencing	It is a system in which persons sitting at coloured television screens. See and talk to each other via a computer communication network.
Communication satellite	Used to relay radio and television programmers.
Global positioning system (GPS)	It is a navigation system based on a network of earth orbiting satellites. The users can find their positions within an accuracy of 100 *m* by receiving.

(3) **According to the transmission channel**

(i) Line communication (ii) Space communication

(4) **According to the type of modulation**

(i) Amplitude modulation (AM)

(ii) Frequency modulation (FM)

(iii) Phase modulation (PM)

(iv) Pulse amplitude modulation (PAM)

(v) Pulse time modulation (PTM)

(vi) Pulse code modulation (PCM)

Analog and Digital Signals

In communication system, a signal means a time varying electrical signal containing informations.

(1) **Analog signals :** It is a continuous wave form which changes smoothly over time.

(i) Such signals can be easily generated from the source of information by using an appropriate transducer *e.g.* pressure variations in the sound waves can be converted into corresponding current or voltage pulses with the help of a microphone.

(ii) A simple analog signal is represented by a sine wave

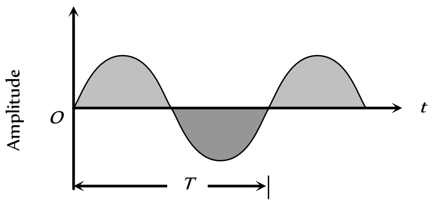


Fig. 28.4

(iii) The frequency of analog signals associated with speed or music varies over a range between 20 *Hz* to 20 *KHz*.

(iv) The range over which the frequencies of a signal vary is called band width.

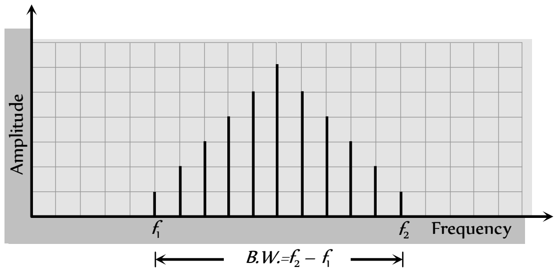


Fig. 28.5

(v) The term base band designates the band of frequencies representing the signal supplied by the source of information.

(vi) A signal consist of two or more waves of different frequencies is known as a complex analog signal.

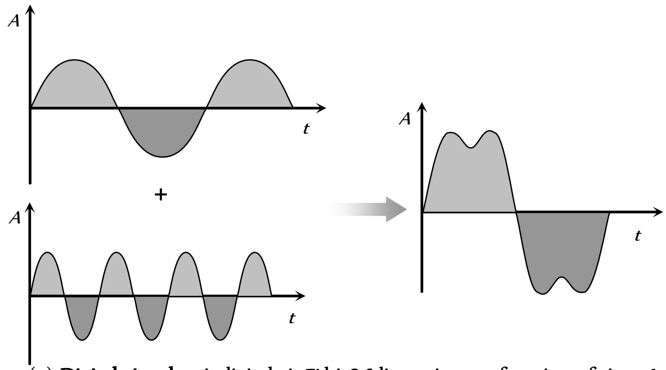


Fig. 28.6

(2) **Digital signals :** A digital signal is a discontinuous function of time. It has only two voltage level *i.e.* either low (0) or high (1).

Either of 0 and 1 is known as bit. A group of bit is called byte.

A byte comprising of 2 bits can give on the four code combination *i.e.* 00, 01, 10 and 11.

The number of code combination increase with number of bits in a byte is given by $N = 2^x$, where x = number of bits in a byte.

The number of binary digits (bits) per second, which describe a digital signal is called it's bit rate. Bit rate is expressed in bits per second (bps).

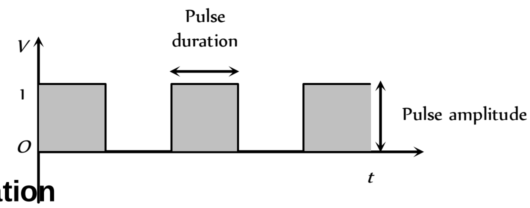


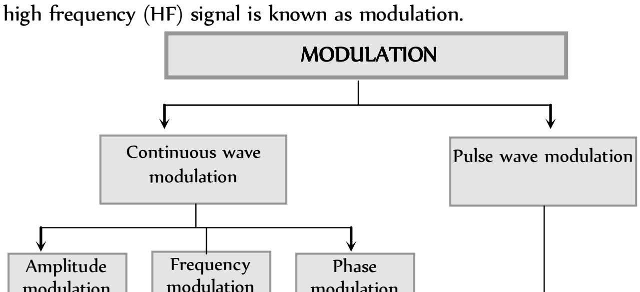
Fig. 28.7

Modulation

(1) Digital and analog signals to be transmitted are usually of low frequency and hence cannot be transmitted as such.

(2) These signals require some carrier to be transported. These carriers are known as carrier waves or high frequency signals.

(3) The process of placement of a low frequency (LF) signal over the high frequency (HF) signal is known as modulation.



(4) **Need for modulation :** The sound wave (20 *Hz* to 20 *KHz*) cannot be transmitted directly from one place to another for the following reasons.

(i) **Height of anteena :** For efficient radiation and reception , the height of transmitting and receiving antennas should be comparable to a quarter of wavelength of the frequency used. For 15 *KHz* it is 5000 *m* (too large) and for 1 *MHz* it is 75 *m*.

The energy radiated from an anteena is practically zero, when the frequency of the signal to be transmitted is below 15 *Hz*.

(ii) **Detecting signals :** All audible signals are in the range of 20 *Hz* to 20 *KHz* so the signals from all sources remains heavily mixed up in air. It will be very difficult to differentiate or detect the broadcast signal at the receiving station.

Thus modulation is necessary for a low frequency signal. When it is to be sent to a distant place so that the information may not die out in the way it self as well as for the proper identification of a signal and to keep the height of anteena small also

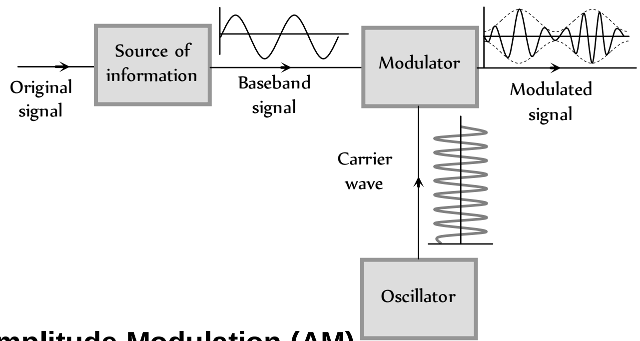


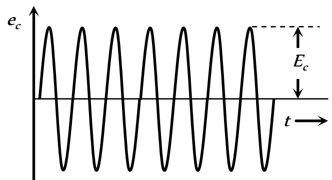
Fig. 28.9

Amplitude Modulation (AM)

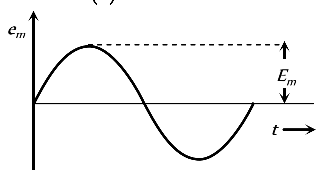
The process of changing the amplitude of a carrier wave in accordance with the amplitude of the audio frequency (AF) signal is known as amplitude modulation (AM).

In AM frequency of the carrier wave remains unchanged.

The amplitude of modulated wave is varied in accordance with the amplitude of modulating wave.



(A) HF carrier wave



(B) LF modulating wave

(1) **Modulation index :** The ratio of change of amplitude of carrier wave to the amplitude of original carrier wave is called the modulation factor or degree of modulation or modulation index (m).

$$m_a = \frac{\text{Change in amplitude of carrier wave}}{\text{Amplitude of original carrier wave}} = \frac{kE_m}{E_c}$$

where k = A factor which determines the maximum change in the amplitude for a given amplitude E_m of the modulating signal. If k = 1 then

$$m_a = \frac{E_m}{E_c} = \frac{E_{\max} - E_{\min}}{E_{\max} + E_{\min}}$$

If a carrier wave is modulated by several sine waves the total modulated index m_t is given by $m_t = \sqrt{m_1^2 + m_2^2 + m_3^2 + ...}$

(2) **Voltage equation for AM wave :** Suppose voltage equations for carrier wave and modulating wave are $e_c = E_c \cos \omega_c t$ and $e_m = E_m \sin \omega_m t = mE_c \sin \omega_m t$

where e_c = Instantaneous voltage of carrier wave, E_c = Amplitude of carrier wave, $\omega_c = 2\pi f_c =$ Angular velocity at carrier frequency f_c, $e_m =$ Instantaneous voltage of modulating, $E_m =$ Amplitude of modulating wave, $\omega_m = 2\pi f_m =$ Angular velocity of modulating frequency f_m

Voltage equation for AM wave is

$$e = E \sin \omega_c t = (E_c + e_m) \sin \omega_c t = (E_c + e_m \sin \omega_m t) \sin \omega_c t$$

$$= E_c \sin \omega_c t + \frac{m_a E_c}{2} \cos(\omega_c - \omega_m)t - \frac{m_a E_c}{2} \cos(\omega_c + \omega_m)t$$

The above AM wave indicated that the AM wave is equivalent to summation of three sinusoidal wave, one having amplitude E_c and the other two having amplitude $\frac{m_a E_c}{2}$.

(3) **Side band frequencies and band width in AM wave**

(i) **Side band frequencies :** The AM wave contains three frequencies $f_c, (f_c + f_m)$ and $(f_c - f_m)$, f_c is called carrier frequency, $(f_c + f_m)$ and $(f_c - f_m)$ are called side band frequencies.

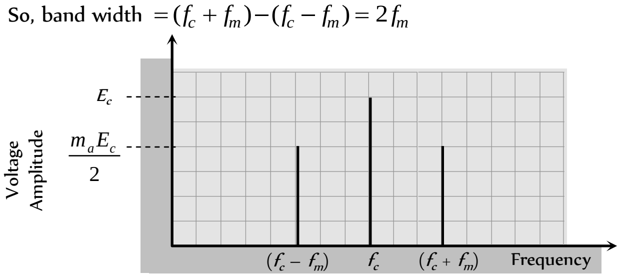
$(f_c + f_m)$: Upper side band (USB) frequency

$(f_c - f_m)$: Lower side band (LSB) frequency

Side band frequencies are generally close to the carrier frequency.

(ii) **Band width :** The two side bands lie on either side of the carrier frequency at equal frequency interval f_m.

So, band width $= (f_c + f_m) - (f_c - f_m) = 2f_m$



(4) **Power in AM waves** : Power dissipated in any circuit $P = \frac{V_{rms}^2}{R}$.

Hence (i) carrier power $P_c = \frac{\left(\frac{E_c}{\sqrt{2}}\right)^2}{R} = \frac{E_c^2}{2R}$

(ii) Total power of side bands $P_{sb} = \frac{\left(\frac{m_a E_c}{2\sqrt{2}}\right)^2}{R} + \frac{\left(\frac{m_a E_c}{2\sqrt{2}}\right)^2}{R} = \frac{m_a^2 E_c^2}{4R}$

(iii) Total power of AM wave $P_{Total} = P_c + P_{sb} = \frac{E_c^2}{2R}\left(1 + \frac{m_a^2}{2}\right)$

(iv) $\frac{P_t}{P_c} = \left(1 + \frac{m_a^2}{2}\right)$ and $\frac{P_{sb}}{P_t} = \frac{m_a^2/2}{\left(1 + \frac{m_a^2}{2}\right)}$

(v) Maximum power in the AM (without distortion) will occur when $m_a = 1$ *i.e.* $P_t = 1.5P = 3P_{sb}$

(vi) If I_c = Unmodulated current and I_t = total or modulated current $\Rightarrow$ $\frac{P_t}{P_c} = \frac{I_t^2}{I_c^2} \Rightarrow \frac{I_t}{I_c} = \sqrt{\left(1 + \frac{m_a^2}{2}\right)}$

(5) **Limitation of amplitude modulation**

(i) Noisy reception (ii) Low efficiency

(iii) Small operating range (iv) Poor audio quality

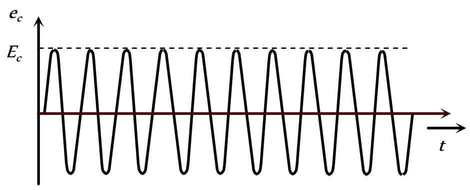
Frequency Modulation (FM)

The process of changing the frequency of a carrier wave in accordance with the audio frequency signal is known as frequency modulation

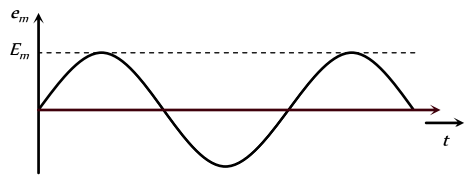
(1) Audio quality of AM transmission is poor. There are need to eliminate amplitude sensitive noise. This is possible if we eliminate amplitude variation. (*i.e.* a need to keep the amplitude of the carrier constant). This is precisely what we do in FM.

(2) In FM the overall amplitude of FM wave remains constant at all times.

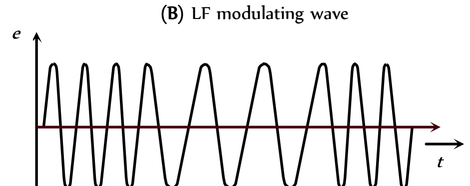
(3) In FM, the total transmitted power remains constant.



(A) HF carrier wave



(B) LF modulating wave



(4) **Frequency deviation :** The maximum change in frequency from mean value (ν_c) is known as frequency deviation. This is also the change or shift either above or below the frequency ν_c and is called as frequency deviation.

$$\therefore \quad \delta = (f_{\max} - f_c) = f_c - f_{\min} = k_f \cdot \frac{E_m}{2\pi}$$

k_f = Constant of proportionality. It determines the maximum variation in frequency of the modulated wave for a given modulating signal.

(5) **Carrier swing (CS) :** The total variation in frequency from the lowest to the highest is called the carrier swing

i.e. $CS = 2 \times \Delta f$

(6) **Frequency modulation index (m_f) :** The ratio of maximum frequency deviation to the modulating frequency is called modulation index.

$$m_f = \frac{\delta}{f_m} = \frac{f_{\max} - f_c}{f_m} = \frac{f_c - f_{\min}}{f_m} = \frac{k_f E_m}{f_m}$$

(7) **Frequency spectrum :** FM side band modulated signal consist of infinite number of side bands whose frequencies are

$$(f_c \pm f_m), (f_c \pm 2f_m), (f_c \pm 3f_m) ....$$

The number of side bands depends on the modulation index m_f.

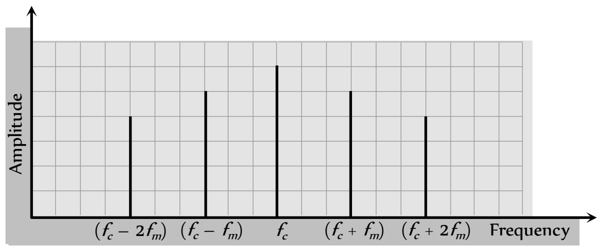


Fig. 28.13

In FM signal, the information (audio signal) is contained in the side bands. Since the side bands are separated from each other by the frequency of modulating signal f_m so

Band width $= 2n \times f_m$; where n = number of significant side band pairs

(8) **Deviation ratio :** The ratio of maximum permitted frequency deviation to the maximum permitted audio frequency is known as deviation ratio. Thus, deviation ratio $= \frac{(\Delta f)_{\max}}{(f_m)_{\max}}$

(9) **Percent modulation :** The ratio of actual frequency deviation to the maximum allowed frequency deviation is defined as percent modulation. Thus, percent modulation, $m = \frac{(\Delta f)_{\text{actual}}}{(\Delta f)_{\max}}$

Table 28.4 : Range of frequency allotted for FM radio/TV broadcast

Type of broadcast	Frequency band
FM radio	88 to 108 *MHz*
UHF TV	47 to 230 *MHz*
UHF TV	470 to 960 *MHz*

Pulse Modulation

Here the carrier wave is in the form of pulses.

(1) **Pulse amplitude modulation (PAM) :** The amplitude of the pulse varies in accordance with the modulating signal.

(2) **Pulse width modulation (PWM) :** The pulse duration varies in accordance with the modulating signal.

(3) **Pulse position modulation (PPM) :** In PPM, the position of the pulses of the carrier wave train is varied in accordance with the instantaneous value of the modulating signal.

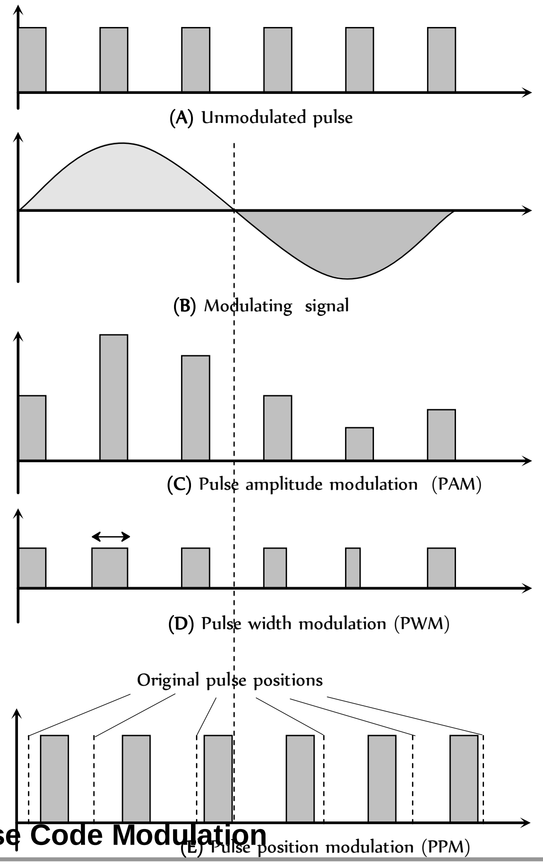


Fig. 28.14

Pulse Code Modulation

The pulse amplitude, pulse width and pulse position modulations are not completely digital.

A completely digital modulation is obtained by pulse code modulation (PCM).

An analog signal is pulse code modulated by following three operation.

(1) **Sampling :** It is the process of generating pulses of zero width and of amplitude equal to the instantaneous amplitude of the analog signal.

The number of samples taken per second is called sampling rate.

(2) **Quantisation :** The process of dividing the maximum amplitude of the analog voltage signal into a fixed number of levels is called quantisation.

e.g. amplitude 5 *V* of the analog voltage signal divides into six. Quantisation level viz 0, 1, 2, 3, 4, 5.

Pulses having amplitude between – 0.5 *V* to 0.5 *V* are approximated (quantised) to a value 0 *V*, amplitude between 0.5 *V* to 1.5 *V* are approximated to a value of 1 *V* and so on.

(3) **Coding :** The process of digitising the quantised pulses according to some code is called coding.

Table 28.5 : Coding

Quantisation level	0	1	2	3	4	5	6	7
Binary code	000	001	010	011	100	101	110	111

For example consider that voltage amplitude of an analog signal varies between 0 and 7 *V*.

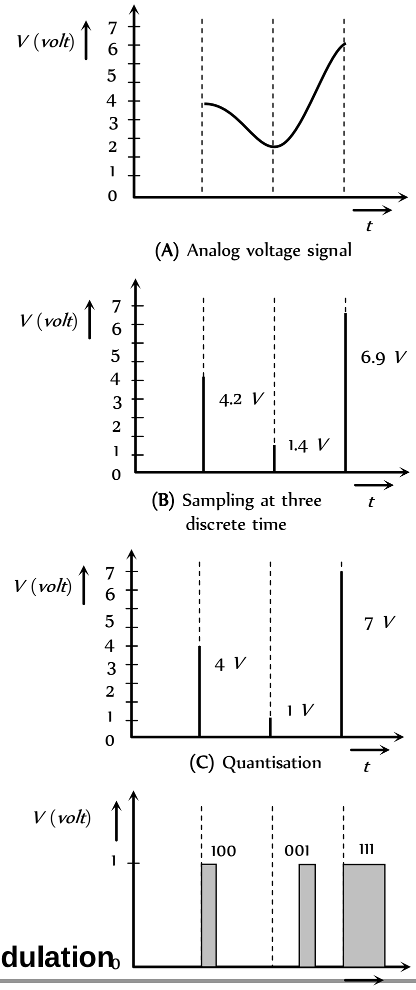


Fig. 28.15

Demodulation

The process of extracting the audio signal from the modulated wave is known as demodulation or detection.

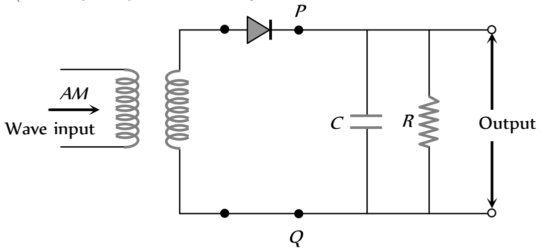
The wireless signals consist of radio frequency (high frequency) carrier wave modulated by audio frequency (low frequency). The diaphragm of a telephone receiver or a loud speaker cannot vibrate with high frequency. So it is necessary to separate the audio frequencies from the radio frequency carrier wave.

Simple demodulator circuit : A diode can be used to detect or demodulate an amplitude modulated (AM) wave. A diode basically acts as a rectifier *i.e.* it reduces the modulated carrier wave into positive envelope only.

The AM wave input is shown in figure. It appears at the output of the diode across *PQ* as a rectified wave (since a diode conducts only in the positive half cycle). This rectified wave after passing through the *RC* network does not contain the radio frequency carrier component. Instead, it has only the envelope of the modulated wave.

In the actual circuit the value of RC is chosen such that $\frac{1}{f_c} << RC$;

where f_c = frequency of carrier signal.



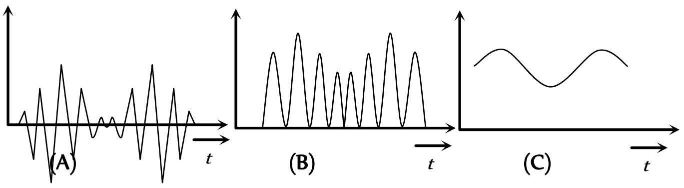


Fig. 28.16

Data Transmission and Retrieval

The term data is applied to a representation of facts, concepts or instructions suitable for communication, interpretation or processing by human beings or by automatic means. Data in most cases consists of pulse type of signals.

The pulse code modulated (PCM) signal is a series of 1's and 0's. The following three modulation techniques are used to transmit a PCM signal.

(1) **Amplitude shift keying (ASK) :** Two different amplitudes of the carrier represent the two binary values of the PCM signal. This method is also known as on-off keying (OOK)

1 : Presence of carrier of same constant amplitude.

0 : Carrier of zero amplitude.

(2) **Frequency shift keying (FSK) :** The binary values of the PCM signal are represented by two frequencies.

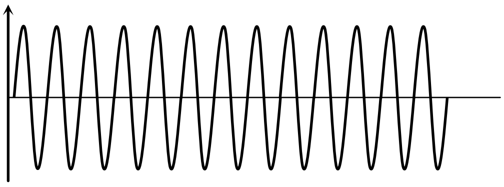
1 : Increase in frequency

0 : Frequency unaffected

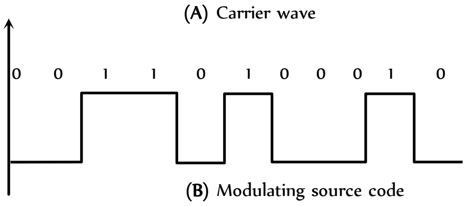
(3) **Phase shift keying (PSK) :** The phase of the carrier wave is changed in accordance with modulating data signal.

1 : Phase changed by π

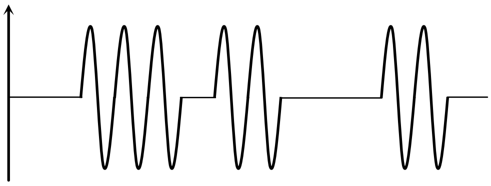
0 : Phase remains unchanged.



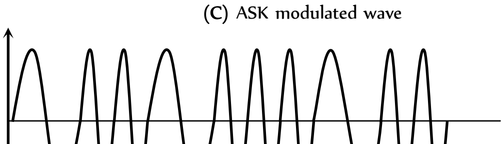
(A) Carrier wave



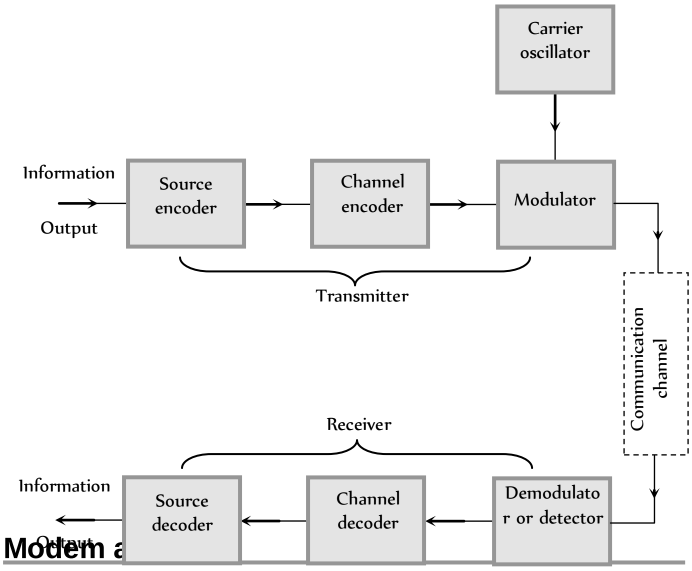
(B) Modulating source code



(C) ASK modulated wave



The analog signal is sampled by the sampler. The sampled pulses are then quantised. The encoder codes the quantised pulses according to the binary codes. After modulating the PCM signal (by ASK, FSK or PSK method) the modulated signal is, then transmitted into free space in the form of bits.



Modem a

Fig. 28.18

(1) **Modem :** Modems are used to interface two digital sources/receivers.

(i) Word modem has been obtained from the words modulator and demodulator. As the name implies both the functions (modulation) and demodulation) are included in a signal unit.

(ii) Modems are placed at both ends of the communication circuit as shown.

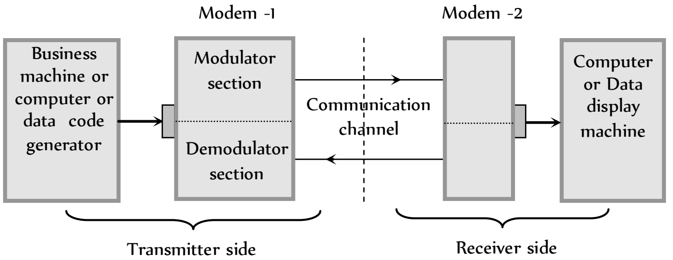


Fig. 28.19

(iii) The modem at the transmitting station changes the digital output from a computer (or any other business machine) to a from (analog signal) which can be easily sent via a communication channel (Telephone line *etc.*). While the receiving modem reverses the process.

(iv) There are three modes of operation of a modem.

(a) Simplex mode : In this mode data is transmitted in only one direction.

(b) Half duplex : In this mode data is transmitted between the transmitter and the receiver in both direction, but only in one direction at a time.

(c) Full duplex : In this mode, the data are transmitted between the transmitter and receiver in both directions at the same time.

Table 28.6 : Modem data transmission speed

Types	Speed in bits per sec and (bps)
Low speed modem	600 bps
Medium speed modem	600 to 2400 bps
High speed modem	2400 to 10,800 bps

(2) **Fax (Facsimile transmission) :** The electronic reproduction of a document at a distance place is known as facsimile transmission (FAX).

The original written document is converted into transmittable codes at the sending end. These codes are converted back into a copy of the original document at the receiving end.

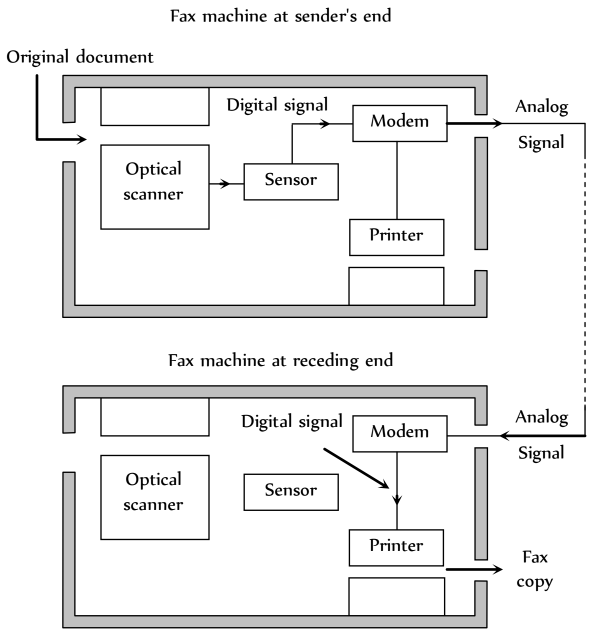


Fig. 28.20

The original written document is put into the machine. A scanner scans the whole document.

The scanned written document is then moved on a glass plate. A beam of light from a given source is projected through the glass and is reflected from the surface of the document.

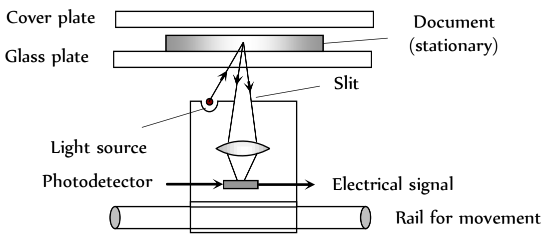


Fig. 28.21

The reflected light is focussed on a device known as photo detector which converts the optical signals (carrying the information regarding the patterns/writings/signatures *etc.*) in to electrical signals.

These electrical signals are then modulated and transmitted on to the telephone lines to the receiving end.

Space Communication

The communication process utilising the physical space around the earth is termed as space communication.

Electromagnetic waves which are used in Radio, Television and other communication system are radio waves and microwaves.

The radio waves emitted from a transmitter anteena can reach the receiver antenna by the following mode of operation.

+ Ground wave propagation

+ Sky wave propagation.

+ Space wave propagation.

(1) **Ground wave propagation**

(i) In ground wave propagation, radio waves travel along the surface of the earth (following the curvature of earth).

(ii) These waves induce currents in the ground as they propagate due to which some energy is lost.

(iii) The decrease in the value of energy (*i.e.* attenuation) increases with the increase in the frequency of radiowave.

(iv) As the ground wave propagates over the earth, it tilts over more and more due to diffraction. (This is another cause of attenuation of ground wave). After covering some distance, the wave just lie down which means it's death.

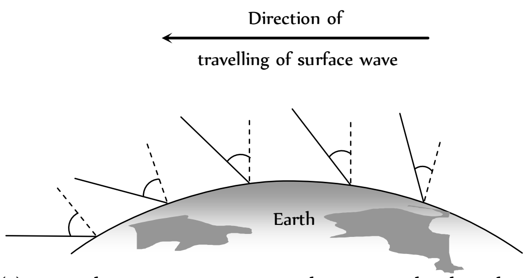


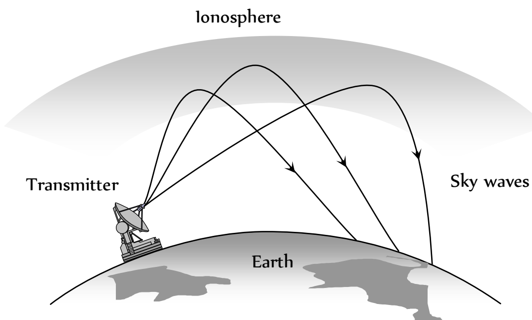
Fig. 28.22

(v) Ground wave propagation can be sustained only at low frequencies (~ 500 kHz to 1500 kHz) or for radio broadcast at long wavelengths.

(2) **Sky wave propagation**

(i) These are the waves which are reflected back to the earth by ionosphere.

Ionosphere is a layer of atmosphere having charged particles, ions and electrons and extended above 80 km – 300 km from the earth's surface.



(ii) These are the radio waves of frequency range 2 *MHz* to 30 *MHz*.

(iii) Sky waves are used for very long distance radio communication at medium and high frequencies (*i.e.* at medium waves and short waves).

(iv) The sky waves being electromagnetic in nature, changes the dielectric constant and refractive index of the ionosphere. The effective refractive index of the ionosphere is

$$n_{eff}=n_0\left[1-\frac{Ne^2}{\varepsilon_0 m\omega^2}\right]^{1/2}=n_0\left[1-\frac{80.5N}{f^2}\right]^{1/2}$$

where n_0 = refractive index of free space, N = electron density of ionosphere, ε_0 = dielectric constant of free space, e = charge on electron, m = mass of electron ω = angular frequency of EM wave.

(v) As we go deep into the ionosphere, N increases so n_{eff} decreases. The refractions or bending of the beam will continue and finally it reflects back.

(vi) **Critical frequency (f_c) :** It is defined as the highest frequency of radio wave, which gets reflected to earth by the ionosphere after having been sent straight to it.

If maximum electron density of the ionosphere is N_{max} per m^3, then $f_c\approx 9(N_{\max})^{1/2}$. Above f_c a wave will penetrate the ionosphere and is not reflected by it.

(vii) **Maximum usable frequency (MUF) :** It is the highest frequency of radio waves which when sent at some angle of incidence θ, towards the ionosphere, get reflected and return to the earth. $MUF=\frac{f_c}{\cos\theta}$

(viii) **Skip distance :** It is the smallest distance from a transmitter along the earth's surface at which a sky wave of a fixed frequency but more than f_c is sent back to the earth.

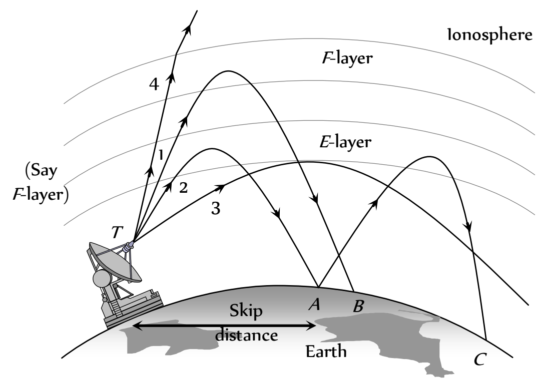


Fig. 28.24

(ix) **Fading :** It is defined as the fluctuation in the strength of a signal at a receiver due to interference of two waves.

Fading is more at high frequencies. It results into errors in data transmission and retrieval.

Space Wave Propagation

The space waves are the radio waves of very high frequency (30 *MHz* to 300 *MHz*) ultra high frequency (300 *MHz* to 3000 *MHz*) and microwave (more than 3000 *MHz*). At such high frequencies, the sky wave as well as ground wave propagation both fails.

These waves can be transmitted from transmitting to receiving antenna either directly or after reflection from the ground or in troposphere, the wave propagation is called space wave propagation.

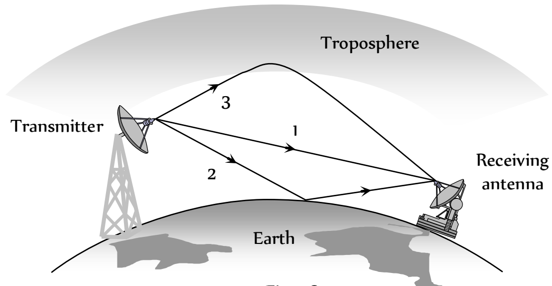


Fig. 28.25

The space wave propagation is also called as line of sight propagation. The line of sight distance is the distance between transmitting antenna and receiving antenna at which they can see each other.

Space wave propagation can be utilised for transmitting high frequency TV and FM signals.

(1) **Television signal propagation :** Frequency range for propagation is 80 *MHz* to 200 *MHz*

Height of transmitting antenna : $h=\frac{d^2}{2R}$ (d = distance covered by the signal, R = Radius of the Earth)

Area covered : $A=\pi d^2=2\pi Rh$

Population cover : Population density × Area covered

(2) **Microwave communication :** Microwave communication systems are used for long distance communication. Since at microwave frequencies, electromagnetic waves cannot bend across the obstacles, such as the top of the buildings, mountains *etc*., it is therefore necessary that microwave transmission is in line-of-sight.

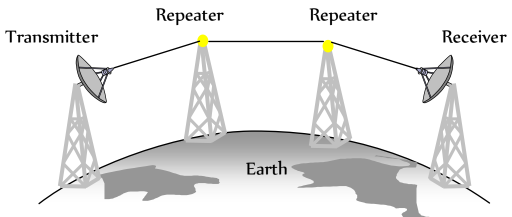


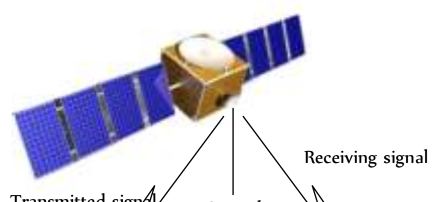
Fig. 28.26

Due to curvature in the surface of earth, the range of microwave transmission is very small (≈ 50 *km*). The range of microwave transmission is also limited by the fact that signals gets weaker and weaker as it propagates. However, these problems are overcome by using repeaters (A repeater is basically an amplifier, which amplifies the attenuated signal and then retransmits it.) at intervals between the transmitter and the receiver. Due to this, the cost of transmission of signal between the two stations increases.

The problems faced in a microwave communication system are solved to a large extent by using a geostationary satellite as a communication satellite.

Satellite Communication

(1) Satellite communication is like the line-of-sight microwave transmission. In this case, a beam of modulated microwave is projected towards the satellite.



(2) Satellite communication is mainly done through geostationary satellites (for steady reliable transmission and reception)

(3) A geostationary satellite has the same time period of revolution of earth (*i.e.* 24 *hrs.*). It appears stationary *w.r.t.* earth. It locates at the height of 36000 *km* above the earth's surface (well above the ionosphere).

(4) A communication satellite is a spacecraft placed in an orbit around the earth which carriers a transmitting and receiving equipment called radio transponder. It amplifies the microwave signals emitted by the transmitter from the surface of earth and send then to the receiving station on earth.

(5) The transmitted signal is UP-LINKED and received by the satellite station which DOWN-LINKS it with the ground station through it's transmitter.

The up-link and down-link frequencies are kept different (both frequencies being in the regions of UHF/microwave).

(6) A single satellite cannot cover the entire surface of the earth. At least three geo-stationary satellites are required which are 120° apart from each other to have the communication link over the entire globe of earth.

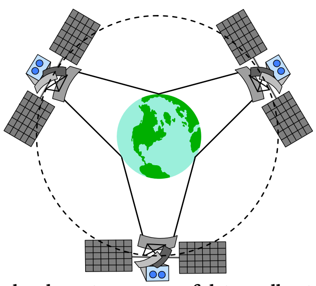


Fig. 28.28

(7) Satellite technology is very useful in collecting information about various factors of the atmosphere which governs the weather and climatic conditions.

The satellite communication can be used for establishing mobile communication with great use the communication satellites are now being used in Global Positioning System (GPS). The ordinary users can find their positions within an accuracy of 100 *m*.

There are two types of satellites used for long distance transmission.

(i) Passive satellite : It act as reflector only for the signals transmitted from earth. Moon the natural satellite of earth is a passive satellite.

(ii) Active satellite : It carries all the equipment used for receiving signals sent from the earth, processing them and then re-transmitting them to the earth. Now a days active satellites are in use.

(8) The Indian communication satellites INSAT-2B and INSAT-2C are positioned in such away in the outer space that they are accessible from any place in India.

Remote Sensing

Remote sensing is the technique to obtain information about an object (in respect of it's size, colour nature, location, temperature *etc.*) by observing it from a distance and without coming to actual contact with it.

(1) There are two types of remote sensing instruments : active and passive. Active instruments provide their own energy to illuminate the object of interest, as radar does. They send an energy pulse to the object and then receive and process the pulse reflected from the object. Passive instruments sense only radiations emitted by the object or solar radiation reflected from the object.

(2) The remote sensing is done through a satellite. The satellite used in remote sensing should move in an orbit around the earth in such away that it always passes over the particular area of the earth at the same local time.

The orbit of such a satellite is known as sun-synchronous orbit. A remote sensing orbit can be circular polar orbit or in highly inclined elliptical orbit.

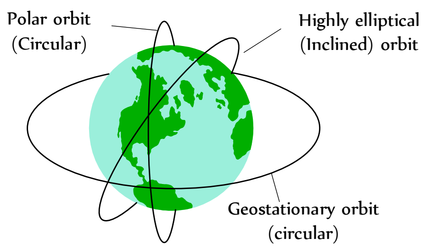


Fig. 28.29

(3) A remote sensing satellite takes, photographs of a particular region which nearly the same illumination every time it passes through that region.

(4) The most useful remote sensing technology is that it makes possible the repetitive surveys of vast areas in a very short time, even if the areas are inaccessible.

(5) Space based remote sensing is a new technology. It has high potential for applications in nearly all aspects of resource management.

(6) The Indian remote sensing satellites are IRS-1A, JRS-1B, and IRS-1C.

(7) **Some applications of remote sensing includes**

(i) Meteorology : (development of weather systems and weather for casting).

(ii) Climatology : Monitoring climatic changes.

(iii) Oceanography : (Sea surface temperatures, mapping of sea-ice and oil pollution monitoring).

(iv) Archaeology, geological surveys.

(v) Water resource surveys,

(vi) Urban land use surveys.

(vii) Agriculture and forestry and natural disaster.

(viii) In the field of spying to detect movements of enemy army an their positions.

(ix) It is used to locate the place where under ground nuclear explosion has carried out.

Line Communication

Line communication means interconnection of two points that are at some distance from each other with the help of wires for exchange of information *e.g.* interconnection between a transmitter and receiver or a transmitter and anteena or an anteena and receiver.

Two Wire Transmission Line

The most commonly used two wire lines are : Parallel wire, twisted pair wires and co-axial cable.

(1) **Parallel wire line :** In a two wire transmission line, two metallic wires are arranged parallel to each other inside a protective insulation coating.

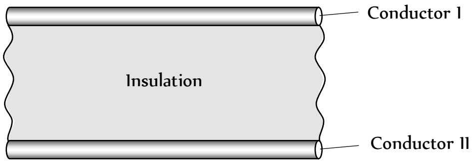


Fig. 28.30

(i) The wires may be hard or flexible depending on the power to be handled.

(ii) It is commonly used to connect an anteena with TV receiver.

(iii) Such wires can suffer from interferences and losses.

(2) **Twisted pair wire :** It consists of two insulated copper wires twisted around each other at regular intervals to minimize electrical interference.

(i) Twisted pair wires are used to connect telephone systems. It works well up to a distance of about 10 *cm*. They cannot transmit signals over very large distances.

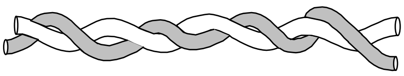


Fig. 28.31

(ii) They can be used for transmitting both, the analog and digital signals.

(iii) They are easy to install and cost effective.

(3) **Coaxial wire lines :** It consist of a central copper wire (which transmits the signals) surrounded by a PVC insulation over which a sleeve of copper mesh (outer conductor) is placed. The outer conductor is normally connected to ground and thus it provides an electrical shield to the signals carried by the central conductor. The outer conductor is externally covered with a polymer jacket for protection.

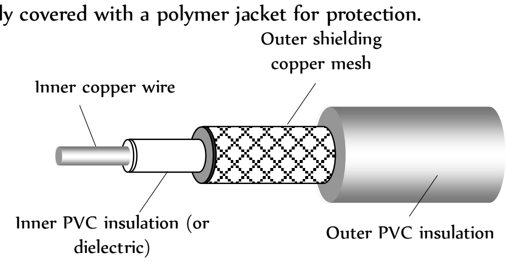


Fig. 28.32

(i) Co-axial line wires can be used for microwaves and ultra high frequency waves.

(ii) The communication through co-axial lines is more efficient than through a twisted pair wire lines.

(iii) Co-axial cables can be gas filled also. To reduce flash over between the conductor handling high power, N_2-gas is used in the cable.

Impedance of Line

Each portion of the transmission line can be considered as a small inductor, resistor and capacitor as shown.

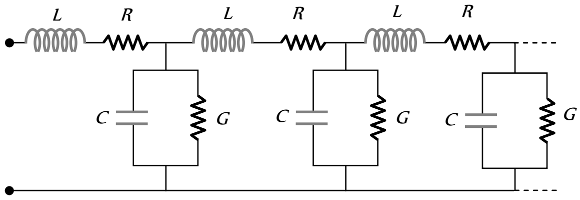


Fig. 28.33

(1) Such inductors, resistors and capacitors are distributed throughout the transmission line.

As a result each length of transmission line has a characteristic impedance.

(2) In case of co-axial cable, the dielectric can be represented by a shunt resistance *G*.

(3) When co-axial cable is used to transmit a radio frequency signal, X_L and X_C are large as compared to *R* and *G* respectively. Hence *R* and *G* can be neglected.

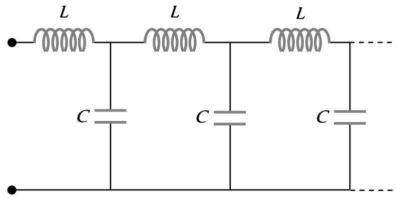


Fig. 28.34

(4) In co-axial cable, *R* is zero, so no loss of energy and hence no attenuation of frequency signal occurs when transmitted along it. That's why co-axial cables are specially used in cable TV network.

(5) **Characteristic impedance (Z_0) :** It is defined as the impedance measured at the input of a line of infinite length.

For parallel line $Z_0 = \dfrac{276}{\sqrt{k}} \log \dfrac{2s}{d}$

d = Diameter of each wire

s = Separation between the two wires

k = Dielectric constant of the insulating medium

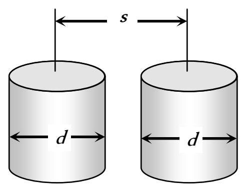


Fig. 28.35

For co-axial line wire $Z_0 = \dfrac{138}{\sqrt{k}} \log \dfrac{D}{d}$

d = Diameter of inner conductor

D = Diameter of outer conductor

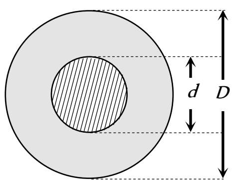


Fig. 28.36

At radio frequency $Z_0 = \sqrt{\dfrac{L}{C}}$

the usual range of characteristic impedance for parallel wire lines is 150Ω to 600 Ω and for co-axial wire it is 40 Ω to 150 Ω.

(6) **Velocity factor of a line (*v. f.*) :** It is the ratio of reduction of speed of light in the dielectric of the cable

$$v.f. = \frac{v}{c} = \frac{\text{Speed of light in medium}}{\text{Speed of light in vacuum}} = \frac{1}{\sqrt{K}}$$

For a line *v.f.* is generally of the order of 0.6 to 0.9.

Telephone Links

(1) Telephone is the most common means of communication. Now a days, the telephone system is required to converse from earth to another heavenly bodies like moon *etc*.

(2) A telephone link can be established with the help of co-axial cables, ground waves, sky waves, microwaves or optical fibre cables.

(3) Simultaneous transmission of a number of messages over a single channel without their interfering with one another is called multiplexing. Two types of multiplexing techniques are in use :

(i) Frequency division multiplexing uses analog modulation of message signals.

(ii) Time division multiplexing makes use of pulse modulation of message signals.

(4) Twisted pair wire lines provide a band width of 2 *MHz*, while co-axial cable provides a band width of 20 *MHz*. For further increase in band width, we use

(i) microwave link

(ii) communication satellite link.

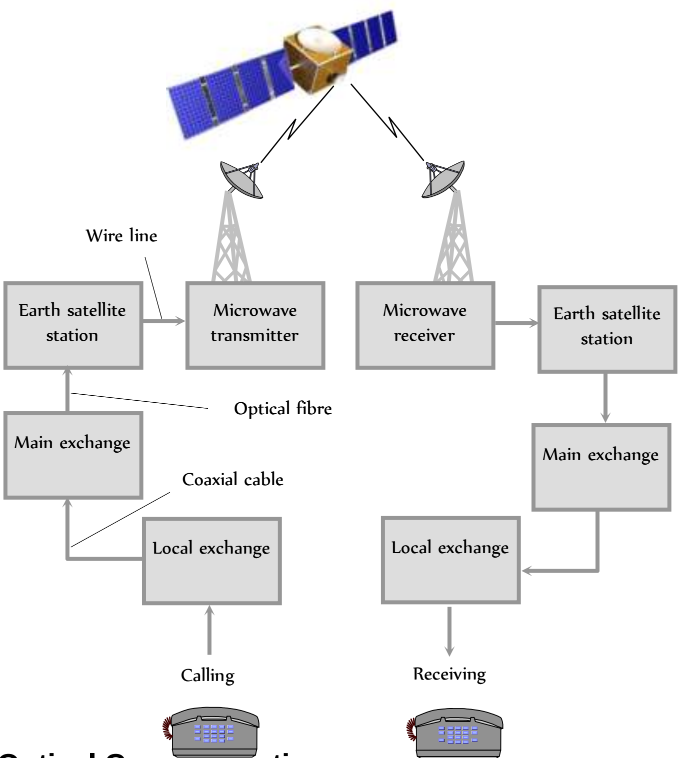


Fig. 28.37

Optical Communication

(1) The use of optical carrier waves for transmission of information from one place to another is called optical communication.

(2) The useful optical frequency range is 10^{12} *Hz* to 10^{16} *Hz* which is very high as compared to radio and microwave frequencies (10^{6} *Hz* – 10^{11} *Hz*).

(3) The information carrying capacity ∝ bandwidth ∝ frequency of carrier wave. So optical communication is better than others. (because of high frequency).

(4) Basic setup of optical communication shown below

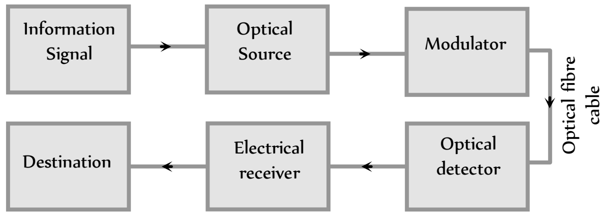


Fig. 28.38

(5) Light emitting diodes (LED) and diode lasers are preferred for optical source. LED's are used for small distance transmission while diode laser is used for very large distance transmission.

(6) In order to transmit information signal via an optical communication system, it is necessary to modulate light with the information signal.

(7) The optical signal reaching the receiving end has to be detected by a detector which converts light into electrical signals, So that the transmitted information may be decoded. Semiconductor based photo-electors are used

Optical Fibre

The optical fibres are used to transmit light signals from one place to another without any practical loss in the intensity of light signal.

(1) **Design :** Optical fibre is made of a thin glass core (diameter 10 to 100 μm) surrounded by a glass coating called cladding are protected by a jacket of plastic.

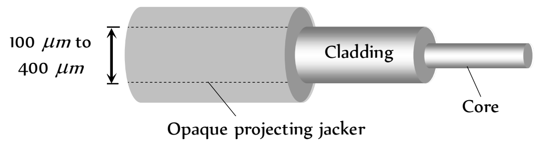


Fig. 28.39

(2) **Principle :** It works on the principle of total internal reflection.

(3) **Action :** The refractive index of the glass used for making core (μ_1 ≈ 1.7) is a little more than the refractive index of the glass (μ_2 ≈ 1.5) used for making the cladding *i.e.* $\mu_1 > \mu_2$.

The core dimension is so small (≈ 10 μm) that the light entering will almost essentially be having incident angle (θ_i) more than the critical angle (θ) and will suffer total internal reflection at the core. Cladding boundary such successive total reflections at opposite boundaries will confine the light to the core as shown in figure.

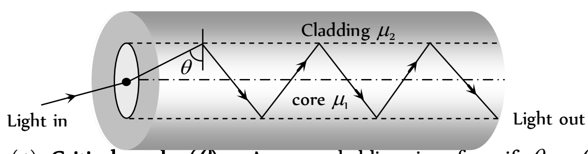


Fig. 28.40

(4) **Critical angle (θ) :** At core-cladding interface if $\theta = \theta_c$ then

$$\cos\theta_c = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_1} \Rightarrow \theta_c = \cos^{-1}\left(\frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_1}\right)$$

(5) **Acceptance angle (θ) :** The value of maximum angle of incidence with the axis of fibre in air for which all the incident light is totally reflected is known as acceptance angle.

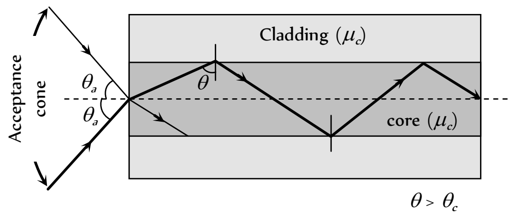


Fig. 28.41

If θ_a = Acceptance angle then μ_1 = refractive index of core, μ_2 = refractive index of cladding.

$$\sin\theta_a = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0} \Rightarrow \theta_a = \sin^{-1}\sqrt{\mu_1^2 - \mu_2^2} \quad \text{(for air } \mu_0 = 1)$$

(6) **Numerical aperture :** Light gathering capability of a fibre is related to numerical aperture. This is defined as the sine of acceptance angle *i.e.*

$$NA = \sin i = \sqrt{\mu_1^2 - \mu_2^2}$$

The numerical aperture can also be given in terms of relative core-cladding index difference (Δ), where $\Delta = \dfrac{\mu_1^2 - \mu_2^2}{2\mu_1^2}$

Thus, $NA = \sqrt{\mu_1^2 - \mu_2^2} = \mu_1\sqrt{2\Delta}$

(7) **Fibre attenuation :** In practice a very small part of light energy is lost from an optical fibre. This reduction in energy of the light is called attenuation and is described by $I = I_0 e^{-\alpha x}$

where I_0 = Intensity of light when it enters the fibre

I = Intensity of light at a distance x along the fibre

α = Absorption co-efficient or attenuation co-efficient

Also attenuation (in dB) $= 10 \log_{10} \dfrac{I}{I_0}$

(8) **Types of optical fibre**

(i) **Monomode optical fibre :** It has a very narrow core of diameter about 5μm or less, cladding is relatively big.

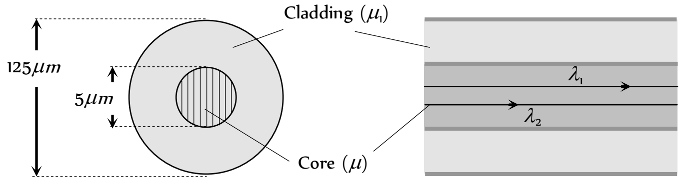


Fig. 28.42

(ii) **Multimode optical fibre :** It is again of two types

(a) Step index multimode fibre :

The diameter of the core is about 50 μm.

Core has constant R.I μ_1 from it's centre to boundary.

The refractive index then changes to a lower value of μ_2 which remains constant through the cladding.

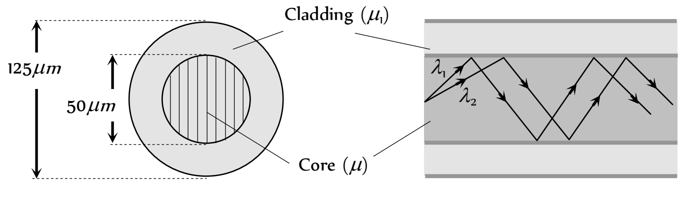


Fig. 28.43

Since refractive index of a material depend on the wavelength of light. The wavelength fellow diff. paths.

The overall time difference between two wavelengths reaches the other end is of the order of 33×10^{-9} *sec/cm* length of the fibre.

(iii) **Graded index multimode fibre :** Refractive index decreases smoothly from it's centre to the outer surface of the fibre (cladding). There is no notieable boundary between core and cladding.

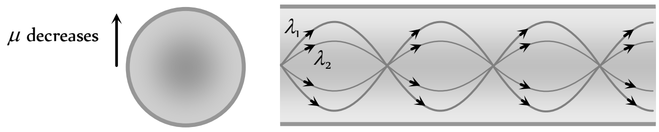


Fig. 28.44

Advantages of Optical Fibres Over Wires

(1) Lower cost in the long run.

(2) Low loss of signal typically less than 0.3 *dB/km*), so repeater-less transmission over long distances is possible.

(3) Large data-carrying capacity (thousands of times greater, reaching speeds of up to 1.6 *Tb/s* in field deployed systems and up to 10 *Tb/s* in lab systems).

(4) No electromagnetic radiation; difficult to eavesdrop.

(5) High electrical resistance, so safe to use near high-voltage equipment or between areas with different earth potentials.

(6) Low weight.

(7) Signals contain very little power.

(8) No cross talk between cables.

(9) No sparks (*e.g.* in automobile applications)

(10) Difficult to place a tap or listening device on the line, providing better physical network security.

Laser

Laser is a process by which we get a beam which is coherent, highly monochromatic and almost perfectly parallel. Such a beam is also called laser.

Coherent : Because all the photons in the light beam, emitted by different atoms, at different instant are in phase.

Monochromatic : Because, the spread $\Delta\lambda$ in wavelength is very small, of the order of 10^{-6} *nm*.

Perfectly parallel : Because, a laser beam can be sent to a far off place and returns back without any practical loss of intensity.

The term LASER stands for **L**ight **A**mplification by **S**timulated **E**mission **R**adiation.

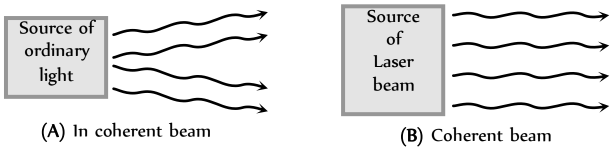


Fig. 28.45

Concepts Related to Production of LASER

(1) **Stimulated absorption :** Consider an atom which has an allowed state at energy E_1 and another allowed state at a higher energy E_2. Suppose the atom is in the lower energy state E_1. If a photon of light having energy $E_2 - E_1$ is incident on this atom, the atom may absorb the photon and jump to the higher energy state E_2. This process is called stimulated absorption of light photon. The incident photon has stimulated the atom to absorb the energy.

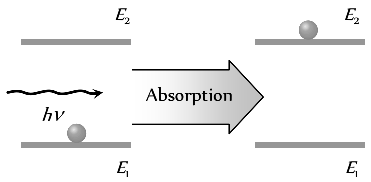


Fig. 28.46

(2) **Spontaneous emission :** If an atom is present in the higher energy state, it tends to return to the lower energy state within a time of 10^{-8} *sec* by emitting a photon of energy $h\nu = E_2 - E_1$. We call this process spontaneous emission. Spontaneous because the event was not triggered by any outside influence.

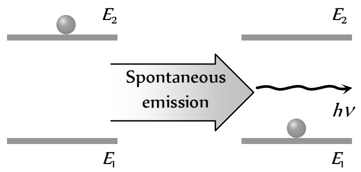


Fig. 28.47

(3) **Stimulated emission :** Suppose a photon of energy $h\nu = E_2 - E_1$ interacts with an atom that is already in the excited state E_2.

The incident photon may stimulate the atom to emit a photon, the energy, phase, and direction of travel of this second photon are exactly the same as those of the incident photon. That is the quantum state of the stimulated photon is identical to that of the incident photon. This process is called stimulated emission.

If these two photons then interact with two more excited state atoms, two more photons are produced, and soon. Therefore, the stimulation process leads to photon amplification.

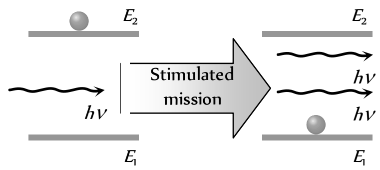


Fig. 28.48

(4) **Population inversion :** Usually the number of atoms in the lower energy state is more than the number of atoms in the excited state. To emit photons which are coherent (*i.e.* in phase), the number of atoms in the higher state must be greater than the number of atoms in the lower energy state. In other words, population of atoms in the higher energy state must be larger than the population of atoms in the lower energy state. The process of making the population of atoms in the higher energy state more than that of lower energy state is known as population inversion.

The method used to invert the population of atoms is known as pumping.

(5) **Metastable states :** A metastable state is one, which has a mean life time of the order of 10^{-3} *s* or more *i.e.* much larger than 10^{-8} *s*, the life time of a higher energy state. Some atomic systems, such as chromium, neon, *etc* possess metastable states. The atom of such an atomic system, when in higher energy state, does not come down to lower energy state directly. It first returns to metastable state and then after a finite lapse of time of the order of 10^{-3} *s*, returns to the lower energy state. Since such atom stays in metastable state for a sufficiently long time, the population inversion can sustain in such atomic system.

A system in which population inversion is achieved is called the active system.

Principle of Laser

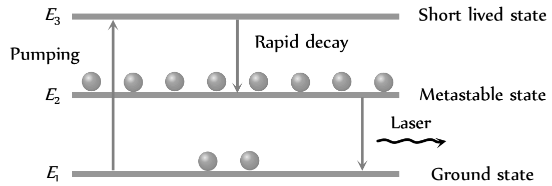


Fig. 28.49

Atoms from the ground state E_1 are 'pumped' up to an excited state E_3. From E_3 the atom decay rapidly to state of energy E_2. For lasing (lasing means laser action) to occur, this state must be metastable. If conditions are right, state E_2 can then become more heavily populated than state E_1, thus providing the needed population inversion.

When photon of energy $h\nu = E_2 - E_1$ is incident on one of the atoms present in the metastable state, the atom will drop to lower energy state E_1, emitting a photon of same energy as that of the incident photon, which is in phase with it and is emitted in the same direction. The two photons, then interact with two more atoms present in metastable state and so on. This process is called amplification of light.

For smooth process two conditions are necessary

(1) The metastable state should all the time have larger number of atoms than the number of atoms in lower energy state.

(It is achieved by pumping)

(2) The photons emitted due to stimulated emission should stimulate other atoms to multiply the photons inside the system.

(It is achieved by two mirrors are fixed at the ends of the system containing lasing material. The mirrors reflect the photons back and forth to keep them inside the region for a long time.)

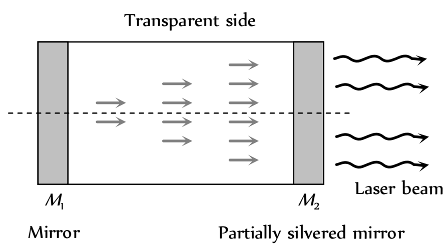


Fig. 28.50

Helium-Neon Laser

This laser contains a mixture of helium (≈ 90%) and neon (≈ 10%) at low pressure in a cylindrical tube with mirrors at each end. The energy level diagram in figure shows the important energy levels for the helium and neon atoms. A large electric field is established in the tube by electrodes connected to a high-voltage power supply, Electrons from ionized atoms are accelerated by the field and collide with atoms.

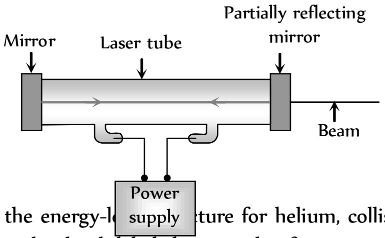


Fig. 28.51

Because of the energy-level structure for helium, collisions often excite helium atoms to the level labeled E_3 in the figure. In a process called collision transfer, energy is transferred from excited helium atoms to neon atoms during collisions, thus producing a population of neon atoms in the E_2 level. The transition from level E_2 to E_1 in neon is forbidden, but the transition out of the E_1 level is allowed. This means that the population of atoms in the E_2 level builds up, and that of the E_1 level is rapidly depleted.

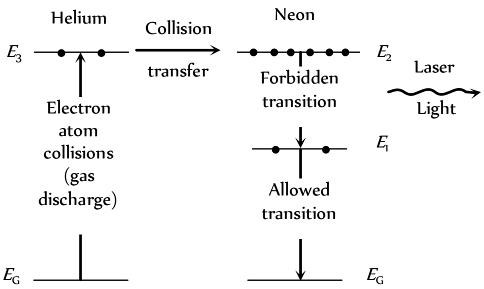


Fig. 28.52

Stimulated emission from E_2 to E_1 predominates and laser light is generated.

The mirrors at each end of the tube encourage emissions along the tube axis by reflecting the light back and forth inside the tube. One of the mirrors is slightly leaky, transmitting about 1 percent of the incident light. This transmitted light forms the laser beam which we find so useful.

T Tips & Tricks

✍ Parallel wire lines are never used for transmission of microwaves. This is because at the frequency of microwaves, separation between the two wires approaches half a wavelength ($\lambda/2$). Therefore radiation loss of energy becomes maximum.

✍ Number of channel accommodated for

$$\text{Transmission} = \frac{\text{Total band width of channel}}{\text{Band width needed per channel}}$$

✍ Bit rate = Sampling rate × no. of bits per sample.

✍ Modulation factor determines the strength and quality of the transmitted signal.

✍ A Hertz antenna is a straight conductor of length equal to half the wavelength of radio signals to be transmitted or received. A Marconi antenna is a straight conductor of length $l = \lambda/4$

✍ In a digital signal, information is carried by the pattern of pulses and not by the shape of pulses.

✍ Sampling converts an analog signal into digital. For example when an analog signal is sampled at interval of 125 μ-*sec* the number of samples taken per second $= \dfrac{1}{125 \times 10^{-6}} = 8000.$

✍ AGC stands for automatic gain control. It is used in receive.

✍ Sputinik-I launched by Russia in 1957 was the first active satellite.

✍ First communication satellite was put in an orbit by USA in 1958.

✍ The first India experimental satellite *i.e.* Apple was launched on June 19, 1981.

✍ The national information cenre at Delhi has linked computers at all head quarters through INSAT 2B.

✍ First communication satellite was put in an orbit by USA in 1958.

✍ Just as $\sqrt{\dfrac{Z}{Y}}$ represents characteristic impedance (Z_0) of a transmission line, $\sqrt{ZY}$ represents propagation constant of the line.

✍ Glass-core and glass cladding (often called SCS fire *i.e.* silica clad silica fibre) have the best propagation characteristics.

✍ The dish type antenna's used for satellite communication are generally of cassegrain type

✍ Ground waves propagate along the surface of the earth. These are vertically polarised to prevent short circuiting of the electric field at a distance d is given by $E = \dfrac{120\,\pi h_t l}{\lambda d}$ and signal received by an antenna of height h_r is given by $V\,(\text{volts}) = \dfrac{120\pi h_t h_r I}{\lambda d}$

✍ Receivers may be of two types, tuned radio frequency (TRF) receivers and superheterodyne receivers. Super heterodyne receivers use local oscillators and intermediate frequency amplifiers before the signal is detected. In this way the reception becomes free of signal frequency but depends only on intermediate frequency which is fixed.

✍ A rectifier with peak detection is used in the AM wave detection and FM detection is achieved by an *LC* circuit tuned at off resonant frequency.

✍ APDs (Avalanche photodiodes) are best suited for detection in fiber optic communication.

✍ **MASER** is microwave amplification by stimulated emission of radiation. It is used as a microwave amplifier or oscillator. The principle of MASER is identical to that of LASER. Only frequency range is $\leq 10^{11}$ *Hz* in masers.

✍ In frequency modulation m_f (frequency modulation index) is inversely proportional to modulating frequency f_m. While in PM it does not vary with modulating frequency. Moreover, FM is more noise immune.

✍ AM with single side band suppressed carrier is better as it contains maximum modulating power.

Ordinary Thinking

Objective Questions

Communication

1. In short wave communication waves of which of the following frequencies will be reflected back by the ionospheric layer, having electron density 10^{11} per m [AIIMS 2003]
   (a) 2 *MHz* (b) 10 *MHz*
   (c) 12 *MHz* (d) 18 *MHz*

2. In an amplitude modulated wave for audio frequency of 500 *cycle/second*, the appropriate carrier frequency will be [AMU 1996]
   (a) 50 *cycles/sec* (b) 100 *cycles/sec*
   (c) 500 *cycles/sec* (d) 50,000 *cycles/sec*

3. AM is used for broadcasting because
   (a) It is more noise immune than other modulation systems
   (b) It requires less transmitting power compared with other systems
   (c) Its use avoids receiver complexity
   (d) No other modulation system can provide the necessary bandwidth faithful transmission

4. Range of frequencies allotted for commercial FM radio broadcast is
   (a) 88 to 108 *MHz* (b) 88 to 108 *kHz*
   (c) 8 to 88 *MHz* (d) 88 to 108 *GHz*

5. The velocity factor of a transmission line *x*. If dielectric constant of the medium is 2.6, the value of *x* is [AFMC 1995]
   (a) 0.26 (b) 0.62
   (c) 2.6 (d) 6.2

6. The process of superimposing signal frequency (*i.e.* audio wave) on the carrier wave is known as [AIIMS 1987]
   (a) Transmission (b) Reception
   (c) Modulation (d) Detection

7. Long distance short-wave radio broadcasting uses [AFMC 1996]
   (a) Ground wave (b) Ionospheric wave
   (c) Direct wave (d) Sky wave

8. A step index fibre has a relative refractive index of 0.88%. What is the critical angle at the corecladding interface [Manipal 2003]
   (a) 60° (b) 75°
   (c) 45° (d) None of these

9. The characteristic impedance of a coaxial cable is of the order of
   (a) 50 Ω (b) 200 Ω
   (c) 270 Ω (d) None of these

10. In which frequency range, space waves are normally propagated
    (a) HF (b) VHF
    (c) UHF (d) SHF

11. If μ_1 and μ_2 are the refractive indices of the materials of core and cladding of an optical fibre, then the loss of light due to its leakage can be minimised by having [BVP 2003]
    (a) $\mu_1 > \mu_2$ (b) $\mu_1 < \mu_2$
    (c) $\mu_1 = \mu_2$ (d) None of these

12. Through which mode of propagation, the radio waves can be sent from one place to another [JIPMER 2003]
    (a) Ground wave propagation
    (b) Sky wave propagation
    (c) Space wave propagation
    (d) All of them

13. A laser beam of pulse power 10^{12} *watt* is focussed on an object are 10^{-4} cm^2. The energy flux in $watt/cm^2$ at the point of focus is
    (a) 10^{20} (b) 10^{16}
    (c) 10^{8} (d) 10^{4}

14. The carrier frequency generated by a tank circuit containing 1 *nF* capacitor and 10 μH inductor is [AFMC 2003]
    (a) 1592 *Hz* (b) 1592 *MHz*
    (c) 1592 *kHz* (d) 159.2 *Hz*

15. Broadcasting antennas are generally [AFMC 2003]
    (a) Omnidirectional type (b) Vertical type
    (c) Horizontal type (d) None of these

16. For television broadcasting, the frequency employed is normally
    (a) 30-300 *MHz* (b) 30-300 *GHz*
    (c) 30-300 *KHz* (d) 30-300 *Hz*

17. The radio waves of frequency 300 *MHz* to 3000 *MHz* belong to [MNR 1997]
    (a) High frequency band
    (b) Very high frequency band
    (c) Ultra high frequency band
    (d) Super high frequency band

18. An antenna behaves as resonant circuit only when its length is
    (a) $\frac{\lambda}{2}$
    (b) $\frac{\lambda}{4}$
    (c) λ
    (d) $\frac{\lambda}{2}$ or integral multiple of $\frac{\lambda}{2}$

19. Maximum useable frequency (MUF) in *F*-region layer is *x*, when the critical frequency is 60 *MHz* and the angle of incidence is 70°. Then *x* is [Himachal PMT 2003]
    (a) 150 *MHz* (b) 170 *MHz*
    (c) 175 *MHz* (d) 190 *MHz*

20. The electromagnetic waves of frequency 2 *MHz* to 30 *MHz* are [CPMT 2003]
    (a) In ground wave propagation
    (b) In sky wave propagation
    (c) In microwave propagation
    (d) In satellite communication

21. A laser is a coherent source because it contains [JIPMER 2003] [EAMCET 2002]
    (a) Many wavelengths
    (b) Uncoordinated wave of a particular wavelength

(c) Coordinated wave of many wavelengths

(d) Coordinated waves of a particular wavelength

22. The attenuation in optical fibre is mainly due to [AFMC 2003]

(a) Absorption

(b) Scattering

(c) Neither absorption nor scattering

(d) Both (a) and (b)

23. The maximum distance upto which TV transmission from a TV tower of height h can be received is proportional to [AIIMS 2003]

(a) $h^{1/2}$ (b) h

(c) $h^{3/2}$ (d) h^2

24. A laser beam is used for carrying out surgery because it [AIIMS 2003]

(a) Is highly monochromatic (b) Is highly coherent

(c) Is highly directional (d) Can be sharply focussed

25. Laser beams are used to measure long distances because [DCE 2002, 03]

(a) They are monochromatic

(b) They are highly polarised

(c) They are coherent

(d) They have high degree of parallelism

26. An oscillator is producing FM waves of frequency 2 *kHz* with a variation of 10 *kHz*. What is the modulating index [DCE 2004]

(a) 0.20 (b) 5.0

(c) 0.67 (d) 1.5

27. The maximum peak to peak voltage of an AM wire is 24 *mV* and the minimum peak to peak voltage is 8 *mV*. The modulation factor is

(a) 10% (b) 20%

(c) 25% (d) 50%

28. Sinusoidal carrier voltage of frequency 1.5 *MHz* and amplitude 50 *V* is amplitude modulated by sinusoidal voltage of frequency 10 *kHz* producing 50% modulation. The lower and upper side-band frequencies in *kHz* are

(a) 1490, 1510 (b) 1510, 1490

(c) $\frac{1}{1490}, \frac{1}{1510}$ (d) $\frac{1}{1510}, \frac{1}{1490}$

29. What is the modulation index of an over modulated wave

(a) 1 (b) Zero

(c) < 1 (d) > 1

30. Basically, the product modulator is

(a) An amplifier (b) A mixer

(c) A frequency separator (d) A phase separator

31. If f_a and f_f represent the carrier wave frequencies for amplitude and frequency modulations respectively, then

(a) $f_a > f_f$ (b) $f_a < f_f$

(c) $f_a \approx f_f$ (d) $f_a \geq f_f$

32. Which of the following is the disadvantage of FM over AM

(a) Larger band width requirement

(b) Larger noise

(c) Higher modulation power

(d) Low efficiency

33. If a number of sine waves with modulation indices n_1, n_2, n_3 .... modulate a carrier wave, then total modulation index (n) of the wave is

(a) $n_1 + n_2 .... + 2(n_1 + n_2.....)$

(b) $\sqrt{n_1 - n_2 + n_3..........}$

(c) $\sqrt{n_1^2 + n_2^2 + n_3^2.........}$

(d) None of these

34. An AM wave has 1800 *watt* of total power content, For 100% modulation the carrier should have power content equal to

(a) 1000 *watt* (b) 1200 *watt*

(c) 1500 *watt* (d) 1600 *watt*

35. The frequency of a FM transmitter without signal input is called

(a) Lower side band frequency

(b) Upper side band frequency

(c) Resting frequency

(d) None of these

36. What type of modulation is employed in India for radio transmission

(a) Amplitude modulation (b) Frequency modulation

(c) Pulse modulation (d) None of these

37. When the modulating frequency is doubled, the modulation index is halved and the modulating voltage remains constant, the modulation system is

(a) Amplitude modulation (b) Phase modulation

(c) Frequency modulation (d) All of the above

38. An antenna is a device

(a) That converts electromagnetic energy into radio frequency signal

(b) That converts radio frequency signal into electromagnetic energy

(c) That converts guided electromagnetic waves into free space electromagnetic waves and vice-versa

(d) None of these

39. While tuning in a certain broadcast station with a receiver, we are actually

(a) Varying the local oscillator frequency

(b) Varying the frequency of the radio signal to be picked up

(c) Tuning the antenna

(d) None of these

40. Indicate which one of the following system is digital

(a) Pulse position modulation

(b) Pulse code modulation

(c) Pulse width modulation

(d) Pulse amplitude modulation

41. In a communication system, noise is most likely to affect the signal

(a) At the transmitter

(b) In the channel or in the transmission line

(c) In the information source

(d) At the receiver

42. The waves used in telecommunication are

(a) IR (b) UV

(c) Microwave (d) Cosmic rays

43. In an FM system a 7 *kHz* signal modulates 108 *MHz* carrier so that frequency deviation is 50 *kHz*. The carrier swing is

(a) 7.143 (b) 8

(c) 0.71 (d) 350

44. Consider telecommunication through optical fibres. Which of the following statements is not true **[AIEEE 2003]**

(a) Optical fibres may have homogeneous core with a suitable cladding

(b) Optical fibres can be of graded refractive index

(c) Optical fibres are subject to electromagnetic interference from outside

(d) Optical fibres have extremely low transmission loss

45. The phenomenon by which light travels in an optical fibres is

(a) Reflection (b) Refraction

(c) Total internal reflection (d) Transmission

46. Television signals on earth cannot be received at distances greater than 100 *km* from the transmission station. The reason behind this is that **[DCE 1995]**

(a) The receiver antenna is unable to detect the signal at a distance greater than 100 *km*

(b) The TV programme consists of both audio and video signals

(c) The TV signals are less powerful than radio signals

(d) The surface of earth is curved like a sphere

47. Advantage of optical fibre **[DCE 2005]**

(a) High bandwidth and EM interference

(b) Low bandwidth and EM interference

(c) High band width, low transmission capacity and no EM interference

(d) High bandwidth, high data transmission capacity and no EM interference

48. In frequency modulation **[Kerala PMT 2005]**

(a) The amplitude of modulated wave varies as frequency of carrier wave

(b) The frequency of modulated wave varies as amplitude of modulating wave

(c) The amplitude of modulated wave varies as amplitude of carrier wave

(d) The frequency of modulated wave varies as frequency of modulating wave

(e) The frequency of modulated wave varies as frequency of carrier wave

49. Audio signal cannot be transmitted because **[Kerala PMT 2005]**

(a) The signal has more noise

(b) The signal cannot be amplified for distance communication

(c) The transmitting antenna length is very small to design

(d) The transmitting antenna length is very large and impracticable

(e) The signal is not a radio signal

50. In which of the following remote sensing technique is not used

(a) Forest density (b) Pollution

(c) Wetland mapping (d) Medical treatment

51. For sky wave propagation of a 10 *MHz* signal, what should be the minimum electron density in ionosphere **[AIIMS 2005]**

(a) $\sim 1.2 \times 10^{12}\ m^{-3}$ (b) $\sim 10^{6}\ m^{-3}$

(c) $\sim 10^{14}\ m^{-3}$ (d) $\sim 10^{22}\ m^{-3}$

52. What should be the maximum acceptance angle at the aircore interface of an optical fibre if n_1 and n_2 are the refractive indices of the core and the cladding, respectively **[AIIMS 2005]**

(a) $\sin^{-1}(n_2/n_1)$ (b) $\sin^{-1}\sqrt{n_1^2 - n_2^2}$

(c) $\left[\tan^{-1}\dfrac{n_2}{n_1}\right]$ (d) $\left[\tan^{-1}\dfrac{n_1}{n_2}\right]$

[DCE 2001]

Critical Thinking

Objective Questions

1. A sky wave with a frequency 55 *MHz* is incident on *D*-region of earth's atmosphere at 45°. The angle of refraction is (electron density for *D*-region is 400 *electron*/cm^3) **[Haryana PMT 2003]**

(a) 60° (b) 45°

(c) 30° (d) 15°

2. In a diode AM-detector, the output circuit consist of $R = 1k\Omega$ and $C = 10\ pF$. A carrier signal of 100 *kHz* is to be detected. Is it good

(a) Yes

(b) No

(c) Information is not sufficient

(d) None of these

3. Consider an optical communication system operating at $\lambda \sim 800\ nm$. Suppose, only 1% of the optical source frequency is the available channel bandwidth for optical communication. How many channels can be accommodated for transmitting audio signals requiring a bandwidth of 8 *kHz*

(a) 4.8×10^{8} (b) 48

(c) 6.2×10^{8} (d) 4.8×10^{5}

4. A photodetector is made from a semiconductor $In_{0.53}Ga_{0.47}As$ with E_g = 0.73 *eV*. What is the maximum wavelength, which it can detect

(a) 1000 *nm* (b) 1703 *nm*

(c) 500 *nm* (d) 173 *nm*

5. A transmitter supplies 9 kW to the aerial when unmodulated. The power radiated when modulated to 40% is

(a) 5 kW (b) 9.72 kW
(c) 10 kW (d) 12 kW

6. The antenna current of an AM transmitter is 8 A when only carrier is sent but increases to 8.96 A when the carrier is sinusoidally modulated. The percentage modulation is

(a) 50% (b) 60%
(c) 65% (d) 71%

7. The total power content of an AM wave is 1500 W. For 100% modulation, the power transmitted by the carrier is

(a) 500 W (b) 700 W
(c) 750 W (d) 1000 W

8. The total power content of an AM wave is 900 W. For 100% modulation, the power transmitted by each side band is

(a) 50 W (b) 100 W
(c) 150 W (d) 200 W

9. The modulation index of an FM carrier having a carrier swing of 200 kHz and a modulating signal 10 kHz is

(a) 5 (b) 10
(c) 20 (d) 25

10. A 500 Hz modulating voltage fed into an FM generator produces a frequency deviation of 2.25 kHz. If amplitude of the voltage is kept constant but frequency is raised to 6 kHz then the new deviation will be

(a) 4.5 kHz (b) 54 kHz
(c) 27 kHz (d) 15 kHz

11. The audio signal used to modulate 60 sin $(2\pi \times 10^6\ t)$ is $15\sin 300\pi t$. The depth of modulation is

(a) 50% (b) 40%
(c) 25% (d) 15%

12. The bit rate for a signal, which has a sampling rate of 8 kHz and where 16 quantisation levels have been used is

(a) 32000 *bits/sec* (b) 16000 *bits/sec*
(c) 64000 *bits/sec* (d) 72000 *bits/sec*

13. An amplitude modulated wave is modulated to 50%. What is the saving in power if carrier as well as one of the side bands are suppressed

(a) 70% (b) 65.4%
(c) 94.4% (d) 25.5%

14. In AM, the centpercent modulation is achieved when

(a) Carrier amplitude = signal amplitude
(b) Carrier amplitude ≠ signal amplitude
(c) Carrier frequency = signal frequency
(d) Carrier frequency ≠ signal frequency

AR Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) *If both assertion and reason are true and the reason is the correct explanation of the assertion.*
(b) *If both assertion and reason are true but reason is not the correct explanation of the assertion.*
(c) *If assertion is true but reason is false.*
(d) *If the assertion and reason both are false.*
(e) *If assertion is false but reason is true.*

1. Assertion : Diode lasers are used as optical sources in optical communication.
Reason : Diode lasers consume less energy. [AIIMS 2005]

2. Assertion : Television signals are received through sky-wave propagation.
Reason : The ionosphere reflects electromagnetic waves of frequencies greater than a certain critical frequency. [AIIMS 2005]

3. Assertion : In high latitude one sees colourful curtains of light hanging down from high altitudes.
Reason : The high energy charged particles from the sun are deflected to polar regions by the magnetic field of the earth. [AIIMS 2003]

4. Assertion : Short wave bands are used for transmission of radio waves to a large distance.
Reason : Short waves are reflected by ionosphere [AIIMS 1994]

5. Assertion : The electrical conductivity of earth's atmosphere decreases with altitude.
Reason : The high energy particles (*i.e.* γ-rays and cosmic rays) coming from outer space and entering our earth's atmosphere cause ionisation of the atoms of the gases present there and the pressure of gases decreases with increase in altitude.

6. Assertion : The electromagnetic waves of shorter wavelength can travel longer distances on earth's surface than those of longer wavelengths.
Reason : Shorter the wavelength, the larger is the velocity of wave propagation.

7. Assertion : The surface wave propagation is used for medium wave band and for television broadcasting.
Reason : The surface waves travel directly from transmitting antenna to receiver antenna through atmosphere.

8. Assertion : The television broadcasting becomes weaker with increasing distance.
Reason : The power transmitted from TV transmitter varies inversely as the distance of the receiver

9. Assertion : Microwave propagation is better than the sky wave propagation.
Reason : Microwaves have frequencies 100 to 300 GHz, which have very good directional properties.

10. Assertion : Satellite is an ideal platform for remote sensing.
Reason : Satellite in polar orbit can provide global coverage or continuous coverage of the fixed area in geostationary configuration.

11. Assertion : Fax is a modulating and demodulating device.

Reason : It is necessary for exact reproduction of a document.

12. Assertion : A dish antenna is highly directional.

Reason : This is because a dipole antenna is omni directional.

Answers

Communication

1	a	2	d	3	c	4	a	5	b
6	c	7	c	8	d	9	c	10	c
11	a	12	d	13	b	14	c	15	b
16	a	17	c	18	d	19	c	20	b
21	d	22	d	23	a	24	d	25	d
26	b	27	d	28	a	29	d	30	b
31	b	32	a	33	c	34	b	35	c
36	a	37	c	38	c	39	a	40	b
41	b	42	c	43	a	44	c	45	c
46	d	47	d	48	b	49	d	50	d
51	a	52	b						

Critical Thinking Questions

1	b	2	b	3	a	4	b	5	b
6	d	7	d	8	c	9	b	10	b
11	c	12	a	13	c	14	a		

Assertion and Reason

1	b	2	d	3	a	4	a	5	e
6	c	7	a	8	c	9	a	10	a
11	e	12	b						

Answers and Solutions

Communication

1. (a) By using $f_c \approx 9(N_{\max})^{1/2} \Rightarrow f_c \approx 2\,MHz$
2. (d) Carrier frequency > audio frequency
3. (c)
4. (a) A maximum frequency deviation of 75 kHz is permitted for commercial FM broadcast stations in the 88 to 108 MHz VHF band.
5. (b) $v.f. = \frac{1}{\sqrt{k}} = \frac{1}{\sqrt{2.6}} = 0.62$
6. (c) Carrier + signal → modulation.
7. (c)
8. (d) Here $\frac{n_1 - n_2}{n_1} = \frac{0.88}{100} \Rightarrow \frac{n_2}{n_1} = 0.9912$

   ∴ Critical angle $\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}(0.9912) = 84°24'$
9. (c)
10. (c)
11. (a)
12. (d) Radio waves can be transmitted from one place to another as grand wave or sky wave or space wave propagation.
13. (b) The energy flux $\phi = \frac{\text{Pulsepower}}{\text{Area}} = \frac{10^{12}}{10^{-4}} = 10^{16}\ \frac{W}{cm^2}$
14. (c) $\nu = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\times 3.14\sqrt{10\times 10^{-6}\times 1\times 10^{-9}}}$ = 1592 kHz
15. (b)
16. (a) VHF (Very High Frequency) band having frequency range 30 MHz to 300 MHz is typically used for TV and radar transmission.
17. (c)
18. (d)
19. (c) $MUF = \frac{f_c}{\cos\theta} = \frac{60}{\cos 70°} = 175\,MHz$
20. (b)
21. (d)
22. (d) A very small part of light energy is lost from an optical fibre due to absorption or due to light leaving the fibre as a result of scattering of light sideways by impurities in the glass fibre.
23. (a) $d = \sqrt{2hR} \Rightarrow d \propto h^{1/2}$
24. (d) Surgery needs sharply focused beam of light and laser can be sharply focused.
25. (d) Laser beams are perfectly parallel. So that they are very narrow and can travel a long distance without spreading. This is the feature of laser while they are monochromatic and coherent these are characteristics only.
26. (b) The formula for modulating index is given by

   $$m_f = \frac{\delta}{v_m} = \frac{\text{Frequency variation}}{\text{Modulatingfrequency}} = \frac{10\times 10^3}{2\times 10^3} = 5$$
27. (d) Here, $V_{\max} = \frac{24}{2} = 12\,mV$ and $V_{\min} = \frac{8}{2} = 2\,mV$

   Now, $m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{12-4}{12+4} = \frac{8}{16} = \frac{1}{2} = 0.5 = 50\%$
28. (a) Here, $f_c = 1.5\,MHz = 1500\,kHz,\ f_m == 10\,kHz$

   ∴ Low side band frequency

   $= f_c - f_m = 1500\,kHz - 10\,kHz = 1490\,kHz$

   Upper side band frequency

   $= f_c + f_m = 1500\,kHz + 10\,kHz = 1510\,kHz$
29. (d) When m > 1 then carrier is said to be over modulated.
30. (b) It mix weak signals with carrier signals.
31. (b)
32. (a) Frequency modulation requires much wider channel (7 to 15 times) as compared to AM.
33. (c)
34. (b) $P_t = P_c\left(1 + \frac{m_a^2}{2}\right)$; Here m = 1

   $\Rightarrow 1800 = P_c\left(1 + \frac{(1)^2}{2}\right) \Rightarrow P_c = 1200W$
35. (c)
36. (a)
37. (c)
38. (c) An antenna is a metallic structure used to radiate or receive EM waves.
39. (a)
40. (b) Pulse code modulation is a digital system.
41. (b)
42. (c) In telecommunication, microwaves are used.

43. (a) Carrier swing $=\dfrac{\text{Frequency deviation}}{\text{Modulating frequency}}=\dfrac{50}{7}=7.143$

44. (c) Optical fibres are not subjected to electromagnetic interference from outside.

45. (c) In optical fibre, light travels inside it, due to total internal reflection.

46. (d)

47. (d) Few advantages of optical fibres are that the number of signals carried by optical fibres is much more than that carried by the *Cu* wire or radio waves. Optical fibres are practically free from electromagnetic interference and problem of cross talks whereas ordinary cables and microwave links suffer a lot from it.

48. (b) The process of changing the frequency of a carrier wave (modulated wave) in accordance with the audio frequency signal (modulating wave) is known as frequency modulation (FM).

49. (d) Following are the problems which are faced while transmitting audio signals directly.

(i) These signals are relatively of short range.

(ii) If every body started transmitting these low frequency signals directly, mutual interference will render all of them ineffective.

(iii) Size of antenna required for their efficient radiation would be larger *i.e.* about 75 *km*.

50. (d) Remote sensing is the technique to collect information about an object in respect of its size, colour, nature, location, temperature *etc.* without physically touching it. There are some areas or location which are inaccessible. So to explore these areas or locations, a technique known as remote sensing is used. Remote sensing is done through a satellite.

51. (a) The critical frequency of a sky wave for reflection from a layer of atmosphere is given by $f_c=9(N_{\max})^{1/2}$

$$\Rightarrow\ 10\times10^6=9(N_{\max})^{1/2}$$

$$\Rightarrow\ N_{\max}=\left(\frac{10\times10^6}{9}\right)^2\simeq1.2\times10^{12}m^{-3}$$

52. (b) Core of acceptance angle $\theta=\sin^{-1}\sqrt{n_1^2-n_2^2}$

Critical Thinking Questions

1. (b) $n_{eff}=n_0\sqrt{1-\left(\dfrac{80.5\,N}{\nu^2}\right)}=1\sqrt{1-\dfrac{80.5\times(400\times10^6)}{(55\times10^6)^2}}\approx1$

Also $n_{eff}=\dfrac{\sin i}{\sin r}\Rightarrow\sin r=\sin i\Rightarrow r=i=45°$

2. (b) For demodulation $\dfrac{1}{f_c}<<RC$

$$\frac{1}{f_c}=\frac{1}{100\times10^3}=10^{-5}\,s$$

$RC=10^3\times10\times10^{-12}\,s=10^{-8}\ s$

We see that $\dfrac{1}{f_c}$ here is not less than *RC* as required by the above condition. Hence, this is not good.

3. (a) Optical source frequency $f=\dfrac{c}{\lambda}$

$=3\times10^8/(800\times10^{-9})=3.8\times10^{14}\ Hz$

Bandwidth of channel (1% of above) $=3.8\times10^{12}\ Hz$

Number of channels =(Total bandwidth of channel)/(Bandwidth needed per channel)

(a) Number of channels for audio signal

$=(3.8\times10^{12})/(8\times10^3)\sim4.8\times10^8$

4. (b) Limiting value of $h\nu$ is E_g such that $h\nu=\dfrac{hc}{\lambda}=E_g$

or $\lambda=\dfrac{hc}{E_g}=\dfrac{6.63\times10^{-34}\,J\text{-}s\times3\times10^8\,ms^{-1}}{0.73\times1.6\times10^{-19}\,J}$

$=1703\ nm$

5. (b) $P_t=P_c\left[1+\dfrac{m^2}{2}\right]=9\left[1+\dfrac{(0.4)^2}{2}\right]$

$=9\left[1+\dfrac{0.16}{2}\right]$ $(\because\ m=40\%=0.4)$

$=9\ (1.08)=9.72\ kW$

6. (d) We know that $\left(\dfrac{I_t}{I_c}\right)^2=1+\dfrac{m^2}{2}$

Here, $I_t=8.96A$ and $I_c=8A$

$\therefore\ \left(\dfrac{8.96}{8}\right)^2=1+\dfrac{m^2}{2}$ or $1.254=1+\dfrac{m^2}{2}$

or $\dfrac{m^2}{2}=0.254$ or $m^2=0.508$

or $m=0.71=71\%$

7. (d) $\dfrac{P_t}{P_c}=1+\dfrac{m^2}{2}$ or $P_c=P_t\left[\dfrac{2}{2+m^2}\right]$

$\therefore\ P_c=1500\left[\dfrac{2}{2+1}\right]$ $\because\ m=100\%=1$

$=1000\ W$

8. (c) $P_c=P_t\left[\dfrac{2}{2+m^2}\right]=900\left[\dfrac{2}{2+1}\right]=600W$

Now, $P_{LSB}=\dfrac{m^2}{4}\times P_c=\dfrac{1}{4}\times600=150\,W$

9. (b) CS $=2\times\Delta f$ or $\Delta f=CS/2$

$\therefore\ \Delta f=\dfrac{200}{2}=100\,kHz$

Now $m_f=\dfrac{\Delta f}{f_m}=\dfrac{100}{10}=10$

10. (b) $m_f = \frac{\delta}{f_m} = \frac{2250}{500} = 4.5$

$\therefore$ New deviation $= 2(m_f\, f_m) = 2 \times 4.5 \times 6 = 54\, kHz.$

11. (c) $m_a = \frac{E_m}{E_c} = \frac{15}{60} \times 100 = 25\%$

12. (a) If n is the number of bits per sample, then number of quantisation level = 2^n

Since the number of quantisation level is 16

$\Rightarrow 2^n = 16 \Rightarrow n = 4$

$\therefore$ bit rate = sampling rate × no. of bits per sample

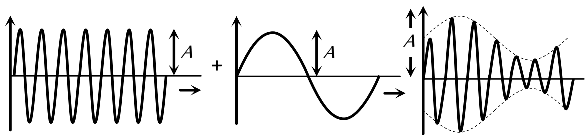
= 8000 × 4 = 32,000 *bits/sec*.

13. (c) $P_{sb} = P_c \left(\frac{m_a}{2}\right)^2 = P_c \frac{(0.5)^2}{4} = 0.0625\, P_c$

Also $P = P_c\left(1 + \frac{m_a^2}{2}\right) = P_c\left(1 + \frac{(0.5)^2}{2}\right) = 1.125 P_c$

$\therefore$ % saving $= \frac{(1.125 P_c - 0.0625 P_c)}{1.125 P_c} \times 100 = 94.4\%.$

14. (a) When signal amplitude is equal to the carrier amplitude, the amplitude of carrier wave varies between $2A$ and zero.



$m_a = \frac{\text{Amplitude charge of carrier}}{\text{Amplitude of normal carrier}} = \frac{2A - A}{A} \times 100 = 100\%$

Assertion and Reason

1. (b) In optical communication, diode laser is used to generate analog signals or digital pulses for transmission or digital pulses for transmission trough optical fibres. The advantage of diode lasers are their small size and low power input.

2. (d) TV signals (frequency greater than 30 *MHz*) cannot be propagated through sky wave propagation.

Above critical frequency, an electromagnetic wave penetratates the ionosphere and is not reflected by it.

3. (a) Microwave communication is preferred over optical communication because microwaves provide large number of channels and wider band width compared to optical signals as information carrying capacity is directly proportional to band width. So, wider the band width, greater the information carrying capacity.

4. (a) Having the range of wavelength from 30 *km* to 30 *cm* are known as short wave. These waves are used for radio transmission and for general communication purpose to a longer distance from ionosphere. Ionosphere is the outermost region of atmosphere extending from height of 80 *km* to 400 *km* approximately, above the surface of earth. Therefore, both the assertion and reason are true and reason is the correct explanation of assertion.

5. (e) The electrical conductivity of earth's atmosphere increases with height so assertion is false.

When high energy particles enters in earth's atmosphere. They ionises the gases present in atmosphere. Also as we go up, the air thins out gradually and air pressure decreases.

6. (c) The electromagnetic waves of shorter wavelength do not suffer much diffraction from the obstacles of earth's atmosphere so they can travel long distance.

Also, shorter the wavelength, shorter is the velocity of wave propagation.

7. (a) Both assertion and reason are true and reason is the correct explanation of assertion. (For more detail, refer theory).

8. (c) As the distance increases, TV signals becomes weaker. So assertion is true. The power transmitted from TV transmitter is inversely proportional to the square of the distance of the receiver. That's why reason is false.

9. (a) Microwaves have got good directional properties. Due to it, the microwaves can be directed as beam signals in a particular direction, much better than radio waves, because microwaves do not bend around the corners of any obstacle coming in their way.

10. (a) The remote sensing is done through a satellite. A remote sensing satellite files in a polar orbit at an altitude of 918 *km*, around the earth, in such away that it passes over a given location on the earth at the same local time.

11. (e) The electronic reproduction of a document at a distance plane is known as FAX modulation and demodulation is done by modem.

12. (b) A dish antenna is a directional antenna because it can transmit or sec.

Communication

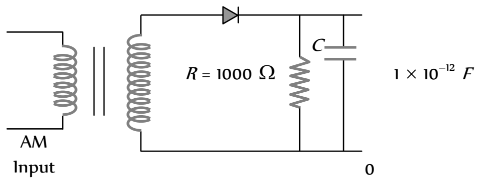
SET Self Evaluation Test - 28

1. A ground receiver station is receiving a signal at (i) 5 MHz and transmitted from a ground transmitter at a height of 300 m, located at a distance of 100 km from the receiver station. The signal is coming via. Radius of earth = 6.4×10^{6} m. N_{max} of isosphere = 10^{12} m

(a) Space wave (b) Sky wave propagation

(c) Satellite transponder (d) All of these

2. In the given detector circuit, the suitable value of carrier frequency is



(a) $<< 10^{9}$ Hz (b) $<< 10^{5}$ Hz

(c) $>> 10^{9}$ Hz (d) None of these

3. The impedance of coaxial cable, when its inductance is 0.40 μH and capacitance is 1×10^{-11} F, can be

(a) $2 \times 10^{2}\ \Omega$ (b) $100\ \Omega$

(c) $3 \times 10^{3}\ \Omega$ (d) $3 \times 10^{-2}\ \Omega$

4. A wave is represented as

$$e = 10\sin(10^{8}t + 6\sin 1250t)$$

then the modulating index is

(a) 10 (b) 1250

(c) 10^{8} (d) 6

5. An optical fibre communication system works on a wavelength of 1.3 μm. The number of subscribers it can feed if a channel requires 20 kHz are

(a) 2.3×10^{10} (b) 1.15×10^{10}

(c) 1×10^{5} (d) None of these

6. In an FM system a 7 kHz signal modulates 108 MHz carrier so that frequency deviation is 50 kHz. The carrier swing is

(a) 7.143 (b) 8

(c) 0.71 (d) 350

7. In a radio receiver, the short wave and medium wave stations are tuned by using the same capacitor but coils of different inductance L_s and L_m respectively then

(a) $L_s > L_m$ (b) $L_s < L_m$

(c) $L_s = L_m$ (d) None of these

8. The electron density of E, F_1, F_2 layers of ionosphere is 2×10^{11}, 5×10^{11} and 8×10^{11} m^{-3} respectively. What is the ratio of critical frequency for reflection of radiowaves

(a) 2 : 4 : 3 (b) 4 : 3 : 2

(c) 2 : 3 : 4 (d) 3 : 2 : 4

9. A carrier is simultaneously modulated by two sine waves with modulation indices of 0.4 and 0.3. The resultant modulation index will be

(a) 1.0 (b) 0.7

(c) 0.5 (d) 0.35

10. Mean optical power launched into an 8 km fibre is 120 μW and mean output power is 4 μW, then the overall attenuation is (Given log 30 = 1.477)

(a) 14.77 dB (b) 16.77 dB

(c) 3.01 dB (d) None of these

11. A antenna current of an AM broadcast transmitter modulated by 50% is 11 A. The carrier current is

(a) 10.35 A (b) 9.25 A

(c) 10 A (d) 5.5 A

12. Because of tilting which waves finally disappear

(a) Microwaves (b) Surface waves

(c) Sky waves (d) Space waves

13. A transmitter transmits a power of 10 kW when modulation is 50%. Power of carrier wave is

(a) 5 kW (b) 8.89 kW

(c) 14 kW (d) 5.7 kW

14. A telephone link operating at a central frequency of 10 GHz is established. If 1% of this is available then how many telephone channel can be simultaneously given when each telephone covering a band width of 5 kHz

(a) 2×10^{4} (b) 2×10^{6}

(c) 5×10^{4} (d) 5×10^{6}

AS Answers and Solutions

(SET -28)

1. (b) Maximum distance covered by space wave communication $\sqrt{2Rh} = 62\,km$

Critical frequency = $f_c = 9(N_{\max})^{1/2} \simeq 9\,MHz$

5 MHz < f, sky wave propagation (ionospheric propagation)

2. (a) Using $\frac{1}{f_{carrier}} << RC$

We get time constant, $RC = 1000 \times 10^{-12} = 10^{-9}\,s$

Now $\nu = \frac{1}{T} = \frac{1}{10^{-9}} = 10^9\,Hz$

Thus the value of carrier frequency should be much less than $10^9\,Hz$, say 100 kHz.

3. (a) Using $Z = \sqrt{\frac{L}{C}}$ we get $Z = \sqrt{\frac{0.40 \times 10^{-6}}{10^{-11}}} = 2 \times 10^2\,\Omega$

4. (d) Comparing with standard equation.

5. (b) Optical source frequency $f = \frac{c}{\lambda} = \frac{3 \times 10^8}{1.3 \times 10^{-6}} = 2.3 \times 10^{14}\,Hz$

$\therefore$ Number of channels or subscribers $= \frac{2.3 \times 10^{14}}{20 \times 10^3}$

$= 1.15 \times 10^{10}$

6. (a) Carrier swing $= \frac{\text{Frequency deviation}}{\text{Modulating frequency}} = \frac{50}{7} = 7.143$

7. (b) As $\nu = \frac{c}{\lambda} \Rightarrow \nu_m = \frac{c}{\lambda_m}$ and $\nu_s = \frac{c}{\lambda_s}$

$\because \lambda_m > \lambda_s \Rightarrow \nu_m < \nu_s$

Also $\nu_m = \frac{1}{2\pi\sqrt{L_m C}}$ and $\nu_s = \frac{1}{2\pi\sqrt{L_s C}}$

$\Rightarrow \frac{\nu_m}{\nu_s} = \sqrt{\frac{L_s}{L_m}} \Rightarrow L_s < L_m$.

8. (c) $f_c \propto (N)^{1/2} \Rightarrow (f_c)_E : (f_c)_{F_1} : (f_c)_{F_2}$

$= (2 \times 10^{11})^{1/2} : (5 \times 10^{11})^{1/2} : (8 \times 10^{11})^{1/2}$ = 2 : 3 : 4

9. (c) $m = \sqrt{m_1^2 + m_2^2} = \sqrt{(0.16) + (0.09)} = 0.5$

10. (a) Attenuation $= 10\log\frac{120}{4} = 10\log 30$

$= 10 \times 1.4771 = 14.77\,dB.$

11. (a) $I_{Carrier} = \frac{I_{rms}}{\sqrt{1 + \frac{m_a^2}{2}}} = \frac{11}{\sqrt{1 + \frac{(0.5)^2}{2}}} = 10.35\,A$

12. (b)

13. (b) $P_c = \frac{P}{\left(1 + \frac{m_a^2}{2}\right)} = \frac{10000}{\left(1 + \frac{(0.5)^2}{2}\right)} = \frac{10000}{1.125} = 8.89\,kW$

14. (a) 1% of 10 GHz = $10 \times 10^9 \times \frac{1}{100} = 10^8\,Hz$

Number of channels $= \frac{10^8}{5 \times 10^3} = 2 \times 10^4$

***

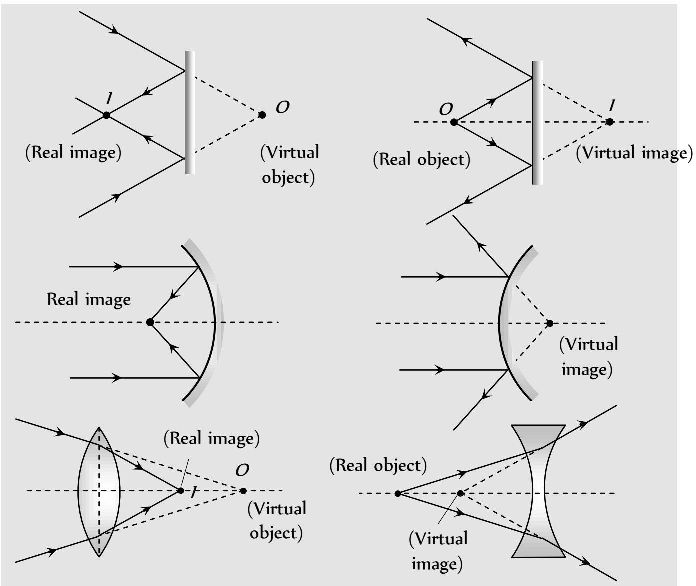
Chapter

29

Ray Optics

Real and Virtual Images

If light rays, after reflection or refraction, actually meets at a point then real image is formed and if they appears to meet virtual image is formed.



Reflection of Light

When a ray of light after incidenting on a boundary separating two media comes back into the same media, then this phenomenon, is called reflection of light.

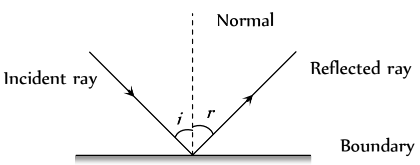


Fig. 29.1

(1) $\angle i = \angle r$

(2) After reflection, velocity, wave length and frequency of light remains same but intensity decreases.

(3) There is a phase change of π if reflection takes place from denser medium.

Reflection From a Plane Surface (Plane Mirror)

The image formed by a plane mirror is virtual, erect, laterally inverted, equal in size that of the object and at a distance equal to the distance of the object in front of the mirror.

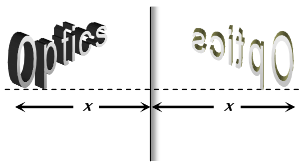


Fig. 29.2

(1) Deviation produced by a plane mirror and by two inclined plane mirrors.

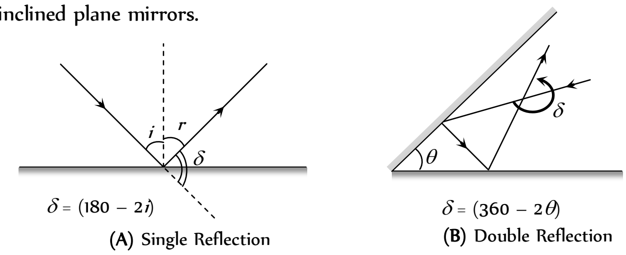


Fig. 29.3

(2) **Images by two inclined plane mirrors :** When two plane mirrors are inclined to each other at an angle θ, then number of images (n) formed of an object which is kept between them.

(i) $n = \left(\dfrac{360^o}{\theta} - 1\right)$; If $\dfrac{360^o}{\theta}$ = even integer

(ii) If $\dfrac{360^o}{\theta}$ = odd integer then there are two possibilities

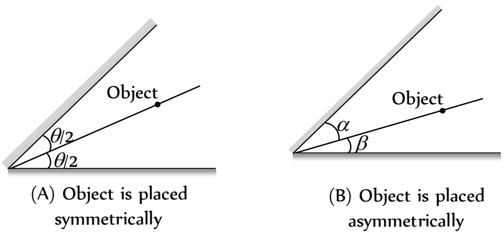


Fig. 29.4

$$n = \left(\frac{360}{\theta} - 1\right) \qquad\qquad n = \frac{360}{\theta}$$

(3) **Other important informations**

(i) When the object moves with speed u towards (or away) from the plane mirror then image also moves towards (or away) with speed u. But relative speed of image *w.r.t.* object is $2u$.

(ii) When mirror moves towards the stationary object with speed u, the image will move with speed $2u$ in same direction as that of mirror.

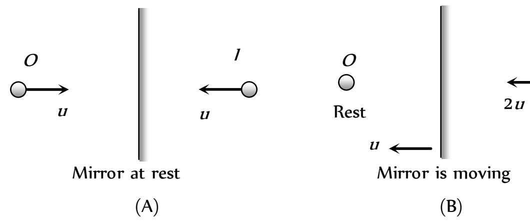


Fig. 29.5

(iii) A man of height h requires a mirror of length at least equal to $h/2$, to see his own complete image.

(iv) To see complete wall behind himself a person requires a plane mirror of at least one third the height of wall. It should be noted that person is standing in the middle of the room.

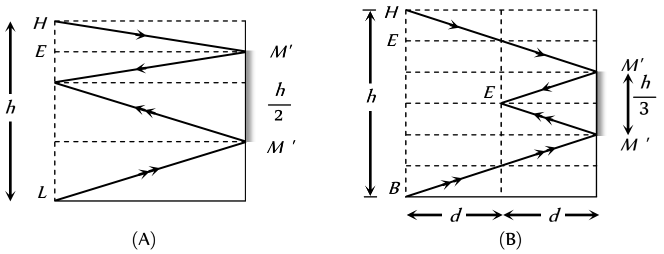


Fig. 29.6

Curved Mirror

It is a part of a transparent hollow sphere whose one surface is polished.

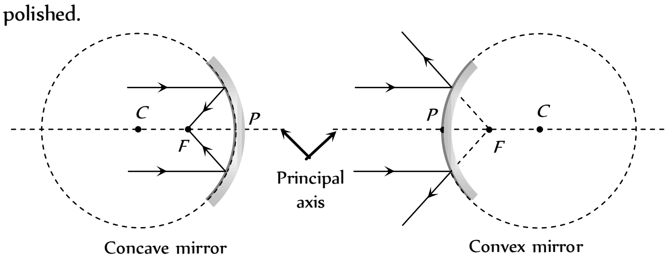


Fig. 29.7

Concave mirror converges the light rays and used as a shaving mirror, In search light, in cinema projector, in telescope, by E.N.T. specialists etc.

Convex mirror diverges the light rays and used in road lamps, side mirror in vehicles *etc*.

(1) **Terminology**

(i) Pole (P) **:** Mid point of the mirror

(ii) Centre of curvature (C) : Centre of the sphere of which the mirror is a part.

(iii) Radius of curvature (R): Distance between pole and centre of curvature. ($R_{concave} = -ve$, $R_{convex} = +ve$, $R_{plane} = \infty$)

(iv) Principle axis : A line passing through P and C.

(v) Focus (F) : An image point on principle axis for an object at ∞.

(vi) Focal length (f) : Distance between P and F.

(vii) Relation between f and R : $f = \frac{R}{2}$

($f_{concave} = -ve$, $f_{convex} = +ve$, $f_{plane} = \infty$)

(viii) Power : The converging or diverging ability of mirror

(ix) Aperture : Effective diameter of light reflecting area. Intensity of image $\propto$ Area $\propto$ (Aperture)^2

(x) Focal plane : A plane passing from focus and perpendicular to principle axis.

(2) **Sign conventions :**

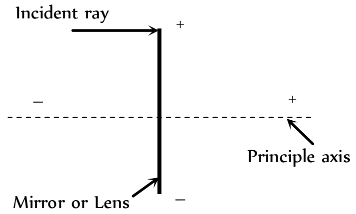


Fig. 29.8

(i) All distances are measured from the pole.

(ii) Distances measured in the direction of incident rays are taken as positive while in the direction opposite of incident rays are taken negative.

(iii) Distances above the principle axis are taken positive and below the principle axis are taken negative.

Table 29.1 : Useful sign

Concave mirror		Convex mirror
Real image ($u \geq f$)	Virtual image ($u< f$)	
Distance of object u $\rightarrow$ –	u $\rightarrow$ –	u $\rightarrow$ –
Distance of image v $\rightarrow$ –	v $\rightarrow$ +	v $\rightarrow$ +
Focal length f $\rightarrow$ –	f $\rightarrow$ –	f $\rightarrow$ +
Height of object O $\rightarrow$ +	O $\rightarrow$ +	O $\rightarrow$ +
Height of image I $\rightarrow$ –	I $\rightarrow$ +	I $\rightarrow$ +
Radius of curvature R $\rightarrow$ –	R $\rightarrow$ –	R $\rightarrow$ +
Magnification m $\rightarrow$ –	m $\rightarrow$ +	m $\rightarrow$ +

Image Formation by Curved Mirrors

Concave mirror : Image formed by concave mirror may be real or virtual, may be inverted or erect, may be smaller, larger or equal in size of object.

(1) When object is placed at infinite (*i.e.* $u = \infty$)

Image

⟶ At F

⟶ Real

⟶ Inverted

⟶ Very small in size

⟶

⟶

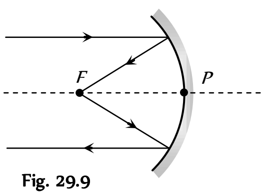


Fig. 29.9

Magnification $m << -1$

(2) When object is placed between infinite and centre of curvature (*i.e.* $u > 2f$)

Image

⟶ Between F and C

⟶ Real

⟶ Inverted

⟶ Small in size

⟶ $m < -1$

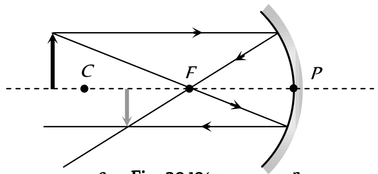


Fig. 29.10

(3) When object is placed at centre of curvature (*i.e.* $u = 2f$)

Image

⟶ At C

⟶ Real

⟶ Inverted

⟶ Equal in size

⟶ $m = -1$

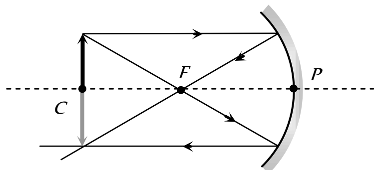


Fig. 29.11

(4) When object is placed between centre of curvature and focus (*i.e.* $f < u < 2f$)

Image

⟶ Between $2f$ and ∞

⟶ Real

⟶ Inverted

⟶ Large in size

⟶ $m > -1$

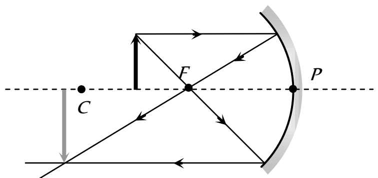


Fig. 29.12

(5) When object is placed at focus (*i.e.* $u = f$)

Image

⟶ At ∞

⟶ Real

⟶ Inverted

⟶ Very large in size

⟶ $m >> -1$

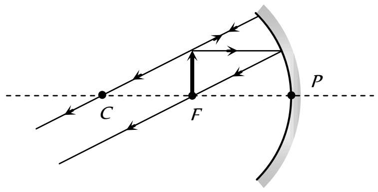


Fig. 29.13

(6) When object is placed between focus and pole (*i.e.* $u < f$)

Image

⟶ Behind the mirror

⟶ Virtual

⟶ Erect

⟶ Large in size

⟶ $m > +1$

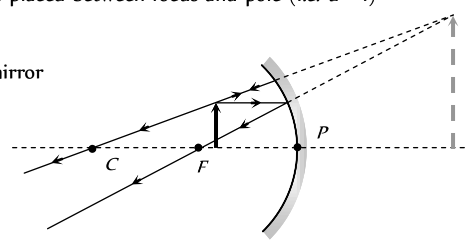


Fig. 29.14

Convex mirror : Image formed by convex mirror is always virtual, erect and smaller in size.

(1) When object is placed at infinite (*i.e.* $u = \infty$)

Image

⟶ At F

⟶ Virtual

⟶ Erect

⟶ Very small in size

⟶ Magnification $m << +1$

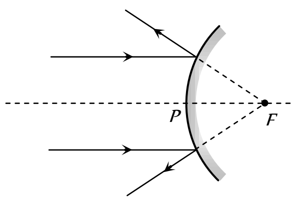


Fig. 29.15

(2) When object is placed any where on the principal axis

Image

⟶ Between P and F

⟶ Virtual

⟶ Erect

⟶ Small in size

⟶ Magnification $m < +1$

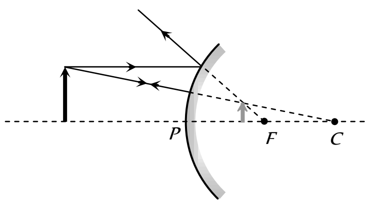


Fig. 29.16

Mirror Formula and Magnification

For a spherical mirror if u = Distance of object from pole, v = distance of image from pole, f = Focal length, R = Radius of curvature, O = Size of object, I = size of image

(1) **Mirror formula :** $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

(2) **Lateral magnification :** When an object is placed perpendicular to the principle axis, then linear magnification is called lateral or transverse magnification.

$$m = \frac{I}{O} = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f}$$

(* Always use sign convention while solving the problems)

Axial magnification : When object lies along the principle axis then its axial magnification $m = \frac{I}{O} = \frac{-(v_2 - v_1)}{(u_2 - u_1)}$

If object is small; $m = -\frac{dv}{du} = \left(\frac{v}{u}\right)^2 = \left(\frac{f}{f-u}\right)^2 = \left(\frac{f-v}{f}\right)^2$

Areal magnification : If a $2D$-object is placed with it's plane perpendicular to principle axis. It's Areal magnification

$$m_s = \frac{\text{Area of image } (A_i)}{\text{Area of object} (A_o)} \Rightarrow m_s = m^2 = \frac{A_i}{A_o}$$

Refraction of Light

The bending of the ray of light passing from one medium to the other medium is called refraction.

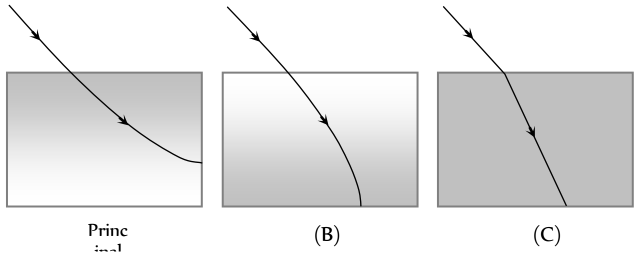


Fig. 29.17

(1) The refraction of light takes place on going from one medium to another because the speed of light is different in the two media.

(2) Greater the difference in the speeds of light in the two media, greater will be the amount of refraction.

(3) A medium in which the speed of light is more is known as optically rarer medium and a medium is which the speed of light is less, is known as optically denser medium.

(4) When a ray of light goes from a rarer medium to a denser medium, it bends towards the normal.

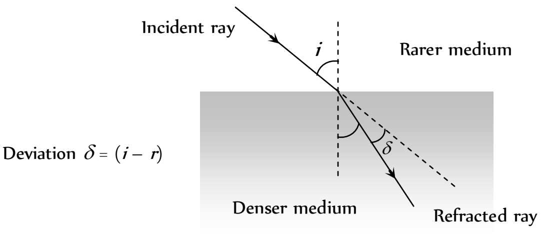


Fig. 29.18

(5) When a ray of light goes from a denser medium to a rarer medium, it bends away from the normal.

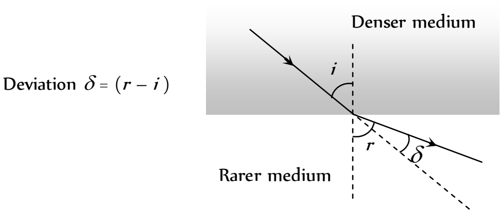


Fig. 29.19

(6) **Snell's law :** The ratio of sine of the angle of incidence to the angle of refraction (r) is a constant called refractive index

i.e. $\frac{\sin i}{\sin r} = \mu$ (a constant). For two media, Snell's law can be written as ${}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{\sin i}{\sin r}$

$\Rightarrow \ \mu_1 \times \sin i = \mu_2 \times \sin r$ *i.e.* $\mu \sin\theta =$ constant

Also in vector form : $\hat{\boldsymbol{i}} \times \hat{\boldsymbol{n}} = \mu(\hat{\boldsymbol{r}} \times \hat{\boldsymbol{n}})$

Refractive Index

(1) Refractive index of a medium is that characteristic which decides speed of light in it.

(2) It is a scalar, unit less and dimensionless quantity.

(3) **Absolute refractive index :** When light travels from vacuum to any transparent medium then refractive index of medium *w.r.t.* vacuum is called it's absolute refractive index *i.e.* ${}_{\text{vacuum}}\mu_{\text{medium}} = \frac{c}{v}$

Absolute refractive indices for glass, water and diamond are respectively $\mu_g = \frac{3}{2} = 1.5$, $\mu_w = \frac{4}{3} = 1.33$ and $\mu_D = \frac{12}{5} = 2.4$

(4) **Relative refractive index :** When light travels from medium (1) to medium (2) then refractive index of medium (2) *w.r.t.* medium (1) is called it's relative refractive index *i.e.* ${}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2}$ (where v_1 and v_2 are the speed of light in medium 1 and 2 respectively).

(5) When we say refractive index we mean absolute refractive index.

(6) The minimum value of absolute refractive index is 1. For air it is very near to 1. ($\cong 1.003$)

(7) Cauchy's equation : $\mu = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} + ...$

($\lambda_{\text{Red}} > \lambda_{\text{violet}}$ so $\mu_{\text{Red}} < \mu_{\text{violet}}$)

(8) If a light ray travels from medium (1) to medium (2), then

$${}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2}$$

(9) **Dependence of Refractive index**

(i) Nature of the media of incidence and refraction.

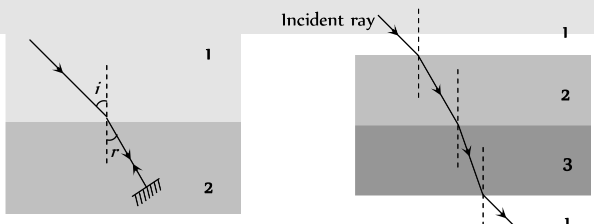
(ii) Colour of light or wavelength of light.

(iii) Temperature of the media : Refractive index decreases with the increase in temperature.

Table 29.2 : Indices of refraction for various substances, Measured with light of vacuum wavelength λ_0 = 589 *nm*

Substance	Refractive index	Substance	Refractive index
Solids at 20°*C*		**Liquids at 20°*C***	
Diamond (C)	2.419	Benzene	1.501
Fluorite (CaF_2)	1.434	Carbon disulfide	1.628
Flused quartz (SiO_2)	1.458	Carbon tetrachloride	1.461
Glass, crown	1.52	Ethyl alcohol	1.361
Glass, flint	1.66	Glycerine	1.473
Ice (H_2O) (at 0°*C*)	1.309	Water	1.333
Polystyrene	1.49	**Gases at 0°*C*, 1 *atm***	
Sodium chloride	1.544	Air	1.000293
Zircon	1.923	Carbon dioxide	1.00045

(10) **Reversibility of light and refraction through several media**



(A) ${}_1\mu_2 = \frac{1}{{}_2\mu_1}$

(B) ${}_1\mu_2 \times {}_2\mu_3 \times {}_3\mu_1 = 1$

or ${}_2\mu_3 = \frac{{}_1\mu_3}{{}_1\mu_2}$

Fig. 29.20

Real and Apparent Depth

If object and observer are situated in different medium then due to refraction, object appears to be displaced from it's real position.

(1) **When object is in denser medium and observer is in rarer medium**

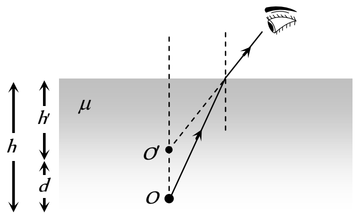


Fig. 29.21

(i) $\mu = \dfrac{\text{Real depth}}{\text{Apparent depth}} = \dfrac{h}{h'}$

(ii) Real depth > Apparent depth

(iii) Shift $d = h - h' = \left(1 - \dfrac{1}{\mu}\right)h$. For water $\mu = \dfrac{4}{3} \Rightarrow d = \dfrac{h}{4}$;

For glass $\mu = \dfrac{3}{2} \Rightarrow d = \dfrac{h}{3}$

(iv) Lateral magnification : consider an object of height x placed vertically in a medium μ such that the lower end (B) is a distance h from the interface and the upper end (A) at a distance ($h - x$) from the interface.

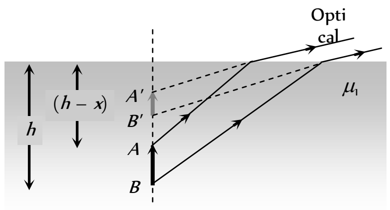


Fig. 29.22

Distance of image of B (*i.e.* B') from the interface = $\dfrac{\mu_2}{\mu_1}h$

Distance of image of A (*i.e.* A') from the interface $= \dfrac{\mu_2}{\mu_1}(h - x)$

Therefore, length of the image $\dfrac{\mu_2}{\mu_1}x$

or, the lateral magnification of the object $m = \dfrac{\mu_2}{\mu_1} = \dfrac{1}{\mu}$

(v) If a beaker contains various immiscible liquids as shown then

Apparent depth of bottom $= \dfrac{d_1}{\mu_1} + \dfrac{d_2}{\mu_2} + \dfrac{d_3}{\mu_3} + ...$

$$\mu = \frac{d_{AC}}{d_{App.}} = \frac{d_1 + d_2 + ...}{\dfrac{d_1}{\mu_1} + \dfrac{d_2}{\mu_2} + ...}$$

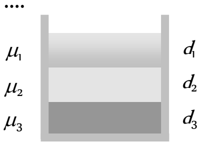
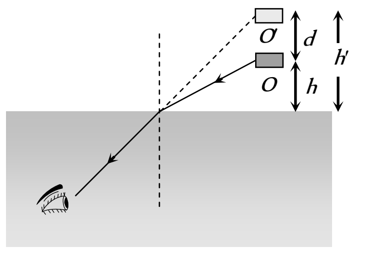


Fig. 29.23

(In case of two liquids if $d_1 = d_2$ than $\mu = \dfrac{2\mu_1\mu_2}{\mu_1 + \mu_2}$)

(2) **Object is in rarer medium and observer is in denser medium**



(i) $\mu = \dfrac{h'}{h}$

(ii) Real depth < Apparent depth.

(iii) $d = (\mu - 1)h$

(iv) Shift for water $d_w = \dfrac{h}{3}$; Shift for glass $d_g = \dfrac{h}{2}$

Refraction Through a Glass Slab

(1) **Lateral shift :** The refracting surfaces of a glass slab are parallel to each other. When a light ray passes through a glass slab it is refracted twice at the two parallel faces and finally emerges out parallel to it's incident direction *i.e.* the ray undergoes no deviation $\delta = 0$. The angle of emergence (e) is equal to the angle of incidence (i)

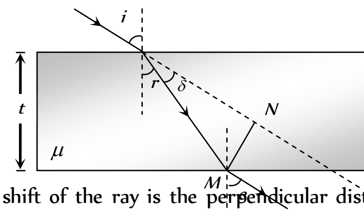


Fig. 29.25

The Lateral shift of the ray is the perpendicular distance between the incident and the emergent ray, and it is given by

$MN = t \sec r \sin (i - r)$

(2) **Normal shift :** If a glass slab is placed in the path of a converging or diverging beam of light then point of convergence or point of divergence appears to be shifted as shown

Normal shift

$OO' = x = \left(1 - \dfrac{1}{\mu}\right)t$

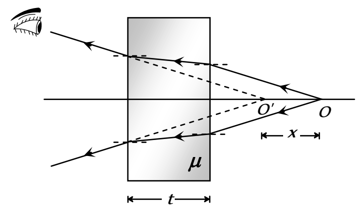


Fig. 29.26

(3) **Optical path :** It is defined as distance travelled by light in vacuum in the same time in which it travels a given path length in a medium.

Time taken by light ray to pass through the medium $= \dfrac{\mu x}{c}$; where x = geometrical path and μx = optical path

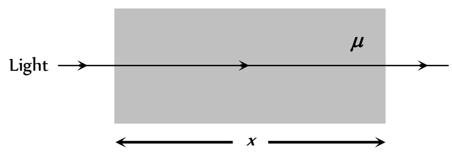
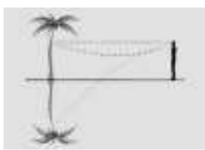


Fig. 29.27

Total Internal Reflection (TIR)


When a ray of light goes from denser to rarer medium it bends away from the normal and as the angle of incidence in denser medium increases, the angle of refraction in rarer medium also increases and at a certain angle, angle of refraction becomes 90°, this angle of incidence is called critical angle (C).

When Angle of incidence exceeds the critical angle than light ray comes back in to the same medium after reflection from interface. This phenomenon is called Total internal reflection (TIR).

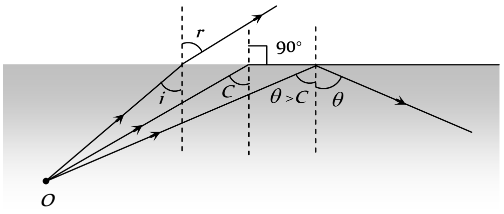


Fig. 29.28

(1) $\mu = \dfrac{1}{\sin C} = \operatorname{cosec} C$ where $\mu \rightarrow {}_{\text{Rarer}}\mu_{\text{Denser}}$

(2) **Conditions for TIR**

(i) The ray must travel from denser medium to rarer medium.

(ii) The angle of incidence i must be greater than critical angle C

(3) **Dependence of critical angle**

(i) Colour of light (or wavelength of light) : Critical angle depends upon wavelength as $\lambda \propto \dfrac{1}{\mu} \propto \sin C$

(a) $\lambda_R > \lambda_V \Rightarrow C_R > C_V$

(b) Sin $C = \dfrac{1}{{}_R\mu_D} = \dfrac{\mu_R}{\mu_D} = \dfrac{\lambda_D}{\lambda_R} = \dfrac{v_D}{v_R}$ (for two media)

(ii) Nature of the pair of media : Greater the refractive index lesser will be the critical angle.

(a) For (glass- air) pair $\rightarrow C_{\text{glass}} = 42^o$

(b) For (water-air) pair $\rightarrow C_{\text{water}} = 49^o$

(c) For (diamond-air) pair $\rightarrow C_{\text{diamond}} = 24^o$

(iii) Temperature : With temperature rise refractive index of the material decreases therefore critical angle increases.

Common Examples of TIR

(1) **Looming :** An optical illusion in cold countries

(2) **Mirage :** An optical illusion in deserts

Fig. 29.29

(3) **Brilliance of diamond :** Due to repeated internal reflections diamond sparkles.

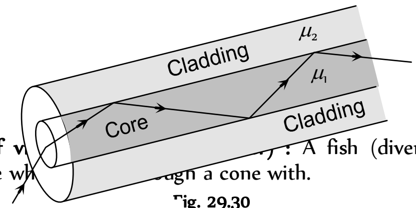
(4) **Optical fibre :** Optical fibres consist of many long high quality composite glass/quartz fibres. Each fibre consists of a core and cladding.

(i) The refractive index of the material of the core (μ) is higher than that of the cladding (μ).

(ii) When the light is incident on one end of the fibre at a small angle, the light passes inside, undergoes repeated total internal reflections along the fibre and finally comes out. The angle of incidence is always larger than the critical angle of the core material with respect to its cladding.

(iii) Even if the fibre is bent, the light can easily travel through along the fibre

(iv) A bundle of optical fibres can be used as a 'light pipe' in medical and optical examination. It can also be used for optical signal transmission. Optical fibres have also been used for transmitting and receiving electrical signals which are converted to light by suitable transducers.



(5) **Field of v**... : A fish (diver) inside the water can see the wh... ugh a cone with.

Fig. 29.30

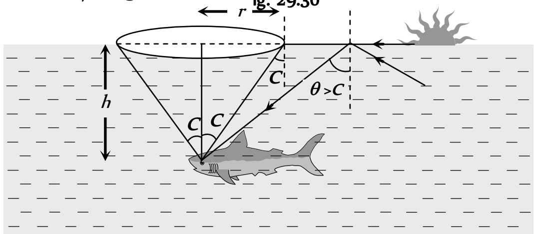


Fig. 29.31

(a) Apex angle $= 2C = 98^o$

(b) Radius of base $r = h\tan C = \dfrac{h}{\sqrt{\mu^2 - 1}}$; for water $r = \dfrac{3h}{\sqrt{7}}$

(c) Area of base $A = \dfrac{\pi h^2}{(\mu^2 - 1)}$; for water $a = \dfrac{9\pi}{7}h^2$

(6) **Porro prism :** A right angled isosceles prism, which is used in periscopes or binoculars. It is used to deviate light rays through 90^o and 180^o and also to erect the image.

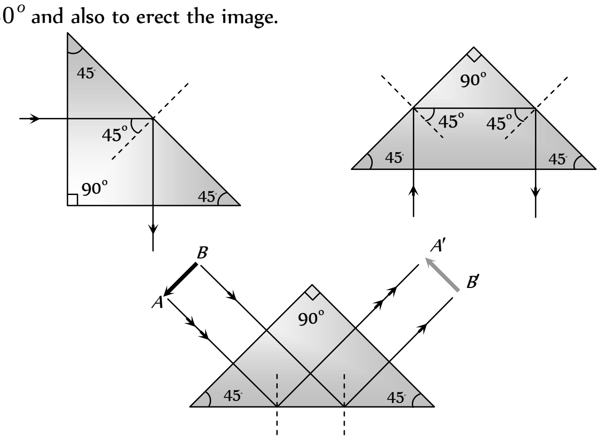


Fig. 29.32

Refraction From Spherical Surface

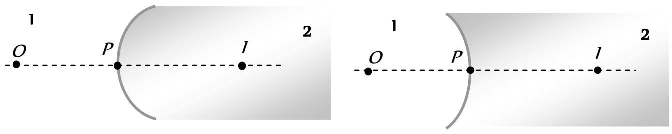


Fig. 29.33

(1) **Refraction formula :** $\dfrac{\mu_2 - \mu_1}{R} = \dfrac{\mu_2}{v} - \dfrac{\mu_1}{u}$

Where μ_1 = Refractive index of the medium from which light rays are coming (from object).

μ_2 = Refractive index of the medium in which light rays are entering.

u = Distance of object, v = Distance of image, R = Radius of curvature

(2) **Lateral magnification :** The lateral magnification m is the ratio of the image height to the object height

or $m = \left(\dfrac{h_i}{h_0}\right) = \left(\dfrac{\mu_1}{\mu_2}\right)\left(\dfrac{v}{u}\right)$

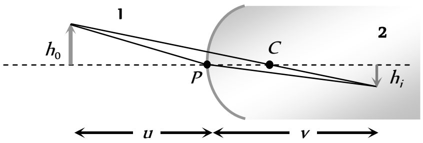


Fig. 29.34

Lens

(1) Lens is a transparent medium bounded by two refracting surfaces, such that at least one surface is curved. Curved surface can be spherical, cylindrical etc.

(2) Lenses are of two basic types convex which are thicker in the middle than at the edges and concave for which the reverse holds.

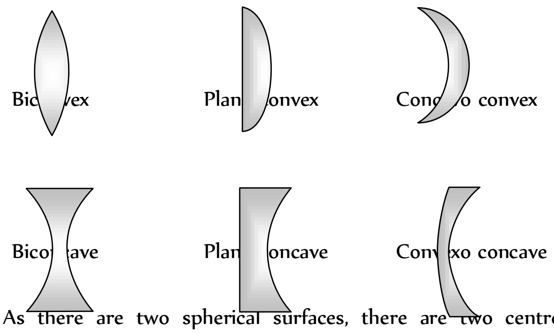


Fig. 29.35

(3) As there are two spherical surfaces, there are two centres of curvature C_1 and C_2 and correspondingly two radii of curvature R_1 and R_2

(4) The line joining C_1 and C_2 is called the principal axis of the lens. The centre of the thin lens which is on the principal axis, is called the optical centre.

(5) A ray passing through optical centre proceeds undeviated through the lens.

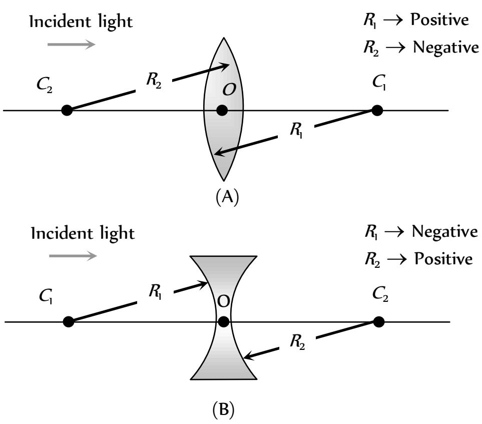


Fig. 29.36

(6) **Principal focus :** We define two principal focus for the lens. We are mainly concerned with the second principal focus (F). Thus wherever we write the focus, it means the second principal focus.

First principal focus : An object point for which image is formed at infinity.

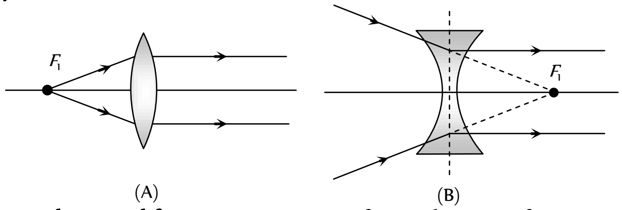


Fig. 29.37

Second principal focus : An image point for an object at infinity.

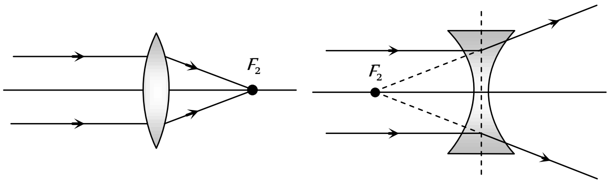


Fig. 29.38

Focal Length, Power and Aperture of Lens

(1) **Focal length (f) :** Distance of second principle focus from optical centre is called focal length

$f_{\text{convex}} \rightarrow$ positive, $f_{\text{concave}} \rightarrow$ negative, $f_{\text{plane}} \rightarrow \infty$

(2) **Aperture :** Effective diameter of light transmitting area is called aperture. $\text{Intensity of image} \propto (\text{Aperture})^2$

(3) **Power of lens (P) :** Means the ability of a lens to deviate the path of the rays passing through it. If the lens converges the rays parallel to the principal axis its power is positive and if it diverges the rays it is negative.

Power of lens $P = \dfrac{1}{f(m)} = \dfrac{100}{f(cm)}$; Unit of power is Diopter (D)

$P_{\text{convex}} \rightarrow \text{positive}, P_{\text{concave}} \rightarrow \text{negative}, P_{\text{plane}} \rightarrow \text{zero}$.

Rules of Image Formation by Lens

Convex lens : The image formed by convex lens depends on the position of object.

(1) When object is placed at infinite (*i.e.* $u = \infty$)

Image

$\longrightarrow$ At F

$\longrightarrow$ Real

$\longrightarrow$ Inverted

$\longrightarrow$ Very small in size

$\longrightarrow$ Magnification $m << -1$

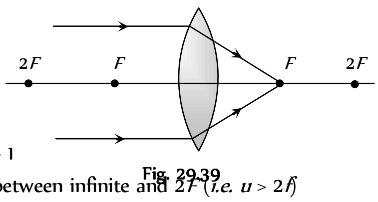


Fig. 29.39

(2) When object is placed between infinite and $2F$ (*i.e.* $u > 2f$)

Image

$\longrightarrow$ Between F and $2F$

$\longrightarrow$ Real

$\longrightarrow$ Inverted

$\longrightarrow$ Very small in size

$\longrightarrow$ Magnification $m < -1$

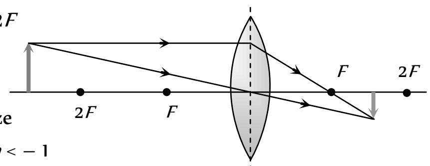


Fig. 29.40

(3) When object is placed at $2F$ (*i.e.* $u = 2f$)

Image

$\longrightarrow$ At $2F$

$\longrightarrow$

$\longrightarrow$

$\longrightarrow$

$\longrightarrow$

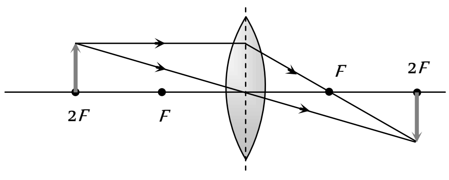


Fig. 29.41

- Real
- Inverted
- Equal in size
- Magnification $m = -1$

(4) When object is placed between F and $2F$ (*i.e.* $f < u < 2f$)

Image

- Beyond $2F$
- Real
- Inverted
- Large in size
- Magnification $m > -1$

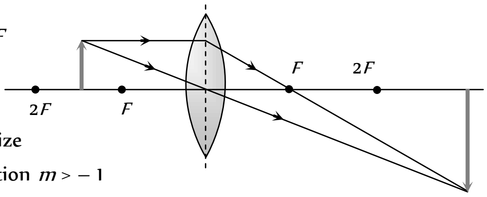


Fig. 29.42

(5) When object is placed at F (*i.e.* $u = f$)

Image

- At ∞
- Real
- Inverted
- Very large in size
- Magnification $m >> -1$

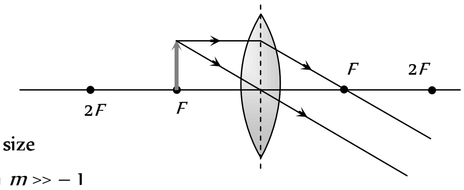


Fig. 29.43

(6) When object is placed between F and optical center (*i.e.* $u < f$)

Image

- Same side as that of object
- Virtual
- Erect
- large in size
- Magnification $m > 1$

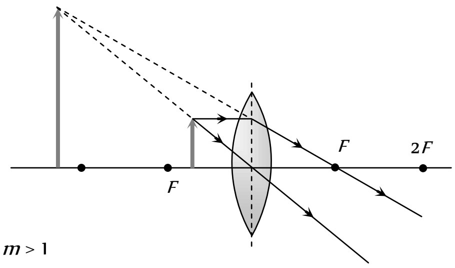


Fig. 29.44

Concave lens : The image formed by a concave lens is always virtual, erect and diminished (like a convex mirror)

(1) When object is placed at ∞

Image

- At F
- Virtual
- Erect
- Point size
- Magnification $m << +1$

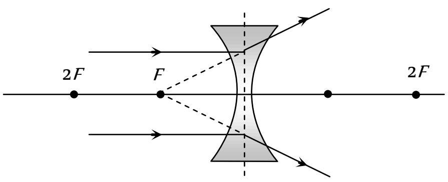


Fig. 29.45

(2) When object is placed any where on the principal axis

Image

- Between optical centre and focus
- Virtual
- Erect
- Smaller in size
- Magnification $m < +1$

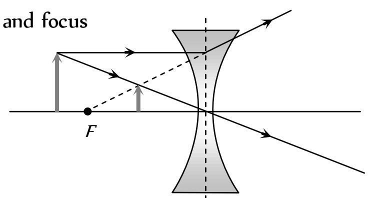


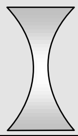
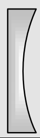
Fig. 29.46

Lens Maker's Formula and Lens Formula

(1) **Lens maker's formula :** If R_1 and R_2 are the radii of curvature of first and second refracting surfaces of a thin lens of focal length f and refractive index μ (*w.r.t.* surrounding medium) then the relation between f, μ, R_1 and R_2 is known as lens maker's formula.

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Table 29.3 : Focal length of different lenses

Lens	Focal length	For $\mu = 1.5$
Biconvex lens $R_1 = R$ $R_2 = -R$ 	$f = \frac{R}{2(\mu - 1)}$	$f = R$
Plano-convex lens $R_1 = \infty$ $R_2 = -R$ 	$f = \frac{R}{(\mu - 1)}$	$f = 2R$
Biconcave $R_1 = -R$ $R_2 = +R$ 	$f = -\frac{R}{2(\mu - 1)}$	$f = -R$
Plano-concave $R_1 = \infty$ $R_2 = R$ 	$f = \frac{-R}{(\mu - 1)}$	$f = -2R$

(2) **Lens formula :** The expression which shows the relation between u, v and f is called lens formula.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Magnification

The ratio of the size of the image to the size of object is called magnification.

(1) Transverse magnification : $m = \frac{I}{O} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$ (use sign convention while solving the problem)

(2) Longitudinal magnification : $m = \frac{I}{O} = \frac{v_2 - v_1}{u_2 - u_1}$. For very small object $m = \frac{dv}{du} = \left(\frac{v}{u}\right)^2 = \left(\frac{f}{f+u}\right)^2 = \left(\frac{f-v}{f}\right)^2$

(3) Areal magnification : $m_s = \frac{A_i}{A_o} = m^2 = \left(\frac{f}{f+u}\right)^2$,

(A_i = Area of image, A_o = Area of object)

(4) **Relation between object and image speed :** If an object moves with constant speed (V_o) towards a convex lens from infinity to focus, the image will move slower in the beginning and then faster. Also $V_i = \left(\frac{f}{f+u}\right)^2 . V_o$

Newton's Formula

If the distance of object (x) and image (x) are not measured from optical centre, but from first and second principal foci then Newton's formula states $f^2 = x_1 x_2$

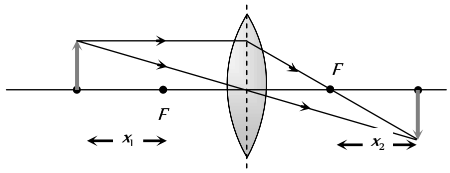


Fig. 29.47

Lens Immersed in a Liquid

If a lens (made of glass) of refractive index μ is immersed in a liquid of refractive index μ_l then its focal length in liquid, f_l is given by

$$\frac{1}{f_l} = ({}_l\mu_g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{......(i)}$$

If f_a is the focal length of lens in air, then

$$\frac{1}{f_a} = ({}_a\mu_g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{......(ii)}$$

$$\Rightarrow \quad \frac{f_l}{f_a} = \frac{({}_a\mu_g - 1)}{({}_l\mu_g - 1)}$$

(1) If $\mu_g > \mu_l$, then f_l and f_a are of same sign and $f_l > f_a$.

That is the nature of lens remains unchanged, but it's focal length increases and hence power of lens decreases.

(2) If $\mu_g = \mu_l$, then $f_l = \infty$. It means lens behaves as a plane glass plate and becomes invisible in the medium.

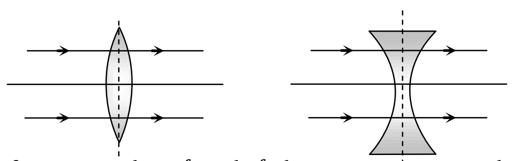


Fig. 29.48

(3) If $\mu_g < \mu_l$, then f_l and f_a have opposite signs and the nature of lens changes *i.e.* a convex lens diverges the light rays and concave lens converges the light rays.

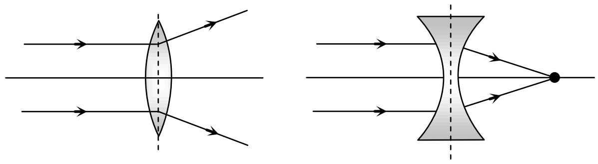


Fig. 29.49

Displacement Method

By this method focal length of convex lens is determined.

Consider an object and a screen placed at a distance D (> $4f$) apart. Let a lens of focal length f be placed between the object and the screen.

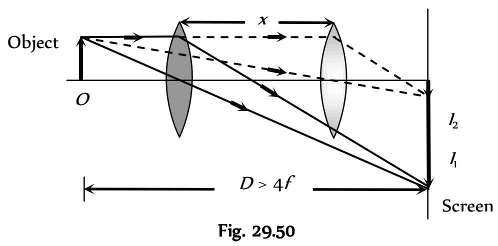


Fig. 29.50

(1) For two different positions of lens two images (I_1 and I_2) of an object are formed at the screen.

(2) Focal length of the lens $f = \frac{D^2 - x^2}{4D} = \frac{x}{m_1 - m_2}$

where $m_1 = \frac{I_1}{O}$; $m_2 = \frac{I_2}{O}$ and $m_1 m_2 = 1$.

(3) Size of object $O = \sqrt{I_1 . I_2}$

Cutting of Lens

(1) A symmetric lens is cut along optical axis in two equal parts. Intensity of image formed by each part will be same as that of complete lens. Focal length is double the original for each part.

(2) A symmetric lens is cut along principle axis in two equal parts. Intensity of image formed by each part will be less compared as that of complete lens.(aperture of each part is $\frac{1}{\sqrt{2}}$ times that of complete lens). Focal length remains same for each part.

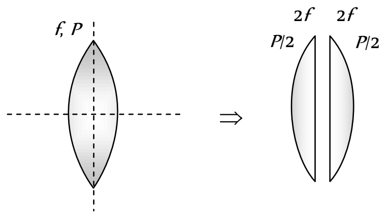


Fig. 29.51

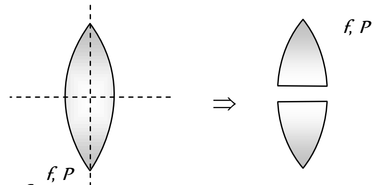


Fig. 29.52

Combination of Lens

(1) For a system of lenses, the net power, net focal length and magnification are given as follows :

$$P = P_1 + P_2 + P_3 ...., \qquad \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + ...,$$

$$m = m_1 \times m_2 \times m_3 \times ....$$

(2) In case when two thin lens are in contact : Combination will behave as a lens, which have more power or lesser focal length.

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \Rightarrow F = \frac{f_1 f_2}{f_1 + f_2} \quad \text{and} \quad P = P_1 + P_2$$

(3) If two lens of equal focal length but of opposite nature are in contact then combination will behave as a plane glass plate and $F_{\text{combination}} = \infty$

(4) When two lenses are placed co-axially at a distance d from each other then equivalent focal length (F).

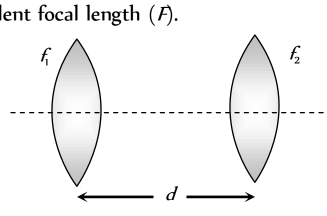


Fig. 29.53

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}-\frac{d}{f_1 f_2} \text{ and } P=P_1+P_2-dP_1P_2$$

(5) Combination of parts of a lens :

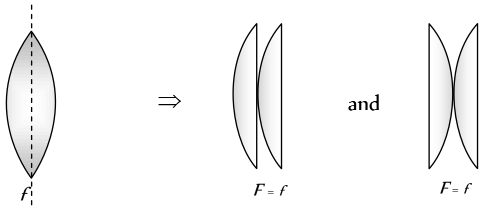


Fig. 29.54

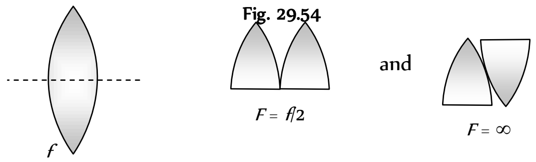


Fig. 29.55

Silvering of Lens

On silvering the surface of the lens it behaves as a mirror. The focal length of the silvered lens is $\frac{1}{F}=\frac{2}{f_l}+\frac{1}{f_m}$

where $f_l =$ focal length of lens from which refraction takes place (twice)

$f_m =$ focal length of mirror from which reflection takes place.

(i) Plano convex is silvered

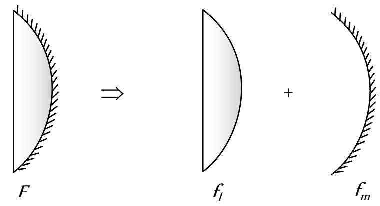


Fig. 29.56

$$f_m=\frac{R}{2},\ f_l=\frac{R}{(\mu-1)} \text{ so } F=\frac{R}{2\mu}$$

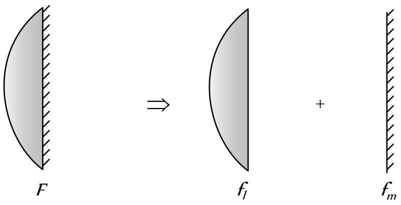


Fig. 29.57

$$f_m=\infty,\ f_l=\frac{R}{(\mu-1)} \text{ so } F=\frac{R}{2(\mu-1)}$$

(ii) Double convex lens is silvered

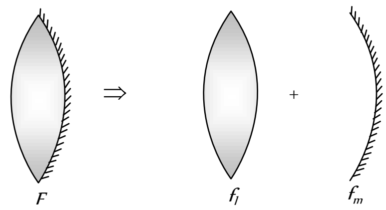


Fig. 29.58

Since $f_l=\frac{R}{2(\mu-1)}, f_m=\frac{R}{2}$ so $F=\frac{R}{2(2\mu-1)}$

Defects in Lens

(1) **Chromatic aberration :** Image of a white object is coloured and blurred because μ (hence f) of lens is different for different colours. This defect is called chromatic aberration.

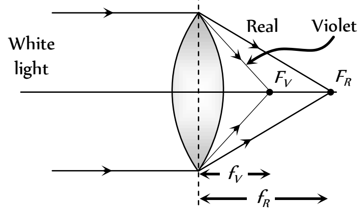


Fig. 29.59

$\mu_V > \mu_R$ so $f_R > f_V$

Mathematically chromatic aberration = $f_R - f_V = \omega f_y$

ω = Dispersive power of lens.

f = Focal length for mean colour $=\sqrt{f_R f_V}$

Removal : To remove this defect *i.e.* for Achromatism we use two or more lenses in contact in place of single lens.

Mathematically condition of Achromatism is : $\frac{\omega_1}{f_1}+\frac{\omega_2}{f_2}=0$ or $\omega_1 f_2=-\omega_2 f_1$

(2) **Spherical aberration :** Inability of a lens to form the point image of a point object on the axis is called Spherical aberration.

In this defect all the rays passing through a lens are not focussed at a single point and the image of a point object on the axis is blurred.

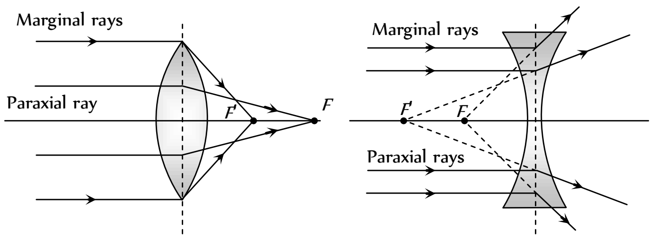


Fig. 29.60

Removal : A simple method to reduce spherical aberration is to use a stop before and infront of the lens. (but this method reduces the intensity of the image as most of the light is cut off). Also by using plano-convex lens, using two lenses separated by distance $d = F - F'$, using crossed lens.

(3) **Coma :** When the point object is placed away from the principle axis and the image is received on a screen perpendicular to the axis, the shape of the image is like a comet. This defect is called Coma.

It refers to spreading of a point object in a plane $\perp$ to principle axis.

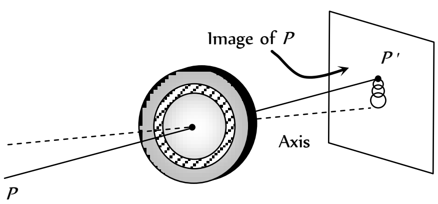


Fig. 29.61

Removal : It can be reduced by properly designing radii of curvature of the lens surfaces. It can also be reduced by appropriate stops placed at appropriate distances from the lens.

(4) **Curvature :** For a point object placed off the axis, the image is spread both along and perpendicular to the principal axis. The best image

is, in general, obtained not on a plane but on a curved surface. This defect is known as Curvature.

Removal : Astigmatism or the curvature may be reduced by using proper stops placed at proper locations along the axis.

(5) **Distortion :** When extended objects are imaged, different portions of the object are in general at different distances from the axis. The magnification is not the same for all portions of the extended object. As a result a line object is not imaged into a line but into a curve.

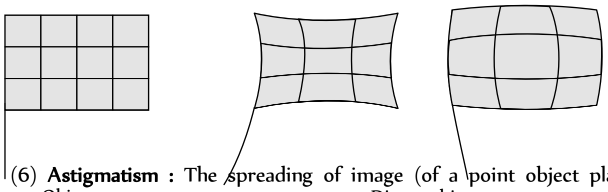


Fig. 29.62

(6) **Astigmatism :** The spreading of image (of a point object placed away from the principal axis) along the principal axis is called Astigmatism.

Prism

Prism is a transparent medium bounded by refracting surfaces, such that the incident surface (on which light ray is incidenting) and emergent surface (from which light rays emerges) are plane and non parallel.

(1) **Refraction through a prism**

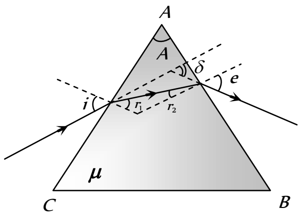


Fig. 29.63

i – Angle of incidence,
e – Angle of emergence,
A – Angle of prism or refracting angle of prism,
r_1 and r_2 – Angle of refraction,
δ – Angle of deviation

$A = r_1 + r_2$ and $i + e = A + \delta$

For surface AC $\mu = \frac{\sin i}{\sin r_1}$; For surface AB $\frac{1}{\mu} = \frac{\sin r_2}{\sin e}$

(2) **Deviation through a prism :** For thin prism $\delta = (\mu - 1)A$. Also deviation is different for different colour light *e.g.* $\mu_R < \mu_V$ so $\delta_R < \delta_V$.

$\mu_{\text{Flint}} > \mu_{\text{Crown}}$ so $\delta_F > \delta_C$

(i) Maximum deviation : Condition of maximum deviation is $\angle i = 90^o \Rightarrow r_1 = C,\ r_2 = A - C$ and from Snell's law on emergent surface

$$e = \sin^{-1}\left[\frac{\sin(A-C)}{\sin C}\right]$$

$$\delta_{\max} = \frac{\pi}{2} + \sin^{-1}\left[\frac{\sin(A-C)}{\sin C}\right] - A$$

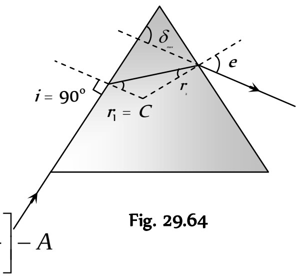


Fig. 29.64

(ii) Minimum deviation : It is observed if $\angle i = \angle e$ and $\angle r_1 = \angle r_2 = r$, deviation produced is minimum.

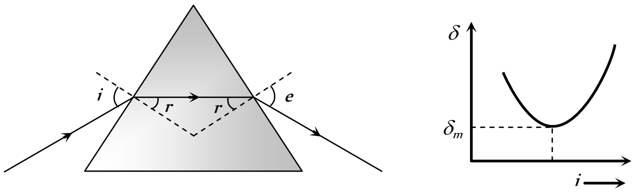


Fig. 29.65

(a) Refracted ray inside the prism is parallel to the base of the prism for equilateral and isosceles prisms.

(b) $r = \frac{A}{2}$ and $i = \frac{A + \delta_m}{2}$

(c) $\mu = \frac{\sin i}{\sin A/2}$ or $\mu = \frac{\sin \frac{A+\delta_m}{2}}{\sin A/2}$ (Prism formula).

(3) **Condition of no emergence :** For no emergence of light, TIR must takes place at the second surface

For TIR at second surface $r_2 > C$

So $A > r_1 + C$ (From $A = r_1 + r_2$)

As maximum value of $r_1 = C$

So, $A \geq 2C$. for any angle of incidence.

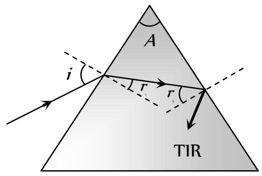


Fig. 29.66

If light ray incident normally on first surface *i.e.* $\angle i = 0°$ it means $\angle r_1 = 0°$. So in this case condition of no emergence from second surface is $A > C$.

$$\Rightarrow \sin A > \sin C \Rightarrow \sin A > \frac{1}{\mu} \Rightarrow \mu > \text{cosec } A$$

Dispersion Through a Prism

The splitting of white light into it's constituent colours is called dispersion of light.

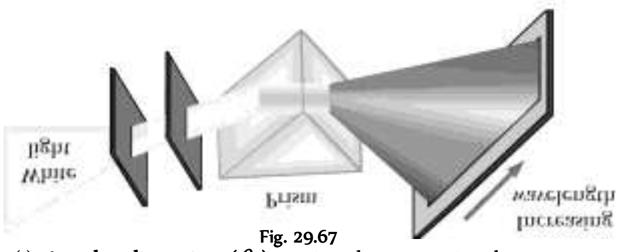


Fig. 29.67

(1) **Angular dispersion (θ) :** Angular separation between extreme colours *i.e.* $\theta = \delta_V - \delta_R = (\mu_V - \mu_R)A$. It depends upon μ and *A*.

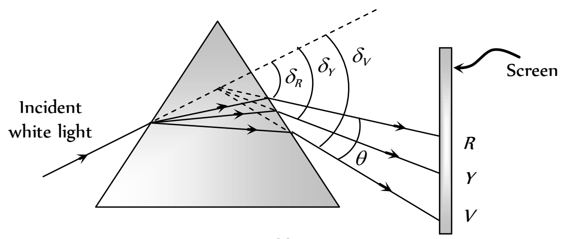


Fig. 29.68

(2) **Dispersive power (ω) :**

$$\omega = \frac{\theta}{\delta_y} = \frac{\mu_V - \mu_R}{\mu_y - 1} \quad \text{where } \left\{\mu_y = \frac{\mu_V + \mu_R}{2}\right\}$$

$\Rightarrow$ It depends only upon the material of the prism *i.e.* μ and it doesn't depends upon angle of prism *A*

(3) **Combination of prisms :** Two prisms (made of crown and flint material) are combined to get either dispersion only or deviation only.

(i) Dispersion without deviation (chromatic combination)

$$\frac{A'}{A} = -\frac{(\mu_y - 1)}{(\mu'_y - 1)}$$

$$\theta_{net} = \theta\left(1 - \frac{\omega'}{\omega}\right) = (\omega\delta - \omega'\delta')$$

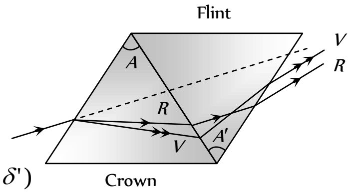


Fig. 29.69

(ii) Deviation without dispersion (Achromatic combination)

$$\frac{A'}{A} = -\frac{(\mu_V - \mu_R)}{(\mu'_V - \mu'_R)}$$

$$\delta_{net} = \delta\left(1 - \frac{\omega}{\omega'}\right)$$

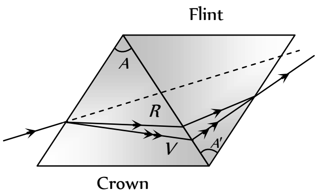


Fig. 29.70

Scattering of Light

Molecules of a medium after absorbing incoming light radiations, emits them in all direction. This phenomenon is called Scattering.

(1) **According to scientist Rayleigh :** Intensity of scattered light $\propto \frac{1}{\lambda^4}$

(2) **Some phenomenon based on scattering :** (i) Sky looks blue due to scattering.

(ii) At the time of sunrise or sunset sun looks reddish.

(iii) Danger signals are made of red colour.

(3) **Elastic scattering :** When the wavelength of radiation remains unchanged, the scattering is called elastic.

(4) **Inelastic scattering (Raman's effect) :** Under specific condition, light can also suffer inelastic scattering from molecules in which it's wavelength changes.

Rainbow

Rainbow is formed due to the dispersion of light suffering refraction and TIR in the droplets present in the atmosphere. Observer should stand with its back towards sun to observe rainbow.

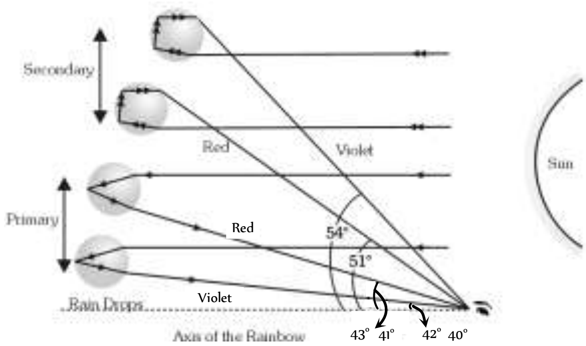


Fig. 29.71

(1) **Primary rainbow :** (i) Two refraction and one TIR. (ii) Innermost arc is violet and outermost is red. (iii) Subtends an angle of 42^o at the eye of the observer. (iv) More bright

(2) **Secondary rainbow :** (i) Two refraction and two TIR. (ii) Innermost arc is red and outermost is violet. (iii) It subtends an angle of 52.5^o at the eye. (iv) Comparatively less bright.

Colours of Objects

Colour is defined as the sensation received by the eye (rod cells of the eye) due to light coming from an object.

(1) **Colours of opaque object :** The colours of opaque bodies are due to selective reflection. *e.g.*

(i) A rose appears red in white light because it reflects red colour and absorbs all remaining colours.

(ii) When yellow light falls on a bunch of flowers, then yellow and white flowers looks yellow. Other flowers looks black.

(2) **Colours of transparent object :** The colours of transperent bodies are due to selective transmission..

(i) A red glass appears red because it absorbs all colours, except red which it transmits.

(ii) When we look on objects through a green glass or green filter then green and white objects will appear green while other black.

(3) **Colour of the sky :** Light of shorter wavelength is scattered much more than the light of longer wavelength. Since blue colour has relatively shorter wavelength, it predominates the sky and hence sky appears bluish.

(4) **Colour of clouds :** Large particles like water droplets and dust do not have this selective scattering power. They scatter all wavelengths alomost equally. Hence clouds appear to the white.

(5) **Colour triangle for spectral colours :** Red, Green and blue are primary colours.

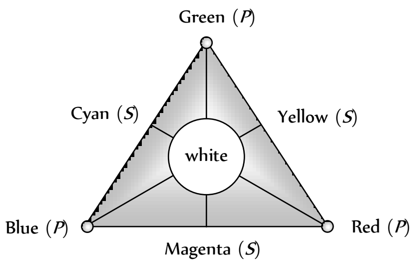


Fig. 29.72

(i) Complementary colours : Green and Magenta, Blue and Yellow, Red and Cyan.

(ii) Combination : Green + Red + Blue = White, Blue + Yellow = White, Red + Cyan = White, Green + Magenta = White

(6) **Colour triangle for pigment and dyes :** Red, Yellow and Blue are the primary colours.

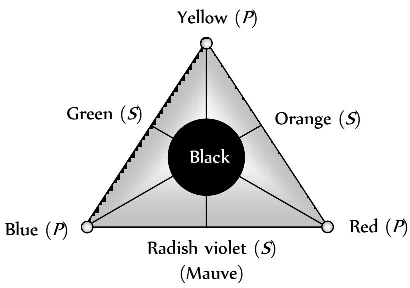


Fig. 29.73

(i) Complementary colours : Yellow and Mauve, Red and Green, Blue and Orange.

(ii) Combination : Yellow + Red + Blue = Black, Blue + Orange = Black, Red + Green = Black, Yellow + Mauve = Black

Spectrum

The ordered arrangements of radiations according to wavelengths or frequencies is called Spectrum. Spectrum can be divided in two parts Emission spectrum and Absorption spectrum.

(1) **Emission spectrum :** When light emitted by a self luminous object is dispersed by a prism to get the spectrum, the spectrum is called emission spectra.

Continuous emission spectrum

(i) It consists of continuously varying wavelengths in a definite wavelength range.

(ii) It is produced by solids, liquids and highly compressed gases heated to high temperature.

(iii) *e.g.* Light from the sun, filament of incandescent bulb, candle flame *etc.*

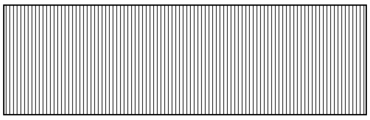


Fig. 29.74

Line emission spectrum

(i) It consist of distinct bright lines.

(ii) It is produced by an excited source in atomic state.

(iii) *e.g.* Spectrum of excited helium, mercury vapours, sodium vapours or atomic hydrogen.

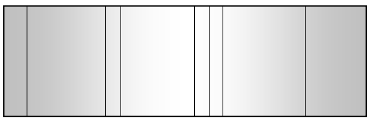


Fig. 29.75

Band emission spectrum

(i) It consist of district bright bands.

(ii) It is produced by an excited source in molecular state.

(iii) *e.g.* Spectra of molecular H_2, *CO*, NH_3 *etc.*

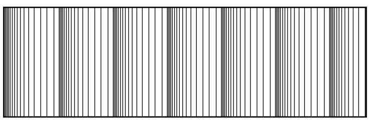


Fig. 29.76

(2) **Absorption spectrum :** When white light passes through a semi-transparent solid, or liquid or gas, it's spectrum contains certain dark lines or bands, such spectrum is called absorption spectrum (of the substance through which light is passed).

(i) Substances in atomic state produces line absorption spectra. Polyatomic substances such as H_2, CO_2 and $KMnO_4$ produces band absorption spectrum.

(ii) Absorption spectra of sodium vapour have two (yellow lines) wavelengths $D_1(5890\,Å)$ and $D_2(5896\,Å)$

(3) **Fraunhoffer's lines :** The central part (photosphere) of the sun is very hot and emits all possible wavelengths of the visible light. However, the outer part (chromosphere) consists of vapours of different elements. When the light emitted from the photosphere passes through the chromosphere, certain wavelengths are absorbed. Hence, in the spectrum of sunlight a large number of dark lines are seen called Fraunhoffer lines.

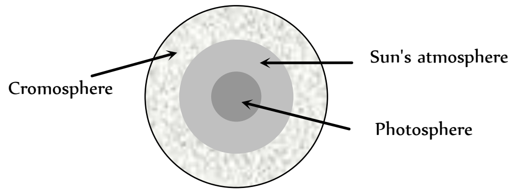


Fig. 29.77

(i) The prominent lines in the yellow part of the visible spectrum were labelled as *D*-lines, those in blue part as *F*-lines and in red part as *C*-line.

(ii) From the study of Fraunhoffer's lines the presence of various elements in the sun's atmosphere can be identified *e.g.* abundance of hydrogen and helium.

(iii) In the event of a solar eclipse, dark lines become bright. This is because of the reason that the presence of an opaque obstacle in between sun and earth cuts the light off from the central region (photo-sphere), while light from corner portion (cromosphere) is still being received. The bright lines appear exactly at the places where dark lines were present.

(4) **Spectrometer :** A spectrometer is used for obtaining pure spectrum of a source in laboratory and calculation of μ of material of prism and μ of a transparent liquid.

It consists of three parts : Collimator which provides a parallel beam of light; Prism Table for holding the prism and Telescope for observing the spectrum and making measurements on it.

The telescope is first set for parallel rays and then collimator is set for parallel rays. When prism is set in minimum deviation position, the spectrum seen is pure spectrum. Angle of prism (*A*) and angle of minimum deviation (δ_m) are measured and μ of material of prism is calculated using prism formula. For μ of a transparent liquid, we take a hollow prism with thin glass sides. Fill it with the liquid and measure (δ_m) and *A* of liquid prism. μ of liquid is calculated using prism formula.

(5) **Direct vision spectroscope :** It is an instrument used to observe pure spectrum. It produces dispersion without deviation with the help of *n* crown prisms and $(n-1)$ flint prisms alternately arranged in a tabular structure.

For no deviation $n(\mu-1)A=(n-1)\,(\mu'-1)A'$.

Human Eye

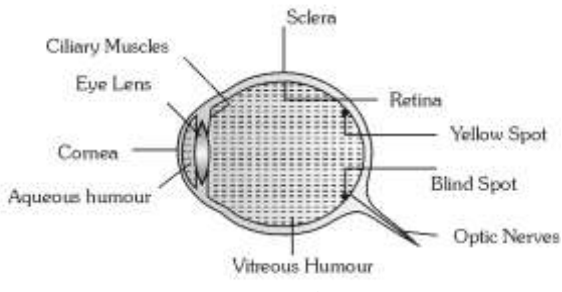


Fig. 29.78

(1) **Eye lens :** Over all behaves as a convex lens of $\mu=1.437$

(2) **Retina :** Real and inverted image of an object, obtained at retina, brain sense it erect.

(3) **Yellow spot :** It is the most sensitive part, the image formed at yellow spot is brightest.

(4) **Blind spot :** Optic nerves goes to brain through blind spot. It is not sensitive for light.

(5) **Ciliary muscles :** Eye lens is fixed between these muscles. It's both radius of curvature can be changed by applying pressure on it through ciliary muscles.

(6) **Power of accomodation :** The ability of eye to see near objects as well as far objects is called power of accomodation.

(7) **Range of vision :** For healthy eye it is 25 *cm* (near point) to ∞ (far point).

A normal eye can see the objects clearly, only if they are at a distance greater than 25 *cm*. This distance is called Least distance of distinct vision and is represented by *D*.

(8) **Persistence of vision :** Is 1/10 *sec. i.e.* if time interval between two consecutive light pulses is lesser than 0.1 *sec.*, eye cannot distinguish them separately.

(9) **Binocular vision :** The seeing with two eyes is called binocular vision.

(10) **Resolving limit :** The minimum angular separation between two objects, so that they are just resolved is called resolving limit. For eye it is $1' = \left(\frac{1}{60}\right)^o$.

Defects in Eye

(1) **Myopia (short sightness) :** A short-sighted eye can see only nearer objects. Distant objects are not seen clearly.

(i) In this defect image is formed before the retina and Far point comes closer.

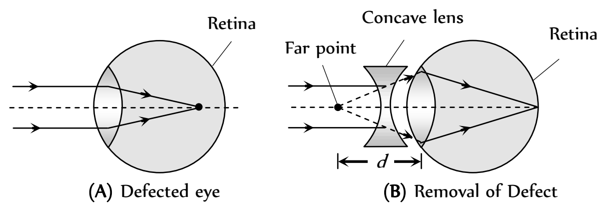


Fig. 29.79

(ii) In this defect focal length or radii of curvature of lens reduced or power of lens increases or distance between eye lens and retina increases.

(iii) This defect can be removed by using a concave lens of suitable focal length.

(iv) If defected far point is at a distance *d* from eye then

Focal length of used lens $f = -d = -$ (defected far point)

(v) A person can see upto distance → *x*, wants to see distance → *y* ($y > x$) so $f = \frac{xy}{x-y}$ or power of the lens $P = \frac{x-y}{xy}$

(2) **Hypermetropia (long sightness) :** A long-sighted eye can see distant objects clearly but nearer object are not clearly visible.

(i) Image formed behind the retina and near point moves away

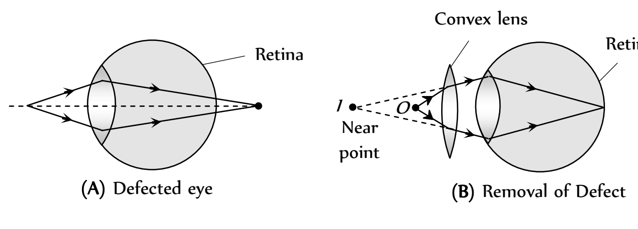


Fig. 29.80

(ii) In this defect focal length or radii of curvature of lens increases or power of lens decreases or distance between eye lens and retina decreases.

(iii) This defect can be removed by using a convex lens.

(iv) If a person cannot see before distance d but wants to see the object placed at distance *D* from eye so $f = \frac{dD}{d-D}$ and power of the lens $P = \frac{d-D}{dD}$

(3) **Presbyopia :** In this defect both near and far objects are not clearly visible. It is an old age disease and it is due to the loosing power of accommodation. It can be removed by using bifocal lens.

(4) **Astigmatism :** In this defect eye cannot see horizontal and vertical lines clearly, simultaneously. It is due to imperfect spherical nature of eye lens. This defect can be removed by using cylindrical lens (Torric lenses).

Lens Camera

(1) In lens camera a converging lens of adjustable aperture is used.

(2) Distance of film from lens is also adjustable.

(3) In photographing an object, the image is first focused on the film by adjusting the distance between lens and film. It is called focusing. After focusing, aperture is set to a specific value and then film is exposed to light for a given time through shutter.

(4) ***f*-number :** The ratio of focal length to the aperture of lens is called *f*-number of the camera.

2, 2.8, 4, 5.6, 8, 11, 22, 32 are the *f*-numbers marked on aperture.

$$f\text{-number} = \frac{\text{Focal length}}{\text{Aperture}} \Rightarrow \text{Aperture} \propto \frac{1}{f\text{-number}}$$

(5) **Time of exposure :** It is the time for which the shutter opens and light enters the camera to expose film.

(i) If intensity of light is kept fixed then for proper exposure

$$\text{Time of exposure } (t) \propto \frac{1}{(\text{Aperture})^2}$$

(ii) If aperture is kept fixed then for proper exposure

$$\text{Time of exposure } (t) \propto \frac{1}{[\text{Intensity}(I)]^2}$$

$$\Rightarrow I\,t = \text{constant} \Rightarrow I_1 t_1 = I_2 t_2$$

(iii) Smaller the *f*-number larger will be the aperture and lesser will be the time of exposure and faster will be the camera.

(6) **Depth of focus :** It refers to the range of distance over which the object may lie so as to form a good quality image. Large *f*-number increase the depth of focus.

Microscope

It is an optical instrument used to see very small objects. It's magnifying power is given by

$$m = \frac{\text{Visual angle with instrument}(\beta)}{\text{Visual angle when object is placed at least distance of distinct vision}(\alpha)}$$

(1) **Simple microscope**

(i) It is a single convex lens of lesser focal length.

(ii) Also called magnifying glass or reading lens.

(iii) Magnification's, when final image is formed at D and ∞ (*i.e.* m_D and m_∞) $m_D = \left(1+\frac{D}{f}\right)_{max}$ and $m_\infty = \left(\frac{D}{f}\right)_{min}$

(iv) If lens is kept at a distance a from the eye then $m_D = 1 + \frac{D-a}{f}$ and $m_\infty = \frac{D-a}{f}$

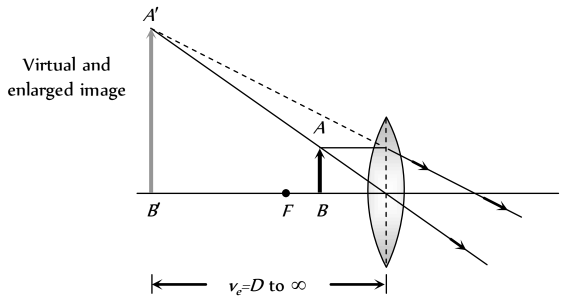


Fig. 29.81

(2) **Compound microscope**

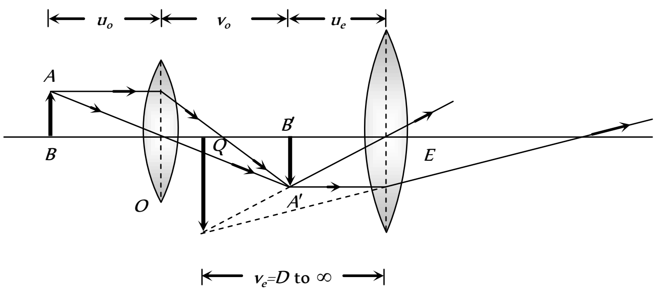


Fig. 29.82

(i) Consist of two converging lenses called objective and eye lens.

(ii) $f_{\text{eye lens}} > f_{\text{objective}}$ and $(\text{diameter})_{\text{eye lens}} > (\text{diameter})_{\text{objective}}$

(iii) Intermediate image is real and enlarged.

(iv) Final image is magnified, virtual and inverted.

(v) u_o = Distance of object from objective (o), v_o = Distance of image ($A'B'$) formed by objective from objective, u_e = Distance of $A'B'$ from eye lens, v_e = Distance of final image from eye lens, f_o = Focal length of objective, f_e = Focal length of eye lens.

(vi) **Final image is formed at D :** Magnification $m_D = -\frac{v_o}{u_o}\left(1+\frac{D}{f_e}\right)$ and length of the microscope tube (distance between two lenses) is $L_D = v_o + u_e$.

Generally object is placed very near to the principal focus of the objective hence $u_o \cong f_o$. The eye piece is also of small focal length and the image formed by the objective is also very near to the eye piece.

So $v_o \cong L_D$, the length of the tube.

Hence, we can write $m_D = \frac{-L}{f_o}\left(1+\frac{D}{f_e}\right)$

(vii) **Final image is formed at ∞ :** Magnification

$m_\infty = -\frac{v_0}{u_0}\cdot\frac{D}{f_e}$ and length of tube $L_\infty = v_0 + f_e$

In terms of length $m_\infty = \frac{(L_\infty - f_o - f_e)D}{f_o f_e}$

(viii) For large magnification of the compound microscope, both f_o and f_e should be small.

(ix) If the length of the tube of microscope increases, then its magnifying power increases.

(x) The magnifying power of the compound microscope may be expressed as $M = m_o \times m_e$; where m_o is the magnification of the objective and m_e is magnifying power of eye piece.

Astronomical Telescope (Refracting Type)

By astronomical telescope heavenly bodies are seen.

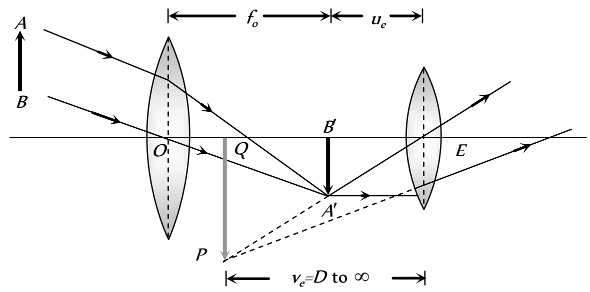


Fig. 29.83

(1) $f_{\text{objective}} > f_{\text{eyelens}}$ and $d_{\text{objective}} > d_{\text{eye lens}}$.

(2) Intermediate image is real, inverted and small.

(3) Final image is virtual, inverted and small.

(4) Magnification : $m_D = -\frac{f_0}{f_e}\left(1+\frac{f_e}{D}\right)$ and $m_\infty = -\frac{f_o}{f_e}$

(5) Length : $L_D = f_0 + u_e$ and $L_\infty = f_0 + f_e$

Terrestrial Telescope

It is used to see far off object on the earth.

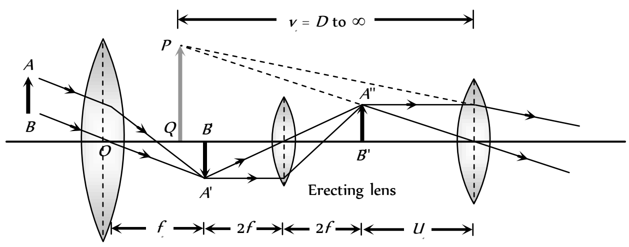


Fig. 29.84

(1) It consists of three converging lens : objective, eye lens and erecting lens.

(2) It's final image is virtual, erect and smaller.

(3) Magnification : $m_D = \frac{f_0}{f_e}\left(1+\frac{f_e}{D}\right)$ and $m_\infty = \frac{f_0}{f_e}$

(4) Length : $L_D = f_0 + 4f + u_e$ and $L_\infty = f_0 + 4f + f_e$

Galilean Telescope

It is also type of terrestrial telescope but of much smaller field of view.

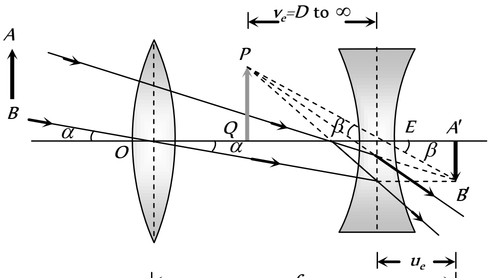


Fig. 29.85

(1) Objective is a converging lens while eye lens is diverging lens.

(2) Magnification : $m_D = \frac{f_0}{f_e}\left(1-\frac{f_e}{D}\right)$ and $m_\infty = \frac{f_0}{f_e}$

(3) Length : $L_D = f_0 - u_e$ and $L_\infty = f_0 - f_e$

Reflecting Telescope

Reflecting telescopes are based upon the same principle except that the formation of images takes place by reflection instead of by refraction.

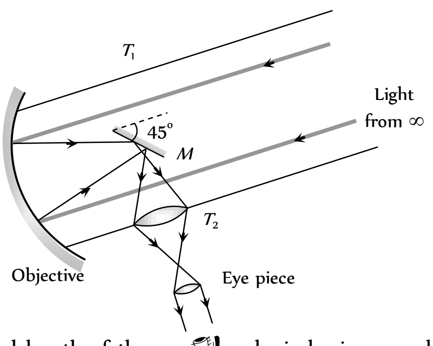


Fig. 29.86

If f_o is focal length of the concave spherical mirror used as objective and f_e the focal length of the eye-piece, the magnifying power of the reflecting telescope is given by $m = \frac{f_o}{f_e}$

Further, if D is diameter of the objective and d, the diameter of the pupil of the eye, then brightness ratio (β) is given by $\beta = \frac{D^2}{d^2}$

Resolving Limit and Resolving Power

(1) **Microscope :** In reference to a microscope, the minimum distance between two lines at which they are just distinct is called Resolving limit (RL) and it's reciprocal is called Resolving power (RP)

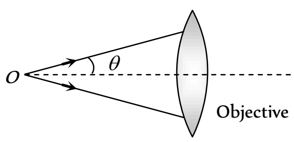


Fig. 29.87

$$R.L. = \frac{\lambda}{2\mu\sin\theta} \text{ and } R.P. = \frac{2\mu\sin\theta}{\lambda} \Rightarrow R.P. \propto \frac{1}{\lambda}$$

λ = Wavelength of light used to illuminate the object,

μ = Refractive index of the medium between object and objective,

θ = Half angle of the cone of light from the point object, $\mu\sin\theta$ = Numerical aperture.

(2) **Telescope :** Smallest angular separations ($d\theta$) between two distant objects, whose images are separated in the telescope is called resolving limit. So resolving limit $d\theta = \frac{1.22\lambda}{a}$

and resolving power $(RP) = \frac{1}{d\theta} = \frac{a}{1.22\lambda} \Rightarrow R.P. \propto \frac{1}{\lambda}$ where a = aperture of objective.

Binocular

If two telescopes are mounted parallel to each other so that an object can be seen by both the eyes simultaneously, the arrangement is called 'binocular'. In a binocular, the length of each tube is reduced by using a set of totally reflecting prisms which provide intense, erect image free from lateral inversion. Through a binocular we get two images of the same object from different angles at same time. Their superposition gives the perception of depth along with length and breadth, *i.e.*, binocular vision gives proper three-dimensional (3D) image.

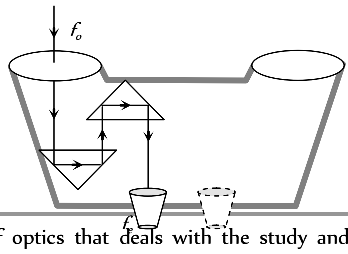


Fig. 29.88

Photometry

The branch of optics that deals with the study and measurement of the light energy is called photometry.

(1) **Radiant flux (R) :** The total energy radiated by a source per second is called radiant flux. It's S.I. unit is **Watt (W)**.

(2) **Luminous flux (ϕ) :** The total light energy emitted by a source per second is called luminous flux. It represents the total brightness producing capacity of the source. It's S.I. unit is **Lumen (lm)**.

(3) **Luminous efficiency (η) :** The Ratio of luminous flux and radiant flux is called luminous efficiency *i.e.* $\eta = \frac{\phi}{R}$.

Table 29.4 : Luminous flux and efficiency

Light source	Flux (lumen)	Efficiency (lumen/watt)
40 W tungsten bulb	465	12
60 W tungsten bulb	835	14
500 W tungsten bulb	9950	20
30 W fluorescent tube	1500	50

(4) **Luminous Intensity (L) :** In a given direction it is defined as luminous flux per unit solid angle *i.e.*

$$L = \frac{\phi}{\omega} \rightarrow \frac{\text{Light energy}}{\sec \times \text{solid angle}} \xrightarrow{\text{S.I. unit}} \frac{\text{lumen}}{\text{steradian}} = \text{candela}\,(Cd)$$

The luminous intensity of a point source is given by : $L = \frac{\phi}{4\pi} \Rightarrow$ $\phi = 4\pi \times (L)$

(5) **Illuminance or intensity of illumination (*I*) :** The luminous flux incident per unit area of a surface is called illuminance. $I = \frac{\phi}{A}$. It's S.I. unit is $\frac{\text{Lumen}}{m^2}$ or Lux (*lx*) and it's C.G.S. unit is Phot. $1\,\text{Phot} = 10^4\,\text{Lux} = \frac{1\,\text{Lumen}}{cm^2}$

(i) Intensity of illumination at a distance *r* from a point source is $I = \frac{\phi}{4\pi r^2} \Rightarrow I \propto \frac{1}{r^2}$.

(ii) Intensity of illumination at a distance *r* from a line source is $I = \frac{\phi}{2\pi r l} \Rightarrow I \propto \frac{1}{r}$

(iii) In case of a parallel beam of light $I \propto r^0$.

(iv) The illuminance represents the luminous flux incident on unit area of the surface, while luminance represents the luminous flux reflected from a unit area of the surface.

(6) **Relation Between Luminous Intensity (*L*) and Illuminance (*I*) :** If S is a unidirectional point source of light of luminous intensity L and there is a surface at a distance r from source, on which light is falling normally.

(i) Illuminance of surface is given by : $I = \frac{L}{r^2}$

(ii) For a given source L = constant so $I \propto \frac{1}{r^2}$; This is called. Inverse square law of illuminance.

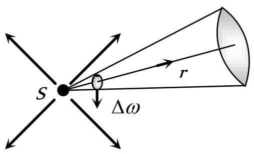


Fig. 29.89

(7) **Lambert's Cosine Law of Illuminance :** In the above discussion if surface is so oriented that light from the source falls, on it obliquely and the central ray of light makes an angle θ with the normal to the surface, then

(i) Illuminance of the surface $I = \frac{L\cos\theta}{r^2}$

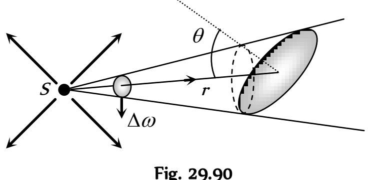


Fig. 29.90

(ii) For a given light source and point of illumination (*i.e.* L and r = constant) $I \propto \cos\theta$ this is called Lambert's cosine law of illuminance. $\Rightarrow I_{\max} = \frac{L}{r^2} = I_o (\text{at}\,\theta = 0^\circ)$

(iii) For a given source and plane of illuminance (*i.e.* L and h = constant)

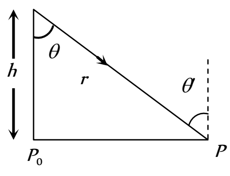


Fig. 29.91

$\cos\theta = \frac{h}{r}$ so $I = \frac{L}{h^2}\cos^3\theta$

or $I = \frac{Lh}{r^3}$ *i.e.* $I \propto \cos^3\theta$ or $I \propto \frac{1}{r^3}$

(8) **Photometer and Principle of Photometry :** A photometer is a device used to compare the illuminance of two sources.

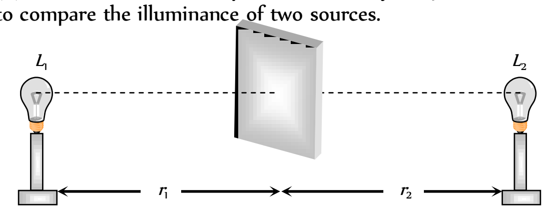


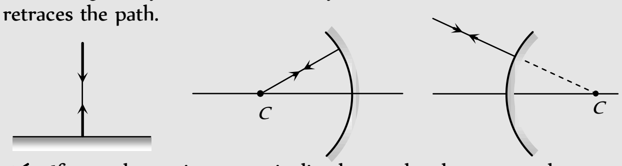
Fig. 29.92

Two sources of luminous intensity L_1, and L_2 are placed at distances r_1 and r_2 from the screen so that their flux are perpendicular to the screen. The distance r_1 and r_2 are adjusted till $I_1 = I_2$. So $\frac{L_1}{r_1^2} = \frac{L_2}{r_2^2} \Rightarrow \frac{L_1}{L_2} = \left(\frac{r_1}{r_2}\right)^2$; This is called principle of photometry.

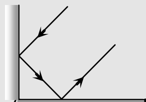
Tips & Tricks

✍ After reflection velocity, wavelength and frequency of light remains same but intensity decreases.

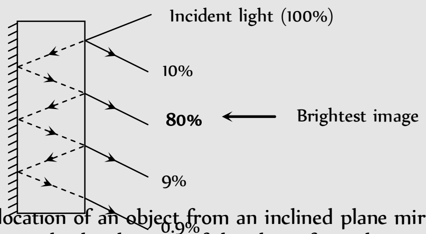
✍ If light ray incident normally on a surface, after reflection it retraces the path.



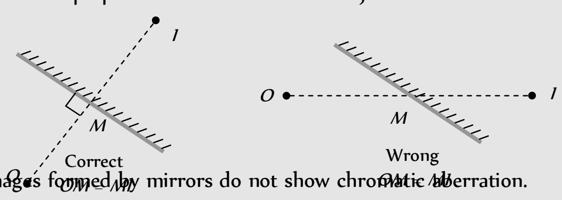
✍ If two plane mirrors are inclined to each other at 90°, the emergent ray is anti-parallel to incident ray, if it suffers one reflection from each. Whatever be the angle to incidence.



✍ We observe number of images in a thick plane mirror, out of them only second is brightest.



✍ To find the location of an object from an inclined plane mirror, you have to see the perpendicular distance of the object from the mirror.



✍ Images formed by mirrors do not show chromatic aberration.

✍ In concave mirror, minimum distance between a real object and it's real image is zero. (*i.e.* when $u = v = 2f$)

✍ If a spherical mirror produces an image '*m*' times the size of the object (m = magnification) then u, v and f are given by the followings

$$u = \left(\frac{m-1}{m}\right)f, \quad v = -(m-1)f \text{ and } f = \left(\frac{m}{m-1}\right)u$$

✍ Focal length of a mirror is independent of material of mirror and medium in which it is placed and wavelength of incident light

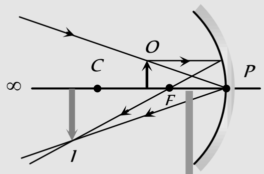
✍ Divergence or Convergence power of a mirror does not change with the change in medium.

✍ If an object is moving at a speed v towards a spherical mirror along it's axis then speed of image away from mirror is

$$v_i = -\left(\frac{f}{u-f}\right)^2 . v_o$$

✍ When object is moved from focus to infinity at constant speed, the image will move faster in the beginning till object moves from f to $2f$, and slower later on, towards the mirror.

✍ As every part of mirror forms a complete image, if a part of the mirror is obstructed, full image will be formed but intensity will be reduced.



✍ In case of refraction of light frequency (and hence colour) and phase do not change (while wavelength and velocity will change).

✍ *In the refraction intensity of incident light decreases as it goes from one medium to another medium.*

✍ A transparent solid is invisible in a liquid of same refractive index (Because of No refraction).

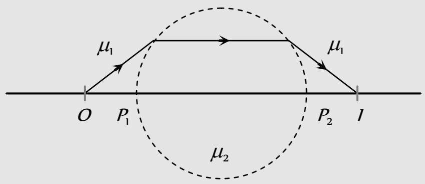
✍ When a glass slab is kept over various coloured letters and seen from the top, the violet colour letters appears closer (Because $\lambda_V < \lambda_R$ so $\mu_V > \mu_R$ and from $\mu = \frac{h}{h'}$ if μ increases then h' decreases i.e. Letter appears to be closer)

✍ Minimum distance between an object and it's real image formed by a convex lens is $4f$.

✍ Component lenses of an achromatic doublet cemented by canada blasam because it is transparent and has a refractive index almost equal to the refractive index of the glass.

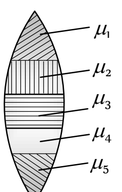
✍ Parabolic mirrors are free from spherical aberration.

✍ If a sphere of radius R made of material of refractive index μ_2 is placed in a medium of refractive index μ_1, then if the object is placed at a distance $\left(\frac{\mu_1}{\mu_2 - \mu_1}\right)R$ from the pole, the real image formed is equidistant from the sphere



✍ The lens doublets used in telescope are achromatic for blue and red colours, while these used in camera are achromatic for violet and green colours. The reason for this is that our eye is most sensitive between blue and red colours, while the photographic plates are most sensitive between violet and green colours.

✍ **Composite lens :** If a lens is made of several materials then



Number of images formed = Number of materials used

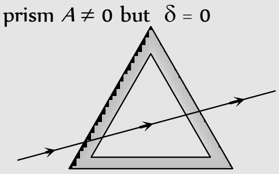
Here no. of images = 5

✍ For the condition of grazing emergence through a prism. Minimum angle of incidence $i_{min} = \sin^{-1}\left[\sqrt{\mu^2 - 1}\,\sin A - \cos A\right]$.

✍ If a substance emits spectral lines at high temperature then it absorbs the same lines at low temperature. This is Kirchoff's law.

✍ When a ray of white light passes through a glass prism red light is deviated less than blue light.

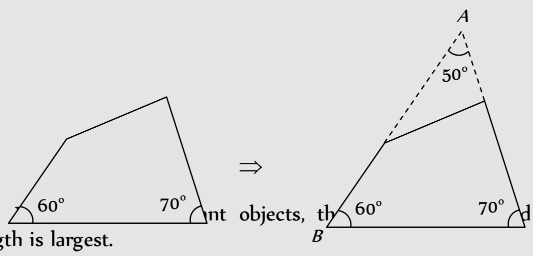
✍ For a hollow prism $A \neq 0$ but $\delta = 0$



✍ If an opaque coloured object or crystal is crushed to fine powder it will appear white (in sun light) as it will lose it's property of selective reflection.

✍ Our eye is most sensitive to that part of the spectrum which lies between the F line (sky green) and the C-line (red) of hydrogen, and the mean refractive index of this part is nearly equal to the refractive index for the D line (yellow) of sodium. Hence for the dispersive power, the following formula is internationally accepted $\omega = \frac{\mu_F - \mu_C}{\mu_D - 1}$

✍ Sometimes a part of prism is given and we keep on thinking whether how should we proceed ? To solve such problems first complete the prism then solve as the problems of prism are solved



✍ nt objects, th d it's focal length is largest.

✍ Minimum separation (d) between objects, so they can just resolved by a telescope is : $d = \frac{r}{R.P.}$

Where r = distance of objects from telescope.

✍ As magnifying power astronomical telescope is negative, the image seen in astronomical telescope is truly inverted, i.e., left is turned right with upside down simultaneously. However, as most of the astronomical objects are symmetrical this inversion does not affect the observations.

✍ If objective and eye lens of a telescope are interchanged, it will not behave as a microscope but object appears very small.

✍ In a telescope, if field and eye lenses are interchanged magnification will change from (f_o/f_e) to (f_e/f_o), *i.e.*, it will change from m to $(1/m)$, i.e., will become $(1/m)$ times of its initial value.

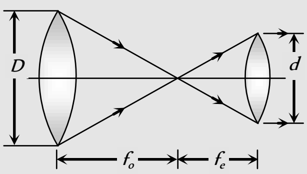
✍ As magnification produced by telescope for normal setting is (f_o/f_e), so to have large magnification, f_o must be as large as practically possible and f_e small. This is why in a telescope, objective is of large focal length while eye piece of small.

✍ In a telescope, aperture of the field lens is made as large as practically possible to increase its resolving power as resolving power of a telescope ∝ (D/λ). Large aperture of objective also helps in improving the brightness of image by gathering more light from distant object. However, it increases aberrations particularly spherical.

✍ For a telescope with increase in length of the tube, magnification decreases.

✍ In case of a telescope if object and final image are at infinity then :

$$m = \frac{f_o}{f_e} = \frac{D}{d}$$



✍ If we are given four convex lenses having focal lengths $f_1 > f_2 > f_3 > f_4$. For making a good telescope and microscope. We choose the following lenses respectively.

Telescope $f_1(o), f_4(e)$ Microscope $f_4(o), f_3(e)$

✍ If a parrot is sitting on the objective of a large telescope and we look towards (or take a photograph)of distant astronomical object (say moon) through it, the parrot will not be seen but the intensity of the image will be slightly reduced as the parrot will act as obstruction to light and will reduce the aperture of the objective.

✍ The luminous flux of a source of (1/685) watt emitting monochromatic light of wavelength 5500 Å is called 1 lumen.

✍ While solving the problems of photometry keep in mind.

$R \propto \phi \propto L$ (As $\phi = \eta R = 4\pi L$)

$$\Rightarrow \frac{R_1}{R_2} = \frac{\phi_1}{\phi_2} = \frac{L_1}{L_2}$$

Ordinary Thinking

Objective Questions

Plane Mirror

1. Two vertical plane mirrors are inclined at an angle of $60°$ with each other. A ray of light travelling horizontally is reflected first from one mirror and then from the other. The resultant deviation is
(a) $60°$ (b) $120°$
(c) $180°$ (d) $240°$

2. A plane mirror reflects a pencil of light to form a real image. Then the pencil of light incident on the mirror is [MP PMT 1997; DCE 2001, 03]
(a) Parallel (b) Convergent
(c) Divergent (d) None of the above

3. What should be the angle between two plane mirrors so that whatever be the angle of incidence, the incident ray and the reflected ray from the two mirrors be parallel to each other [KCET 1994; SCRA 1994]
(a) $60°$ (b) $90°$
(c) $120°$ (d) $175°$

4. A plane mirror reflecting a ray of incident light is rotated through an angle θ about an axis through the point of incidence in the plane of the mirror perpendicular to the plane of incidence, then
(a) The reflected ray does not rotate
(b) The reflected ray rotates through an angle θ
(c) The reflected ray rotates through an angle 2θ
(d) The incident ray is fixed

5. A plane mirror is approaching you at a speed of $10\,cm/sec$ You can see your image in it. At what speed will your image approach you [CPMT 1974]
(a) $10cm/sec$ (b) $5cm/sec$
(c) $20cm/sec$ (d) $15cm/sec$

6. A light bulb is placed between two plane mirrors inclined at an angle of $60°$. The number of images formed are
SCRA 1994; AIIMS 1997; RPMT 1999; AIEEE 2002; Orissa JEE 2003; MP PMT 2004; MP PET 2004]
(a) 6 (b) 2
(c) 5 (d) 4

7. It is desired to photograph the image of an object placed at a distance of $3m$ from the plane mirror. The camera which is at a distance of $4.5m$ from the mirror should be focussed for a distance of [NCERT 1971]
(a) $3m$ (b) $4.5m$
(c) $6m$ (d) $7.5m$

8. A thick plane mirror shows a number of images of the filament of an electric bulb. Of these, the brightest image is the
(a) First (b) Second
(c) Fourth (d) Last

9. A man is $180cm$ tall and his eyes are $10cm$ below the top of his head. In order to see his entire height right from toe to head, he uses a plane mirror kept at a distance of $1m$ from him. The minimum length of the plane mirror required is [MP PMT 1993; DPMT 2001]
(a) $180cm$ (b) $90cm$
(c) $85cm$ (d) $170cm$

10. A person is in a room whose ceiling and two adjacent walls are mirrors. How many images are formed [AFMC 2002]
(a) 5 (b) 6
(c) 7 (d) 8

11. When a plane mirror is placed horizontally on a level ground at a distance of $60m$ from the foot of a tower, the top of the tower and its image in the mirror subtend an angle of $90°$ at the eye. The height of the tower will be [CPMT 1984]
(a) $30m$ (b) $60m$
(c) $90m$ (d) $120m$

12. A ray of light incidents on a plane mirror at an angle of $30°$. The deviation produced in the ray is [NCERT 1978; CPMT 1991]
(a) $30°$ (b) $60°$
(c) $90°$ (d) $120°$

13. A ray of light is incidenting normally on a plane mirror. The angle of reflection will be [MP PET 2000]

(a) $0°$ (b) $90°$

(c) Will not be reflected (d) None of the above

14. When light wave suffers reflection at the interface from air to glass, the change in phase of the reflected wave is equal to [CPMT 1991; J & KCET 2004]

(a) 0 (b) $\frac{\pi}{2}$

(c) π (d) 2π

15. A ray is reflected in turn by three plain mirrors mutually at right angles to each other. The angle between the incident and the reflected rays is [Roorkee 1995]

(a) $90°$ (b) $60°$

(c) $180°$ (d) None of these

16. Two plane mirrors are at right angles to each other. A man stands between them and combs his hair with his right hand. In how many of the images will he be seen using his right hand [MP PMT 1995; UPSEAT 2001]

(a) None (b) 1

(c) 2 (d) 3

17. When a plane mirror is rotated through an angle θ then the reflected ray turns through the angle 2θ then the size of the image

(a) Is doubled (b) Is halved

(c) Remains the same (d) Becomes infinite

18. A plane mirror produces a magnification of [MP PET/PMT 1997]

(a) -1 (b) $+1$

(c) Zero (d) Between 0 and $+\infty$

19. A plane mirror makes an angle of $30°$ with horizontal. If a vertical ray strikes the mirror, find the angle between mirror and reflected ray [RPET 1997]

(a) $30°$ (b) $45°$

(c) $60°$ (d) $90°$

20. A watch shows time as $3:25$ when seen through a mirror, time appeared will be [RPMT 1997; JIPMER 2001, 02]

(a) $8:35$ (b) $9:35$

(c) $7:35$ (d) $8:25$

21. If an observer is walking away from the plane mirror with $6m/sec$. Then the velocity of the image with respect to observer will be [RPMT 1999]

(a) $6m/sec$ (b) $-6m/sec$

(c) $12m/sec$ (d) $3m/sec$

22. A man runs towards mirror at a speed of $15m/s$. What is the speed of his image [CBSE PMT 2000]

(a) 7.5 *m/s* (b) 15 *m/s*

(c) 30 *m/s* (d) 45 *m/s*

23. A small object is placed 10 *cm* infront of a plane mirror. If you stand behind the object 30 *cm* from the mirror and look at its image, the distance focused for your eye will be [KCET (Engg.) 2001]

(a) 60 *cm* (b) 20 *cm*

(c) 40 *cm* (d) 80 *cm*

24. An object is at a distance of 0.5 *m* in front of a plane mirror. Distance between the object and image is [CPMT 2002]

(a) 0.5 *m* (b) 1 *m*

(c) 0.25 *m* (d) 1.5 *m*

25. A man runs towards a mirror at a speed 15 *m/s* The speed of the image relative to the man is [Kerala PET 2002]

(a) 15 ms^{-1} (b) 30 ms^{-1}

(c) 35 ms^{-1} (d) 20 ms^{-1}

26. The light reflected by a plane mirror may form a real image [KCET (Engg. & Med.) 2002]

(a) If the rays incident on the mirror are diverging

(b) If the rays incident on the mirror are converging

(c) If the object is placed very close to the mirror

(d) Under no circumstances

27. Two plane mirrors are inclined at an angle of 72^0. The number of images of a point object placed between them will be [KCET (Engg. & Med.)1999;

(a) 2 (b) 3

(c) 4 (d) 5

28. To get three images of a single object, one should have two plane mirrors at an angle of [AIEEE 2003]

(a) 30° (b) 60°

(c) 90° (d) 150°

29. A man of length h requires a mirror, to see his own complete image of length at least equal to [MP PET 2003]

(a) $\frac{h}{4}$ (b) $\frac{h}{3}$

(c) $\frac{h}{2}$ (d) h

30. Two plane mirrors are at $45°$ to each other. If an object is placed between them, then the number of images will be [MP PMT 2003]

(a) 5 (b) 9

(c) 7 (d) 8

31. A man having height 6 *m*. He observes image of 2 *m* height erect, then mirror used is [BCECE 2004]

(a) Concave (b) Convex

(c) Plane (d) None of these

32. A light beam is being reflected by using two mirrors, as in a periscope used in submarines. If one of the mirrors rotates by an angle θ, the reflected light will deviate from its original path by the angle [UPSEAT 2004]

(a) 2θ (b) 0^o

(c) θ (d) 4θ

33. Focal length of a plane mirror is [RPMT 2000]

(a) Zero (b) Infinite

(c) Very less (d) Indefinite

34. A ray of light is incident at 50° on the middle of one of the two mirrors arranged at an angle of 60° between them. The ray then touches the second mirror, get reflected back to the first mirror, making an angle of incidence of [MP PET 2005]

(a) 50° (b) 60°

(c) 70° (d) 80°

Spherical Mirror

1. A convex mirror of focal length *f* forms an image which is $\frac{1}{n}$ times the object. The distance of the object from the mirror is

(a) $(n-1)f$ (b) $\left(\frac{n-1}{n}\right)f$

(c) $\left(\frac{n+1}{n}\right)f$ (d) $(n+1)f$

2. A diminished virtual image can be formed only in [MP PMT 2002]

(a) Plane mirror (b) A concave mirror

(c) A convex mirror (d) Concave-parabolic mirror

3. Which of the following could not produce a virtual image

(a) Plane mirror

(b) Convex mirror

(c) Concave mirror

(d) All the above can produce a virtual image

4. An object $5cm$ tall is placed $1m$ from a concave spherical mirror which has a radius of curvature of $20cm$ The size of the image is

(a) $0.11cm$ (b) $0.50cm$

(c) $0.55cm$ (d) $0.60cm$

5. The focal length of a concave mirror is $50cm$. Where an object be placed so that its image is two times and inverted

(a) 75 *cm* (b) 72 *cm*

(c) 63 *cm* (d) 50 *cm*

6. An object of size $7.5cm$ is placed in front of a convex mirror of radius of curvature $25cm$ at a distance of $40cm$. The size of the image should be

(a) $2.3cm$ (b) $1.78cm$

(c) $1cm$ (d) $0.8cm$

7. The field of view is maximum for

(a) Plane mirror (b) Concave mirror

(c) Convex mirror (d) Cylindrical mirror

8. The focal length of a concave mirror is *f* and the distance from the object to the principle focus is *x*. The ratio of the size of the image to the size of the object is [Kerala PET 2005]

(a) $\frac{f+x}{f}$ (b) $\frac{f}{x}$

(c) $\sqrt{\frac{f}{x}}$ (d) $\frac{f^2}{x^2}$

9. Image formed by a convex mirror is [MP PET 1993]

(a) Virtual (b) Real

(c) Enlarged (d) Inverted

10. In a concave mirror experiment, an object is placed at a distance x_1 from the focus and the image is formed at a distance x_2 from the focus. The focal length of the mirror would be

(a) x_1x_2 (b) $\sqrt{x_1x_2}$

(c) $\frac{x_1+x_2}{2}$ (d) $\sqrt{\frac{x_1}{x_2}}$

11. A convex mirror is used to form the image of an object. Then which of the following statements is wrong [CPMT 1973]

(a) The image lies between the pole and the focus

(b) The image is diminished in size

(c) The image is erect

(d) The image is real

12. Given a point source of light, which of the following can produce a parallel beam of light [CPMT 1974; KCET 2005]

(a) Convex mirror

(b) Concave mirror

(c) Concave lens

(d) Two plane mirrors inclined at an angle of $90°$

13. The image formed by a convex mirror of focal length $30cm$ is a quarter of the size of the object. The distance of the object from the mirror is [MP PET 1993]

(a) $30cm$ (b) $90cm$

(c) $120cm$ (d) $60cm$

14. A boy stands straight infront of a mirror at a distance of $30cm$ away from it. He sees his erect image whose height is $\frac{1}{5}th$ of his real height. The mirror he is using is [MP PMT 1993]

(a) Plane mirror (b) Convex mirror

(c) Concave mirror (d) Plano-convex mirror

15. A person sees his virtual image by holding a mirror very close to the face. When he moves the mirror away from his face, the image becomes inverted. What type of mirror he is using

(a) Plane mirror (b) Convex mirror

(c) Concave mirror (d) None of these

16. Which one of the following statements is true

(a) An object situated at the principle focus of a concave lens will have its image formed at infinity

(b) Concave mirror can give diminished virtual image

(c) Given a point source of light, a convex mirror can produce a parallel beam of light

(d) The virtual image formed in a plane mirror can be photographed

17. The relation between the linear magnification m, the object distance u and the focal length f is

(a) $m = \frac{f-u}{f}$ (b) $m = \frac{f}{f-u}$

(c) $m = \frac{f+u}{f}$ (d) $m = \frac{f}{f+u}$

18. While using an electric bulb, the reflection for street lighting should be from

(a) Concave mirror (b) Convex mirror

(c) Cylindrical mirror (d) Parabolic mirror

19. A concave mirror is used to focus the image of a flower on a nearby well $120 cm$ from the flower. If a lateral magnification of 16 is desired, the distance of the flower from the mirror should be

(a) $8 cm$ (b) $12 cm$

(c) $80 cm$ (d) $120 cm$

20. A virtual image larger than the object can be obtained by [MP PMT 1986]

(a) Concave mirror (b) Convex mirror

(c) Plane mirror (d) Concave lens

21. An object is placed $40 cm$ from a concave mirror of focal length $20 cm$. The image formed is [MP PET 1986; MP PMT/PET 1998]

(a) Real, inverted and same in size

(b) Real, inverted and smaller

(c) Virtual, erect and larger

(d) Virtual, erect and smaller

22. A virtual image three times the size of the object is obtained with a concave mirror of radius of curvature $36 cm$. The distance of the object from the mirror is [MP PET 1986]

(a) $5 cm$ (b) $12 cm$

(c) $10 cm$ (d) $20 cm$

23. Radius of curvature of concave mirror is $40 cm$ and the size of image is twice as that of object, then the object distance is [AFMC 1995]

(a) $60 cm$ (b) $20 cm$

(c) $40 cm$ (d) $30 cm$

24. All of the following statements are correct except [Manipal MEE 1995]

(a) The magnification produced by a convex mirror is always less than one

(b) A virtual, erect, same-sized image can be obtained using a plane mirror

(c) A virtual, erect, magnified image can be formed using a concave mirror

(d) A real, inverted, same-sized image can be formed using a convex mirror

25. If an object is placed $10 cm$ infront of a concave mirror of focal length $20 cm$, the image will be [MP PMT 1995]

(a) Diminished, upright, virtual

(b) Enlarged, upright, virtual

(c) Diminished, inverted, real

(d) Enlarged, upright, real

26. Which of the following form(s) a virtual and erect image for all positions of the object [IIT-JEE 1996]

(a) Convex lens (b) Concave lens

(c) Convex mirror (d) Concave mirror

27. A convex mirror has a focal length f. A real object is placed at a distance f in front of it from the pole produces an image at [MP PET 1986]

(a) Infinity (b) f

(c) $f/2$ (d) $2f$

28. An object $1 cm$ tall is placed $4 cm$ infront of a mirror. In order to produce an upright image of $3 cm$ height one needs a

(a) Convex mirror of radius of curvature $12 cm$

(b) Concave mirror of radius of curvature $12 cm$

(c) Concave mirror of radius of curvature $4 cm$

(d) Plane mirror of height $12 cm$

29. Match List I with List II and select the correct answer using the codes given below the lists : [SCRA 1998]

List I	**List II**
(Position of the object)	(Magnification)
(I) An object is placed at focus before a convex mirror	(A) Magnification is $-\infty$
(II) An object is placed at centre of curvature before a concave mirror	(B) Magnification is 0.5
(III) An object is placed at focus before a concave mirror	(C) Magnification is $+1$
(IV) An object is placed at centre of curvature before a convex mirror	(D) Magnification is – 1
	(E) Magnification is 0.33

Codes :

(a) I-B, II-D, III-A, IV-E (b) I-A, II-D, III-C, IV-B

(c) I-C, II-B, III-A, IV-E (d) I-B, II-E, III-D, IV-C

30. A concave mirror gives an image three times as large as the object placed at a distance of $20 cm$ from it. For the image to be real, the focal length should be [SCRA 1998; JIPMER 2000]

(a) $10 cm$ (b) $15 cm$

(c) $20 cm$ (d) $30 cm$

31. The minimum distance between the object and its real image for concave mirror is [RPMT 1999]

(a) f (b) $2f$

(c) $4f$ (d) Zero

32. An object is placed at $20 cm$ from a convex mirror of focal length $10\,cm$. The image formed by the mirror is [JIPMER 1999]

(a) Real and at $20\,cm$ from the mirror

(b) Virtual and at $20\,cm$ from the mirror

(c) Virtual and at $20/3\,cm$ from the mirror

(d) Real and at $20/3\,cm$ from the mirror

33. A point object is placed at a distance of $10\,cm$ and its real image is formed at a distance of $20\,cm$ from a concave mirror. If the object is moved by $0.1 cm$ towards the mirror, the image will shift by about [MP PMT 2000]

(a) $0.4 cm$ away from the mirror

(b) $0.4 cm$ towards the mirror

(c) $0.8 cm$ away from the mirror

(d) $0.8 cm$ towards the mirror

34. Under which of the following conditions will a convex mirror of focal length f produce an image that is erect, diminished and virtual [AMU (Engg.) 2001]

(a) Only when $2f > u > f$ (b) Only when $u = f$

(c) Only when $u < f$ (d) Always

35. The focal length of a convex mirror is 20 *cm* its radius of curvature will be [MP PMT 2001]

(a) 10 *cm* (b) 20 *cm*

(c) 30 *cm* (d) 40 *cm*

36. A concave mirror of focal length 15 *cm* forms an image having twice the linear dimensions of the object. The position of the object when the image is virtual will be

(a) 22.5 *cm* (b) 7.5 *cm*

(c) 30 *cm* (d) 45 *cm*

37. A point object is placed at a distance of 30 *cm* from a convex mirror of focal length 30*cm*. The image will form at [JIPMER 2002]

(a) Infinity

(b) Focus

(c) Pole

(d) 15 *cm* behind the mirror

38. An object 2.5 *cm* high is placed at a distance of 10 *cm* from a concave mirror of radius of curvature 30 *cm* The size of the image is [BVP 2003]

(a) 9.2 *cm* (b) 10.5 *cm*

(c) 5.6 *cm* (d) 7.5 *cm*

39. For a real object, which of the following can produced a real image

(a) Plane mirror (b) Concave lens

(c) Convex mirror (d) Concave mirror

40. An object of length 6 *cm* is placed on the principle axis of a concave mirror of focal length f at a distance of $4f$. The length of the image will be [MP PET 2003]

(a) 2 *cm* (b) 12 *cm*

(c) 4 *cm* (d) 1.2 *cm*

41. Convergence of concave mirror can be decreased by dipping in

(a) Water (b) Oil

(c) Both (d) None of these

42. What will be the height of image when an object of 2 *mm* is placed on the axis of a convex mirror at a distance 20 *cm* of radius of curvature 40 *cm* [Orissa PMT 2004]

(a) 20 *mm* (b) 10 *mm*

(c) 6 *mm* (d) 1 *mm*

43. Image formed by a concave mirror of focal length 6 *cm*, is 3 times of the object, then the distance of object from mirror is [RPMT 2000]

(a) – 4 *cm* (b) 8 *cm*

(c) 6 *cm* (d) 12 *cm*

44. A concave mirror of focal length f (in air) is immersed in water ($\mu = 4/3$). The focal length of the mirror in water will be

(a) f (b) $\frac{4}{3}f$

(c) $\frac{3}{4}f$ (d) $\frac{7}{3}f$

Refraction of Light at Plane Surfaces

1. To an observer on the earth the stars appear to twinkle. This can be ascribed to [CPMT 1972, 74; AFMC 1995]

(a) The fact that stars do not emit light continuously

(b) Frequent absorption of star light by their own atmosphere

(c) Frequent absorption of star light by the earth's atmosphere

(d) The refractive index fluctuations in the earth's atmosphere

2. The ratio of the refractive index of red light to blue light in air is

(a) Less than unity

(b) Equal to unity

(c) Greater than unity

(d) Less as well as greater than unity depending upon the experimental arrangement

3. The refractive index of a piece of transparent quartz is the greatest for [MP PET 1985, 94]

(a) Red light (b) Violet light

(c) Green light (d) Yellow light

4. The refractive index of a certain glass is 1.5 for light whose wavelength in vacuum is 6000 Å. The wavelength of this light when it passes through glass is [NCERT 1979; CBSE PMT 1993; MP PET 1985, 89] [Orissa JEE 2003]

(a) 4000 Å (b) 6000 Å

(c) 9000 Å (d) 15000 Å

5. When light travels from one medium to the other of which the refractive index is different, then which of the following will change [MP PMT 1986; AMU 2001; BVP 2003]

(a) Frequency, wavelength and velocity

(b) Frequency and wavelength

(c) Frequency and velocity

(d) Wavelength and velocity

6. A light wave has a frequency of 4×10^{14} Hz and a wavelength of 5×10^{-7} meters in a medium. The refractive index of the medium is [MP PMT 1989]

(a) 1.5 (b) 1.33

(c) 1.0 (d) 0.66

7. How much water should be filled in a container 21 *cm* in height, so that it appears half filled when viewed from the top of the container (given that ${}_a\mu_\omega = 4/3$) [MP PMT 1989]

(a) 8.0 *cm* (b) 10.5 *cm*

(c) 12.0 *cm* (d) None of the above

8. Light of different colours propagates through air

(a) With the velocity of air

(b) With different velocities

(c) With the velocity of sound

(d) Having the equal velocities

9. Monochromatic light is refracted from air into the glass of refractive index μ. The ratio of the wavelength of incident and refracted waves is [JIPMER 2000; MP PMT 1996, 2003]

(a) $1:\mu$ (b) $1:\mu^2$

(c) $\mu:1$ (d) $1:1$

10. A monochromatic beam of light passes from a denser medium into a rarer medium. As a result [CPMT 1972]

(a) Its velocity increases (b) Its velocity decreases

(c) Its frequency decreases (d) Its wavelength decreases

11. Refractive index for a material for infrared light is [CPMT 1984]

(a) Equal to that of ultraviolet light

(b) Less than for ultraviolet light

(c) Equal to that for red colour of light

(d) Greater than that for ultraviolet light

12. The index of refraction of diamond is 2.0, velocity of light in diamond in *cm/second* is approximately [CPMT 1975; MNR 1987; UPSEAT 2000]

(a) 6×10^{10} (b) 3.0×10^{10}

(c) 2×10^{10} (d) 1.5×10^{10}

13. A beam of light propagating in medium *A* with index of refraction *n* (*A*) passes across an interface into medium *B* with index of refraction *n*(*B*). The angle of incidence is greater than the angle of refraction; *v*(*A*) and *v*(*B*) denotes the speed of light in *A* and *B*. Then which of the following is true

(a) $v(A) > v(B)$ and $n(A) > n(B)$

(b) $v(A) > v(B)$ and $n(A) < n(B)$

(c) $v(A) < v(B)$ and $n(A) > n(B)$

(d) $v(A) < v(B)$ and $n(A) < n(B)$

14. A rectangular tank of depth 8 meter is full of water ($\mu = 4/3$), the bottom is seen at the depth [MP PMT 1987]

(a) 6 *m* (b) 8/3 *m*

(c) 8 *cm* (d) 10 *cm*

15. A vessel of depth 2d *cm* is half filled with a liquid of refractive index μ_1 and the upper half with a liquid of refractive index μ_2. The apparent depth of the vessel seen perpendicularly is

(a) $d\left(\frac{\mu_1\mu_2}{\mu_1+\mu_2}\right)$ (b) $d\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)$

(c) $2d\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)$ (d) $2d\left(\frac{1}{\mu_1\mu_2}\right)$

16. A beam of light is converging towards a point I on a screen. A plane glass plate whose thickness in the direction of the beam = t, refractive index = μ, is introduced in the path of the beam. The convergence point is shifted by [MNR 1987]

(a) $t\left(1-\frac{1}{\mu}\right)$ away (b) $t\left(1+\frac{1}{\mu}\right)$ away

(c) $t\left(1-\frac{1}{\mu}\right)$ nearer (d) $t\left(1+\frac{1}{\mu}\right)$ nearer

17. Light travels through a glass plate of thickness *t* and having refractive index *n*. If *c* is the velocity of light in vacuum, the time taken by the light to travel this thickness of glass is [NCERT 1976; MP PET 1994; CBSE PMT 1996; KCET 1994; MP PMT 1999, 2001]

(a) $\frac{t}{nc}$ (b) tnc

(c) $\frac{nt}{c}$ (d) $\frac{tc}{n}$

18. When a light wave goes from air into water, the quality that remains unchanged is its [AMU 1995; MNR 1985, 95; KCET 1993; CPMT 1990, 97; MP PET 1991, 2000, 02; UPSEAT 1999, 2000; AFMC 1993, 98, 2003; RPET 1996, 2000, 03; RPMT 1999, 2000; DCE 2001; BHU 2001]

(a) Speed (b) Amplitude

(c) Frequency (d) Wavelength

19. Light takes 8 *min* 20 *sec* to reach from sun on the earth. If the whole atmosphere is filled with water, the light will take the time (${}_a\mu_w = 4/3$)

(a) 8 *min* 20 *sec* (b) 8 *min*

(c) 6 *min* 11 *sec* (d) 11 *min* 6 *sec*

20. The length of the optical path of two media in contact of length d_1 and d_2 of refractive indices μ_1 and μ_2 respectively, is

(a) $\mu_1 d_1+\mu_2 d_2$ (b) $\mu_1 d_2+\mu_2 d_1$

(c) $\frac{d_1 d_2}{\mu_1\mu_2}$ (d) $\frac{d_1+d_2}{\mu_1\mu_2}$

21. Immiscible transparent liquids A, B, C, D and E are placed in a rectangular container of glass with the liquids making layers according to their densities. The refractive index of the liquids are shown in the adjoining diagram. The container is illuminated from the side and a small piece of glass having refractive index 1.61 is

gently dropped into the liquid layer. The glass piece as it descends downwards will not be visible in [CPMT 1986]

A	1.51
B	1.53
C	1.61
D	1.52
E	1.65

(a) Liquid *A* and *B* only
(b) Liquid *C* only
(c) Liquid *D* and *E* only
(d) Liquid *A*, *B*, *D* and *E*

22. The refractive indices of glass and water *w.r.t.* air are 3/2 and 4/3 respectively. The refractive index of glass *w.r.t.* water will be

[MNR 1990; JIPMER 1997, 2000; MP PET 2000]

(a) 8/9 (b) 9/8
(c) 7/6 (d) None of these

23. If ${}_i\mu_j$ represents refractive index when a light ray goes from medium *i* to medium *j*, then the product ${}_2\mu_1 \times {}_3\mu_2 \times {}_4\mu_3$ is equal to [CBSE PMT 1990]

(a) ${}_3\mu_1$ (b) ${}_3\mu_2$
(c) $\dfrac{1}{{}_1\mu_4}$ (d) ${}_4\mu_2$

24. The wavelength of light diminishes μ times ($\mu = 1.33$ for water) in a medium. A diver from inside water looks at an object whose natural colour is green. He sees the object as

[CPMT 1990; MNR 1998]

(a) Green (b) Blue
(c) Yellow (d) Red

25. Ray optics fails when

(a) The size of the obstacle is 5 *cm*
(b) The size of the obstacle is 3 *cm*
(c) The size of the obstacle is less than the wavelength of light
(d) (a) and (b) both

26. When light travels from air to water and from water to glass, again from glass to CO_2 gas and finally through air. The relation between their refractive indices will be given by

(a) ${}_an_w \times {}_wn_{gl} \times {}_{gl}n_{gas} \times {}_{gas}n_a = 1$
(b) ${}_an_w \times {}_wn_{gl} \times {}_{gas}n_{gl} \times {}_{gl}n_a = 1$
(c) ${}_an_w \times {}_wn_{gl} \times {}_{gl}n_{gas} = 1$
(d) There is no such relation

27. For a colour of light the wavelength for air is 6000 $\mathring{A}$ and in water the wavelength is 4500 $\mathring{A}$. Then the speed of light in water will be

(a) $5.\times 10^{14}\ m/s$ (b) 2.25×10^{8} *m/s*
(c) 4.0×10^{8} *m/s* (d) Zero

28. A ray of light travelling inside a rectangular glass block of refractive index $\sqrt{2}$ is incident on the glass–air surface at an angle of incidence of 45°. The refractive index of air is 1. Under these conditions the ray [CPMT 1972]

(a) Will emerge into the air without any deviation
(b) Will be reflected back into the glass
(c) Will be absorbed
(d) Will emerge into the air with an angle of refraction equal to 90°

29. If ε_0 and μ_0 are respectively, the electric permittivity and the magnetic permeability of free space, ε and μ the corresponding quantities in a medium, the refractive index of the medium is

[IIT-JEE 1982; MP PET 1995; CBSE PMT 1997]

(a) $\sqrt{\dfrac{\mu\varepsilon}{\mu_0\varepsilon_0}}$ (b) $\dfrac{\mu\varepsilon}{\mu_0\varepsilon_0}$
(c) $\sqrt{\dfrac{\mu_0\varepsilon_0}{\mu\varepsilon}}$ (d) $\sqrt{\dfrac{\mu\mu_0}{\varepsilon\varepsilon_0}}$

30. A beam of monochromatic blue light of wavelength 4200Å in air travels in water ($\mu = 4/3$). Its wavelength in water will be

(a) 2800 $\mathring{A}$ (b) 5600 $\mathring{A}$
(c) 3150 $\mathring{A}$ (d) 4000 $\mathring{A}$

31. If μ_0 be the relative permeability and K_0 the dielectric constant of a medium, its refractive index is given by

[MNR 1995]

(a) $\dfrac{1}{\sqrt{\mu_0 K_0}}$ (b) $\dfrac{1}{\mu_0 K_0}$
(c) $\sqrt{\mu_0 K_0}$ (d) $\mu_0 K_0$

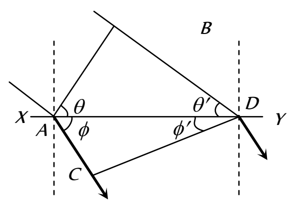
32. If the speed of light in vacuum is $C\ m/\sec$, then the velocity of light in a medium of refractive index 1.5

[NCERT 1977; MP PMT 1984; CPMT 2002]

(a) Is $1.5 \times C$ (b) Is C
(c) Is $\dfrac{C}{1.5}$ (d) Can have any velocity

33. In the adjoining diagram, a wavefront *AB*, moving in air is incident on a plane glass surface *XY*. Its position *CD* after refraction through a glass slab is shown also along with the normals drawn at *A* and *D*. The refractive index of glass with respect to air ($\mu = 1$) will be equal to

[CPMT 1988; DPMT 1999]



(a) $\dfrac{\sin\theta}{\sin\theta'}$
(b) $\dfrac{\sin\theta}{\sin\phi'}$
(c) $\dfrac{\sin\phi'}{\sin\theta}$
(d) $\dfrac{AB}{CD}$

34. When light enters from air to water, then its

[MP PMT 1994; MP PET 1996]

(a) Frequency increases and speed decreases

(b) Frequency is same but the wavelength is smaller in water than in air

(c) Frequency is same but the wavelength in water is greater than in air

(d) Frequency decreases and wavelength is smaller in water than in air

35. On a glass plate a light wave is incident at an angle of 60°. If the reflected and the refracted waves are mutually perpendicular, the refractive index of material is

[MP PMT 1994; Haryana CEE 1996; KCET 1994; 2000]

(a) $\frac{\sqrt{3}}{2}$ (b) $\sqrt{3}$

(c) $\frac{3}{2}$ (d) $\frac{1}{\sqrt{3}}$

36. Refractive index of glass is $\frac{3}{2}$ and refractive index of water is $\frac{4}{3}$. If the speed of light in glass is 2.00×10^8 m/s, the speed in water will be [MP PMT 1994; RPMT 1997]

(a) 2.67×10^8 *m/s* (b) 2.25×10^8 *m/s*

(c) 1.78×10^8 *m/s* (d) 1.50×10^8 *m/s*

37. Monochromatic light of frequency 5×10^{14} *Hz* travelling in vacuum enters a medium of refractive index 1.5. Its wavelength in the medium is

[MP PET/ PMT 1995; Pb. PET 2003]

(a) 4000 Å (b) 5000 Å

(c) 6000 Å (d) 5500 Å

38. Light of wavelength is 7200 Å in air. It has a wavelength in glass ($\mu = 1.5$) equal to [DCE 1999]

(a) 7200 Å (b) 4800 Å

(c) 10800 Å (d) 7201.5 Å

39. Which of the following is *not* a correct statement

[MP PET 1997]

(a) The wavelength of red light is greater than the wavelength of green light

(b) The wavelength of blue light is smaller than the wavelength of orange light

(c) The frequency of green light is greater than the frequency of blue light

(d) The frequency of violet light is greater than the frequency of blue light

40. Which of the following is a *correct* relation [MP PET 1997]

(a) ${}_a\mu_r = {}_a\mu_w \times {}_r\mu_\omega$ (b) ${}_a\mu_r \times {}_r\mu_w = {}_w\mu_a$

(c) ${}_a\mu_r \times {}_r\mu_a = 0$ (d) ${}_a\mu_r / {}_w\mu_r = {}_a\mu_w$

41. The time taken by sunlight to cross a 5 *mm* thick glass plate ($\mu = 3/2$) is [MP PMT/PET 1998; BHU 2005]

(a) 0.25×10^{-10} *s* (b) 0.167×10^{-7} *s*

(c) 2.5×10^{-10} *s* (d) 1.0×10^{-10} *s*

42. The distance travelled by light in glass (refractive index =1.5) in a nanosecond will be [MP PET 1999]

(a) 45 *cm* (b) 40 *cm*

(c) 30 *cm* (d) 20 *cm*

43. When light is refracted from air into glass

[IIT 1980; CBSE PMT 1992; MP PET 1999; MP PMT 1999; RPMT 1997, 2000, 03; MH CET 2004]

(a) Its wavelength and frequency both increase

(b) Its wavelength increases but frequency remains unchanged

(c) Its wavelength decreases but frequency remains unchanged

(d) Its wavelength and frequency both decrease

44. A mark at the bottom of a liquid appears to rise by 0.1 *m*. The depth of the liquid is 1 *m*. The refractive index of the liquid is

(a) 1.33 (b) $\frac{9}{10}$

(c) $\frac{10}{9}$ (b) 1.5

45. A man standing in a swimming pool looks at a stone lying at the bottom. The depth of the swimming pool is *h*. At what distance from the surface of water is the image of the stone formed (Line of vision is normal; Refractive index of water is *n*)

(a) h / n (b) n / h

(c) h (d) hn

46. On heating a liquid, the refractive index generally

[KCET 1994]

(a) Decreases

(b) Increases or decreases depending on the rate of heating

(c) Does not change

(d) Increases

47. If $\hat{i}$ denotes a unit vector along incident light ray, $\hat{r}$ a unit vector along refracted ray into a medium of refractive index μ and $\hat{n}$ unit vector normal to boundary of medium directed towards incident medium, then law of refraction is

[EAMCET (Engg.) 1995]

(a) $\hat{i} . \hat{n} = \mu(\hat{r} . \hat{n})$ (b) $\hat{i} \times \hat{n} = \mu(\hat{n} \times \hat{r})$

(c) $\hat{i} \times \hat{n} = \mu(\hat{r} \times \hat{n})$ (d) $\mu(\hat{i} \times \hat{n}) = \hat{r} \times \hat{n}$

48. The bottom of a container filled with liquid appear slightly raised because of [RPMT 1997]

(a) Refraction (b) Interference

(c) Diffraction (d) Reflection

49. The speed of light in air is 3×10^8 m/s. What will be its speed in diamond whose refractive index is 2.4

[KCET 1993]

(a) 3×10^8 m/s (b) 332 m/s

(c) 1.25×10^8 *m/s* (d) 7.2×10^8 m/s

50. Time taken by the sunlight to pass through a window of thickness 4 mm whose refractive index is 1.5 is

[CBSE PMT 1993]

(a) 2×10^{-8} *sec* (b) 2×10^{8} *sec*

(c) 2×10^{-11} *sec* (d) 2×10^{11} *sec*

51. Ray optics is valid, when characteristic dimensions are [CBSE PMT 1994; CPMT 2001]

(a) Of the same order as the wavelength of light

(b) Much smaller than the wavelength of light

(c) Of the order of one millimetre

(d) Much larger than the wavelength of light

52. The refractive index of water is 1.33. What will be the speed of light in water [CBSE PMT 1996; KCET 1998]

(a) $3\times10^{8}\ m/s$ (b) $2.25\times10^{8}\ m/s$

(c) $4\times10^{8}\ m/s$ (d) 1.33×10^{8} *m/s*

53. The time required to pass the light through a glass slab of 2 *mm* thick is ($\mu_{glass}=1.5$) [AFMC 1997; MH CET 2002, 04]

(a) $10^{-5}\ s$ (b) $10^{-11}\ s$

(c) $10^{-9}\ s$ (d) $10^{-13}\ s$

54. The refractive index of water with respect to air is 4 / 3 and the refractive index of glass with respect to air is 3/2. The refractive index of water with respect to glass is [BHU 1997; JIPMER 2000]

(a) $\frac{9}{8}$ (b) $\frac{8}{9}$

(c) $\frac{1}{2}$ (d) 2

55. Electromagnetic radiation of frequency *n*, wavelength λ, travelling with velocity *v* in air, enters a glass slab of refractive index μ. The frequency, wavelength and velocity of light in the glass slab will be respectively [CBSE PMT 1997]

(a) $\frac{n}{\mu},\frac{\lambda}{\mu},\frac{v}{\mu}$ (b) $n,\frac{\lambda}{\mu},\frac{v}{\mu}$

(c) $n,\lambda,\frac{v}{\mu}$ (d) $\frac{n}{\mu},\ \frac{\lambda}{\mu},v$

56. What is the time taken (in *seconds*) to cross a glass of thickness 4 *mm* and μ = 3 by light [BHU 1998; Pb. PMT 1999, 2001; MH CET 2000; MP PET 2001]

(a) 4×10^{-11} (b) 2×10^{-11}

(c) 16×10^{-11} (d) 8×10^{-10}

57. A plane glass slab is kept over various coloured letters, the letter which appears least raised is [J & K CET 2004; BHU 1998, 05]

(a) Blue (b) Violet

(c) Green (d) Red

58. A ray of light is incident on the surface of separation of a medium at an angle 45° and is refracted in the medium at an angle 30°. What will be the velocity of light in the medium [AFMC 1998; MH CET (Med.) 1999]

(a) 1.96×10^{8} *m/s* (b) $2.12\times10^{8}\ m/s$

(c) $3.18\times10^{8}\ m/s$ (d) $3.33\times18^{8}\ m/s$

59. Absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$. The ratio of velocity of light in glass and water will be [UPSEAT 1999]

(a) 4 : 3 (b) 8 : 7

(c) 8 : 9 (d) 3 : 4

60. The ratio of thickness of plates of two transparent mediums *A* and *B* is 6 : 4. If light takes equal time in passing through them, then refractive index of *B* with respect to *A* will be [UPSEAT 1999]

(a) 1.4 (b) 1.5

(c) 1.75 (d) 1.33

61. The refractive index of water and glass with respect to air is 1.3 and 1.5 respectively. Then the refractive index of glass with respect to water is [MH CET (Med.) 1999]

(a) $\frac{2.6}{1.5}$ (b) $\frac{1.5}{2.6}$

(c) $\frac{1.3}{1.5}$ (d) $\frac{1.5}{1.3}$

62. A tank is filled with benzene to a height of 120 *mm*. The apparent depth of a needle lying at a bottom of the tank is measured by a microscope to be 80 *mm*. The refractive index of benzene is

(a) 1.5 (b) 2.5

(c) 3.5 (d) 4.5

63. Each quarter of a vessel of depth *H* is filled with liquids of the refractive indices *n*, *n*, *n* and *n* from the bottom respectively. The apparent depth of the vessel when looked normally is

(a) $\frac{H(n_1+n_2+n_3+n_4)}{4}$ (b) $\frac{H\left(\frac{1}{n_1}+\frac{1}{n_2}+\frac{1}{n_3}+\frac{1}{n_4}\right)}{4}$

(c) $\frac{(n_1+n_2+n_3+n_4)}{4H}$ (d) $\frac{H\left(\frac{1}{n_1}+\frac{1}{n_2}+\frac{1}{n_3}+\frac{1}{n_4}\right)}{2}$

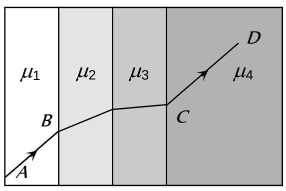
64. A ray of light passes through four transparent media with refractive indices $\mu_1 . \mu_2 \mu_3$, and μ_4 as shown in the figure. The surfaces of all media are parallel. If the emergent ray *CD* is parallel to the incident ray *AB*, we must have [IIT-JEE (Screening) 2001]

(a) $\mu_1=\mu_2$

(b) $\mu_2=\mu_3$

(c) $\mu_3=\mu_4$

(d) $\mu_4=\mu_1$



65. The reason of seeing the Sun a little before the sunrise is [MP PMT 2001; Orissa JEE 2003]

(a) Reflection of the light (b) Refraction of the light

(c) Scattering of the light (d) Dispersion of the light

66. An under water swimmer is at a depth of 12 *m* below the surface of water. A bird is at a height of 18 *m* from the surface of water, directly above his eyes. For the swimmer the bird appears to be at a distance from the surface of water equal to (Refractive Index of water is 4/3)

[KCET (Engg.) 2001]

(a) 24 *m* (b) 12 *m*

(c) 18 *m* (d) 9 *m*

67. The optical path of a monochromatic light is same if it goes through 4.0 *cm* of glass or 4.5 *cm* of water. If the refractive index of glass is 1.53, the refractive index of the water is

[UPSEAT 2002]

(a) 1.30 (b) 1.36

(c) 1.42 (d) 1.46

68. Which of the following statement is true [Orissa JEE 2002]

(a) Velocity of light is constant in all media

(b) Velocity of light in vacuum is maximum

(c) Velocity of light is same in all reference frames

(d) Laws of nature have identical form in all reference frames

69. A ray of light is incident on a transparent glass slab of refractive index 1.62. The reflected and the refracted rays are mutually perpendicular. The angle of incidence is

[MP PET 2002]

(a) 58.3° (b) 50°

(c) 35° (d) 30°

70. A microscope is focussed on a coin lying at the bottom of a beaker. The microscope is now raised up by 1 *cm*. To what depth should the water be poured into the beaker so that coin is again in focus ? (Refractive index of water is $\frac{4}{3}$)

[BHU 2003]

(a) 1 *cm* (b) $\frac{4}{3}$ *cm*

(c) 3 *cm* (d) 4 *cm*

71. Velocity of light in glass whose refractive index with respect to air is 1.5 is 2×10^{8} *m/s* and in certain liquid the velocity of light found to be 2.5×10^{8} *m/s*. The refractive index of the liquid with respect to air is [CPMT 1978; MP PET/PMT 1988]

(a) 0.64 (b) 0.80

(c) 1.20 (d) 1.44

72. Stars are twinkling due to [CPMT 1997]

(a) Diffraction (b) Reflection

(c) Refraction (d) Scattering

73. A thin oil layer floats on water. A ray of light making an angle of incidence of 40° shines on oil layer. The angle of refraction of light ray in water is ($\mu_{oil} = 1.45, \mu_{water} = 1.33$)

[MP PMT 1993]

(a) 36 .1° (b) 44.5°

(c) 26. 8° (d) 28.9°

74. An object is immersed in a fluid. In order that the object becomes invisible, it should [AIIMS 2004]

(a) Behave as a perfect reflector

(b) Absorb all light falling on it

(c) Have refractive index one

(d) Have refractive index exactly matching with that of the surrounding fluid

75. When light travels from glass to air, the incident angle is θ_1 and the refracted angle is θ_2. The true relation is

[Orissa PMT 2004]

(a) $\theta_1 = \theta_2$ (b) $\theta_1 < \theta_2$

(c) $\theta_1 > \theta_2$ (d) Not predictable

76. Velocity of light in a medium is $1.5 \times 10^8 \, m/s$. Its refractive index will be [Pb. PET 2000]

(a) 8 (b) 6

(c) 4 (d) 2

77. The frequency of a light ray is $6 \times 10^{14} \, Hz$. Its frequency when it propagates in a medium of refractive index 1.5, will be

[MP PMT 2000; DPMT 2003; Pb PMT 2003; MH CET 2004]

(a) $1.67 \times 10^{14} \, Hz$ (b) $9.10 \times 10^{14} \, Hz$

(c) $6 \times 10^{14} \, Hz$ (d) $4 \times 10^{14} \, Hz$

78. The refractive indices of water and glass with respect to air are 1.2 and 1.5 respectively. The refractive index of glass with respect to water is [Pb. PET 2002]

(a) 0.6 (b) 0.8

(c) 1.25 (d) 1.75

79. The wavelength of sodium light in air is 5890 *Å*. The velocity of light in air is $3 \times 10^8 ms^{-1}$. The wavelength of light in a glass of refractive index 1.6 would be close to

[DCE 2003]

(a) 5890 *Å* (b) 3681 *Å*

(c) 9424 *Å* (d) 15078 *Å*

80. The mean distance of sun from the earth is $1.5 \times 10^8 \, Km$ (nearly). The time taken by the light to reach earth from the sun is

(a) 0.12 *min* (b) 8.33 *min*

(c) 12.5 *min* (d) 6.25 *min*

81. Refractive index of air is 1.0003. The correct thickness of air column which will have one more wavelength of yellow light (6000 Å) than in the same thickness in vacuum is

[RPMT 1995]

(a) 2 *mm* (b) 2 *cm*

(c) 2 *m* (d) 2 *km*

82. The wavelength of light in air and some other medium are respectively λ_a and λ_m. The refractive index of medium is

[RPMT 2003]

(a) λ_a / λ_m (b) λ_m / λ_a

(c) $\lambda_a \times \lambda_m$ (d) None of these

83. An astronaut in a spaceship see the outer space as

[CPMT 1990, MP PMT 1991; JIPMER 1997]

(a) White (b) Black

(c) Blue (d) Red

84. Speed of light is maximum in

[CPMT 1990; MP PMT 1994; AFMC 1996]

(a) Water (b) Air

(c) Glass (d) Diamond

85. Which one of the following statements is correct

[KCET 1994]

(a) In vacuum, the speed of light depends upon frequency

(b) In vacuum, the speed of light does not depend upon frequency

(c) In vacuum, the speed of light is independent of frequency and wavelength

(d) In vacuum, the speed of light depends upon wavelength

86. If the wavelength of light in vacuum be λ, the wavelength in a medium of refractive index n will be

[UPSEAT 2001; MP PET 2001]

(a) $n\lambda$ (b) $\frac{\lambda}{n}$

(c) $\frac{\lambda}{n^2}$ (d) $n\lambda$

87. In vacuum the speed of light depends upon [MP PMT 2001]

(a) Frequency

(b) Wave length

(c) Velocity of the source of light

(d) None of these

88. A transparent cube of 15 *cm* edge contains a small air bubble. Its apparent depth when viewed through one face is 6 *cm* and when viewed through the opposite face is 4 *cm*. Then the refractive index of the material of the cube is

[CPMT 2004; MP PMT 2005]

(a) 2.0 (b) 2.5

(c) 1.6 (d) 1.5

89. A glass slab of thickness 3 *cm* and refractive index 3/2 is placed on ink mark on a piece of paper. For a person looking at the mark at a distance 5.0 *cm* above it, the distance of the mark will appear to be [Kerala PMT 2005]

(a) 3.0 *cm* (b) 4.0 *cm*

(c) 4.5 *cm* (d) 5.0 *cm*

90. A fish at a depth of 12 *cm* in water is viewed by an observer on the bank of a lake. To what height the image of the fish is raised.

(a) 9 *cm* (b) 12 *cm*

(c) 3.8 *cm* (d) 3 *cm*

Total Internal Reflection

1. A cut diamond sparkles because of its

[NCERT 1974; RPET 1996; AFMC 2005]

(a) Hardness

(b) High refractive index

(c) Emission of light by the diamond

(d) Absorption of light by the diamond

2. A diver in a swimming pool wants to signal his distress to a person lying on the edge of the pool by flashing his water proof flash light [NCERT 1972]

(a) He must direct the beam vertically upwards

(b) He has to direct the beam horizontally

(c) He has to direct the beam at an angle to the vertical which is slightly less than the critical angle of incidence for total internal reflection

(d) He has to direct the beam at an angle to the vertical which is slightly more than the critical angle of incidence for the total internal reflection

3. Finger prints on a piece of paper may be detected by sprinkling fluorescent powder on the paper and then looking it into

(a) Mercury light (b) Sunlight

(c) Infrared light (d) Ultraviolet light

4. Critical angle of light passing from glass to air is minimum for

(a) Red (b) Green

(c) Yellow (d) Violet

5. The wavelength of light in two liquids '*x*' and '*y*' is 3500 Å and 7000 Å, then the critical angle of x relative to y will be

(a) 60° (b) 45°

(c) 30° (d) 15°

6. A fish is a little away below the surface of a lake. If the critical angle is 49°, then the fish could see things above the water surface within an angular range of θ^o where

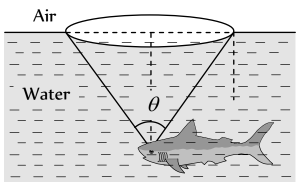
[MP PMT 1986]

(a) $\theta = 49^o$

(b) $\theta = 90^o$

(c) $\theta = 98^o$

(d) $\theta = 24\frac{1}{2}^o$



7. If the critical angle for total internal reflection from a medium to vacuum is 30°, the velocity of light in the medium is

KCET 2000; BCECE 2003; RPMT 2003]

(a) 3×10^8 *m/s* (b) 1.5×10^8 *m/s*

(c) 6×10^8 *m/s* (d) $\sqrt{3}\times10^8$ *m/s*

8. A ray of light is incident at an angle *i* from denser to rare medium. The reflected and the refracted rays are mutually perpendicular. The angle of reflection and the angle of refraction are respectively *r* and *r'*, then the critical angle will be

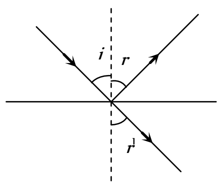
CBSE PMT 1996; MP PMT 1985, 99; Pb. PET 2002]

(a) $\sin^{-1}(\sin r)$

(b) $\sin^{-1}(\tan r')$ [MP PET 2005]

(c) $\sin^{-1}(\tan i)$

(d) $\tan^{-1}(\sin i)$



9. For total internal reflection to take place, the angle of incidence i and the refractive index μ of the medium must satisfy the inequality **[MP PET 1994]**

(a) $\frac{1}{\sin i} < \mu$ (b) $\frac{1}{\sin i} > \mu$

(c) $\sin i < \mu$ (d) $\sin i > \mu$

10. Total internal reflection of light is possible when light enters from

(a) Air to glass (b) Vacuum to air

(c) Air to water (d) Water to air

11. Total internal reflection of a ray of light is possible when the (i_c = critical angle, i = angle of incidence)

[NCERT 1977; MP PMT 1994]

(a) Ray goes from denser medium to rarer medium and $i < i_c$

(b) Ray goes from denser medium to rarer medium and $i > i_c$

(c) Ray goes from rarer medium to denser medium and $i > i_c$

(d) Ray goes from rarer medium to denser medium and $i < i_c$

12. A diver at a depth of $12m$ in water ($\mu = 4/3$) sees the sky in a cone of semi-vertical angle

[KCET 1999; Pb. PMT 2002; MP PMT 1995, 2003]

(a) $\sin^{-1}(4/3)$ (b) $\tan^{-1}(4/3)$

(c) $\sin^{-1}(3/4)$ (d) $90°$

13. Critical angle is that angle of incidence in the denser medium for which the angle of refraction in rarer medium is

[MP PMT 1996]

(a) $0°$ (b) $57°$

(c) $90°$ (d) $180°$

14. The critical angle for diamond (refractive index = 2) is

[MP PET 2003]

(a) About $20°$ (b) $60°$

(c) $45°$ (d) $30°$

15. The reason for shining of air bubble in water is

[MP PET 1997; KCET 1999]

(a) Diffraction of light

(b) Dispersion of light

(c) Scattering of light

(d) Total internal reflection of light

16. With respect to air critical angle in a medium for light of red colour $[\lambda_1]$ is θ. Other facts remaining same, critical angle for light of yellow colour $[\lambda_2]$ will be **[MP PET 1999]**

(a) θ (b) More than θ

(c) Less than θ (d) $\frac{\theta\lambda_1}{\lambda_2}$

17. 'Mirage' is a phenomenon due to

[AIIMS 1998; MP PET 2002; AFMC 2003]

(a) Reflection of light

(b) Refraction of light

(c) Total internal reflection of light

(d) Diffraction of light

18. A ray of light travelling in a transparent medium falls on a surface separating the medium from air at an angle of incidence of 45°. The ray undergoes total internal reflection. If n is the refractive index of the medium with respect to air, select the possible value (s) of n from the following

[IIT-JEE 1998]

[CPMT 1973; MP PMT 1994]

(a) 1.3 (b) 1.4

(c) 1.5 (d) 1.6

19. When a ray of light emerges from a block of glass, the critical angle is **[KCET 1994]**

(a) Equal to the angle of reflection

(b) The angle between the refracted ray and the normal

(c) The angle of incidence for which the refracted ray travels along the glass-air boundary

(d) The angle of incidence

20. The phenomenon utilised in an optical fibre is

[KCET 1994; AMU 1995; CBSE PMT 2001; DCE 1999, 2000, 01, 02; AIEEE 2002]

(a) Refraction (b) Interference

(c) Polarization (d) Total internal reflection

21. The refractive index of water is $4/3$ and that of glass is 5/3. What will be the critical angle for the ray of light entering water from the glass **[RPMT 1996]**

(a) $\sin^{-1}\frac{4}{5}$ (b) $\sin^{-1}\frac{5}{4}$

(c) $\sin^{-1}\frac{1}{2}$ (d) $\sin^{-1}\frac{2}{1}$

22. Total internal reflection is possible when light rays travel

[RPMT 1999]

(a) Air to water (b) Air to glass

(c) Glass to water (d) Water to glass

23. The velocity of light in a medium is half its velocity in air. If ray of light emerges from such a medium into air, the angle of incidence, at which it will be totally internally reflected, is

[Roorkee 1999]

(a) 15° (b 30°

(c) 45° (d) 60°

24. A ray of light propagates from glass (refractive index = 3/2) to water (refractive index = 4/3). The value of the critical angle **[JIPMER 1999; UPSEAT 20**

(a) sin (1/2) (b) $\sin^{-1}\left(\frac{\sqrt{8}}{9}\right)$

(c) $\sin^{-1}(8/9)$ (d) $\sin^{-1}(5/7)$

25. Relation between critical angles of water and glass is

[CBSE PMT 2000; Pb. PET 2000; CPMT 2001]

(a) $C_w > C_g$ (b) $C_w < C_g$

(c) $C_w = C_g$ (d) $C_w = C_g = 0$

26. If critical angle for a material to air is 30°, the refractive index of the material will be **[MP PET 2001]**

(a) 1.0 (b) 1.5

(c) 2.0 (d) 2.5

27. The refractive index of water is 1.33. The direction in which a man under water should look to see the setting sun is

[MP PET 1991; Kerala PET 2002; Pb. PET 2003]

(a) 49° to the horizontal (b) 90° with the vertical

(c) 49° to the vertical (d) Along the horizontal

28. Optical fibres are related with [AFMC 2002]

(a) Communication (b) Light

(c) Computer (d) None of these

29. Brilliance of diamond is due to

[AIIMS 2002; MP PMT 2003]

(a) Shape (b) Cutting

(c) Reflection (d) Total internal reflection

30. A light ray from air is incident (as shown in figure) at one end of a glass fiber (refractive index $\mu = 1.5$) making an incidence angle of 60° on the lateral surface, so that it undergoes a total internal reflection. How much time would it take to traverse the straight fiber of length 1 *km*

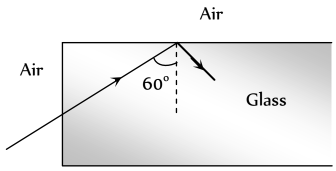
[Orissa JEE 2002]

(a) 3.33 μ *sec*

(b) 6.67 μ *sec*

(c) 5.77 μ *sec*

(d) 3.85 μ *sec*



31. Light wave enters from medium 1 to medium 2. Its velocity in 2nd medium is double from 1st. For total internal reflection the angle of incidence must be greater than [CPMT 2002]

(a) 30° (b) 60°

(c) 45° (d) 90°

32. Consider telecommunication through optical fibres. Which of the following statements is not true

[AIEEE 2003]

(a) Optical fibres may have homogeneous core with a suitable cladding

(b) Optical fibres can be of graded refractive index

(c) Optical fibres are subject to electromagnetic interference from outside

(d) Optical fibres have extremely low transmission loss

33. The critical angle for a medium is 60^o. The refractive index of the medium is [MP PMT 2004]

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{2}}{3}$

(c) $\sqrt{3}$ (d) $\frac{\sqrt{3}}{2}$

34. Glass has refractive index μ with respect to air and the critical angle for a ray of light going from glass to air is θ. If a ray of light is incident from air on the glass with angle of incidence θ, the corresponding angle of refraction is

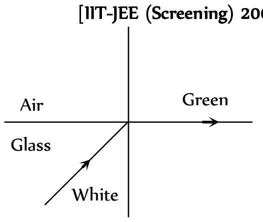
[MP PMT 2004]

(a) $\sin^{-1}\left(\frac{1}{\sqrt{\mu}}\right)$ (b) 90^o

(c) $\sin^{-1}\left(\frac{1}{\mu^2}\right)$ (d) $\sin^{-1}\left(\frac{1}{\mu}\right)$

35. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected then the emerging ray in air contains

[IIT-JEE (Screening) 2004]



(a) Yellow, orange, red

(b) Violet, indigo, blue

(c) All colours

(d) All colours except green

36. Material *A* has critical angle i_A, and material *B* has critical angle $i_B (i_B > i_A)$. Then which of the following is true

(i) Light can be totally internally reflected when it passes from *B* to *A*

(ii) Light can be totally internally reflected when it passes from *A* to *B*

(iii) Critical angle for total internal reflection is $i_B - i_A$

(iv) Critical angle between *A* and *B* is $\sin^{-1}\left(\frac{\sin i_A}{\sin i_B}\right)$

[UPSEAT 2004]

(a) (i) and (iii) (b) (i) and (iv)

(c) (ii) and (iii) (d) (ii) and (iv)

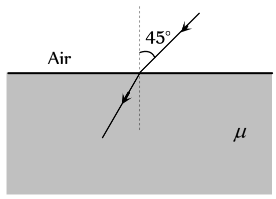
37. In the figure shown, for an angle of incidence 45^o, at the top surface, what is the minimum refractive index needed for total internal reflection at vertical face [DCE 2002]

(a) $\frac{\sqrt{2}+1}{2}$

(b $\sqrt{\frac{3}{2}}$

(c) $\sqrt{\frac{1}{2}}$

(d) $\sqrt{2}+1$



38. Critical angle for light going from medium (i) to (ii) is θ. The speed of light in medium (i) is v then speed in medium (ii) is

[DCE 2002]

(a) $v(1-\cos\theta)$ (b $v/\sin\theta$

(c) $v/\cos\theta$ (d) $v(1-\sin\theta)$

39. If light travels a distance *x* in t_1 *sec* in air and 10 *x* distance in t_2 *sec* in a medium, the critical angle of the medium will be

(a) $\tan^{-1}\left(\frac{t_1}{t_2}\right)$ (b) $\sin^{-1}\left(\frac{t_1}{t_2}\right)$

(c) $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$ (d) $\tan^{-1}\left(\frac{10t_1}{t_2}\right)$

40. The critical angle of a medium with respect to air is 45^o. The refractive index of medium is [MH CET 2003]

(a) 1.41 (b) 1.2

(c) 1.5 (d) 2

41. An endoscope is employed by a physician to view the internal parts of a body organ. It is based on the principle of

[AIIMS 2004]

(a) Refraction (b) Reflection

(c) Total internal reflection (d) Dispersion

42. A normally incident ray reflected at an angle of 90^o. The value of critical angle is [RPMT 1996]

(a) 45^o (b 90^o

(c) 65^o (d) 43.2^o

43. The phenomena of total internal reflection is seen when angle of incidence is [RPMT 2001]

(a) 90^o (b Greater than critical angle

(c) Equal to critical angle (d) 0^o

44. A fish looking up through the water sees the outside world contained in a circular horizon. If the refractive index of water is $\frac{4}{3}$ and the fish is 12 *cm* below the surface, the radius of this circle in *cm* is

[NCERT 1980; KCET 2002; AIEEE 2005; CPMT 2005]

(a) $36\sqrt{5}$ (b) $4\sqrt{5}$

(c) $36\sqrt{7}$ (d) $36/\sqrt{7}$

45. A point source of light is placed 4 *m* below the surface of water of refractive index 5 / 3. The minimum diameter of a disc which should be placed over the source on the surface of water to cut–off all light coming out of water is

[CBSE PMT 1994; JIPMER 2001, 02]

(a) 2 *m* (b 6 *m*

(c) 4 *m* (d) 3 *m*

46. A fist looking from within water sees the outside world through a circular horizon. If the fish $\sqrt{7}$ *cm* below the surface of water, what will be the radius of the circular horizon

(a) 3.0 *cm* (b) 4.0 *cm*

(c) 4.5 *cm* (d) 5.0 *cm*

Refraction at Curved Surface

1. The radius of curvature for a convex lens is 40 *cm*, for each surface. Its refractive index is 1.5. The focal length will be

[MP PMT 1989]

(a) 40 *cm* (b) 20 *cm*

(c) 80 *cm* (d) 30 *cm*

2. A convex lens of focal length f is placed somewhere in between an object and a screen. The distance between the object and the screen is x. If the numerical value of the magnification produced by the lens is m, , then the focal length of the lens is

(a) $\frac{mx}{(m+1)^2}$ (b) $\frac{mx}{(m-1)^2}$

(c) $\frac{(m+1)^2}{m}x$ (d) $\frac{(m-1)^2}{m}x$

3. A thin lens focal length f_1 and its aperture has diameter *d*. It forms an image of intensity *I*. Now the central part of the aperture upto diameter $\frac{d}{2}$ is blocked by an opaque paper. The focal length and image intensity will change to

[CPMT 1989; MP PET 1997; KCET 1998]

(a) $\frac{f}{2}$ and $\frac{I}{2}$ (b) f and $\frac{I}{4}$

(c) $\frac{3f}{4}$ and $\frac{I}{2}$ (d) f and $\frac{3I}{4}$

4. A lens of power + 2 diopters is placed in contact with a lens of power – 1 diopter. The combination will behave like

(a) A convergent lens of focal length 50 *cm*

(b) A divergent lens of focal length 100 *cm*

(c) A convergent lens of focal length 100 *cm*

(d) A convergent lens of focal length 200 *cm*

5. A convex lens of focal length 40 cm is in contact with a concave lens of focal length 25 cm. The power of combination is

[IIT-JEE 1982; AFMC 1997; CBSE PMT 2000; RPMT 2003]

(a) – 1.5 *D* (b) – 6.5 *D*

(c) + 6.5 *D* (d) + 6.67 *D*

6. Two lenses are placed in contact with each other and the focal length of combination is 80 *cm*. If the focal length of one is 20 *cm*, then the power of the other will be

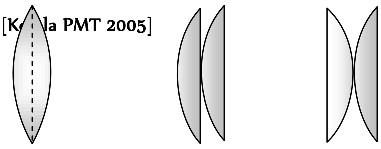
[NCERT 1981]

(a) 1.66 *D* (b) 4.00 *D*

(c) –1.00 *D* (d) – 3.75 *D*

7. Two similar plano-convex lenses are combined together in three different ways as shown in the adjoining figure. The ratio of the focal lengths in three cases will be

[Kerala PMT 2005]



(a) 2 : 2 : 1 (b 1 : 1 : 1

(c) 1 : 2 : 2 (d) 2 : 1 : 1

8. Two lenses of power +12 and – 2 diopters are placed in contact. What will the focal length of combination

[MP PET 1990; MNR 1987;

MH CET (Med.) 2001; UPSEAT 2000; Pb. PMT 2003]

(a) 10 *cm* (b) 12.5 *cm*

(c) 16.6 *cm* (d) 8.33 *cm*

9. A concave and convex lens have the same focal length of 20 cm and are put into contact to form a lens combination. The combination is used to view an object of 5 *cm* length kept at 20 cm from the lens combination. As compared to the object, the image will be

(a) Magnified and inverted

(b) Reduced and erect

(c) Of the same size as the object and erect

(d) Of the same size as the object but inverted

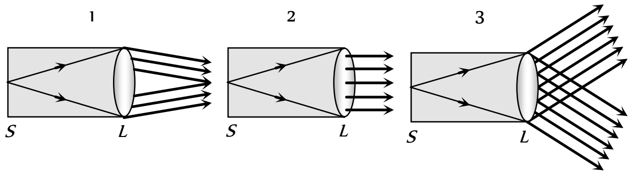
10. If in a plano-convex lens, the radius of curvature of the convex surface is 10 *cm* and the focal length of the lens is 30 *cm*, then the refractive index of the material of lens will be

[CPMT 1986; MNR 1988; MP PMT 2002; UPSEAT 2000]

(a) 1.5 (b) 1.66

(c) 1.33 (d) 3

11. The slit of a collimator is illuminated by a source as shown in the adjoining figures. The distance between the slit *S* and the collimating lens *L* is equal to the focal length of the lens. The correct direction of the emergent beam will be as shown in figure



(a) 1 (b) 3

(c) 2 (d) None of the figures

12. A converging lens is used to form an image on a screen. When upper half of the lens is covered by an opaque screen

[IIT-JEE 1986; SCRA 1994; MP PET 1996; MP PMT 2004; BHU 1998, 05]

(a) Half the image will disappear

(b) Complete image will be formed of same intensity

(c) Half image will be formed of same intensity

(d) Complete image will be formed of decreased intensity

13. A thin convex lens of focal length 10 *cm* is placed in contact with a concave lens of same material and of same focal length. The focal length of combination will be

[CPMT 1972; 1988]

(a) Zero (b) Infinity

(c) 10 *cm* (d) 20 *cm*

14. A convex lens of focal length 84 *cm* is in contact with a concave lens of focal length 12 *cm*. The power of combination (in diopters) is

(a) 25/24 (b) 25/18

(c) – 50/7 (d) + 50/7

15. A convex lens makes a real image 4 *cm* long on a screen. When the lens is shifted to a new position without disturbing the object, we again get a real image on the screen which is 16 *cm* tall. The length of the object must be [MP PET 1991]

(a) 1/4 *cm* (b) 8 *cm*

(c) 12 *cm* (d) 20 *cm*

16. A glass convex lens $(\mu_g = 1.5)$ has a focal length of $8\,cm$ when placed in air. What would be the focal length of the lens when it is immersed in water ($\mu_w = 1.33$)

[BHU 1994; MP PMT 1996]

(a) 2 *m* (b) 4 *cm*

(c) 16 *cm* (d) 32 *cm*

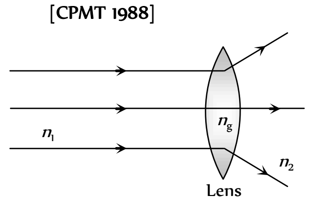
17. The ray diagram could be correct [CPMT 1988]

(a) If $n_1 = n_2 = n_g$

(b) If $n_1 = n_2$ and $n_1 < n_g$

(c) If $n_1 = n_2$ and $n_1 > n_g$

(d) Under no circumstances



18. A thin convex lens of refractive index 1.5 has a focal length of 15 *cm* in air. When the lens is placed in liquid of refractive index 4/3 , its focal length will be

[CPMT 1974, 77; MP PMT 1992]

(a) 15 *cm* (b) 10 *cm*

(c) 30 *cm* (d) 60 *cm*

19. A glass lens is placed in a medium in which it is found to behave like a glass plate. Refractive index of the medium will be

(a) Greater than the refractive index of glass

(b) Smaller than the refractive index of glass

(c) Equal to refractive index of glass [CPMT 1986]

(d) No case will be possible from above

20. If I_1 and I_2 be the size of the images respectively for the two positions of lens in the displacement method, then the size of the object is given by [CPMT 1988]

(a) I_1 / I_2 (b) $I_1 \times I_2$

(c) $\sqrt{I_1 \times I_2}$ (d) $\sqrt{I_1 / I_2}$

21. A convex lens of crown glass (n =1.525) will behave as a divergent lens if immersed in [CPMT 1984]

(a) Water (n =1.33)

(b) In a medium of n = 1.525

(c) Carbon disulphide n =1.66

(d) It cannot act as a divergent lens

22. A divergent lens will produce [CPMT 1984]

(a) Always a virtual image

(b) Always real image

(c) Sometimes real and sometimes virtual

(d) None of the above

23. The minimum distance between an object and its real image formed by a convex lens is [CPMT 1973; JIPMER 1997]

(a) 1.5 f (b) 2 f

(c) 2.5 f [MP PET 1991] (d) 4 f

24. An object is placed at a distance of 20 *cm* from a convex lens of focal length 10 *cm*. The image is formed on the other side of the lens at a distance [CPMT 1971; RPET 2003]

(a) 20 *cm* (b) 10 *cm*

(c) 40 *cm* (d) 30 *cm*

25. Two thin lenses, one of focal length + 60 *cm* and the other of focal length – 20 *cm* are put in contact. The combined focal length is [CPMT 1973, 8

(a) + 15 *cm* (b) – 15 *cm*

(c) + 30 *cm* (d) –30 *cm*

26. A double convex lens of focal length 20 *cm* is made of glass of refractive index 3 / 2. When placed completely in water $({}_a\mu_w = 4/3)$, its focal length will be

[CBSE PMT 1990; MP PMT/PET 1998]

(a) 80 *cm* (b) 15 *cm*

(c) 17.7 *cm* (d) 22.5 *cm*

27. Two thin lenses of focal lengths 20 *cm* and 25 *cm* are placed in contact convex. The effective power of the combination is

[CBSE PMT 1990; RPMT 2001]

(a) 45 *dioptres* (b) 9 *dioptres*

(c) 1/9 *dioptre* (d) 6 *dioptres*

28. An object is placed at a distance of $f/2$ from a convex lens. The image will be [CPMT 1974, 89]

(a) At one of the foci, virtual and double its size

(b) At $3f/2$, real and inverted

(c) At $2f$, virtual and erect

(d) None of these

29. A double convex thin lens made of glass (refractive index μ = 1.5) has both radii of curvature of magnitude 20 *cm*. Incident light rays parallel to the axis of the lens will converge at a distance L such that [MNR 1991; MP PET 1996; UPSEAT 2000; Pb PET 2004]

(a) $L = 20$ *cm* (b) $L = 10$ *cm*

(c) $L = 40$ *cm* (d) $L = 20/3$ *cm*

30. A lens behaves as a converging lens in air and a diverging lens in water. The refractive index of the material is [CPMT 1991; NCERT 1979; BHU 2005]

(a) Equal to unity (b) Equal to 1.33

(c) Between unity and 1.33 (d) Greater than 1.33

31. A biconvex lens forms a real image of an object placed perpendicular to its principal axis. Suppose the radii of curvature of the lens tend to infinity. Then the image would [MP PET 1994]

(a) Disappear

(b) Remain as real image still

(c) Be virtual and of the same size as the object

(d) Suffer from aberrations

32. The radius of curvature of convex surface of a thin plano-convex lens is 15 *cm* and refractive index of its material is 1.6. The power of the lens will be [MP PMT 1994]

(a) $+1\,D$ (b) $-2\,D$

(c) $+3\,D$ (d) $+4\,D$

33. Focal length of a convex lens will be maximum for [MP PMT 1994]

(a) Blue light (b) Yellow light

(c) Green light (d) Red light

34. A lens is placed between a source of light and a wall. It forms images of area A_1 and A_2 on the wall for its two different positions. The area of the source or light is [CBSE PMT 1995]

(a) $\dfrac{A_1 + A_2}{2}$ (b) $\left[\dfrac{1}{A_1} + \dfrac{1}{A_2}\right]^{-1}$

(c) $\sqrt{A_1 A_2}$ (d) $\left[\dfrac{\sqrt{A_1} + \sqrt{A_2}}{2}\right]^2$

35. Two lenses of power $6D$ and $-2D$ are combined to form a single lens. The focal length of this lens will be [MP PET 2003]

(a) $\dfrac{3}{2}\,m$ (b $\dfrac{1}{4}\,m$

(c) 4 m (d) $\dfrac{1}{8}\,m$

36. A combination of two thin lenses with focal lengths f_1 and f_2 respectively forms an image of distant object at distance 60 *cm* when lenses are in contact. The position of this image shifts by 30 *cm* towards the combination when two lenses are separated by 10 *cm*. The corresponding values of f_1 and f_2 are

(a) $30\ cm, -60\ cm$ (b) $20\ cm, -30\ cm$

(c) $15\ cm, -20\ cm$ (d) $12\ cm, -15\ cm$

37. An achromatic combination of lenses is formed by joining [BHU 1995; Pb. PMT 2000, 04]

(a) 2 convex lenses

(b) 2 concave lenses

(c) 1 convex lens and 1 concave lens

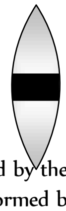
(d) Convex lens and plane mirror

38. A plano convex lens $(f = 20cm)$ is silvered at plane surface. Now f will be [BHU 1995; DPMT 2001; MP PMT 2005]

(a) 20 *cm* (b) 40 *cm*

(c) 30 cm (d) 10 *cm*

39. If the central portion of a convex lens is wrapped in black paper as shown in the figure [Manipal MEE 1995; KCET 2001]



(a) No image will be formed by the remaining portion of the lens

(b) The full image will be formed but it will be less bright

(c) The central portion of the image will be missing

(d) There will be two images each produced by one of the exposed portions of the lens

40. A diminished image of an object is to be obtained on a screen 1.0 m from it. This can be achieved by appropriately placing

(a) A convex mirror of suitable focal length

(b) A concave mirror of suitable focal length

(c) A concave lens of suitable focal length

(d) A convex lens of suitable focal length less than 0.25 m

41. The focal length of convex lens is 30 *cm* and the size of image is quarter of the object, then the object distance is [AFMC 1995]

(a) 150 *cm* (b) 60 *cm*

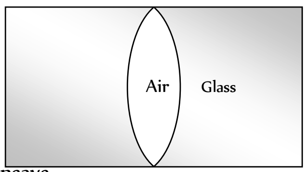
(c) 30 *cm* (d) 40 *cm*

42. A convex lens forms a real image of a point object placed on its principal axis. If the upper half of the lens is painted black, the image will [MP PET 1995]

(a) Be shifted downwards (b) Be shifted upwards

(c) Not be shifted (d) Shift on the principal axis

43. In the figure, an air lens of radii of curvature 10 *cm* ($R_1 = R_2$ = 10 *cm*) is cut in a cylinder of glass $(\mu = 1.5)$. The focal length and the nature of the lens is [MP PET 1995; Pb. PET 2000]



(a) 15 *cm*, concave

(b) 15 *cm*, convex

(c) ∞, neither concave nor convex

(d) 0, concave

44. A lens (focal length 50 *cm*) forms the image of a distant object which subtends an angle of 1 *milliradian* at the lens. What is the size of the image [MP PMT 1995]

(a) 5 *mm* (b) 1 *mm*

(c) 0.5 *mm* [AIIMS 1995] (d) 0.1 *mm*

45. A convex lens of focal length 12 *cm* is made of glass of $\mu = \dfrac{3}{2}$. What will be its focal length when immersed in liquid of $\mu = \dfrac{5}{4}$

(a) 6 *cm* (b) 12 *cm*

(c) 24 *cm* (d) 30 *cm*

46. Two thin lenses of focal lengths f_1 and f_2 are in contact and coaxial. The combination is equivalent to a single lens of power [MP PET 1996, 98; MP PMT 1998; DCE 2000; UP SEAT 2005]

(a) $f_1 + f_2$ (b) $\frac{f_1 f_2}{f_1 + f_2}$

(c) $\frac{1}{2}(f_1 + f_2)$ (d) $\frac{f_1 + f_2}{f_1 f_2}$

47. A plano convex lens is made of glass of refractive index 1.5. The radius of curvature of its convex surface is *R*. Its focal length is

(a) *R*/ 2 (b) *R*

(c) 2*R* (d) 1.5 *R*

48. Two lenses have focal lengths f_1 and f_2 and their dispersive powers are ω_1 and ω_2 respectively. They will together form an achromatic combination if

(a) $\omega_1 f_1 = \omega_2 f_2$ (b) $\omega_1 f_2 + \omega_2 f_1 = 0$

(c) $\omega_1 + f_1 = \omega_2 + f_2$ (d) $\omega_1 - f_1 = \omega_2 - f_2$

49. The dispersive powers of glasses of lenses used in an achromatic pair are in the ratio 5 : 3. If the focal length of the concave lens is 15 *cm*, then the nature and focal length of the other lens would be

(a) Convex, 9 *cm* (b) Concave, 9 *cm*

(c) Convex, 25 *cm* (d) Concave, 25 *cm*

50. A thin double convex lens has radii of curvature each of magnitude 40 *cm* and is made of glass with refractive index 1.65. Its focal length is nearly [MP PMT 1997]

(a) 20 *cm* (b) 31 *cm*

(c) 35 *cm* (d) 50 *cm*

51. The plane surface of a plano-convex lens of focal length *f* is silvered. It will behave as [MP PMT/PET 1998]

(a) Plane mirror

(b) Convex mirror of focal length 2 f

(c) Concave mirror of focal length $f / 2$

(d) None of the above

52. An equiconvex lens of glass of focal length 0.1 metre is cut along a plane perpendicular to principle axis into two equal parts. The ratio of focal length of new lenses formed is

[MP PET 1999; DPMT 2000]

(a) 1 : 1 (b) 1 : 2

(c) 2 : 1 (d) $2 : \frac{1}{2}$

53. A lens of refractive index n is put in a liquid of refractive index n' of focal length of lens in air is f, its focal length in liquid will be

(a) $-\frac{fn'(n-1)}{n'-n}$ (b) $-\frac{f(n'-n)}{n'(n-1)}$

(c) $-\frac{n'(n-1)}{f(n'-n)}$ (d) $\frac{fn'\ n}{n-n'}$

54. An object of height 1.5 *cm* is placed on the axis of a convex lens of focal length 25 *cm*. A real image is formed at a distance of 75 *cm* from the lens. The size of the image will be

(a) 4.5 *cm* (b) 3.0 *cm*

(c) 0.75 *cm* (d) 0.5 *cm*

55. A symmetric double convex lens is cut in two equal parts by a plane perpendicular to the principal axis. If the power of the original lens was 4 *D*, the power of a cut lens will be

[MP PMT 1999]

(a) 2 *D* (b) 3 *D*

(c) 4 *D* (d) 5 *D*

56. A plane convex lens is made of refractive index 1.6. The radius of curvature of the curved surface is 60 *cm*. The focal length of the lens is [CBSE PMT 1999; Pb. PMT 1999; BHU 2001; Very Similar to BHU 2003]

(a) 50[RPET 2003] (b) 100 *cm*

(c) 200 *cm* (d) 400 *cm*

57. A concave lens of glass, refractive index 1.5, has both surfaces of same radius of curvature *R*. On immersion in a medium of refractive index 1.75, it will behave as a

[IIT-JEE 1999]

(a) Convergent lens of focal length 3.5 *R*

(b) Convergent lens of focal length 3.0 *R*

(c) Divergent lens of focal length 3.5 *R*

(d) Divergent lens of focal length 3.0 *R*

58. A convex lens of focal length 0.5 *m* and concave lens of focal length 1 *m* are combined. The power of the resulting lens will be

(a) 1 *D*[MP PET 1997] (b) – 1 *D*

(c) 0.5 *D* (d) – 0.5 *D*

59. A double convex lens is made of glass of refractive index 1.5. If its focal length is 30 cm, then radius of curvature of each of its curved surface is [Bihar CEET 1995]

(a) 10 *cm* (b) 15 *cm*

(c) 18 *cm* (d) None of these

60. A thin lens made of glass of refractive index 1.5 has a front surface + 11 *D* power and back surface – 6 *D*. If this lens is submerged in a liquid of refractive index 1.6, the resulting power of the lens is

(a) – 0.5 *D* (b) + 0.5 *D*

(c) – 0.625 *D* (d) + 0.625 *D*

61. An object is placed first at infinity and then at 20 *cm* from the object side focal plane of the convex lens. The two images thus formed are 5 *cm* apart. The focal length of the lens is

(a) 5 *cm* (b 10 *cm*

(c) 15 *cm* (d) 20 *cm*

62. The distance between an object and the screen is 100 *cm*. A lens produces an image on the screen when placed at either of the positions 40 *cm* apart. The power of the lens is

[SCRA 1994]

(a) ≈ 3 *dioptres* (b) ≈ 5 *dioptres*

(c) ≈ 7 *diopters* (d) ≈ 9 *diopters*

63. The image distance of an object placed 10 *cm* in front of a thin lens of focal length 5 *cm* is [MP PET 1999] [SCRA 1994]

(a) 6.5 *cm* (b) 8.0 *cm*

(c) 9.5 *cm* (d) 10.0 *cm*

64. A achromatic combination is made with a lens of focal length f and dispersive power ω with a lens having dispersive power of 2ω. The focal length of second will be

[RPET 1997]

(a) 2 f [MP PET 1999] (b) $f / 2$

(c) $-f / 2$ (d) – 2 f

65. A biconvex lens with equal radii curvature has refractive index 1.6 and focal length 10 *cm*. Its radius of curvature will be

(a) 20 *cm* (b) 16 *cm*
(c) 10 *cm* (d) 12 *cm*

66. A convex lens [RPMT 1997]

(a) Converges light rays
(b) Diverges light rays
(c) Form real images always
(d) Always forms virtual images

67. The focal length of a combination of lenses formed with lenses having powers of + 2.50 D and – 3.75 D will be [RPMT 1997]

(a) – 20 *cm* (b) – 40 *cm*
(c) – 60 *cm* (d) – 80 *cm*

68. Focal length of a converging lens in air is R. If it is dipped in water of refractive index 1.33, then its focal length will be around (Refractive index of lens material is 1.5) [RPMT 1997; EAMCET (Med.) 1995]

(a) R (b) $2R$
(c) $4R$ (d) $R/2$

69. Focal length of a convex lens of refractive index 1.5 is 2 *cm*. Focal length of lens when immersed in a liquid of refractive index of 1.25 will be [CBSE PMT 1993]

(a) 10 *cm* (b) 2.5 *cm*
(c) 5 *cm* (d) 7.5 *cm*

70. The focal length of a convex lens depends upon [AFMC 1994]

(a) Frequency of the light ray
(b) Wavelength of the light ray
(c) Both (a) and (b)
(d) None of these

71. If a convex lens of focal length 80 *cm* and a concave lens of focal length 50 *cm* are combined together, what will be their resulting power [CBSE PMT 1996; AFMC 2002]

(a) + 6.5D (b) – 6.5 D
(c) + 7.5 D (d) – 0.75 D

72. f_v and f_r are the focal lengths of a convex lens for violet and red light respectively and F_v and F_r are the focal lengths of a concave lens for violet and red light respectively, then

(a) $f_v < f_r$ and $F_v > F_r$ (b $f_v < f_R$ and $F_v < F_r$
(c) $f_c > f_r$ and $F_v > F_r$ (d) $f_v > f_r$ and $F_v < F_r$

73. If a lens is cut into two pieces perpendicular to the principal axis and only one part is used, the intensity of the image [CPMT 1996]

(a) Remains same (b) $\frac{1}{2}$ times
(c) 2 times (d) Infinite

74. A convex lens of focal length f produces an image $\frac{1}{n}$ times than that of the size of the object. The distance of the object from the lens is [BHU 1997; JIPMER 2001, 02]

(a) nf [MP PET 2003] (b) $\frac{f}{n}$
(c) $(n+1)f$ (d) $(n-1)f$

75. Two thin lenses whose powers are +2D and –4D respectively combine, then the power of combination is [AFMC 1998; CPMT 1996; Very Similar to BHU 2004]

(a) – 2D (b) + 2D
(c) – 4D (d) + 4D

76. A substance is behaving as convex lens in air and concave in water, then its refractive index is [BHU 1998]

(a) Smaller than air
(b) Greater than both air and water
(c) Greater than air but less than water
(d) Almost equal to water

77. A concave lens of focal length 20 *cm* placed in contact with a plane mirror acts as a [SCRA 1998]

(a) Convex mirror of focal length 10 *cm*
(b) Concave mirror of focal length 40 *cm*
(c) Concave mirror of focal length 60 *cm*
(d) Concave mirror of focal length 10 *cm*

78. A convex lens is used to form real image of an object on a screen. It is observed that even when the positions of the object and that screen are fixed there are two positions of the lens to form real images. If the heights of the images are 4 *cm* and 9 *cm* respectively, the height of the object is [AMU (Med.) 1999]

(a) 2.25 *cm* (b 6.00 *cm*
(c) 6.50 *cm* (d) 36.00 *cm*

79. A convex lens of power + 6D is placed in contact with a concave lens of power – 4D. What is the nature and focal length of the combination [AMU (Engg.) 1999]

(a) Concave, 25 *cm* (b) Convex, 50 *cm*
(c) Concave, 20 *cm* (d) Convex, 100 *cm*

80. A double convex lens of glass of μ = 1.5 has radius of curvature of each of its surface is 0.2 *m*. The power of the lens is [CBSE PMT 1996]

(a) + 10 *dioptres* (b) – 10 *dioptres*
(c) – 5 *dioptres* (d) +5 *dioptres*

81. A lens of focal power 0.5 D is [JIPMER 1999]

(a) A convex lens of focal length 0.5 *m*
(b) A concave lens of focal length 0.5 *m*
(c) A convex lens of focal length 2 *m*
(d) A concave lens of focal length 2 *m*

82. A lens which has focal length of 4 *cm* and refractive index of 1.4 is immersed in a liquid of refractive index 1.6, then the focal length will be [RPMT 1999]

(a) – 12.8 *cm* (b) 32 *cm*
(c) 12.8 *cm* (d) – 32 *cm*

83. A convex lens has 9 *cm* focal length and a concave lens has – 18 *cm* focal length. The focal length of the combination in contact will be

(a) 9 *cm* (b) – 18 *cm*
(c) – 9 *cm* (d) 18 *cm*

84. A double convex thin lens made of glass of refractive index 1.6 has radii of curvature 15 *cm* each. The focal length of this lens when immersed in a liquid of refractive index 1.63 is

(a) – 407 *cm* (b) 250 *cm*
(c) 125 *cm* (d) 25 *cm*

85. A lens of power + 2 *diopters* is placed in contact with a lens of power – 1 *diopoter*. The combination will behave like

[UPSEAT 2000]

(a) A divergent lens of focal length 50 *cm*
(b) A convergent lens of focal length 50 *cm*
(c) A convergent lens of focal length 100 *cm*
(d) A divergent lens of focal length 100 *cm*

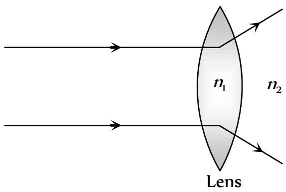
86. Chromatic aberration of lens can be corrected by

[AFMC 2000]

(a) Reducing its aperature
(b) Proper polishing of its two surfaces
(c) Suitably combining it with another lens
(d) Providing different suitable curvature to its two surfaces

87. The relation between n_1 and n_2, if behaviour of light rays is as shown in figure is [KCET 2000]

(a) $n_1 >> n_2$
(b $n_2 > n_1$
(c) $n_1 > n_2$
(d) $n_1 = n_2$



88. A candle placed 25 *cm* from a lens, forms an image on a screen placed 75 *cm* on the other end of the lens. The focal length and type of the lens should be [KCET 2000]

(a) + 18.75 *cm* and convex lens
(b) – 18.75 *cm* and concave lens
(c) + 20.25 *cm* and convex lens
(d) – 20.25 *cm* and concave lens

89. We combined a convex lens of focal length f_1 and concave lens of focal lengths f_2 and their combined focal length was *F*. The combination of these lenses will behave like a concave lens, if

(a) $f_1 > f_2$ (b) $f_1 < f_2$
(c) $f_1 = f_2$ (d) $f_1 \le f_2$

90. In a plano-convex lens the radius of curvature of the convex lens is 10 *cm*. If the plane side is polished, then the focal length will be (Refractive index = 1.5)

[CBSE PMT 2000; BHU 2004]

(a) 10.5 *cm* (b 10 *cm*
(c) 5.5 *cm* (d) 5 *cm*

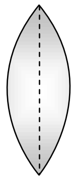
91. The focal length of a convex lens is 10 *cm* and its refractive index is 1.5. If the radius of curvature of one surface is 7.5 *cm*, the radius of curvature of the second surface will be [RPMT 1999]

[MP PMT 2000]

(a) 7.5 *cm* (b) 15.0 *cm*
(c) 75 *cm* (d) 5.0 *cm*

92. [UPSEAT 2000; Pb. PET 2004] A convex lens has a focal length *f*. It is cut into two parts along the dotted line as shown in the figure. The focal length of each part will be [MP PET 2000]

(a) $\frac{f}{2}$
(b) f
(c) $\frac{3}{2}f$
(d) $2f$



93. An object has image thrice of its original size when kept at 8 *cm* and 16 *cm* from a convex lens. Focal length of the lens is

(a) 8 *cm*
(b) 16 *cm*
(c) Between 8 *cm* and 16 *cm*
(d) Less than 8 *cm*

94. The combination of a convex lens (*f* = 18 *cm*) and a thin concave lens (*f* = 9 *cm*) is [AMU (Engg.) 2001]

(a) A concave lens (*f* = 18 *cm*)
(b) A convex lens (*f* = 18 *cm*)
(c) A convex lens (*f* = 6 *cm*)
(d) A concave lens (*f* = 6 *cm*)

95. A convex lens forms a real image of an object for its two different positions on a screen. If height of the image in both the cases be 8 *cm* and 2 *cm*, then height of the object is

[KCET 2000, 01]

(a) 16 *cm* (b) 8 *cm*
(c) 4 *cm* (d) 2 *cm*

96. A convex lens of focal length 25 *cm* and a concave lens of focal length 10 *cm* are joined together. The power of the combination will be [MP PMT 2001]

(a) – 16 *D* (b) + 16 *D*
(c) – 6 *D* (d) + 6 *D*

97. The unit of focal power of a lens is [KCET 2001]

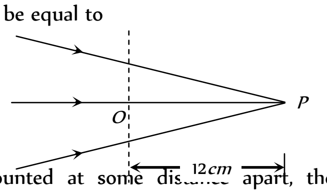
(a) Watt (b) Horse power
(c) Dioptre (d) Lux

98. A thin lens made of glass of refractive index μ = 1.5 has a focal length equal to 12 *cm* in air. It is now immersed in water $\left(\mu = \frac{4}{3}\right)$. Its new focal length is [UPSEAT 2002]

(a) 48 *cm* [KCET 2000] (b) 36 *cm*
(c) 24 *cm* (d) 12 *cm*

99. Figure given below shows a beam of light converging at point *P*. When a convex lens of focal length 16 *cm* is introduced in the path of the beam at a place *O* shown by dotted line such that *OP* becomes the axis of the lens, the beam converges at a distance *x* from the lens. The value *x* will be equal to

(a) 12 *cm*
(b) 24 *cm*
(c) 36 *cm*
(d) 48 *cm*



100. If two + 5 *D* lenses are mounted at some distance apart, the equivalent power will always be negative if the distance is

[UPSEAT 2002]

(a) Greater than 40 *cm* (b) Equal to 40 *cm*

(c) Equal to 10 *cm* (d) Less than 10 *cm*

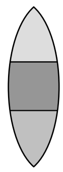
101. A convex lens produces a real image *m* times the size of the object. What will be the distance of the object from the lens

[JIPMER 2002]

(a) $\left(\frac{m+1}{m}\right)f$ (b) $(m-1)f$

(c) $\left(\frac{m-1}{m}\right)f$ (d) $\frac{m+1}{f}$

102. A convex lens is made up of three different materials as shown in the figure. For a point object placed on its axis, the number of images formed are [KCET 2002]



(a) 1

(b) 5

(c) 4

(d) 3

103. An object is placed 12 *cm* to the left of a converging lens of focal length 8 *cm*. Another converging lens of 6 *cm* focal length is placed at a distance of 30 *cm* to the right of the first lens. The second lens will produce [KCET 2002]

(a) No image (b) A virtual enlarged image

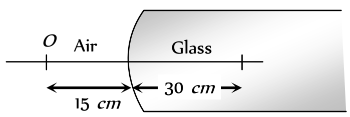
(c) A real enlarged image (d) A real smaller image

104. If convex lens of focal length 80*cm* and a concave lens of focal length 50 *cm* are combined together, what will be their resulting power [AFMC 2002]

(a) + 6.5 *D* (b) – 6.5 *D*

(c) + 7.5 *D* (d) – 0.75 *D*

105. A point object *O* is placed in front of a glass rod having spherical end of radius of curvature 30 *cm*. The image would be formed at



(a) 30 *cm* left

(b) Infinity

(c) 1 *cm* to the right

(d) 18 *cm* to the left

106. The focal length of lens of refractive index 1.5 in air is 30 *cm*. When it is immersed in a liquid of refractive index $\frac{4}{3}$, then its focal length in liquid will be [BHU 2002]

(a) 30 *cm* (b) 60 *cm*

(c) 120 *cm* (d) 240 *cm*

107. Two thin lenses of focal lengths f_1 and f_2 are in contact. The focal length of this combination is [MP PET 2002]

(a) $\frac{f_1 f_2}{f_1 - f_2}$ (b) $\frac{f_1 f_2}{f_1 + f_2}$

(c) $\frac{2 f_1 f_2}{f_1 - f_2}$ (d) $\frac{2 f_1 f_2}{f_1 + f_2}$

108. A convex lens is dipped in a liquid whose refractive index is equal to the refractive index of the lens. Then its focal length will

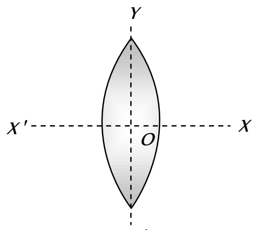
(a) Become infinite

(b) Become small, but non–zero

(c) Remain unchanged

(d) Become zero

109. An equiconvex lens is cut into two halves along (i) *XOX′* and (ii) *YOY′* as shown in the figure. Let f, f', f'' be the focal lengths of the complete lens, of each half in case (i), and of each half in case (ii), respectively



Choose the correct statement from the following

[CBSE PMT 2003]

(a) $f' = 2f, f'' = f$ (b) $f' = f, f'' = f$

(c) $f' = 2f, f'' = 2f$ (d) $f' = f, f'' = 2f$

110. The sun makes 0.5° angle on earth surface. Its image is made by convex lens of 50 *cm* focal length. The diameter of the image will be [CPMT 2003]

(a) 5 *mm* (b) 4.36 *mm*

(c) 7 *mm* (d) None of these

111. The chromatic Aberration in lenses becomes due to

[CPMT 2003]

(a) Disimilarity of main axis of rays

(b) Disimilarity of radii of curvature

(c) Variation of focal length of lenses with wavelength

(d) None of these

112. If aperture of lens is halved then image will be [AFMC 2003]

[Orissa JEE 2002]

(a) No effect on size

(b) Intensity of image decreases

(c) Both (a) and (b)

(d) None of these

113. When the convergent nature of a convex lens will be less as compared with air [AFMC 2003]

(a) In water (b) In oil

(c) In both (a) and (b) (d) None of these

114. An achromatic combination of lenses produces

[KCET 1993; JIPMER 1997]

(a) Coloured images

(b) Highly enlarged image

(c) Images in black and white

(d) Images unaffected by variation of refractive index with wavelength

115. In a parallel beam of white light is incident on a converging lens, the colour which is brought to focus nearest to the lens is

(a) Violet (b) Red

(c) The mean colour (d) All the colours together

116. A magnifying glass is to be used at the fixed object distance of 1 *inch*. If it is to produce an erect image magnified 5 *times* its focal length should be [MP PMT 1990]

(a) 0.2 *inch* (b) 0.8 *inch*

(c) 1.25 *inch* (d) 5 *inch*

117. A film projector magnifies a 100 *cm* film strip on a screen. If the linear magnification is 4, the area of magnified film on the screen is CPMT 1977, 91; MP PET 1985, 89; RPMT 2001; BCEC 2005]

(a) 1600 *cm* (b) 400 *cm*

(c) 800 *cm* (d) 200 *cm*

118. An object placed 10 *cm* in front of a lens has an image 20 *cm* behind the lens. What is the power of the lens (in *dioptres*) [MP PMT 1995]

(a) 1.5 (b) 3.0

(c) – 15.0 (d) + 15.0

119. A beam of parallel rays is brought to a focus by a plano-convex lens. A thin concave lens of the same focal length is joined to the first lens. The effect of this is [KCET 2004]

(a) The focal point shifts away from the lens by a small distance

(b) The focus remains undisturbed

(c) The focus shifts to infinity

(d) The focal point shifts towards the lens by a small distance

120. A thin plano-convex lens acts like a concave mirror of focal length 0.2 *m* when silvered from its plane surface. The refractive index of the material of the lens is 1.5. The radius of curvature of the convex surface of the lens will be [KCET 2004]

(a) 0.4 *m* (b 0.2 *m*

(c) 0.1 *m* (d) 0.75 *m*

121. A point object is placed at the center of a glass sphere of radius 6 *cm* and refractive index 1.5. The distance of the virtual image from the surface of the sphere is [IIT-JEE (Screening) 2004]

(a) 2 *cm* (b) 4 *cm*

(c) 6 *cm* (d) 12 *cm*

122. In order to obtain a real image of magnification 2 using a converging lens of focal length 20 *cm*, where should an object be placed [AFMC 2004]

(a) 50 *cm* (b) 30 *cm*

(c) – 50 *cm* (d) – 30 *cm*

123. A plano-convex lens of refractive index 1.5 and radius of curvature 30 *cm* is silvered at the curved surface. Now this lens has been used to form the image of an object. At what distance from this lens an object be placed in order to have a real image of the size of the object [AIEEE 2004]

(a) 20 *cm* (b) 30 *cm*

(c) 60 *cm* (d) 80 *cm*

124. A double convex lens $(R_1 = R_2 = 10\,cm)$ $(\mu = 1.5)$ having focal length equal to the focal length of a concave mirror. The radius of curvature of the concave mirror is [Orssia PMT 2004]

(a) 10 *cm* (b 20 *cm*

(c) 40 *cm* (d) 15 *cm*

125. At what distance from a convex lens of focal length 30 *cm*, an object should be placed so that the size of the image be 1/2 of the object

(a) 30 *cm* (b) 60 *cm*

(c) 15 *cm* (d) 90 *cm*

126. A plano-convex lens is made of refractive index of 1.6. The radius of curvature of the curved surface is 60 *cm*. The focal length of the lens is [NCERT 1980; [Pb. PET 2000]

(a) 400 *cm* (b) 200 *cm*

(c) 100 *cm* (d) 50 *cm*

127. The radius of the convex surface of plano-convex lens is 20 *cm* and the refractive index of the material of the lens is 1.5. The focal length of the lens is [CPMT 2004]

(a) 30 *cm* (b) 50 *cm*

(c) 20 *cm* (d) 40 *cm*

128. A combination of two thin convex lenses of focal length 0.3 *m* and 0.1 *m* will have minimum spherical and chromatic aberrations if the distance between them is [UPSEE 2004]

(a) 0.1 *m* (b 0.2 *m*

(c) 0.3 *m* (d) 0.4 *m*

129. A bi-convex lens made of glass (refractive index 1.5) is put in a liquid of refractive index 1.7. Its focal length will [UPSEAT 2004]

(a) Decrease and change sign

(b Increase and change sign

(c) Decrease and remain of the same sign

(d) Increase and remain of the same sign

130. Spherical aberration in a lens [UPSEAT 2004]

(a) Is minimum when most of the deviation is at the first surface

(b) Is minimum when most of the deviation is at the second surface

(c) Is minimum when the total deviation is equally distributed over the two surface

(d) Does not depend on the above consideration

131. The focal lengths of convex lens for red and blue light are 100 *cm* and 96.8 *cm* respectively. The dispersive power of material of lens is [Pb. PET 2003]

(a) 0.325 (b 0.0325

(c) 0.98 (d) 0.968

132. The power of an achromatic convergent lens of two lenses is + 2*D*. The power of convex lens is + 5*D*. The ratio of dispersive power of convex and concave lens will be [Pb. PET 2003]

(a) 5 : 3 (b 3 : 5

(c) 2 : 5 (d) 5 : 2

133. The focal lengths for violet, green and red light rays are f_V, f_G and f_R respectively. Which of the following is the true relationship [BHU 2004; CBS

(a) $f_R < f_G < f_V$ (b $f_V < f_G < f_R$

(c) $f_G < f_R < f_V$ (d) $f_G < f_V < f_R$

134. Two lenses of power + 12 and – 2 diopters are placed in contact. The combined focal length of the combination will be

(a) 8.33 *cm* (b) 1.66 *cm*

(c) 12.5 *cm* (d) 10 *cm*

135. When light rays from the sun fall on a convex lens along a direction parallel to its axis [MP PMT 2004]

(a) Focal length for all colours is the same

(b Focal length for violet colour is the shortest

(c) Focal length for yellow colour is the longest

(d) Focal length for red colour is the shortest

136. A convex lens is in contact with concave lens. The magnitude of the ratio of their focal length is 2/3. Their equivalent focal length is 30 *cm*. What are their individual focal lengths

(a) – 75, 50 (b) – 10, 15

(c) 75, 50 (d) – 15, 10

137. A thin glass (refractive index 1.5) lens has optical power of $-5D$ in air. It's optical power in a liquid medium with refractive index 1.6 will be [AIEEE 2005]

(a) 25 D (b) – 25 D

(c) 1 D (d) None of these

138. The plane faces of two identical plano-convex lenses each having focal length of 40 *cm* are pressed against each other to form a usual convex lens. The distance from this lens, at which an object must be placed to obtain a real, inverted image with magnification one is [NCERT 1980; CPMT 1981; MP PMT 1999; UPSEAT 1999]

(a) 80 *cm* (b 40 *cm*

(c) 20 *cm* (d) 162 *cm*

139. If two lenses of +5 diopters are mounted at some distance apart, the equivalent power will always be negative if the distance is

(a) Greater than 40 *cm* (b) Equal to 40 *cm*

(c) Equal to 10 *cm* (d) Less than 10 *cm*

140. A concave lens and a convex lens have same focal length of 20 *cm* and both put in contact this combination is used to view an object 5 *cm* long kept at 20 *cm* from the lens combination. As compared to object the image will be

[CPMT 2005]

(a) Magnified and inverted

(b Reduced and erect

(c) Of the same size and erect

(d) Of the same size and inverted

141. The focal length of the field lens (which is an achromatic combination of two lenses) of telescope is 90 *cm*. The dispersive powers of the two lenses in the combination are 0.024 and 0.036. The focal lengths of two lenses are

[CPMT 2005]

(a) 30 *cm* and 60 *cm* (b 30 *cm* and – 45 *cm*

(c) 45 *cm* and 90 *cm* (d) 15 *cm* and 45 *cm*

142. A combination of two thin lenses of the same material with focal lengths f_1 and f_2, arranged on a common axis minimizes chromatic aberration, if the distance between them is

(a) $\frac{(f_1+f_2)}{4}$ (b) $\frac{(f_1+f_2)}{2}$

(c) (f_1+f_2) (d) $2(f_1+f_2)$

143. If the focal length of a double convex lens for red light is f_R, its focal length for the violet light is [EAMCET 2005]

(a) f_R (b) Greater than f_R

(c) Less than f_R (d) $2f_R$

144. A thin equiconvex lens is made of glass of refractive index 1.5 and its focal length is 0.2 *m*, if it acts as a concave lens of 0.5 *m* focal length when dipped in a liquid, the refractive index of the liquid is

(a) $\frac{17}{8}$ (b) $\frac{15}{8}$

(c) $\frac{13}{8}$ (d) $\frac{9}{8}$

145. The dispersive power of the material of lens of focal length 20 *cm* is 0.08. The longitudinal chromatic aberration of the lens is

(a) 0.08 *cm* (b) 0.08/20 *cm*

(c) 1.6 *cm* (d) 0.16 *cm*

[IIT-JEE (Screening) 2005]

Prism Theory & Dispersion of Light

1. Which source is associated with a line emission spectrum

[MP PET/PMT 1988; CBSE PMT 1993]

(a) Electric fire (b) Neon street sign

(c) Red traffic light (d) Sun

2. Formula for dispersive power is (where symbols have their usual meanings) [MP PMT/PET 1988]

or

If the refractive indices of crown glass for red, yellow and violet colours are respectively μ_r, μ_y and μ_v, then the dispersive power of this glass would be [MP PMT 1996]

(a) $\frac{\mu_v-\mu_y}{\mu_r-1}$ (b) $\frac{\mu_v-\mu_r}{\mu_y-1}$

(c) $\frac{\mu_v-\mu_y}{\mu_y-\mu_r}$ [BCECE 2005] (d) $\frac{\mu_v-\mu_r}{\mu_y}-1$

3. The critical angle between an equilateral prism and air is 45°. If the incident ray is perpendicular to the refracting surface, then

(a) After deviation it will emerge from the second refracting surface

(b) It is totally reflected on the second surface and emerges out perpendicularly from third surface in air

(c) It is totally reflected from the second and third refracting surfaces and finally emerges out from the first surface

(d) It is totally reflected from all the three sides of prism and never emerges out

4. When white light passes through a glass prism, one gets spectrum on the other side of the prism. In the emergent beam, the ray which is deviating least is or

Deviation by a prism is lowest for [MP PMT 1997]

(a) Violet ray (b) Green ray

(c) Red ray (d) Yellow ray

5. We use flint glass prism to disperse polychromatic light because light of different colours

[EAMCET 2005]

[MP PET 1993]

(a) Travel with same speed

(b) Travel with same speed but deviate differently due to the shape of the prism

(c) Have different anisotropic properties while travelling through the prism

(d) Travel with different speeds

6. A prism ($\mu = 1.5$) has the refracting angle of 30°. The deviation of a monochromatic ray incident normally on its one surface will be (sin 48° 36′ = 0.75)

[MP PMT/PET 1988]

(a) 18° 36′ (b) 20° 30′

(c) 18° (d) 22°1′

7. Fraunhofer lines are obtained in

[CPMT 1973; MP PMT 1989; MP PMT 2004]

(a) Solar spectrum

(b) The spectrum obtained from neon lamp

(c) Spectrum from a discharge tube

(d) None of the above

8. When light rays are incident on a prism at an angle of 45°, the minimum deviation is obtained. If refractive index of the material of prism is $\sqrt{2}$, then the angle of prism will be

[MP PMT 1986]

(a) 30° (b) 40°

(c) 50° (d) 60°

9. A spectrum is formed by a prism of dispersive power $'\omega'$. If the angle of deviation is $'\delta'$, then the angular dispersion is

[MP PMT 1989]

(a) ω / δ (b) δ / ω

(c) $1/\omega\delta$ (d) $\omega\delta$

10. Light from sodium lamp is passed through cold sodium vapours, the spectrum of transmitted light consists of

[MP PET 1989; RPMT 2001]

(a) A line at 5890 Å (b) A line at 5896 Å

(c) Sodium doublet lines (d) No spectral features

11. Angle of minimum deviation for a prism of refractive index 1.5 is equal to the angle of prism. The angle of prism is (cos 41° = 0.75)

(a) 62° (b) 41°

(c) 82° (d) 31°

12. In the formation of primary rainbow, the sunlight rays emerge at minimum deviation from rain-drop after

[MP PET 1989]

(a) One internal reflection and one refraction

(b) One internal reflection and two refractions

(c) Two internal reflections and one refraction

(d) Two internal reflections and two refractions

13. Dispersive power depends upon [RPMT 1997]

(a) The shape of prism (b) Material of prism

(c) Angle of prism (d) Height of the prism

14. When white light passes through the achromatic combination of prisms, then what is observed

[MP PMT 1989]

(a) Only deviation (b) Only dispersion

(c) Deviation and dispersion (d) None of the above

15. The dispersion for a medium of wavelength λ is *D*, then the dispersion for the wavelength 2λ will be

[MP PET 1989]

(a) *D*/8 (b) *D*/4

(c) *D*/2 (d) *D*

16. The refractive index of a prism for a monochromatic wave is $\sqrt{2}$ and its refracting angle is 60°. For minimum deviation, the angle of incidence will be

[MNR 1998; MP PMT 1989, 92, 2002; CPMT 1993, 2004]

(a) 30° (b) 45°

(c) 60° (d) 75°

17. The ratio of angle of minimum deviation of a prism in air and when dipped in water will be (${}_a\mu_g = 3/2$ and ${}_a\mu_w = 4/3$)

(a) 1/8 (b) 1/2

(c) 3/4 (d) 1/4

18. The respective angles of the flint and crown glass prisms are *A*′ and *A*. They are to be used for dispersion without deviation, then the ratio of their angles *A*′ /*A* will be

[MP PMT 1989]

(a) $-\dfrac{(\mu_y - 1)}{(\mu_y' - 1)}$ (b) $\dfrac{(\mu_y' - 1)}{(\mu_y - 1)}$

(c) $(\mu_y' - 1)$ (d) $(\mu_y - 1)$

19. The number of wavelengths in the visible spectrum

[MP PMT 1989]

(a) 4000 (b) 6000

(c) 2000 (d) Infinite

20. The black lines in the solar spectrum during solar eclipse can be explained by [MP PMT 1989]

(a) Planck's law (b) Kirchoff's law

(c) Boltzmann's law (d) Solar disturbances

21. The dispersive power is maximum for the material

(a) Flint glass (b) Crown glass

(c) Mixture of both [MP PET/PMT 1988] (d) None of the above

22. A light ray is incident by grazing one of the face of a prism and after refraction ray does not emerge out, what should be the angle of prism while critical angle is *C*

(a) Equal to 2*C* (b) Less than 2*C*

(c) More than 2*C* (d) None of the above

23. A parallel beam of monochromatic light is incident at one surface of a equilateral prism. Angle of incidence is 55° and angle of emergence is 46°. The angle of minimum deviation will be

(a) Less than 41° (b) Equal to 41°

(c) More than 41° (d) None of the above

24. The spectrum of light emitted by a glowing solid is

(a) Continuous spectrum (b) Line spectrum

(c) Band spectrum (d) Absorption spectrum

25. Light rays from a source are incident on a glass prism of index of refraction μ and angle of prism α . At near normal incidence, the angle of deviation of the emerging rays is

[MP PMT 1993]

(a) $(\mu - 2)\alpha$ (b) $(\mu - 1)\alpha$

(c) $(\mu + 1)\alpha$ (d) $(\mu + 2)\alpha$

26. Which of the following element was discovered by study of Fraunhofer lines

(a) Hydrogen (b) Oxygen

(c) Helium (d) Ozone

27. By placing the prism in minimum deviation position, images of the spectrum

(a) Becomes inverted (b) Becomes broader

(c) Becomes distinct (d) Becomes intensive

28. Our eye is most sensitive for which of the following wavelength

(a) 4500 $\mathring{A}$

(b) 5500 $\mathring{A}$

(c) 6500 $\mathring{A}$

(d) Equally sensitive for all wave lengths of visible spectrum

29. Three prisms of crown glass, each have angle of prism 9° and two prisms of flint glass are used to make direct vision spectroscope. What will be the angle of flint glass prisms if μ for flint is 1.60 and μ for crown glass is 1.53

(a) 11.9° (b) 16.0°

(c) 15.3° (d) 9.11°

30. If the refractive indices of crown glass for red, yellow and violet colours are 1.5140, 1.5170 and 1.5318 respectively and for flint glass these are 1.6434, 1.6499 and 1.6852 respectively, then the dispersive powers for crown and flint glass are respectively

(a) 0.034 and 0.064 (b) 0.064 and 0.034

(c) 1.00 and 0.064 (d) 0.034 and 1.0

31. The minimum temperature of a body at which it emits light is

(a) 1200°C (b) 1000°C

(c) 500°C (d) 200°C

32. Band spectrum is obtained when the source emitting light is in the form of **or**

Band spectrum is characteristic of

[CPMT 1988; MP PET 1994; DCE 2004; MP PET 2005]

(a) Atoms (b Molecules

(c) Plasma (d) None of the above

33. Flint glass prism is joined by a crown glass prism to produce dispersion without deviation. The refractive indices of these for mean rays are 1.602 and 1.500 respectively. Angle of prism of flint prism is 10°, then the angle of prism for crown prism will be

(a) 12°2.4' (b) 12°4'

(c) 1.24° (d) 12°

34. The angle of minimum deviation for a prism is 40° and the angle of the prism is 60°. The angle of incidence in this position will be

[EAMCET (Engg.) 1995; MH CET 1999; CPMT 2000]

(a) 30° (b) 60°

(c) 50° (d) 100°

35. In the position of minimum deviation when a ray of yellow light passes through the prism, then its angle of incidence is

[MP PMT 1989; RPMT 1997]

(a) Less than the emergent angle

(b) Greater than the emergent angle

(c) Sum of angle of incidence and emergent angle is 90°

(d) Equal to the emergent angle

36. A circular disc of which 2/3 part is coated with yellow and 1/3 part is with blue. It is rotated about its central axis with high velocity, then it will be seen as

(a) Green (b) Brown

(c) White (d) Violet

37. The fine powder of a coloured glass is seen as

(a) Coloured (b) White

(c) That of the glass colour (d) Black

38. When a white light passes through a hollow prism, then

[MP PMT 1987]

(a) There is no dispersion and no deviation

(b) Dispersion but no deviation

(c) Deviation but no dispersion

(d) There is dispersion and deviation both

39. The light ray is incidence at angle of 60° on a prism of angle 45°. When the light ray falls on the other surface at 90°, the refractive index of the material of prism μ and the angle of deviation δ are given by [MP PET/PMT 1988] [DPMT 2001]

(a) $\mu = \sqrt{2}\,, \delta = 30^o$ (b) $\mu = 1.5\,, \delta = 15^o$

(c) $\mu = \dfrac{\sqrt{3}}{2}, \delta = 30^o$ (d) $\mu = \sqrt{\dfrac{3}{2}}, \delta = 15^o$

40. In dispersion without deviation

(a) The emergent rays of all the colours are parallel to the incident ray

(b) Yellow coloured ray is parallel to the incident ray

(c) Only red coloured ray is parallel to the incident ray

(d) All the rays are parallel, but not parallel to the incident ray

41. Deviation of 5° is observed from a prism whose angle is small and whose refractive index is 1.5. The angle of prism is

[DPMT 2001]

(a) 7.5° (b) 10°

(c) 5° (d) 3.3°

42. The refractive indices of violet and red light are 1.54 and 1.52 respectively. If the angle of prism is 10°, then the angular dispersion is [MP PMT 1990]

(a) 0.02 (b) 0.2

(c) 3.06 (d) 30.6

43. The angle of minimum deviation measured with a prism is 30° and the angle of prism is 60°. The refractive index of prism material is

(a) $\sqrt{2}$ (b) 2

(c) 3/2 (d) 4/3

44. If the refractive indices of a prism for red, yellow and violet colours be 1.61, 1.63 and 1.65 respectively, then the dispersive power of the prism will be

[MP PET 1991; DPMT 1999]

(a) $\frac{1.65-1.62}{1.61-1}$ (b) $\frac{1.62-1.61}{1.65-1}$

(c) $\frac{1.65-1.61}{1.63-1}$ (d) $\frac{1.65-1.63}{1.61-1}$

45. The minimum deviation produced by a hollow prism filled with a certain liquid is found to be 30°. The light ray is also found to be refracted at angle of 30°. The refractive index of the liquid is

(a) $\sqrt{2}$ (b) $\sqrt{3}$

(c) $\sqrt{\frac{3}{2}}$ (d) $\frac{3}{2}$

46. Minimum deviation is observed with a prism having angle of prism *A*, angle of deviation δ, angle of incidence *i* and angle of emergence *e*. We then have generally

[MP PET 1991]

(a) $i > e$ (b) $i < e$

(c) $i = e$ (d) $i = e = \delta$

47. A thin prism P_1 with angle 4° and made from glass of refractive index 1.54 is combined with another thin prism P_2 made from glass of refractive index 1.72 to produce dispersion without deviation. The angle of prism P_2 is

[MP PMT 1991, 92; IIT-JEE 1990; MP PET 1995, 99; UPSEAT 2001; RPMT 2004]

(a) 2.6° (b) 3°

(c) 4° (d) 5.33°

48. An achromatic prism is made by combining two prisms $P_1(\mu_v = 1.523, \mu_r = 1.515)$ and $P_2(\mu_v = 1.666, \mu_r = 1.650)$; where μ represents the refractive index. If the angle of the prism P_1 is 10°, then the angle of the prism P_2 will be

[MP PMT 1991]

(a) 5° (b) 7.8°

(c) 10.6° (d) 20°

49. Angle of a prism is 30° and its refractive index is $\sqrt{2}$ and one of the surface is silvered. At what angle of incidence, a ray should be incident on one surface so that after reflection from the silvered surface, it retraces its path

[MP PMT 1991; UPSEAT 2001; CBSE PMT 2004]

(a) 30° (b) 60°

(c) 45° (d) $\sin^{-1}\sqrt{1.5}$

50. For a material, the refractive indices for red, violet and yellow colour light are respectively 1.52, 1.64 and 1.60. The dispersive power of the material is [MP PMT 1991]

(a) 2 (b) 0.45

(c) 0.2 (d) 0.045

51. Band spectrum is produced by [CPMT 1978]

(a) *H* (b) *He*

(c) H_2 (d) *Na*

52. The band spectra (characteristic of molecular species) is due to emission of radiation [CPMT 1982, 90]

(a) Gaseous state (b) Liquid state

(c) Solid state (d) All of three states

53. Line spectrum was first of all theoretically explained by

(a) Swan (b) Fraunhofer

(c) Kirchoff (d) Bohr

54. The spectrum of iodine gas under white light will be [MP PET 1991]

(a) Only violet

(b) Bright lines

(c) Only red lines

(d) Some black bands in continuous spectrum

55. Continuous spectrum is not due to

(a) Hydrogen flame (b) Electric bulb

(c) Kerosene oil lamp flame (d) Candle flame

56. Fraunhofer lines are produced by

(a) The element present in the photosphere of sun

(b) The elements present in the chromosphere of the sun

(c) The vapour of the element present in the chromosphere of the sun

(d) The carbon dioxide present in the atmosphere

57. A medium is said to be dispersive, if [MP PMT 1990]

(a) Light of different wavelengths propagate at different speeds

(b) Light of different wavelengths propagate at same speed but has different frequencies

(c) Light is gradually bent rather than sharply refracted at an interface between the medium and air

(d) Light is never totally internally reflected

58. A ray of light is incident at an angle of 60° on one face of a prism of angle 30°. The ray emerging out of the prism makes an angle of 30° with the incident ray. The emergent ray is

[EAMCET 1990; MP PMT 1990]

(a) Normal to the face through which it emerges

(b) Inclined at 30° to the face through which it emerges

(c) Inclined at 60° to the face through which it emerges

(d) None of these

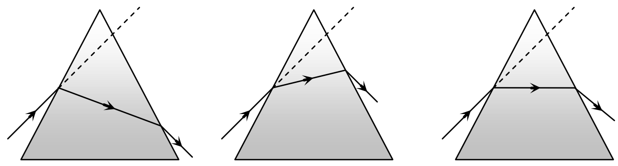
59. In a thin prism of glass (refractive index 1.5), which of the following relations between the angle of minimum deviations δ_m and angle of refraction r will be correct

[MP PMT 1990]

(a) $\delta_m = r$ (b) $\delta_m = 1.5\,r$

(c) $\delta_m = 2r$ (d) $\delta_m = \frac{r}{2}$

60. The figures represent three cases of a ray passing through a prism of angle *A*. The case corresponding to minimum deviation is



(1) (2) (3)

(a) 1 (b) 2

(c) 3 (d) None of these

61. Dispersion can take place for [MP PET 1992]

(a) Transverse waves only but not for longitudinal waves

(b) Longitudinal waves only but not for transverse waves

(c) Both transverse and longitudinal waves

(d) Neither transverse nor longitudinal waves

62. Emission spectrum of CO_2 gas [MP PET 1992]

(a) Is a line spectrum

(b Is a band spectrum

(c) Is a continuous spectrum

(d) Does not fall in the visible region

63. A ray of light passes through an equilateral glass prism in such a manner that the angle of incidence is equal to the angle of emergence and each of these angles is equal to 3/4 of the angle of the prism. The angle of deviation is

[MNR 1988; MP PMT 1999; Roorkee 2000; UPSEAT 2000; MP PET 2005]

(a) 45° (b) 39°

(c) 20° (d) 30°

64. The true statement is

(a) The order of colours in the primary and the secondary rainbows is the same

(b) The intensity of colours in the primary and the secondary rainbows is the same

(c) The intensity of light in the primary rainbow is greater and the order of colours is the same than the secondary rainbow

(d) The intensity of light for different colours in primary rainbow is greater and the order of colours is reverse than the secondary rainbow

65. What will be the colour of sky as seen from the earth, if there were no atmosphere [MP PMT 1992]

(a) Black (b) Blue

(c) Orange (d) Red

66. When light emitted by a white hot solid is passed through a sodium flame, the spectrum of the emergent light will show

[MP PMT 1992]

(a) The D_1 and D_2 bright yellow lines of sodium

(b) Two dark lines in the yellow region

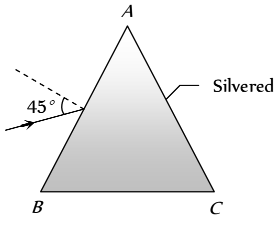
(c) All colours from violet to red

(d) No colours at all

67. A prism *ABC* of angle 30° has its face *AC* silvered. A ray of light incident at an angle of 45° at the face *AB* retraces its path after refraction at face *AB* and reflection at face *AC*. The refractive index of the material of the prism is

[MP PMT 1992; EAMCET 2001]

(a) 1.5

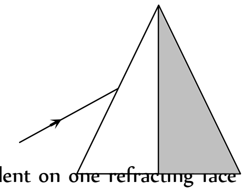


(b) $\frac{3}{\sqrt{2}}$

(c) $\sqrt{2}$

(d) $\frac{4}{3}$

68. A light ray is incident upon a prism in minimum deviation position and suffers a deviation of 34°. If the shaded half of the prism is knocked off, the ray will [MP PMT 1992]



(a) Suffer a deviation of 34°

(b) Suffer a deviation of 68°

(c) Suffer a deviation of 17°

(d) Not come out of the prism

69. A ray of monochromatic light is incident on one refracting face of a prism of angle 75°. It passes through the prism and is incident on the other face at the critical angle. If the refractive index of the material of the prism is $\sqrt{2}$, the angle of incidence on the first face of the prism is

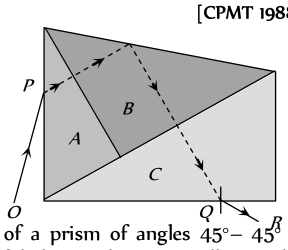
[EAMCET 1983]

(a) 30° (b) 45°

(c) 60° (d) 0°

70. Three glass prisms *A*, *B* and *C* of same refractive index are placed in contact with each other as shown in figure, with no air gap between the prisms. Monochromatic ray of light *OP* passes through the prism assembly and emerges as *QR*. The conditions of minimum deviation is satisfied in the prisms

[CPMT 1988]



(a) *A* and *C*

(b) *B* and *C*

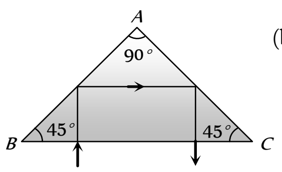
(c) *A* and *B*

(d) In all prisms *A*, *B* and *C*

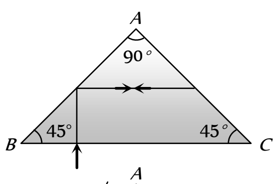
71. The refractive index of a material of a prism of angles 45°– 45° – 90° is 1.5. The path of the ray of light incident normally on the hypotenuse side is shown in

[EAMCET 1985]

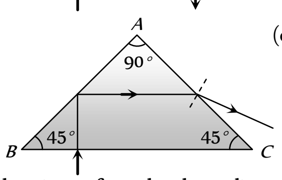
(a)



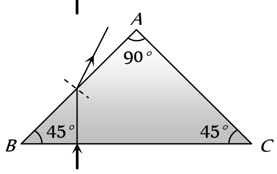
(b)



(c)



(d)



72. At the time of total solar eclipse, the spectrum of solar radiation would be [MP PMT 1990; RPMT 2004]

(a) A large number of dark Fraunhofer lines

(b) A less number of dark Fraunhofer lines

(c) No lines at all

(d) All Fraunhofer lines changed into brilliant colours

73. Angle of deviation (δ) by a prism (refractive index = μ and supposing the angle of prism A to be small) can be given by

(a) $\delta = (\mu - 1)A$ (b) $\delta = (\mu + 1)A$

(c) $\delta = \dfrac{\sin\frac{A+\delta}{2}}{\sin\frac{A}{2}}$ (d) $\delta = \dfrac{\mu - 1}{\mu + 1}A$

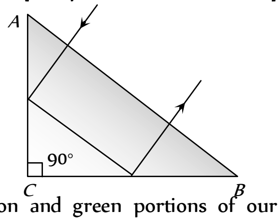
74. Angle of prism is A and its one surface is silvered. Light rays falling at an angle of incidence 2A on first surface return back through the same path after suffering reflection at second silvered surface. Refractive index of the material of prism is

(a) 2 sin A (b) 2 cos A

(c) $\frac{1}{2}\cos A$ (d) $\tan A$

75. A ray of light incident normally on an isosceles right angled prism travels as shown in the figure. The least value of the refractive index of the prism must be

[Manipal MEE 1995; BHU 2003]



(a) $\sqrt{2}$

(b) $\sqrt{3}$

(c) 1.5

(d) 2.0

76. When seen in green light, the saffron and green portions of our National Flag will appear to be [Manipal MEE 1995]

(a) Black

(b) Black and green respectively

(c) Green

(d) Green and yellow respectively

77. At sun rise or sunset, the sun looks more red than at mid-day because [AFMC 1995; Similar to DCE 2003]

(a) The sun is hottest at these times

(b) Of the scattering of light

(c) Of the effects of refraction

(d) Of the effects of diffraction

78. Line spectrum contains information about [MP PET 1995]

(a) The atoms of the prism

(b) The atoms of the source

(c) The molecules of the source

(d) The atoms as well as molecules of the source

79. Missing lines in a continuous spectrum reveal

[MP PET 1995]

(a) Defects of the observing instrument

(b) Absence of some elements in the light source

(c) Presence in the light source of hot vapours of some elements

(d) Presence of cool vapours of some elements around the light source

80. A source emits light of wavelength 4700Å, 5400 Å and 6500 Å. The light passes through red glass before being tested by a spectrometer. Which wavelength is seen in the spectrum

[MP PMT 1995]

(a) 6500 Å (b) 5400 Å

(c) 4700 Å (d) All the above

81. A ray passes through a prism of angle 60° in minimum deviation position and suffers a deviation of 30°. What is the angle of incidence on the prism [MP PMT 1994]

[MP PMT 1995; Pb. PMT 2001; RPMT 2003]

(a) 30° (b) 45°

(c) 60° (d) 90°

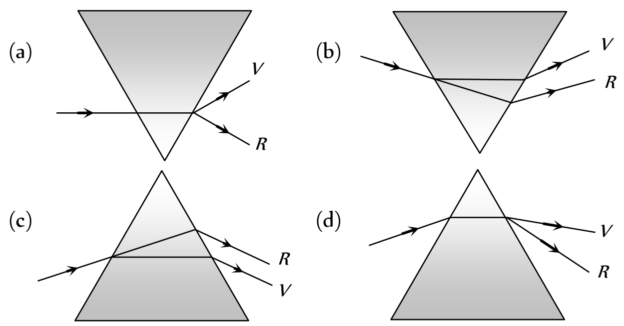
82. When light of wavelength λ is incident on an equilateral prism kept in its minimum deviation position, it is found that the angle of deviation equals the angle of the prism itself. The refractive index of the material of the prism for the wavelength λ is, then [AIIMS 1995]

(a) $\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$

(c) 2 (d) $\sqrt{2}$

83. Which of the following diagrams, shows correctly the dispersion of white light by a prism

[NSEP 1994; MP PET 1996]



84. A neon sign does not produce

[MP PET 1996; UPSEAT 2004]

(a) Line spectrum

(b) An emission spectrum

(c) An absorption spectrum

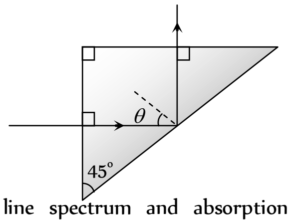
(d) Photons

85. The refractive index of flint glass for blue F line is 1.6333 and red C line is 1.6161. If the refractive index for yellow D line is 1.622, the dispersive power of the glass is

(a) 0.0276 (b) 0.276

(c) 2.76 (d) 0.106

86. A triangular prism of glass is shown in the figure. A ray incident normally to one face is totally reflected, if $\theta = 45^o$. The index of refraction of glass is [AIEEE 2004]



(a) Less than 1.41

(b) Equal to 1.41

(c) Greater than 1.41

(d) None of the above

87. The wavelength of emission line spectrum and absorption line spectrum of a substance are related as

(a) Absorption has larger value

(b) Absorption has smaller value

(c) They are equal

(d) No relation

88. White light is passed through a prism whose angle is 5°. If the refractive indices for rays of red and blue colour are respectively 1.64 and 1.66, the angle of deviation between the two colours will be

(a) 0.1 *degree* (b) 0.2 *degree*

(c) 0.3 *degree* (d) 0.4 *degree*

89. From which source a continuous emission spectrum and a line absorption spectrum are simultaneously obtained [MP PMT 1997]

(a) Bunsen burner flame

(b) The sun

(c) Tube light

(d) Hot filament of an electric bulb

90. A thin prism P_1 with angle 6° and made from glass of refractive index 1.54 is combined with another thin prism P of refractive index 1.72 to produce dispersion without deviation. The angle of prism P_2 will be [MP PMT 1999]

(a) 5° 24′ (b) 4° 30′

(c) 6° (d) 8°

91. If the refractive index of a material of equilateral prism is $\sqrt{3}$, then angle of minimum deviation of the prism is [CBSE PMT 1999; Pb. PMT 2004; MH CET 2004]

(a) 30° (b) 45°

(c) 60° (d) 75°

92. The splitting of white light into several colours on passing through a glass prism is due to [CPMT 1999]

(a) Refraction (b) Reflection

(c) Interference (d) Diffraction

93. A white screen illuminated by green and red light appears to be

(a) Green (b) Red

(c) Yellow (d) White

94. Dark lines on solar spectrum are due to [EAMCET (Engg.) 1995]

(a) Lack of certain elements

(b) Black body radiation

(c) Absorption of certain wavelengths by outer layers

(d) Scattering

95. Line spectra are due to [EAMCET (Med.) 1995]

(a) Hot solids

(b) Atoms in gaseous state

(c) Molecules in gaseous state

(d) Liquid at low temperature

96. The path of a refracted ray of light in a prism is parallel to the base of the prism only when the [SCRA 1994]

(a) Light is of a particular wavelength

(b) Ray is incident normally at one face

(c) Ray undergoes minimum deviation

(d) Prism is made of a particular type of glass

97. For a medium, refractive indices for violet, red and yellow are 1.62, 1.52 and 1.55 respectively, then dispersive power of medium will be [RPET 1997]

(a) 0.65 (b) 0.22

(c) 0.18 [MP PET 1997] (d) 0.02

98. Two lenses having $f_1 : f_2 = 2 : 3$ has combination to make no dispersion. Find the ratio of dispersive power of glasses used

(a) 2 : 3 (b) 3 : 2

(c) 4 : 9 (d) 9 : 4

99. If refractive index of red, violet and yellow lights are 1.42, 1.62 and 1.50 respectively for a medium. Its dispersive power will be

(a) 0.4 (b) 0.3

(c) 0.2 (d) 0.1

100. A ray is incident at an angle of incidence i on one surface of a prism of small angle A and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ , the angle of incidence i is nearly equal to [CBSE PMT 1992]

(a) A / μ (b) $A / 2\mu$

(c) μA (d) $\mu A / 2$

101. Fraunhofer spectrum is a [KCET 1993, 94; RPET 1997; MP PET 1997, 2001; JIPMER 2000; AIIMS 2001]

(a) Line absorption spectrum

(b) Band absorption spectrum

(c) Line emission spectrum

(d) Band emission spectrum

102. The angle of a prism is 60° and its refractive index is $\sqrt{2}$. The angle of minimum deviation suffered by a ray of light in passing through it is [MP PET 2003]

(a) About 20° [KCET 1994; RPMT 1997] (b) 30°

(c) 60° (d) 45°

103. Colour of the sky is blue due to [CPMT 1996, 99; AFMC 1993; AIIMS 1999; AIEEE 2002; BCECE 2003; BHU 2004]

(a) Scattering of light (b) Total internal reflection

(c) Total emission (d) None of the above

104. Which of the following spectrum have all the frequencies from high to low frequency range [CPMT 1996]

(a) Band spectrum (b) Continuous spectrum

(c) Line spectrum (d) Discontinuous spectrum

105. Stars are not visible in the day time because [JIPMER 1997]

(a) Stars hide behind the sun

(b) Stars do not reflect sun rays during day

(c) Stars vanish during the day

(d) Atmosphere scatters sunlight into a blanket of extreme brightness through which faint stars cannot be visible

106. Which of the following colours suffers maximum deviation in a prism [KCET 1998; DPMT 2000]

(a) Yellow (b) Blue

(c) Green (d) Orange

107. If a thin prism of glass is dipped into water then minimum deviation (with respect to air) of light produced by prism will be left $\left({}_a\mu_g = \dfrac{3}{2} \text{ and } {}_a\mu_w = \dfrac{4}{3}\right)$ **[UPSEAT 1999]**

(a) $\dfrac{1}{2}$ (b) $\dfrac{1}{4}$

(c) 2 (d) $\dfrac{1}{5}$

108. The refractive indices for the light of violet and red colours of any material are 1.66 and 1.64 respectively. If the angle of prism made of this material is 10°, then angular dispersion will be

(a) 0.20° (b) 0.10°

(c) 0.40° (d) 1°

109. The refractive index of the material of the prism for violet colour is 1.69 and that for red is 1.65. If the refractive index for mean colour is 1.66, the dispersive power of the material of the prism

(a) 0.66 (b) 0.06

(c) 0.65 (d) 0.69

110. The deviation caused in red, yellow and violet colours for crown glass prism are 2.84°, 3.28° and 3.72° respectively. The dispersive power of prism material is **[KCET (Engg.) 1999]**

(a) 0.268 (b) 0.368

(c) 0.468 (d) 0.568

111. Dispersion of light is due to **[DCE 1999]**

(a) Wavelength (b) Intensity of light

(c) Density of medium (d) None of these

112. A prism of refracting angle 60° is made with a material of refractive index μ. For a certain wavelength of light, the angle of minimum deviation is 30°. For this, wavelength the value of refractive index of the material is **[CPMT 1999, MH CET 2000]**

(a) 1.231 (b) 1.820

(c) 1.503 (d) 1.414

113. Which of the prism is used to see infrared spectrum of light **[RPMT 2000]**

(a) Rock Salt (b) Nicol

(c) Flint (d) Crown

114. When white light enters a prism, it gets split into its constituent colours. This is due to **[DCE 2000]**

(a) High density of prism material

(b) Because μ is different for different λ

(c) Diffraction of light

(d) Velocity changes for different frequencies

115. The dispersive powers of crown and flint glasses are 0.02 and 0.04 respectively. In an achromatic combination of lenses the focal length of flint glass lens is 40 *cm*. The focal length of crown glass lens will be **[DCE 2000]**

(a) – 20 *cm* (b) + 20 *cm*

(c) – 10 *cm* (d) + 10 *cm*

116. When a ray of light is incident normally on one refracting surface of an equilateral prism (Refractive index of the material of the prism = 1.5 **[EAMCET (Med.) 2000]**

(a) Emerging ray is deviated by 30°

(b) Emerging ray is deviated by 45°

(c) Emerging ray just grazes the second refracting surface

(d) The ray undergoes total internal reflection at the second refracting surface

117. Consider the following two statements A and B and identify the correct choice in the given answers **[EAMCET (Engg.) 2000]**

A : Line spectra is due to atoms in gaseous state

B : Band spectra is due to molecules

(a) Both A and B are false

[UPSEAT 1999]

(b) A is true and B is false

(c) A is false and B is true

(d) Both A and B are true

118. Under minimum deviation condition in a prism, if a ray is incident at an angle 30°, the angle between the emergent ray and the second refracting surface of the prism is **[JIPMER 1999]**

[EAMCET (Engg.) 2000]

(a) 0° (b) 30°

(c) 45° (d) 60°

119. The angle of prism is 5° and its refractive indices for red and violet colours are 1.5 and 1.6 respectively. The angular dispersion produced by the prism is **[MP PMT 2000]**

(a) 7.75° (b) 5°

(c) 0.5° (d) 0.17°

120. If the refractive angles of two prisms made of crown glass are 10° and 20° respectively, then the ratio of their colour deviation powers will be **[KCET 1999; AFMC 2001]**

(a) 1 : 1 (b) 2 : 1

(c) 4 : 1 (d) 1 : 2

121. The nature of sun's spectrum is **[MP PET 2000; MP PMT 2001]**

(a) Continuous spectrum with absorption lines

(b) Line spectrum

(c) The spectrum of the helium atom

(d) Band spectrum

122. A ray of light is incident normally on one of the face of a prism of angle 30° and refractive index $\sqrt{2}$. The angle of deviation will be **[KCET 2001]**

(a) 26° (b) 0°

(c) 23° (d) 15°

123. For a prism of refractive index 1.732, the angle of minimum deviation is equal to the angle of the prism. The angle of the prism is **[CBSE PMT 2001]**

(a) 80° (b) 70°

(c) 60°   (d) 50°

124. The spectrum obtained from an electric lamp or red hot heater is

(a) Line spectrum   (b) Band spectrum

(c) Absorption spectrum   (d) Continuous spectrum

125. When a glass prism of refracting angle 60° is immersed in a liquid its angle of minimum deviation is 30°. The critical angle of glass with respect to the liquid medium is [EAMCET 2001]

(a) 42°   (b) 45°

(c) 50°   (d) 52°

126. Three prisms 1, 2 and 3 have the prism angle A = 60°, but their refractive indices are respectively 1.4, 1.5 and 1.6. If δ_1, δ_2, δ_3 be their respective angles of deviation then

[MP PMT 2001]

(a) $\delta_3 > \delta_2 > \delta_1$ (b) $\delta_1 > \delta_2 > \delta_3$

(c) $\delta_1 = \delta_2 = \delta_3$ (d) $\delta_2 > \delta_1 > \delta_3$

127. Which one of the following alternative is FALSE for a prism placed in a position of minimum deviation [MP PET 2001]

(a) $i_1 = i_2$ (b) $r_1 = r_2$

(c) $i_1 = r_1$ (d) All of these

128. In the visible region the dispersive powers and the mean angular deviations for crown and flint glass prisms are ω, ω' and d, d' respectively. The condition for getting deviation without dispersion when the two prisms are combined is

[EAMCET 2001]

(a) $\sqrt{\omega d} + \sqrt{\omega' d'} = 0$ (b) $\omega' d + \omega d' = 0$

(c) $\omega d + \omega' d' = 0$ (d) $(\omega d)^2 + (\omega' d')^2 = 0$

129. A ray of light passes through the equilateral prism such that angle of incidence is equal to the angle of emergence if the angle of incidence is 45°. The angle of deviation will be

[Pb. PMT 2002]

(a) 15°   (b) 75°

(c) 60°   (d) 30°

130. The solar spectrum during a complete solar eclipse is

[Kerala PET 2002]

(a) Continuous   (b) Emission line

(c) Dark line   (d) Dark band

131. Why sun has elliptical shape on the time when rising and sun setting ? It is due to [AFMC 2002]

(a) Refraction   (b) Reflection

(c) Scattering   (d) Dispersion

132. In the formation of a rainbow light from the sun on water droplets undergoes [CBSE PMT 2000; Orissa JEE 2002; MP PET 2003; KCET 2004]

(a) Dispersion only

(b) Only total internal reflection

(c) Dispersion and total internal reflection

(d) None of these

[BHU 2001; Pb. PET 2003]

133. The Cauchy's dispersion formula is [AIIMS 2002]

(a) $n = A + B\lambda^{-2} + C\lambda^{-4}$ (b) $n = A + B\lambda^{2} + C\lambda^{-4}$

(c) $n = A + B\lambda^{-2} + C\lambda^{4}$ (d) $n = A + B\lambda^{2} + C\lambda^{4}$

134. A prism of refractive index μ and angle A is placed in the minimum deviation position. If the angle of minimum deviation is A, then the value of A in terms of μ is

[EAMCET 2003]

(a) $\sin^{-1}\left(\frac{\mu}{2}\right)$ (b) $\sin^{-1}\sqrt{\frac{\mu-1}{2}}$

(c) $2\cos^{-1}\left(\frac{\mu}{2}\right)$ (d) $\cos^{-1}\left(\frac{\mu}{2}\right)$

135. A given ray of light suffers minimum deviation in an equilateral prism P. Additional prisms Q and R of identical shape and material are now added to P as shown in the figure. The ray will suffer

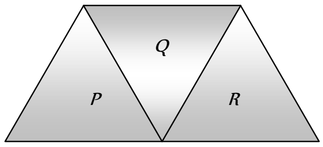
[IIT-JEE (Screening) 2001; KCET 2003]

(a) Greater deviation

(b) Same deviation

(c) No deviation

(d) Total internal reflection



136. In the given figure, what is the angle of prism

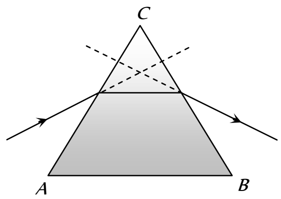
[Orissa JEE 2003]

(a) A

(b) B

(c) C

(d) D



137. A prism of refractive index $\sqrt{2}$ has a refracting angle of 60°. At what angle a ray must be incident on it so that it suffers a minimum deviation [BHU 2003; MP PMT 2005]

(a) 45°   (b) 60°

(c) 90°   (d) 180°

138. A convex lens, a glass slab, a glass prism and a solid sphere all are made of the same glass, the dispersive power will be

[CPMT 1986]

(a) In the glass slab and prism

(b) In the lens and solid sphere

(c) Only in prism

(d) In all the four

139. A parallel beam of white light falls on a convex lens. Images of blue, yellow and red light are formed on other side of the lens at a distance of 0.20 m, 0.205 m and 0.214 m respectively. The dispersive power of the material of the lens will be

(a) 619/1000   (b) 9/200

(c) 14/205   (d) 5/214

140. The refractive index of the material of the prism for violet colour is 1.69 and that for red is 1.65. If the refractive index for mean colour is 1.66, the dispersive power of the material of the prism [JIPMER 1999]

(a) 0.66 (b 0.06

(c) 0.65 (d) 0.69

141. If the angle of prism is 60^o and the angle of minimum deviation is 40^o, the angle of refraction will be [MP PMT 2004]

(a) 30^o (b) 60^o

(c) 100^o (d) 120^o

142. The refractive index of a particular material is 1.67 for blue light, 1.65 for yellow light and 1.63 for red light. The dispersive power of the material is .... [KCET 2004]

(a) 0.0615 (b) 0.024

(c) 0.031 (d) 1.60

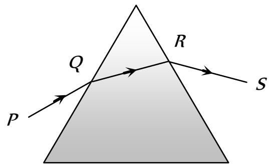
143. A ray of light is incident on an equilateral glass prism placed on a horizontal table. For minimum deviation which of the following is true [IIT-JEE (Screening) 2004]

(a) *PQ* is horizontal

(b) *QR* is horizontal

(c) *RS* is horizontal

(d) Either *PQ* or *RS* is horizontal



144. A beam of light composed of red and green ray is incident obliquely at a point on the face of rectangular glass slab. When coming out on the opposite parallel face, the red and green ray emerge from [CBSE PMT 2004]

(a) Two points propagating in two different directions

(b) Two points propagating in two parallel directions

(c) One point propagating in two different directions

(d) One point propagating in the same directions

145. White light is passed through a prism .... colour shows minimum deviation [Orissa PMT 2004]

(a) Red (b) Violet

(c) Yellow (d) Green

146. A ray of monochromatic light suffers minimum deviation of 38^o while passing through a prism of refracting angle 60^o. Refractive index of the prism material is [Pb. PET 2001]

(a) 1.5 (b) 1.3

(c) 0.8 (d) 2.4

147. A ray incident a 15^o on one refracting surface of a prism of angle 60^o, suffers a deviation of 55^o. What is the angle of emergence [DCE 2002]

(a) 95^o (b) 45^o

(c) 30^o (d) None of these

148. The spectrum obtained from a sodium vapour lamp is an example of [MH CET 2003]

(a) Absorption spectrum (b) Emission spectrum
(c) Continuous spectrum (d) Band spectrum

149. The sky would appear red instead of blue if [DCE 2004]

(a) Atmospheric particles scatter blue light more than red light
(b) Atmospheric particles scatter all colours equally
(c) Atmospheric particles scatter red light more than the blue light
(d) The sun was much hotter

150. Sir C.V. Raman was awarded Nobel Prize for his work connected with which of the following phenomenon of radiation

(a) Scattering (b) Diffraction
(c) Interference (d) Polarisation

151. In absorption spectrum of *Na* the missing wavelength (*s*) are [BCECE 2005]

(a) 589 *nm* (b) 589.6 *nm*
(c) Both (d) None of these

Human Eye and Lens Camera

1. A far sighted man who has lost his spectacles, reads a book by looking through a small hole (3-4 *mm*) in a sheet of paper. The reason will be [CPMT 1977]

(a) Because the hole produces an image of the letters at a longer distance
(b) Because in doing so, the focal length of the eye lens is effectively increased
(c) Because in doing so, the focal length of the eye lens is effectively decreased
(d) None of these

2. For a normal eye, the least distance of distinct vision is [CPMT 1984]

(a) 0.25 *m* (b) 0.50 *m*
(c) 25 *m* (d) Infinite

3. For the myopic eye, the defect is cured by [CPMT 1990; KCET (Engg.) 2000]

(a) Convex lens (b) Concave lens
(c) Cylindrical lens (d) Toric lens

4. Lens used to remove long sightedness (hypermetropia) is

or

A person suffering from hypermetropia requires which type of spectacle lenses [MP PMT 1995]

(a) Concave lens (b) Plano-concave lens
(c) Convexo-concave lens (d) Convex lens

5. Substance on the choroid is

(a) Japan black (b) Nigrim pigment
(c) Carbon black (d) Platinum black

6. Astigmatism (for a human eye) can be removed by using [CPMT 1972; MP PET/PMT 1988; CBSE PMT 1990]

(a) Concave lens (b) Convex lens
(c) Cylindrical lens (d) Prismatic lens

7. Circular part in the centre of retina is called [MP PET/PMT 1988]

(a) Blind spot (b) Yellow spot
(c) Red spot (d) None of the above

8. Image formed on the retina is

(a) Real and inverted (b) Virtual and erect
(c) Real and erect (d) Virtual and inverted

9. If there had been one eye of the man, then

(a) Image of the object would have been inverted
(b) Visible region would have decreased
(c) Image would have not been seen three dimensional
(d) (b) and (c) both

10. A person cannot see distinctly at the distance less than one metre. Calculate the power of the lens that he should use to read a book at a distance of 25 *cm*

[CPMT 1983; AFMC 2005] [CPMT 1977; MP PET 1985, 88; MP PMT 1990]

(a) + 3.0 *D* (b) + 0.125 *D*
(c) – 3.0 *D* (d) + 4.0 *D*

11. How should people wearing spectacles work with a microscope

(a) They cannot use the microscope at all
(b) They should keep on wearing their spectacles
(c) They should take off spectacles
(d) (b) and (c) is both way

12. A man who cannot see clearly beyond 5 *m* wants to see stars clearly. He should use a lens of focal length [MP PET/PMT 1988; Pb. PET 2003]

(a) – 100 *m* (b) + 5 *m*
(c) – 5 *m* (d) Very large

13. A man can see only between 75 *cm* and 200 *cm*. The power of lens to correct the near point will be

(a) + 8/3 *D* (b) + 3 *D*
(c) – 3 *D* (d) – 8/3 *D*

14. Image is formed for the short sighted person at [AFMC 1988]

(a) Retina (b) Before retina
(c) Behind the retina (d) Image is not formed at all

15. A man can see the objects upto a distance of one metre from his eyes. For correcting his eye sight so that he can see an object at infinity, he requires a lens whose power is

or

A man can see upto 100 *cm* of the distant object. The power of the lens required to see far objects will be [MP PMT 1993, 2003]

(a) + 0.5 *D* (b) + 1.0 *D*
(c) + 2.0 *D* (d) – 1.0 *D*

16. A man can see the object between 15 *cm* and 30 *cm*. He uses the lens to see the far objects. Then due to the lens used, the near point will be at

(a) $\frac{10}{3}$ *cm* (b) 30 *cm*
(c) 15 *cm* (d) $\frac{100}{3}$ *cm*

17. The far point of a myopia eye is at 40 *cm*. For removing this defect, the power of lens required will be [MP PMT 1987]

(a) 40 *D* (b) – 4 *D*
(c) – 2.5 *D* (d) 0.25 *D*

18. A man suffering from myopia can read a book placed at 10 *cm* distance. For reading the book at a distance of 60 *cm* with relaxed vision, focal length of the lens required will be [MP PMT 1989]

(a) 45 *cm* (b) – 20 *cm*

(c) – 12 *cm* (d) 30 *cm*

19. If the distance of the far point for a myopia patient is doubled, the focal length of the lens required to cure it will become

(a) Half

(b) Double

(c) The same but a convex lens

(d) The same but a concave lens

20. A presbyopic patient has near point as 30 *cm* and far point as 40 *cm*. The dioptric power for the corrective lens for seeing distant objects is

(a) 40 *D* (b) 4 *D*

(c) – 2.5 *D* (d) 0.25 *D*

21. An imaginary line joining the optical centre of the eye lens and the yellow point is called as

(a) Principal axis (b) Vision axis

(c) Neutral axis (d) Optical axis

22. The light when enters the human eye experiences most of the refraction while passing through

(a) Cornea (b) Aqueous humour

(c) Vitrous humour (d) Crystalline lens

23. The impact of an image on the retina remains for

(a) 0.1 *sec* (b) 0.5 *sec*

(c) 10 *sec* (d) 15 *sec*

24. A person is suffering from myopic defect. He is able to see clear objects placed at 15 *cm*. What type and of what focal length of lens he should use to see clearly the object placed 60 *cm* away

(a) Concave lens of 20 *cm* focal length

(b) Convex lens of 20 *cm* focal length

(c) Concave lens of 12 *cm* focal length

(d) Convex lens of 12 *cm* focal length

25. The sensation of vision in the retina is carried to the brain by

(a) Ciliary muscles (b) Blind spot

(c) Cylindrical lens (d) Optic nerve

26. When the power of eye lens increases, the defect of vision is produced. The defect is known as

(a) Shortsightedness (b) Longsightedness

(c) Colourblindness (d) None of the above

27. A man is suffering from colour blindness for green colour. To remove this defect, he should use goggles of

(a) Green colour glasses (b) Red colour glasses

(c) Smoky colour glasses (d) None of the above

28. In human eye the focussing is done by [CPMT 1983]

(a) To and fro movement of eye lens

(b) To and fro movement of the retina

(c) Change in the convexity of the lens surface

(d) Change in the refractive index of the eye fluids

29. A short sighted person can see distinctly only those objects which lie between 10 *cm* and 100 *cm* from him. The power of the spectacle lens required to see a distant object is

[MP PET 1992]

(a) + 0.5 *D* (b) – 1.0 *D*

(c) – 10 *D* (d) + 4.0 *D*

30. A person can see clearly only upto a distance of 25 *cm*. He wants to read a book placed at a distance of 50 *cm*. What kind of lens does he require for his spectacles and what must be its power [MP PET 1989]

(a) Concave, – 1.0 *D* (b) Convex, + 1.5 *D*

(c) Concave, – 2.0 *D* (d) Convex, + 2.0 *D*

31. The human eye has a lens which has a [MP PET 1994]

(a) Soft portion at its centre

(b) Hard surface

(c) Varying refractive index

(d) Constant refractive index

32. A man with defective eyes cannot see distinctly object at the distance more than 60 *cm* from his eyes. The power of the lens to be used will be [MP PMT 1994]

(a) + 60 *D* (b) – 60 *D*

(c) – 1.66 *D* (d) $\frac{1}{1.66}D$

33. A person's near point is 50 *cm* and his far point is 3 *m*. Power of the lenses he requires for

(i) reading and

(ii) for seeing distant stars are [MP PMT 1994]

(a) – 2 *D* and 0.33 *D* (b) 2 *D* and – 0.33 *D*

(c) – 2 *D* and 3 *D* (d) 2 *D* and – 3 *D*

34. A person wears glasses of power – 2.5 *D*. The defect of the eye and the far point of the person without the glasses are respectively [MP PMT 1991]

(a) Farsightedness, 40 *cm* (b) Nearsightedness, 40 *cm*

(c) Astigmatism, 40 *cm* (d) Nearsightedness, 250 *cm*

35. Myopia is due to [AFMC 1996]

(a) Elongation of eye ball

(b) Irregular change in focal length

(c) Shortening of eye ball

(d) Older age

36. A person is suffering from the defect astigmatism. Its main reason is

(a) Distance of the eye lens from retina is increased

(b) Distance of the eye lens from retina is decreased

(c) The cornea is not spherical

(d) Power of accommodation of the eye is decreased

37. A person cannot see objects clearly beyond 2.0 *m*. The power of lens required to correct his vision will be

[MP PMT/PET 1998; JIPMER 2000; KCET 2000; Pb. PET 2001]

(a) + 2.0 *D* (b) – 1.0 *D*

(c) + 1.0 *D* (d) – 0.5 *D*

38. The resolving limit of healthy eye is about

[MP PET 1999; RPMT 1999; AIIMS 2001]

(a) $1'$ or $\left(\frac{1}{60}\right)^\circ$ (b) $1''$

(c) 1° (d) $\frac{1}{60}''$

39. When objects at different distances are seen by the eye, which of the following remains constant **[MP PMT 1999]**

(a) The focal length of the eye lens

(b) The object distance from the eye lens

(c) The radii of curvature of the eye lens

(d) The image distance from the eye lens

40. A person wears glasses of power – 2.0 *D*. The defect of the eye and the far point of the person without the glasses will be

(a) Nearsighted, 50 *cm* (b) Farsighted, 50 *cm*

(c) Nearsighted, 250 *cm* (d) Astigmatism, 50 *cm*

41. An eye specialist prescribes spectacles having a combination of convex lens of focal length 40 *cm* in contact with a concave lens of focal length 25 *cm*. The power of this lens combination in diopters is

[IIT 1997 Cancelled; DPMT 2000]

(a) + 1.5 (b) – 1.5

(c) + 6.67 (d) – 6.67

42. Match the List *I* with the List *II* from the combinations shown

(I)	Presbiopia	(A)	Sphero-cylindrical lens
(II)	Hypermetropia	(B)	Convex lens of proper power may be used close to the eye
(III)	Astigmatism	(C)	Concave lens of suitable focal length
(IV)	Myopia	(D)	Bifocal lens of suitable focal length

(a) I-A; II-C; III-B; IV-D (b) I-B; II-D; III-C; IV-A

(c) I-D; II-B; III-A; IV-C (d) I-D; II-A; III-C; IV-B

43. Near and far points of a human eye are

[EAMCET (Med.) 1995; MP PET 2001; BCECE 2004]

(a) 0 and 25 *cm* (b) 0 and ∞

(c) 25 *cm* and 100 *cm* (d) 25 *cm* and ∞

44. Two parallel pillars are 11 *km* away from an observer. The minimum distance between the pillars so that they can be seen separately will be **[RPET 1997; RPMT 2000]**

(a) 3.2 *m* (b) 20.8 *m*

(c) 91.5 *m* (d) 183 *m*

45. Retina of eye acts like .... of camera **[AFMC 2003]**

(a) Shutter (b) Film

(c) Lens (d) None of these

46. The hyper-metropia is a **[CBSE PMT 2000]**

(a) Short-side defect (b) Long- side defect

(c) Bad vision due to old age (d) None of these

47. Amount of light entering into the camera depends upon

[DCE 2000]

(a) Focal length of the objective lens

(b) Product of focal length and diameter of the objective lens

(c) Distance of the object from camera

(d) Aperture setting of the camera

48. A man cannot see clearly the objects beyond a distance of 20 *cm* from his eyes. To see distant objects clearly he must use which kind of lenses and of what focal length

[MP PMT 2000]

(a) 100 *cm* convex (b) 100 *cm* concave

(c) 20 *cm* convex (d) 20 *cm* concave

49. A person uses spectacles of power +2*D*. He is suffering from

[MP PMT 1999] **[MP PET 2000]**

(a) Short sightedness or myopia

(b) Long sightedness or hypermetropia

(c) Presbyopia

(d) Astigmatism

50. To remove myopia (short sightedness) a lens of power 0.66 *D* is required. The distant point of the eye is approximately

[MP PMT 2001]

(a) 100 *cm* (b) 150 *cm*

(c) 50 *cm* (d) 25 *cm*

51. A person suffering from 'presbyopia' (myopia and hyper metropia both defects) should use **[ISM Dhanbad 1994]** **[MP PET 2001]**

(a) A concave lens

(b) A convex lens

(c) A bifocal lens whose lower portion is convex

(d) A bifocal lens whose upper portion is convex

52. A person who can see things most clearly at a distance of 10 *cm*. Requires spectacles to enable to him to see clearly things at a distance of 30 *cm*. What should be the focal length of the spectacles

[BHU 2003; CPMT 2004; PM PMT 2005]

(a) 15 *cm* (Concave) (b) 15 *cm* (Convex)

(c) 10 *cm* (d) 0

53. Far points of myopic eye is 250 *cm*, then the focal length of the lens to be used will be **[DPMT 2002]**

(a) – 250 *cm* (b) – 250/9 *cm*

(c) + 250 *cm* (d) + 250/9 *cm*

54. A man can see clearly up to 3 *metres*. Prescribe a lens for his spectacles so that he can see clearly up to 12 *metres*

[DPMT 2002]

(a) – 3/4 *D* (b) 3 *D*

(c) – 1/4 *D* (d) – 4 *D*

55. A satisfactory photographic print is obtained when the exposure time is 10 *sec* at a distance of 2 *m* from a 60 *cd* lamp. The time of exposure required for the same quality print at a distance of 4 *m* from a 120 *cd* lamp is

[Kerala PMT 2002]

(a) 5 *sec* (b) 10 *sec*

(c) 15 *sec* (d) 20 *sec*

56. A person can not see the objects clearly placed at a distance more than 40 *cm*. He is advised to use a lens of power

[DCE 2002; MP PMT 2002, 03]

(a) – 2.5 *D* (b) + 2.5 *D*

(c) – 6.25 *D* (d) + 1.5 *D*

57. A person uses a lens of power + 3*D* to normalise vision. Near point of hypermetropic eye is **[CPMT 2002]**

(a) 1 *m* (b) 1.66 *m*

(c) 2 *m* (d) 0.66 *m*

58. A defective eye cannot see close objects clearly because their image is formed [MP PET 2003]

(a) On the eye lens

(b) Between eye lens and retina

(c) On the retina

(d) Beyond retina

59. Image formed on retina of eye is proportional to [RPMT 2001]

(a) Size of object (b) Area of object

(c) $\frac{\text{Size of object}}{\text{Size of image}}$ (d) $\frac{\text{size of image}}{\text{size of object}}$

60. A student can distinctly see the object upto a distance 15 *cm*. He wants to see the black board at a distance of 3 *m*. Focal length and power of lens used respectively will be [Pb. PMT 2003]

(a) $-4.8\,cm, -3.3\,D$ (b) $-5.8\,cm, -4.3\,D$

(c) $-7.5\,cm, -6.3\,D$ (d) $-15.8\,cm, -6.3\,D$

61. A camera objective has an aperture diameter *d*. If the aperture is reduced to diameter $d/2$, the exposure time under identical conditions of light should be made [Kerala PMT 2004]

(a) $\sqrt{2}$ fold (b) 2 fold

(c) $2\sqrt{2}$ fold (d) 4 fold

62. The light gathering power of a camera lens depends on [DCE 2003]

(a) Its diameter only

(b) Ratio of focal length and diameter

(c) Product of focal length and diameter

(d) Wavelength of light used

63. The exposure time of a camera lens at the $\frac{f}{2.8}$ setting is $\frac{1}{200}$ second. The correct time of exposure at $\frac{f}{5.6}$ is [DCE 2003]

(a) 0.4 *sec* (b) 0.02 *sec*

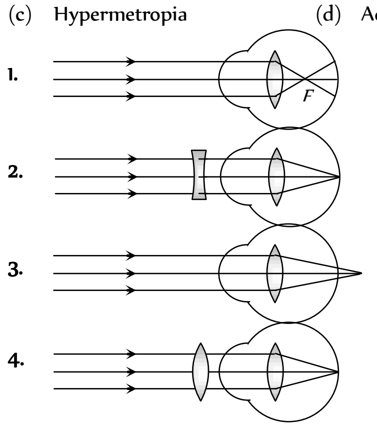
(c) 0.002 *sec* (d) 0.04 *sec*

64. Ability of the eye to see objects at all distances is called [AFMC 2005]

(a) Binocular vision (b) Myopia

(c) Hypermetropia (d) Accommodation

65. [KCET 2005]



Identify the wrong description of the above figures

(a) 1 represents far-sightedness

(b) 2 correction for short sightedness

(c) 3 represents far sightedness

(d) 4 correction for far-sightedness

Microscope and Telescope

1. The focal lengths of the objective and eye-lens of a microscope are 1 *cm* and 5 *cm* respectively. If the magnifying power for the relaxed eye is 45, then the length of the tube is

(a) 30 *cm* (b) 25 *cm*

(c) 15 *cm* (d) 12 *cm*

2. In a compound microscope magnification will be large, if the focal length of the eye piece is [CPMT 1984]

(a) Large (b) Smaller

(c) Equal to that of objective (d) Less than that of objective

3. The focal length of the objective lens of a compound microscope is [CPMT 198

(a) Equal to the focal length of its eye piece

(b) Less than the focal length of eye piece

(c) Greater than the focal length of eye piece

(d) Any of the above three

4. Microscope is an optical instrument which

(a) Enlarges the object

(b) Increases the visual angle formed by the object at the eye

(c) Decreases the visual angle formed by the object at the eye

(d) Brings the object nearer

5. Magnifying power of a simple microscope is (when final image is formed at D = 25 *cm* from eye) [MP PET 1996; BVP 2003]

(a) $\frac{D}{f}$ (b) $1+\frac{D}{f}$

(c) $1+\frac{f}{D}$ (d) $1-\frac{D}{f}$

6. If in compound microscope m_1 and m_2 be the linear magnification of the objective lens and eye lens respectively, then magnifying power of the compound microscope will be [CPMT 1985; KCET 1994]

(a) $m_1 - m_2$ (b) $\sqrt{m_1 + m_2}$

(c) $(m_1 + m_2)/2$ (d) $m_1 \times m_2$

7. For which of the following colour, the magnifying power of a microscope will be maximum

(a) White colour (b) Red colour

(c) Violet colour (d) Yellow colour

8. The length of the compound microscope is 14 *cm*. The magnifying power for relaxed eye is 25. If the focal length of eye lens is 5 *cm*, then the object distance for objective lens will be

(a) 1.8 *cm* (b) 1.5 *cm*

(c) 2.1 *cm* (d) 2.4 *cm*

9. If the focal length of objective and eye lens are 1.2 *cm* and 3 *cm* respectively and the object is put 1.25 *cm* away from the objective lens and the final image is formed at infinity. The magnifying power of the microscope is

(a) 150 (b) 200

(c) 250 (d) 400

10. The focal length of objective and eye lens of a microscope are 4 *cm* and 8 *cm* respectively. If the least distance of distinct vision is 24 *cm* and object distance is 4.5 *cm* from the objective lens, then the magnifying power of the microscope will be

(a) 18 (b) 32
(c) 64 (d) 20

11. When the length of a microscope tube increases, its magnifying power [MNR 1986]

(a) Decreases (b) Increases
(c) Does not change (d) May decrease or increase

12. In a compound microscope, if the objective produces an image I_o and the eye piece produces an image I_e, then [MP PET 1990]

(a) I_o is virtual but I_e is real
(b) I_o is real but I_e is virtual
(c) I_o and I_e are both real
(d) I_o and I_e are both virtual

13. The magnifying power of a simple microscope can be increased, if we use eye-piece of [MP PMT 1986]

(a) Higher focal length (b) Smaller focal length
(c) Higher diameter (d) Smaller diameter

14. An electron microscope is superior to an optical microscope in

(a) Having better resolving power
(b) Being easy to handle
(c) Low cost
(d) Quickness of observation

15. The magnifying power of a microscope with an objective of 5 *mm* focal length is 400. The length of its tube is 20 *cm*. Then the focal length of the eye-piece is [MP PMT 1991]

(a) 200 *cm* (b) 160 *cm*
(c) 2.5 *cm* (d) 0.1 *cm*

16. The maximum magnification that can be obtained with a convex lens of focal length 2.5 *cm* is (the least distance of distinct vision is 25 *cm*) [MP PET 2003]

(a) 10 (b) 0.1
(c) 62.5 (d) 11

17. When the object is self-luminous, the resolving power of a microscope is given by the expression

(a) $\frac{2\mu \sin\theta}{1.22\,\lambda}$ (b) $\frac{\mu \sin\theta}{\lambda}$

(c) $\frac{2\mu \cos\theta}{1.22\,\lambda}$ (d) $\frac{2\mu}{\lambda}$

18. The power of two convex lenses A and B are 8 *diopters* and 4 *diopters* respectively. If they are to be used as a simple microscope, the magnification of

(a) B will be greater than A
(b) A will be greater than B
(c) The information is incomplete
(d) None of the above

19. Finger prints are observed by the use of

(a) Telescope (b) Microscope
(c) Gallilean telescope (d) Concave lens

20. To produce magnified erect image of a far object, we will be required along with a convex lens, is [MNR 1983]

(a) Another convex lens (b) Concave lens
(c) A plane mirror (d) A concave mirror

21. In order to increase the magnifying power of a compound microscope [JIPMER 1986; MP PMT 1997]

(a) The focal lengths of the objective and the eye piece should be small
(b) Objective should have small focal length and the eye piece large
(c) Both should have large focal lengths
(d) The objective should have large focal length and eye piece should have small

22. If the focal length of the objective lens is increased then [MP PMT 1994]

(a) Magnifying power of microscope will increase but that of telescope will decrease
(b) Magnifying power of microscope and telescope both will increase
(c) Magnifying power of microscope and telescope both will decrease
(d) Magnifying power of microscope will decrease but that of telescope will increase

23. The magnification produced by the objective lens and the eye lens of a compound microscope are 25 and 6 respectively. The magnifying power of this microscope is [CPMT 1984]

[Manipal MEE 1995; DPMT 2002]

(a) 19 (b) 31
(c) 150 (d) $\sqrt{150}$

24. The focal lengths of the objective and the eye-piece of a compound microscope are 2.0 *cm* and 3.0 *cm* respectively. The distance between the objective and the eye-piece is 15.0 *cm*. The final image formed by the eye-piece is at infinity. The two lenses are thin. The distances in *cm* of the object and the image produced by the objective measured from the objective lens are respectively [IIT 1995]

(a) 2.4 and 12.0 (b) 2.4 and 15.0
(c) 2.3 and 12.0 (d) 2.3 and 3.0

25. Resolving power of a microscope depends upon [MP PET 1995]

(a) The focal length and aperture of the eye lens
(b) The focal lengths of the objective and the eye lens
(c) The apertures of the objective and the eye lens
(d) The wavelength of light illuminating the object

26. The objective lens of a compound microscope produces magnification of 10. In order to get an overall magnification of 100 when image is formed at 25 *cm* from the eye, the focal length of the eye lens should be

(a) 4 *cm* (b) 10 *cm*
(c) $\frac{25}{9}$ *cm* (d) 9 *cm*

27. A person using a lens as a simple microscope sees an

(a) Inverted virtual image
(b) Inverted real magnified image
(c) Upright virtual image
(d) Upright real magnified image

28. Least distance of distinct vision is 25 *cm*. Magnifying power of simple microscope of focal length 5 *cm* is [EAMCET (Engg.) 1995; Pb. PMT 1999]

(a) 1 / 5 (b) 5
(c) 1 / 6 (d) 6

29. The objective of a compound microscope is essentially

[SCRA 1998]

(a) A concave lens of small focal length and small aperture

(b) Convex lens of small focal length and large aperture

(c) Convex lens of large focal length and large aperture

(d) Convex lens of small focal length and small aperture

30. Resolving power of a microscope depends upon

[DCE 1999]

(a) Wavelength of light used, directly

(b) Wavelength of light used, inversely

(c) Frequency of light used

(d) Focal length of objective

31. In a compound microscope cross-wires are fixed at the point

[EAMCET (Engg.) 2000]

(a) Where the image is formed by the objective

(b) Where the image is formed by the eye-piece

(c) Where the focal point of the objective lies

(d) Where the focal point of the eye-piece lies

32. In a compound microscope, the focal lengths of two lenses are 1.5 *cm* and 6.25 *cm* an object is placed at 2 *cm* form objective and the final image is formed at 25 *cm* from eye lens. The distance between the two lenses is

[EAMCET (Med.) 2000]

(a) 6.00 *cm* (b) 7.75 *cm*

(c) 9.25 *cm* (d) 11.00 *cm*

33. The length of the tube of a microscope is 10 *cm*. The focal lengths of the objective and eye lenses are 0.5 *cm* and 1.0 *cm*. The magnifying power of the microscope is about

[MP PMT 2000]

(a) 5 (b) 23

(c) 166 (d) 500

34. In a compound microscope, the intermediate image is

[IIT-JEE (Screening) 2000; MP PET 2005]

(a) Virtual, erect and magnified

(b) Real, erect and magnified

(c) Real, inverted and magnified

(d) Virtual, erect and reduced

35. The magnifying power of a compound microscope increases when

(a) The focal length of objective lens is increased and that of eye lens is decreased

(b) The focal length of eye lens is increased and that of objective lens is decreased

(c) Focal lengths of both objective and eye-piece are increased

(d) Focal lengths of both objective and eye-piece are decreased

36. If the red light is replaced by blue light illuminating the object in a microscope the resolving power of the microscope

(a) Decreases (b) Increases

(c) Gets halved (d) Remains unchanged

37. The magnifying power of a simple microscope is 6. The focal length of its lens in *metres* will be, if least distance of distinct vision is 25 *cm* [MP PMT 2001]

(a) 0.05 (b) 0.06

(c) 0.25 (d) 0.12

38. Two points separated by a distance of 0.1 *mm* can just be resolved in a microscope when a light of wavelength 6000 Å is used. If the light of wavelength 4800Å is used this limit of resolution becomes

(a) 0.08 *mm* (b) 0.10 *mm*

(c) 0.12 *mm* (d) 0.06 *mm*

39. A compound microscope has two lenses. The magnifying power of one is 5 and the combined magnifying power is 100. The magnifying power of the other lens is

[Kerala PMT 2002]

(a) 10 (b) 20

(c) 50 (d) 25

40. The angular magnification of a simple microscope can be increased by increasing [Orissa JEE 2002]

(a) Focal length of lens (b) Size of object

(c) Aperture of lens (d) Power of lens

41. Wavelength of light used in an optical instrument are $\lambda_1 = 4000$ Å and $\lambda_2 = 5000$ Å, then ratio of their respective resolving power (corresponding to λ_1 and λ_2) is

[AIEEE 2002]

(a) 16 : 25 (b) 9 : 1

(c) 4 : 5 (d) 5 : 4

42. The separation between two microscopic particles is measured P_A and P_B by two different lights of wavelength 2000 Å and 3000 Å respectively, then [AIEEE 2002]

(a) $P_A > P_B$ (b) $P_A < P_B$

(c) $P_A < 3/2P_B$ (d) $P_A = P_B$

43. The image formed by an objective of a compound microscope is

(a) Virtual and enlarged (b) Virtual and diminished

(c) Real and diminished (d) Real and enlarged

44. An achromatic telescope objective is to be made by combining the lenses of flint and crown glasses. This proper choice is

(a) Convergent of crown and divergent of flint

(b) Divergent of crown and convergent of flint

[MP PET 2000]

(c) Both divergent

(d) Both convergent

45. If F_o and F_e are the focal length of the objective and eye-piece respectively of a telescope, then its magnifying power will be [CPMT 1977, 82, 97

SCRA 1994; KCET 1999; Pb. PMT 2000; BHU 2001;

DCE 2002; RPMT 2003; BCECE 2003, 04]

(a) $F_o + F_e$ (b) $F_o \times F_e$

[DCE 2001]

(c) F_o / F_e (d) $\frac{1}{2}(F_o + F_e)$

46. The magnifying power of a telescope can be increased by

[CPMT 1979]

(a) Increasing focal length of the system

(b) Fitting eye piece of high power

(c) Fitting eye piece of low power

(d) Increasing the distance of objects

47. A simple telescope, consisting of an objective of focal length 60 *cm* and a single eye lens of focal length 5 *cm* is focussed on a distant object is such a way that parallel rays comes out from the eye lens. If the object subtends an angle 2° at the objective, the angular width of the image

[CPMT 1979; NCERT 1980; MP PET 1992; JIPMER 1997; UPSEAT 2001]

(a) 10° (b) 24°

(c) 50° (d) 1/6°

48. The diameter of the objective of the telescope is 0.1 *metre* and wavelength of light is 6000 Å. Its resolving power would be approximately [MP PET 1997]

(a) $7.32\times10^{-6} rad$ (b) $1.36\times10^{6} rad$

(c) $7.32\times10^{-5} rad$ (d) $1.36\times10^{5} rad$

49. A photograph of the moon was taken with telescope. Later on, it was found that a housefly was sitting on the objective lens of the telescope. In photograph

[NCERT 1970; MP PET 1999]

(a) The image of housefly will be reduced

(b) There is a reduction in the intensity of the image

(c) There is an increase in the intensity of the image

(d) The image of the housefly will be enlarged

50. For a telescope to have large resolving power the

[CPMT 1980, 81, 85; MP PET 1994; DCE 2001; AFMC 2005]

(a) Focal length of its objective should be large

(b) Focal length of its eye piece should be large

(c) Focal length of its eye piece should be small

(d) Aperture of its objective should be large

51. An observer looks at a tree of height 15 *m* with a telescope of magnifying power 10. To him, the tree appears

[CPMT 1975]

(a) 10 times taller (b) 15 times taller

(c) 10 times nearer (d) 15 times nearer

52. The focal length of objective and eye lens of a astronomical telescope are respectively 2 *m* and 5 *cm*. Final image is formed at (i) least distance of distinct vision (ii) infinity. The magnifying power in both cases will be [MP PMT/PET 1988]

(a) – 48, – 40 (b) – 40, – 48

(c) – 40, 48 (d) – 48, 40

53. For observing a cricket match, a binocular is preferred to a terrestrial telescope because

(a) The binocular gives the proper three dimensional view

(b) The binocular has shorter length

(c) The telescope does not give erect image

(d) Telescope have chromatic aberrations

54. To increase the magnifying power of telescope (f_o = focal length of the objective and f_e = focal length of the eye lens)

[MP PET/PMT 1988; MP PMT 1992, 94]

(a) f_o should be large and f_e should be small

(b) f_o should be small and f_e should be large

(c) f_o and f_e both should be large

(d) f_o and f_e both should be small

55. Relative difference of focal lengths of objective and eye lens in the microscope and telescope is given as

[MH CET 2001]

(a) It is equal in both (b) It is more in telescope

(c) It is more in microscope (d) It may be more in any one

56. If the telescope is reversed *i.e.* seen from the objective side

(a) Object will appear very small

(b) Object will appear very large

(c) There will be no effect on the image formed by the telescope

(d) Image will be slightly greater than the earlier one

57. The focal length of the objective of a terrestrial telescope is 80 *cm* and it is adjusted for parallel rays, then its magnifying power is 20. If the focal length of erecting lens is 20 *cm*, then full length of telescope will be

(a) 84 *cm* (b) 100 *cm*

(c) 124 *cm* (d) 164 *cm*

58. An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and the eye piece is 36 *cm* and the final image is formed at infinity. The focal length f_o of the objective and the focal length f_e of the eye piece are

[IIT 1989; MP PET 1995; JIPMER 2000]

(a) f_o= 45 *cm* and f_e = – 9 *cm*

(b) f_o = 7.2 *cm* and f_e = 5 *cm*

(c) f_o = 50 *cm* and f_e = 10 *cm*

(d) f_o = 30 *cm* and f_e = 6 *cm*

59. In an astronomical telescope, the focal lengths of two lenses are 180 *cm* and 6 *cm* respectively. In normal adjustment, the magnifying power will be [MP PET 1990]

(a) 1080 (b) 200

(c) 30 (d) 186

60. The magnifying power of an astronomical telescope for relaxed vision is 16. On adjusting, the distance between the objective and eye lens is 34 *cm*. Then the focal length of objective and eye lens will be respectively [MP PMT 1989]

(a) 17 *cm*, 17 *cm* (b) 20 *cm*, 14 *cm*

(c) 32 *cm*, 2 *cm* (d) 30 *cm*, 4 *cm*

61. In Gallilean telescope, if the powers of an objective and eye lens are respectively +1.25 *D* and – 20 *D*, then for relaxed vision, the length and magnification will be

(a) 21.25 *cm* and 16 (b) 75 *cm* and 20

(c) 75 *cm* and 16 (d) 8.5 *cm* and 21.25

62. The aperture of a telescope is made large, because

[DPMT 1999]

(a) To increase the intensity of image

(b) To decrease the intensity of image

(c) To have greater magnification

(d) To have lesser resolution

63. In Gallilean telescope, the final image formed is

(a) Real, erect and enlarged

(b) Virtual, erect and enlarged

(c) Real, inverted and enlarged

(d) Virtual, inverted and enlarged

64. The magnifying power of a telescope is 9. When it is adjusted for parallel rays, the distance between the objective and the eye-piece is found to be 20 *cm*. The focal length of the two lenses are

(a) 18 *cm*, 2 *cm* (b) 11 *cm*, 9 *cm*

(c) 10 *cm*, 10 *cm* (d) 15 *cm*, 5 *cm*

65. The focal length of the objective and eye piece of a telescope are respectively 60 *cm* and 10 *cm*. The magnitude of the magnifying power when the image is formed at infinity is

(a) 50 (b) 6

(c) 70 (d) 5

66. The magnifying power of an astronomical telescope is 8 and the distance between the two lenses is 54 *cm*. The focal length of eye lens and objective lens will be respectively

[MP PMT 1991; CPMT 1991; Pb. PMT 2001]

(a) 6 *cm* and 48 *cm* (b) 48 *cm* and 6 *cm*

(c) 8 *cm* and 64 *cm* (d) 64 *cm* and 8 *cm*

67. An opera glass (Gallilean telescope) measures 9 *cm* from the objective to the eyepiece. The focal length of the objective is 15 *cm*. Its magnifying power is [DPMT 1988]

(a) 2.5 (b) 2/5

(c) 5/3 (d) 0.4

68. When a telescope is adjusted for parallel light, the distance of the objective from the eye piece is found to be 80 *cm*. The magnifying power of the telescope is 19. The focal lengths of the lenses are

[MP PMT 1992; Very similar to DPMT 2004]

(a) 61 *cm*, 19 *cm* (b) 40 *cm*, 40 *cm*

(c) 76 *cm*, 4 *cm* (d) 50 *cm*, 30 *cm*

69. A reflecting telescope utilizes [CPMT 1983]

(a) A concave mirror (b) A convex mirror

(c) A prism (d) A plano-convex lens

70. The aperture of the objective lens of a telescope is made large so as to [AIEEE 2003; KCET 2003]

(a) Increase the magnifying power of the telescope

(b) Increase the resolving power of the telescope

(c) Make image aberration less

(d) Focus on distant objects

71. On which of the following does the magnifying power of a telescope depends [MP PET 1992]

(a) The focal length of the objective only

(b) The diameter of aperture of the objective only

(c) The focal length of the objective and that of the eye piece

(d) The diameter of aperture of the objective and that of the eye piece

72. Large aperture of telescope are used for

[CPMT 1981; MP PMT 1995; AFMC 2000]

(a) Large image (b) Greater resolution

(c) Reducing lens aberration (d) Ease of manufacture

73. Two convex lenses of focal lengths 0.3 *m* and 0.05 *m* are used to make a telescope. The distance kept between the two is

(a) 0.35 *m* (b) 0.25 *m*

(c) 0.175 *m* (d) 0.15 *m*

74. The diameter of the objective lens of a telescope is 5.0 *m* and wavelength of light is 6000 Å. The limit of resolution of this telescope will be [MP PMT 1994]

(a) 0.03 *sec* (b) 3.03 *sec*

(c) 0.06 *sec* (d) 0.15 *sec*

75. All of the following statements are correct except

[Manipal MEE 1995]

(a) The total length of an astronomical telescope is the sum of the focal lengths of its two lenses

(b) The image formed by the astronomical telescope is always erect because the effect of the combination of the two lenses is divergent [MP PET 1991]

(c) The magnification of an astronomical telescope can be increased by decreasing the focal length of the eye-piece

(d) The magnifying power of the refracting type of astronomical telescope is the ratio of the focal length of the objective to that of the eye-piece

76. A terrestrial telescope is made by introducing an erecting lens of focal length f between the objective and eye piece lenses of an astronomical telescope. This causes the length of the telescope tube to increase by an amount equal to

[KCEE 1996]

(a) f (b) $2f$

(c) $3f$ (d) $4f$

77. The length of an astronomical telescope for normal vision (relaxed eye) (f_o = focal length of objective lens and f_e = focal length of eye lens) is

[EAMCET (Med.) 1995; CPMT 1999; BVP 2003]

(a) $f_o \times f_e$ (b) $\frac{f_o}{f_e}$

(c) $f_o + f_e$ (d) $f_o - f_e$

78. A Gallilean telescope has objective and eye-piece of focal lengths 200 *cm* and 2 *cm* respectively. The magnifying power of the telescope for normal vision is

[MP PMT 1996]

(a) 90 (b) 100

(c) 108 (d) 198

79. In an astronomical telescope, the focal length of the objective lens is 100 *cm* and of eye-piece is 2 *cm*. The magnifying power of the telescope for the normal eye is

[MP PET 1997]

(a) 50 (b) 10

(c) 100 (d) $\frac{1}{50}$

80. When diameter of the aperture of the objective of an astronomical telescope is increased, its [MP PMT 1997]

(a) Magnifying power is increased and resolving power is decreased

(b) Magnifying power and resolving power both are increased

(c) Magnifying power remains the same but resolving power is increased [MNR 1994]

(d) Magnifying power and resolving power both are decreased

81. The focal lengths of the objective and eye lenses of a telescope are respectively 200 *cm* and 5 *cm*. The maximum magnifying power of the telescope will be

[MP PMT/PET 1998; JIPMER 2001, 02]

(a) – 40 (b) – 48

(c) – 60 (d) – 100

82. The minimum magnifying power of a telescope is *M*, If the focal length of its eye lens is halved, the magnifying power will become

(a) $M/2$ (b) $2M$

(c) $3M$ (d) $4M$

83. The astronomical telescope consists of objective and eye-piece. The focal length of the objective is

[AIIMS 1998; BHU 2000]

(a) Equal to that of the eye-piece

(b) Greater than that of the eye-piece

(c) Shorter than that of the eye-piece

(d) Five times shorter than that of the eye-piece

84. Four convergent lenses have focal lengths 100 *cm*, 10 *cm*, 4 *cm* and 0.3 *cm*. For a telescope with maximum possible magnification, we choose the lenses of focal length

[KCET 1994]

(a) 100 *cm*, 0.3 *cm* (b) 10 *cm*, 0.3 *cm*

(c) 10 *cm*, 4 *cm* (d) 100 *cm*, 4 *cm*

85. The focal length of objective and eye-piece of a telescope are 100 *cm* and 5 *cm* respectively. Final image is formed at least distance of distinct vision. The magnification of telescope is

(a) 20 (b) 24

(c) 30 (d) 36

86. A planet is observed by an astronomical refracting telescope having an objective of focal length 16 *m* and an eye-piece of focal length 2 *cm* [IIT-JEE 1992; Roorkee 2000]

(a) The distance between the objective and the eye-piece is 16.02 *m*

(b) The angular magnification of the planet is 800

(c) The image of the planet is inverted

(d) The objective is larger than the eye-piece

87. If tube length of astronomical telescope is 105 *cm* and magnifying power is 20 for normal setting, calculate the focal length of objective

(a) 100 *cm* (b) 10 *cm*

(c) 20 *cm* (d) 25 *cm*

88. The length of a telescope is 36 *cm*. The focal lengths of its lenses can be [Bihar MEE 1995]

(a) 30 *cm*, 6 *cm* (b) – 30 *cm*, – 6 *cm*

(c) 30 *cm*, – 6 *cm* (d) – 30 *cm*, 6 *cm*

89. An astronomical telescope of ten-fold angular magnification has a length of 44 *cm*. The focal length of the objective is

[CBSE PMT 1997]

(a) 4 *cm* (b) 40 *cm*

(c) 44 *cm* (d) 440 *cm*

90. If both the object and image are at infinite distances form a refracting telescope its magnifying power will be equal to

[AMU (Engg.) 1999]

(a) The sum of the focal lengths of the objective and the eyepiece

(b) The difference of the focal lengths of the two lenses

(c) The ratio of the focal length of the objective and eyepiece

(d) The ratio of the focal length of the eyepiece and objective

91. The number of lenses in a terrestrial telescope is

[KCET 1999; MH CET 2003]

(a) Two (b) Three

(c) Four (d) Six

92. The focal lengths of the lenses of an astronomical telescope are 50 *cm* and 5 *cm*. The length of the telescope when the image is formed at the least distance of distinct vision is

[EAMCET (Engg.) 2000]

(a) $45\,cm$ (b) $55\,cm$

(c) $\frac{275}{6}cm$ (d) $\frac{325}{6}cm$

93. The focal lengths of the objective and eye-piece of a telescope are respectively 100 *cm* and 2 *cm*. The moon subtends an angle of 0.5^o at the eye. If it is looked through the telescope, the angle subtended by the moon's image will be

(a) 100^o (b) 50^o

(c) 25^o (d) 10^o

94. The diameter of the objective of a telescope is a, its magnifying power is m and wavelength of light is λ. The resolving power of the telescope is [MP PMT 2000]

(a) $(1.22\lambda)/a$ (b) $(1.22a)/\lambda$

(c) $\lambda m/(1.22a)$ [RPET 1997] (d) $a/(1.22\lambda)$

95. The sun's diameter is $1.4\times10^9\,m$ and its distance from the earth is $10^{11}\,m$. The diameter of its image, formed by a convex lens of focal length $2\,m$ will be [MP PET 2000]

(a) 0.7 *cm* (b) 1.4 *cm*

(c) 2.8 *cm* (d) Zero (*i.e.* point image)

96. In a terrestrial telescope, the focal length of objective is 90 *cm*, of inverting lens is $5\,cm$ and of eye lens is $6\,cm$. If the final image is at 30 *cm*, then the magnification will be

[DPMT 2001]

(a) 21 (b) 12

(c) 18 [AFMC 1994] (d) 15

97. The resolving power of a telescope depends on

[MP PET 2000, 01; DCE 2003]

(a) Focal length of eye lens

(b) Focal length of objective lens

(c) Length of the telescope

(d) Diameter of the objective lens

98. Four lenses of focal length + 15 *cm*, + 20*cm*, + 150*cm* and + 250 *cm* are available for making an astronomical telescope. To produce the largest magnification, the focal length of the eye-piece should be

[CPMT 2001; AIIMS 2001]

(a) + 15 *cm* (b) + 20 *cm*

(c) +150 *cm* (d) + 250 *cm*

99. In an astronomical telescope, the focal length of objective lens and eye-piece are 150 *cm* and 6 *cm* respectively. In case when final image is formed at least distance of distinct vision. the magnifying power is [KCET 2001]

(a) 20 (b) 30

(c) 60 (d) 15

100. In a laboratory four convex lenses L_1, L_2, L_3 and L_4 of focal lengths 2, 4, 6 and 8 *cm* respectively are available. Two of these lenses form a telescope of length 10 *cm* and magnifying power 4. The objective and eye lenses are

[MP PMT 2001]

(a) L_2, L_3 (b) L_1, L_4

(c) L_3, L_2 (d) L_4, L_1

101. A telescope has an objective of focal length 50 *cm* and an eye piece of focal length 5 *cm*. The least distance of distinct vision is 25 *cm*.

The telescope is focussed for distinct vision on a scale 200 *cm* away. The separation between the objective and the eye-piece is

(a) 75 *cm* (b) 60 *cm*

(c) 71 *cm* (d) 74 *cm*

102. The resolving power of a telescope whose lens has a diameter of 1.22 *m* for a wavelength of 5000 *Å* is [Kerala PMT 2002]

(a) 2×10^5 (b) 2×10^6

(c) 2×10^2 (d) 2×10^4

103. To increase both the resolving power and magnifying power of a telescope [Kerala PET 2002; KCET 2002]

(a) Both the focal length and aperture of the objective has to be increased

(b) The focal length of the objective has to be increased

(c) The aperture of the objective has to be increased

(d) The wavelength of light has to be decreased

104. A Galileo telescope has an objective of focal length $100 cm$ and magnifying power 50. The distance between the two lenses in normal adjustment will be [BHU 2002; Pb. PET 2002]

(a) 96 *cm* (b) 98 *cm*

(c) 102 *cm* (d) 104 *cm*

105. An astronomical telescope has a magnifying power 10. The focal length of eyepiece is 20 *cm*. The focal length of objective is [MP PMT 2002, 03; Pb. PET 2004]

(a) 2 *cm* (b) 200 *cm*

(c) $\frac{1}{2} cm$ (d) $\frac{1}{200} cm$

106. A telescope of diameter 2*m* uses light of wavelength 5000 Å for viewing stars. The minimum angular separation between two stars whose image is just resolved by this telescope is [MP PET 2003]

(a) 4×10^{-4} *rad* (b) 0.25×10^{-6} *rad*

(c) 0.31×10^{-6} *rad* (d) 5.0×10^{-3} *rad*

107. A simple magnifying lens is used in such a way that an image is formed at 25 *cm* away from the eye. In order to have 10 *times* magnification, the focal length of the lens should be

(a) 5 *cm* (b) 2 *cm*

(c) 25 *mm* (d) 0.1 *mm*

108. In a simple microscope, if the final image is located at infinity then its magnifying power is [MP PMT 2004]

(a) $\frac{25}{f}$ (b) $\frac{D}{26}$

(c) $\frac{f}{25}$ (d) $\frac{f}{D+1}$

109. In a compound microscope the objective of f_o and eyepiece of f_e are placed at distance L such that L equals [Kerala PMT 2004]

(a) $f_o + f_e$

(b) $f_o - f_e$

(c) Much greater than f_o or f_e [Kerala PET 2002]

(d) Much less than f_o or f_e

(e) Need not depend either value of focal lengths

110. For a compound microscope, the focal lengths of object lens and eye lens are f_o and f_e respectively, then magnification will be done by microscope when [RPMT 2001]

(a) $f_o = f_e$ (b) $f_o > f_e$

(c) $f_o < f_e$ (d) None of these

111. The angular resolution of a 10 *cm* diameter telescope at a wavelength of 5000 Å is of the order [CBSE PMT 2005]

(a) $10^6 rad$ (b) $10^{-2} rad$

(c) $10^{-4} rad$ (d) $10^{-6} rad$

112. The resolving power of an astronomical telescope is 0.2 seconds. If the central half portion of the objective lens is covered, the resolving power will be [MP PMT 2004]

(a) 0.1 *sec* (b) 0.2 *sec*

(c) 1.0 *sec* (d) 0.6 *sec*

113. An astronomical telescope has objective and eye-piece lens of powers 0.5 *D* and 20 *D* respectively, its magnifying power will be

(a) 8 (b) 20

(c) 30 (d) 40

114. Which of the following is not correct regarding the radio telescope

(a) It can not work at night

(b) It can detect a very faint radio signal

(c) It can be operated even in cloudy weather

(d) It is much cheaper than optical telescope

115. The diameter of objective of a telescope is 1*m*. Its resolving limit for the light of wave length 4538 Å, will be [Pb. PET 2003]

(a) $5.54\times10^{-7} rad$ [MP PET 1990] (b) $2.54\times10^{-4} rad$

(c) $6.54\times10^{-7} rad$ (d) None of these

116. A telescope has an objective lens of focal length 200 *cm* and an eye piece with focal length 2 *cm*. If this telescope is used to see a 50 *meter* tall building at a distance of 2 *km*, what is the height of the image of the building formed by the objective lens

(a) 5 *cm* (b) 10 *cm*

(c) 1 *cm* (d) 2 *cm*

117. Magnification of a compound microscope is 30. Focal length of eye-piece is 5 *cm* and the image is formed at a distance of distinct vision of 25 *cm*. The magnification of the objective lens is

(a) 6 (b) 5

(c) 7.5 (d) 10

118. At Kavalur in India, the astronomers using a telescope whose objective had a diameter of one meter started using a telescope of diameter 2.54 *m*. This resulted in [KCET 2005]

(a) The increase in the resolving power by 2.54 times for the same λ

(b) The increase in the limiting angle by 2.54 times for the same λ

(c) Decrease in resolving power

(d) No effect on the limiting angle

119. A Galileo telescope has an objective of focal length 100 *cm* and magnifying power 50. The distance between the two lenses in normal adjustment will be [BCECE 2005]

(a) 98 *cm* (b) 100 *cm*

(c) 150 *cm* (d) 200 *cm*

120. A compound microscope has an eye piece of focal length 10 *cm* and an objective of focal length 4 *cm*. Calculate the magnification, if an object is kept at a distance of 5 *cm* from the objective so that final image is formed at the least distance vision (20 *cm*)

(a) 12 (b) 11

(c) 10 (d) 13

Photometry

1. If luminous efficiency of a lamp is 2 lumen/watt and its luminous intensity is 42 candela, then power of the lamp is

[AFMC 1998]

(a) 62 *W* (b) 76 *W*

(c) 138 *W* (d) 264 *W*

2. An electric bulb illuminates a plane surface. The intensity of illumination on the surface at a point 2*m* away from the bulb is 5×10^{-4} phot (*lumen/cm*). The line joining the bulb to the point makes an angle of 60° with the normal to the surface. The intensity of the bulb in *candela* is

[IIT-JEE 1980; CPMT 1991]

(a) $40\sqrt{3}$ (b) 40

(c) 20 (d) 40×10^{-4}

3. In a movie hall, the distance between the projector and the screen is increased by 1% illumination on the screen is

[CPMT 1990]

(a) Increased by 1% (b) Decreased by 1%

(c) Increased by 2% (d) Decreased by 2%

4. Correct exposure for a photographic print is 10 *seconds* at a distance of one metre from a point source of 20 *candela*. For an equal fogging of the print placed at a distance of 2 *m* from a 16 *candela* source, the necessary time for exposure is

(a) 100 *sec* (b) 25 *sec*

(c) 50 *sec* (d) 75 *sec*

5. A bulb of 100 *watt* is hanging at a height of one meter above the centre of a circular table of diameter 4 *m*. If the intensity at a point on its rim is I_0, then the intensity at the centre of the table will be

[CPMT 1996]

(a) I_0 (b) $2\sqrt{5}I_0$

(c) $2I_0$ (d) $5\sqrt{5}I_0$

6. A movie projector forms an image 3.5*m* long of an object 35 *mm*. Supposing there is negligible absorption of light by aperture then illuminance on slide and screen will be in the ratio of

(a) 100 : 1 (b) 10^4 : 1

(c) 1 : 100 (d) 1 : 10^4

7. A 60 *watt* bulb is hung over the center of a table 4 *m* × 4 *m* at a height of 3 *m*. The ratio of the intensities of illumination at a point on the centre of the edge and on the corner of the table is

(a) $(17/13)^{3/2}$ (b) 2 / 1

(c) 17 / 13 (d) 5 / 4

8. "Lux" is a unit of [Kerala PMT 2001]

(a) Luminous intensity of a source

(b) Illuminance on a surface

(c) Transmission coefficient of a surface

(d) Luminous efficiency of source of light

9. Total flux produced by a source of 1 *cd* is [UP SEAT 2005] [CPMT 2001]

(a) $\frac{1}{4\pi}$ (b) 8π

(c) 4π (d) $\frac{1}{8\pi}$

10. If the luminous intensity of a 100 *W* unidirectional bulb is 100 candela, then total luminous flux emitted from the bulb is

(a) 861 *lumen* (b) 986 *lumen*

(c) 1256 *lumen* (d) 1561 *lumen*

11. The maximum illumination on a screen at a distance of 2 *m* from a lamp is 25 *lux*. The value of total luminous flux emitted by the lamp is [JIMPER 1997]

(a) 1256 *lumen* (b) 1600 *lumen*

(c) 100 *candela* (d) 400 *lumen*

12. A small lamp is hung at a height of 8 feet above the centre of a round table of diameter 16 *feet*. The ratio of intensities of illumination at the centre and at points on the circumference of the table will be [CPMT 1984, 1996]

(a) 1 : 1 (b) 2 : 1

(c) $2\sqrt{2} : 1$ (d) 3 : 2

13. Lux is equal to [CPMT 1993]

(a) 1 *lumen/m* (b) 1 *lumen/cm*

(c) 1 *candela/m* (d) 1 *candela/cm*

14. Five *lumen/watt* is the luminous efficiency of a lamp and its luminous intensity is 35 *candela*. The power of the lamp is

[CPMT 1992]

(a) 80 *W* (b) 176 *W*

(c) 88 *W* (d) 36 *W*

15. A lamp rated at 100 *cd* hangs over the middle of a round table with diameter 3 *m* at a height of 2 *m*. It is replaced by a lamp of 25 *cd* and the distance to the table is changed so that the illumination at the centre of the table remains as before. The illumination at edge of the table becomes *X* times the original. Then *X* is

(a) $\frac{1}{3}$ (b) $\frac{16}{27}$

(c) $\frac{1}{4}$ (d) $\frac{1}{9}$

16. The distance between a point source of light and a screen which is 60 *cm* is increased to 180 *cm*. The intensity on the screen as compared with the original intensity will be

[CPMT 1982] [CPMT 1888]

(a) (1 / 9) times (b) (1 / 3) times

(c) 3 times (d) 9 times

17. A source of light emits a continuous stream of light energy which falls on a given area. Luminous intensity is defined as [CPMT 1986]

(a) Luminous energy emitted by the source per second

(b) Luminous flux emitted by source per unit solid angle

(c) Luminous flux falling per unit area of a given surface

(d) Luminous flux coming per unit area of an illuminated surface

18. Venus looks brighter than other stars because [MNR 1985]

(a) It has higher density than other stars

(b) It is closer to the earth than other stars

(c) It has no atmosphere

(d) Atomic fission takes place on its surface

19. To prepare a print the time taken is 5 sec due to lamp of 60 watt at 0.25 *m* distance. If the distance is increased to 40 *cm* then what is the time taken to prepare the similar print [CPMT 1982]

(a) 3.1 *sec* (b) 1 *sec*

(c) 12.8 *sec* (d) 16 *sec*

20. A lamp is hanging 1 *m* above the centre of a circular table of diameter 1*m*. The ratio of illuminaces at the centre and the edge is

(a) $\frac{1}{2}$ (b) $\left(\frac{5}{4}\right)^{\frac{3}{2}}$

(c) $\frac{4}{3}$ (d) $\frac{4}{5}$

21. Two stars situated at distances of 1 and 10 light years respectively from the earth appear to possess the same brightness. The ratio of their real brightness is [NCERT 1981]

(a) 1 : 10 (b) 10 : 1

(c) 1 : 100 (d) 100 : 1

22. The intensity of direct sunlight on a surface normal to the rays is I_0. What is the intensity of direct sunlight on a surface, whose normal makes an angle of 60° with the rays of the sun

(a) I_0 (b) $I_0\left(\frac{\sqrt{3}}{2}\right)$

(c) $\frac{I_0}{2}$ (d) $2I_0$

23. Inverse square law for illuminance is valid for [CPMT 1978]

(a) Isotropic point source (b) Cylindrical source

(c) Search light (d) All types of sources

24. 1% of light of a source with luminous intensity 50 *candela* is incident on a circular surface of radius 10 *cm*. The average illuminance of surface is

(a) 100 *lux* (b) 200 *lux*

(c) 300 *lux* (d) 400 *lux*

25. Two light sources with equal luminous intensity are lying at a distance of 1.2 *m* from each other. Where should a screen be placed between them such that illuminance on one of its faces is four times that on another face

(a) 0.2 *m* (b) 0.4 *m*

(c) 0.8 *m* (d) 1.6 *m*

26. Two lamps of luminous intensity of 8 *Cd* and 32 *Cd* respectively are lying at a distance of 1.2 *m* from each other. Where should a screen be placed between two lamps such that its two faces are equally illuminated due to two sources

(a) 10 *cm* from 8 *Cd* lamp (b) 10 *cm* from 32*Cd* lamp

(c) 40 *cm* from 8 *Cd* lamp (d) 40 *cm* from 32 *Cd* lamp

27. A lamp is hanging along the axis of a circular table of radius *r*. At what height should the lamp be placed above the table, so that the illuminance at the edge of the table is $\frac{1}{8}$ of that at its center

(a) $\frac{r}{2}$ (b) $\frac{r}{\sqrt{2}}$

(c) $\frac{r}{3}$ (d) $\frac{r}{\sqrt{3}}$

[NCERT 1982]

28. A point source of 100 *candela* is held 5*m* above a sheet of blotting paper which reflects 75% of light incident upon it. The illuminance of blotting paper is

(a) 4 *phot* (b) 4 *lux*

(c) 3 *phot* (d) 3 *lux*

29. A lamp is hanging at a height 40 *cm* from the centre of a table. If its height is increased by 10 *cm* the illuminance on the table will decrease by

(a) 10 % (b) 20%

(c) 27% (d) 36%

30. Which has more luminous efficiency

(a) A 40 *W* bulb (b) A 40 *W* fluorescent tube

(c) Both have same (d) Cannot say

31. An electric lamp is fixed at the ceiling of a circular tunnel as shown is figure. What is the ratio the intensities of light at base *A* and a point *B* on the wall

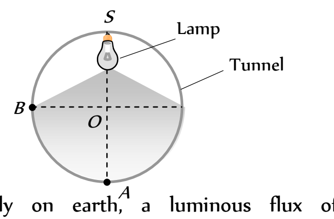
[CPMT 1981]

(a) $1:2$

(b) $2:\sqrt{3}$

(c) $\sqrt{3}:1$

(d) $1:\sqrt{2}$



32. When sunlight falls normally on earth, a luminous flux of $1.57\times10^5\ lumen/m^2$ is produced on earth. The distance of earth from sun is $1.5\times10^8\ Km$. The luminous intensity of sun in candela will be

(a) 3.53×10^{27} (b) 3.53×10^{25}

(c) 3.53×10^{29} (d) 3.53×10^{21}

33. In the above problem, the luminous flux emitted by sun will be

(a) $4.43\times10^{25}\ lm$ (b) $4.43\times10^{26}\ lm$

(c) $4.43\times10^{27}\ lm$ (d) $4.43\times10^{28}\ lm$

34. A screen receives 3 *watt* of radiant flux of wavelength 6000 Å. One lumen is equivalent to $1.5\times10^{-3} watt$ of monochromatic light of wavelength 5550 Å. If relative luminosity for 6000 Å is 0.685 while that for 5550 Å is 1.00, then the luminous flux of the source is

(a) $4\times10^{3} lm$ (b) $3\times10^{3} lm$

(c) $2\times10^{3} lm$ (d) $1.37\times10^{3} lm$

35. A point source of 3000 *lumen* is located at the centre of a cube of side length $2m$. The flux through one side is

(a) 500 *lumen* (b) 600 *lumen*

(c) 750 *lumen* (d) 1500 *lumen*

36. Light from a point source falls on a small area placed perpendicular to the incident light. If the area is rotated about the incident light by an angle of 60°, by what fraction will the illuminance change

(a) It will be doubled (b) It will be halved

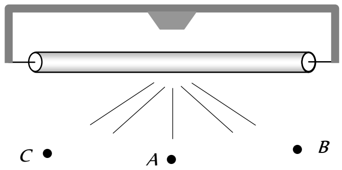
(c) It will not change (d) It will become one-fourth

37. A point source of light moves in a straight line parallel to a plane table. Consider a small portion of the table directly below the line of movement of the source. The illuminance at this portion varies with its distance r from the source as

(a) $E \propto \frac{1}{r}$ (b) $E \propto \frac{1}{r^2}$

(c) $E \propto \frac{1}{r^3}$ (d) $E \propto \frac{1}{r^4}$

38. Figure shows a glowing mercury tube. The illuminances at point A, B and C are related as



(a) $B > C > A$

(b) $A > C > B$

(c) $B = C > A$

(d) $B = C < A$

39. The relative luminosity of wavelength 600 *nm* is 0.6. Find the radiant flux of 600 *nm* needed to produce the same brightness sensation as produced by 120 *W* of radiant flux at 555 *nm*

(a) 50 *W* (b) 72 *W*

(c) $120 \times (0.6)^2 W$ (d) 200 *W*

40. Find the luminous intensity of the sun if it produces the same illuminance on the earth as produced by a bulb of 10000 *candela* at a distance 0.3 *m*. The distance between the sun and the earth is $1.5 \times 10^{11} m$

(a) $25 \times 10^{22} cd$ (b) $25 \times 10^{18} cd$

(c) $25 \times 10^{26} cd$ (d) $25 \times 10^{36} cd$

41. A lamp is hanging at a height of $4m$ above a table. The lamp is lowered by $1m$. The percentage increase in illuminace will be

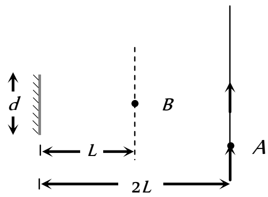
(a) 40 % (b) 64%

(c) 78% (d) 92%

Critical Thinking

Objective Questions

1. A point source of light B is placed at a distance L in front of the centre of a mirror of width d hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance $2L$ from it as shown. The greatest distance over which he can see the image of the light source in the mirror is



(a) $d/2$

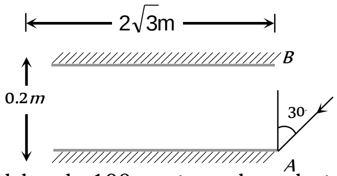
(b) d

(c) $2d$

(d) $3d$

2. Two plane mirrors. A and B are aligned parallel to each other, as shown in the figure. A light ray is incident at an angle of $30°$ at a point just inside one end of A. The plane of incidence coincides with the plane of the figure. The maximum number of times the ray undergoes reflections (including the first one) before it emerges out is

[IIT-JEE (Screening) 2002]



(a) 28

(b) 30

(c) 32

(d) 34

3. A concave mirror of focal length $100 cm$ is used to obtain the image of the sun which subtends an angle of $30°$. The diameter of the image of the sun will be

(a) $1.74 cm$ (b) $0.87 cm$

(c) $0.435 cm$ (d) $100 cm$

4. A square of side $3cm$ is placed at a distance of $25cm$ from a concave mirror of focal length $10cm$. The centre of the square is at the axis of the mirror and the plane is normal to the axis. The area enclosed by the image of the square is

(a) $4cm^2$ (b) $6cm^2$

(c) $16cm^2$ (d) $36cm^2$

5. A short linear object of length l lies along the axis of a concave mirror of focal length f at a distance u from the pole of the mirror. The size of the image is approximately equal to [IIT-JEE 1988; BHU 2003; CPMT 2

(a) $l\left(\frac{u-f}{f}\right)^{1/2}$ (b) $l\left(\frac{u-f}{f}\right)^{2}$

(c) $l\left(\frac{f}{u-f}\right)^{1/2}$ (d) $l\left(\frac{f}{u-f}\right)^{2}$

6. A thin rod of length $f/3$ lies along the axis of a concave mirror of focal length f. One end of its magnified image touches an end of the rod. The length of the image is

[MP PET 1995]

(a) f (b) $\frac{1}{2} f$

(c) $2f$ (d) $\frac{1}{4} f$

7. A ray of light falls on the surface of a spherical glass paper weight making an angle α with the normal and is refracted in the medium at an angle β. The angle of deviation of the emergent ray from the direction of the incident ray

[IIT-JEE (Screening) 2000] [NCERT 1982]

(a) $(\alpha - \beta)$ (b) $2(\alpha - \beta)$

(c) $(\alpha - \beta)/2$ (d) $(\beta - \alpha)$

8. Light enters at an angle of incidence in a transparent rod of refractive index n. For what value of the refractive index of the material of the rod the light once entered into it will not leave it through its lateral face whatsoever be the value of angle of incidence

[CBSE PMT 1998]

(a) $n > \sqrt{2}$ (b) $n = 1$

(c) $n = 1.1$ (d) $n = 1.3$

9. A glass hemisphere of radius 0.04 m and *R.I.* of the material 1.6 is placed centrally over a cross mark on a paper (i) with the flat face; (ii) with the curved face in contact with the paper. In each case the cross mark is viewed directly from above. The position of the images will be

[ISM Dhanbad 1994]

(a) (i) 0.04 *m* from the flat face; (ii) 0.025 *m* from the flat face

(b) (i) At the same position of the cross mark; (ii) 0.025 *m* below the flat face

(c) (i) 0.025 *m* from the flat face; (ii) 0.04 *m* from the flat face

(d) For both (i) and (ii) 0.025 *m* from the highest point of the hemisphere

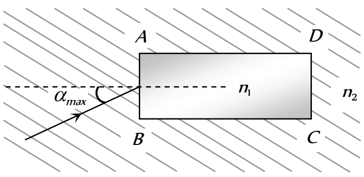
10. One face of a rectangular glass plate 6 *cm* thick is silvered. An object held 8 *cm* in front of the first face, forms an image 12 *cm* behind the silvered face. The refractive index of the glass is

(a) 0.4 (b) 0.8

(c) 1.2 (d) 1.6

11. A rectangular glass slab *ABCD*, of refractive index n_1 is immersed in water of refractive index n_2 ($n_1>n_2$). A ray of light in incident at the surface *AB* of the slab as shown. The maximum value of the angle of incidence α_{max}, such that the ray comes out only from the other surface *CD* is given by

[IIT-JEE (Screening) 2000]



(a) $\sin^{-1}\left[\frac{n_1}{n_2}\cos\left(\sin^{-1}\frac{n_2}{n_1}\right)\right]$ (b) $\sin^{-1}\left[n_1\cos\left(\sin^{-1}\frac{1}{n_2}\right)\right]$

(c) $\sin^{-1}\left(\frac{n_1}{n_2}\right)$ (d) $\sin^{-1}\left(\frac{n_2}{n_1}\right)$

12. A diverging beam of light from a point source *S* having divergence angle α, falls symmetrically on a glass slab as shown. The angles of incidence of the two extreme rays are equal. If the thickness of the glass slab is *t* and the refractive index *n*, then the divergence angle of the emergent beam is

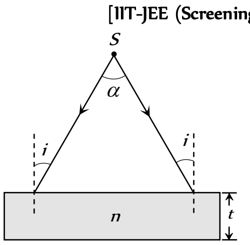
[IIT-JEE (Screening) 2000]

(a) Zero

(b) α

(c) $\sin^{-1}(1/n)$

(d) $2\sin^{-1}(1/n)$



13. A concave mirror is placed at the bottom of an empty tank with face upwards and axis vertical. When sunlight falls normally on the mirror, it is focussed at distance of 32 *cm* from the mirror. If the tank filled with water $\left(\mu=\frac{4}{3}\right)$ upto a height of 20 *cm*, then the sunlight will now get focussed at

[UPSEAT 2002]

(a) 16 *cm* above water level

(b) 9 *cm* above water level

(c) 24 *cm* below water level

(d) 9 *cm* below water level

14. An air bubble in sphere having 4 *cm* diameter appears 1 *cm* from surface nearest to eye when looked along diameter. If μ_a = 1.5, the distance of bubble from refracting surface is

[CPMT 2002]

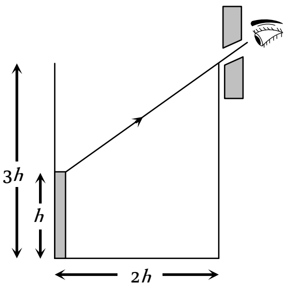
(a) 1.2 *cm* (b) 3.2 *cm*

(c) 2.8 *cm* (d) 1.6 *cm*

[CPMT 1999]

15. An observer can see through a pin–hole the top end of a thin rod of height *h*, placed as shown in the figure. The beaker height is 3*h* and its radius *h*. When the beaker is filled with a liquid up to a height 2*h*, he can see the lower end of the rod. Then the refractive index of the liquid is

[IIT-JEE (Screening) 2002]



(a) 5/2 (b) $\sqrt{(5/2)}$

(c) $\sqrt{(3/2)}$ (d) 3/2

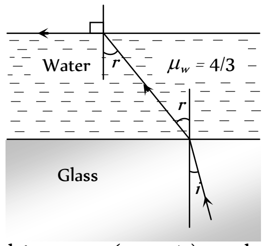
16. A ray of light is incident at the glass–water interface at an angle *i*, it emerges finally parallel to the surface of water, then the value of μ_g would be [IIT-JEE (Screening) 2003]

(a) (4/3) sin *i*

(b) 1/sin *i*

(c) 4/3

(d) 1



17. A glass prism (μ = 1.5) is dipped in water (μ = 4/3) as shown in figure. A light ray is incident normally on the surface *AB*. It reaches the surface BC after totally reflected, if

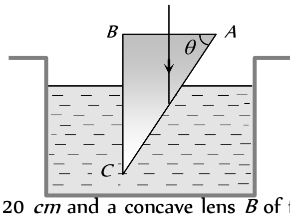
[IIT JEE 1981; MP PMT 1997]

(a) sin θ ≥ 8/9

(b) 2/3 < sin θ < 8/9

(c) sin θ ≤ 2/3

(d) It is not possible



18. A convex lens *A* of focal length 20 *cm* and a concave lens *B* of focal length 5 *cm* are kept along the same axis with the distance *d* between them. If a parallel beam of light falling on *A* leaves *B* as a parallel beam, then distance *d* in *cm* will be

(a) 25 (b) 15

(c) 30 (d) 50

19. Diameter of a plano–convex lens is 6 *cm* and thickness at the centre is 3 *mm*. If the speed of light in the material of the lens is 2×10^8 *m/sec*, the focal length of the lens is

[CPMT 1989]

(a) 15 *cm* (b) 20 *cm*

(c) 30 *cm* (d) 10 *cm*

20. A point object *O* is placed on the principal axis of a convex lens of focal length 20 *cm* at a distance of 40 cm to the left of it. The diameter of the lens is 10 *cm*. If the eye is placed 60 *cm* to the right of the lens at a distance *h* below the principal axis, then the maximum value of *h* to see the image will be

(a) 0 (b) 5 *cm*

(c) 2.5 *cm* (d) 10 *cm*

21. A luminous object is placed at a distance of 30 *cm* from the convex lens of focal length 20 *cm*. On the other side of the lens, at what distance from the lens a convex mirror of radius of curvature 10 *cm* be placed in order to have an upright image of the object coincident with it

[CBSE PMT 1998; JIPMER 2001, 02]

(a) 12 *cm* (b) 30 *cm*

(c) 50 *cm* (d) 60 *cm*

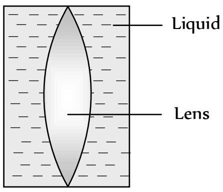
22. Shown in the figure here is a convergent lens placed inside a cell filled with a liquid. The lens has focal length + 20 *cm* when in air and its material has refractive index 1.50. If the liquid has refractive index 1.60, the focal length of the system is [NSEP 1994; DPMT 2000]

(a) + 80 *cm*

(b) – 80 *cm*

(c) – 24 *cm*

(d) –100 *cm*



23. A hollow double concave lens is made of very thin transparent material. It can be filled with air or either of two liquids L_1 and L_2 having refractive indices n_1 and n_2 respectively ($n_1 > n_2 > 1$). The lens will diverge a parallel beam of light if it is filled with

(a) Air and placed in air

(b) Air and immersed in L_1

(c) L_1 and immersed in L_2

(d) L_2 and immersed in L_1

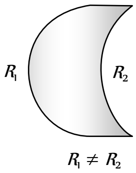
24. The object distance *u*, the image distance *v* and the magnification *m* in a lens follow certain linear relations. These are

(a) $\frac{1}{u}$ versus $\frac{1}{v}$ (b) *m* versus *u*

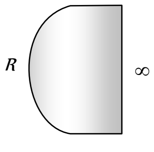
(c) *u* versus *v* (d) *m* versus *v*

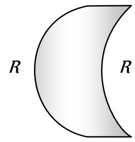
25. Which one of the following spherical lenses does not exhibit dispersion? The radii of curvature of the surfaces of the lenses are as given in the diagrams

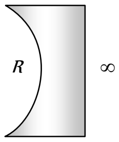
[IIT-JEE (Screening) 2002]

(a) 

$R_1 \neq R_2$

(b) 

(c) 

(d) 

26. The size of the image of an object, which is at infinity, as formed by a convex lens of focal length 30*cm* is 2 *cm*. If a concave lens of focal length 20 *cm* is placed between the convex lens and the image at a distance of 26 *cm* from the convex lens, calculate the new size of the image

[MP PMT 1999] [IIT-JEE (Screening) 2003]

(a) 1.25 *cm* (b) 2.5 *cm*

(c) 1.05 *cm* (d) 2 *cm*

27. An achromatic prism is made by crown glass prism $(A_c = 19^o)$ and flint glass prism $(A_F = 6^o)$. If ${}^C\mu_v = 1.5$ and ${}^F\mu_v = 1.66$, then resultant deviation for red coloured ray will be

(a) 1.04° (b) 5°

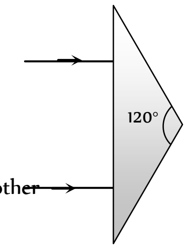
(c) 0.96° (d) 13.5°

28. The refracting angle of prism is *A* and refractive index of material of prism is $\cot\frac{A}{2}$. The angle of minimum deviation is

(a) 180°– 3A (b) 180° + 2A

(c) 90° –A (d) 180° – 2A

29. An isosceles prism of angle 120° has a refractive index of 1.44. Two parallel monochromatic rays enter the prism parallel to each other in air as shown. The rays emerging from the opposite faces



[IIT-JEE (Screening) 2000]

(a) Are parallel to each other

(b) Are diverging

(c) Make an angle $2\sin^{-1}(0.72)$ with each other

(d) Make an angle $2\{\sin^{-1}(0.72) - 30^o\}$ with each other

30. A ray of light is incident on the hypotenuse of a right-angled prism after travelling parallel to the base inside the prism. If μ is the refractive index of the material of the prism, the maximum value of the base angle for which light is totally reflected from the hypotenuse is [Roorkee 2000] [EAMCET 2003]

(a) $\sin^{-1}\left(\frac{1}{\mu}\right)$ (b) $\tan^{-1}\left(\frac{1}{\mu}\right)$

(c) $\sin^{-1}\left(\frac{\mu - 1}{\mu}\right)$ (d) $\cos^{-1}\left(\frac{1}{\mu}\right)$

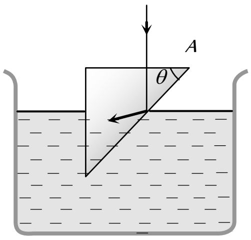
31. The refractive index of the material of the prism and liquid are 1.56 and 1.32 respectively. What will be the value of θ for the following refraction [BHU 2003; CPMT 2004]

(a) $\sin\theta \geq \frac{13}{11}$

(b) $\sin\theta \geq \frac{11}{13}$

(c) $\sin\theta \geq \frac{\sqrt{3}}{2}$

(d) $\sin\theta \geq \frac{1}{\sqrt{2}}$



32. A spherical surface of radius of curvature R separates air (refractive index 1.0) from glass (refractive index 1.5). The centre of curvature is in the glass. A point object P placed in air is found to have a real image Q in the glass. The line PQ cuts the surface at a point O, and PO = OQ. The distance PO is equal to

(a) 5 R (b) 3 R

(c) 2 R (d) 1.5 R

33. A plano-convex lens when silvered in the plane side behaves like a concave mirror of focal length 30cm. However, when silvered on the convex side it behaves like a concave mirror of focal length 10 cm. Then the refractive index of its material will be

(a) 3.0 (b) 2.0

(c) 2.5 (d) 1.5

34. A ray of light travels from an optically denser to rarer medium. The critical angle for the two media is C. The maximum possible deviation of the ray will be

[KCET (Engg./Med.) 2002]

(a) $\left(\frac{\pi}{2} - C\right)$ (b) $2C$

(c) $\pi - 2C$ (d) $\pi - C$

35. An astronaut is looking down on earth's surface from a space shuttle at an altitude of $400\,km$. Assuming that the astronaut's pupil diameter is $5\,mm$ and the wavelength of visible light is $500\,nm$. The astronaut will be able to resolve linear object of the size of about [AIIMS 2003]

(a) 0.5 m (b) $5\,m$

(c) 50 m (d) 500 m

36. The average distance between the earth and moon is 38.6×10^4 km. The minimum separation between the two points on the surface of the moon that can be resolved by a telescope whose objective lens has a diameter of 5 m with $\lambda = 6000\ \mathring{A}$ is

(a) 5.65 m (b) 28.25 m

(c) 11.30 m (d) 56.51 m

37. The distance of the moon from earth is $3.8 \times 10^5\,km$. The eye is most sensitive to light of wavelength 5500 Å. The separation of two points on the moon that can be resolved by a 500 cm telescope will be [AMU (Med.) 2002]

(a) 51 m (b) 60 m

(c) 70 m (d) All the above

38. A small source of light is to be suspended directly above the centre of a circular table of radius R. What should be the height of the light source above the table so that the intensity of light is maximum at the edges of the table compared to any other height of the source

(a) $\frac{R}{2}$ (b) $\frac{R}{\sqrt{2}}$

(c) R (d) $\sqrt{2}R$

39. A light source is located at P_1 as shown in the figure. All sides of the polygon are equal. The intensity of illumination at P_2 is I_0. What will be the intensity of illumination at P_3

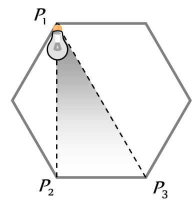
[IIT JEE 1998; DPMT 2000]

(a) $\frac{3\sqrt{3}}{8}I_0$

(b) $\frac{I_0}{8}$

(c) $\frac{3}{8}I_0$

(d) $\frac{\sqrt{3}}{8}I_0$



[BHU 1997; UPSEAT 1995]

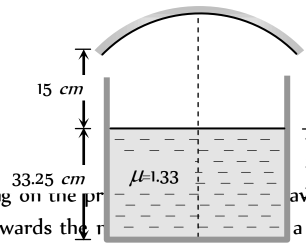
40. A container is filled with water (μ = 1.33) upto a height of 33.25 cm. A concave mirror is placed 15 cm above the water level and the image of an object placed at the bottom is formed 25 cm below the water level. The focal length of the mirror is

(a) 10

(b) 15

(c) 20

(d) 25



41. A point object is moving on the pr ave mirror of focal length $24cm$ towards th a distance of $60cm$ from the mirror, its velocity is $9cm/sec$. What is the velocity of the image at that instant

[MP PMT 1997]

(a) $5cm/sec$ towards the mirror

(b) $4cm/sec$ towards the mirror

(c) $4cm/sec$ away from the mirror

(d) $9cm/sec$ away from the mirror

42. A concave mirror is placed on a horizontal table with its axis directed vertically upwards. Let O be the pole of the mirror and C its centre of curvature. A point object is placed at C. It has a real image, also located at C. If the mirror is now filled with water, the image will be [IIT-JEE 1998]

[MP PMT 1993]

(a) Real, and will remain at C

(b) Real, and located at a point between C and ∞

(c) Virtual and located at a point between C and O

(d) Real, and located at a point between C and O

43. The diameter of moon is 3.5×10^3 *km* and its distance from the earth is 3.8×10^5 *km*. If it is seen through a telescope whose focal length for objective and eye lens are 4 *m* and 10 *cm* respectively, then the angle subtended by the moon on the eye will be approximately

[NCERT 1982; CPMT 1991]

(a) 15° (b) 20°
(c) 30° (d) 35°

44. The focal length of an objective of a telescope is 3 *metre* and diameter 15 *cm*. Assuming for a normal eye, the diameter of the pupil is 3 *mm* for its complete use, the focal length of eye piece must be [MP PET 1989]

(a) 6 *cm* (b) 6.3 *cm*
(c) 20 *cm* (d) 60 *cm*

45. We wish to see inside an atom. Assuming the atom to have a diameter of 100 *pm*, this means that one must be able to resolved a width of say 10 *p.m.* If an electron microscope is used, the minimum electron energy required is about

[AIIMS 2004]

(a) 1.5 *KeV* (b) 15 *KeV*
(c) 150 *KeV* (d) 1.5 *KeV*

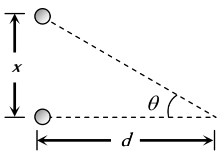
46. A telescope has an objective lens of 10 *cm* diameter and is situated at a distance of one kilometre from two objects. The minimum distance between these two objects, which can be resolved by the telescope, when the mean wavelength of light is 5000 Å, is of the order of [CBSE PMT 2004]

(a) 0.5 *m* (b) 5 *m*
(c) 5 *mm* (d) 5 *cm*

47. Two point white dots are 1*mm* apart on a black paper. They are viewed by eye of pupil diameter 3 *mm*. Approximately, what is the maximum distance at which dots can be resolved by the eye ? [Take wavelength of light = 500 *nm*]

[AIEEE 2005]

(a) 6 *m*
(b) 3 *m*
(c) 5 *m*
(d) 1 *m*



48. A convex lens of focal length 30 *cm* and a concave lens of 10 *cm* focal length are placed so as to have the same axis. If a parallel beam of light falling on convex lens leaves concave lens as a parallel beam, then the distance between two lenses will be

(a) 40 *cm* (b) 30 *cm*
(c) 20 *cm* (d) 10 *cm*

49. A small plane mirror placed at the centre of a spherical screen of radius *R*. A beam of light is falling on the mirror. If the mirror makes *n* revolution. per second, the speed of light on the screen after reflection from the mirror will be

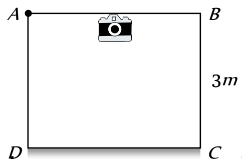
(a) $4\pi nR$ (b) $2\pi nR$
(c) $\frac{nR}{2\pi}$ (d) $\frac{nR}{4\pi}$

50. A room (cubical) is made of mirrors. An insect is moving along the diagonal on the floor such that the velocity of image of insect on two adjacent wall mirrors is 10 *cms*. The velocity of image of insect in ceiling mirror is

(a) 10 *cms* (b) 20 *cms*
(c) $\frac{10}{\sqrt{2}}$ *cms* (d) $10\sqrt{2}$ *cms*

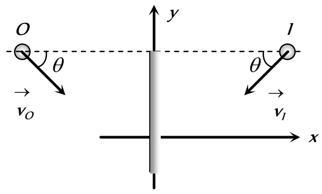
51. Figure shows a cubical room *ABCD* with the wall *CD* as a plane mirror. Each side of the room is 3*m*. We place a camera at the midpoint of the wall *AB*. At what distance should the camera be focussed to photograph an object placed at *A*

(a) 1.5 *m*
(b) 3 *m*
(c) 6 *m*
(d) More than 6 *m*



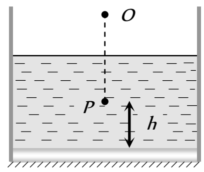
52. If an object moves towards a plane mirror with a speed *v* at an angle θ to the perpendicular to the plane of the mirror, find the relative velocity between the object and the image

(a) *v*
(b) 2*v*
(c) $2v\cos\theta$
(d) $2v\sin\theta$



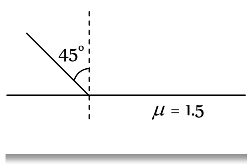
53. A plane mirror is placed at the bottom of the tank containing a liquid of refractive index μ . *P* is a small object at a height *h* above the mirror. An observer *O*-vertically above *P* outside the liquid see *P* and its image in the mirror. The apparent distance between these two will be

(a) $2\mu h$
(b) $\frac{2h}{\mu}$
(c) $\frac{2h}{\mu - 1}$
(d) $h\left(1 + \frac{1}{\mu}\right)$

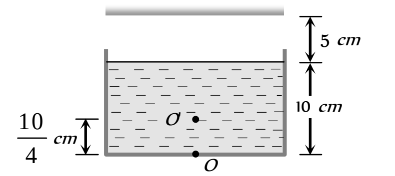


54. One side of a glass slab is silvered as shown. A ray of light is incident on the other side at angle of incidence $i = 45^o$. Refractive index of glass is given as 1.5. The deviation of the ray of light from its initial path when it comes out of the slab is

(a) 90°
(b) 180°
(c) 120°
(d) 45°



55. Consider the situation shown in figure. Water $\left(\mu_w = \frac{4}{3}\right)$ is filled in a breaker upto a height of 10 *cm*. A plane mirror fixed at a height of 5 *cm* from the surface of water. Distance of image from the mirror after reflection from it of an object *O* at the bottom of the beaker is

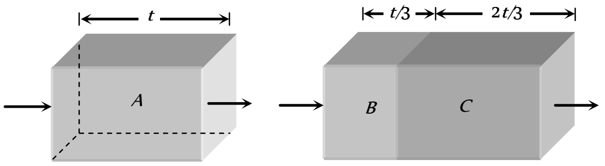


(a) 15 *cm*
(b) 12.5 *cm*
(c) 7.5 *cm*
(d) 10 *cm*

56. A person runs with a speed *u* towards a bicycle moving away from him with speed *v*. The person approaches his image in the mirror fixed at the rear of bicycle with a speed of

(a) $u - v$ (b) $u - 2v$
(c) $2u - v$ (d) $2(u - v)$

57. Two transparent slabs have the same thickness as shown. One is made of material *A* of refractive index 1.5. The other is made of two materials *B* and *C* with thickness in the ratio 1 : 2. The refractive index of *C* is 1.6. If a monochromatic parallel beam passing through the slabs has the same number of waves inside both, the refractive index of *B* is

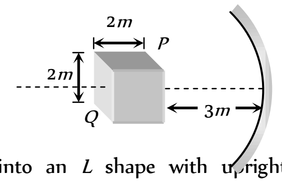


(a) 1.1 (b) 1.2
(c) 1.3 (d) 1.4

58. An object is placed infront of a convex mirror at a distance of 50 *cm*. A plane mirror is introduced covering the lower half of the convex mirror. If the distance between the object and plane mirror is 30 *cm*, it is found that there is no parallax between the images formed by two mirrors. Radius of curvature of mirror will be

(a) 12.5 *cm* (b) 25 *cm*
(c) $\frac{50}{3}$ *cm* (d) 18 *cm*

59. A cube of side 2 *m* is placed in front of a concave mirror focal length 1*m* with its face *P* at a distance of 3 *m* and face *Q* at a distance of 5 *m* from the mirror. The distance between the images of face *P* and *Q* and height of images of *P* and *Q* are

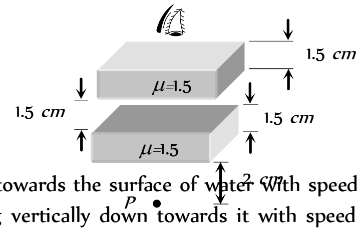


(a) 1 *m*, 0.5 *m*, 0.25 *m*
(b) 0.5 *m*, 1 *m*, 0.25 *m*
(c) 0.5 *m*, 0.25 *m*, 1*m*
(d) 0.25 *m*, 1*m*, 0.5 *m*

60. A small piece of wire bent into an *L* shape with upright and horizontal portions of equal lengths, is placed with the horizontal portion along the axis of the concave mirror whose radius of curvature is 10 *cm*. If the bend is 20 *cm* from the pole of the mirror, then the ratio of the lengths of the images of the upright and horizontal portions of the wire is

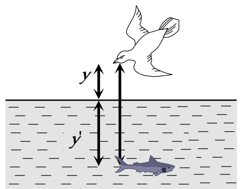
(a) 1 : 2 (b) 3 : 1
(c) 1 : 3 (d) 2 : 1

61. The image of point *P* when viewed from top of the slabs will be



(a) 2.0 *cm* above *P*
(b) 1.5 *cm* above *P*
(c) 2.0 *cm* below *P*
(d) 1 *cm* above *P*

62. A fish rising vertically up towards the surface of water with speed 3 *ms* observes a bird diving vertically down towards it with speed 9 *ms*. The actual velocity of bird is



(a) 4.5 *ms*
(b) 5. *ms*
(c) 3.0 *ms*
(d) 3.4 *ms*

63. A beaker containing liquid is placed on a table, underneath a microscope which can be moved along a vertical scale. The microscope is focussed, through the liquid onto a mark on the table when the reading on the scale is *a*. It is next focussed on the upper surface of the liquid and the reading is *b*. More liquid is added and the observations are repeated, the corresponding readings are *c* and *d*. The refractive index of the liquid is

(a) $\frac{d-b}{d-c-b+a}$ (b) $\frac{b-d}{d-c-b+a}$
(c) $\frac{d-c-b+a}{d-b}$ (d) $\frac{d-b}{a+b-c-d}$

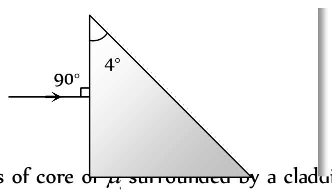
64. Two point light sources are 24 *cm* apart. Where should a convex lens of focal length 9 *cm* be put in between them from one source so that the images of both the sources are formed at the same place

(a) 6 *cm* (b) 9 *cm*
(c) 12 *cm* (d) 15 *cm*

65. There is an equiconvex glass lens with radius of each face as *R* and ${}_a\mu_g = 3/2$ and ${}_a\mu_w = 4/3$. If there is water in object space and air in image space, then the focal length is

(a) 2*R* (b) *R*
(c) 3 *R*/2 (d) R^2

66. A prism having an apex angle 4· and refraction index 1.5 is located in front of a vertical plane mirror as shown in figure. Through what total angle is the ray deviated after reflection from the mirror



(a) 176·
(b) 4·
(c) 178·
(d) 2·

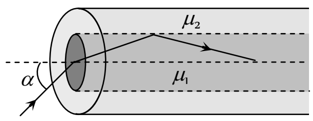
67. An optical fibre consists of core of μ surrounded by a cladding of μ < μ. A beam of light enters from air at an angle α with axis of fibre. The highest α for which ray can be travelled through fibre is

(a) $\cos^{-1}\sqrt{\mu_2^2-\mu_1^2}$

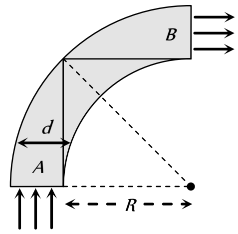
(b) $\sin^{-1}\sqrt{\mu_1^2-\mu_2^2}$

(c) $\tan^{-1}\sqrt{\mu_1^2-\mu_2^2}$

(d) $\sec^{-1}\sqrt{\mu_1^2-\mu_2^2}$



68. A rod of glass (μ = 1.5) and of square cross section is bent into the shape shown in the figure. A parallel beam of light falls on the plane flat surface A as shown in the figure. If d is the width of a side and R is the radius of circular arc then for what maximum value of $\frac{d}{R}$ light entering the glass slab through surface A emerges from the glass through B



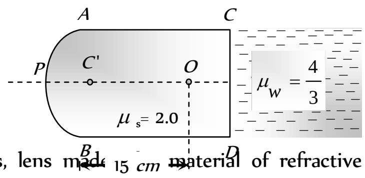
(a) 1.5

(b) 0.5

(c) 1.3

(d) None of these

69. The slab of a material of refractive index 2 shown in figure has curved surface APB of radius of curvature 10 cm and a plane surface CD. On the left of APB is air and on the right of CD is water with refractive indices as given in figure. An object O is placed at a distance of 15 cm from pole P as shown. The distance of the final image of O from P, as viewed from the left is



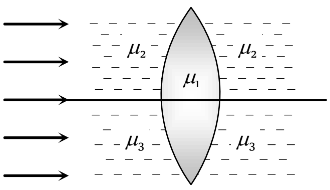
(a) 20 cm

(b) 30 cm

(c) 40 cm

(d) 50 cm

70. A double convex lens, lens made of a material of refractive index μ_1, is placed inside two liquids of refractive indices μ_2 and μ_3, as shown. $\mu_2 > \mu_1 > \mu_3$. A wide, parallel beam of light is incident on the lens from the left. The lens will give rise to



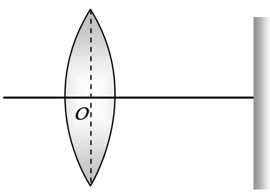
(a) A single convergent beam

(b) Two different convergent beams

(c) Two different divergent beams

(d) A convergent and a divergent beam

71. The distance between a convex lens and a plane mirror is 10 cm. The parallel rays incident on the convex lens after reflection from the mirror form image at the optical centre of the lens. Focal length of lens will be



(a) 10 cm

(b) 20 cm

(c) 30 cm

(d) Cannot be determined

72. A compound microscope is used to enlarge an object kept at a distance 0.03m from it's objective which consists of several convex lenses in contact and has focal length 0.02m. If a lens of focal length 0.1m is removed from the objective, then by what distance the eye-piece of the microscope must be moved to refocus the image

(a) 2.5 cm (b) 6 cm

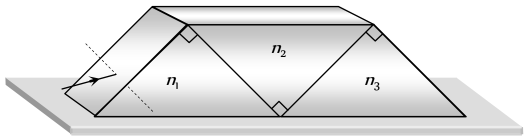
(c) 15 cm (d) 9 cm

73. If the focal length of the objective lens and the eye lens are 4 mm and 25 mm respectively in a compound microscope. The length of the tube is 16 cm. Find its magnifying power for relaxed eye position

(a) 32.75 (b) 327.5

(c) 0.3275 (d) None of the above

74. Three right angled prisms of refractive indices n_1, n_2 and n_3 are fixed together using an optical glue as shown in figure. If a ray passes through the prisms without suffering any deviation, then



(a) $n_1 = n_2 = n_3$ (b) $n_1 = n_2 \neq n_3$

(c) $1 + n_1 = n_2 + n_3$ (d) $1 + n_2^2 = n_1^2 + n_3^2$

75. In a compound microscope, the focal length of the objective and the eye lens are 2.5 cm and 5 cm respectively. An object is placed at 3.75 cm before the objective and image is formed at the least distance of distinct vision, then the distance between two lenses will be (*i.e.* length of the microscopic tube)

(a) 11.67 cm (b) 12.67 cm

(c) 13.00 cm (d) 12.00 cm

76. In a grease spot photometer light from a lamp with dirty chimney is exactly balanced by a point source distance 10 cm from the grease spot. On clearing the chimney, the point source is moved 2 cm to obtain balance again. The percentage of light absorbed by dirty chimney is nearly

(a) 56% (b) 44%

(c) 36% (d) 64%

77. The separation between the screen and a plane mirror is 2r. An isotropic point source of light is placed exactly midway between the mirror and the screen. Assume that mirror reflects 100% of incident light. Then the ratio of illuminances on the screen with and without the mirror is

(a) 10 : 1 (b) 2 : 1

(c) 10 : 9 (d) 9 : 1

78. The separation between the screen and a concave mirror is 2r. An isotropic point source of light is placed exactly midway between the mirror and the point source. Mirror has a radius of curvature r and reflects 100% of the incident light. Then the ratio of illuminances on the screen with and without the mirror is

(a) 10 : 1 (b) 2 : 1

(c) 10 : 9 (d) 9 : 1

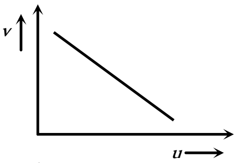
79. The apparent depth of water in cylindrical water tank of diameter 2R cm is reducing at the rate of x cm/minute when water is being drained out at a constant rate. The amount of water drained in *c.c.* per minute is (n_1 = refractive index of air, n_2 = refractive index of water) **[AIIMS 2005]**

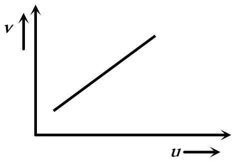
(a) $x\pi R^2 n_1/n_2$ (b) $x\pi R^2 n_2/n_1$

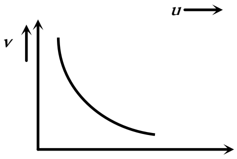
(c) $2\pi R n_1/n_2$ (d) $\pi R^2 x$

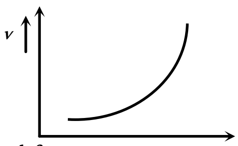
Graphical Questions

1. In an experiment of find the focal length of a concave mirror a graph is drawn between the magnitudes of u and v. The graph looks like [AIIMS 2003]

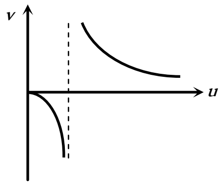
(a) 

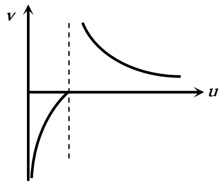
(b) 

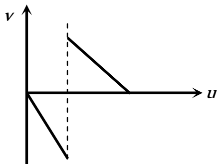
(c) 

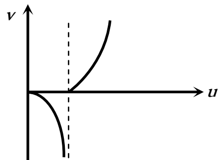
(d) 

2. As the position of an object (u) reflected from a concave ~~mirror~~ is varied, the position of the image (v) also varies. By letting the u changes from 0 to $+\infty$ the graph between v versus u will be

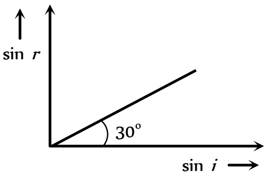
(a) 

(b) 

(c) 

(d) 

3. When light is incident on a medium at angle i and refracted into a second medium at an angle r, the graph of sin i vs sin$r$ is as shown in the graph. From this, one can conclude that



(a) Velocity of light in the second medium is 1.73 times the velocity of light in the I medium

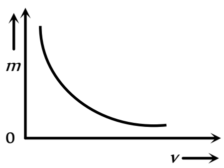
(b) Velocity of light in the I medium is 1.73 times the velocity in the II medium

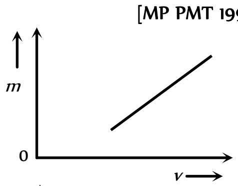
(c) The critical angle for the two media is given by

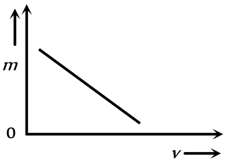
$$\sin\, i_c = \frac{1}{\sqrt{3}}$$

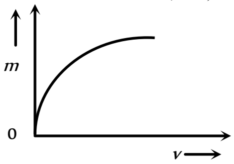
(d) $\sin\, i_c = \frac{1}{2}$

4. The graph between the lateral magnification (m) produced by a lens and the distance of the image (v) is given by [MP PMT 1994]

(a) 

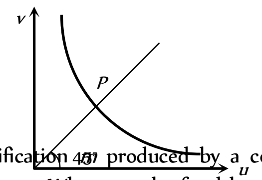
(b) 





(c) (d)

5. The graph shows variation of v with change in u for a mirror. Points plotted above the point P on the curve are for values of v



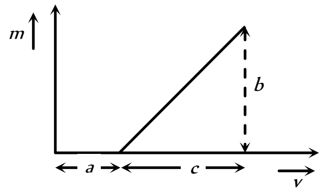
(a) Smaller then f

(b) Smaller then $2f$

(c) Larger then $2f$

(d) Larger than f

6. The graph shows how the magnification m produced by a convex thin lens varies with image distance v. What was the focal length of the used [DPMT 1995]



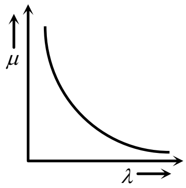
(a) $\frac{b}{c}$

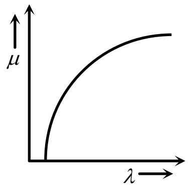
(b) $\frac{b}{ca}$

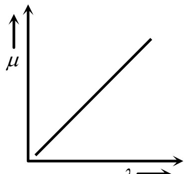
(c) $\frac{bc}{a}$

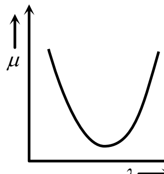
(d) $\frac{c}{b}$

7. Which of the following graphs shows appropriate variation of refractive index μ with wavelength λ

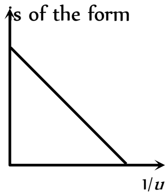
(a) 

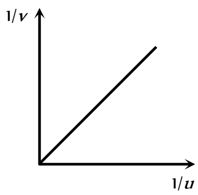
(b) 

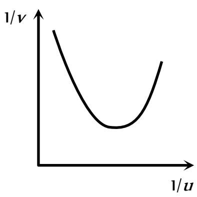
(c) 

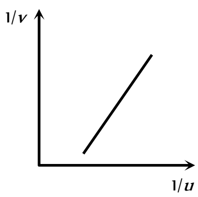
(d) 

8. For a concave mirror, if real image is formed the graph between $\frac{1}{u}$ and $\frac{1}{v}$ is of the form

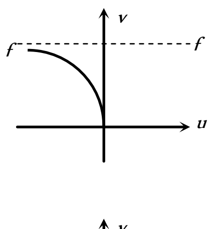
(a) 

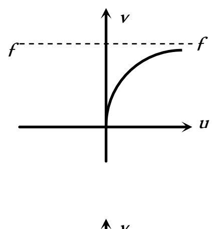
(b) 

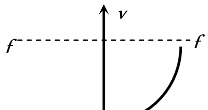
(c) 

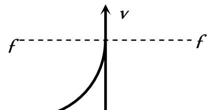
(d) 

9. The graph between u and v for a convex mirror is

(a) 

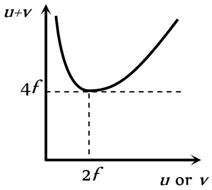
(b) 

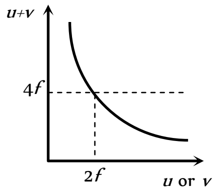


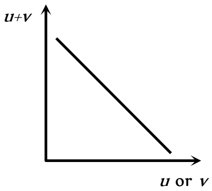


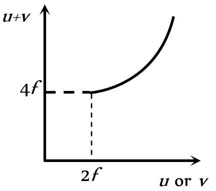
(c) (d)

10. For a convex lens, if real image is formed the graph between $(u + v)$ and u or v is as follows

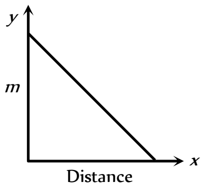
(a)


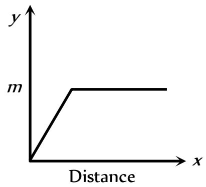
(b)


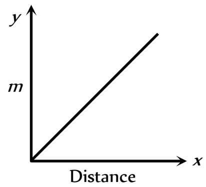
(c)


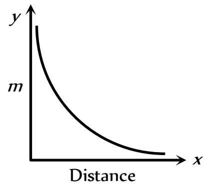
(d)


11. Which of the following graphs is the magnification of a real image against the distance from the focus of a concave mirror

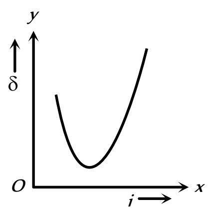
(a)


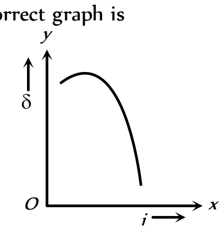
(b)


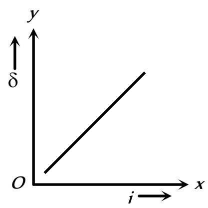
(c)


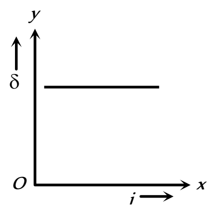
(d)


12. A graph is plotted between angle of deviation (δ) and angle of incidence (i) for a prism. The nearly correct graph is

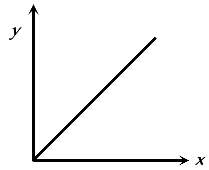
(a)


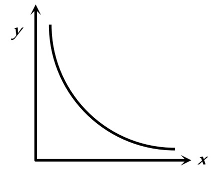
(b)


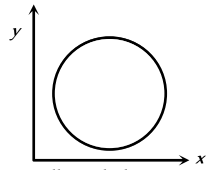
(c)


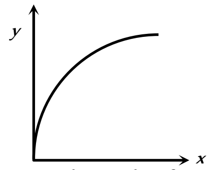
(d)


13. If x is the distance of an object from the focus of a concave mirror and y is the distance of image from the focus, then which of the following graphs is correct between x and y

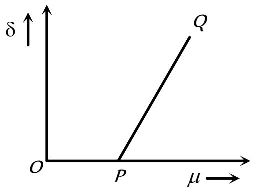
(a)


(b)


(c)


(d)


14. For a small angled prism, angle of prism A, the angle of minimum deviation (δ) varies with the refractive index of the prism as shown in the graph



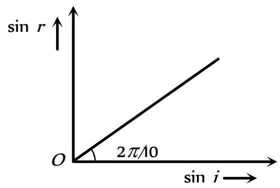
(a) Point P corresponds to $\mu = 1$

(b) Slope of the line $PQ = A/2$

(c) Slope $= A$

(d) None of the above statements is true

15. The graph between sine of angle of refraction (sin r) in medium 2 and sine of angle of incidence (sin i) in medium 1 indicates that $(\tan 36° \approx \dfrac{3}{4})$



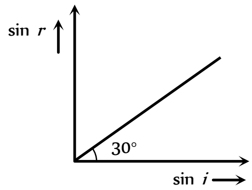
(a) Total internal reflection can take place

(b) Total internal reflection cannot take place

(c) Any of (a) and (b)

(d) Data is incomplete

16. A medium shows relation between i and r as shown. If speed of light in the medium is nc then value of n is



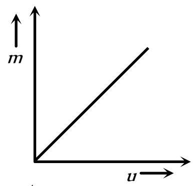
(a) 1.5

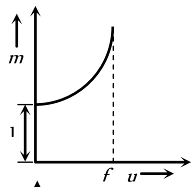
(b) 2

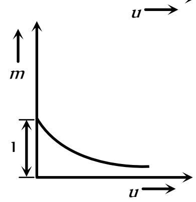
(c) 2

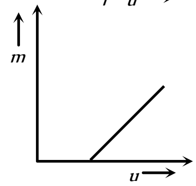
(d) 3

17. For a concave mirror, if virtual image is formed, the graph between m and u is of the form

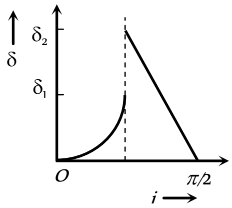
(a)


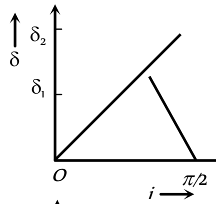
(b)


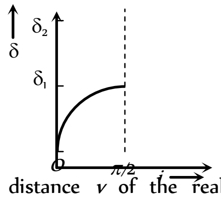
(c)


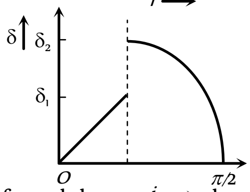
(d)


18. A ray of light travels from a medium of refractive index μ to air. Its angle of incidence in the medium is i, measured from the normal to the boundary, and its angle of deviation is δ. δ is plotted against i which of the following best represents the resulting curve

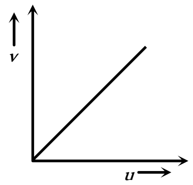
(a) 

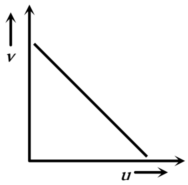
(b) 

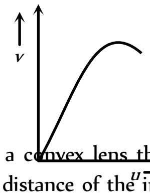
(c) 

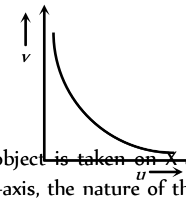
(d) 

19. The distance v of the real image formed by a convex lens is measured for various object distance u. A graph is plotted between v and u, which one of the following graphs is correct

(a) 

(b) 

(c) 

(d) 

20. For a convex lens the distance of the object is taken on X-axis and the distance of the image is taken on Y-axis, the nature of the graph so obtained is [BVP 2003]

(a) Straight line (b) Circle

(c) Parabola (d) Hyperbola

AR Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) *If both assertion and reason are true and the reason is the correct explanation of the assertion.*
(b) *If both assertion and reason are true but reason is not the correct explanation of the assertion.*
(c) *If assertion is true but reason is false.*
(d) *If the assertion and reason both are false.*
(e) *If assertion is false but reason is true.*

1. Assertion : A red object appears dark in the yellow light

Reason : A red colour is scattered less [AIIMS 2004]

2. Assertion : The stars twinkle while the planets do not.

Reason : The stars are much bigger in size than the planets. [AIIMS 2003]

3. Assertion : Owls can move freely during night.

Reason : They have large number of rods on their retina. [AIIMS 2003]

4. Assertion : The air bubble shines in water.

Reason : Air bubble in water shines due to refraction of light [AIIMS 2002]

5. Assertion : In a movie, ordinarily 24 frames are projected per second from one end to the other of the complete film.

Reason : The image formed on retina of eye is sustained upto 1/10 second after the removal of stimulus. [AIIMS 2001]

6. Assertion : Blue colour of sky appears due to scattering of blue colour.

Reason : Blue colour has shortest wave length in visible spectrum. [AIIMS 2001]

7. Assertion : The refractive index of diamond is $\sqrt{6}$ and that of liquid is $\sqrt{3}$. If the light travels from diamond to the liquid, it will totally reflected when the angle of incidence is 30°.

Reason [BVP 2003] : $\mu = \dfrac{1}{\sin C}$, where μ is the refractive index of diamond with respect to liquid. [AIIMS 2000]

8. Assertion : The setting sun appears to be red.

Reason : Scattering of light is directly proportional to the wavelength. [AIIMS 2000]

9. Assertion : A double convex lens (μ = 1.5) has focal length 10 *cm*. When the lens is immersed in water (μ = 4/3) its focal length becomes 40 *cm*.

Reason : $\dfrac{1}{f} = \dfrac{\mu_l - \mu_m}{\mu_m}\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)$ [AIIMS 1999]

10. Assertion : Different colours travel with different speed in vacuum.

Reason : Wavelength of light depends on refractive index of medium. [AIIMS 1998]

11. Assertion : The colour of the green flower seen through red glass appears to be dark.

Reason : Red glass transmits only red light. [AIIMS 1997]

12. Assertion : The focal length of the mirror is f and distance of the object from the focus is u, the magnification of the mirror is f/u.

Reason : Magnification $= \dfrac{\text{Size of image}}{\text{Size of object}}$ [AIIMS 1994]

13. Assertion : If a plane glass slab is placed on the letters of different colours all the letters appear to be raised up to the same height.

Reason : Different colours have different wavelengths.

14. Assertion : The fluorescent tube is considered better than an electric bulb.

Reason : Efficiency of fluorescent tube is more than the efficiency of electric bulb.

15. Assertion : The polar caps of earth are cold in comparison to equatorial plane.

Reason : The radiation absorbed by polar caps is less than the radiation absorbed by equatorial plane.

16. Assertion : The illumination of earth's surface from sun is more at noon than in the morning.

Reason : Luminance of a surface refers to brightness of the surface.

17. Assertion : When an object is placed between two plane parallel mirrors, then all the images found are of equal intensity.

Reason : In case of plane parallel mirrors, only two images are possible.

18. Assertion : The mirrors used in search lights are parabolic and not concave spherical.

Reason : In a concave spherical mirror the image formed is always virtual.

19. Assertion : The size of the mirror affect the nature of the image.

Reason : Small mirrors always forms a virtual image.

20. Assertion : Just before setting, the sun may appear to be elliptical. This happens due to refraction.

Reason : Refraction of light ray through the atmosphere may cause different magnification in mutually perpendicular directions.

21. Assertion : Critical angle of light passing from glass to air is minimum for violet colour.

Reason : The wavelength of blue light is greater than the light of other colours.

22. Assertion : We cannot produce a real image by plane or convex mirrors under any circumstances.

Reason : The focal length of a convex mirror is always taken as positive.

23. Assertion : A piece of red glass is heated till it glows in dark. The colour of glowing glass would be orange.

Reason : Red and orange is complementary colours.

24. Assertion : Within a glass slab, a double convex air bubble is formed. This air bubble behaves like a converging lens.

Reason : Refractive index of air is more than the refractive index of glass.

25. Assertion : The images formed by total internal reflections are much brighter than those formed by mirrors or lenses.

Reason : There is no loss of intensity in total internal reflection.

26. Assertion : The focal length of lens does not change when red light is replaced by blue light.

Reason : The focal length of lens does not depends on colour of light used.

27. Assertion : There is no dispersion of light refracted through a rectangular glass slab.

Reason : Dispersion of light is the phenomenon of splitting of a beam of white light into its constituent colours.

28. Assertion : All the materials always have the same colour, whether viewed by reflected light or through transmitted light.

Reason : The colour of material does not depend on nature of light.

29. Assertion : A beam of white light gives a spectrum on passing through a hollow prism.

Reason : Speed of light outside the prism is different from the speed of light inside the prism.

30. Assertion : By increasing the diameter of the objective of telescope, we can increase its range.

Reason : The range of a telescope tells us how far away a star of some standard brightness can be spotted by telescope.

31. Assertion : For the sensitivity of a camera, its aperture should be reduced.

Reason : Smaller the aperture, image focussing is also sharp.

32. Assertion : If objective and eye lenses of a microscope are interchanged then it can work as telescope.

Reason : The objective of telescope has small focal length.

33. Assertion : The illuminance of an image produced by a convex lens is greater in the middle and less towards the edges.

Reason : The middle part of image is formed by undeflected rays while outer part by inclined rays.

34. Assertion : Although the surfaces of a goggle lens are curved, it does not have any power.

Reason : In case of goggles, both the curved surfaces have equal radii of curvature.

35. Assertion : The resolving power of an electron microscope is higher than that of an optical microscope.

Reason : The wavelength of electron is more than the wavelength of visible light.

36. Assertion : If the angles of the base of the prism are equal, then in the position of minimum deviation, the refracted ray will pass parallel to the base of prism.

Reason : In the case of minimum deviation, the angle of incidence is equal to the angle of emergence.

37. Assertion : Dispersion of light occurs because velocity of light in a material depends upon its colour.

Reason : The dispersive power depends only upon the material of the prism, not upon the refracting angle of the prism.

38. Assertion : An empty test tube dipped into water in a beaker appears silver, when viewed from a suitable direction.

Reason : Due to refraction of light, the substance in water appears silvery.

39. Assertion : Spherical aberration occur in lenses of larger aperture.

Reason : The two rays, paraxial and marginal rays focus at different points.

40. Assertion : It is impossible to photograph a virtual image.

Reason : The rays which appear diverging from a virtual image fall on the camera and a real image is captured.

41. Assertion : The speed of light in a rarer medium is greater than that in a denser medium

Reason : One light year equals to 9.5×10^{12} *km*

[AIIMS 1999]

42. Assertion : The frequencies of incident, reflected and refracted beam of monochromatic light incident from one medium to another are same

Reason : The incident, reflected and refracted rays are coplanar [EAMCET (Engg.) 2000]

43. Assertion : The refractive index of a prism depends only on the kind of glass of which it is made of and the colour of light

Reason : The refractive index of a prism depends upon the refracting angle of the prism and the angle of minimum deviation [AIIMS 2000]

44. Assertion : The resolving power of a telescope is more if the diameter of the objective lens is more.

Reason : Objective lens of large diameter collects more light.[AIIMS

45. Assertion : By roughening the surface of a glass sheet its transparency can be reduced.

Reason : Glass sheet with rough surface absorbs more light.[AIIMS

46. Assertion : Diamond glitters brilliantly.

Reason : Diamond does not absorb sunlight.

[AIIMS 2005]

47. Assertion : The cloud in sky generally appear to be whitish.

Reason : Diffraction due to cloud is efficient in equal measure at all wavelengths. [AIIMS 2005]

Answers

Plane Mirror

1	d	2	b	3	b	4	c,d	5	c
6	c	7	d	8	b	9	b	10	c
11	b	12	d	13	a	14	c	15	c
16	b	17	c	18	b	19	c	20	a
21	c	22	b	23	c	24	b	25	b
26	b	27	c	28	c	29	c	30	c
31	b	32	a	33	b	34	c		

Spherical Mirror

1	a	2	c	3	d	4	c	5	a
6	b	7	c	8	b	9	a	10	b
11	d	12	b	13	b	14	b	15	c
16	d	17	b	18	b	19	a	20	a
21	a	22	b	23	d	24	d	25	b
26	bc	27	c	28	b	29	a	30	b
31	d	32	c	33	a	34	d	35	d
36	b	37	d	38	d	39	d	40	a
41	d	42	d	43	a	44	a		

Refraction of Light at Plane Surfaces

1	d	2	a	3	b	4	a	5	d
6	a	7	c	8	d	9	c	10	a
11	b	12	d	13	b	14	a	15	b
16	a	17	c	18	c	19	d	20	a
21	b	22	b	23	c	24	a	25	c
26	a	27	b	28	d	29	a	30	c
31	c	32	c	33	b	34	b	35	b
36	b	37	a	38	b	39	c	40	d
41	a	42	d	43	c	44	c	45	a
46	a	47	c	48	a	49	c	50	c
51	d	52	b	53	b	54	b	55	b
56	a	57	d	58	b	59	c	60	b
61	d	62	a	63	b	64	d	65	b
66	a	67	b	68	b	69	a	70	d
71	c	72	c	73	d	74	d	75	b
76	d	77	c	78	c	79	b	80	b
81	a	82	a	83	b	84	b	85	c
86	b	87	d	88	d	89	b	90	d

Total Internal Reflection

1	b	2	c	3	d	4	d	5	c
6	c	7	b	8	c	9	a	10	d
11	b	12	c	13	c	14	d	15	d
16	c	17	c	18	cd	19	c	20	d
21	a	22	c	23	b	24	c	25	a
26	c	27	c	28	a	29	d	30	d
31	a	32	c	33	a	34	c	35	a
36	d	37	b	38	b	39	c	40	a
41	c	42	b	43	b	44	d	45	B
46	a								

Refraction at Curved Surface

1	a	2	a	3	d	4	c	5	a
6	d	7	b	8	a	9	c	10	c
11	c	12	d	13	b	14	c	15	b
16	d	17	c	18	d	19	c	20	c
21	c	22	a	23	d	24	a	25	d
26	a	27	b	28	a	29	a	30	c
31	c	32	d	33	d	34	c	35	b
36	b	37	c	38	d	39	b	40	d
41	a	42	c	43	a	44	c	45	d
46	d	47	c	48	b	49	a	50	b
51	c	52	a	53	a	54	b	55	a

56	b	57	a	58	a	59	d	60	c
61	b	62	b	63	d	64	d	65	d
66	a	67	d	68	c	69	c	70	b
71	d	72	b	73	a	74	c	75	a
76	c	77	a	78	b	79	b	80	d
81	c	82	a	83	d	84	a	85	c
86	c	87	b	88	a	89	a	90	b
91	b	92	d	93	c	94	a	95	c
96	c	97	c	98	a	99	d	100	a
101	a	102	d	103	c	104	d	105	a
106	c	107	b	108	a	109	d	110	b
111	c	112	c	113	c	114	d	115	a
116	c	117	a	118	d	119	c	120	b
121	c	122	d	123	a	124	b	125	d
126	c	127	d	128	b	129	b	130	c
131	b	132	b	133	b	134	d	135	b
136	d	137	d	138	b	139	a	140	c
141	b	142	b	143	c	144	b	145	c

Prism Theory & Dispersion of Light

1	b	2	b	3	b	4	c	5	c
6	a	7	a	8	d	9	d	10	d
11	c	12	b	13	b	14	a	15	a
16	b	17	d	18	a	19	d	20	b
21	a	22	c	23	a	24	a	25	b
26	c	27	c	28	b	29	a	30	a
31	c	32	b	33	a	34	c	35	d
36	a	37	b	38	a	39	d	40	b
41	b	42	b	43	a	44	c	45	a
46	c	47	b	48	a	49	c	50	c
51	c	52	a	53	d	54	d	55	a
56	c	57	a	58	a	59	a	60	c
61	c	62	b	63	d	64	d	65	a
66	b	67	c	68	c	69	b	70	c
71	a	72	d	73	a	74	b	75	a
76	b	77	b	78	b	79	d	80	a
81	b	82	a	83	b	84	c	85	a
86	c	87	c	88	a	89	b	90	b
91	c	92	a	93	c	94	c	95	b
96	c	97	c	98	a	99	a	100	c
101	a	102	b	103	a	104	b	105	d
106	b	107	b	108	a	109	b	110	a
111	a	112	d	113	a	114	b	115	a
116	d	117	d	118	d	119	c	120	d
121	a	122	d	123	c	124	d	125	b
126	a	127	c	128	c	129	d	130	a

131	a	132	c	133	a	134	c	135	b
136	c	137	a	138	d	139	c	140	b
141	a	142	a	143	b	144	b	145	a
146	a	147	d	148	b	149	c	150	a
151	c								

Human Eye and Lens Camera

1	c	2	a	3	b	4	d	5	b
6	c	7	b	8	a	9	d	10	a
11	c	12	c	13	a	14	b	15	d
16	b	17	c	18	c	19	b	20	c
21	b	22	a	23	a	24	a	25	d
26	a	27	d	28	c	29	b	30	c
31	c	32	c	33	b	34	b	35	a
36	c	37	d	38	a	39	d	40	a
41	b	42	c	43	d	44	a	45	b
46	b	47	d	48	d	49	b	50	b
51	c	52	a	53	a	54	c	55	d
56	a	57	a	58	d	59	a	60	d
61	d	62	a	63	b	64	d	65	a

Microscope and Telescope

1	c	2	b	3	b	4	b	5	b
6	d	7	c	8	a	9	b	10	b
11	a	12	b	13	b	14	a	15	c
16	d	17	a	18	b	19	b	20	b
21	a	22	d	23	c	24	a	25	d
26	c	27	c	28	d	29	d	30	b
31	a	32	d	33	d	34	c	35	d
36	b	37	a	38	a	39	b	40	d
41	d	42	b	43	d	44	a	45	c
46	b	47	b	48	d	49	b	50	d
51	c	52	a	53	a	54	a	55	b
56	a	57	d	58	d	59	c	60	c
61	c	62	a	63	b	64	a	65	b
66	a	67	a	68	c	69	a	70	b
71	c	72	b	73	a	74	a	75	b
76	d	77	c	78	b	79	a	80	c
81	b	82	b	83	b	84	a	85	b
86	abcd	87	a	88	a	89	b	90	c
91	b	92	d	93	c	94	d	95	c
96	c	97	d	98	a	99	b	100	d
101	c	102	b	103	a	104	b	105	b
106	c	107	c	108	a	109	c	110	c
111	d	112	a	113	d	114	a	115	a

116	a	117	b	118	a	119	a	120	a

Photometry

1	d	2	b	3	d	4	c	5	d
6	b	7	a	8	b	9	c	10	c
11	a	12	c	13	c	14	c	15	a
16	a	17	b	18	b	19	c	20	b
21	c	22	c	23	a	24	b	25	bc
26	c	27	d	28	b	29	d	30	b
31	d	32	a	33	d	34	d	35	a
36	c	37	c	38	d	39	d	40	c
41	c								

Critical Thinking Questions

1	d	2	b	3	b	4	a	5	d
6	b	7	b	8	a	9	b	10	c
11	a	12	b	13	b	14	a	15	b
16	b	17	a	18	b	19	c	20	c
21	c	22	d	23	d	24	ad	25	c
26	b	27	d	28	d	29	d	30	d
31	b	32	a	33	d	34	c	35	c
36	d	37	a	38	b	39	a	40	c
41	c	42	d	43	b	44	a	45	b
46	c	47	c	48	c	49	a	50	d
51	d	52	c	53	b	54	a	55	b
56	d	57	c	58	b	59	d	60	b
61	d	62	a	63	a	64	a	65	c
66	c	67	b	68	b	69	b	70	d
71	b	72	d	73	b	74	d	75	a
76	c	77	c	78	b	79	b		

Graphical Questions

1	c	2	a	3	bc	4	c	5	c
6	d	7	a	8	a	9	a	10	a
11	d	12	a	13	b	14	ac	15	b
16	d	17	b	18	a	19	d	20	d

Assertion and Reason

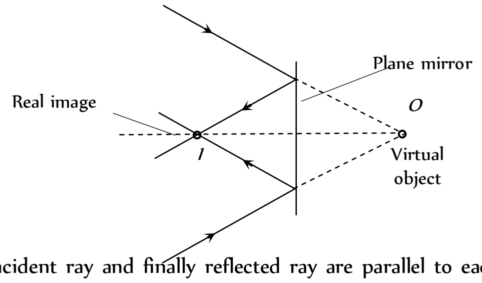
1	b	2	b	3	c	4	c	5	c
6	a	7	e	8	c	9	a	10	e
11	a	12	a	13	e	14	a	15	c
16	b	17	d	18	c	19	d	20	a
21	c	22	e	23	d	24	d	25	a
26	d	27	b	28	d	29	d	30	b
31	c	32	d	33	a	34	a	35	c
36	a	37	b	38	c	39	a	40	e
41	b	42	b	43	c	44	a	45	c
46	b	47	c						

Answers and Solutions

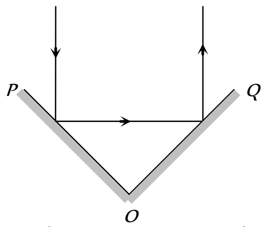
Plane Mirror

1. (d) $\delta = (360 - 2\theta) = (360 - 2 \times 60) = 240^o$

2. (b) When converging beam incident on plane mirror, real image is formed as shown

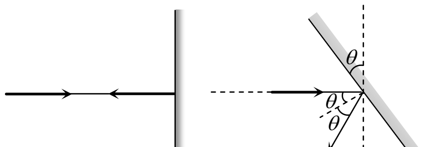


3. (b) Incident ray and finally reflected ray are parallel to each other means $\delta = 180^o$

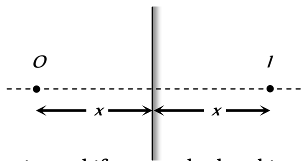


From $\delta = 360 - 2\theta \Rightarrow 180 = 360 - 2\theta \Rightarrow \theta = 90^o$

4. (c, d) By keeping the incident ray is fixed, if plane mirror rotates through an angle θ reflected ray rotates through an angle 2θ.

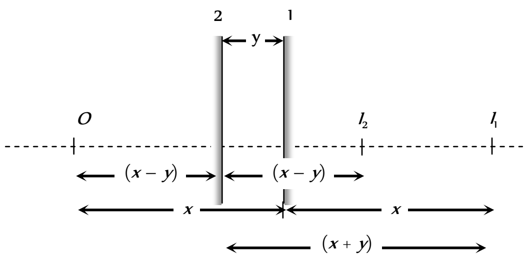


5. (c) Suppose at any instant, plane mirror lies at a distance x from object. Image will be formed behind the mirror at the same distance x.



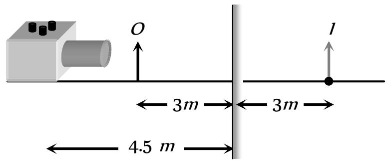
When the mirror shifts towards the object by distance 'y' the image shifts $= x + y - (x - y) = 2y$

So speed of image = 2 × speed of mirror

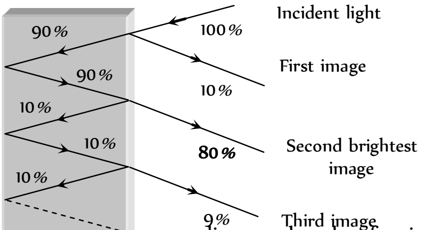


6. (c) Number of images $=\left(\frac{360}{\theta}-1\right)=\left(\frac{360}{60}-1\right)=5$

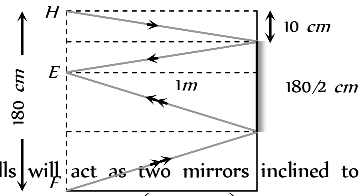
7. (d) F using distance of image = 4.5 m + 3 m = 7.5 m.



8. (b) Several images will be formed but second image will be brightest



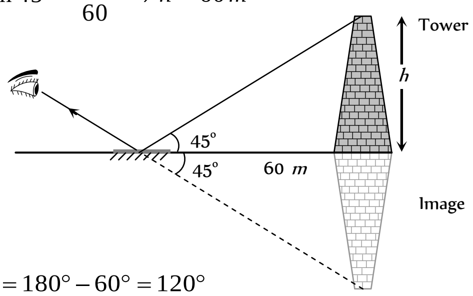
9. (b) According to g ray diagram length of mirror $=\frac{1}{2}(10+170)=90\,cm$



10. (c) The walls will act as two mirrors inclined to each other at $90°$ and so will form $\left(\frac{360}{90}-1\right)=4-1$ *i.e.* 3 images of the person. Now these images with person will act as objects for the ceiling mirror and so ceiling mirror will form 4 images further. Therefore total number of images formed $=3+3+1=7$

Note : He can see. 6 images of himself.

11. (b) $\tan 45°=\frac{h}{60}\Rightarrow h=60\,m$



12. (d) $\delta=180°-60°=120°$

13. (a) $<i=<r=0°$

14. (c) When light is reflected from denser medium, a phase difference of π always occurs.

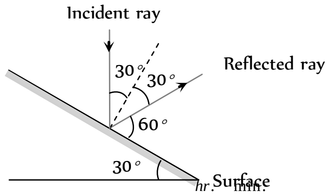
15. (c) Ray after reflection from three mutually perpendicular mirrors becomes anti-parallel.

16. (b) In two images man will see himself using left hand.

17. (c) In plane mirror, size of the image is independent of the angle of incidence.

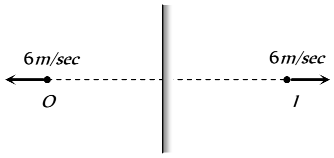
18. (b) Size of image formed by a plane mirror is same as that of the object. Hence its magnification will be 1.

19. (c)



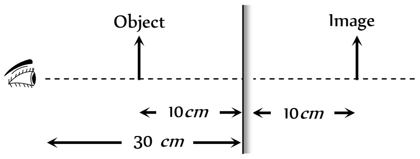
20. (a) Subtract the given time from 11 : 60

21. (c) Relative velocity of image *w.r.t.* object $=6-(-6)=12\,m/\sec$



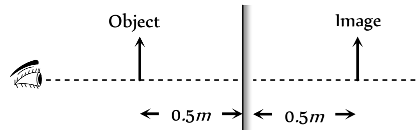
22. (b)

23. (c) See following ray diagram

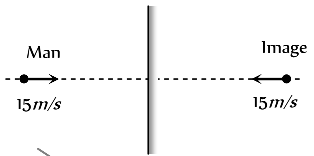


The distance focussed for eye $=30+10=40\,cm$

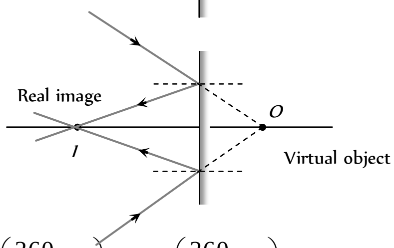
24. (b) Distance between object and image $=0.5+0.5=1\,m$



25. (b) Relative velocity of image *w.r.t.* man $=15-(-15)=30\,m/s$



26. (b)



27. (c) $n=\left(\frac{360}{\theta}-1\right)\Rightarrow n=\left(\frac{360}{72}-1\right)=4$

28. (c) $n=\left(\frac{360}{\theta}-1\right)\Rightarrow 3=\left(\frac{360}{\theta}-1\right)\Rightarrow \theta=90°$

29. (c)

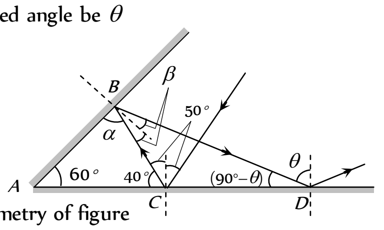
30. (c) $n=\frac{360}{45}-1=7$

31. (b) Diminished, erect image is formed by convex mirror.

32. (a) When a mirror is rotated by an angle θ, the reflected ray deviate from its original path by angle 2θ.

33. (b) $f = \dfrac{R}{2}$, and $R = \infty$ for plane mirror.

34. (c) Let required angle be θ



From geometry of figure

In Δ ABC; α = 180° – (60° + 40°) = 80°

$\Rightarrow \beta$ = 90° – 80° = 10°

In Δ ABD; $\angle A$ = 60°, $\angle B = (\alpha + 2\beta)$

= (80 + 2 × 10) = 100° and $\angle D$ = (90° – θ)

$\because$ $\angle A + \angle B + \angle D$ = 180° $\Rightarrow$ 60° + 100° + (90° – θ) = 180° $\Rightarrow \theta$ = 70°

Spherical Mirror

1. (a) $m = +\dfrac{1}{n} = -\dfrac{v}{u} \Rightarrow v = -\dfrac{u}{n}$

By using mirror formula $\dfrac{1}{f} = \dfrac{1}{-\dfrac{u}{n}} + \dfrac{1}{u} \Rightarrow u = -(n-1)f$

2. (c)

3. (d)

4. (c) $\dfrac{I}{O} = \dfrac{f}{(f-u)} \Rightarrow \dfrac{I}{+5} = \dfrac{-10}{-10-(-100)} \Rightarrow I = 0.55\ cm$

5. (a) For real image $m = -2$, so by using $m = \dfrac{f}{f-u}$

$\Rightarrow -2 = \dfrac{-50}{-50-u} \Rightarrow u = -75\ cm$

6. (b) By using $\dfrac{I}{O} = \dfrac{f}{f-u}$

$\Rightarrow \dfrac{I}{+(7.5)} = \dfrac{(25/2)}{\left(\dfrac{25}{2}\right) - (-40)} \Rightarrow I = 1.78\ cm$

7. (c)

8. (b) $\dfrac{I}{O} = \dfrac{f}{f-u}$; where $u = f + x$ $\therefore \dfrac{I}{O} = -\dfrac{f}{x}$

9. (a) Image formed by convex mirror is virtual for real object placed anywhere.

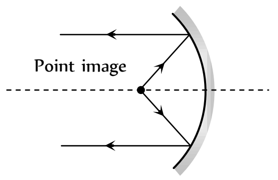
10. (b) Given $u = (f + x_1)$ and $v = (f + x_2)$

The focal length $f = \dfrac{uv}{u+v} = \dfrac{(f+x_1)(f+x_2)}{(f+x_1)+(f+x_2)}$

On solving, we get $f^2 = x_1 x_2$ or $f = \sqrt{x_1 x_2}$

11. (d) The image formed by a convex mirror is always virtual.

12. (b) Object should be placed on focus of concave mirror.



13. (b) $m = \dfrac{f}{(f-u)} \Rightarrow \left(+\dfrac{1}{4}\right) = \dfrac{(+30)}{(+30)-u} \Rightarrow u = -90\ cm$

14. (b) Size is $\dfrac{1}{5}$. It can't be plane and concave mirror, because both conditions are not satisfied in plane or concave mirror. Convex mirror can meet all the requirements.

15. (c) Plane mirror and convex mirror always forms erect images. Image formed by concave mirror may be erect or inverted depending on position of object.

16. (d) Virtual image is seen on the photograph.

17. (b) $\because m = -\dfrac{v}{u}$ also $\dfrac{1}{f} = \dfrac{1}{v} + \dfrac{1}{u} \Rightarrow \dfrac{u}{f} = \dfrac{u}{v} + 1$

$\Rightarrow -\dfrac{u}{v} = 1 - \dfrac{u}{f} \Rightarrow \dfrac{-v}{u} = \dfrac{f}{f-u}$ so $m = \dfrac{f}{f-u}$.

18. (b) To make the light diverging as much as possible.

19. (a) Let distance $= u$. Now $\dfrac{v}{u} = 16$ and $v = u + 120$

$\therefore \dfrac{120+u}{u} = 16 \Rightarrow 15u = 120 \Rightarrow u = 8\ cm$.

20. (a) Virtual image formed is larger in size in case of concave mirror.

21. (a) Real, inverted and same in size because object is at the centre of curvature of the mirror.

22. (b) Image is virtual so $m = +3$. and $f = \dfrac{R}{2} = 18\ cm$

So from $m = \dfrac{f}{f-u} \Rightarrow 3 = \dfrac{(-18)}{(-18)-u} \Rightarrow u = -12\ cm$.

23. (d) $f = \dfrac{R}{2} = 20\,cm$, $m = 2$ For real image; $m = -2$,

By using $m = \dfrac{f}{f-u}$, $-2 = \dfrac{-20}{-20-u} \Rightarrow u = -30\,cm$

For virtual image; $m = +2$

So, $+2 = \dfrac{-20}{-20-u} \Rightarrow u = -10\,cm$

24. (d) Convex mirror always forms, virtual, erect and smaller image.

25. (b) When object is placed. Between focus and pole, image formed is erect, virtual and enlarged.

26. (b, c) Convex mirror and concave lens form virtual image for all positions of object.

27. (c) Here focal length $= f$ and $u = -f$

On putting these values in $\dfrac{1}{f} = \dfrac{1}{u} + \dfrac{1}{v}$

$\Rightarrow \dfrac{1}{f} = -\dfrac{1}{f} + \dfrac{1}{v} \Rightarrow v = \dfrac{f}{2}$

28. (b) Erect and enlarged image can produced by concave mirror.

$\dfrac{I}{O} = \dfrac{f}{f-u} \Rightarrow \dfrac{+3}{+1} = \dfrac{f}{f-(-4)} \Rightarrow f = -6\,cm$

$\Rightarrow R = 2f = -12\,cm$

29. (a)

30. (b) $m = \dfrac{f}{f-u} \Rightarrow -3 = \dfrac{f}{f-(-20)} \Rightarrow f = -15\,cm$

31. (d) When object is kept at centre of curvature. It's real image is also formed at centre of curvature.

32. (c) $u = -20\,cm$, $f = +10\,cm$ also $\dfrac{1}{f} = \dfrac{1}{v} + \dfrac{1}{u}$

$\Rightarrow \dfrac{1}{+10} = \dfrac{1}{v} + \dfrac{1}{(-20)} \Rightarrow v = \dfrac{20}{3}\ cm$; virtual image.

33. (a) Mirror formula

$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{f}=\frac{1}{-20}+\frac{1}{(-10)}\Rightarrow f=\frac{20}{3}\ cm$. If object moves towards the mirror by $0.1\,cm$ then.

$u=(10-0.1)=9.9\,cm$. Hence again from mirror formula $\frac{1}{-20/3}=\frac{1}{v'}+\frac{1}{-9.9}\Rightarrow v'=20.4\ cm$ i.e. image shifts away from the mirror by $0.4\,cm$.

34. (d) Image formed by convex mirror is always. Erect diminished and virtual.

35. (d) $f=\frac{R}{2}\Rightarrow R=40\,cm$

36. (b) $f=-15\,cm,\ \ m=+2$ (Positive because image is virtual)

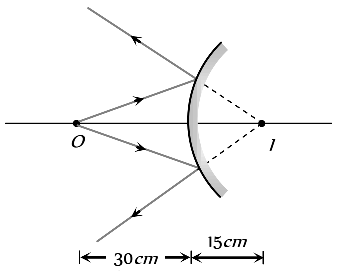
$\because m=-\frac{v}{u}\Rightarrow v=-2u$. By using mirror formula

$\frac{1}{-15}=\frac{1}{(-2u)}+\frac{1}{u}\Rightarrow u=-7.5\,cm$

37. (d) $u=-30\,cm,\ \ f=+30\,cm,$ by using mirror formula

$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{+30}=\frac{1}{v}+\frac{1}{(-30)}$

$v=15\,cm,$ behind the mirror



38. (d) $R=-30\,cm\Rightarrow f=-15\,cm$

$O=+2.5\,cm,\ \ u=-10\,cm$

By mirror formula $\frac{1}{-15}=\frac{1}{v}+\frac{1}{(-10)}\Rightarrow v=30\,cm.$

Also $\frac{I}{O}=-\frac{v}{u}\Rightarrow\frac{I}{(+2.5)}=-\frac{30}{(-10)}\Rightarrow I=+7.5\,cm.$

39. (d)

40. (a) $\frac{I}{O}=\frac{f}{f-u}\Rightarrow\frac{I}{+6}=\frac{-f}{-f-(-4f)}\Rightarrow I=-2\,cm.$

41. (d) Convergence (or power) is independent of medium for mirror.

42. (d) $\frac{I}{O}=\frac{f}{f-u}\Rightarrow\frac{I}{2}=\frac{20}{20+20}=\frac{1}{2}\Rightarrow\ I=1\,mm$

43. (a) $m=\pm 3$ and $f=-6\ cm$

Now $m=\frac{f}{f-u}\ \Rightarrow\pm 3=\frac{-6}{-6-u}$

For real image $-3=\frac{-6}{-6-u}\ \Rightarrow u=-8\ cm$

For virtual image $3=\frac{-6}{-6-u}\ \Rightarrow u=-4\ cm$

44. (a) Focal length of the mirror remains unchanged.

Refraction of Light at Plane Surfaces

1. (d)

2. (a) $\mu_{blue}>\mu_{red}$

3. (b) $\mu\propto\frac{1}{\lambda},\lambda_r>\lambda_v$

4. (a) $\lambda_{medium}=\frac{\lambda_{air}}{\mu}=\frac{6000}{1.5}=4000\ \mathring{A}$

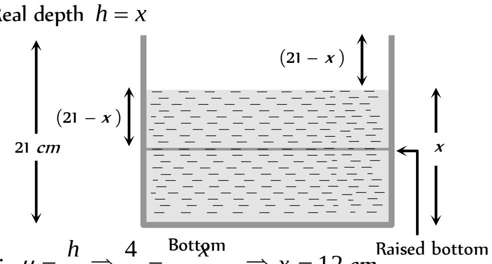
5. (d) Velocity and wavelength change but frequency remains same.

6. (a) $\mu=\frac{c}{v}=\frac{c}{v\lambda}=\frac{3\times10^{8}}{4\times10^{14}\times5\times10^{-7}}=1.5$

7. (c) To see the container half-filled from top, water should be filled up to height x so that bottom of the container should appear to be raised upto height (21–x).

As shown in figure apparent depth $h'=(21-x)$

Real depth $h=x$



$\therefore\ \mu=\frac{h}{h'}\Rightarrow\frac{4}{3}=\frac{x}{21-x}\Rightarrow x=12\ cm$

8. (d) In vacuum, the speed of light is independent of wave length. Thus vacuum (or air) is a non dispersive medium in which all colours travel with the same speed.

9. (c) $\lambda\propto\frac{1}{\mu}\Rightarrow\frac{\lambda_1}{\lambda_2}=\frac{\mu_2}{\mu_1}=\frac{\mu}{1}$

10. (a) $v\propto\frac{1}{\mu},\mu_{\text{rarer}}<\mu_{\text{denser}}$

11. (b) $\mu\propto\frac{1}{\lambda}$

12. (d) $v=\frac{c}{\mu}=\frac{3\times10^{8}}{2}=1.5\times10^{8}\,m/s=1.5\times10^{10}\,cm/s$

13. (b) $\because\ \angle i>\angle r,$ it means light ray is going from rarer medium (A) to denser medium.

So $v(A)>v(B)$ and $n(A)<n(B)$

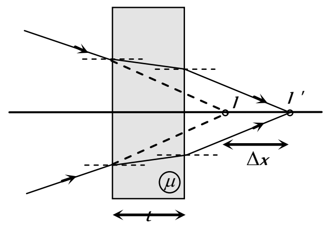
14. (a) $\mu=\frac{h}{h'}\Rightarrow h'=\frac{8}{4/3}=6\ m$

15. (b) $h'=\frac{d_1}{\mu_1}+\frac{d_2}{\mu_2}=d\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)$

16. (a) Normal

shift $\Delta x=\left(1-\frac{1}{\mu}\right)t$

and shift takes place in direction of ray.



17. (c)

$\text{time}=\frac{\text{distance}}{\text{speed}}=\frac{t}{c/x}=\frac{nt}{c}$

18. (c) Let ν' and λ' represents frequency and wavelength of light in medium respectively.

so $\nu' = \frac{v}{\lambda'} = \frac{c/\mu}{\lambda/\mu} = \frac{c}{\lambda} = \nu$

19. (d) $\mu = \frac{c_a}{c_w} = \frac{t_w}{t_a} \Rightarrow t_w = \frac{25}{3} \times \frac{4}{9} = 11\frac{1}{9} = 11\ min\ 6\ sec$

20. (a) Optical path = μt

In medium (1), optical path = $\mu_1 d_1$

In medium (2), optical path = $\mu_2 d_2$

$\therefore$ Total path = $\mu_1 d_1 + \mu_2 d_2$

21. (b) Refractive index of liquid C is same as that of glass piece. So it will not be visible in liquid C.

22. (b) ${}_a\mu_g = \frac{3}{2}, {}_a\mu_w = \frac{4}{3}$

$\therefore\ {}_w\mu_g = \frac{{}_a\mu_g}{{}_a\mu_w} = \frac{3/2}{4/3} = \frac{9}{8}$

23. (c) ${}_2\mu_1 \times {}_3\mu_2 \times {}_4\mu_3 = \frac{\mu_1}{\mu_2} \times \frac{\mu_2}{\mu_3} \times \frac{\mu_3}{\mu_4} = \frac{\mu_1}{\mu_4} = {}_4\mu_1 = \frac{1}{{}_1\mu_4}$

24. (a) Colour of light is determined by its frequency and as frequency does not change, colour will also not change and will remains green.

25. (c) Ray optics fails if the size of the object is of the order of the wavelength.

26. (a) ${}_an_w \times {}_wn_{gl} \times {}_{gl}n_{gas} \times {}_{gas}n_a = \frac{n_w}{n_a} \times \frac{n_{gl}}{n_w} \times \frac{n_{gas}}{n_{gl}} \times \frac{n_a}{n_{gas}} = 1$

27. (b) $v \propto \lambda \Rightarrow \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$

$\therefore\ v_2 = \frac{v_1}{\lambda_1} \times \lambda_2 = 3 \times 10^8 \times \frac{4500}{6000} = 2.25 \times 10^8\ m/s$

28. (d) Since ${}_a\mu_g = \sqrt{2}$, so ${}_g\mu_a = \frac{\sin i}{\sin r} = \frac{1}{\sqrt{2}}$

$\therefore\ \sin r = 1 \Rightarrow r = 90^o$

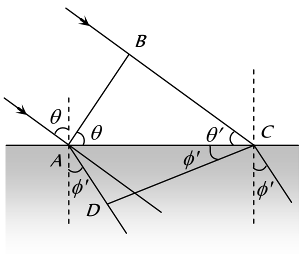
29. (a) $\mu = \frac{c}{v} = \frac{1/\sqrt{\mu_o \varepsilon_o}}{1/\sqrt{\mu\varepsilon}} = \sqrt{\frac{\mu\varepsilon}{\mu_o \varepsilon_o}}$

30. (c) $\mu \propto \frac{1}{\lambda} \Rightarrow \frac{1}{4/3} = \frac{x}{4200} \Rightarrow x = 3150\ Å$

31. (c) $\mu = \sqrt{\frac{\mu\varepsilon}{\mu_0 \varepsilon_0}} = \sqrt{\mu_r K}$

32. (c) $\mu = \frac{C}{C_m} \Rightarrow C_m = \frac{C}{1.5}$

33. (b) In the case of refraction if CD is the refracted wave front and v and v are the speed of light in the two media, then in the time the wavelets from B reaches C, the wavelet from A will reach D, such that



$t = \frac{BC}{v_a} = \frac{AD}{v_g} \Rightarrow \frac{BC}{AD} = \frac{v_a}{v_g}$

But in ΔACB, $BC = AC \sin\theta$ .....(ii)

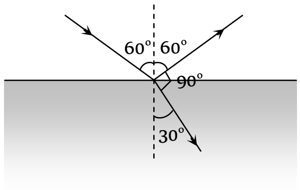
while in ΔACD, $AD = AC \sin\phi'$ .....(iii)

From equations (i), (ii) and (iii) $\frac{v_a}{v_g} = \frac{\sin\theta}{\sin\phi'}$

Also $\mu \propto \frac{1}{v} \Rightarrow \frac{v_a}{v_g} = \frac{\mu_g}{\mu_a} = \frac{\sin\theta}{\sin\phi'} \Rightarrow \mu_g = \frac{\sin\theta}{\sin\phi'}$

34. (b)

35. (b) From figure



$< i = 60^o, < r = 30^o$

so $\mu = \frac{\sin 60}{\sin 30} = \sqrt{3}$

36. (b) $\mu \propto \frac{1}{v} \Rightarrow \frac{\mu_g}{\mu_w} = \frac{v_w}{v_g} \Rightarrow \frac{3/2}{4/3} = \frac{v_w}{2 \times 10^8}$

$\Rightarrow v_w = 2.25 \times 10^8\ m/s$

37. (a) $\lambda_m = \frac{\lambda_a}{\mu} = \frac{c}{\nu\mu} = \frac{3 \times 10^8}{5 \times 10^{14} \times 1.5} = 4000\ Å$

38. (b) $\lambda_{glass} = \frac{\lambda_{air}}{\mu} = \frac{7200}{1.5} = 4800\ Å$

39. (c)

40. (d) $\frac{{}_a\mu_r}{{}_w\mu_r} = \frac{\mu_r/\mu_a}{\mu_r/\mu_w} = \frac{\mu_w}{\mu_a} = {}_a\mu_w$

41. (a) $t = \frac{\mu x}{c} = \frac{\frac{3}{2} \times 5 \times 10^{-3}}{3 \times 10^8} = 0.25 \times 10^{-10}\ s$

42. (d) Distance $= v \times t = \frac{c}{\mu} \times t = \frac{3 \times 10^8}{1.5} \times 10^{-9}$

$= 0.2\ m = 20\ cm.$

43. (c) $f \propto \frac{1}{\lambda}$. As $\lambda_b < \lambda_g \Rightarrow f_b > f_g$

44. (c) Real depth = $1\ m$

Apparent depth = $1 - 0.1 = 0.9\ m$

Refractive index $\mu = \frac{\text{Real depth}}{\text{Apparent depth}} = \frac{1}{0.9} = \frac{10}{9}$

45. (a) $\mu = \frac{h}{h'} \Rightarrow h' = \frac{h}{n}$

46. (a) Refractive index $\propto \frac{1}{(\text{Temperatu re})}$

47. (c) Snell's law in vector form is $\hat{i} \times \hat{n} = \mu(\hat{r} \times \hat{n})$

48. (a)

49. (c) $v = \frac{c}{\mu} = \frac{3 \times 10^8}{2.4} = 1.25 \times 10^8\ m/s$

50. (c) Velocity of light in the window

$= \frac{3 \times 10^8}{1.5} ms^{-1} = 2 \times 10^8\ ms^{-1}$

Hence $t = \dfrac{4\times10^{-3}}{2\times10^{8}} s = 2\times10^{-11} s$

51. (d) Ray optics is valid when size of the objects is much larger than the order of wavelength of light.

52. (b) $v = \dfrac{c}{\mu} = \dfrac{3\times10^{8}}{1.33} = 2.25\times10^{8}$ *m/s*

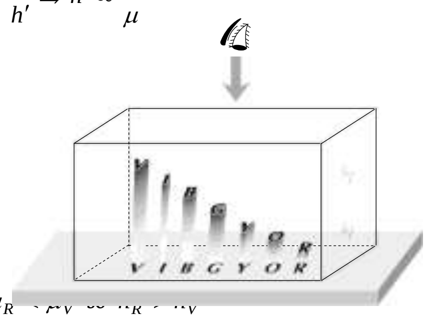
53. (b) $t = \dfrac{\mu x}{c} = \dfrac{1.5\times2\times10^{-3}}{3\times10^{8}} = 10^{-11}$ *sec*

54. (b) ${}_{g}\mu_{w} = \dfrac{\mu_{w}}{\mu_{g}} = \dfrac{4/3}{3/2} = \dfrac{8}{9}$

55. (b) Frequency does not change with medium but wavelength and velocity decrease with the increase in refractive index.

56. (a) $t = \dfrac{\mu x}{c} = \dfrac{3\times4\times10^{-3}}{3\times10^{8}} = 4\times10^{-11}$ *sec*

57. (d) $\mu = \dfrac{h}{h'} \Rightarrow h' \propto \dfrac{1}{\mu}$



$\because \mu_R < \mu_V$ so $h_R > h_V$

i.e. Red colour letter appears least raised.

58. (b) $\mu = \dfrac{c}{v} = \dfrac{\sin i}{\sin r} = \dfrac{\sin 45^{o}}{\sin 30^{o}}$

$\Rightarrow \ v = \dfrac{3\times10^{8}}{\sqrt{2}} = 2.12\times10^{8}\ m/s$

59. (c) $v \propto \dfrac{1}{\mu} \Rightarrow \dfrac{v_1}{v_2} = \dfrac{\mu_2}{\mu_1} \Rightarrow \dfrac{v_g}{v_w} = \dfrac{\mu_w}{\mu_g} = \dfrac{4/3}{3/2} = \dfrac{8}{9}$

60. (b) Time taken by light to travel distance x through a medium of refractive index μ is

$t = \dfrac{\mu x}{c} \Rightarrow \dfrac{\mu_B}{\mu_A} = \dfrac{x_A}{x_B} = \dfrac{6}{4} \Rightarrow {}_{A}\mu_{B} = \dfrac{3}{2} = 1.5$

61. (d) ${}_{w}\mu_{g} = \dfrac{{}_{a}\mu_{g}}{{}_{a}\mu_{w}} = \dfrac{1.5}{1.3}$

62. (a) $\mu = \dfrac{\text{Real depth}}{\text{apparent depth}} = \dfrac{120}{80} = 1.5$

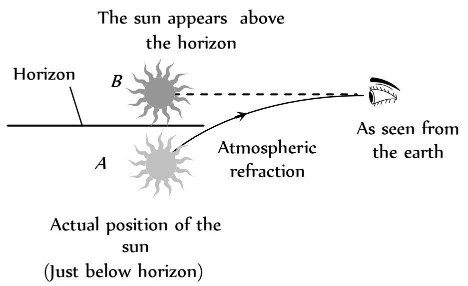
63. (b) Apparent depth of bottom

$= \dfrac{H/4}{\mu_1} + \dfrac{H/4}{\mu_2} + \dfrac{H/4}{\mu_3} + \dfrac{H/4}{\mu_4}$

$= \dfrac{H}{4}\left(\dfrac{1}{\mu_1} + \dfrac{1}{\mu_2} + \dfrac{1}{\mu_3} + \dfrac{1}{\mu_4}\right)$

64. (d) For successive refraction through different media $\mu \sin\theta =$ constant. Here as θ is same in the two extreme media, $\mu_1 = \mu_4$.

65. (b)



66. (a) $\mu = \dfrac{h'}{h} \Rightarrow h' = \mu h = \dfrac{4}{3}\times18 = 24\ cm$

67. (b) Optical path $\mu x =$ constant *i.e.* $\mu_1 x_1 = \mu_2 x_2$

$\Rightarrow\ 1.53\times4 = \mu_2\times4.5 \Rightarrow \mu_2 = 1.36$

68. (b) Velocity of light is maximum in vacuum.

69. (a) $\mu = \tan i \Rightarrow i = \tan^{-1}\mu = \tan^{-1}1.62 = 58.3°$

70. (d) Suppose water is poured up to the height h,

So $h\left(1 - \dfrac{1}{\mu}\right) = 1 \ \Rightarrow h = 4\ cm$

71. (c) $\mu \propto \dfrac{1}{v} \Rightarrow \dfrac{\mu_l}{\mu_g} = \dfrac{v_g}{v_l} \Rightarrow \dfrac{\mu_l}{1.5} = \dfrac{2\times10^{8}}{2.5\times10^{8}} \Rightarrow \mu_l = 1.2$

72. (c) Stars twinkle due to variation in *R.I.* of atmosphere.

73. (d) Refraction at air-oil point $\mu_{oil} = \dfrac{\sin i}{\sin r_1}$

$\therefore\ \sin r_1 = \dfrac{\sin 40}{1.45} = 0.443$

Refraction at oil-water point ${}_{oil}\mu_{water} = \dfrac{\sin r_1}{\sin r}$

$\therefore\ \dfrac{1.33}{1.45} = \dfrac{0.443}{\sin r}$ or $\sin r = \dfrac{0.443\times1.45}{1.33} \Rightarrow r = 28.9^{o}$

74. (d) Objects are invisible in liquid of R.I. equal to that of object.

75. (b) When light ray travels from denser to rarer, it deviates away from the normal.

76. (d) $\mu = \dfrac{c}{v} = \dfrac{3\times10^{8}}{1.5\times10^{8}} = 2.$

77. (c) Frequency remain unchanged.

78. (c) ${}_{w}\mu_{g} = \dfrac{{}_{a}\mu_{g}}{{}_{a}\mu_{w}} = \dfrac{1.5}{1.2} = \dfrac{5}{4} = 1.25$.

79. (b) $\lambda_g = \dfrac{\lambda_a}{\mu_g} = \dfrac{5890}{1.6} = 3681\,Å$.

80. (b) $t = \dfrac{s}{v} = \dfrac{1.5\times10^{8}\times10^{3}}{3\times10^{8}} = 500\ \text{sec} = 8.33$ *min.*

81. (a) For vacuum $t = n\lambda_o$ .....(i)

For air $t = (n+1)\lambda_a$ .....(ii)

From equation (i) and (ii)

$t = \dfrac{\lambda}{\mu - 1} = \dfrac{6\times10^{-7}}{1.0003 - 1}\ \left(\mu = \dfrac{\lambda_o}{\lambda_a}\right)$

$= 2\times10^{-3} m = 2$*mm.*

82. (a) $\mu_m = \dfrac{c}{v} = \dfrac{n\lambda_a}{n\lambda_m} = \dfrac{\lambda_a}{\lambda_m}$

83. (b) As no scattering of light occurs. Space appears black.

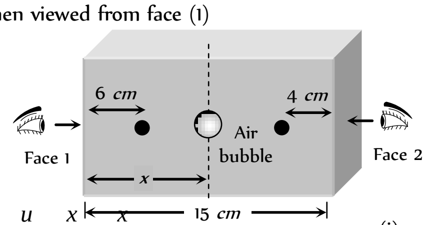
84. (b) $v \propto \dfrac{1}{\mu}$, μ is smaller for air than water, glass and diamond.

85. (c) In vacuum speed of light is constant and it is equal to 3×10^{8} *m/sec*

86. (b) $\lambda_{medium} = \dfrac{\lambda_{vacuum}}{\mu}$

87. (d) In vacuum speed of light is constant and is equal to $3\times10^8 m/s$.

88. (d) When viewed from face (1)



$\mu = \dfrac{u}{v} = \dfrac{x}{v} = \dfrac{x}{6}$ .....(i)

Now when viewed from face (2)

$\mu = \dfrac{15-x}{v} = \dfrac{15-x}{4}$ .....(ii)

From equation (i) and (ii) $\mu = \dfrac{15-6\mu}{4} \Rightarrow \mu = 1.5$.

89. (b) The apparent depth of ink mark

$= \dfrac{\text{real depth}}{\mu} = \dfrac{3}{3/2} = 2\,cm$

Thus person views mark at a distance $= 2+2 = 4\,cm$.

90. (d) Apparent rise $= d\left(1 - \dfrac{1}{{}_a\mu_w}\right) = 12\times\left(1-\dfrac{3}{4}\right) = 3$ *cm.*

Total Internal Reflection

1. (b) Due to high refractive index its critical angle is very small so that most of the light incident on the diamond is total internally reflected repeatedly and diamond sparkles.

2. (c) When incident angle is greater than critical angle, then total internal reflection takes place and will come back in same medium.

3. (d)

4. (d) ${}_a\mu_g = \dfrac{1}{\sin C} \Rightarrow \sin C = \dfrac{1}{{}_a\mu_g}$

As μ for violet colour is maximum, so sin C is minimum and hence critical angle C is minimum for voilet colour.

5. (c) The critical angle C is given by

$\sin C = \dfrac{n_2}{n_1} = \dfrac{\lambda_1}{\lambda_2} = \dfrac{3500}{7000} = \dfrac{1}{2} \Rightarrow C = 30^o$

6. (c) From figure given in question $\theta = 2c = 98°$.

7. (b) $\mu = \dfrac{1}{\sin C} = \dfrac{1}{\sin 30} = 2$

$\therefore v = \dfrac{3\times10^8}{2} = 1.5\times10^8 m/s$

8. (c) ${}_D\mu_R = \dfrac{\sin i}{\sin r'} \Rightarrow {}_R\mu_D = \dfrac{\sin r'}{\sin i} = \dfrac{1}{\sin C}$

$\Rightarrow \sin C = \dfrac{\sin i}{\sin(90-r)} = \dfrac{\sin i}{\cos r} = \dfrac{\sin i}{\cos i}$ (as $\angle i = \angle r$)

$\Rightarrow \sin C = \tan i \Rightarrow C = \sin^{-1}(\tan i)$

9. (a) For total internal reflection $i > C$

$\Rightarrow \sin i > \sin C \Rightarrow \sin i > \dfrac{1}{\mu} \Rightarrow \dfrac{1}{\sin i} < \mu.$

10. (d) For total internal reflection light must travel from denser medium to rarer medium.

11. (b)

12. (c) Semi vertical angle $= C = \sin^{-1}\left(\dfrac{1}{\mu}\right) = \sin^{-1}\left(\dfrac{3}{4}\right)$

13. (c)

14. (d) $\mu = \dfrac{1}{\sin C} \Rightarrow C = \sin^{-1}\left(\dfrac{1}{2}\right) = 30^o$

15. (d)

16. (c) Critical angle $= \sin^{-1}\left(\dfrac{1}{\mu}\right)$

$\therefore \theta = \sin^{-1}\left(\dfrac{1}{\mu_{\lambda_1}}\right)$ and $\theta' = \sin^{-1}\left(\dfrac{1}{\mu_{\lambda_2}}\right)$

Since $\mu_{\lambda_2} > \mu_{\lambda_1}$, hence $\theta' < \theta$

17. (c)

18. (c, d) For TIR $i > C$

$\Rightarrow \sin i > \sin C \Rightarrow \sin 45° > \dfrac{1}{n} \Rightarrow n > \sqrt{2} \Rightarrow n > 1.4$

19. (c)

20. (d)

21. (a) ${}_w\mu_g = \dfrac{1}{\sin C} \Rightarrow \dfrac{\mu_g}{\mu_w} = \dfrac{5/3}{4/3} = \dfrac{1}{\sin C}$

$\Rightarrow \sin C = \dfrac{4}{5} \Rightarrow C = \sin^{-1}\left(\dfrac{4}{5}\right)$

22. (c) Total internal reflection occurs when light ray travels from denser medium to rarer medium.

23. (b) $\mu = \dfrac{c}{v} \Rightarrow \mu = \dfrac{c}{c/2} = 2$ also for total internal reflection

$i > c \Rightarrow \sin i \ge \sin c \Rightarrow \sin i \ge \dfrac{1}{\mu}$

Hence $i \ge \sin^{-1}\left(\dfrac{1}{\mu}\right)$ or $i \ge 30^o$

24. (c) $C = \sin^{-1}\left(\dfrac{1}{{}_w\mu_g}\right) = \sin^{-1}\left(\dfrac{\mu_w}{\mu_g}\right) = \sin^{-1}\left(\dfrac{8}{9}\right)$

25. (a) $\mu_w < \mu_g \Rightarrow c_w > c_g$.

26. (c) $\mu = \dfrac{1}{\sin C} = \dfrac{1}{\sin 30} = 2$

27. (c) Ray from setting sum will be refracted at angle equal to critical angle.

28. (a) Optical fibres are used to send signals from one place to another.

29. (d)

30. (d) When total internal reflection just takes place from lateral surface $i = C$ i.e. $60^o = C$

$\Rightarrow \sin 60^o = \sin C = \dfrac{1}{\mu} \Rightarrow \mu = \dfrac{2}{\sqrt{3}}$

Time taken by light to traverse some distance in a medium

$t = \dfrac{\mu x}{c} = \dfrac{\dfrac{2}{\sqrt{3}}\times10^3}{3\times10^8} = 3.85\,\mu$ *sec.*

31. (a) $\dfrac{\mu_2}{\mu_1} = \dfrac{v_1}{v_2} = \dfrac{1}{2} \Rightarrow \dfrac{\mu_1}{\mu_2} = 2(\mu_1 > \mu_2)$

For total internal reflection ${}_2\mu_1 = \frac{1}{\sin C} \Rightarrow \frac{\mu_1}{\mu_2}$

$= \frac{1}{\sin C} \Rightarrow 2 = \frac{1}{\sin C} \Rightarrow C = 30^o$

So, for total (Internal reflection angle of incidence must be greater than 30°.

32. (c)

33. (a) $\mu = \frac{1}{\sin C} = \frac{1}{\sin 60^o} = \frac{2}{\sqrt{3}}$

34. (c) ${}_a\mu_g = \frac{1}{\sin\theta} \Rightarrow \mu = \frac{1}{\sin\theta}$ .....(i)

Now from Snell's law $\mu = \frac{\sin i}{\sin r} = \frac{\sin\theta}{\sin r}$

$\Rightarrow \sin r = \frac{\sin\theta}{\mu}$ .....(ii)

From equation (i) and (ii)

$\sin r = \frac{1}{\mu^2} \Rightarrow r = \sin^{-1}\left(\frac{1}{\mu^2}\right)$

35. (a) $C = \sin^{-1}\left(\frac{1}{\mu}\right)$ and $\mu \propto \frac{1}{\lambda}$

Yellow, orange and red have higher wavelength than green, so μ will be less for these rays, consequently critical angle for these rays will be high, hence if green is just totally internally reflected then yellow, orange and red rays will emerge out.

36. (d) We know $C = \sin^{-1}\left(\frac{1}{\mu}\right)$

Given critical angle $i_B > i_A$

So $\mu_B < \mu_A$ *i.e.* *B* is rarer and *A* is denser.

Hence light can be totally internally reflected when it passes from *A* to *B*

Now critical angle for *A* to *B*

$C_{AB} = \sin^{-1}\left(\frac{1}{{}_B\mu_A}\right) = \sin^{-1}\left[{}_A\mu_B\right]$

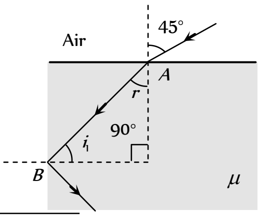
$= \sin^{-1}\left[\frac{\mu_B}{\mu_A}\right] = \sin^{-1}\left[\frac{\sin i_A}{\sin i_B}\right]$

37. (b) At point *A*, by Snell's law

$\mu = \frac{\sin 45}{\sin r} \Rightarrow \sin r = \frac{1}{\mu\sqrt{2}}$ .....(i)

At point *B*, for total internal reflection $\sin i_1 = \frac{1}{\mu}$

From figure, $i_1 = 90 - r$



$\therefore \sin(90° - r) = \frac{1}{\mu}$

$\Rightarrow \cos r = \frac{1}{\mu}$ .....(ii)

Now $\cos r = \sqrt{1 - \sin^2 r} = \sqrt{1 - \frac{1}{2\mu^2}}$

$= \sqrt{\frac{2\mu^2 - 1}{2\mu^2}}$ .....(iii)

From equation (ii) and (iii) $\frac{1}{\mu} = \sqrt{\frac{2\mu^2 - 1}{2\mu^2}}$

Squaring both side and then solving we get $\mu = \sqrt{\frac{3}{2}}$

38. (b) ${}_2\mu_1 = \frac{1}{\sin\theta} \Rightarrow \frac{\mu_1}{\mu_2} = \frac{1}{\sin\theta} \Rightarrow \frac{v_2}{v_1} = \frac{1}{\sin\theta} \Rightarrow \frac{v_2}{v} = \frac{1}{\sin\theta}$

$\Rightarrow v_2 = \frac{v}{\sin\theta}$

39. (c) From the formula $\sin C = \frac{1}{{}_1\mu_2} \Rightarrow \sin C = {}_2\mu_1$

$= \frac{u_1}{u_2} = \frac{v_2}{v_1} \Rightarrow \sin C = \frac{10x / t_2}{x / t_1}$

$\Rightarrow \sin C = \frac{10\,t_1}{t_2} \Rightarrow C = \sin^{-1}\left(\frac{10\,t_1}{t_2}\right)$

40. (a) $\sin 45^o = \frac{1}{\mu} \Rightarrow \mu = \sqrt{2} = 1.41$

41. (c)

42. (b) Critical angle *C* is equal to incident angle if ray reflected normally $\therefore C = 90^o$

43. (b)

44. (d) $r = \frac{3h}{\sqrt{7}} = \frac{3 \times 12}{\sqrt{7}} = \frac{36}{\sqrt{7}}$.

45. (b) Here $\sin i = \frac{1}{\mu} = \frac{3}{5}$ and hence $\tan i = \frac{3}{4} = \frac{r}{4}$

This gives $r = 3\,m$, hence diameter = $6m$

46. (a) Radius of horizon circle $= \frac{3h}{\sqrt{7}} = \frac{3\sqrt{7}}{\sqrt{7}} = 3\,cm$.

Refraction at Curved Surface

1. (a) By formula $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$= (1.5 - 1)\left(\frac{1}{40} + \frac{1}{40}\right) = 0.5 \times \frac{1}{20} = \frac{1}{40}$

$\therefore\ f = 40\ cm$

2. (a) $\frac{v}{-u} = -m$ and $v + u = x \Rightarrow u = \frac{x}{1+m}$

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow f = \frac{mx}{(m+1)^2}$.

3. (d) $I \propto A^2 \Rightarrow \frac{I_2}{I_1} = \left(\frac{A_2}{A_1}\right)^2 = \frac{\pi r^2 - \frac{\pi r^2}{4}}{\pi r^2} = \frac{3}{4}$

$\Rightarrow I_2 = \frac{3}{4} I_1$ and focal length remains unchanged.

4. (c) $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{P_1}{100}+\frac{P_2}{100}=\frac{1}{100}\Rightarrow f=100\ cm$

$\therefore$ A convergent lens of focal length 100 *cm*.

5. (a) Focal length of the combination can be calculated as

$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}\Rightarrow\frac{1}{F}=\frac{1}{(+40)}+\frac{1}{(-25)}\Rightarrow F=-\frac{200}{3}cm$

$\therefore P=\frac{100}{F}=\frac{100}{-200/3}=-1.5\ D$

6. (d) $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}\Rightarrow\frac{1}{80}=\frac{1}{20}+\frac{1}{f_2}\Rightarrow f_2=-\frac{80}{3}cm$

$\therefore$ Power of second lens

$P_2=\frac{100}{f_2}=\frac{100}{-80/3}=-3.75\ D$

7. (b) In each case two plane-convex lens are placed close to each other, and $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}$.

8. (a) Power of the combination $P=P_1+P_2=12-2=10\,D$

$\therefore$ Focal length of the combination

$F=\frac{100}{P}=\frac{100}{10}=10\,cm$

9. (c) Resultant focal length $=\infty$

$\therefore$ It behaves as a plane slab of glass.

10. (c) $f=\frac{R}{(\mu-1)}\Rightarrow 30=\frac{10}{(\mu-1)}\Rightarrow\mu=1.33$.

11. (c) In case of convex lens if rays are coming from the focus, then the emergent rays after refraction are parallel to principal axis.

12. (d) Because to form the complete image only two rays are to be passed through the lens and moreover, since the total amount of light released by the object is not passing through the lens, therefore image is faint (intensity is decreased).

13. (b) $f=\frac{f_1f_2}{f_1+f_2}=\frac{10(-10)}{10+(-10)}=\frac{-100}{10-10}=\infty$

14. (c) Focal length of the combination

$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{1}{(+84)}+\frac{1}{(-12)}\Rightarrow F=-14\ cm$

$\therefore P=\frac{100}{F}=\frac{100}{-14}=-\frac{50}{7}D$

15. (b) $O=\sqrt{I_1I_2}=\sqrt{4\times16}=8\,cm$

16. (d) $\frac{f_l}{f_a}=\frac{({}_a\mu_g-1)}{({}_l\mu_g-1)}\Rightarrow\frac{f_w}{f_a}=\frac{(1.5-1)}{\left(\frac{1.5}{1.33}-1\right)}\Rightarrow f_w=32\,cm$

17. (c) If $n_l>n_g$ then the lens will be in more denser medium. Hence its nature will change and the convex lens will behave like a concave lens.

18. (d) $\frac{f_l}{f_a}=\frac{({}_a\mu_g-1)}{({}_l\mu_g-1)}\Rightarrow\frac{f_l}{15}=\frac{(1.5-1)}{\left(\frac{1.5}{4/3}-1\right)}\Rightarrow f_l=60\,cm$

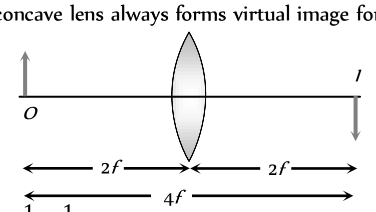
19. (c) $\frac{f_l}{f_a}=\frac{({}_a\mu_g-1)}{({}_l\mu_g-1)}\Rightarrow f_l=\infty$ if ${}_l\mu_g=1\Rightarrow{}_a\mu_l={}_a\mu_g$.

20. (c) $\frac{I_1}{O}=\frac{v}{u}$ and $\frac{I_2}{O}=\frac{u}{v}\Rightarrow O^2=I_1I_2$

21. (c) A lens shows opposite behaviour if $\mu_{\text{medium}}>\mu_{\text{lens}}$

22. (a) A concave lens always forms virtual image for real objects.

23. (d)



24. (a) $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$ (Given $u=-20\ cm,\ f=10\ cm,\ v=?$)

$\therefore\ \frac{1}{10}=\frac{1}{v}-\frac{1}{(-20)}\Rightarrow v=20\,cm$

25. (d) $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{1}{60}+\frac{1}{(-20)}\Rightarrow F=-30$

26. (a) $f_{\text{water}}=4\times f_{\text{air}}$, air lens is made up of glass.

27. (b) $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{1}{20}+\frac{1}{25}\Rightarrow F=\frac{100}{9}\ cm=\frac{1}{9}\ metre$

$\therefore\ P=\frac{1}{1/9}D=9D$

28. (a) $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$ (Given $u=\frac{-f}{2}$)

$\Rightarrow\frac{1}{f}=\frac{1}{v}+\left(\frac{1}{f/2}\right)\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{2}{f}$

$\Rightarrow\frac{1}{v}=\frac{-1}{f}$ and $m=\frac{v}{u}=\frac{f}{f/2}=2$

So virtual at the focus and of double size.

29. (a) $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

Given $R_1=+20cm, R_2=-20\,cm$, $\mu=1.5$

$\Rightarrow f=20\ cm$. Parallel rays converge at focus. So *L=f.*

30. (c) $\mu_{air}<\mu_{lens}<\mu_{water}$ *i.e.,* $1<\mu_{lens}<1.33$

31. (c) $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

For biconvex lens $R_2=-R_1$ $\quad\therefore\ \frac{1}{f}=(\mu-1)\left(\frac{2}{R}\right)$

Given $R=\infty\ \therefore\ f=\infty$, so no focus at real distance.

32. (d) $f=\frac{R}{(\mu-1)}=\frac{15}{(1.6-1)}=25\,cm$

$\therefore P=\frac{100}{f}=\frac{100}{25}=+4\ D$

33. (d) $f\propto\frac{1}{(\mu-1)}$ and $\mu\propto\frac{1}{\lambda}$. Hence $f\propto\lambda$ and $\lambda_r>\lambda_v$

34. (c) $m_1=\frac{A_1}{O}$ and $m_2=\frac{A_2}{O}\Rightarrow m_1m_2=\frac{A_1A_2}{O_2}$

Also it can be proved that $m_1m_2=1$

So $O = \sqrt{A_1 A_2}$

35. (b) Combined power $P = P_1 + P_2 = 6 - 2 = 4D$. So focal length of combination $F = \frac{1}{P} = \frac{1}{4}m$

36. (b) $\frac{1}{60} = \frac{1}{f_1} + \frac{1}{f_2}$ ...(i)

and $\frac{1}{30} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{10}{f_1 f_2}$ ...(ii)

On solving (i) and (ii) $f_1 f_2 = -600$ and $f_1 + f_2 = -10$

Hence $f_1 = 20\ cm$ and $f_2 = -30\ cm$

37. (c) For an achromatic combination $\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$

i.e. 1 convex lens and 1 concave lens.

38. (d) $\frac{1}{F} = \frac{2}{f_l} + \frac{1}{f_m} \Rightarrow \frac{1}{F} = \frac{2}{20} + \frac{1}{\infty} \Rightarrow F = 10\ cm$

39. (b) Since aperture of lens reduces so brightness will reduce but their will be no effect on size of image.

40. (d) Convex mirror and concave lens do not form real image. For concave mirror $v > u$, so image will be enlarged, hence only convex lens can be used for the purpose.

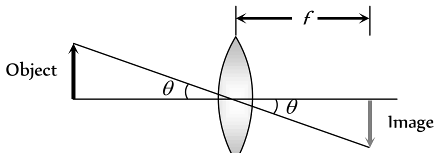
41. (a) $m = \frac{f}{f+u} \Rightarrow -\frac{1}{4} = \frac{30}{30+u} \Rightarrow u = -150\,cm$

42. (c) Covering a portion of lens does not effect position and size of image.

43. (a) $\frac{1}{f} = ({}_g\mu_a - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \left(\frac{2}{3} - 1\right)\left(\frac{2}{10}\right)$

$\Rightarrow f = -15\,cm$, so behaves as concave lens.

44. (c) Size of image = $f\theta = 0.5 \times (1 \times 10^{-3})$ = 0.5 *mm*



45. (d) $\frac{f_l}{f_a} = \frac{({}_a\mu_g - 1)}{({}_l\mu_g - 1)} = \frac{\left(\frac{3}{2} - 1\right)}{\left(\frac{3/2}{5/4} - 1\right)} = \frac{5}{2}$

$\therefore\ f_l = f_a\left(\frac{5}{2}\right) = \frac{12 \times 5}{2} = 30\ cm$

46. (d) $P = \frac{1}{F} = \frac{f_1 + f_2}{f_1 f_2}$

47. (c) $f = \frac{R}{(\mu - 1)} = \frac{R}{(1.5 - 1)} = 2R$

48. (b) For achromatic combination, $\frac{w_1}{f_1} + \frac{w_2}{f_2} = 0$

$\Rightarrow w_1 f_2 + w_2 f_1 = 0$

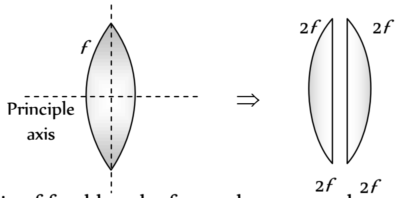
49. (a) $\frac{\omega_1}{\omega_2} = -\frac{f_1}{f_2} \Rightarrow \frac{5}{3} = \frac{-(-15)}{f_2} \Rightarrow f_2 = 9\ cm$

50. (b) $f = \frac{R}{2(\mu - 1)} \Rightarrow f = \frac{40}{2(1.65 - 1)} \approx 31\,cm$

51. (c) Focal length of effective lens

$\frac{1}{F} = \frac{2}{f_l} + \frac{1}{f_m} = \frac{2}{f_l} + \frac{1}{\infty} \Rightarrow F = \frac{f_l}{2}$

52. (a)



Ratio of focal length of new plano convex lenses is 1 : 1

53. (a) $\frac{1}{f} = \left(\frac{n-1}{1}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ and $\frac{1}{f'} = \left(\frac{n - n'}{n'}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\therefore\ \frac{f'}{f} = \frac{n-1}{1} \times \frac{n'}{n - n'} \Rightarrow f' = -\frac{fn'(n-1)}{n' - n}$

54. (b) $\frac{I}{O} = \frac{f - v}{f} \Rightarrow \frac{I}{+1.5} = \frac{(25 - 75)}{25} = -2 \Rightarrow I = -3\,cm$

55. (a) $P = P_1 + P_2$, if $P_1 = P_2 = P' \Rightarrow P' = P/2 = 2D.$

56. (b) $f = \frac{R}{(\mu - 1)} = \frac{60}{(1.6 - 1)} = 100\ cm.$

57. (a) $\frac{f_l}{f_a} = \frac{{}_a\mu_g - 1}{{}_l\mu_g - 1} = \frac{1.5 - 1}{\frac{1.5}{1.75} - 1} = -\frac{1.75 \times 0.50}{0.25} = -3.5$

$\therefore f_l = -3.5\ f_a \Rightarrow f_l = +3.5\ R$ $\quad(\because f_a = R)$

Hence on immersing the lens in the liquid, it behaves as a converging lens of focal length 3.5 *R*.

58. (a) $P = P_1 + P_2 = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{(0.5)} + \frac{1}{(-1)} = 1\,D$

59. (d) $f = \frac{R}{2(\mu - 1)} \Rightarrow 30 = \frac{R}{2(1.5 - 1)} \Rightarrow R = 30\ cm$

60. (c) Total power $P = P_1 + P_2 = 11 - 6 = 5D$

Also $\frac{f_l}{f_a} = \frac{({}_a\mu_g - 1)}{({}_l\mu_g - 1)} \Rightarrow \frac{P_a}{P_l} = \frac{({}_a\mu_g - 1)}{({}_l\mu_g - 1)}$

$\Rightarrow \frac{5}{P_l} = \frac{(1.5 - 1)}{(1.5/1.6 - 1)} \Rightarrow P_l = -0.625\,D$

61. (b) For first case : $\frac{1}{f} = \frac{1}{v} - \frac{1}{\infty} \Rightarrow f = v$

For second case $\frac{1}{f} = \frac{1}{(f+5)} - \frac{1}{-(f+20)} \Rightarrow f = 10\ cm$

Alternative sol. – $f^2 = x_1 x_2 \Rightarrow f = 10cm.$

62. (b) $f = \frac{D^2 - x^2}{4D}$ (Focal length by displacement method)

$\Rightarrow f = \frac{(100)^2 - (40)^2}{4 \times 40} = 21\ cm$

$\therefore P = \frac{100}{f} = \frac{100}{21} \approx 5\,D$

63. (d) $\frac{1}{f}=\frac{1}{v}-\frac{1}{u} \Rightarrow \frac{1}{+5}=\frac{1}{v}-\frac{1}{(-10)} \Rightarrow v=10\ cm$

64. (d) $\omega / f=-2\omega / f' \Rightarrow f'=-2f$

65. (d) $f=\frac{R}{2(\mu-1)} \Rightarrow 10=\frac{R}{2(1.6-1)} \Rightarrow R=12\ cm$

66. (a)

67. (d) $P=P_1+P_2=2.50-3.75=-1.25D$

So $f=\frac{100}{1.25}=-80\ cm$

68. (c) $\frac{f_l}{f_a}=\frac{{}_a\mu_g-1}{{}_l\mu_g-1} \Rightarrow f_l=4R$

69. (c) $\frac{f_l}{f_a}=\frac{{}_a\mu_g-1}{{}_l\mu_g-1}=\frac{{}_a\mu_g-1}{\frac{{}_a\mu_g}{{}_a\mu_l}-1} \Rightarrow \frac{f_l}{2}=\frac{1.5-1}{\frac{1.5}{1.25}-1} \Rightarrow f_l=5\ cm$

70. (b) $f \propto \frac{1}{\mu-1}$ and $\mu \propto \frac{1}{\lambda}$

71. (d) $P=\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{1}{(+0.8)}+\frac{1}{(-0.5)}=-0.75\ D$

72. (b) According to lens makers formula

$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right) \Rightarrow \frac{1}{f} \propto (\mu-1)$

Since $\mu_{\text{Red}}<\mu_{\text{violet}} \Rightarrow f_v<f_r$ and $F_v<F_r$

Always keep in mind that whenever you are asked to compare (greater than or less than) *u, v* or *f* you must not apply sign conventions for comparison.

73. (a) Since light transmitting area is same, there is no effect on intensity.

74. (c) $m=\frac{f}{(f+u)} \Rightarrow -\frac{1}{n}=\frac{f}{(f+u)} \Rightarrow u=-(n+1)f$

75. (a) $P=P_1+P_2=2D-4D=-2D.$

76. (c)

77. (a) $\frac{1}{F}=\frac{2}{f}+\frac{1}{f_m}$. Here $f_m=\infty$, hence $F=\frac{f}{2}=10\,cm$

78. (b) $O=\sqrt{I_1I_2} \Rightarrow O=\sqrt{4\times 9}=6\ cm$

79. (b) $P=P_1+P_2 \Rightarrow P=+6-4=+2\,D$. So focal length

$f=\frac{100}{2}=+50\ cm$, convex lens

80. (d) $f=\frac{R}{2(\mu-1)} \Rightarrow P=\frac{2(\mu-1)}{R}=\frac{2(1.5-1)}{0.2}=+5\,D$

81. (c) $P=\frac{1}{f} \Rightarrow f=\frac{1}{0.5}=2m$

82. (a) $\frac{f_l}{f_a}=\left(\frac{{}_a\mu_g-1}{{}_l\mu_g-1}\right) \Rightarrow \frac{f_l}{4}=\frac{(1.4-1)}{\frac{1.4}{1.6}-1} \Rightarrow f_l=-12.8\,cm$

83. (d) $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2} \Rightarrow \frac{1}{F}=\frac{1}{(+18)} \Rightarrow F=18\,cm$

84. (a) $\frac{f_l}{f_a}=\frac{({}_a\mu_g-1)}{({}_l\mu_g-1)}$; $f_a=\frac{R}{2(\mu_g-1)}=\frac{15}{2(1.6-1)}=12.5$

$\Rightarrow \frac{f_l}{12.5}=\frac{(1.6-1)}{\left(\frac{1.6}{1.63}-1\right)} \Rightarrow f_l=-407.5\ cm$

85. (c) $P=P_1+P_2 \Rightarrow P=+2+(-1)=+1D,$

$f=\frac{+100}{P}=\frac{+100}{1}=100\,cm$

86. (c)

87. (b) Nature of lens changes, if $\mu_{\text{mediume}}>\mu_{\text{lens}}$

88. (a) $u=-25\ cm, v=+75\,cm$

$\Rightarrow \frac{1}{f}=\frac{1}{+75}-\frac{1}{-25} \Rightarrow f=+18.75\ cm$; convex lens.

89. (a) $F=\frac{f_1f_2}{f_2-f_1}$, *F* will be negative if $f_1>f_2$

90. (b) $f=\frac{R}{2(\mu-1)}=\frac{10}{2(1.5-1)}=10\,cm$

91. (b) $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

$\Rightarrow \frac{1}{+10}=(1.5-1)\left(\frac{1}{+7.5}-\frac{1}{R_2}\right) \Rightarrow R_2=-15\,cm$

92. (d) $f=\frac{R}{2(\mu-1)}$, $f'=\frac{R}{(\mu-1)} \Rightarrow f'=2f$

93. (c) $m=\pm 3$, using $m=\frac{f}{f+u}$

For virtual image $3=\frac{f}{f-8}$ .....(i)

For real image $-3=\frac{f}{f-16}$ .....(ii)

Solving (i) and (ii) we get $f=12\,cm$

94. (a) $\frac{1}{F}=\frac{1}{+18}+\frac{1}{(-9)} \Rightarrow F=-18\ cm$ (*i.e.* concave lens)

95. (c) $O=\sqrt{I_1I_2}=\sqrt{8\times 2}=4cm$

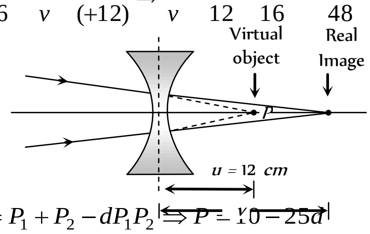
96. (c) $P=\frac{100}{f_1}+\frac{100}{f_2}=\frac{100}{(+25)}+\frac{100}{(-10)}=-6D$

97. (c)

98. (a) $f_w=4\times f_a=4\times 12=48\ cm.$

99. (d) By using lens formula

$\frac{1}{-16}=\frac{1}{v}-\frac{1}{(+12)} \Rightarrow \frac{1}{v}=\frac{1}{12}-\frac{1}{16}=\frac{4-3}{48} \Rightarrow v=48\ cm$



100. (a) $P=P_1+P_2-dP_1P_2 \Rightarrow P=10-25d$

$\Rightarrow$ For P to be negative $25d>10$

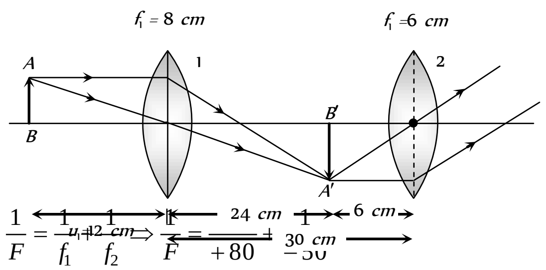
$\Rightarrow d>0.4\,m$ or $d>40\ cm$

101. (a) $m=\frac{f}{f+u} \Rightarrow -m=\frac{f}{f+u} \Rightarrow u=-\left(\frac{m+1}{m}\right)f$

102. (d) Number of images = (Number of materials)

103. (c) For lens (1) $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}\Rightarrow\frac{1}{(+8)}=\frac{1}{v}-\frac{1}{(-12)}$

$\Rightarrow v=24\ cm$ *i.e.* Image $A'B'$ is obtained 6 *cm* before the lens 2 or at the focus of lens 2. Hence final image formed by lens 2 will be real enlarged and it is obtained at ∞.



104. (d) $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}\Rightarrow\frac{1}{F}=\frac{1}{+80}+\frac{1}{-50}$

$\Rightarrow\ F=-\frac{400}{3}\Rightarrow P=\frac{-3}{4}D$

105. (a) By using formula $\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$

$\Rightarrow\frac{1.5}{v}-\frac{1}{(-15)}=\frac{(1.5-1)}{+30}\Rightarrow v=-30\,cm$.

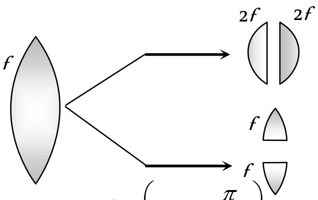
Negative sign shows that, image is obtained on the same side of object *i.e.* towards left.

106. (c) By using $\frac{f_l}{f_a}=\frac{({}_a\mu_g-1)}{({}_l\mu_g-1)}\Rightarrow f_w=4f_a=4\times 30$ =120 *cm.*

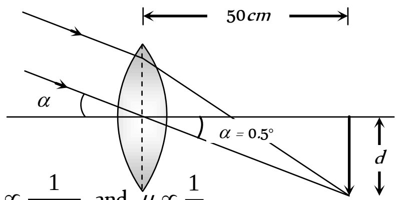
107. (b)

108. (a)

109. (d)



110. (b) Diameter of image $d=\left(0.5\times\frac{\pi}{180}\right)\times 500$ = 4.36 *mm*



111. (c) $f\propto\frac{1}{\mu-1}$ and $\mu\propto\frac{1}{\lambda}$.

112. (c) Since intensity $\propto$ (Aperature)², so intensity of image will decrease but no change in the size occurs.

113. (c) In liquids converging ability (power) of convex lens decreases.

114. (d) Since $f\propto\frac{1}{\mu}\propto\lambda$, so voilet colour is focused nearer to the lens.

115. (a) Focal length for voilet is minimum.

116. (c) $m=\frac{v}{u}=5\Rightarrow v=5$ *inch* (Given u = 1 *inch*)

Using sign convention $u=-1\ inch,\ v=-5\ inch$

$\therefore\frac{1}{f}=\frac{1}{v}-\frac{1}{u}=\frac{1}{-5}-\frac{1}{-1}\Rightarrow f=1.25\ inch$

117. (a) $m_L=4$

$m_A=(m_1)^2$ so that $A'=A_0\times 16=1600\ cm^2$

118. (d) $u=-10\ cm,\ v=20\ cm$

$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}=\frac{1}{20}-\left(-\frac{1}{10}\right)=\frac{3}{20}\Rightarrow f=\frac{20}{3}\ cm$

Now $P=\frac{100}{f}=\frac{100}{20/3}=+15\ D$

119. (c) $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}$

120. (b) $f=\frac{R}{2(\mu-1)}\Rightarrow R=2f(\mu-1)=2\times 0.2(1.5-1)$ =0.2*m.*

121. (c) Using refraction formula $\frac{{}_1\mu_2-1}{R}=\frac{{}_1\mu_2}{v}-\frac{1}{u}$

in given case, medium (1) is glass and (2) is air

So $\frac{{}_g\mu_a-1}{R}=\frac{{}_g\mu_a}{v}-\frac{1}{u}\ \Rightarrow\frac{\frac{1}{1.5}-1}{-6}=\frac{1}{1.5v}-\frac{1}{-6}$

$\Rightarrow\frac{1-1.5}{-6}=\frac{1}{v}+\frac{1.5}{6}\Rightarrow\frac{0.5}{6}=\frac{1}{v}+\frac{1}{4}$

$\Rightarrow\frac{1}{v}=\frac{1}{12}-\frac{1}{4}=-\frac{2}{12}=-\frac{1}{6}\Rightarrow$ v = 6 *cm.*

122. (d) For real image $m=-2$

$\therefore m=\frac{f}{u+f}\Rightarrow -2=\frac{f}{u+f}=\frac{20}{u+20}\Rightarrow$ u = – 30 *cm.*

123. (a) Focal length of the system (concave mirror)

$F=\frac{R}{2\mu}=\frac{30}{2\times 1.5}$ =10 *cm*

In order to have a real image of the same size of the object, object must be placed at centre of curvature u = (2*f*).

124. (b) $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

= $(1.5-1)\left(\frac{1}{10}+\frac{1}{10}\right)=\frac{1}{10}\ \Rightarrow f$= 10 *cm*

$\therefore$ Radius of curvature of concave mirror = 2*f* = 20 *cm.*

125. (d) $m=-\frac{1}{2}$

$\therefore m=\frac{f}{u+f}\Rightarrow -\frac{1}{2}=\frac{30}{u+30}\Rightarrow u=-90\,cm$

126. (c) $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

$\frac{1}{f}=(1.6-1)\left(\frac{1}{60}-\frac{1}{\infty}\right)=\frac{1}{100}\Rightarrow f=100\,cm$

127. (d) $\frac{1}{F}=(1.5-1)\left(\frac{1}{20}-\frac{1}{\infty}\right)\Rightarrow F=40\,cm$.

128. (b) For minimum spherical and chromatic aberration distance between lenses.

$d = f_1 - f_2 = 0.3 - 0.1 = 0.2\,m$.

129. (b) $\dfrac{f_l}{f_a} = \dfrac{{}_a\mu_g - 1}{{}_l\mu_g - 1} = \dfrac{(1.5-1)\times 1.7}{(1.5-1.7)}$

$\Rightarrow f_l = \dfrac{0.85}{-0.2} f_a = -4.25 f_a$.

130. (c)

131. (b) $\omega = \dfrac{f_R - f_V}{f_y} = \dfrac{f_R - f_V}{\sqrt{f_V f_R}}$

Putting value of f_V and f_R we get $\omega = 0.0325$.

132. (b) $P_1 + P_2 = 2D$ and $P_1 = 5D$, so $P_2 = -3D$

For achromatic combination

$\dfrac{\omega_1}{\omega_2} = \left(-\dfrac{p_2}{p_1}\right) = -\left(\dfrac{-3}{5}\right) = \dfrac{3}{5}$

133. (b) $f \propto \dfrac{1}{\mu - 1}$ and $\mu \propto \dfrac{1}{\lambda}$

134. (d) $P = P_1 + P_2 = +12 - 2 = 10D$

Now $F = \dfrac{1}{P} = \dfrac{1}{10}m = 10\,cm$.

135. (b) Focal length for voilet colour is minimum

136. (d) $\dfrac{f_1}{f_2} = \dfrac{2}{3}$ .....(i)

$\dfrac{1}{f_1} - \dfrac{1}{f_2} = \dfrac{1}{30}$ .....(ii)

Solving equation (i) and (ii)

$f_2 = -15\,cm$ (Concave)

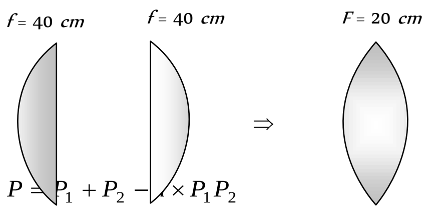
$f_1 = 10\,cm$ (Convex)

137. (d) $\dfrac{f_l}{f_a} = \dfrac{({}_a\mu_g - 1)}{({}_l\mu_g - 1)}$

$\Rightarrow \dfrac{f_l}{f_a} = \dfrac{{}_a\mu_g - 1}{{}_l\mu_g - 1} = \dfrac{1.5 - 1}{\dfrac{1.5}{1.6} - 1} = \dfrac{0.5 \times 1.6}{-0.1} = -8$

$\Rightarrow P_l = \dfrac{P_a}{8} = \dfrac{5}{8}$

138. (b) To obtain, an inverted and equal size image, object must be paced at a distance of $2f$ from lens, *i.e.* 40 *cm* in this case.



139. (a) Using $P = P_1 + P_2 - d \times P_1 P_2$

for equivalent power to be negative

$d \times P_1 P_2 > P_1 + P_2 \Rightarrow d \times 25 > 10$

$\Rightarrow d > \dfrac{10}{25} m \Rightarrow d > \dfrac{10 \times 100}{25} \Rightarrow d > 40cm.$

140. (c) Combination of lenses will act as a simple glass plate.

141. (b) For achromatic combination $\dfrac{f_1}{f_2} = -\dfrac{\omega_2}{\omega_1} = -\dfrac{0.036}{0.024} = -\dfrac{3}{2}$

and $\dfrac{1}{f_1} - \dfrac{1}{f_2} = \dfrac{1}{90}$

solving above equations be get $f_1 = 30cm,\ f_2 = -45cm$

142. (b)

143. (c) $f \propto \dfrac{1}{\mu - 1}$ and $\mu \propto \dfrac{1}{\lambda}$.

144. (b) $\dfrac{f_l}{f_a} = \dfrac{{}_a\mu_g - 1}{{}_l\mu_g - 1} \Rightarrow \dfrac{-0.5}{0.2} = \dfrac{1.5 - 1}{{}_l\mu_g - 1} \Rightarrow {}_l\mu_g - 1 = -0.2$

$\Rightarrow {}_l\mu_g = 0.8 = \dfrac{4}{5} \Rightarrow \dfrac{{}_a\mu_g}{{}_a\mu_l} = \dfrac{4}{5} \Rightarrow \dfrac{1.5}{{}_a\mu_l} = \dfrac{4}{5}$

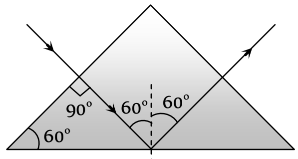
$\Rightarrow {}_a\mu_l = \dfrac{15}{8}$.

145. (c) Longitudinal chromatic aberration

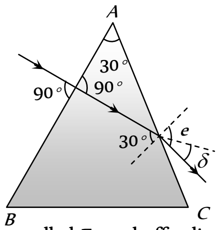
$= \omega f = 0.08 \times 20 = 1.6cm$.

Prism Theory & Dispersion of Light

1. (b) Neon street sign emits light of specific wavelengths.
2. (b)
3. (b)



4. (c) $\delta \propto (\mu - 1) \Rightarrow \mu_R$ is least so δ_R is least.
5. (c)
6. (a) For surface AC $\dfrac{1}{\mu} = \dfrac{\sin 30°}{\sin e} \Rightarrow \sin e = \mu \sin 30°$



$\Rightarrow \sin e = 1.5 \times \dfrac{1}{2} = 0.75$

$\Rightarrow e = \sin^{-1}(0.75) = 48^o 36'$

From figure $\delta = e - 30^o$

$= 48^o\ 36' - 30^o = 18^o 36'$

7. (a) The black lines in solar spectrum are called Fraunhoffer lines.

8. (d) $\dfrac{\sin\dfrac{A + \delta m}{2}}{\sin\dfrac{A}{2}} = \mu$, But $\dfrac{A + \delta_m}{2} = i = 45^o$

So $\dfrac{\sin 45^o}{\sin(A/2)} = \sqrt{2} \Rightarrow \dfrac{1}{2} = \sin\dfrac{A}{2} \Rightarrow A = 60^o$

9. (d) We know that $\dfrac{\delta_v - \delta_r}{\delta_{mean}} = \omega$

$\Rightarrow$ Angular dispersion $= \delta_v - \delta_r = \theta = \omega\delta_{mean}$

10. (d) According to Kirchhoff's law, a substance in unexcited state will absorb these wavelength which it emits in de-excitation.

11. (c) By prism formula $n = \dfrac{\sin\dfrac{A + A}{2}}{\sin\dfrac{A}{2}} = \dfrac{2\sin\dfrac{A}{2}\cos\dfrac{A}{2}}{\sin\dfrac{A}{2}}$

$\therefore \cos\frac{A}{2} = \frac{n}{2} = \frac{1.5}{2} = 0.75 = \cos 41^o \Rightarrow A = 82^o$

12. (b)

13. (b) ω depend only on nature of material.

14. (a) Because achromatic combination has same μ for all wavelengths.

15. (a) $\because \mu = a + \frac{b}{\lambda^2}$ (Cauchy's equation)

and dispersion $D = -\frac{d\mu}{d\lambda} \Rightarrow D = -(-2\lambda^{-3})b = \frac{2b}{\lambda^3}$

$\Rightarrow D \propto \frac{1}{\lambda^3} \Rightarrow \frac{D'}{D} = \left(\frac{\lambda}{\lambda'}\right)^3 = \left(\frac{\lambda}{2\lambda}\right) = \frac{1}{8} \Rightarrow D' = \frac{D}{8}$

16. (b) $\mu = \frac{\sin i}{\sin A/2} \Rightarrow \sqrt{2} = \frac{\sin i}{\sin\left(\frac{60}{2}\right)}$

$\Rightarrow \sqrt{2} \times \sin 30 = \sin i \Rightarrow i = 45^o$

17. (d) $\frac{\delta_w}{\delta_a} = \frac{({}_w\mu_g - 1)}{({}_a\mu_g - 1)} = \frac{\left(\frac{9}{8} - 1\right)}{\left(\frac{3}{2} - 1\right)} = \frac{1}{4}$

18. (a) Since $A(\mu_y - 1) + A'(\mu_{y'} - 1) = 0 \Rightarrow \frac{A'}{A} = -\left(\frac{\mu_y - 1}{\mu_{y'} - 1}\right)$

19. (d)

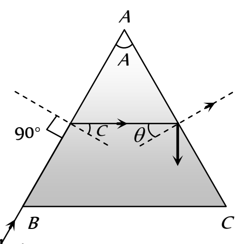
20. (b)

21. (a)

22. (c) From ray diagram

$A = C + \theta$ for TIR at AC

$\theta > C$ so $A > 2C$



23. (a) By the hypothesis, we know that

$i_1 + i_2 = A + \delta \Rightarrow 55^o + 46^o = 60^o + \delta \Rightarrow \delta = 41^o$

But $\delta_m < \delta$, so $\delta_m < 41^o$

24. (a)

25. (b) $\delta_m = (\mu - 1)A$. $A =$ angle of prism.

26. (c)

27. (c)

28. (b)

29. (a) Total deviation = 0

$\delta_1 + \delta_2 + \delta_3 + \delta_4 + \delta_5 = (\mu_1 - 1)A_1 - (\mu_2 - 1)A_2$

$+(\mu_3 - 1)A_3 - (\mu_4 - 1)A_4 + (\mu_5 - 1)A_5 = 0$

$\Rightarrow 2 \times A_2(1.6 - 1) = 3(1.53 - 1)9$

$\Rightarrow A_2 = 3\left(\frac{0.53 \times 9}{1.2}\right) = 11.9^o$

30. (a) The dispersive power for crown glass $\omega = \frac{n_v - n_r}{n_y - 1}$

$= \frac{1.5318 - 1.5140}{(1.5170 - 1)} = \frac{0.0178}{0.5170} = 0.034$

and for flint glass $\omega' = \frac{1.6852 - 1.6434}{(1.6499 - 1)} = 0.064$

31. (c)

32. (b)

33. (a) For dispersion without deviation $\frac{A}{A'} = \left(\frac{\mu'_y - 1}{\mu_y - 1}\right)$

$\therefore \frac{A}{10} = \frac{(1.602 - 1)}{(1.500 - 1)} = \frac{0.602}{0.500} \Rightarrow A = 12°2.4'$

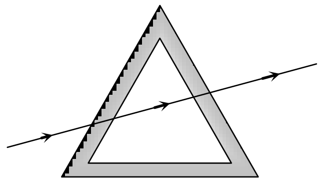
34. (c) $i = \frac{A + \delta_m}{2}$ =50°

35. (d) In minimum deviation position $\angle i = \angle e$

36. (a) $\underset{\text{(Primary)}}{\text{Yellow}} + \underset{\text{(Primary)}}{\text{Blue}} = \underset{\text{(Secondary)}}{\text{Green}}$

37. (b) All colours are reflected.

38. (a) Effectively there is no deviation or dispersion.

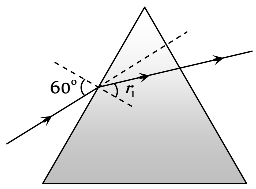


39. (d) From figure it is clear that $\angle e = \angle r_2 = 0$

From $A = r_1 + r_2$

$\Rightarrow r_1 = A = 45^o$

$\therefore \mu = \frac{\sin i}{\sin r_1} = \frac{\sin 60}{\sin 45} = \sqrt{\frac{3}{2}}$



Also from $i + e = A + \delta \Rightarrow 60 + 0 = 45 + \delta \Rightarrow \delta = 15^o$

40. (b) Deviation is zero only for a particular colour, it is generally taken to be yellow.

41. (b) $5 = (\mu - 1)A = (1.5 - 1)A \Rightarrow A = 10^o$

42. (b) $\delta = (\mu_v - \mu_r)A = 0.02 \times 10 = 0.2$

43. (a) $\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin(A/2)} = \frac{\sin 45^o}{\sin 30^o} = \sqrt{2}$

44. (c) $\omega = \frac{\mu_V - \mu_R}{\mu_Y - 1} = \frac{1.65 - 1.61}{1.63 - 1}$

45. (a) For minimum angle of deviation for a prism

$A = 2r$, $\therefore A = 60^o$

Now $\mu = \frac{\sin\frac{60 + 30}{2}}{\sin\frac{60}{2}} = \frac{\sin 45^o}{\sin 30^o} = \frac{1}{\sqrt{2}} \times \frac{2}{1} = \sqrt{2}$

46. (c) In minimum deviation condition $\angle i = \angle e$, $\angle r_1 = \angle r_2$

47. (b) For dispersion without deviation $\frac{A}{A'} = \frac{(\mu' - 1)}{(\mu - 1)}$

$$\frac{4}{A_F} = \frac{(1.72-1)}{(1.54-1)} = \frac{0.72}{0.54} \text{ or } A_F = \frac{4\times 0.54}{0.72} = 3^o$$

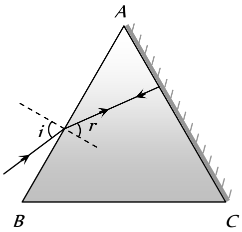
48. (a) $A(\mu_v - \mu_r) + A'(\mu'_v - \mu'_r) = 0^o \Rightarrow A' = 5^o$

49. (c) $A = r + 0 \Rightarrow r = 30^o$

From Snell's law at surface *AB*

$$\mu = \frac{\sin i}{\sin r}$$

$$\Rightarrow \sqrt{2} = \frac{\sin i}{\sin 30^o} \Rightarrow i = 45^o$$



50. (c) $\omega = \dfrac{1.64-1.52}{1.6-1} = \dfrac{0.12}{0.6} = 0.2$

51. (c) Because band spectrum can be found in case of molecules (generally gas).

52. (a) Solids and liquids give continuous and line spectra. Only gases are known to give band spectra.

53. (d)

54. (d)

55. (a) Hydrogen is molecular, therefore it gives band spectrum but not continuous spectrum.

56. (c)

57. (a) Dispersion take place because the refractive index of medium for different colour is different, for example, red light bends less than violet, refractive index of the material of the prism for red light is less than that for violet light. Equivalently, we can say that red light travels faster than violet light in a glass prism.

58. (a) We know that $\delta = i + e - A \Rightarrow e = \delta + A - i$

$= 30^o + 30^o - 60^o = 0^o$

$\therefore$ Emergent ray will be perpendicular to the face.

Therefore it will make an angle of 90° with the face through which it emerges.

59. (a) $\delta_m = (\mu - 1)(2r) = (1.5-1)2r = 0.5\times 2r = r$

60. (c)

61. (c)

62. (b)

63. (d) Given $i = e = \dfrac{3}{4}A = \dfrac{3}{4}\times 60 = 45^o$

In the position of minimum deviation

$2i = A + \delta_m$ or $\delta_m = 2i - A = 90 - 60 = 30^o$

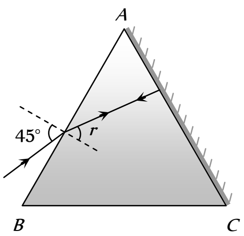
64. (d)

65. (a) Sky appears white due to scattering. In absence of atmosphere no scattering will occur.

66. (b)

67. (c) $A = r + 0 \Rightarrow r = 30°$

$$\therefore \mu = \frac{\sin i}{\sin r} = \frac{\sin 45°}{\sin 30°} = \sqrt{2}$$



68. (c) By formula $\delta = (n-1)A \Rightarrow 34 = (n-1)A$ and in the second position $\delta' = (n-1)\dfrac{A}{2}$

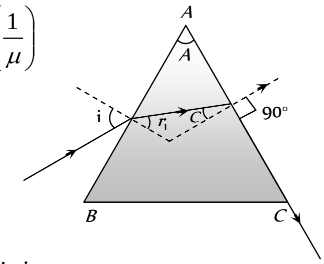
$$\therefore \frac{34}{\delta'} = \frac{(n-1)A}{(n-1)\frac{A}{2}} \text{ or } \delta' = \frac{34}{2} = 17^o$$

69. (b) From figure

$$A = r_1 + c = r_1 + \sin^{-1}\left(\frac{1}{\mu}\right)$$

$$\Rightarrow r_1 = 75 - \sin^{-1}\left(\frac{1}{\mu}\right)$$

$\Rightarrow 75 - 45 = 30^o$



From Snell's law At *B*

$$\mu = \frac{\sin i}{\sin r_1} \Rightarrow \sqrt{2} = \frac{\sin i}{\sin 30^o} \Rightarrow i = 45^o$$

70. (c) In both *A* and *B*, the refracted ray is parallel to the base of prism.

71. (a) According to given conditions TIR must take place at both the surfaces *AB* and *AC*. Hence only option (*a*) is correct.

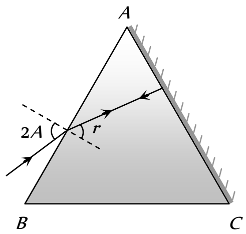
72. (d)

73. (a)

74. (b) $A = r + 0$ and $\mu = \dfrac{\sin i}{\sin r}$

$$\Rightarrow \mu = \frac{\sin 2A}{\sin A}$$

$$= \frac{2\sin A\cos A}{\sin A} = 2\cos A$$



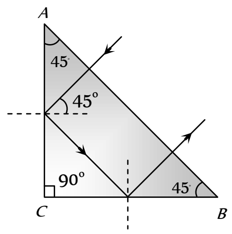
75. (a) From figure it is clear that TIR takes place at surface *AC* and *BC*

i.e. $45^o > C$

$\Rightarrow \sin 45^o > \sin C$

$$\Rightarrow \frac{1}{\sqrt{2}} > \frac{1}{\mu} \Rightarrow \mu > \sqrt{2}$$

Hence $\mu_{\text{least}} = \sqrt{2}$



76. (b)

77. (b) According to Rayleigh's law of scattering, intensity scattered is inversely proportional to the forth power of wavelength. So red is least scattered and sun appears Red.

78. (b)

79. (d)

80. (a) Only red colour will be seen in spectrum.

81. (b) $i = \dfrac{A+\delta_m}{2} = \dfrac{60° + 30°}{2} = 45°$

82. (a) $$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}} = \frac{\sin\left(\frac{60°+60°}{2}\right)}{\sin\left(\frac{60°}{2}\right)} = \sqrt{3}$$

83. (b) Because in dispersion of white light, the rays of different colours are not parallel to each other. Also deviation takes place in same direction.

84. (c)

85. (a) $\omega = \dfrac{\mu_F - \mu_C}{(\mu_D - 1)} = \dfrac{(1.6333 - 1.6161)}{(1.622 - 1)} = 0.0276$

86. (c) For total internal reflection $\theta > C$

$\Rightarrow \sin\theta > \sin C \Rightarrow \sin\theta > \dfrac{1}{\mu}$

or $\mu > \dfrac{1}{\sin\theta} \Rightarrow \mu > \dfrac{1}{\sin 45^o} \Rightarrow \mu > \sqrt{2} \Rightarrow \mu > 1.41$

87. (c)

88. (a) $\theta = (\mu_v - \mu_r)A = 0.02 \times 5^o = 0.1^o$

89. (b)

90. (b) $\dfrac{A'}{A} = \dfrac{(\mu_y - 1)}{(\mu_{y'} - 1)} \Rightarrow \dfrac{A'}{6} = -\dfrac{(1.54 - 1)}{(1.72 - 1)}$

$\Rightarrow A' = -4.5^o = 4^o 30'$

91. (c) $\mu = \dfrac{\sin\left(\dfrac{A + \delta_m}{2}\right)}{\sin\dfrac{A}{2}} \Rightarrow \sqrt{3} = \dfrac{\sin\left(\dfrac{60^\circ + \delta_m}{2}\right)}{\sin\dfrac{60^\circ}{2}}$

$\Rightarrow \dfrac{\sqrt{3}}{2} = \sin\left(30^\circ + \dfrac{\delta_m}{2}\right) \Rightarrow \delta_m = 60^o$

92. (a) Dispersion is caused due to refraction as μ depends on λ.

93. (c) From colour triangle

94. (c) Due to the absorption of certain wavelengths by the elements in outer layers of sun.

95. (b)

96. (c)

97. (c) $\omega = \dfrac{\mu_v - \mu_R}{\mu_y - 1} = \dfrac{1.62 - 1.52}{1.55 - 1} = 0.18$

98. (a) $\dfrac{\omega_1}{\omega_2} = -\dfrac{f_1}{f_2} = -\dfrac{2}{3}$.

99. (a) $\omega = \dfrac{\mu_V - \mu_R}{\mu_Y - 1} = \dfrac{1.62 - 1.42}{1.5 - 1} = 0.4$

100. (c) Since the ray emerges normally, therefore $e = 0$.

According to relation $A + \delta = i + e$, we get $i = A + \delta$.

Hence by $\delta = (\mu - 1)A$, we get $i = \mu A$.

101. (a) The atoms in the chromosphere absorb certain wavelengths of light coming from the photosphere. This gives rise to absorption lines.

102. (b) $\mu = \dfrac{\sin\left(\dfrac{A + \delta_m}{2}\right)}{\sin\left(\dfrac{A}{2}\right)} \Rightarrow \sqrt{2}\mu = \dfrac{\sin\left(\dfrac{60 + \delta_m}{2}\right)}{\sin\left(\dfrac{60}{2}\right)}$

$\Rightarrow \sqrt{2} \times \sin 30 = \sin\left(\dfrac{60 + \delta_m}{2}\right) \Rightarrow \sin 45^o$

$= \sin\left(\dfrac{60 + \delta_m}{2}\right) \Rightarrow \delta_m = 30^o$

103. (a) Intensity of scattered light $I \propto \dfrac{1}{\lambda^4}$, since λ_{blue} is least that's why sky looks blue.

104. (b) In continuos spectrum all wavelength are present.

105. (d)

106. (b) Deviation is greater for lower wavelengths.

107. (b) $\dfrac{\delta_a}{\delta_w} = \dfrac{({}_a\mu_g - 1)}{({}_w\mu_g - 1)} = \dfrac{\left(\dfrac{3}{2} - 1\right)}{\left(\dfrac{3/2}{4/3} - 1\right)} = 4 \Rightarrow \delta_w = \dfrac{\delta_a}{4}$

108. (a) $\theta = (\mu_v - \mu_r)A = (1.66 - 1.64) \times 10^o = 0.2^o$

109. (b) $\omega = \dfrac{(\mu_v - \mu_R)}{(\mu_y - 1)} \Rightarrow \dfrac{(1.69 - 1.65)}{(1.66 - 1)} = 0.06$

110. (a) $\omega = \dfrac{\delta_V - \delta_R}{\delta_Y} = \dfrac{3.72 - 2.84}{3.28} = 0.268$

111. (a)

112. (d) $\mu = \dfrac{\sin\left(\dfrac{A + \delta_m}{2}\right)}{\sin\left(\dfrac{A}{2}\right)} = \dfrac{\sin\left(\dfrac{60^\circ + 30^\circ}{2}\right)}{\sin\dfrac{60^o}{2}} = \dfrac{\sin 45^o}{\sin 30^o} = 1.414$

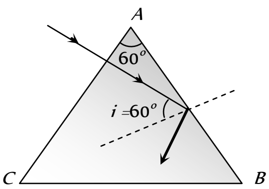
113. (a) Rock salt prism is used to see infrared radiations.

114. (b) For different colours μ changes so deviation of different colour is also different.

115. (a) By using $\dfrac{\omega_1}{f_1} + \dfrac{\omega_2}{f_2} = 0 \Rightarrow \dfrac{0.02}{f_1} + \dfrac{0.04}{40} = 0$

$f_1 = -20\ cm$

116. (d) Critical angle for the material of prism $C = \sin^{-1}\left(\dfrac{1}{\mu}\right) = \sin^{-1} = 42^o$ since angle of incidence at surface $AB\,(60^o)$ is greater then the critical angle (42^o) so total internal reflection takes place.



117. (d) Line and band spectrum are also known as atomic and molecular spectra respectively.

118. (d) In minimum deviation $i = e = 30^o$, so angle between emergent ray and second refracting surface is $90^o - 30^o = 60^o$

119. (c) $\theta = (\mu_v - \mu_R)A = (1.6 - 1.5) \times 5 = 0.5^o$

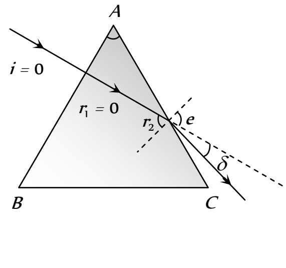
120. (d) $\dfrac{\delta_1}{\delta_2} = \dfrac{A_1}{A_2}$

121. (a) Sunlight consists of all the wavelength with some black lines.

122. (d) $A = 30^o, \mu = \sqrt{2}$. As we know

$A = r_1 + r_2 = 0 + r_2 \Rightarrow A = r_2$.

Applying Snell's law for the surface AC



$\frac{1}{\mu} = \frac{\sin r_2}{\sin e} = \frac{\sin A}{\sin e}$

$\Rightarrow \frac{1}{\sqrt{2}} = \frac{\sin 30^o}{\sin e} \Rightarrow e = 45^o$

$\delta = e - r_2 = 45^o - 30^o = 15°$

123. (c) $\mu = \frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}} = \frac{\sin\frac{A+A}{2}}{\sin\frac{A}{2}} = \frac{\sin A}{\sin\frac{A}{2}}$

$= \frac{2\sin\frac{A}{2}\cos\frac{A}{2}}{\sin\frac{A}{2}} = 2\cos\frac{A}{2}$

So, $\sqrt{3} = 2\cos\frac{A}{2} \Rightarrow \frac{\sqrt{3}}{2} = \cos\frac{A}{2} \Rightarrow A = 60°$

124. (d) Light from lamp or electric heater gives continuos spectrum.

125. (b) $A = 60^o, \delta_m = 30^o$ so $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

$\mu = \frac{\sin\left(\frac{60° + 30°}{2}\right)}{\sin\left(\frac{60°}{2}\right)} = \frac{\sin 45°}{\sin 30°} = \sqrt{2}$

Also $\mu = \frac{1}{\sin C} \Rightarrow C = \sin^{-1}\left(\frac{1}{\mu}\right) \Rightarrow C = 45^o$

126. (a) $\delta \propto (\mu - 1)$

127. (c) In minimum deviation position $\angle i_1 = \angle i_2$ and $\angle r_1 = \angle r_2$.

128. (c) $\theta_{net} = \theta + \theta' = 0 \Rightarrow \omega d + \omega' d' = 0$

(θ = Angular dispersion = $\omega.\ \delta_y$)

129. (d) $A = 60^o, i = e = 45^o$ By $i + e = A + \delta$

$\Rightarrow\ 45 + 45 = 60 + \delta \Rightarrow \delta = 30^o$

130. (a) At the time of solar eclipse light received from chromosphere. The bright lines appear exactly at the places where dark lines were there. Hence at the time of solar eclipse continuos spectrum is obtained.

131. (a) In the morning or evening, the sun is at the horizon and refractive index in the atmosphere of the earth decreases with height. Due to this, the light reaching the earth's atmosphere, bends unequally, and the image of the sun get's distorted and it appears elliptical and larger.

132. (c) In Rainbow formation dispersion and TIR both takes place.

133. (a)

134. (c) Given $\delta_m = A$, as $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

$\Rightarrow \mu = \frac{\sin\left(\frac{A+A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = 2\cos\frac{A}{2} \Rightarrow A = 2\cos^{-1}\left(\frac{\mu}{2}\right)$

135. (b) As the prisms Q and R are of the same material and have identical shape they combine to form a slab with parallel faces. Such a slab does not cause any deviation.

136. (c) Angle of prism is the angle between incident and emergent surfaces.

137. (a) $\mu = \frac{\sin i}{\sin\frac{A}{2}} \Rightarrow \sqrt{2} = \frac{\sin i}{\sin\left(\frac{60}{2}\right)} \Rightarrow i = 45^o$

138. (d) Convex lens, glass slab, prism and glass sphere they all disperse the light.

139. (c) For a lens $f_r - f_v = \omega f_y$

$\Rightarrow \omega = \frac{f_r - f_v}{f_y} = \frac{0.214 - 0.200}{0.205} = \frac{14}{205}$.

140. (b) $\omega = \frac{(\mu_v - \mu_R)}{(\mu_y - 1)} \Rightarrow \frac{(1.69 - 1.65)}{(1.66 - 1)} = 0.06$

141. (a) In minimum deviation condition $r = \frac{A}{2} = \frac{60}{2} = 30^o$

142. (a) $\omega = \frac{\mu_v - \mu_r}{\mu_y - 1} = \frac{1.67 - 1.63}{1.65 - 1} = 0.615$.

143. (b) In minimum deviation position refracted ray inside the prism is parallel to the base of the prism.

144. (b) Angle of refraction will be different, due to which red and green emerge from different points and will be parallel.

145. (a) Deviation $\delta \propto \mu \propto \frac{1}{\lambda}$

146. (a) $\mu = \frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}} = \frac{\sin\frac{60+38}{2}}{\sin\frac{60}{2}}$

$= \frac{\sin 49^o}{\sin 30^o} = \frac{0.7547}{0.5} = 1.5$.

147. (d) Using $\delta = i_1 + i_2 - A \Rightarrow 55 = 15 + i_2 - 60 \Rightarrow i_2 = 100^o$

148. (b) Sodium light gives emission spectrum having two yellow lines.

149. (c) Colour of the sky is highly scattered light (colour).

150. (a)

151. (c)

Human Eye and Lens Camera

1. (c) Man is suffering from hypermetropia. The hole works like a convex lens.

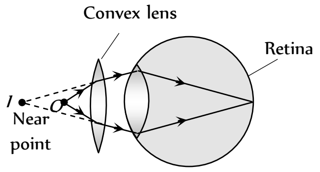
2. (a)

3. (b) In myopia, $u = \infty$, $v = d$ = distance of far point

By $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$, we get $f = -d$

Since f is negative, hence the lens used is concave.

4. (d) Hypermetropia is removed by convex lens.



5. (b)

6. (c) Cylindrical lens are used for removing astigmatism.

7. (b)

8. (a) Image formed at retina is real and inverted.

9. (d) Visible region decreases, so the depth of image will not be seen.

10. (a) $P=\frac{1}{f}=-\frac{1}{v}+\frac{1}{u}=-\frac{1}{100}+\frac{1}{25}=\frac{3}{100}=+3\,D$

11. (c) If eye is kept at a distance d, then $MP=\frac{L(D-d)}{f_0 f_e}$, MP decreases.

12. (c) For lens u = want's to see = ∞

v = can see = $-5\ m$

$\therefore$ From $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}\Rightarrow\frac{1}{f}=\frac{1}{-5}-\frac{1}{\infty}\Rightarrow f=-5\,m.$

13. (a) For improving near point, convex lens is required and for this convex lens

$u=-25\ cm,\ \ v=-75\ cm$

$\therefore\frac{1}{f}=\frac{1}{-75}-\frac{1}{-25}\Rightarrow f=\frac{75}{2}cm$

So power $P=\frac{100}{f}=\frac{100}{75/2}=+\frac{8}{3}D$

14. (b) In short sightedness, the focal length of eye lens decreases, so image is formed before retina.

15. (d) The image of object at infinity should be formed at 100 cm from the eye

$\frac{1}{f}=\frac{1}{\infty}-\frac{1}{100}=-\frac{1}{100}$

So the power $=\frac{-100}{100}=-1\,D$

(Distance is given in *cm* but $P=\frac{1}{f}$ in *metres*)

16. (b) For improving far point, concave lens is required and for this concave lens $u=\infty,\ \ v=-30\ cm$

So $\frac{1}{f}=\frac{1}{-30}-\frac{1}{\infty}\Rightarrow f=-30\ cm$

for near point $\frac{1}{-30}=\frac{1}{-15}-\frac{1}{u}\Rightarrow u=-30\ cm$

17. (c) For myopic eye f = – (defected far point)

$\Rightarrow f=-40\,cm\ \ \Rightarrow\ P=\frac{100}{-40}=-2.5\,D$

18. (c) For lens u = want's to see $=-60\,cm$

v = can see $=-10\,cm$

$\therefore\frac{1}{f}=\frac{1}{v}-\frac{1}{u}\Rightarrow\frac{1}{f}=\frac{1}{-10}-\frac{1}{(-60)}\Rightarrow f=-12\,cm$

19. (b) Focal length = – (Detected far point)

20. (c) In this case, for seeing distant objects the far point is 40 $cm.$ Hence the required focal length is

$f=-d$ (distance of far point) = – 40 cm

Power $P=\frac{100}{f}cm=\frac{100}{-40}=-2.5\ D$

21. (b)

22. (a)

23. (a)

24. (a) For viewing far objects, concave lenses are used and for concave lens

u = wants to see $=-60\,cm$; v = can see $=-15\,cm$

so from $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}\Rightarrow f=-20\,cm$.

25. (d)

26. (a) In short sightedness, the focal length of eye lens decreases and so the power of eye lens increases.

27. (d) Colour blindness is a genetic disease and still cannot be cured.

28. (c) Convexity to lens changes by the pressure applied by ciliary muscles.

29. (b) $f=-d=-100\ cm=-1\ m$

$\therefore P=\frac{1}{f}=\frac{1}{-1}=-1\ D$

30. (c) For correcting myopia, concave lens is used and for lens.

u = wants to see $=-50\,cm$

v = can see $=-25\,cm$

From $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}\Rightarrow\frac{1}{f}=\frac{1}{-25}-\frac{1}{(-50)}\Rightarrow f=-50\,cm$

So power $P=\frac{100}{f}=\frac{100}{-50}=-2\,D$

31. (c)

32. (c) $f=-d=-60\ cm$

$\therefore P=\frac{100}{f}=-\frac{100}{60}=-\frac{10}{6}=-1.66\ D$

33. (b) For correcting the near point, required focal length

$f=\frac{50\times 25}{(50-25)}=50\,cm$

So power $P=\frac{100}{50}=+2\,D$

For correcting the far point, required focal length

$f=-(\text{defected far point})=-3\,m$

$\therefore P=-\frac{1}{3}D=-0.33\,D$

34. (b) Negative power is given, so defect of eye is nearsigntedness

Also defected far point $=-f=-\frac{1}{p}=-\frac{100}{(-2.5)}=40\,cm$

35. (a) In myopia, eye ball may be elongated so, light rays focussed before the retina.

36. (c)

37. (d) $P=\frac{1}{f}=\frac{1}{-(\text{defected far point})}=-\frac{1}{2}=-0.5\,D$

38. (a) Resolving limit of eye is one minute (1').

39. (d) Because for healthy eye image is always formed at retina.

40. (a) The defect is myopia (nearsightness)

As we know for myopic person f = – (defected far point)

$\Rightarrow$ Defected far point = – f = $-\frac{1}{P}=-\frac{1}{(-2)}=0.5\,m$

= 50 cm

41. (b) Power of convex lens $P_1 = \frac{100}{40} = 2.5\ D$

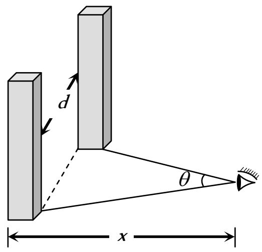
Power of concave lens $P_2 = -\frac{100}{25} = -4\ D$

Now $P = P_1 + P_2 = 2.5\ D - 4\ D = -1.5\ D$

42. (c)

43. (d)

44. (a) As limit of resolution of eye is $\left(\frac{1}{60}\right)^o$, the pillars will be seen distinctly if $\theta > \left(\frac{1}{60}\right)^o$



i.e., $\frac{d}{x} > \left(\frac{1}{60}\right) \times \frac{\pi}{180}$

$\Rightarrow d > \frac{\pi \times x}{60 \times 180}$

$\Rightarrow d > \frac{3.14 \times 11 \times 10^3}{60 \times 180} \Rightarrow d > 3.2\ m$

45. (b)

46. (b)

47. (d)

48. (d) f = – (defected far point) = – 20 cm

49. (b) Power of the lens given positive so defect is hypermetropia.

50. (b) Far point of the eye = focal length of the lens

$= \frac{100}{P} = \frac{100}{0.66} = 151\ cm$

51. (c) A bifocal lens consist of both convex or concave lenses with lower part is convex.

52. (a) For lens u = wants to see = – 30 cm

and v = can see = – 10 cm

$\therefore \frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{-10} - \frac{1}{(-30)} \Rightarrow f = -15\,cm$

53. (a) Focal length = – (far point)

54. (c) For lens u = wants to see = – 12 cm

v = can see = – 3 m

$\therefore P = \frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow P = \frac{1}{-3} - \frac{1}{(-12)} = -\frac{1}{4} D$

55. (d) $I_1 D_1^2 t_1 = I_2 D_2^2 t_2$

Here D is constant and $I = \frac{L}{r^2}$

So $\frac{L_1}{r_1^2} \times t_1 = \frac{L_2}{r_2^2} \times t_2 \Rightarrow \frac{60}{(2)^2} \times 10 = \frac{120}{(4)^2} \times t \Rightarrow 20\ sec$

56. (a) $f = -40\ cm$ and $P = \frac{100}{-40} = -2.5\ D$

57. (a) Focal length of the lens $f = \frac{100}{3} cm$

By lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\Rightarrow \frac{1}{+100/3} = \frac{1}{v} - \frac{1}{-25} \Rightarrow v = -100\ cm = -1\ m$

58. (d) This is the defect of hypermetropia.

59. (a) For large objects, large image is formed on retina.

60. (d) $v = -15cm,\ u = -300cm,$

From lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\Rightarrow \frac{1}{f} = \frac{1}{-15} - \frac{1}{-300} = \frac{-19}{300} \Rightarrow f = \frac{-300}{19} = -15.8\,cm$

and power $P = \frac{100}{f} cm = \frac{-100 \times 19}{300} = -6.33\ D.$

61. (d) Time of exposure $\propto \frac{1}{(\text{Aperture})^2}$

62. (a) Light gathering power $\propto$ Area of lens aperture or d

63. (b) Time of exposure $\propto (f.\text{number})^2 \Rightarrow \frac{t_2}{t_1} = \left(\frac{5.6}{2.8}\right)^2 = 4$

$t_2 = 4\,t_1 = 4 \times \frac{1}{200} = \frac{1}{50}\ sec = 0.02\ sec.$

64. (d)

65. (a)

Microscope and Telescope

1. (c) By using $m_\infty = \frac{(L_\infty - f_o - f_e).D}{f_o f_e}$

$\Rightarrow 45 = \frac{(L_\infty - 1 - 5) \times 25}{1 \times 5} \Rightarrow L_\infty = 15\,cm.$

2. (b) For a compound microscope $m \propto \frac{1}{f_o f_e}$

3. (b) For a compound microscope $f_{objective} < f_{eye\ piece}$

4. (b) In microscope final image formed is enlarged which in turn increases the visual angle.

5. (b)

6. (d) Magnification of a compound microscope is given by

$m = -\frac{v_o}{u_o} \times \frac{D}{u_e} \Rightarrow |m| = m_o \times m_e.$

7. (c) Magnifying power of a microscope $m \propto \frac{1}{f}$

Since $f_{\text{violet}} < f_{\text{red}}$; $\therefore m_{\text{violet}} > m_{\text{red}}$

8. (a) $L_\infty = v_o + f_e \Rightarrow 14 = v_o + 5 \Rightarrow v_o = 9\ cm$

Magnifying power of microscope for relaxed eye

$m = \frac{v_o}{u_o} . \frac{D}{f_e}$ or $25 = \frac{9}{u_o} . \frac{25}{5}$ or $u_o = \frac{9}{5} = 1.8\ cm$

9. (b) $m_\infty = -\frac{v_o}{u_o} \times \frac{D}{f_e}$

From $\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$

$\Rightarrow \frac{1}{(+1.2)} = \frac{1}{v_o} - \frac{1}{(-1.25)} \Rightarrow v_o = 30\ cm$

$\therefore |m_\infty| = \frac{30}{1.25} \times \frac{25}{3} = 200$

10. (b) For objective lens $\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$

$\Rightarrow \frac{1}{(+4)} = \frac{1}{v_o} - \frac{1}{(-4.5)} \Rightarrow v_o = 36\ cm$

$\therefore |m_D| = \frac{v_o}{u_o}\left(1 + \frac{D}{f_e}\right) = \frac{36}{4.5}\left(1 + \frac{24}{8}\right) = 32$

11. (a) For a microscope $|m| = \frac{v_o}{u_o} \times \frac{D}{u_e}$ and $L = v_o + u_e$

For a given microscope, with increase in L, u will increase and hence magnifying power (m) will decrease.

12. (b) In compound microscope objective forms real image while eye piece forms virtual image.

13. (b) $m = 1 + \frac{D}{f}$

Smaller the focal length, higher the magnifying power.

14. (a) In electron microscope, electron beam $(\lambda \approx 1Å)$ is used so it's $R.P.$ is approx. 5000 times more than that of ordinary microscope $(\lambda \approx 5000Å)$.

15. (c) If nothing is said then it is considered that final image is formed at infinite and $m_\infty = \frac{(L_\infty - f_o - f_e).D}{f_o f_e} \simeq \frac{LD}{f_0 f_e}$

$\Rightarrow 400 = \frac{20 \times 25}{0.5 \times f_e} \Rightarrow f_e = 2.5\ cm.$

16. (d) $m_{\max} = 1 + \frac{D}{f} = 1 + \frac{25}{2.5} = 11$.

17. (a)

18. (b) $m = 1 + \frac{D}{f} = 1 + DP$ (m increases with P)

19. (b)

20. (b) Like Gallilean telescope.

21. (a) $|m| \propto \frac{1}{f_o f_e}$

22. (d) A microscope consists of lens of small focal lengths. A telescope consists of objective lens of large focal length.

23. (c) $m = m_o \times m_e = 25 \times 6 = 150$

24. (a) When final image is formed at infinity,

length of the tube $= v_o + f_e$

$\Rightarrow 15 = v_o + 3 \Rightarrow v_o = 12\ cm$

For objective lens $\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$

$\Rightarrow \frac{1}{(+2)} = \frac{1}{(+12)} - \frac{1}{u_o} \Rightarrow u_o = -2.4\ cm$

25. (d) R.P. of microscope $= \frac{2\mu \sin\theta}{\lambda}$

26. (c) $m = m_o \times m_e \Rightarrow m = m_o \times \left(1 + \frac{D}{f_e}\right)$

$\Rightarrow 100 = 10 \times \left(1 + \frac{25}{f_e}\right) \Rightarrow f_e = \frac{25}{9}\ cm$

27. (c) A simple microscope is just a convex lens with object lying between optical centre and focus of the lens.

28. (d) In general, the simple microscope is used with image at D, hence

$m = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6$

29. (d)

30. (b) Resolving power of microscope $\propto \frac{1}{\lambda}$

31. (a) Cross wire arrangement is used to make measurements.

32. (d) $L = v_o + u_e = \frac{u_o f_o}{(u_o - f_o)} + \frac{f_e D}{f_e + D}$

$\Rightarrow L = \frac{2 \times 1.5}{(2 - 1.5)} + \frac{6.25 \times 25}{(6.25 + 25)} = 11\ cm$

33. (d) $m \simeq \frac{LD}{f_o f_e} \Rightarrow m = \frac{10 \times 25}{0.5 \times 1} = 500$.

34. (c) Intermediate image means the image formed by objective, which is real, inverted and enlarged.

35. (d) $m \propto \frac{1}{f_o f_e}$

36. (b) R.P. $\propto \frac{1}{\lambda}$; $\lambda_{\text{Blue}} < \lambda_{\text{Red}}$ so $(R.P.)_{\text{Blue}} > (R.P.)_{\text{Red}}$

37. (a) $m = 1 + \frac{D}{f} \Rightarrow 6 = 1 + \frac{25}{f} \Rightarrow f = 5\ cm = 0.05\ m$

38. (a) Resolving limit

$x \propto \lambda \Rightarrow \frac{x_1}{x_2} = \frac{\lambda_1}{\lambda_2} \Rightarrow \frac{0.1}{x_2} = \frac{6000}{4800} \Rightarrow x_2 = 0.08\ mm$

39. (b) $m = m_o \times m_e \Rightarrow 100 = 5 \times m_e \Rightarrow m_e = 20$

40. (d) $m \propto \frac{1}{f} \propto P$

41. (d) R.P. $\propto \frac{1}{\lambda} \Rightarrow \frac{(R.P.)_1}{(R.P.)_2} = \frac{\lambda_2}{\lambda_1} = \frac{5}{4}$

42. (b) Resolving limit (minimum separation) $\propto \lambda$

$\Rightarrow \frac{P_A}{P_B} = \frac{2000}{3000} \Rightarrow P_A < P_B$

43. (d) Similar to Q.No. 34

44. (a) For achromatic telescope objective lens, convergent of crown and divergent of flint is the best combination because $\mu_{\text{crown}} < \mu_{\text{flint}}$

45. (c)

46. (b) Magnifying power of telescope is $\frac{f_o}{f_e}$, so as $\frac{1}{f_e}$ increases, magnifying power increases.

47. (b) Since $m = \frac{f_o}{f_e}$

Also $m = \frac{\text{Angle subtended by the image}}{\text{Angle subtended by the object}}$

$\therefore \frac{f_o}{f_e} = \frac{\alpha}{\beta} \Rightarrow \alpha = \frac{f_o \times \beta}{f_e} = \frac{60 \times 2}{5} = 24^o$

48. (d) Resolving power $= \frac{d}{1.22\,\lambda} = \frac{0.1}{1.22 \times 6000 \times 10^{-10}}$

$\cong 1.36 \times 10^5\, radian$

49. (b) Because size of the aperture decreases.

50. (d) Resolving power $\propto$ aperture.

51. (c) Telescope is used to see the distant objects. More magnifying power means more nearer image.

52. (a) When the final image is at the least distance of distinct vision, then

$m = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right) = \frac{200}{5}\left(1 + \frac{5}{25}\right) = \frac{200 \times 6}{5 \times 5} = -\,48$

When the final image is at infinity, then

$m = \frac{-f_o}{f_e} = \frac{200}{5} = -\,40$

53. (a) In terrestrial telescope erecting lens absorbs a part of light, so less constant image. But binocular lens gives the proper three dimensional image.

54. (a) By formula $m = \frac{f_o}{f_e}$

55. (b) In telescope $f_o >> f_e$ as compared to microscope.

56. (a) Because magnification in this case becomes reciprocal of initial magnification.

57. (d) $m = \frac{f_o}{f_e} \Rightarrow \frac{80}{f_e} = 20 \Rightarrow f_e = 4\ cm$

Hence length of terrestrial telescope

$= f_o + f_e + 4f = 80 + 4 + 4 \times 20 = 164\ cm$

58. (d) In this case $|\,m\,| = \frac{f_o}{f_e} = 5$ .... (i)

and length of telescope $= f_o + f_e = 36$ .... (ii)

Solving (i) and (ii), we get $f_e = 6\ cm$, $f_o = 30\,cm$.

59. (c) $|\,m\,| = \frac{f_o}{f_e} = \frac{180}{6} = 30$

60. (c) Same as Q. No. 58.

61. (c) $f_o = \frac{1}{1.25} = 0.8\,m$ and $f_e = \frac{1}{-20} = -\,0.05\,m$

$\therefore |\,L_\infty| = |\,f_o| - |\,f_e| = 0.8 - 0.05 = 0.75\,m = 75\,cm$

and $|\,m_\infty| = \frac{f_o}{f_e} = \frac{0.8}{0.05} = 16$

62. (a) For greater aperture of lens, light passing through lens is more and so intensity of image increases.

63. (b)

64. (a) Same as Q. No. 58.

65. (b) $m = \frac{f_o}{f_e} = \frac{60}{10} = 6$.

66. (a) $f_o + f_e = 54$ and $\frac{f_o}{f_e} = m = 8 \Rightarrow f_o = 8\,f_e$

$\Rightarrow 8\,f_e + f_e = 54 \Rightarrow f_e = \frac{54}{9} = 6$

$\Rightarrow f_o = 8\,f_e = 8 \times 6 = 48$

67. (a) $f_o - f_e = 9\ cm$ and $f_e = f_o - 9 = 15 - 9 = 6\ cm$

$\Rightarrow m = \frac{f_o}{f_e} = \frac{15}{6} = 2.5$

68. (c) $f_o + f_e = 80$ and $\frac{f_o}{f_e} = 19 \Rightarrow f_o = 76$ and $f_e = 4\ cm.$

69. (a)

70. (b) $R.P. \propto \frac{D}{\lambda}$

71. (c) $m = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$

72. (b) Resolving power $\propto$ Aperture

73. (a) If final image is formed at infinity, then the distance between the two lenses of telescope is equal to length of tube $= f_o + f_e = 0.3 + 0.05 = 0.35\ m$

74. (a) Limit of resolution $= \frac{1.22\,\lambda}{a} \times \frac{180}{\pi}$ (in degree)

$= \left(\frac{1.22 \times (6000 \times 10^{-10})}{5} \times \frac{180}{\pi}\right)^o = 0.03\ sec$

75. (b) Final image formed by astronomical telescope is inverted not erect.

76. (d)

77. (c)

78. (b) For normal vision (relaxed eye), the image is formed at infinity. Hence the magnifying power of Gallilean telescope $= \frac{f_o}{f_e} = \frac{200}{2} = 100$.

79. (a) $m = -\frac{f_o}{f_e} = -\frac{100}{2} = -50.$

80. (c)

81. (b) Magnifying power of astronomical telescope

$m = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right) = -\frac{200}{5}\left(1 + \frac{5}{25}\right) = -\,48$.

82. (b) $m \propto \frac{1}{f_e}$

83. (b) $f_0 > f_e$ for telescope.

84. (a) $m = -\frac{f_0}{f_e}$.

85. (b) $|\,m\,| = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right) = \frac{100}{5}\left(1 + \frac{5}{25}\right) = 24$

86. (a, b, c, d)

87. (a) $|\,m\,| = \frac{f_o}{f_e} = 20$ and $L = f_o + f_e = 105 \Rightarrow f_o = 100\ cm$

88. (a) Total length $L = f_o + f_e$ and both lenses are convex.

89. (b) $L = f_o + f_e = 44$ and $|\,m\,| = \frac{f_o}{f_e} = 10$

This gives $f_o = 40\ cm$

90. (c) In case of a telescope if object and final image are at infinity then $m = \dfrac{f_o}{f_e}$

91. (b) Three lenses are $\rightarrow$ objective, eye piece and erecting lens.

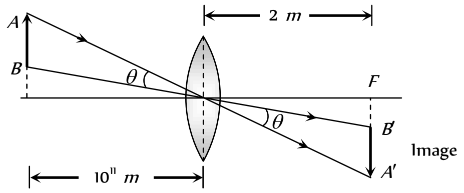
92. (d) Length of the telescope when final image is formed at least distance of distinct vision is

$$L = f_o + u_e = f_o + \frac{f_e D}{f_e + D} = 50 + \frac{5 \times 25}{5 + 25} = \frac{325}{6}\ cm$$

93. (c) $\dfrac{\beta}{\alpha} = \dfrac{f_o}{f_e} \Rightarrow \dfrac{\beta}{0.5^o} = \dfrac{100}{2} \Rightarrow \beta = 25^o$

94. (d)

95. (c) $\theta = \dfrac{AB}{10^{11}} = \dfrac{A'B'}{2} \Rightarrow A'B' = \dfrac{2 \times 1.4 \times 10^9}{10^{11}} = 2.8\ cm$



96. (c) $m = \dfrac{f_o}{f_e}\left(1 + \dfrac{f_e}{D}\right) \Rightarrow m = \dfrac{90}{6}\left(1 + \dfrac{6}{30}\right) \Rightarrow m = 18$

97. (d) Resolving power of telescope $= \dfrac{d}{1.22\ \lambda}$

98. (a) For largest magnification focal length of eye lens should be least.

99. (b) $m = \dfrac{f_o}{f_e}\left(1 + \dfrac{f_e}{D}\right) = \dfrac{150}{6}\left(1 + \dfrac{6}{25}\right) = 30$.

100. (d) To make telescope of higher magnifying power, f_o should be large and f_e should be least.

101. (c) $f_o = 50\ cm$, $f_e = 5\ cm$, $D = 25\ cm$ and $u_o = 200\ cm$.

Separation between the objective and the eye lens is

$$L = \frac{u_o f_o}{(u_o - f_o)} + \frac{f_e D}{(f_e + D)} = \frac{200 \times 50}{(200 - 50)} + \frac{5 \times 25}{(5 + 25)} = 71\ cm$$

102. (b) Resolving power $= \dfrac{d}{1.22\ \lambda} = \dfrac{1.22}{1.22 \times 5000 \times 10^{-10}} = 2 \times 10^6$

103. (a)

104. (b) By using $m = \dfrac{f_o}{f_e} \Rightarrow f_e = \dfrac{100}{50} = 2\ cm$

Also $L = f_o - f_e = 100 - 2 = 98\ cm$

105. (b) $m = \dfrac{f_o}{f_e} \Rightarrow 10 = \dfrac{f_o}{20} \Rightarrow f_o = 200\ cm$

106. (c) Minimum angular separation $\Delta\theta = \dfrac{1}{R.P.} = \dfrac{1.22\ \lambda}{d}$

$$= \frac{1.22 \times 5000 \times 10^{-10}}{2} = 0.3 \times 10^{-6}\ rad$$

107. (c) $m = 1 + \dfrac{D}{f_e} \Rightarrow 10 = 1 + \dfrac{25}{f_e} \Rightarrow f_e = \dfrac{25}{9} \simeq 25\ mm$

108. (a) $\dfrac{D}{F}$ or $\dfrac{25}{F}$

109. (c) $L = v_0 + u_e$ and $v_0 >> f_0$, $u_e \simeq f_e$

110. (c) Magnification will be done by compound microscope only when $f_o < f_e$

111. (d) Angular resolution $d\theta = \dfrac{1.22\ \lambda}{a}$

$$= \frac{1.22 \times 5000 \times 10 \times 10^{-10}}{10 \times 10^{-2}} = 6.1 \times 10^{-6}\ rad.$$

112. (a) Resolving power $= \dfrac{a}{1.22\ \lambda}$

113. (d) $M = \dfrac{f_o}{f_e} = \dfrac{P_e}{P_o} = \dfrac{20}{0.5} = 40$.

114. (a) Radio, waves can pass through dust, clouds, fog, etc, in a radio, telescope. It can detect very faint radio signal due to enormous size of its reflector. So it can be used at night and even in cloudy weather.

115. (a) Resolving limit

$$d\theta = \frac{1.22\ \lambda}{a} = \frac{1.22 \times 4538 \times 10^{-10}}{1} = 5.54 \times 10^{-7}\ rad.$$

116. (a) Magnification of objective lens $m = \dfrac{I}{O} = \dfrac{v_0}{u_0} = \dfrac{f_0}{u_0}$

$\Rightarrow \dfrac{I}{50} = \dfrac{200 \times 10^{-2}}{2 \times 10^3} \Rightarrow I = 5 \times 10^{-2}\ m = 5\ cm.$

117. (b) $m = \dfrac{v_o}{u_o}\left(1 + \dfrac{D}{f_e}\right) = m_o\left(1 + \dfrac{D}{f_e}\right)$

$\Rightarrow 30 = m_o\left(1 + \dfrac{25}{5}\right) = m_0 \times 6 \Rightarrow m_o = 5$.

118. (a)

119. (a) $m = \dfrac{f_o}{f_e} \Rightarrow \dfrac{100}{f_e} = 50 \Rightarrow f_e = 2\ cm$

Normal distance $f_o - f_e = 100 - 2 = 98\ cm.$

120. (a) For objective lens $\dfrac{1}{f_o} = \dfrac{1}{v_o} - \dfrac{1}{u_o}$

$\Rightarrow \dfrac{1}{v_o} = \dfrac{1}{f_o} + \dfrac{1}{u_o} = \dfrac{1}{4} + \dfrac{1}{-5} = \dfrac{1}{20} \Rightarrow v_o = 20cm$

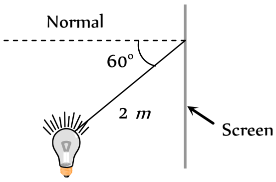
Now $M = \dfrac{v_o}{u_o}\left(1 + \dfrac{D}{f_e}\right) = \dfrac{20}{5}\left(1 + \dfrac{20}{10}\right) = 12.$

Photometry

1. (d) Luminous flux $= 4\pi\ L = 4 \times 3.14 \times 42 = 528$ *Lumen*

Power of lamp $= \dfrac{\text{Luminous flux}}{\text{Luminous efficiency}} = \dfrac{528}{2} = 264\ W$

2. (b) $I = \dfrac{L\cos\theta}{r^2}$



$\Rightarrow L = \dfrac{I \times r^2}{\cos\theta}$

$= \dfrac{5\times10^{-4}\times10^{4}\times2^2}{\cos 60°} = 40\ Candela$

3. (d) $I = \dfrac{L}{r^2} \Rightarrow \dfrac{dI}{I} = -\dfrac{2dr}{r}$ ($\because$ L = constant)

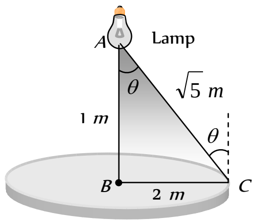
$\Rightarrow \dfrac{dI}{I}\times100 = -\dfrac{2\times dr}{r}\times100 = -2\times1 = -2\%$

4. (c) For equal fogging $I_2 \times t_2 = I_1 \times t_1$

$\Rightarrow \dfrac{L_2}{r_2^2}\times t_2 = \dfrac{L_1}{r_1^2}\times t_1 \Rightarrow \dfrac{16}{4}\times t_2 = \dfrac{20}{1}\times 10$

$\Rightarrow t_2 = 50\ \text{sec}.$

5. (d) The illuminance at B



$I_B = \dfrac{L}{1^2}$ ....(i)

and illuminance at point C

$I_C = \dfrac{L\cos\theta}{(\sqrt{5})^2} = \dfrac{L}{(\sqrt{5})^2}\times\dfrac{1}{\sqrt{5}}$

$\Rightarrow I_C = \dfrac{L}{5\sqrt{5}}$ ...... (ii)

From equation (i) and (ii) $I_B = 5\sqrt{5}\ I_0$

6. (b) $I \propto \dfrac{1}{r^2}$ so,

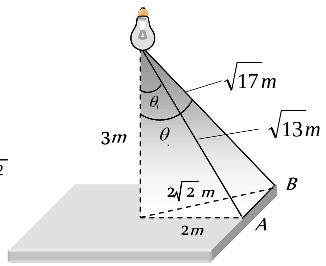
$\dfrac{\text{Illuminance on slide}}{\text{Illuminance on screen}} = \dfrac{(\text{Length of image on screen})^2}{(\text{Length of object on slide})^2}$

$= \left(\dfrac{3.5\ m}{35\ mm}\right)^2 = 10^4 : 1$

7. (a) The illuminance at A is

$I_A = \dfrac{L}{(\sqrt{13})^2}\times\cos\theta_1 = \dfrac{L}{13}\times\dfrac{3}{\sqrt{13}} = \dfrac{3L}{(13)^{3/2}}$

The illuminance at B is



$I_B = \dfrac{L}{(\sqrt{17})^2}\times\cos\theta_2$

$= \dfrac{L}{17}\times\dfrac{3}{\sqrt{17}} = \dfrac{3L}{(17)^{3/2}}$

$\therefore \dfrac{I_A}{I_B} = \left(\dfrac{17}{13}\right)^{3/2}$

8. (b)

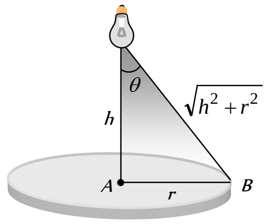
9. (c) Luminous intensity $L = \dfrac{\phi}{4\pi} \Rightarrow 1 = \dfrac{\phi}{4\pi} \Rightarrow \phi = 4\pi$.

10. (c) $\phi = 4\pi L = 4\times3.14\times100 = 1256$ *lumen*.

11. (a) $I = \dfrac{L}{r^2} \Rightarrow L = I.r^2 = 22\times2^2 = 100$

Now $\phi = 4\pi L = 4\times3.14\times100 = 1256$ *lumen*.

12. (c) Illuminance at A,



$I_A = \dfrac{L}{h^2}$

Illuminance at B,

$I_B = \dfrac{L}{\sqrt{(h^2+r^2)^2}}\cos\theta$

$= \dfrac{Lh}{(r^2+h^2)^{3/2}}$

$\therefore \dfrac{I_A}{I_B} = \left(1+\dfrac{r^2}{h^2}\right)^{3/2} = \left(1+\dfrac{8^2}{8^2}\right)^{3/2} = 2^{3/2} = 2\sqrt{2} : 1$

13. (c) $I = \dfrac{L}{r^2}$

14. (c) Efficiency of light source

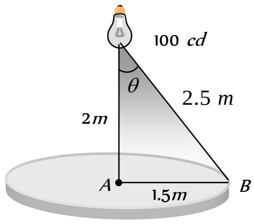
$\eta = \dfrac{\phi}{p}$ ..... (i)

and $L = \dfrac{\phi}{4\pi}$ ..... (ii)

From equation (i) and (ii)

$\Rightarrow p = \dfrac{4\pi L}{\eta} = \dfrac{4\pi\times35}{5} \approx 88\,W$.

15. (a) Case I

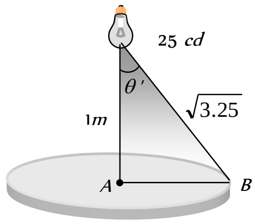


$I_A = \dfrac{100}{2^2} = 25\ cd$

and $I_B = \dfrac{100}{(2.5)^2}\cos\theta$

$= \dfrac{100}{2.5^2}\times\dfrac{2}{2.5} = \dfrac{200}{(2.5)^3}$

Case II,



$I'_B = X\,I_B = \dfrac{25}{(3.25)^{3/2}}$

so $\dfrac{I'_B}{I_B} = \dfrac{25}{200}\times\dfrac{(2.5)^3}{(3.25)^{3/2}}$

$\Rightarrow X = 1/3$

16. (a) $I \propto \dfrac{1}{r^2} \Rightarrow \dfrac{I_2}{I_1} = \dfrac{r_1^2}{r_2^2} = \dfrac{60^2}{180^2} = \dfrac{1}{9}$

17. (b)

18. (b) $I \propto \dfrac{1}{r^2}$

19. (c) To develop a print a fix amount of energy is required. Total light energy incident on photo print

$I\times A\,t = \dfrac{L}{r^2}A\,t \Rightarrow \dfrac{L_1}{r_1^2}A_1t_1 = \dfrac{L_2}{r_2^2}A_2t_2$

$\Rightarrow \dfrac{t_1}{r_1^2} = \dfrac{t_2}{r_2^2}$ ($\because L_1 = L_2$ and $A_1 = A_2$)

$\Rightarrow t_2 = \frac{r_2^2}{r_1^2}.t_1 = \left(\frac{0.40}{0.25}\right) 2 \times 5$ = 12.8 *sec.*

20. (b) $\frac{I_{\text{centre}}}{I_{\text{edge}}} = \frac{(r^2+h^2)^{3/2}}{h^3} = \frac{\left(1+\frac{1}{4}\right)^{3/2}}{1^3} = \left(\frac{5}{4}\right)^{3/2}$

21. (c) $I = \frac{L}{r^2} \Rightarrow \frac{L_1}{r_1^2} = \frac{L_2}{r_2^2}$ (I is same)

$\Rightarrow \frac{L_1}{L_2} = \frac{r_1^2}{r_2^2} = \left(\frac{1}{10}\right)^2 = 1:100$.

22. (c) $I_\theta = I_o \cos\theta = I_o \cos 60^o = \frac{I_o}{2}$

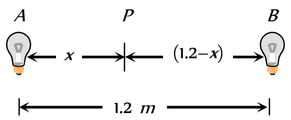
23. (a)

24. (b) $\phi = 4\pi L = 200\,\pi$ *lumen.*

so $I = \frac{\phi}{100\,A} = \frac{200\,\pi}{100 \times \pi r^2} = \frac{2}{(0.1)^2}$ = 200 *lux.*

25. (b,c) According to the problem

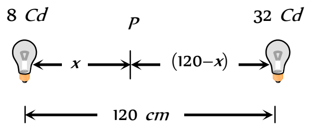
$\frac{I_A}{x^2} = 4\frac{I_B}{(1.2-x)^2}$



$\Rightarrow \frac{1}{x^2} = \frac{4}{(1.2-x)^2}$

$\Rightarrow \frac{1}{x} = \frac{2}{1.2-x} \Rightarrow x = 0.4\,m$ and $1.2 - x = 0.8\,m$.

26. (c) $I = \frac{L}{r^2} \Rightarrow \frac{L_1}{L_2} = \frac{r_1^2}{r_2^2}$



or $\frac{8}{x^2} = \frac{32}{(120-x)^2}$

Solving it we get $x = 40\,cm$.

27. (d) $\frac{I_{\text{center}}}{I_{\text{edge}}} = \frac{(r^2+h^2)^{3/2}}{h^3}$

$\Rightarrow 8 = \frac{(r^2+h^2)^{3/2}}{h^3} \Rightarrow 2h = (r^2+h^2)^{1/2}$

$\Rightarrow 4h^2 = r^2 + h^2 \Rightarrow 3h^2 = r^2 \Rightarrow h = \frac{r}{\sqrt{3}}$

28. (b) $I = \frac{L}{r^2} = \frac{100}{5^2}$ =4 *Lux.*

29. (d) $I_1 = \frac{L}{r_1^2} = \frac{L}{1600}$ and $I_2 = \frac{L}{2500}$

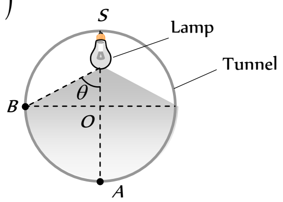
$\therefore$ % decease in illuminance

$= \frac{I_1 - I_2}{I_1} \times 100 = \left(1 - \frac{1600}{2500}\right) \times 100 = \frac{900}{2500} \times 100 = 36$

30. (b)

31. (d) $I_A = \frac{L}{(2r)^2}$ and $I_B = \frac{L}{\left(r\sqrt{2}\right)^2}\cos\theta$

$= \frac{L}{2r^2}.\frac{r}{r\sqrt{2}} = \frac{L}{2\sqrt{2}\,r^2}$



$\therefore \frac{I_A}{I_B} = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}$

32. (a) $I = \frac{L}{r^2} \Rightarrow L = 1.57 \times 10^5 \times (1.5 \times 10^{11})^2 = 3.53 \times 10^{27}\ Cd$

33. (d) $\phi = 4\pi L = 4 \times 3.14 \times 3.53 \times 10^{27} = 4.43 \times 10^{28}$ *lumen.*

34. (d) $\phi = \frac{3}{1.5 \times 10^{-3}} \times 0.685 = 1.37 \times 10^3$ *lumen*

35. (a) $\phi_{\text{surface}} = \frac{3000}{6} = 500$ *lumen.*

36. (c) Rotation of area about incident light doesn't change the inclination of the light ray on the area.

37. (c) $I = \frac{Lh}{r^3}$

38. (d) By the symmetry of the rays and location of the points.

39. (d) If η is the luminous efficiency of the bulb then.

luminous flux by 120 *watt* at 555 *nm* $= \eta \times 120$

Let bulb of P watt at 600 *nm* produces the same luminous flux as by 120 *watt* at 555 *nm* then

$\eta \times 120 = \eta P \times 0.6 \Rightarrow P = \frac{120}{0.6} = 200$ *watt.*

40. (c) Illuminance produce by the sun $= \frac{L}{(1.5 \times 10^{11})^2}$

Illuminance produce by the bulb $= \frac{10000}{(0.3)^2}$

According to problem $\frac{L}{(1.5 \times 10^{11})^2} = \frac{10000}{(0.3)^2}$

$\Rightarrow L = \frac{2.25 \times 10^{22} \times 10^4}{9 \times 10^{-2}} = 25 \times 10^{26}\,Cd$

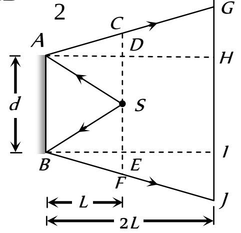
41. (c) $I_1 = \frac{L}{r_1^2} = \frac{L}{16}$ and $I_2 = \frac{L}{r_2^2} = \frac{L}{9}$

% increase in illuminance

$= \frac{I_2 - I_1}{I} \times 100 = \left(\frac{16}{9} - 1\right) \times 100 \approx 78\%$

Critical Thinking Questions

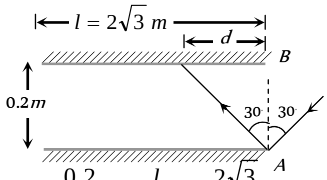
1. (d) According to the following ray diagram $HI = AB = d$ and $DS = CD = \frac{d}{2}$



$$\because AH = 2AD \Rightarrow GH = 2CD = \frac{2d}{2} = d$$

Similarly $IJ = d$ so $GJ = GH + HI + IJ = d + d + d = 3d$

2. (b) From the following ray diagram

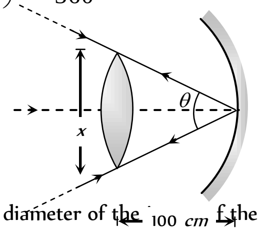


$$d = 0.2 \tan 30° = \frac{0.2}{\sqrt{3}} \Rightarrow \frac{l}{d} = \frac{2\sqrt{3}}{0.2/\sqrt{3}} = 30$$

Therefore maximum number of reflections are 30.

3. (b) The angle subtended by the image of the sun at the mirror

$$= 30' = \left(\frac{1}{2}\right)^o = \frac{\pi}{360}\ rad$$



If x be the diameter of the image of the sun, then

$$\frac{\text{Arc}}{\text{Radius}} = \frac{x}{100} = \frac{1}{2}.\frac{2\pi}{360} = \frac{\pi}{360} \Rightarrow x = \frac{100\pi}{360} = 0.87\ cm$$

4. (a) $m = \frac{I}{O} = \frac{f}{u-f} = \frac{10}{25-10} = \frac{10}{15} = \frac{2}{3}$

$$m^2 = \frac{A_i}{A_o} \Rightarrow A_i = m^2 \times A_o = \left(\frac{2}{3}\right)^2 \times (3)^2 = 4\ cm^2$$

5. (d) From mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ .....(i)

Differentiating equation (i), we obtain

$$0 = -\frac{1}{v^2}dv - \frac{1}{u^2}du \Rightarrow dv = -\left(\frac{v}{u}\right)^2 du \quad .....(ii)$$

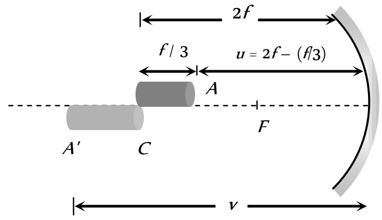
Also from equation (i) $\frac{v}{u} = \frac{f}{u-f}$ .....(iii)

From equation (ii) and (iii) we get $dv = -\left(\frac{f}{u-f}\right)^2 . l$

Therefore size of image is $\left(\frac{f}{u-f}\right)^2 l$.

6. (b) If end A of rod acts an object for mirror then it's image will be A' and if

$$u = 2f - \frac{f}{3} = \frac{5f}{3} \text{ so by using } \frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

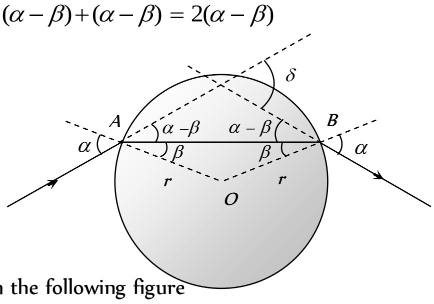


$$\Rightarrow \frac{1}{-f} = \frac{1}{v} + \frac{1}{\frac{-5f}{3}} \Rightarrow v = -\frac{5}{2}f$$

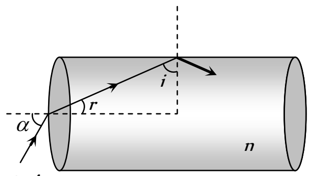
$$\therefore \text{ Length of image } = \frac{5}{2}f - 2f = \frac{f}{2}$$

7. (b) From the following ray diagram it is clear that

$$\delta = (\alpha - \beta) + (\alpha - \beta) = 2(\alpha - \beta)$$



8. (a) From the following figure



$r + i = 90^\circ \Rightarrow i = 90^\circ - r$

For ray not to emerge from curved surface $i > C$

$\Rightarrow \sin i > \sin C \Rightarrow \sin (90^\circ - r) > \sin C \Rightarrow \cos r > \sin C$

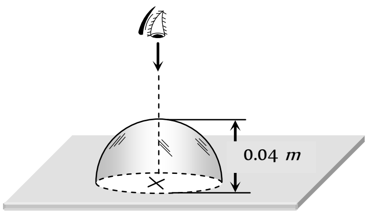
$$\Rightarrow \sqrt{1 - \sin^2 r} > \frac{1}{n} \qquad \left\{\because \sin C = \frac{1}{n}\right\}$$

$$\Rightarrow 1 - \frac{\sin^2 \alpha}{n^2} > \frac{1}{n^2} \Rightarrow 1 > \frac{1}{n^2}(1 + \sin^2 \alpha)$$

$$\Rightarrow n^2 > 1 + \sin^2 \alpha \Rightarrow n > \sqrt{2} \qquad \{\sin i \to 1\}$$

$\Rightarrow$ Least value $= \sqrt{2}$

9. (b) **Case (i)** When flat face is in contact with paper.



$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \text{ where}$$

μ_2 = *R. I.* of medium in which light rays are going = 1

μ_1 = *R. I.* of medium from which light rays are coming = 1.6

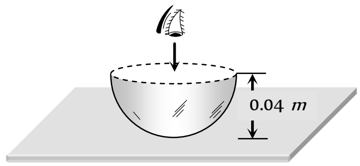
u = distance of object from curved surface = – 0.04 m

R = – 0.04 m.

$$\therefore \frac{1}{v} - \frac{1.6}{(-0.04)} = \frac{1 - 1.6}{(-0.04)} \Rightarrow v = -0.04\,m$$

i.e. the image will be formed at the same position of cross.

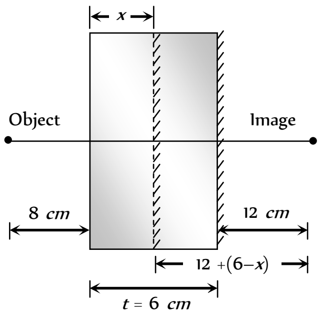
Case (ii) When curved face is in contact with paper



$$\mu = \frac{\text{Real depth } (h)}{\text{Apparent depth } (h')}$$

$$\Rightarrow 1.6 = \frac{0.04}{h'} \Rightarrow h' = 0.025\,m \qquad \text{(Below the flat face)}$$

10. (c) Let x be the apparent position of the silvered surface.

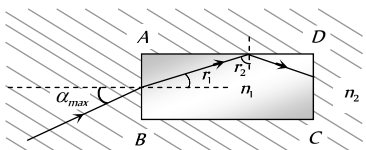


According to property of plane mirror

$x + 8 = 12 + 6 - x \Rightarrow x = 5$ cm

$$\text{Also } \mu = \frac{t}{x} \Rightarrow \mu = \frac{6}{5} = 1.2$$

11. (a) Ray comes out from CD, means rays after refraction from AB get, total internally reflected at AD



$$\frac{n_1}{n_2} = \frac{\sin \alpha_{\max}}{\sin r_1} \Rightarrow \alpha_{\max} = \sin^{-1}\left[\frac{n_1}{n_2}\sin r_1\right] \qquad \text{...(i)}$$

$$\text{Also } r_1 + r_2 = 90^o \Rightarrow r_1 = 90 - r_2 = 90 - C$$

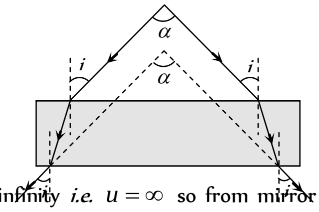
$$\Rightarrow r_1 = 90 - \sin^{-1}\left(\frac{1}{{}_2\mu_1}\right) \Rightarrow r_1 = 90 - \sin^{-1}\left(\frac{n_2}{n_1}\right) \qquad \text{...(ii)}$$

Hence from equation (i) and (ii)

$$\alpha_{\max} = \sin^{-1}\left[\frac{n_1}{n_2}\sin\left\{90 - \sin^{-1}\frac{n_2}{n_1}\right\}\right]$$

$$= \sin^{-1}\left[\frac{n_1}{n_2}\cos\left(\sin^{-1}\frac{n_2}{n_1}\right)\right]$$

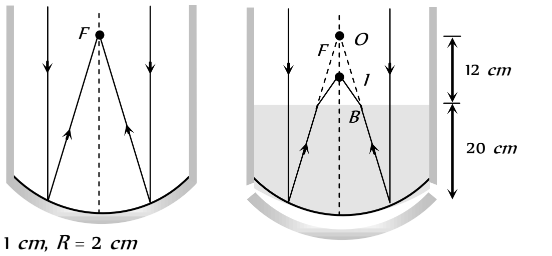
12. (b) Since rays after passing through the glass slab just suffer lateral displacement hence we have angle between the emergent rays as α.



13. (b) Sun is at infinity *i.e.* $u = \infty$ so from mirror formula we have $\frac{1}{f} = \frac{1}{-32} + \frac{1}{(-\infty)} \Rightarrow f = -32\ cm$.

When water is filled in the tank upto a height of 20 *cm*, the image formed by the mirror will act as virtual object for water surface. Which will form it's image at I such that $\frac{\text{Actual height}}{\text{Apperant height}} = \frac{\mu_w}{\mu_a}$ *i.e.* $\frac{BO}{BI} = \frac{4/3}{1}$

$$\Rightarrow BI = BO \times \frac{3}{4} = 12 \times \frac{3}{4} = 9\ cm.$$



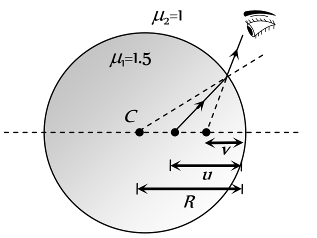
14. (a) $v = 1\ cm$, $R = 2\ cm$

By using

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{-1} - \frac{1.5}{u} = \frac{1 - 1.5}{-2}$$

$$\Rightarrow u = -1.2\,cm$$

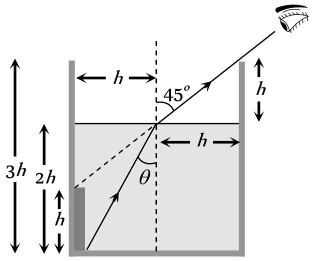


15. (b) The line of sight of the observer remains constant, making an angle of 45° with the normal.

$$\sin\theta = \frac{h}{\sqrt{h^2 + (2h)^2}} = \frac{1}{\sqrt{5}}$$

$$\mu = \frac{\sin 45^o}{\sin\theta}$$

$$= \frac{1/\sqrt{2}}{1/\sqrt{5}} = \sqrt{\left(\frac{5}{2}\right)}$$



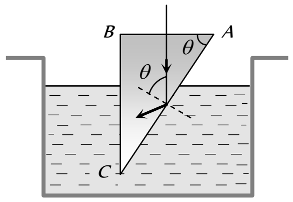
16. (b) For *glass-water* interface ${}_g\mu_w = \frac{\sin i}{\sin r}$ ...(i)

For *water-air* interface ${}_w\mu_a = \frac{\sin r}{\sin 90^o}$ ...(ii)

$$\Rightarrow {}_g\mu_w \times {}_w\mu_a = \frac{\sin i}{\sin r} \times \frac{\sin r}{\sin 90^o} = \sin i$$

$$\Rightarrow \frac{\mu_w}{\mu_g} \times \frac{\mu_a}{\mu_w} = \sin i \Rightarrow \mu_g = \frac{1}{\sin i}$$

17. (a) For TIR at AC



$$\theta > C$$

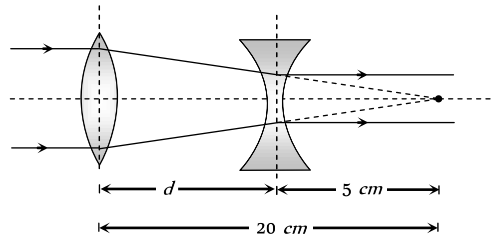
$$\Rightarrow \sin\theta \geq \sin C$$

$$\Rightarrow \sin\theta \geq \frac{1}{{}_w\mu_g}$$

$$\Rightarrow \sin\theta \geq \frac{\mu_w}{\mu_g} \Rightarrow \sin\theta \geq \frac{8}{9}$$

18. (b) From figure it is clear that separation between lenses

$$d = 20 - 5 = 15\,cm$$



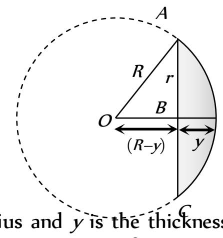
19. (c) According to lens formula $\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$

The lens is plano-convex *i.e.*, $R_1 = R$ and $R_2 = \infty$

Hence $\frac{1}{f} = \frac{\mu - 1}{R} \Rightarrow f = \frac{R}{\mu - 1}$

Speed of light in medium of lens $v = 2 \times 10^8\ m/s$

$$\Rightarrow \mu = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = \frac{3}{2} = 1.5$$



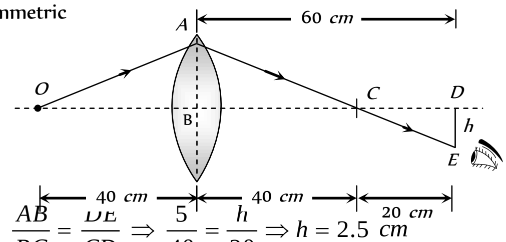
If r is the radius and y is the thickness of lens (at the centre), the radius of curvature R of its curved surface in accordance with the figure is given by

$$R^2 = r^2 + (R - y)^2 \Rightarrow r^2 + y^2 - 2Ry = 0$$

Neglecting y^2; we get $R = \frac{r^2}{2y} = \frac{(6/2)^2}{2 \times 0.3} = 15\ cm$

Hence $f = \frac{15}{1.5 - 1} = 30\ cm$

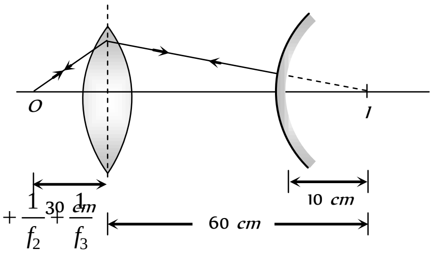
20. (c) In the following ray diagram Δ's, ABC and CDE are symmetric



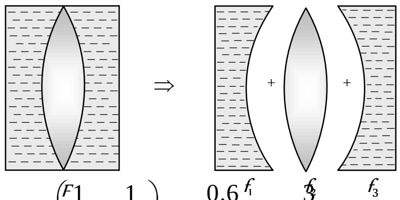
So, $\frac{AB}{BC} = \frac{DE}{CD} \Rightarrow \frac{5}{40} = \frac{h}{20} \Rightarrow h = 2.5\ cm$

21. (c) For lens $u = 30\ cm$, $f = 20\,cm$, hence by using $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{+20} = \frac{1}{v} - \frac{1}{-30} \Rightarrow v = 60\ cm$

The final image will coincide the object, if light ray falls normally on convex mirror as shown. From figure it is seen clear that separation between lens and mirror is 60 – 10 = 50 *cm*.



22. (d) $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}$



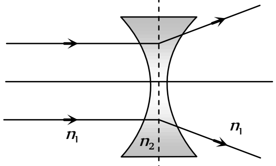
$$\frac{1}{f_1}=(1.6-1)\left(\frac{1}{\infty}-\frac{1}{20}\right)=-\frac{0.6}{20}=-\frac{3}{100} \quad ...(i)$$

$$\frac{1}{f_2}=(1.5-1)\left(\frac{1}{20}-\frac{1}{-20}\right)=\frac{1}{20} \quad ...(ii)$$

$$\frac{1}{f_3}=(1.6-1)\left(\frac{1}{-20}-\frac{1}{\infty}\right)=-\frac{3}{100} \quad ...(iii)$$

$$\Rightarrow \frac{1}{F}=-\frac{3}{100}+\frac{1}{20}-\frac{3}{100}\Rightarrow F=-100\ cm$$

23. (d) $\frac{1}{f}=\left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$ where n_2 and n_1 are the refractive indices of the material of the lens and of the surroundings respectively. For a double concave lens, $\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$ is always negative.



Hence f is negative only when $n_2>n_1$

24. (a, d) For a lens $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}\Rightarrow\frac{1}{v}=\frac{1}{u}+\frac{1}{f}$ .....(i)

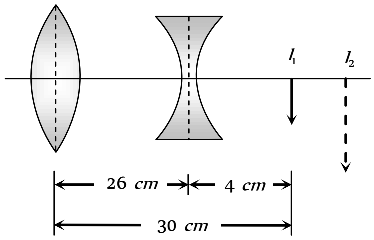
Also $m=\frac{f-v}{f}=1-\frac{v}{f}\Rightarrow m=\left(-\frac{1}{f}\right)v+1$ ....(ii)

On comparing equations (i) and (ii) with $y=mx+c$.

It is clear that relationship between $\frac{1}{v}$ *vs* $\frac{1}{u}$ and *m vs v* is linear.

25. (c) The dispersion produced by a spherical surface depends on it's radius of curvature. Hence, a lens will not exhibit dispersion only if it's two surfaces have equal radii, with one being convex and the other concave.

26. (b) Convex lens will form image I_1 at it's focus which acts like a virtual object for concave lens.



Hence for concave lens $u=+4\,cm$, $f=20\ cm$. So by lens formula $\frac{1}{-20}=\frac{1}{v}-\frac{1}{4}\Rightarrow v=5\,cm$ *i.e.* distance of final image (I_2) from concave lens $v=5\,cm$ by using $\frac{v}{u}=\frac{I}{O}\Rightarrow\frac{5}{4}=\frac{I}{2}\Rightarrow(I_2)=2.5\,cm$

27. (d) For achromatic combination $\omega_C=-\omega_F$

$[(\mu_v-\mu_r)A]_C=-[(\mu_v-\mu_r)A]_F$

$\Rightarrow[\mu_rA]_C+[\mu_rA]_F=[\mu_vA]_C+[\mu_vA]_F$

$=1.5\times19+6\times1.66=38.5$

Resultant $\delta=[(\mu_r-1)Al]_C+[(\mu_r-1)A]_F$

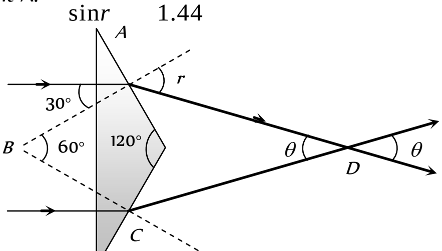
$=[\mu_rA]_C+[\mu_rA]_F-(A_C+A_F)=38.5-(19+6)=13.5^o$

28. (d) By using $\mu=\frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}}\Rightarrow\cot\frac{A}{2}=\frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}}$

$\Rightarrow\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}}=\frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}}$

$\Rightarrow\sin\left(90°-\frac{A}{2}\right)=\sin\left(\frac{A+\delta_m}{2}\right)\Rightarrow\delta_m=180-2A$

29. (d) At point A. $\frac{\sin30^o}{\sin r}=\frac{1}{1.44}$



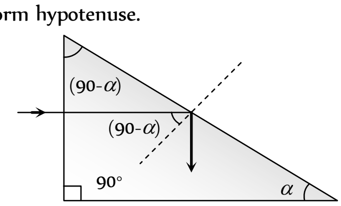
$\Rightarrow r=\sin^{-1}(0.72)$ also $\angle BAD=180^o-\angle r$

In rectangle *ABCD*, ∠ *A* + ∠ *B* + ∠ *C* + ∠ *D* = 360°

$\Rightarrow(180°-r)+60°+(180°-r)+\theta=360°$

$\Rightarrow\theta=2[\sin^{-1}(0.72)-30^o]$

30. (d) If α = maximum value of base angle for which light is totally reflected form hypotenuse.



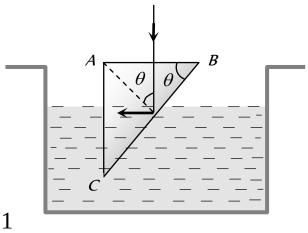
$(90° - \alpha) = C$ = minimum value of angle of incidence at hypotenuse for total internal reflection

$$\sin(90° - \alpha) = \sin C = \frac{1}{\mu} \Rightarrow \cos\alpha = \frac{1}{\mu} \Rightarrow \alpha = \cos^{-1}\left(\frac{1}{\mu}\right)$$

31. (b) For total internal reflection from surface BC

$\theta \geq C \Rightarrow \sin\theta \geq \sin C$

$\Rightarrow \sin\theta \geq \left(\frac{1}{{}_l\mu_g}\right)$



$\Rightarrow \sin\theta \geq \left(\frac{\mu_{\text{Liquid}}}{\mu_{\text{Prism}}}\right)$

$\sin\theta \geq \left(\frac{1.32}{1.56}\right) \Rightarrow \sin\theta \geq \frac{11}{13}$

32. (a) $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \Rightarrow \frac{1.5}{+OQ} - \frac{1}{(-OP)} = \frac{(1.5-1)}{+R}$

On putting $OQ = OP,\ OP = 5R$

33. (d) Here $\frac{1}{F} = \frac{2}{f} + \frac{1}{f_m}$

Plano-convex lens silvered on plane side has $f_m = \infty$.

$\therefore\ \frac{1}{F} = \frac{2}{f} + \frac{1}{\infty} \Rightarrow \frac{1}{30} = \frac{2}{f} \Rightarrow f = 60\ cm$

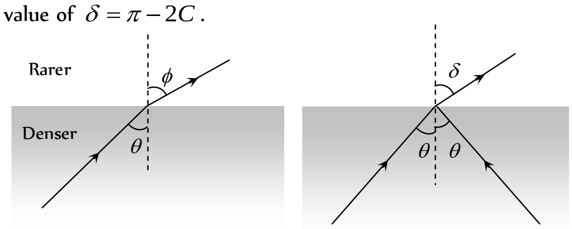
Plano-convex lens silvered on convex side has $f_m = \frac{R}{2}$

$\therefore\ \frac{1}{F} = \frac{2}{f} + \frac{2}{R} \Rightarrow \frac{1}{10} = \frac{2}{60} + \frac{2}{R} \Rightarrow R = 30\ cm$

Now using $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R}\right)$, we get $\mu = 1.5$

34. (c) When the ray passes into the rarer medium, the deviation is $\delta = \phi - 0$. This can have a maximum value of $\left(\frac{\pi}{2} - C\right)$ for $\theta = C$ and $\phi = \frac{\pi}{2}$.

When total internal reflection occurs, the deviation is $\delta = \pi - 2$, the minimum value of θ being C. The maximum value of $\delta = \pi - 2C$.



35. (c) $\frac{}{r} = \frac{}{d} \to x = \frac{}{d}$

$$= \frac{1.22 \times 500 \times 10^{-9} \times 400 \times 10^3}{5 \times 10^{-3}} = 50\ m$$

36. (d) Resolving power $= \frac{1.22\ \lambda}{a} = \frac{1.22 \times 6000 \times 10^{-10}}{5}$

Also resolving power $= \frac{d}{D} = \frac{d}{38.6 \times 10^7}$

$\therefore\ \frac{1.22 \times 6 \times 10^{-7}}{5} = \frac{d}{38.6 \times 10^7}$

$\Rightarrow d = \frac{1.22 \times 6 \times 10^{-7} \times 38.6 \times 10^7}{5} m = 56.51\ m$

37. (a) As limit of resolution

$\Delta\theta = \frac{1}{\text{Resolving Power(RP)}}$; and if x is the distance between points on the surface of moon which is at a distance r from the telescope.

$\Delta\theta = \frac{x}{r}$

So $\Delta\theta = \frac{1}{RP} = \frac{x}{r}$ *i.e.* $x = \frac{r}{RP} = \frac{r}{d/1.22\ \lambda} \Rightarrow x = \frac{1.22\ \lambda r}{d}$

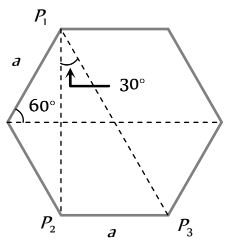
$$= \frac{1.22 \times 5500 \times 10^{-10} \times (3.8 \times 10^8)}{500 \times 10^{-2}} = 51\ m$$

38. (b) $I_{\text{edge}} = \frac{L\cos\theta}{(h^2 + r^2)} = \frac{Lh}{(h^2 + r^2)^{3/2}}$

For maximum extensity $\frac{dI}{dh} = 0$

Applying this condition have get $h = \frac{r}{\sqrt{2}}$

39. (a) From the geometry of the figure



$p_1 p_2 = 2a\sin 60^o$

so, $I_{P_2} = \frac{L}{p_1 p_2^2}$

$= \frac{L}{(2a\sin 60^o)^2} = \frac{L}{3a^2}$

and $I_{P_3} = \frac{L}{(P_1 P_2^2 + a^2)} \cos 30^o$

$= \frac{L}{[(2a\sin 60^o)^2 + a^2]} \frac{\sqrt{3}}{2} = \frac{\sqrt{3}\ L}{8\ a^2}$

$\Rightarrow I_{P_3} = \frac{3\sqrt{3}}{8} I_{P_2} = \frac{3\sqrt{3}}{8} I_0$

All options are wrong.

40. (c) Distance of object from mirror

$= 15 + \frac{33.25}{4} \times 3 = 39.93\ cm$

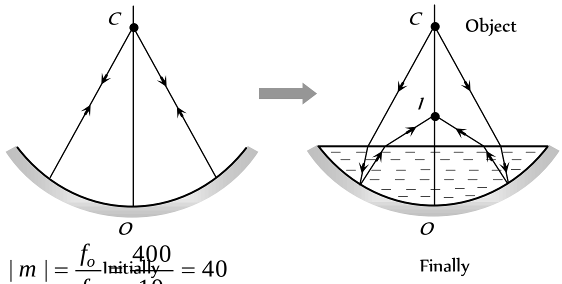
Distance of image from mirror $= 15 + \frac{25}{4} \times 3 = 33.75$

For mirror, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\Rightarrow\ \frac{1}{-33.75} - \frac{1}{39.93} = \frac{1}{f} \Rightarrow f \approx -18.3\ cm.$

41. (c) $v_i = -\left(\frac{f}{f-u}\right)^2 . v_o = -\left(\frac{-24}{-24-(-60)}\right)^2 \times 9 = 4\ cm/sec.$

42. (d) From the following figures it is clear that real image (I) will be formed between C and O



43. (b) $|m| = \frac{f_o}{f_e} = \frac{400}{10} = 40$

Angle subtented by moon on the objective of telescope

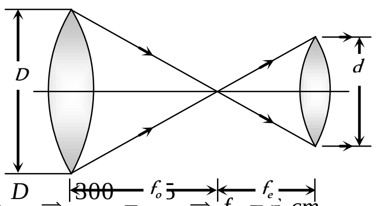
$\propto = \frac{3.5 \times 10^3}{3.8 \times 10^3} = \frac{3.5}{3.8} \times 10^{-2} rad$

Also $|m| = \frac{\beta}{\alpha} \Rightarrow$ Angular size of final image

$\beta = |m| \times \alpha = 40 \times \frac{3.5}{3.8} \times 10^{-2} = 0.36\ rad$

$= 0.3 \times \frac{180}{\pi} \approx 21^o$

44. (a) Full use of resolving power means whole aperture of objective in use. And for relaxed vision.



$\frac{f_o}{f_e} = \frac{D}{d} \Rightarrow \frac{300}{f_e} = \frac{5}{0.3} \Rightarrow f_e = 6\ cm$

45. (b) Wave length of the electron wave be $10 \times 10^{-12} m$,

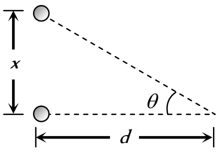
Using $\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow E = \frac{h^2}{\lambda^2 \times 2m}$

$= \frac{(6.63 \times 10^{-34})^2}{(10 \times 10^{-12})^2 \times 2 \times 9.1 \times 10^{-31}} Joule$

$= \frac{(6.63 \times 10^{-34})^2}{(10 \times 10^{-12})^2 \times 2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}} eV$

$= 15.1\ KeV$.

46. (c) $\theta = \frac{x}{d} = \frac{1.22\lambda}{a}$



$\Rightarrow x = \frac{1.22 \times d}{a}$

$= \frac{1.22 \times 5000 \times 10^{-10} \times 10^3}{10 \times 10^{-2}} = 6.1\ mm$

i.e. order will be 5 *mm*.

47. (c) $\frac{1.22\lambda}{a} = \frac{x}{d} \Rightarrow d = \frac{x \times a}{1.22\lambda} = \frac{1 \times 10^{-3} \times 3 \times 10^{-3}}{1.22 \times 500 \times 10^{-9}} = 5m$

48. (c) Let distance between lenses be x. As per the given condition, combination behaves as a plane glass plate, having focal length ∞.

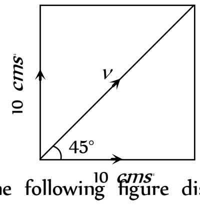
So by using $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{x}{f_1 f_2}$

$\Rightarrow \frac{1}{\infty} = \frac{1}{+30} + \frac{1}{-10} - \frac{x}{(+30)(-10)} \Rightarrow x = 20\ cm$

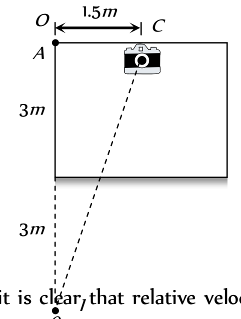
49. (a) When plane mirror rotates through an angle θ, the reflected ray rotates through an angle 2θ. So spot on the screen will make $2n$ revolution per second.

50. (d) $v \cos 45° = 10 \quad v = 10\sqrt{2}\ cms$

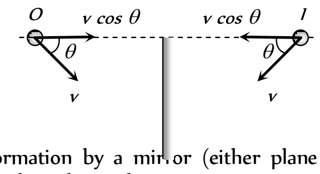
In the ceiling mirror the original velocity will be seen.



51. (d) According to the following figure distance of image I from camera $= \sqrt{(6)^2 + (1.5)^2} = 6.18 m$



52. (c) From figure it is clear, that relative velocity between object and it's image $= 2v \cos\theta$



53. (b) Image formation by a mirror (either plane or spherical) does not depend on the medium.

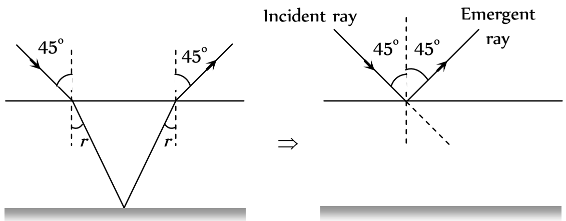
The image of P will be formed at a distance h below the mirror. If d = depth of liquid in the tank.

Apparent depth of $P = x_1 = \frac{d-h}{\mu}$

Apparent depth of the image of $P = x_2 = \frac{d+h}{\mu}$

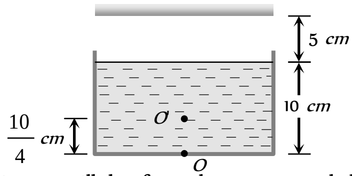
$\therefore$ Apparent distance between P and it's image $= x_2 - x_1 = \frac{2h}{\mu}$

54. (a) From the figure it is clear that the angle between incident ray and the emergent ray is 90°.



55. (b) From figure it is clear that object appears to be raised by $\frac{10}{4} cm\,(2.5\,cm)$

Hence distance between mirror and $O' = 5 + 7.5 = 12.5\,cm$



So final image will be formed at 12.5 cm behind the plane mirror.

56. (d) Velocity of approach of man towards the bicycle = $(u - v)$

Hence velocity of approach of image towards man is $2(u - v)$.

57. (c) For A

Total number of waves = $\frac{(1.5)t}{\lambda}$ ....(i)

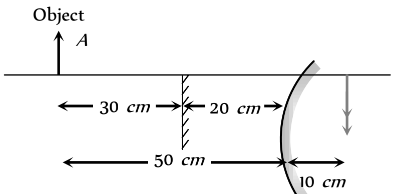
$\because \begin{pmatrix}\text{Total number}\\ \text{of waves}\end{pmatrix} = \left(\frac{\text{optical path length}}{\text{wavelength}}\right)$

For B and C

Total number of waves = $\frac{n_B\left(\frac{t}{3}\right)}{\lambda} + \frac{(1.6)\left(\frac{2t}{3}\right)}{\lambda}$ ....(ii)

Equating (i) and (ii) $n_B = 1.3$

58. (b) Since there is no parallex, it means that both images (By plane mirror and convex mirror) coinciding each other.



According to property of plane mirror it will form image at a distance of 30 cm behind it. Hence for convex mirror $u = -50$ cm, $v = +10$ cm

By using $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ $\Rightarrow \frac{1}{f} = \frac{1}{+10} + \frac{1}{-50} = \frac{4}{50}$

$\Rightarrow f = \frac{25}{2} cm$ $\Rightarrow R = 2f = 25cm.$

59. (d) For surface P, $\frac{1}{v_1} = \frac{1}{f} - \frac{1}{u} = 1 - \frac{1}{3} = \frac{2}{3} \Rightarrow v_1 = \frac{3}{2} m$

For surface Q, $\frac{1}{v_2} = \frac{1}{f} - \frac{1}{u} = 1 - \frac{1}{5} = \frac{4}{5} \Rightarrow v_2 = \frac{5}{4} m$

$\therefore v_1 - v_2 = 0.25m$

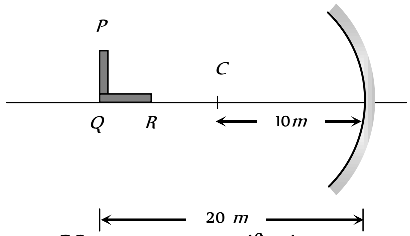
Magnification of $P = \frac{v_1}{u} = \frac{3/2}{3} = \frac{1}{2}$

$\therefore$ Height of $P = \frac{1}{2} \times 2 = 1m$

Magnification of $Q = \frac{v_2}{u} = \frac{5/4}{5} = \frac{1}{4}$

$\therefore$ Height of $Q = \frac{1}{4} \times 2 = 0.5m$

60. (b) Focal length of mirror $f = \frac{R}{2} = \frac{10}{2} = 5cm$



For part PQ : transverse magnification

length of image $L = \left(\frac{f}{f-u}\right) \times L_0$

$= \left(\frac{-5}{-5-(-20)}\right) \times L_0 = \frac{-L_0}{3}$

For part QR : longitudinal magnification

Length of image $L_2 = \left(\frac{f}{f-u}\right)^2 L_0$

$= \left(\frac{-5}{-5-(-20)}\right)^2 \times L_0 = \frac{L_0}{9} \Rightarrow \frac{L_1}{L_2} = \frac{3}{1}$

61. (d) The two slabs will shift the image a distance

$d = 2\left(1 - \frac{1}{\mu}\right)t = 2\left(1 - \frac{1}{1.5}\right)(1.5) = 1\,cm$

Therefore, final image will be 1 cm above point P.

62. (a) Here optical distance between fish and the bird is

$s = y' + \mu y$

Differentiating $w.r.t$ t we get $\frac{ds}{dt} = \frac{dy'}{dt} + \frac{\mu dy}{dt}$

$\Rightarrow 9 = 3 + \frac{4}{3}\frac{dy}{dt} \Rightarrow \frac{dy}{dt} = 4.5\ m/\text{sec}$

63. (a) The real depth $= \mu$ (apparent depth)

$\Rightarrow$ In first case, the real depth $h_1 = \mu(b - a)$

Similarly in the second case, the real depth $h_2 = \mu(d - c)$

Since $h_2 > h_1$, the difference of real depths $= h_2 - h_1 = \mu(d - c - b + a)$

Since the liquid is added in second case, $h_2 - h_1 = (d - b)$

$\Rightarrow \mu = \frac{(d-b)}{(d-c-b+a)}$

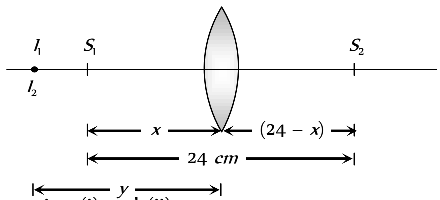
64. (a) The given condition will be satisfied only if one source (S) placed on one side such that $u < f$ ($i.e.$ it lies under the focus). The other source (S) is placed on the other side of the lens such that $u > f$ ($i.e.$ it lies beyond the focus).

If S_1 is the object for lens then $\frac{1}{f} = \frac{1}{-y} - \frac{1}{-x}$

$\Rightarrow \frac{1}{y} = \frac{1}{x} - \frac{1}{f}$ .....(i)

If S_2 is the object for lens then

$$\frac{1}{f}=\frac{1}{+y}-\frac{1}{-(24-x)} \Rightarrow \frac{1}{y}=\frac{1}{f}-\frac{1}{(24-x)} \quad .....(ii)$$



From equation (i) and (ii)

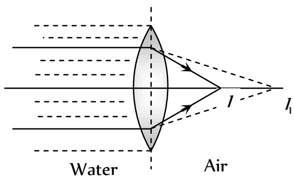
$$\frac{1}{x}-\frac{1}{f}=\frac{1}{f}-\frac{1}{(24-x)} \Rightarrow \frac{1}{x}+\frac{1}{(24-x)}=\frac{2}{f}=\frac{2}{9}$$

$\Rightarrow x^2-24x+108=0$. After solving the equation $x=18\ cm$, 6 *cm*.

65. (c) Consider the refraction of the first surface *i.e.* refraction from rarer medium to denser medium

$$\frac{\mu_2-\mu_1}{R}=\frac{\mu_1}{-u}+\frac{\mu_2}{v_1} \Rightarrow \frac{\left(\frac{3}{2}\right)-\left(\frac{4}{3}\right)}{R}=\frac{\frac{4}{3}}{\infty}+\frac{\frac{3}{2}}{v_1} \Rightarrow v_1=9R$$

Now consider the refraction at the second surface of the lens *i.e.* refraction from denser medium to rarer medium



$$\frac{1-\frac{3}{2}}{-R}=-\frac{\frac{3}{2}}{9R}+\frac{1}{v_2} \Rightarrow v_2=\left(\frac{3}{2}\right)R$$

The image will be formed at a distance of $\frac{3}{2}R$. This is equal to the focal length of the lens.

66. (c) $\delta_{Prism}=(\mu-1)A=(1.5-1)4^o=2^o$

$\therefore\ \delta_{Total}=\delta_{Prism}+\delta_{Mirror}$

$=(\mu-1)A+(180-2i)=2^o+(180-2\times 2)=178^o$

67. (b) Here the requirement is that $i>c$

$\Rightarrow \sin i>\sin c\ \Rightarrow \sin i>\frac{\mu_2}{\mu_1}$ .....(i)

From Snell's law $\mu_1=\frac{\sin\alpha}{\sin r}$ ....(ii)

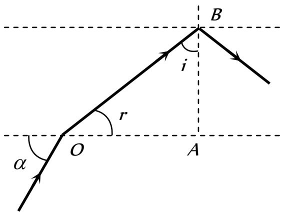
Also in ΔOBA

$r+i=90^o\ \Rightarrow\ r=(90-i)$

Hence from equation (ii)

$\sin\alpha=\mu_1\sin(90-i)$

$\Rightarrow\ \cos i=\frac{\sin\alpha}{\mu_1}$



$$\sin i=\sqrt{1-\cos^2 i}=\sqrt{1-\left(\frac{\sin\alpha}{\mu_1}\right)^2} \quad ....(iii)$$

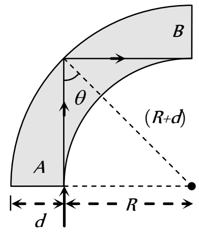
From equation (i) and (iii) $\sqrt{1-\left(\frac{\sin\alpha}{\mu_1}\right)^2}>\frac{\mu_2}{\mu_1}$

$\Rightarrow \sin^2\alpha<(\mu_1^2-\mu_2^2)\ \Rightarrow \sin\alpha<\sqrt{\mu_1^2-\mu_2^2}$

$\alpha_{max}=\sin^{-1}\sqrt{\mu_1^2-\mu_2^2}$

68. (b) Consider the figure if smallest angle of incidence θ is greater than critical angle then all light will emerge out of *B*

$\Rightarrow \theta\ge\sin^{-1}\left(\frac{1}{\mu}\right)\Rightarrow \sin\theta\ge\frac{1}{\mu}$



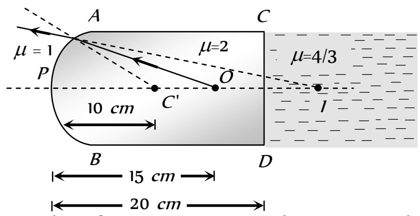
from figure $\sin\theta=\frac{R}{R+d}$

$\Rightarrow \frac{R}{R+d}\ge\frac{1}{\mu}\Rightarrow\left(1+\frac{d}{R}\right)\le\mu$

$\Rightarrow \frac{d}{R}\le\mu-1\Rightarrow\left(\frac{d}{R}\right)_{max}=0.5$

69. (b) In case of refraction from a curved surface, we have

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}\Rightarrow\frac{1}{v}-\frac{2}{(-15)}=\frac{(1-2)}{-10}\Rightarrow v=-30\ cm.$$

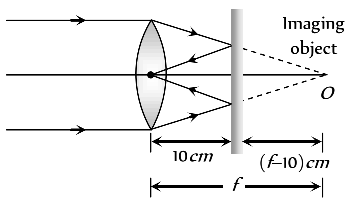


i.e. the curved surface will form virtual image *I* at distance of 30 *cm* from *P*. Since the image is virtual there will be no refraction at the plane surface *CD* (as the rays are not actually passing through the boundary), the distance of final image *I* from *P* will remain 30 *cm*.

70. (d) As $\mu_2>\mu_1$, the upper half of the lens will become diverging.

As $\mu_1>\mu_3$, the lower half of the lens will become converging

71. (b)



From the figure,

Using property of plane mirror

Image distance = Object distance

$f-10=10\Rightarrow f=20\ cm$

72. (d) If initially the objective (focal length *F*) forms the image at distance *v* then $v_o=\frac{u_o f_o}{u_o-f_o}=\frac{3\times 2}{3-2}=6\ cm$

Now as in case of lenses in contact

$$\frac{1}{F_o} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + ... = \frac{1}{f_1} + \frac{1}{F'_o}$$

$$\left\{ \text{where } \frac{1}{F'_o} = \frac{1}{f_2} + \frac{1}{f_3} + ... \right\}$$

So if one of the lens is removed, the focal length of the remaining lens system

$$\frac{1}{F'_o} = \frac{1}{F_0} - \frac{1}{f_1} = \frac{1}{2} - \frac{1}{10} \Rightarrow F'_o = 2.5\,cm$$

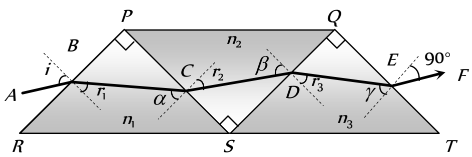
This lens will form the image of same object at a distance v'_o such that $v'_o = \frac{u_o F'_o}{u_o - F'_o} = \frac{3 \times 2.5}{(3 - 2.5)} = 15\,cm$

So to refocus the image, eye-piece must be moved by the same distance through which the image formed by the objective has shifted *i.e.* 15 – 6 = 9 *cm*.

73. (b) By using $m_\infty = \frac{(L_\infty - f_o - f_e)D}{f_o f_e}$

$$= \frac{(16 - 0.4 - 2.5) \times 25}{0.4 \times 2.5} = 327.5$$

74. (d)



$\alpha = 90 - r_1$ $\beta = 90 - r_2$ $\gamma = 90 - r_3$

At B

$\sin i = n_1 \sin r_1 \Rightarrow \sin^2 i = n_1^2 \sin^2 r_1$ .... (i)

At C

$n_1 \sin(90 - r_1) = n_2 \sin r_2 \Rightarrow n_2^2 \sin^2 r_2 = n_1^2 \cos^2 r_1$ ....(ii)

At D

$n_2 \sin(90 - r_2) = n_3 \sin r_3 \Rightarrow n_2^2 \cos^2 r_2 = n_3^2 \sin^2 r_3$ ....(iii)

At E

$n_3 \sin(90 - r_3) = (1)\sin(90 - 1) \Rightarrow \cos^2 i = n_3^2 \cos^2 r_3$ ....(iv)

Adding (i), (ii), (iii) and (iv) we get $1 + n_2^2 = n_1^2 + n_3^2$

75. (a) $L_D = v_o + u_e$ and for objective lens $\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$

Putting the values with proper sign convention.

$$\frac{1}{+2.5} = \frac{1}{v_o} - \frac{1}{(-3.75)} \Rightarrow v_o = 7.5\ cm$$

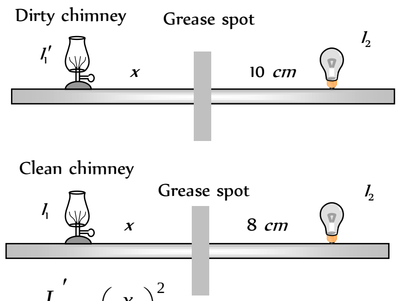
For eye lens $\frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e}$

$$\Rightarrow \frac{1}{+5} = \frac{1}{(-25)} - \frac{1}{u_e} \Rightarrow u_e = -4.16\ cm$$

$$\Rightarrow |u_e| = 4.16\ cm$$

Hence $L_D = 7.5 + 4.16 = 11.67\ cm$

76. (c) The actual luminous intensity of the lamp is I_1 whereas the intensity is I_1' in the dirty state.

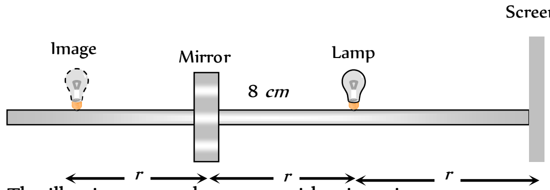


I position, $\frac{I_1'}{I_2} = \left(\frac{x}{10}\right)^2$

II position, $\frac{I_1}{I_2} = \left(\frac{x}{8}\right)^2 \Rightarrow \frac{I_1'}{I_1} = 0.64$

$\Rightarrow I_1' = 0.64\, I_1$. Thus, % of light absorbed = 36%.

77. (c) The illuminance on the screen without mirror is $I_1 = \frac{L}{r^2}$

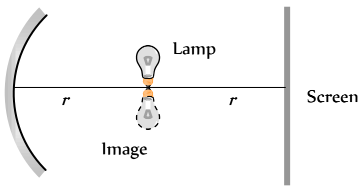


The illuminance on the screen with mirror is

$$I_2 = \frac{L}{r^2} + \frac{L}{(3r)^2} = \frac{10}{9} \times \frac{L}{r^2}$$

$$\therefore \frac{I_2}{I_1} = \frac{10}{9} = 10 : 9$$

78. (b) Illuminance on the screen without mirror is $I_1 = \frac{L}{r^2}$



Illuminance on the screen with mirror

$$I_2 = \frac{L}{r^2} + \frac{L}{r^2} = \frac{2L}{r^2} \Rightarrow \frac{I_2}{I_1} = 2 : 1$$

79. (b) Apparent depth $h' = \frac{h}{{}_{air}\mu_{liquid}}$

$$\Rightarrow \frac{dh'}{dt} = \frac{1}{{}_a\mu_w} = \frac{1}{{}_a\mu_w}\frac{dh}{dt} \Rightarrow x = \frac{1}{{}_a\mu_w}\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = {}_a\mu_w\, x$$

Now volume of water $V = \pi R^2 h$

$$\Rightarrow \frac{dV}{dt} = \pi R^2 \frac{dh}{dt} = \pi R^2 . {}_a\mu_w\, x$$

$$= {}_a\mu_w \pi R^2 x = \frac{\mu_w}{\mu_a}\pi R^2 x = \left(\frac{n_2}{n_1}\right)\pi R^2 x$$

Graphical Questions

1. (c) As $u \to f,\ v \to \infty$; $u \to \infty,\ v \to f$

2. (a) At $u = f$, $v = \infty$

At $u = 0$, $v = 0$ (*i.e.* object and image both lies at pole)

Satisfying these two conditions, only option (a) is correct.

3. (b, c) From graph $\tan 30^o = \frac{\sin r}{\sin i} = \frac{1}{{}_1\mu_2}$

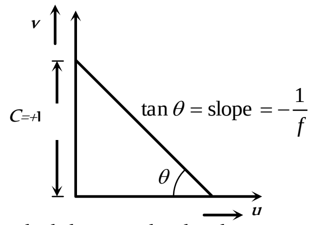
$$\Rightarrow {}_1\mu_2 = \sqrt{3} \Rightarrow \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2} = 1.73 \Rightarrow v_1 = 1.73 v_2$$

Also from $\mu = \frac{1}{\sin C} \Rightarrow \sin C = \frac{1}{{}_{Rarer}\mu_{Denser}}$

$$\Rightarrow \sin C = \frac{1}{{}_1\mu_2} = \frac{1}{\sqrt{3}}.$$

4. (c) For a lens $m = \frac{f-v}{f} \Rightarrow m = \left(-\frac{1}{f}\right)v + 1$

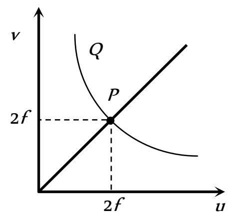
Comparing this equation with $y = mx + c$ (equation of straight line)



5. (c) At P, $u = v$ which happened only when $u = 2f$

At another point Q on the graph (above P)

$v > 2f$



6. (d) For a lens $m = \frac{f-v}{f} = -\frac{1}{f}v + 1$

Comparing it with $y = mx + c$

Slope $= m = -\frac{1}{f}$

From graph, slope of the line $= \frac{b}{c}$

Hence $-\frac{1}{f} = \frac{b}{c} \Rightarrow |f| = \frac{c}{b}$

7. (a) $\mu = A + \frac{B}{\lambda^2}$

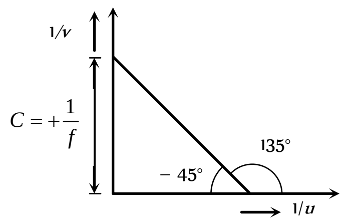
8. (a) Since $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{v} = -\frac{1}{u} + \frac{1}{f}$

Putting the sign convention properly

$$\frac{1}{(-v)} = \frac{-1}{(-u)} + \frac{1}{(-f)} \Rightarrow \frac{1}{v} = -\frac{1}{u} + \frac{1}{f}$$

Comparing this equation with $y = mx + c$

Slope $= m = \tan\theta = -1 \Rightarrow \theta = 135°$ or $-45°$ and intercept $C = +\frac{1}{f}$



9. (a) As u goes from 0 to $-\infty$, v goes from $+0$ to $+f$

10. (a) For convex lens (for real image) $u + v \geq 4f$

For $u = 2f$, v is also equal to $2f$

Hence $u + v = 4f$

11. (d) For concave mirror $m = \frac{f}{f-u}$

For real image $m = -\frac{f}{(u-f)} = -\frac{f}{x}$

$$= -\frac{f}{(\text{Distance of object from focus})} \Rightarrow m \propto \frac{1}{x}.$$

12. (a) For a prism, as the angle of incidence increases, the angle of deviation first decreases, goes to a minimum value and then increases.

13. (b) From Newton's formula $xy = f^2$. This is the equation of a rectangular hyperbola.

14. (a, c) At P, $\delta = 0 = A(\mu - 1) \Rightarrow \mu = 1$.

Also $\delta_m = (\mu - 1)A = A\mu_m - A$

Comparing it with $y = mx + c$

Slope of the line $= m = A$

15. (b) From graph, slope $= \tan\left(\frac{2\pi}{10}\right) = \frac{\sin r}{\sin i}$

Also ${}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{\sin i}{\sin r} = \frac{1}{\tan\left(\frac{2\pi}{10}\right)} = \frac{4}{3} \Rightarrow \mu_2 > \mu_1$

It means that medium 2 is denser medium. So total internal reflection cannot occur.

16. (d) From graph it is clear that $\tan 30^o = \frac{\sin r}{\sin i}$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{\sin r}{\sin i} = \frac{1}{\mu} \Rightarrow \mu = \sqrt{3}$$

Also $v = \frac{c}{\mu} = nc \Rightarrow n = \frac{1}{\mu} = \frac{1}{\sqrt{3}} = (3)^{-1/2}$

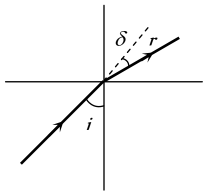
17. (b) In concave mirror, if virtual images are formed, u can have values zero and f

At $u = 0$, $m = \frac{f}{f-u} = \frac{f}{f} = 1$

At $u = f$, $m = \frac{f}{f-u} = -\frac{f}{-f-(-f)} = \infty$

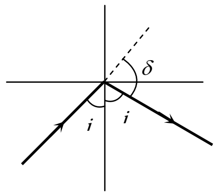
18. (a) The ray of light is refracted at the plane surface. However, since the ray is travelling from a denser to a rarer medium, for an angle of incidence (i) greater then the critical angle (c) the ray will be totally internally reflected.

For $i < c$; deviation $\delta = r - i$ with $\frac{1}{\mu} = \frac{\sin i}{\sin r}$



Hence $\delta = \sin^{-1}(\mu \sin i) - i$

This is a non-linear relation. The maximum value of δ is $\delta_1 = \frac{\pi}{2} - c$; where $i = c$ and $\mu = \frac{1}{\sin c}$



For $i > c$, deviation $\delta = \pi - 2i$

δ decreases linearly with i

$\delta_{,} = \pi - 2c = 2\delta_{,}$

19. (d) For a lens $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

If $u = \infty$, $v = f$ and if $u = f$, $v = \infty$

20. (d)

Assertion and Reason

1. (b)
2. (b) The stars twinkle while the planets do not. It is due to variation in density of atmospheric layer. As the stars are very far and giving light continuously to us. So, the light coming from stars is found to change their intensity continuously. Hence they are seen twinkling. Also stars are much bigger in size than planets but it has nothing to deal with twinkling phenomenon.
3. (c) Owls can move freely during night, because they have large number of cones on their retina which help them to see in night.
4. (c) Shining of air bubble in water is on account of total internal reflection.
5. (c) After the removal of stimulus the image formed on retina is sustained up to 1/6 second.
6. (a) Because of smallest wavelength of blue colour it is scattered to large extent than other colours, so the sky appears blue.
7. (e) For total internal reflection the angle of incidence should be greater than the critical angle. As critical angle is approximately 35°. Therefore, total internal reflection is not possible. So, assertion is not true but reason is true.
8. (c) The sun and its surroundings appears red during sunset or sunrise because of scattering of light. The amount of scattered light is inversely proportional to the fourth power of wavelength of light *i.e.* $I \propto \frac{1}{\lambda^4}$
9. (a) Focal length of lens immersed in water is four times the focal length of lens in air. It means

$$f_w = 4f_a = 4 \times 10 = 40\,cm$$

10. (e) The velocity of light of different colours (all wavelengths) is same in vacuum and $\mu \propto \frac{1}{\lambda}$.
11. (a) The red glass absorbs the radiations emitted by green flowers; so flower appears black.
12. (a) Magnification produced by mirror $m = \frac{I}{O} = \frac{f}{f-u} = \frac{f}{x}$

x is distance from focus.

13. (e) Apparent shift for different coloured letter is $d = h\left(1 - \frac{1}{\mu}\right)$

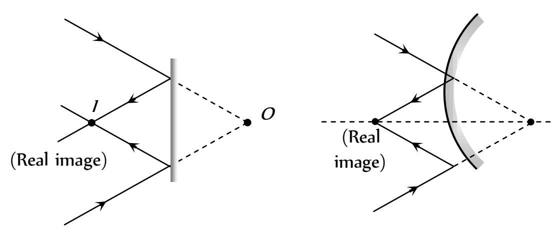
$\Rightarrow \lambda_R > \lambda_V$ so $\mu_R < \mu_V$

Hence $d_R < d_V$ *i.e.* red coloured letter raised least.

14. (a) The efficiency of fluorescent tube is about 50 *lumen/watt*, whereas efficiency of electric bulb is about 12 *lumen/watt*. Thus for same amount of electric energy consumed, the tube gives nearly 4 times more light than the filament bulb.
15. (c) Polar caps receives almost the same amount of radiation as the equatorial plane. For the polar caps angle between sun rays and normal (to polar caps) tends to 90°. As per Lambert's cosine law, $E \propto \cos\theta$, therefore E is zero. For the equatorial plane, $\theta = 0°$, therefore E is maximum. Hence polar caps of earth are so cold. (where E is radiation received).
16. (b) At noon, rays of sun light fall normally on earth. Therefore $\theta = 0°$. According to Lambert's cosine law, $E \propto \cos\theta$, when $\theta = 0°$, $\cos\theta = \cos 0° = 1$ = max. Therefore, E is maximum.
17. (d) When an object is placed between two plane parallel mirrors, then infinite number of images are formed. Images are formed due to multiple reflections. At each reflection, a part of light energy is absorbed. Therefore, distant images get fainter.
18. (c) In search lights, we need an intense parallel beam of light. If a source is placed at the focus of a concave spherical mirror, only paraxial rays are rendered parallel. Due to large aperture of mirror, marginal rays give a divergent beam.

But in case of parabolic mirror, when source is at the focus, beam of light produced over the entire cross-section of the mirror is a parallel beam.

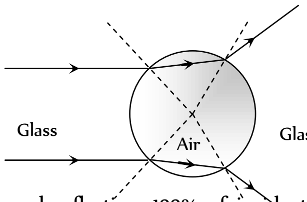
19. (d) The size of the mirror does not affect the nature of the image except that a bigger mirror forms a brighter image.
20. (a) When the sun is close to setting, refraction will effect the top part of the sun differently from the bottom half. The top half will radiate its image truly, while the bottom portion will send an apparent image. Since the bottom portion of sun is being seen through thicker, more dense atmosphere. The bottom image is being bent intensely and gives the impression of being squashed or "flattened" or elliptical shape.
21. (c) $\mu \propto \frac{1}{\lambda} \propto \frac{1}{C}$. λ_V is least so $C_{,}$ is also least. Also the greatest wavelength is for red colour.
22. (e) We can produce a real image by plane or convex mirror.



Focal length of convex mirror is taken positive.

23. (d) The colour of glowing red glass in dark will be green as red and green are complimentary colours.

24. (d) The air bubble would behave as a diverging lens, because refractive index of air is less than refractive index of glass. However, the geometrical shape of the air bubble shall resemble a double convex lens.



25. (a) In total internal reflection, 100% of incident light is reflected back into the same medium, and there is no loss of intensity, while in reflection from mirrors and refraction from lenses, there is always some loss of intensity. Therefore images formed by total internal reflection are much brighter than those formed by mirrors or lenses.

26. (d) Focal length of the lens depends upon it's refractive index as $\frac{1}{f} \propto (\mu - 1)$. Since $\mu_b > \mu_r$ so $f_b < f_r$

Therefore, the focal length of a lens decreases when red light is replaced by blue light.

27. (b) After refraction at two parallel faces of a glass slab, a ray of light emerges in a direction parallel to the direction of incidence of white light on the slab. As rays of all colours emerge in the same direction (of incidence of white light), hence there is no dispersion, but only lateral displacement.

28. (d) It is not necessary for a material to have same colour in reflected and transmitted light. A material may reflect one colour strongly and transmit some other colour. For example, some lubricating oils reflect green colour and transmit red. Therefore, in reflected light, they will appear green and in transmitted light, they will appear red.

29. (d) Dispersion of light cannot occur on passing through air contained in a hollow prism. Dispersion take place because the refractive index of medium for different colour is different. Therefore when white light travels from air to air, refractive index remains same and no dispersion occurs.

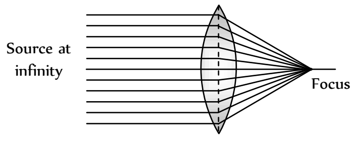
30. (b) The light gathering power (or brightness) of a telescope $\propto$ $(\text{diameter})^2$. So by increasing the objective diameter even far off stars may produce images of optimum brightness.

31. (c) Very large apertures gives blurred images because of aberrations. By reducing the aperture the clear image is obtained and thus the sensitivity of camera increases.

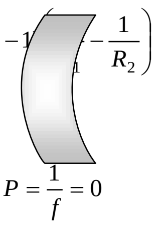
Also the focussing of object at different distance is achieved by slightly altering the separation of the lens from the film.

32. (d) We cannot interchange the objective and eye lens of a microscope to make a telescope. The reason is that the focal length of lenses in microscope are very small, of the order of *mm* or a few *cm* and the difference $(f_o - f_e)$ is very small, while the telescope objective have a very large focal length as compared to eye lens of microscope.

33. (a) Image formed by convex lens



34. (a) The focal length of a lens is given by $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

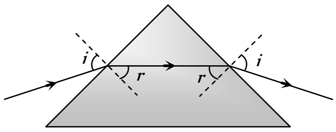


For, goggle, $R_1 = R_2$

$\therefore \ \frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = 0$. Therefore, $P = \frac{1}{f} = 0$

35. (c) The wavelength of wave associated with electrons (de Broglie waves) is less than that of visible light. We know that resolving power is inversely proportional to wavelength of wave used in microscope. Therefore the resolving power of an electron microscope is higher than that of an optical microscope.

36. (a) In case of minimum deviation of a prism $\angle i = \angle e$ so $\angle r_1 = \angle r_2$



37. (b) The velocity of light in a material medium depends upon it's colour (wavelength). If a ray of white light incident on a prism, then on emerging, the different colours are deviated through different angles.

Also dispersive power $\omega = \frac{(\mu_V - \mu_R)}{(\mu_Y - 1)}$

i.e. ω depends upon only μ.

38. (c) The ray of light incident on the water air interface suffers total internal reflections, in that case the angle of incidence is greater than the critical angle. Therefore, if the tube is viewed from suitable direction (so that the angle of incidence is greater than the critical angle), the rays of light incident on the tube undergoes total internal reflection. As a result, the test tube appears as highly polished *i.e.* silvery.

39. (a) In wide beam of light, the light rays of light which travel close to the principal axis are called paraxial rays, while the rays which travel quite away from the principal axis is called marginal rays. In case of lens having large aperture, the behaviour of the paraxial and marginal rays are markedly different from each other. The two types of rays come to focus at different points on the principal axis of the lens, thus the spherical aberration occur. However in case of a lens with small aperture, the two types of rays come to focus quite close to each other.

40. (e)

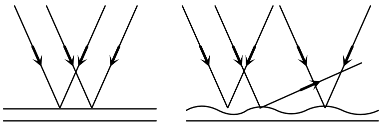
41. (b)

42. (b)

43. (c)

44. (a) Resolving power $= \dfrac{a}{1.22\lambda}$.

45. (c) When glass surface is made rough then the light falling on it is scattered in different direction due to which its transparency decreases.



Smooth surface Rough surface

46. (b) Diamond glitters brilliantly because light enters in diamond suffers total internal reflection. All the light entering in it comes out of diamond after number of reflections and no light is absorb by it.

47. (c) The clouds consist of dust particles and water droplets. Their size is very large as compared to the wavelength of the incident light from the sun. So there is very little scattering of light. Hence the light which we receive through the clouds has all the colours of light. As a result of this, we receive almost white light. Therefore, the cloud are generally white.

Ray Optics

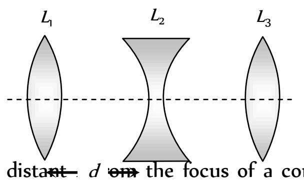
SET Self Evaluation Test - 29

1. In an astronomical telescope in normal adjustment, a straight black line of length L is drawn on the objective lens. The eyepiece forms a real image of this line. The length of this image is l. The magnification of the telescope is

(a) $\frac{L}{l}$ (b) $\frac{L}{l}+1$

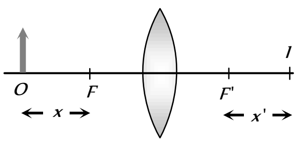
(c) $\frac{L}{l}-1$ (d) $\frac{L+l}{L-l}$

2. Three lenses L_1, L_2, L_3 are placed co-axially as shown in figure. Focal length's of lenses are given 30 cm, 10 cm and 5 cm respectively. If a parallel beam of light falling on lens L_1, emerging L_3 as a convergent beam such that it converges at the focus of L_3. Distance between L_1 and L_2 will be



(a) 40 cm
(b) 30 cm
(c) 20 cm
(d) 10 cm

3. An object is placed at a point distant ~~d cm~~ the focus of a convex lens and its image is formed at I as shown in the figure. The distances x, x' satisfy the relation



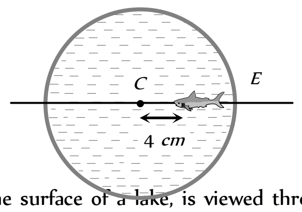
(a) $\frac{x+x'}{2}=f$
(b) $f=xx'$
(c) $x+x'\leq 2f$
(d) $x+x'\geq 2f$

4. The diameter of the eye-ball of a normal eye is about 2.5 cm. The power of the eye lens varies from

(a) 2 D to 10 D (b) 40 D to 32 D

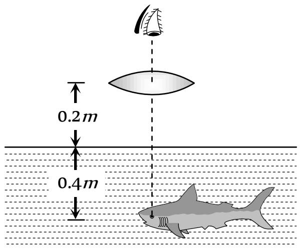
(c) 9 D to 8 D (d) 44 D to 40 D

5. In a thin spherical fish bowl of radius 10 cm filled with water of refractive index 4/3 there is a small fish at a distance of 4 cm from the centre C as shown in figure. Where will the image of fish appears, if seen from E



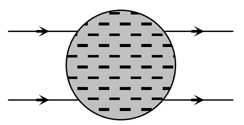
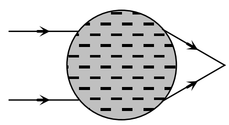
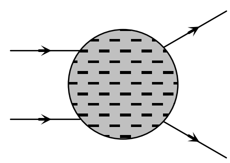
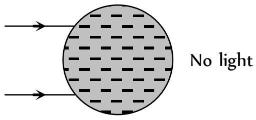
(a) 5.2 cm
(b) 7.2 cm
(c) 4.2 cm
(d) 3.2 cm

6. A small fish 0.4 m below the surface of a lake, is viewed through a simple converging lens of focal length 3 m. The lens is kept at 0.2 m above the water surface such that fish lies on the optical axis of the lens. The image of the fish seen by observer will be at $\left(\mu_{water}=\frac{4}{3}\right)$

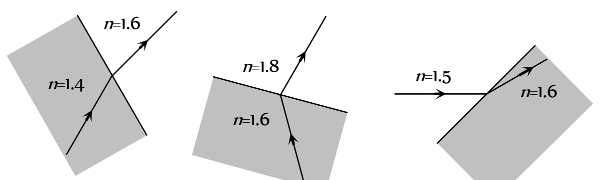


(a) A distance of 0.2 m from the water surface
(b) A distance of 0.6 m from the water surface
(c) A distance of 0.3 m from the water surface
(d) The same location of fish

7. A water drop in air refractes the light ray as

(a)

(b)

(c)

(d)


8. Which of the following ray diagram show physically possible refraction

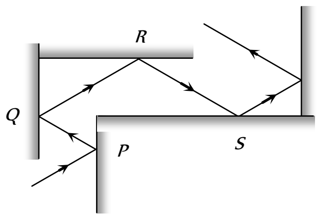


(i) (ii) (iii)

(a) (i) (b) (ii)

(c) (iii) (d) None of these

9. Following figure shows the multiple reflections of a light ray along a glass corridor where the walls are either parallel or perpendicular to one another. If the angle of incidence at point P is 30°, what are the angles of reflection of the light ray at points Q, R, S and T respectively



(a) 30°, 30°, 30°, 30°

(b) 30°, 60°, 30°, 60°

(c) 30°, 60°, 60°, 30°

(d) 60°, 60°, 60°, 60°

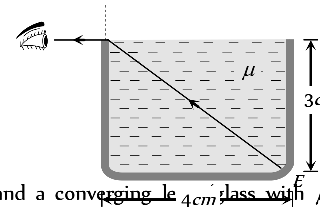
10. When the rectangular metal tank is filled to the top with an unknown liquid, as observer with eyes level with the top of the tank can just see the corner *E*; a ray that refracts towards the observer at the top surface of the liquid is shown. The refractive index of the liquid will be

(a) 1.2

(b) 1.4

(c) 1.6

(d) 1.9



11. A concave mirror and a converging lens (glass with μ = 1.5) both have a focal length of 3 *cm* when in air. When they are in water $\left(\mu = \frac{4}{3}\right)$, their new focal lengths are

(a) f_{Lens} = 12 *cm*, f_{Mirror} = 3*cm*

(b) f_{Lens} = 3 *cm*, f_{Mirror} = 12*cm*

(c) f_{Lens} = 3 *cm*, f_{Mirror} = 3*cm*

(d) f_{Lens} = 12 *cm*, f_{Mirror} = 12*cm*

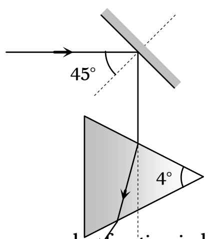
12. A ray of light strikes a plane mirror *M* at an angle of 45° as shown in the figure. After reflection, the ray passes through a prism of refractive index 1.5 whose apex angle is 4°. The total angle through which the ray is deviated is

(a) 90°

(b) 91°

(c) 92°

(d) 93°



13. A slab of glass, of thickness 6 *cm* and refractive index 1.5, is placed in front of a concave mirror, the faces of the slab being perpendicular to the principal axis of the mirror. If the radius of curvature of the mirror is 40 *cm* and the reflected image coincides with the object, then the distance of the object from the mirror is

(a) 30 *cm* (b) 22 *cm*

(c) 42 *cm* (d) 28 *cm*

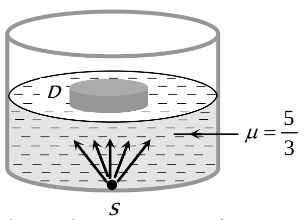
14. A point source of light *S* is placed at the bottom of a vessel containing a liquid of refractive index 5/3. A person is viewing the source from above the surface. There is an opaque disc *D* of radius 1 *cm* floating on the surface of the liquid. The centre of the disc lies vertically above the source *S*. The liquid from the vessel is gradually drained out through a tap. The maximum height of the liquid for which the source cannot be seen at all from above is

(a) 1.50 *cm*

(b) 1.64 *cm*

(c) 1.33 *cm*

(d) 1.86 *cm*



15. A point object is placed mid-way between two plane mirrors distance '*a*' apart. The plane mirror forms an infinite number of images due to multiple reflection. The distance between the *n*th order image formed in the two mirrors is

(a) *na* (b) 2*na*

(c) *na*/2 (d) $n^2 a$

16. A convergent beam of light is incident on a convex mirror so as to converge to a distance 12 *cm* from the pole of the mirror. An inverted image of the same size is formed coincident with the virtual object. What is the focal length of the mirror

(a) 24 *cm* (b) 12 *cm*

(c) 6 *cm* (d) 3 *cm*

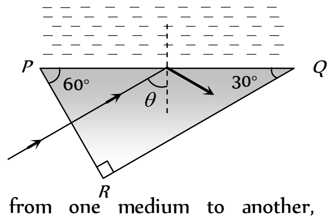
17. *PQR* is a right angled prism with other angles as 60° and 30°. Refractive index of prism is 1.5. *PQ* has a thin layer of liquid. Light falls normally on the face *PR*. For total internal reflection, maximum refractive index of liquid is

(a) 1.4

(b) 1.3

(c) 1.2

(d) 1.6



18. When a ray is refracted from one medium to another, the wavelength changes from 6000 Å to 4000 Å. The critical angle for the interface will be

(a) $\cos^{-1}\left(\frac{2}{3}\right)$ (b) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(c) $\sin^{-1}\left(\frac{2}{3}\right)$ (d) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

19. Two thin lenses, when in contact, produce a combination of power + 10 *D*. When they are 0.25 *m* apart, the power reduces to + 6*D*. The focal lengths of the lenses (in *m*) are

(a) 0.125 and 0.5 (b) 0.125 and 0.125

(c) 0.5 and 0.75 (d) 0.125 and 0.75

20. The plane faces of two identical plano convex lenses, each with focal length *f* are pressed against each other using an optical glue to form a usual convex lens. The distance from the optical centre at which an object must be placed to obtain the image same as the size of object is

(a) $\frac{f}{4}$ (b) $\frac{f}{2}$

(c) *f* (d) 2 *f*

21. A parallel beam of light emerges from the opposite surface of the sphere when a point source of light lies at the surface of the sphere. The refractive index of the sphere is

(a) $\frac{3}{2}$ (b) $\frac{5}{3}$

(c) 2 (d) $\frac{5}{2}$

22. A ray of light makes an angle of 10° with the horizontal above it and strikes a plane mirror which is inclined at an angle θ to the horizontal. The angle θ for which the reflected ray becomes vertical is

(a) 40° (b) 50°

(c) 80° (d) 100°

23. A thin rod of $5cm$ length is kept along the axis of a concave mirror of $10cm$ focal length such that its image is real and magnified and one end touches the rod. Its magnification will be

(a) 1 (b) 2

(c) 3 (d) 4

24. A telescope using light having wavelength 5000 Å and using lenses of focal 2.5 and 30 *cm*. If the diameter of the aperture of the objective is 10 *cm*, then the resolving limit and magnifying power of the telescope is respectively

(a) $6.1 \times 10^{-6} rad$ and 12

(b) $5.0 \times 10^{-6} rad$ and 12

(c) $6.1 \times 10^{-6} rad$ and 8.3×10^{-2}

(d) $5.0 \times 10^{-6} rad$ and 8.3×10^{-2}

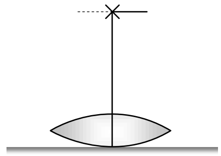
25. A lens when placed on a plane mirror then object needle and its image coincide at 15 *cm*. The focal length of the lens is

(a) 15 *cm*

(b) 30 *cm*

(c) 20 *cm*

(d) ∞



Answers and Solutions

(SET -29)

1. (a) Here we treat the line on the objective as the object and the eyepiece as the lens.

Hence $u = -(f_o + f_e)$ and $f = f_e$

Now $\frac{1}{v} - \frac{1}{-(f_o + f_e)} = \frac{1}{f_e}$

Solving we get $v = \frac{(f_o + f_e)f_e}{f_o}$

Magnification $= \left|\frac{v}{u}\right| = \frac{f_e}{f_o} = \frac{\text{Image size}}{\text{Object size}} = \frac{l}{L}$

$\therefore$ Magnification of telescope in normal adjustment

$= \frac{f_o}{f_e} = \frac{L}{l}$

2. (c) According to the problem, combination of L_1 and L_2 act a simple glass plate. Hence according to formula $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$

$\frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} = 0 \Rightarrow \frac{1}{f_1} + \frac{1}{f_2} = \frac{d}{f_1 f_2}$

$\Rightarrow \frac{1}{30} - \frac{1}{10} = \frac{d}{30 \times -10} \Rightarrow \frac{-20}{30 \times 10} = -\frac{d}{30 \times 10}$

$\Rightarrow d = 20\,cm$

3. (d) From the figure for real image formation
$x + x' + 2f \geq 4f \Rightarrow x + x' \geq 2f.$

4. (d) An eye sees distant objects with full relaxation

So $\frac{1}{2.5 \times 10^{-2}} - \frac{1}{-\infty} = \frac{1}{f}$ or $P = \frac{1}{f} = \frac{1}{25 \times 10^{-2}} = 40D$

An eye sees an object at 25 cm with strain

So $\frac{1}{2.5 \times 10^{-2}} - \frac{1}{-25 \times 10^{-2}} = \frac{1}{f}$

or $P = \frac{1}{f} = 40 + 4 = 44D$

5. (a) By using $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

where $\mu_1 = \frac{4}{3}$, $\mu_2 = 1$, $u = -6\,cm$, $v = ?$

On putting values $v = -5.2\,cm$

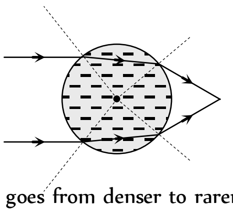
6. (d) Apparent distance of fish from lens $u = 0.2 + \frac{h}{\mu}$

$= 0.2 + \frac{0.4}{4/3} = 0.5\,m$

From $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{(+3)} = \frac{1}{v} - \frac{1}{(-0.5)}$ $v = -0.6\,m$

The image of the fish is still where the fish is 0.4 m below the water surface.

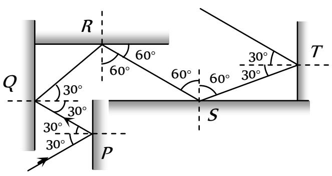
7. (b) A water drop in air behaves as converging lens.



8. (a) When light ray goes from denser to rarer medium (*i.e.* more μ to less μ) it deviates away from the normal while if light ray goes from rarer to denser medium (*i.e.* less μ more μ) it bend towards the normal.

This property is satisfying by the ray diagram (i) only.

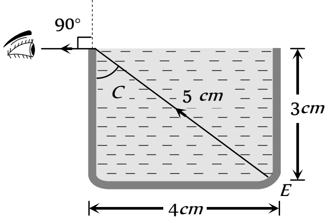
9. (c)



10. (a) Light ray is going from liquid (Denser) to air (Rarer) and angle of refraction is 90^o, so angle of incidence must be equal to critical angle

from figure

$\sin C = \frac{4}{5}$



Also $\mu = \frac{1}{\sin C} = \frac{5}{4} = 1.2$

11. (a) Focal length of lens will increase by four times (*i.e.* 12 *cm)* while focal length of mirror will not affected by medium.

12. (c) $\delta_{\text{net}} = \delta_{\text{mirror}} + \delta_{\text{prism}}$

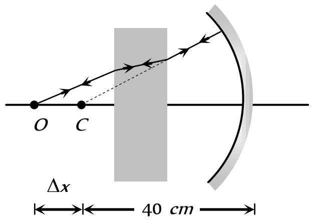
$= (180 - 2i) + (\mu - 1)A$

$= (180 - 2 \times 45) + (1.5 - 1) \times 4 = 92^o$

13. (c) $\Delta x = \left(1 - \frac{1}{\mu}\right)t$

$= \left(1 - \frac{1}{1.5}\right) \times 6$

$= 2\ cm.$



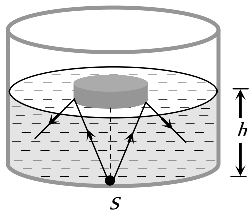
Distance of object from mirror = 42 cm.

14. (c) Suppose the maximum height of the liquid is h for which the source is not visible.

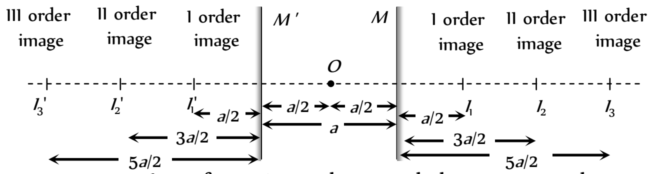
Hence radius of the disc

$r = \frac{h}{\sqrt{\mu^2 - 1}}$

$1 = \frac{h}{\sqrt{\left(\frac{5}{3}\right)^2 - 1}} \Rightarrow h = 1.33\,cm$



15. (b)

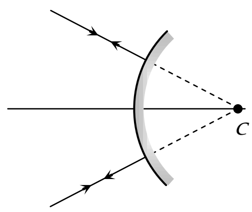


From above figure it can be proved that separation between nth order image formed in the two mirrors $= 2na$

16. (c) Here object and image are at the same position so this position must be centre of curvature

$\therefore\ R = 12\ cm$

$\Rightarrow f = \frac{R}{2}$



17. (b) For TIR at PQ; $\theta < C$

From geometry of figure $\theta = 60$ *i.e.* $60 > C$

$\Rightarrow \sin 60 > \sin C$

$\Rightarrow \frac{\sqrt{3}}{2} > \frac{\mu_{Liquid}}{\mu_{\text{Prism}}} \Rightarrow \mu_{Liquid} < \frac{\sqrt{3}}{2} \times \mu_{\text{Prism}}$

$\Rightarrow \mu_{Liquid} < \frac{\sqrt{3}}{2} \times 1.5 \Rightarrow \mu_{Liquid} < 1.3$.

18. (c) $_1\mu_2 = \dfrac{1}{\sin C} \Rightarrow \dfrac{\mu_2}{\mu_1} = \dfrac{\lambda_1}{\lambda_2} = \dfrac{1}{\sin C}$

$\Rightarrow \dfrac{6000}{4000} = \dfrac{1}{\sin C} \Rightarrow C = \sin^{-1}\left(\dfrac{2}{3}\right)$

19. (a) When lenses are in contact

$P = \dfrac{1}{F} = \dfrac{1}{f_1} + \dfrac{1}{f_2} \Rightarrow 10 = \dfrac{1}{f_1} + \dfrac{1}{f_2}$ ..... (i)

When they are distance d apart

$P' = \dfrac{1}{F'} = \dfrac{1}{f_1} + \dfrac{1}{f_2} - \dfrac{d}{f_1 f_2} \Rightarrow 6 = \dfrac{1}{f_1} + \dfrac{1}{f_2} - \dfrac{0.25}{f_1 f_2}$ ..... (ii)

From equation (i) and (ii) $f_1 f_2 = \dfrac{1}{16}$ ..... (iii)

From equation (i) and (iii) $f_1 + f_2 = \dfrac{5}{8}$ ..... (iv)

Also $(f_1 - f_2)^2 = (f_1 + f_2)^2 - 4 f_1 f_2$

Hence $(f_1 - f_2)^2 = \left(\dfrac{5}{8}\right)^2 - 4 \times \dfrac{1}{16} = \dfrac{9}{64}$

$\Rightarrow f_1 - f_2 = \dfrac{3}{8}$ ..... (v)

On solving (iv) and (v) $f_1 = 0.5\, m$ and $f_2 = 0.125\, m$

20. (c) Two plano-convex lens of focal length *f*, when combined will give rise to a convex lens of focal length $f/2$.

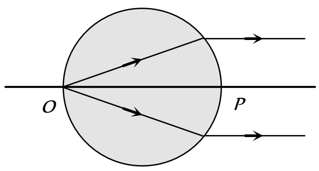
The image will be of same size if object is placed at 2*f* *i.e.* at a distance *f* from optical centre.

21. (c) Considering pole at *P*, we have

$\dfrac{\mu_2}{v} - \dfrac{\mu_1}{u} = \dfrac{\mu_2 - \mu_1}{R}$

$\Rightarrow \dfrac{1}{\infty} - \dfrac{\mu}{(-2R)} = \dfrac{1-\mu}{(-R)}$

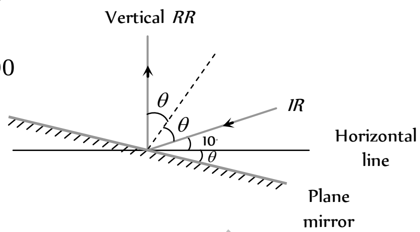
$\Rightarrow \dfrac{\mu}{2R} = \dfrac{1-\mu}{(-R)} \Rightarrow \mu = 2$



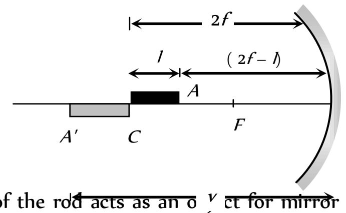
22. (a) From figure

$\theta + \theta + 10 = 90$

$\Rightarrow \theta = 40^o$



23. (b)



End *A* of the rod acts as an object for mirror and *A*' will be its image so $u = 2f - l = 20 - 5 = 15\ cm$

$\because \dfrac{1}{f} = \dfrac{1}{v} + \dfrac{1}{u} \Rightarrow \dfrac{1}{-10} = \dfrac{1}{v} - \dfrac{1}{15} \Rightarrow v = -30\ cm$.

Now $m = \dfrac{\text{Length of image}}{\text{Length of object}} = \dfrac{(30-20)}{5} = 2$

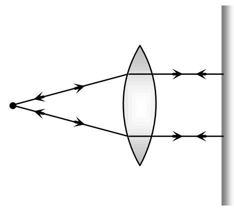
***

24. (a) $m = \dfrac{f_0}{f_e} = \dfrac{30}{2.5} = 12$

Resolving limit $= \dfrac{1.22\,\lambda}{a} = \dfrac{1.22 \times (5000 \times 10^{-10})}{0.1}$

$= 6.1 \times 10^{-6}\, rad$

25. (a) When the object is placed at focus the rays are parallel. The mirror placed normal sends them back. Hence image is formed at the object itself as illustrated in figure.



Chapter 30
Wave Optics

Newton's Corpuscular Theory

(1) Newton thought that light is made up of tiny, light and elastic particles called corpuscles which are emitted by a luminous body.

(2) The corpuscles travel with speed equal to the speed of light in all directions in straight lines.

(3) The corpuscles carry energy with them. When they strike retina of the eye, they produce sensation of vision.

(4) The corpuscles of different colour are of different sizes (red corpuscles larger than blue corpuscles).

(5) The corpuscular theory explains that light carry energy and momentum, light travels in a straight line, Propagation of light in vacuum, Laws of reflection and refraction

(6) The corpuscular theory fails to explain interference, diffraction and polarization.

(7) A major prediction of the corpuscular theory is that the speed of light in a denser medium is more than the speed of light in a rarer medium. The truth is that the speed of the light is smaller in a denser medium. Therefore, the Newton's corpuscular theory is wrong.

Huygen's Wave Theory

(1) Wave theory of light was given by Christian Huygen. According to this, a luminous body is a source of disturbance in a hypothetical medium ether. This medium pervades all space.

(2) It is assumed to be transparent and having zero inertia. The disturbance from the source is propagated in the form of waves through the space.

(3) The waves carry energy and momentum. Huygen assumed that the waves were longitudinal. Further when polarization was discovered, then to explain it, light waves were, assumed to be transverse in nature by Fresnel.

(4) This theory explains successfully, the phenomenon of interference and diffraction apart from other properties of light.

(5) The Huygen's theory fails to explain photo-electric effect, Compton's effect *etc*.

(6) The wave theory introduces the concept of wavefront.

Wavefront

(1) Suggested by Huygens

(2) The locus of all particles in a medium, vibrating in the same phase is called Wave Front (WF)

(3) The direction of propagation of light (ray of light) is perpendicular to the WF.

(4) Every point on the given wave front acts as a source of new disturbance called secondary wavelets which travel in all directions with the velocity of light in the medium.

(5) A surface touching these secondary wavelets tangentially in the forward direction at any instant gives the new wave front at that instant. This is called secondary wave front

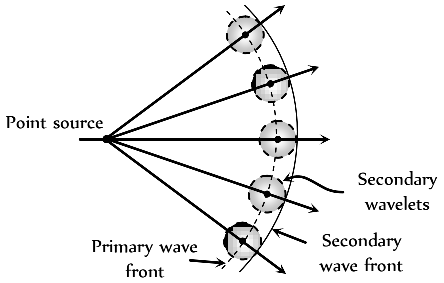
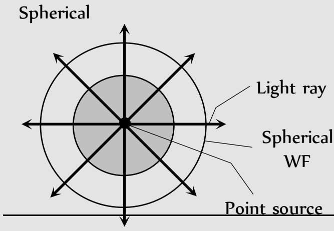
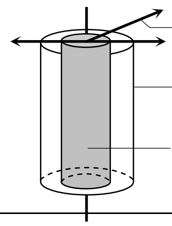
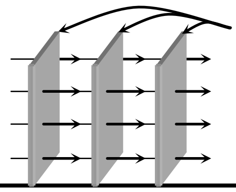


Fig. 30.1

Table 30.1 : Different types of wavefront

Type of wavefront	Intensity	Amplitude
Spherical 	$I \propto \frac{1}{r^2}$	$A \propto \frac{1}{r}$

Cylindrical 	$I \propto \frac{1}{r}$	$A \propto \frac{1}{\sqrt{r}}$
Plane 	$I \propto r^0$	$A \propto r^0$

Reflection and Refraction of Wavefront

Reflection

$BC = AD$

and $\angle i = \angle r$

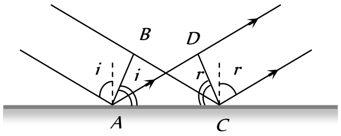


Fig. 30.2

Refraction

$$\frac{BC}{AD} = \frac{v_1}{v_2} = \frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$$

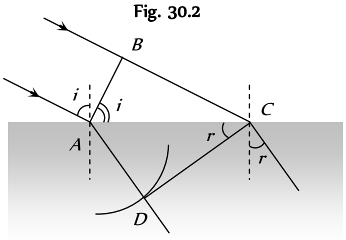


Fig. 30.3

Super Position of Waves

When two or more than two waves superimpose over each other at a common particle of the medium then the resultant displacement (y) of the particle is equal to the vector sum of the displacements (y_1 and y_2) produced by individual waves. *i.e.* $\vec{y} = \vec{y}_1 + \vec{y}_2$

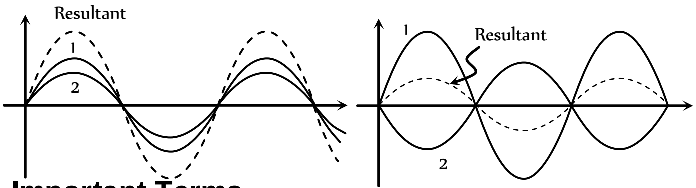


Fig. 30.4

Important Terms

(1) **Phase :** The argument of sine or cosine in the expression for displacement of a wave is defined as the phase. For displacement $y = a \sin \omega t$; term ωt = phase or instantaneous phase.

(2) **Phase difference (ϕ) :** The difference between the phases of two waves at a point is called phase difference *i.e.* if $y_1 = a_1 \sin \omega t$ and $y_2 = a_2 \sin(\omega t + \phi)$ so phase difference = ϕ

(3) **Path difference (Δ) :** The difference in path length's of two waves meeting at a point is called path difference between the waves at that point. Also $\Delta = \frac{\lambda}{2\pi} \times \phi$

(4) **Time difference (*T.D.*) :** Time difference between the waves meeting at a point is $T.D. = \frac{T}{2\pi} \times \phi$

Resultant Amplitude and Intensity

Let us consider two waves that have the same frequency but have a certain fixed (constant) phase difference between them. Their super position shown below

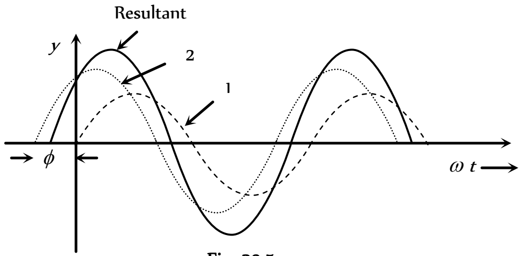


Fig. 30.5

Let the two waves are

$y_1 = a_1 \sin \omega t$ and $y_2 = a_2 \sin(\omega t + \phi)$

where a_1, a_2 = Individual amplitudes,

ϕ = Phase difference between the waves at an instant when they are meeting a point.

(1) **Resultant amplitude :** The resultant wave can be written as $y = A \sin(\omega t + \theta)$

where A = resultant amplitude $= \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos\varphi}$

(2) **Resultant intensity :** As we know intensity $\propto$ (Amplitude)^2

$\Rightarrow$ $I_1 = ka_1^2, I_2 = ka_2^2$ and $I = kA^2$ (k is a proportionality constant). Resultant intensity $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

For two identical source $I_1 = I_2 = I_0 \Rightarrow I = I_0 + I_0 + 2\sqrt{I_0 I_0}\cos\phi$

$= 4 I_0 \cos^2 \frac{\phi}{2}$ $\quad$ $[1 + \cos\theta = 2\cos^2\frac{\theta}{2}]$

Coherence

The phase relationship between two light waves can very from time to time and from point to point in space. The property of definite phase relationship is called coherence.

(1) **Temporal coherence :** In a light source a light wave (photon) is produced when an excited atom goes to the ground state and emits light.

(i) The duration of this transition is about 10^{-} to 10^{-} *sec.* Thus the emitted wave remains sinusoidal for this much time. This time is known as coherence time (τ).

(ii) Definite phase relationship is maintained for a length $L = c\tau_c$ called coherence length. For neon λ = 6328 Å, $\tau \approx 10^{-}$ *sec* and L = 0.03 *m*.

For cadmium λ = 6438 Å, $\tau = 10^{-}$ *sec* and L = 0.3 *m*

For Laser $\tau = 10^{-}$ *sec* and L = 3 *km*

(iii) The spectral lines width $\Delta\lambda$ is related to coherence length L and coherence time τ. $\Delta\lambda \approx \frac{\lambda^2}{c\tau_c}$ or $\Delta\lambda \approx \frac{\lambda^2}{L}$

(2) **Spatial coherence :** Two points in space are said to be spatially coherence if the waves reaching there maintains a constant phase difference

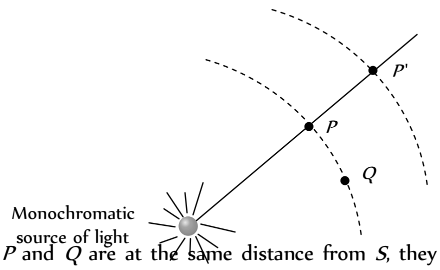


Fig. 30.6

Points P and Q are at the same distance from S, they will always be having the same phase. Points P and P' will be spatially coherent if the distance between P and P' is much less than the coherence length *i.e.* $PP' << c\tau_c$

(3) **Methods of obtaining coherent sources :** Two coherent sources are produced from a single source of light by two methods (i) By division of wavefront and (ii) By division of amplitude

(i) **Division of wave front :** The wave front emitted by a narrow source is divided in two parts by reflection, refraction or diffraction.

The coherent sources so obtained are imaginary. There produced in Fresnel's biprism, Llyod's mirror Youngs' double slit *etc.*

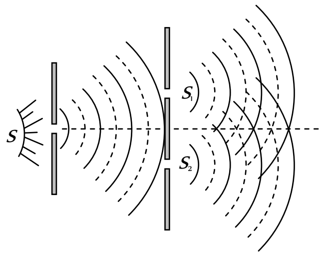


Fig. 30.7

(ii) **Division of amplitude :** In this arrangement light wave is partly reflected (50%) and partly transmitted (50%) to produced two light rays.

The amplitude of wave emitted by an extend source of light is divided in two parts by partial reflection and partial refraction.

The coherent sources obtained are real and are obtained in Newton's rings, Michelson's interferrometer, colours in thin films.

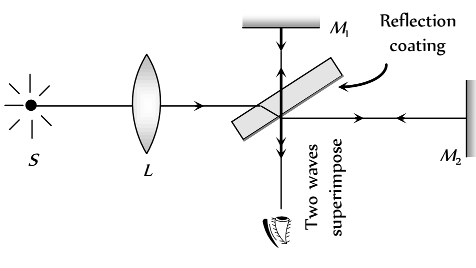
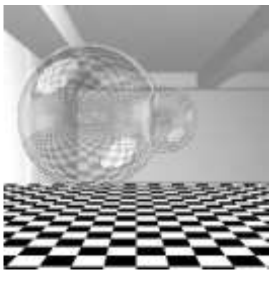


Fig. 30.8

Interference of Light

When two waves of exactly same frequency (coming from two coherent sources) travels in a medium, in the same direction simultaneously then due to their superposition, at some points intensity of light is maximum while at some other points intensity is minimum. This phenomenon is called Interference of light. It is of following two types



(1) **Constructive interference :** When the waves meets a point with same phase, constructive interference is obtained at that point (*i.e.* maximum light)

(i) Phase difference between the waves at the point of observation $\phi = 0^o$ or $2n\pi$

(ii) Path difference between the waves at the point of observation $\Delta = n\lambda$ (*i.e.* even multiple of $\lambda/2$)

(iii) Resultant amplitude at the point of observation will be maximum $A_{max} = a_1 + a_2$

If $a_1 = a_2 = a_0 \Rightarrow A_{\max} = 2a_0$

(iv) Resultant intensity at the point of observation will be maximum $I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

If $I_1 = I_2 = I_0 \Rightarrow I_{\max} = 4I_0$

(2) **Destructive interference :** When the wave meets a point with opposite phase, destructive interference is obtained at that point (*i.e.* minimum light)

(i) Phase difference $\phi = 180^o$ or $(2n-1)\pi$; n = 1, 2, ...

or $(2n+1)\pi$; $n = 0,1,2.....$

(ii) Path difference $\Delta = (2n-1)\frac{\lambda}{2}$ (*i.e.* odd multiple of $\lambda/2$)

(iii) Resultant amplitude at the point of observation will be minimum $A_{\min} = a_1 - a_2$

If $a_1 = a_2 \Rightarrow A_{\min} = 0$

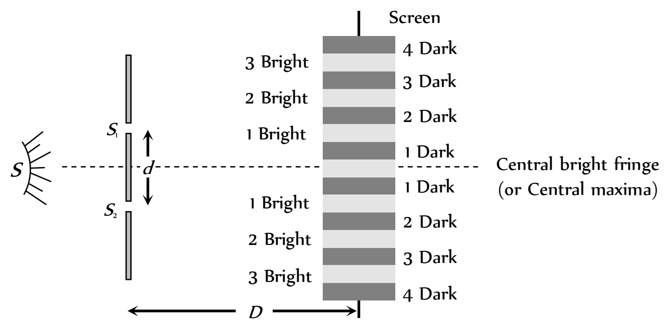
(iv) Resultant intensity at the point of observation will be minimum $I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$

If $I_1 = I_2 = I_0 \Rightarrow I_{\min} = 0$

(3) **Super position of waves of random phase difference :** When two waves (or more waves) having random phase difference between them super impose, then no interference pattern is produced. Then the resultant intensity is just the sum of the two intensities. $I = I_1 + I_2$

Young's Double Slit Experiment (YDSE)

Monochromatic light (single wavelength) falls on two narrow slits S_1 and S_2 which are very close together acts as two coherent sources, when waves coming from two coherent sources (S_1, S_2) superimposes on each other, an interference pattern is obtained on the screen. In YDSE alternate bright and dark bands obtained on the screen. These bands are called Fringes.



d = Distance between slits
D = Distance between slits and screen
λ = Wavelength of monochromatic light emitted from source

(1) Central fringe is always bright, because at central position $\phi = 0^o$ or $\Delta = 0$

(2) The fringe pattern obtained due to a slit is more bright than that due to a point.

(3) If the slit widths are unequal, the minima will not be complete dark. For very large width uniform illumination occurs.

(4) If one slit is illuminated with red light and the other slit is illuminated with blue light, no interference pattern is observed on the screen.

(5) If the two coherent sources consist of object and it's reflected image, the central fringe is dark instead of bright one.

Useful Results

(1) **Path difference :** Path difference between the interfering waves meeting at a point P on the screen is given by

$\Delta = \Delta_i + \Delta_f$; where Δ_i = initial path difference between the waves before the slits and Δ_f = path difference between the waves after emerging from the slits. In this case $\Delta_i = 0$ (Commonly used condition). So $\Delta = \Delta_f = \frac{xd}{D} = d\sin\theta$

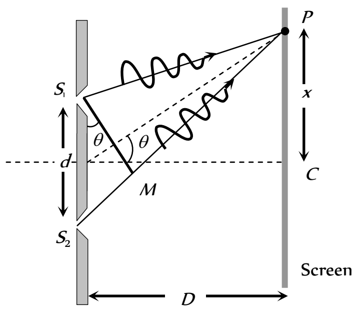


Fig. 30.10

where x is the position of point P from central maxima.

For maxima at P: $\Delta = n\lambda$; where $n = 0, \pm 1, \pm 2, \ldots$

and For minima at P: $\Delta = \frac{(2n-1)\lambda}{2}$; where $n = \pm 1, \pm 2, \ldots$

(2) **Location of fringe :** Position of n^{th} bright fringe from central maxima $x_n = \frac{n\lambda D}{d} = n\beta$; $n = 0, 1, 2 \ldots$

Position of n^{th} dark fringe from central maxima

$$x_n = \frac{(2n-1)\lambda D}{2d} = \frac{(2n-1)\beta}{2} \; ; \; n = 1, 2, 3 \ldots$$

(3) **Fringe width (β) :** The separation between any two consecutive bright or dark fringes is called fringe width. In *YDSE* all fringes are of equal width. Fringe width $\beta = \frac{\lambda D}{d}$.

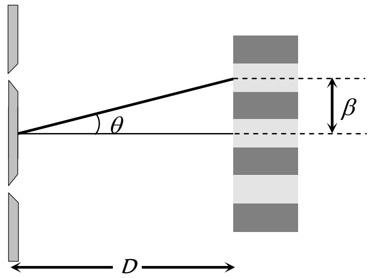


Fig. 30.11

and angular fringe width $\theta = \frac{\lambda}{d} = \frac{\beta}{D}$

(4) In YDSE, if n_1 fringes are visible in a field of view with light of wavelength λ_1, while n_2 with light of wavelength λ_2 in the same field, then $\boldsymbol{n_1\lambda_1 = n_2\lambda_2}$.

(5) **Separation (Δx) between fringes**

(i) Between n^{th} bright and m^{th} bright fringes $(n > m)$

$$\Delta x = (n-m)\beta$$

(ii) Between n^{th} bright and m^{th} dark fringe

(a) If $n > m$ then $\Delta x = \left(n - m + \frac{1}{2}\right)\beta$

(b) If $n < m$ then $\Delta x = \left(m - n - \frac{1}{2}\right)\beta$

(6) **Identification of central bright fringe :** To identify central bright fringe, monochromatic light is replaced by white light. Due to overlapping central maxima will be white with red edges. On the other side of it we shall get a few coloured band and then uniform illumination.

If the whole YDSE set up is taken in another medium then λ changes so β changes

e.g. in water $\lambda_w = \frac{\lambda_a}{\mu_w} \Rightarrow \beta_w = \frac{\beta_a}{\mu_w} = \frac{3}{4}\beta_a$

Condition for Observing Interference

(1) The initial phase difference between the interfering waves must remain constant. Otherwise the interference will not be sustained.

(2) The frequency and wavelengths of two waves should be equal. If not the phase difference will not remain constant and so the interference will not be sustained.

(3) The light must be monochromatic. This eliminates overlapping of patterns as each wavelength corresponds to one interference pattern.

(4) The amplitudes of the waves must be equal. This improves contrast with $I_{max} = 4I_0$ and $I_{min} = 0$.

(5) The sources must be close to each other. Otherwise due to small fringe width $\left(\beta \propto \frac{1}{d}\right)$ the eye can not resolve fringes resulting in uniform illumination.

Shifting of Fringe Pattern in YDSE

If a transparent thin film of mica or glass is put in the path of one of the waves, then the whole fringe pattern gets shifted towards the slit in front of which glass plate is placed.

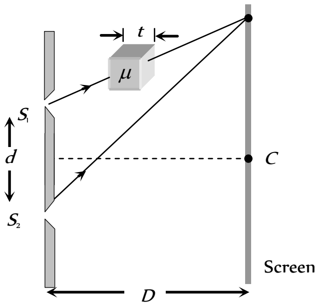


Fig. 30.12

(1) $\text{Fringe shift} = \frac{D}{d}(\mu-1)t = \frac{\beta}{\lambda}(\mu-1)t$

(2) Additional path difference $= (\mu-1)t$

(3) If shift is equivalent to n fringes then $n = \frac{(\mu-1)t}{\lambda}$ or $t = \frac{n\lambda}{(\mu-1)}$

(4) Shift is independent of the order of fringe (*i.e.* shift of zero order maxima = shift of n^{th} order maxima.

(5) Shift is independent of wavelength.

Fringe Visibility (*V*)

With the help of visibility, knowledge about coherence, fringe contrast an interference pattern is obtained.

$V = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} = 2\frac{\sqrt{I_1 I_2}}{(I_1 + I_2)}$ If $I_{min} = 0$, $V = 1$ (maximum) *i.e.*, fringe visibility will be best.

Also if $I_{max} = 0, V = -1$ and If $I_{max} = I_{min}, V = 0$

Missing Wavelength in Front of One Slit in YDSE

Suppose P is a point of observation infront of slit S_1 as shown

Missing wavelength at P

$$\lambda = \frac{d^2}{(2n-1)D}$$

By putting $n = 1, 2, 3....$

Missing wavelengths are

$$\lambda = \frac{d^2}{D}, \frac{d^2}{3D}, \frac{d^2}{5D}....$$

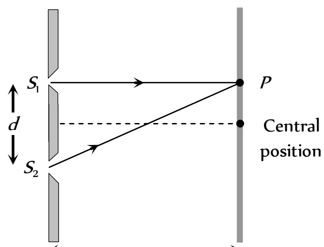
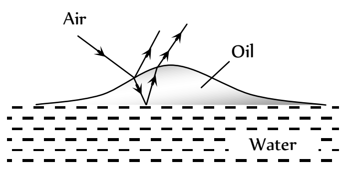


Fig. 30.13

Interference in Thin Films

Interference effects are commonly observed in thin films when their thickness is comparable to wavelength of incident light (If it is too thin as compared to wavelength of light it appears dark and if it is too thick, this will result in uniform illumination of film). Thin layer of oil on water surface and soap bubbles shows various colours in white light due to interference of waves reflected from the two surfaces of the film.



Oil film on water surface

Fig. 30.14

In thin films interference takes place between the waves reflected from it's two surfaces and waves refracted through it.

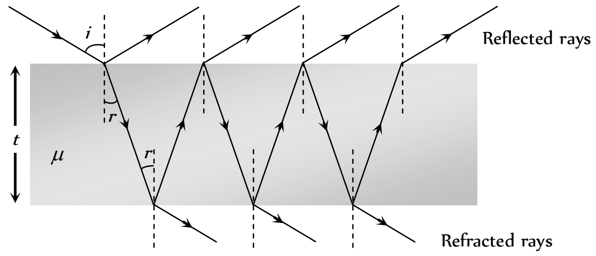


Fig. 30.15

(1) **Interference in reflected light** : Condition of constructive interference (maximum intensity)

$$\Delta = 2\mu\ t\cos r = (2n-1)\frac{\lambda}{2}.$$

For normal incidence $r = 0$ so $2\mu t = (2n-1)\frac{\lambda}{2}$

Condition of destructive interference (minimum intensity)

$\Delta = 2\mu t\cos r = (2n)\frac{\lambda}{2}$. For normal incidence $2\mu t = n\lambda$

(2) **Interference in refracted light** : Condition of constructive interference (maximum intensity)

$\Delta = 2\mu t\cos r = (2n)\frac{\lambda}{2}$. For normal incidence $2\mu t = n\lambda$

Condition of destructive interference (minimum intensity)

$$\Delta = 2\mu t\cos r = (2n-1)\frac{\lambda}{2}$$

For normal incidence $2\mu t = (2n-1)\frac{\lambda}{2}$

Lloyd's Mirror

A plane glass plate (acting as a mirror) is illuminated at almost grazing incidence by a light from a slit S_1. A virtual image S_2 of S_1 is formed closed to S_1 by reflection and these two act as coherent sources. The expression giving the fringe width is the same as for the double slit, but the fringe system differs in one important respect.

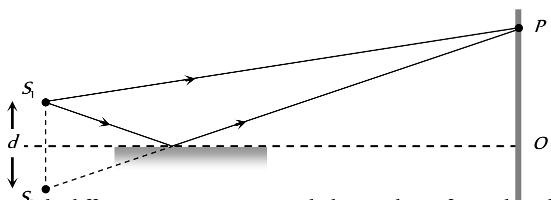


Fig. 30.16

The path difference $S_2P - S_1P$ is a whole number of wavelengths, the fringe at P is dark not bright. This is due to 180° phase change which occurs when light is reflected from a denser medium. At grazing incidence a fringe is formed at O, where the geometrical path difference between the direct and reflected waves is zero and it follows that it will be dark rather than bright.

Thus, whenever there exists a phase difference of a π between the two interfering beams of light, conditions of maximas and minimas are interchanged, *i.e.*, $\Delta x = n\lambda$ (for minimum intensity)

and $\Delta x = (2n-1)\lambda/2$ (for maximum intensity)

Fresnel's Biprims

(1) It is an optical device of producing interference of light Fresnel's biprism is made by joining base to base two thin prism of very small angle

(2) Acute angle of prism is about 1/2° and obtuse angle of prism is about 179°.

(3) When a monochromatic light source is kept in front of biprism two coherent virtual source S_1 and S_2 are produced.

(4) Interference fringes are found on the screen placed behind the biprism interference fringes are formed in the limited region which can be observed with the help eye piece.

(5) Fringe width is measured by a micrometer attached to the eye piece. Fringes are of equal width and its value is $\beta = \frac{\lambda D}{d}$

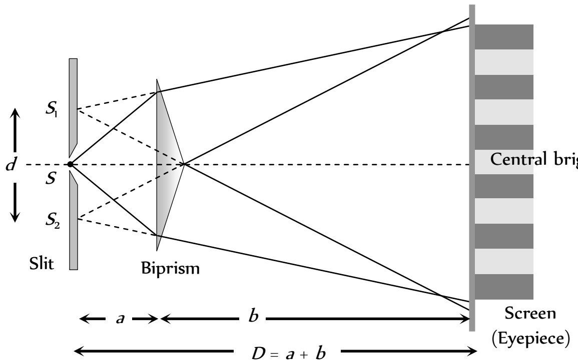


Fig. 30.17

(6) Let the separation between S_1 and S_2 be d and the distance of slits and the screen from the biprism be a and b respectively *i.e.* $D = (a + b)$. If angle of prism is α and refractive index is μ then $d = 2a(\mu - 1)\alpha$

$$\therefore \quad \lambda = \frac{\beta[2a(\mu-1)\alpha]}{(a+b)} \quad \Rightarrow \quad \beta = \frac{(a+b)\lambda}{2a(\mu-1)\alpha}$$

(7) If a convex lens is mounted between the biprism and eye piece. There will be two positions of lens when the sharp images of coherent sources will be observed in the eyepiece. The separation of the images in the two positions are measured. Let these be d_1 and d_2 then $d = \sqrt{d_1 d_2}$

$$\therefore \ \lambda = \frac{\beta d}{D} = \frac{\beta\sqrt{d_1 d_2}}{(a+b)}.$$

Newton's Rings

(1) If we place a plano-convex lens on a plane glass surface, a thin film of air is formed between the curved surface of the lens and plane glass plate.

(2) If we allow monochomatic light to fall normally on the surface of lens, then circular interference fringes of radius r can be seen in the reflected light. This circular fringes are called Newton rings.

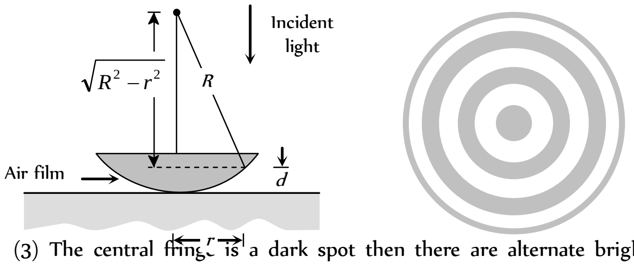


Fig. 30.18

(3) The central fringe is a dark spot then there are alternate bright and dark fringes (Ring shape).

(4) Radius of n^{th} dark ring $r_m \simeq \sqrt{n\lambda R}$

$n = 0, 1, 2, \ldots\ldots$, R = Radius of convex surface

(5) Radius of n^{th} bright ring $r_n = \sqrt{\left(n + \frac{1}{2}\right)\lambda R}$

(6) If a liquid of ref index μ is introduced between the lens and glass plate, the radii of dark ring would be $r_n = \sqrt{\frac{n\lambda R}{\mu}}$

(7) Newton's ring arrangement is used of determining the wavelength of monochromatic light. For this the diameter of n^{th} dark ring (D_n) and $(n + p)^{th}$ dark ring (D_{n+p}) are measured then

$$D^2_{(n+p)} = 4(n+p)\lambda R \text{ and } D_n^2 = 4n\lambda R \quad \Rightarrow \quad \lambda = \frac{D^2_{n+p} - D^2_n}{4pR}$$

Doppler's Effect of Light

The phenomenon of apparent change in frequency (or wavelength) of the light due to relative motion between the source of light and the observer is called Doppler's effect.

If $\nu =$ actual frequency, $\nu' =$ Apparent frequency, v = speed of source *w.r.t* stationary observer, c = speed of light

(1) **Source of light moves towards the stationary observer :** When a light source is moving towards an observer with a relative velocity v then the apparent frequency (ν') is greater than the actual frequency (ν) of light. Thus apparent wavelength (λ') is lesser the actual wavelength (λ).

$$\nu' = \nu\sqrt{\frac{(1+v/c)}{(1-v/c)}} \text{ and } \lambda' = \lambda\sqrt{\frac{(1-v/c)}{(1+v/c)}}$$

For $v << c$:

(i) Apparent frequency $\nu' = \nu\left(1 + \frac{v}{c}\right)$ and

(ii) Apparent wavelength $\lambda' = \lambda\left(1 - \frac{v}{c}\right)$

(iii) Doppler's shift : Apparent wavelength < actual wavelength,

So spectrum of the radiation from the source of light shifts towards the violet end of spectrum. This is called violet shift

Doppler's shift $\Delta\lambda = \lambda . \frac{v}{c}$

(iv) The fraction decrease in wavelength $= \frac{\Delta\lambda}{\lambda} = \frac{v}{c}$

(2) **Source of light moves away from the stationary observer :** In this case $\nu' < \nu$ and $\lambda' > \lambda$

$$\nu' = \nu\sqrt{\frac{(1-v/c)}{(1+v/c)}} \text{ and } \lambda' = \lambda\sqrt{\frac{(1+v/c)}{(1-v/c)}}$$

For $v << c$:

(i) Apparent frequency $\nu' = \nu\left(1 - \frac{v}{c}\right)$ and

(ii) Apparent wavelength $\lambda' = \lambda\left(1 + \frac{v}{c}\right)$

(iii) Doppler's shift : Apparent wavelength > actual wavelength,

So spectrum of the radiation from the source of light shifts towards the red end of spectrum. This is called red shift

Doppler's shift $\Delta\lambda = \lambda.\frac{v}{c}$

(iv) The fractional increase in wavelength $= \frac{\Delta\lambda}{\lambda} = \frac{v}{c}$.

(3) **Doppler broadening :** For a gas in a discharge tube, atoms are moving randomly in all directions. When spectrum of light emitted from these atoms is analyzed, then due to Doppler effect (because some atoms are moving towards detector, some atoms are moving away from detector), the frequency of a spectral line is not observed as having one value, but is spread over a range

$$\pm\Delta\nu = \pm\frac{v}{c}\nu, \quad \pm\Delta\lambda = \pm\frac{v}{c}\lambda$$

This broadens the spectral line by an amount $(2\Delta\lambda)$. It is called Doppler broadening. The Doppler broadening is proportional to *v*, which in turn is proportional to $\sqrt{T}$, where *T* is the temperature in Kelvin.

(4) **Radar :** Radar is a system for locating distant object by means of reflected radio waves, usually of microwave frequencies. Radar is used for navigation and guidance of aircraft, ships *etc*.,.

Radar employs the Doppler effect to distinguish between stationary and moving targets. The change in frequency between transmitted and received waves is measured. If v is the velocity of the approaching target, then the change in frequency is

$\Delta\nu = \frac{2v}{c}\nu$. (The factor of 2 arises due to refection of waves). For a receding target $\Delta\nu = -\frac{2v}{c}\nu$. (The minus sign indicates decrease in frequency).

(5) **Applications of Doppler effect**

(i) Determination of speed of moving bodies (aeroplane, submarine etc) in RADAR and SONAR.

(ii) Determination of the velocities of stars and galaxies by spectral shift.

(iii) Determination of rotational motion of sun.

(iv) Explanation of width of spectral lines.

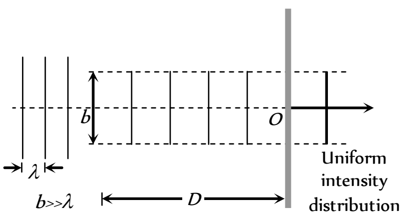
(v) Tracking of satellites.

(vi) In medical sciences in echo cardiogram, sonography *etc*.

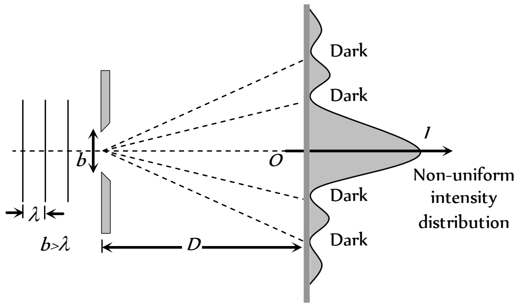
Diffraction of Light

The phenomenon of diffraction was first discovered by Girmaldi. It's experimental study was done by Newton's and young. The theoretical explanation was first given by Fresnel's.

(1) The phenomenon of bending of light around the corners of an obstacle/aperture of the size of the wave length of light is called diffraction.



(A) Size of the slit is very large compared to wavelength



(B) Size of the slit is comparable to wavelength

(2) The phenomenon resulting from the superposition of secondary wavelets originating from different parts of the same wave front is define as diffraction of light.

(3) Diffraction is the characteristic of all types of waves.

(4) Greater the wave length of wave higher will be it's degree of diffraction.

Types of Diffraction

(1) **Fresnel diffraction :** If either source or screen or both are at finite distance from the diffracting device (obstacle or aperture), the diffraction is called Fresnel type.

Common examples : Diffraction at a straight edge, narrow wire or small opaque disc *etc*.

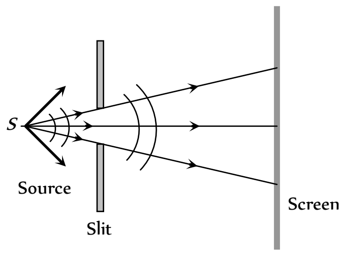


Fig. 30.20

(2) **Fraunhofer diffraction :** In this case both source and screen are effectively at infinite distance from the diffracting device.

Common examples : Diffraction at single slit, double slit and diffraction grating.

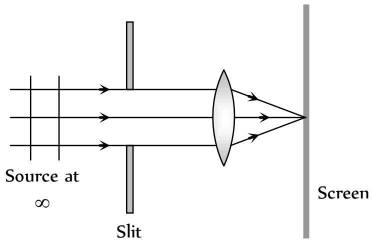
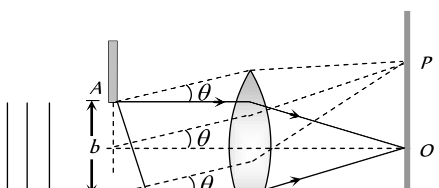


Fig. 30.21

Diffraction at Single Slit (Fraunhoffer Diffraction)

Suppose a plane wave front is incident on a slit *AB* (of width b). Each and every part of the expose part of the plane wave front (i.e. every part of the slit) acts as a source of secondary wavelets spreading in all directions. The diffraction is obtained on a screen placed at a large distance. (In practice, this condition is achieved by placing the screen at the focal plane of a converging lens placed just after the slit).



(1) The diffraction pattern consists of a central bright fringe (central maxima) surrounded by dark and bright lines (called secondary minima and maxima).

(2) At point O on the screen, the central maxima is obtained. The wavelets originating from points A and B meets in the same phase at this point, hence at O, intensity is maximum.

(3) **Secondary minima :** For obtaining n secondary minima at P on the screen, path difference between the diffracted waves $\Delta = b\sin\theta = n\lambda$

(i) Angular position of n secondary minima $\sin\theta \approx \theta = \dfrac{n\lambda}{b}$

(ii) Distance of n secondary minima from central maxima

$x_n = D.\theta = \dfrac{n\lambda D}{b} = \dfrac{n\lambda f}{b}$; where D = Distance between slit and screen. $f \approx D$ = Focal length of converging lens.

(4) **Secondary maxima :** For n secondary maxima at P on the screen.

Path difference $\Delta = b\sin\theta = (2n+1)\dfrac{\lambda}{2}$; where n = 1, 2, 3 .....

(i) Angular position of n secondary maxima

$$\sin\approx\theta \approx \frac{(2n+1)\lambda}{2b}$$

(ii) Distance of n secondary maxima from central maxima

$$x_n = D.\theta = \frac{(2n+1)\lambda D}{2b} = \frac{(2n+1)\lambda f}{2b}$$

(5) **Central maxima :** The central maxima lies between the first minima on both sides.

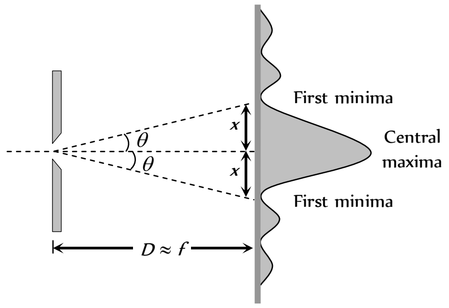
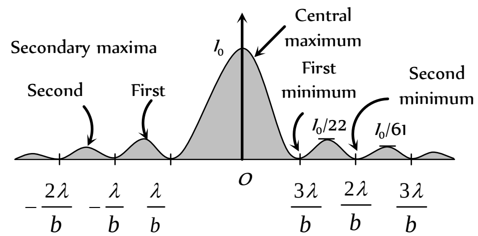


Fig. 30.23

(i) The Angular width d central maxima = $2\theta = \dfrac{2\lambda}{b}$

(ii) Linear width of central maxima $= 2x = 2D\theta = 2f\theta = \dfrac{2\lambda f}{b}$

(6) **Intensity distribution :** If the intensity of the central maxima is I then the intensity of the first and second secondary maxima are found to be $\dfrac{I_0}{22}$ and $\dfrac{I_0}{61}$. Thus diffraction fringes are of unequal width and unequal intensities.



(i) The mathematical expression for in intensity distribution on the screen is given by

$I = I_o\left(\dfrac{\sin\alpha}{\alpha}\right)^2$ where α is just a convenient connection between the angle θ that locates a point on the viewing screening and light intensity I.

ϕ = Phase difference between the top and bottom ray from the slit width b.

Also $\alpha = \dfrac{1}{2}\phi = \dfrac{\pi b}{\lambda}\sin\theta$

(ii) As the slit width increases (relative to wavelength) the width of the control diffraction maxima decreases; that is, the light undergoes less flaring by the slit. The secondary maxima also decreases in width (and becomes weaker).

(iii) If $b >> \lambda$, the secondary maxima due to the slit disappear; we then no longer have single slit diffraction.

(iv) When the slit width is reduced by a factor of 2, the amplitude of the wave at the centre of the screen is reduced by a factor of 2, so the intensity at the centre is reduced by a factor of 4.

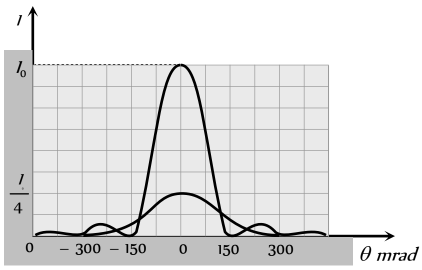


Fig. 30.25

Diffraction Gratings

One of the most useful tools in the study of light and of objects that emit and absorbs light is the diffraction grating.

(1) this device consists parallel slits of equal width and equal spacing called rulings, perhaps as many as several thousand per mm.

(2) The separation (d) between rulings is called grating spacing. (If N-rulings occupy a total width ω, then $d = \dfrac{\omega}{N}$)

(3) For light ray emerging from each slit at an angle θ, there is a path difference $d\sin\theta$, between each ray the one directly above. The d is called the grating element

$d = a + e$

where a = width of the slit

e = opaque part

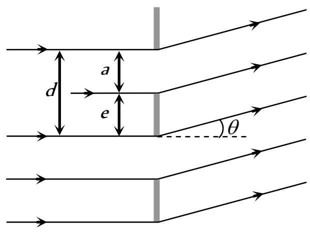


Fig. 30.26

(4) The condition for formation of bright fringe is $d \sin\theta = n\lambda$, where n = 0, 1, 2, ... is called the order of diffraction.

Fresnel's Half Period Zone (HPZ)

According to Fresnel's the entire wave front can be divided into a large number of parts of zones which are known as Fresnel's half period zones (HPZ's).

The resultant effect at any point on screen is due to the combined effect of all the secondary waves from the various zones.

Suppose ABCD is a plane wave front. We desire to find it's effect at point P consider a sphere of radius $\left(d+\frac{\lambda}{2}\right)$ with centre at P, then this sphere will cut the wave front in a circle (circle 1). This circular zone is called Fresnel's first (I) HPZ.

A sphere of radius $b+2\left(\frac{\lambda}{2}\right)$ with centre at P will cut the wave front in circle 2, the annular region between circle 2 and circle 1 is called second (II) HPZ.

The peripheral area enclosed between the n circle and $(n-1)^{\text{th}}$ circle is defined as n HPZ.

(1) **Radius of HPZ :** For n HPZ, it is given by

$$r_n = \sqrt{nd\lambda} \quad \Rightarrow r_n \propto \sqrt{\lambda}$$

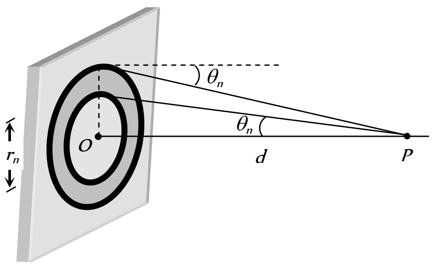


Fig. 30.28

(2) **Area of HPZ :** Area of n HPZ is given by

A_n = Area of n circle – Area of $(n-1)^{\text{th}}$ circle

$$= \pi(r_n^2 - r_{n-1}^2) = \pi d\lambda$$

(3) **Mean distance of the observation point P from n HPZ :** $d_n = \frac{r_n + r_{n-1}}{2} = b + \frac{(2n-1)\lambda}{4}$

(4) **Phase difference between the HPZ :** phase difference between the wavelets originating from two consecutive HPZ's and reaching the point P is π (or path difference is $\frac{\lambda}{2}$, time difference is $\frac{T}{2}$).

The phase difference between any two even or old number HPZ is 2π.

(5) **Amplitude of HPZ :** The amplitude of light at point P due to n HPZ is $R_n \propto \frac{A_n}{d_n}(1+\cos\theta_n)$; where A_n = Area of n HPZ, d_n = Mean distance of n HPZ

$(1+\cos\theta_n)$ = Obliquity factor.

On increasing the value of n, the value of R_n gradually goes on decreasing i.e. $R_1 > R_2 > R_3 > R_4 > ....\ .. > R_{n-1} > R_n$

(6) **Resultant Amplitude :** The wavelets from two consecutive HPZ's meets in opposite phase at P.

Hence Resultant amplitude at P

$$R = R_1 - R_2 + R_3 - R_4 + ...(-1)^{n-1} R_n$$

When $n = \infty$, then $R_{n-1} = R_n = 0$, therefore $R = \frac{R_1}{2}$

i.e. For large number of HPZ, the amplitude of light at point P due to whole wave front is half the amplitude due to first HPZ.

The ratio of amplitudes due to consecutive HPZ's is constant and is less than 1

$$\frac{R_n}{R_{n-1}} .... \frac{R_5}{R_4} = \frac{R_4}{R_3} = \frac{R_3}{R_2} = \frac{R_2}{R_1} = k \qquad (\text{where } k < 1)$$

(7) **Resultant Intensity :** Intensity $\propto$ (amplitude)^2

For $n = \infty$, $I \propto \frac{R_1^2}{4} \propto \frac{I_1}{4}$

i.e. the resultant intensity due to whole wave front is $\frac{1}{4}th$ the intensity due to first HPZ.

Diffraction Due to a Circular Disc

When a disc is placed in the path of a light beam, then diffraction pattern is formed on the screen.

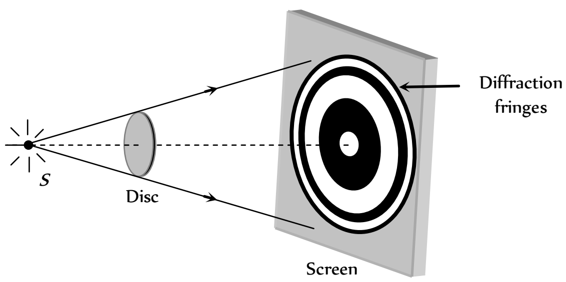


Fig. 30.29

(1) At the centre of the circular shadow of disc, there occurs a bright spot. This spot is called Fresnel's spot or Poisson's spot.

(2) The intensity of bright spot decreases, when the size of the disc is increased or when the screen is moved towards the disc.

(3) Circular alternate bright and dark fringes are formed around the bright spot with fringe width in decreasing order.

(4) Let r be the radius of the disc, d is the distance between screen and the disc and λ is the wavelength of light used.

If n HPZ are covered by disc then $nd\lambda = \pi r^2 \Rightarrow n = \frac{r^2}{d\lambda}$

(5) If the disc obstruct only first HPZ, the resultant amplitude at the central point $R = -R_2 + R_3 + ...\ \approx -\frac{R_2}{2}$.

So intensity is $\frac{kR_2^2}{4}$ which is slightly less than the intensity $\frac{k\,R_1^2}{4}$ due to whole wave front, when no obstacle is placed.

(6) The intensity at bright spot is given by $I = k\left[\frac{R_{n+1}}{2}\right]^2$

where n = Number of obstructed HPZ's

Diffraction Due to a Circular Aperture

When a circular aperture is placed in the path of a light beam, then following diffraction pattern is formed on the screen.

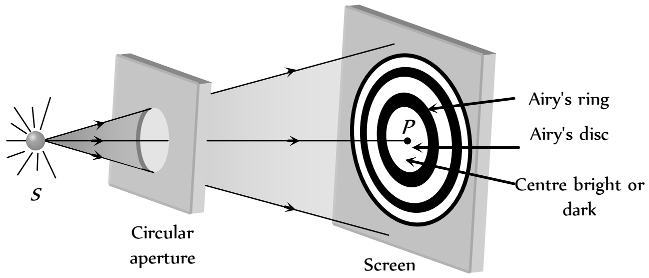
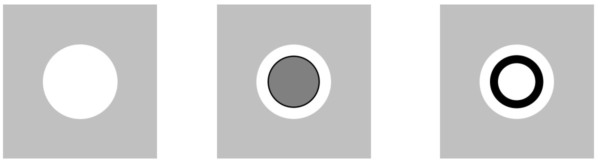


Fig. 30.30

(1) If only one HPZ is allowed by the aperture then the resultant amplitude at P would be R_1 which is twice the value of amplitude for the unobstructed wave front. The intensity would there fore be $4I$, where I represents the intensity at point P, due to unobstructed wave front.

(2) If the first two HPZ's are permitted by aperture than the resultant intensity at the centre point P will be very small (as $R_1 - R_2 \approx 0$). In this case the diffraction pattern consist of a bright circle of light with a dark spot.

(3) In general if number of HPZ's (n) passing through aperture is odd, then the central point will be bright and if n is even, central point will be dark.



(A) n=1, r^2=$b\lambda$ bright centre | (B) n=2, r^2=2$b\lambda$ dark centre | (C) n=3, r^2=3$b\lambda$ bright centre

Fig. 30.31

(4) The central bright disc is known as Airy's disc.

(5) In the non axial region bright and dark diffraction rings are obtained. The intensity of bright diffraction rings gradually goes on decreasing whereas that of dark diffraction goes on increasing.

(6) The first dark ring obtained around the central bright disc is known as Airy's ring.

Zone Plate

It is a diffracting device used to experimentally demonstrate the diffraction effect.

(1) It is formed on a glass plate by drawing a number of concentric circles on it whose radii are in the ratio of

$$\sqrt{1} : \sqrt{2} : \sqrt{3}.........\ .i.e.\ r \propto \sqrt{n}$$

For some specific distance from this plate the circles coincides with the HPZ's of the Fresnel's theory. (Alternate zones are made opaque).

(2) **Positive zone plate :** When odd zones are kept transparent to the light and even zones are made opaque, then it is called positive zone plate.

The resultant amplitude due to this zone plate in

$$R = R_1 + R_3 + R_5 + ...\ >> \frac{R_1}{2}$$

Thus, intensity of light tremendously increases.

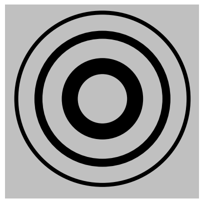


Fig. 30.32

(3) **Negative zone plate :** when even zones are kept transparent to light and odd zones are made opaque, then it is called negative zone plate.

The resultant amplitude due to this zone plate is

$$R = R_2 + R_4 + R_6 + ...\ >> \frac{R_1}{2}$$

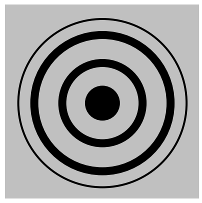


Fig. 30.33

(4) Zone plate behaves like a convex lens. For a plane wave front the image of source is formed at distance d *i.e.* d is equal to the principle focal length or first focal length $f_1 = d = \frac{r^2}{\lambda}$

(5) Multiple focii of zone plate are given by $f_p = \frac{r^2}{(2p-1)\lambda}$ where p = 1, 2, 3,......... represents the order of focii

(6) If the radius of n circle on zone plate is r_n then in terms of r_n.

Principal focal length $f_1 = \frac{r_n^2}{n\lambda}$

Other focal length $f_p = \frac{r_n^2}{(2p-1)n\lambda}$

(7) If a is the distance of the source from the zone plate then the distance b of the point where maximum intensity is observes is given by $\frac{1}{a} + \frac{1}{b} = \frac{n\lambda}{r_n^2}$

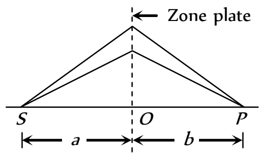


Fig. 30.34

Polarisation of Light

Light propagates as transverse EM waves. The magnitude of electric field is much larger as compared to magnitude of magnetic field. We generally prefer to describe light as electric field oscillations.

(1) **Unpolarised light :** In ordinary light (light from sun, bulb *etc.*) the electric field vectors are distributed in all directions in a light is called unpolarised light. The oscillation of propagation of light wave. This resolved into horizontal and vertical component.

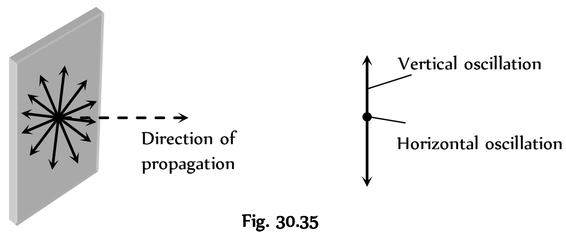


Fig. 30.35

(2) **Polarised light :** The phenomenon of limiting the vibrating of electric field vector in one direction in a plane perpendicular to the direction of propagation of light wave is called polarization of light.

(i) The plane in which oscillation occurs in the polarised light is called plane of oscillation.

(ii) The plane perpendicular to the plane of oscillation is called plane of polarisation.

(iii) Light can be polarised by transmitting through certain crystals such as tourmaline or polaroids.

(3) **Polaroids :** It is a device used to produce the plane polarised light. It is based on the principle of selective absorption and is more effective than the tourmaline crystal. or

It is a thin film of ultramicroscopic crystals of quinine idosulphate with their optic axis parallel to each other.

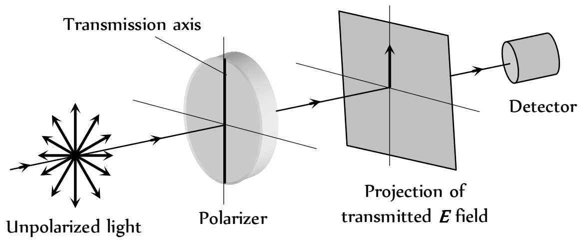
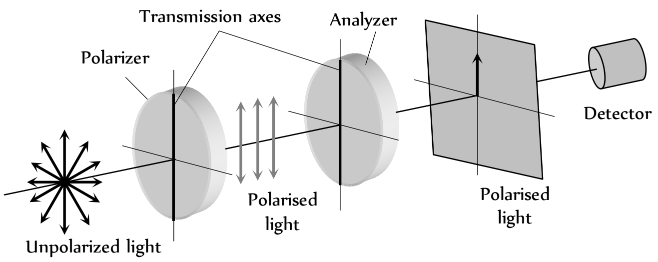


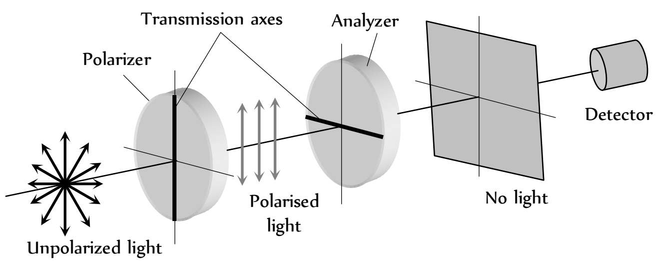
Fig. 30.36

(i) Polaroids allow the light oscillations parallel to the transmission axis pass through them.

(ii) The crystal or polaroid on which unpolarised light is incident is called polariser. Crystal or polaroid on which polarised light is incident is called analyser.



(A) Transmission axes of the polariser and analyser are parallel to each other, so whole of the polarised light passes through analyser



(B) Transmission axis of the analyser is perpendicular to the polariser, hence no light passes through the analyser

Fig. 30.37

(4) **Malus law :** This law states that the intensity of the polarised light transmitted through the analyser varies as the square of the cosine of the angle between the plane of transmission of the analyser and the plane of the polariser.

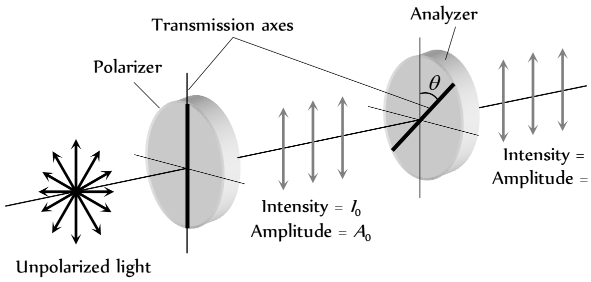


Fig. 30.38

(i) $I = I_0 \cos^2\theta$ and $A^2 = A_0^2\cos^2\theta \Rightarrow A = A_0\cos\theta$

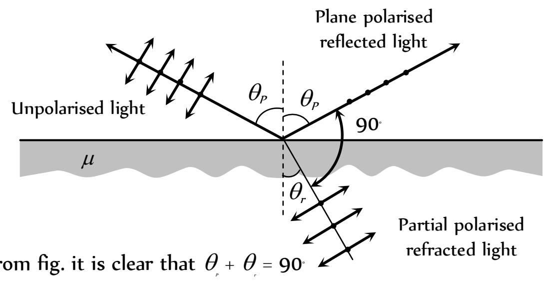
If $\theta = 0^o$, $I = I_0$, $A = A_0$, If $\theta = 90^o$, $I = 0$, $A = 0$

(ii) If $I_i =$ Intensity of unpolarised light.

So $I_0 = \dfrac{I_i}{2}$ *i.e.* if an unpolarised light is converted into plane polarised light (say by passing it through a Polaroid or a Nicol-prism), its intensity becomes half. and $I = \dfrac{I_i}{2}\cos^2\theta$

Methods of Producing Polarised Light

(1) **Polarisation by reflection :** Brewster discovered that when a beam of unpolarised light is reflected from a transparent medium (refractive index $=\mu$), the reflected light is completely plane polarised at a certain angle of incidence (called the angle of polarisation θ_p).



From fig. it is clear that $\theta_r + \theta_p = 90°$

Also $\mu = \tan\theta_p$ Brewster's law

Fig. 30.39

(i) For $i < \theta_p$ or $i > \theta_p$

Both reflected and refracted rays becomes partially polarised

(ii) For glass $\theta_P \approx 57^o$, for water $\theta_P \approx 53^o$

(2) **By Dichroism :** Some crystals such as tourmaline and sheets of iodosulphate of quinine have the property of strongly absorbing the light with vibrations perpendicular to a specific direction (called transmission axis) transmitting the light with vibrations parallel to it. This selective absorption of light is called dichroism.

(3) **By double refraction :** In certain crystals, like calcite, quartz and tourmaline *etc*, incident unpolarized light splits up into two light beams of equal intensities with perpendicular polarization.

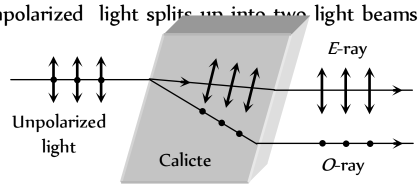


Fig. 30.40

(i) One of the ray is ordinary ray (O-ray) it obey's the Snell's law. Another ray's extra ordinary ray (*E*-ray) it doesn't obey's the Snell's law.

(ii) Along a particular direction (fixed in the crystal, the two velocities (velocity of *O*-ray v_o and velocity of *E*-ray v_e) are equal; this direction is known as the optic axis of the crystal (crystal's known as uniaxial crystal). Optic axis is a direction and not any line in crystal.

(iii) In the direction, perpendicular to the optic axis for negative crystal (calcite) $v_e > v_o$ and $\mu_e < \mu_o$.

For positive crystal $v_e < v_o$, $\mu_e > \mu_o$.

(4) **Nicol prism :** Nicol prism is made up of calcite crystal and in it *E*-ray is isolated from *O*-ray through total internal reflection of *O*-ray at canada balsam layer and then absorbing it at the blackened surface as shown in fig.

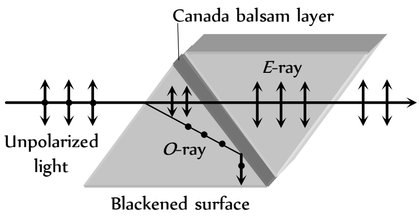


Fig. 30.41

The refractive index for the *O*-ray is more that for the *E*-ray. The refractive index of Canada balsam lies between the refractive indices of calcite for the *O*-ray and *E*-ray

(5) **By Scattering :** It is found that scattered light in directions perpendicular to the direction of incident light is completely plane polarised while transmitted light is unpolarised. Light in all other directions is partially polarised.

(6) **Optical activity and specific rotation :** When plane polarised light passes through certain substances, the plane of polarisation of the light is rotated about the direction of propagation of light through a certain angle. This phenomenon is called optical activity or optical rotation and the substances optically active.

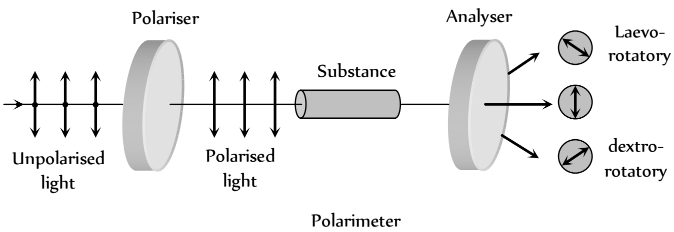


Fig. 30.42

If the optically active substance rotates the plane of polarisation clockwise (looking against the direction of light), it is said to be *dextro-rotatory* or *right-handed.* However, if the substance rotates the plane of polarisation anti-clockwise, it is called *laevo-rotatory* or *left-handed.*

The optical activity of a substance is related to the asymmetry of the molecule or crystal as a whole, *e.g.,* a solution of cane-sugar is dextro-rotatory due to asymmetrical molecular structure while crystals of quartz are dextro or laevo-rotatory due to structural asymmetry which vanishes when quartz is fused.

Optical activity of a substance is measured with help of polarimeter in terms of 'specific rotation' which is defined as the rotation produced by a solution of length 10 *cm* (1 *dm*) and of unit concentration (*i.e.* 1 *g*/cc) for a given wavelength of light at a given temperature. *i.e.* $[\alpha]^{\lambda}_{t^o C} = \dfrac{\theta}{L \times C}$

where θ is the rotation in length L at concentration C.

(7) **Applications and uses of polarisation**

(i) By determining the polarising angle and using Brewster's law, *i.e.* $\mu = \tan\theta$, refractive index of dark transparent substance can be determined.

(ii) It is used to reduce glare.

(iii) In calculators and watches, numbers and letters are formed by liquid crystals through polarisation of light called liquid crystal display (LCD).

(iv) In CD player polarised laser beam acts as needle for producing sound from compact disc which is an encoded digital format.

(v) It has also been used in recording and reproducing three-dimensional pictures.

(vi) Polarisation of scattered sunlight is used for navigation in solar-compass in polar regions.

(vii) Polarised light is used in optical stress analysis known as 'photoelasticity'.

(viii) Polarisation is also used to study asymmetries in molecules and crystals through the phenomenon of 'optical activity'.

(ix) A polarised light is used to study surface of nucleic acids (DNA, RNA)

Electromagnetic Waves

A changing electric field produces a changing magnetic field and vice versa which gives rise to a transverse wave known as electromagnetic wave. The time varying electric and magnetic field are mutually perpendicular to each other and also perpendicular to the direction of propagation of this wave.

The electric vector is responsible for the optical effects of an EM wave and is called the *light vector.*

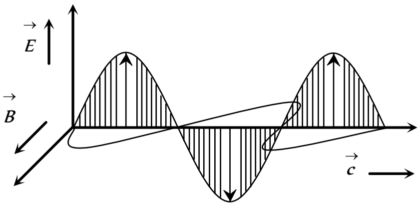


Fig. 30.43

(1) $\vec{E}$ and $\vec{B}$ always oscillates in phase.

(2) $\vec{E}$ and $\vec{B}$ are such that $\vec{E} \times \vec{B}$ is always in the direction of propagation of wave.

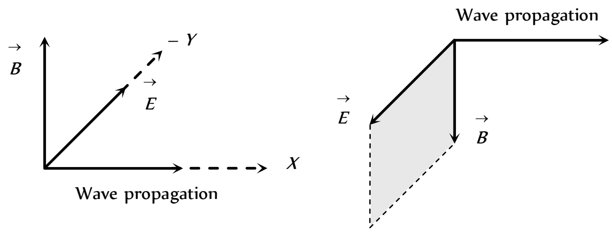


Fig. 30.44

(3) The EM wave propagating in the positive *x*-direction may be represented by

$$E = E_y = E_0 \sin(kx - \omega t)$$

$$B = B_z = B_0 \sin(kx - \omega t)$$

where E (or E_y), B (or B_z) are the instantaneous values of the fields, E_0, B_0 are amplitude of the fields and K = angular wave number $= \dfrac{2\pi}{\lambda}$.

Maxwell's Contribution

(1) **Ampere's Circuital law :** According to this law the line integral of magnetic field along any closed path or circuit is μ_0 times the total current threading the closed circuit *i.e.,* $\oint \vec{B}.\vec{dl} = \mu_0 i$

(2) **Inconsistency of Ampere's law :** Maxwell explained that Ampere's law is valid only for steady current or when the electric field does not change with time. To see this inconsistency consider a parallel plate capacitor being charged by a battery. During the charging time varying current flows through connecting wires.

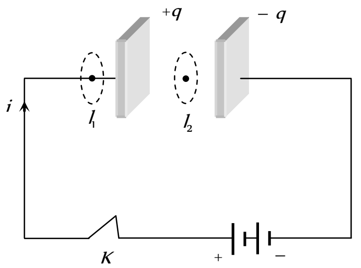


Fig. 30.45

Applying Ampere's law for loop l_1 and l $\oint_{l_1} \vec{B}.\vec{dl} = \mu_0 i$

But $\oint_{l_2} \vec{B}.\vec{dl} = 0$ (Since no current flows through the region between the plates). But practically it is observed that there is a magnetic field between the plates. Hence Ampere's law fails

i.e. $\oint_{l_1} \vec{B}.\vec{dl} \neq \mu_0 i$.

(3) **Modified Ampere's Circuital law or Ampere- Maxwell's Circuital law :** Maxwell assumed that some sort of current must be flowing between the capacitor plates during charging process. He named it displacement current. Hence modified law is as follows

$$\oint \vec{B}.\vec{dl} = \mu_0 (i_c + i_d) \text{ or } \oint \vec{B}.\vec{dl} = \mu_0 (i_c + \varepsilon_0 \frac{d\phi_E}{dt})$$

where i_c = conduction current = current due to flow of charges in a conductor and

i_d = Displacement current = $\varepsilon_0 \frac{d\phi_E}{dt}$ = current due to the changing electric field between the plates of the capacitor

(4) **Maxwell's equations**

(i) $\oint_s \vec{E}.\vec{ds} = \frac{q}{\varepsilon_0}$ (Gauss's law in electrostatics)

(ii) $\oint_s \vec{B}.\vec{ds} = 0$ (Gauss's law in magnetism)

(iii) $\oint \vec{B}.\vec{dl} = -\frac{d\phi_B}{dt}$ (Faraday's law of EMI)

(iv) $\oint \vec{B}\ \vec{dl} = \mu_o (i_c + \varepsilon_o \frac{d\phi_E}{dt}$ (Maxwell- Ampere's Circuital law)

History of EM Waves

(1) **Maxwell :** Was the first to predict the EM wave.

(2) **Hertz :** Produced and detected electromagnetic waves experimentally at wavelengths of 6 *m*.

(3) **J.C. Bose :** Produced EM waves of wavelength ranging from 5*mm* to 25 *mm*.

(4) **Marconi :** Successfully transmitted the EM waves up to a few *kilometer*. Marconi discovered that if one of the spark gap terminals is connected to an antenna and the other terminal is Earthed, the electromagnetic waves radiated could go upto several kilometers.

Experimental Setup for Producing EM Waves

Hertz experiment based on the fact that a oscillating charge is accelerating continuously, it will radiate electromagnetic waves continuously. In the following figure

(1) The metallic plates (P_1 and P_2) acts as a capacitor.

(2) The wires connecting spheres S_1 and S_2 to the plates provide a low inductance.

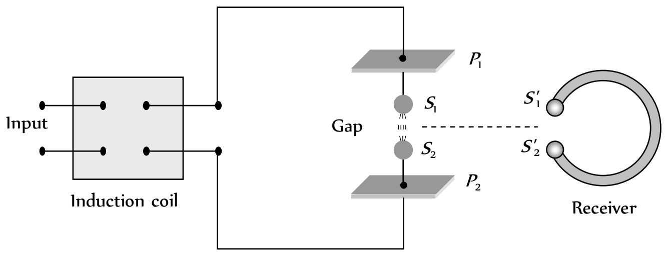


Fig. 30.46

(3) When a high voltage is applied across metallic plates these plates get discharged by sparking across the narrow gap. The spark will give rise to oscillations which in turn send out electromagnetic waves. Frequency of these wave is given by $\nu = \frac{1}{2\pi\sqrt{LC}}$

The succession of sparks send out a train of such waves which are received by the receiver.

Source, Production and Nature of EM Waves

(1) A charge oscillating harmonically is a source of EM waves of same frequency.

(2) A simple *LC* oscillator and energy source can produce waves of desired frequency $\left(\nu = \frac{1}{2\pi\sqrt{LC}}\right)$.

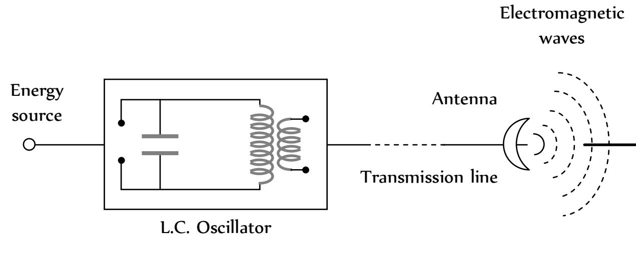


Fig. 30.47

(3) The EM Waves are transverse in nature. They do not require any material medium for their propagation.

Properties of EM Waves

(1) **Speed :** In free space it's speed

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E_0}{B_0} = 3 \times 10^8 m/s.$$

In medium $v = \frac{1}{\sqrt{\mu\varepsilon}}$; where μ_0 = Absolute permeability, ε = Absolute permittivity.

(2) **Energy :** The energy in an EM waves is divided equally between the electric and magnetic fields.

Energy density of electric field $u_e = \frac{1}{2}\varepsilon_0 E^2$, Energy density of magnetic field $u_B = \frac{1}{2}\frac{B^2}{\mu_0}$

The total energy per unit volume is $u = u_e + u_m = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0}$.

Also $u_{av} = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0}$

(3) **Intensity (*I*) :** The energy crossing per unit area per unit time, perpendicular to the direction of propagation of EM wave is called intensity.

i.e. $I = \frac{\text{Total EM energy}}{\text{Surface area} \times \text{Time}} = \frac{\text{Total energy density} \times \text{Volume}}{\text{Surface area} \times \text{Time}}$

$\Rightarrow I = u_{av} \times c = \frac{1}{2}\varepsilon_0 E_0^2 c = \frac{1}{2}\frac{B_0^2}{\mu_0}.c\,\frac{Watt}{m^2}$.

(4) **Momentum :** EM waves also carries momentum, if a portion of EM wave of energy *u* propagating with speed *c*, then linear momentum $= \frac{\text{Energy}(u)}{\text{Speed }(c)}$

If wave incident on a completely absorbing surface then momentum delivered $p = \frac{u}{c}$. If wave incident on a totally reflecting surface then momentum delivered $-p = \frac{2u}{c}$.

(5) **Poynting vector** $(\vec{S})$. **:** In EM waves, the rate of flow of energy crossing a unit area is described by the Poynting vector.

(i) It's unit is $Watt/m^2$ and $\vec{S} = \frac{1}{\mu_o}(\vec{E}\times\vec{B}) = c^2\varepsilon_0(\vec{E}\times\vec{B})$.

(ii) Because in EM waves $\vec{E}$ and $\vec{B}$ are perpendicular to each other, the magnitude of $\vec{S}$ is $|\vec{S}| = \frac{1}{\mu_0}E\,B\sin 90^o = \frac{EB}{\mu_0} = \frac{E^2}{\mu C}$.

(iii) The direction of $\vec{S}$ does not oscillate but it's magnitude varies between zero and a maximum $\left(S_{\max} = \frac{E_0 B_0}{\mu_0}\right)$ each quarter of a period.

(iv) Average value of poynting vector is given by

$$\overline{S} = \frac{1}{2\mu_0}E_0 B_0 = \frac{1}{2}\varepsilon_0 E_0^2 c = \frac{cB_0^2}{2\mu_0}$$

The direction of the poynting vector $\vec{S}$ at any point gives the wave's direction of travel and direction of energy transport the point.

(6) **Radiation pressure :** Is the momentum imparted per second pre unit area. On which the light falls.

For a perfectly reflecting surface $P_r = \frac{2S}{c}$; S = Poynting vector; c = Speed of light

For a perfectly absorbing surface $P_a = \frac{S}{c}$.

(7) **Wave impedance (*Z*) :** The medium offers hindrance to the propagation of wave. Such hindrance is called wave impedance and it is given by $Z = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_r}{\varepsilon_r}}\sqrt{\frac{\mu_0}{\varepsilon_0}}$

For vacuum or free space $Z = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 376.6\,\Omega$.

EM Spectrum

The whole orderly range of frequencies/wavelengths of the EM waves is known as the EM spectrum.

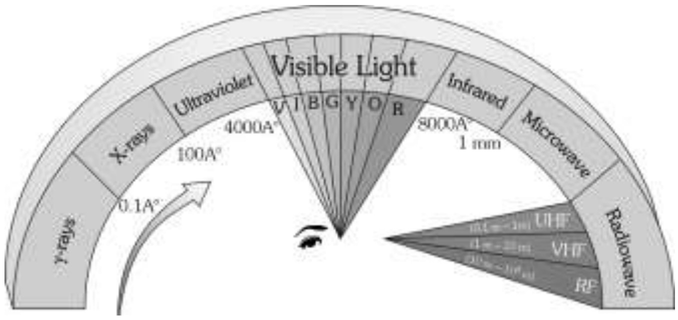


Fig. 30.48

Table 30.2 : Uses of EM spectrum

Radiation	Uses
γ-rays	Gives informations on nuclear structure, medical treatment *etc.*
X-rays	Medical diagnosis and treatment study of crystal structure, industrial radiograph.
UV- rays	Preserve food, sterilizing the surgical instruments, detecting the invisible writings, finger prints etc.
Visible light	To see objects
Infrared rays	To treat, muscular strain for taking photography during the fog, haze etc.
Micro wave and radio wave	In radar and telecommunication.

Earth's Atmosphere

The gaseous envelope surrounding the earth is called it's atmosphere. The atmosphere contains 78% N_2, 21% O_2, and traces of other gases (like helium, krypton, CO_2 etc.)

(1) **Division of earth's atmosphere :** Earth atmosphere has been divided into regions as shown.

(i) Troposphere : In this region, the temperature decreases with height from 290 *K* to 220 *K*.

(ii) Stratosphere : The temperature of stratosphere varies from 220 *K* to 200 *K*.

(iii) Mesosphere : In this region, the temperature falls to 180 *K*.

(iv) Ionosphere : Ionosphere is partly composed of charged particles, ions and electrons, while the rest of the atmosphere contains neutral molecules.

(v) Ozone layer absorbs most of the ultraviolet rays emitted by the sun.

(vi) Kennelly heaviside layer lies at about 110km from the earth's surface. In this layer concentration of electron is very high.

(vii) The ionosphere plays a vital role in the radio communication.

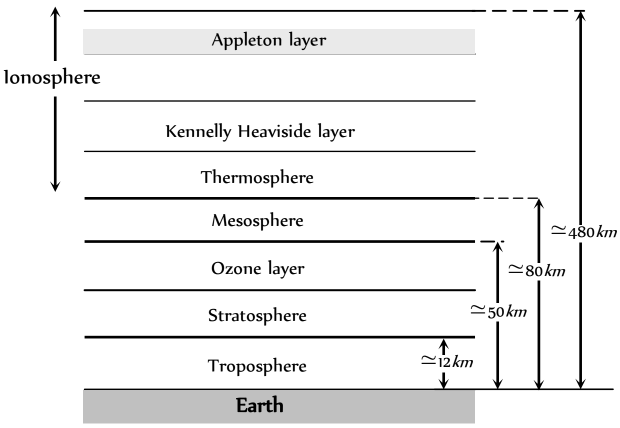


Fig. 30.49

(2) **Green house effect :** The warming of earth's atmosphere due to the infrared radiations reflected by low lying clouds and carbon dioxide in the atmosphere of earth is called green house effect.

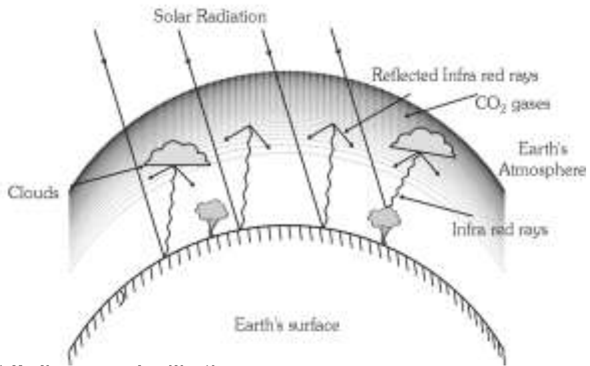


Fig. 30.50

(i) **Radio waves classification**

(a) Very low frequency (VLF) $\rightarrow$ 10 KHz to 30 KHz

(b) Low frequency (LF) $\rightarrow$ 30 KHz to 300 KHz

(c) Medium frequency (MF) or medium wave (MW) $\rightarrow$ 300 KHz to 3000 KHz

(d) High frequency (HF) or short wave (SW) $\rightarrow$ 3 MHz to 30 MHz

(e) Very high frequency (VHF) $\rightarrow$ 30 MHz to 300 MHz

(f) Ultra high frequency (UHF) $\rightarrow$ 300 MHz to 3000 MHz

(g) Super high frequency or micro waves $\rightarrow$ 3000 MHz to 300, 000 MHz

(ii) **Amplitude modulated transmission :** Radio waves having frequency less than or equal to 30 MHz form an amplitude modulation band (or AM band). The signals can be transmitted from one place to another place on earth's surface in two ways

(a) Ground wave propagation : The radio waves following the surface of the earth are called ground waves.

(b) Sky wave propagation : The amplitude modulated radio waves which are reflected back by the ionosphere are called sky waves.

(iii) **Frequency modulated (FM) transmission :** Radio waves having frequencies between 80 MHz and 200 MHz form a frequency modulated bond. T.V. signals are normally frequency modulated.

(4) **T.V. Signals**

(i) T.V. signals are normally frequency modulated. So T.V. signals can be transmitted by using tall antennas.

(ii) Distance covered by the T.V. signals $d = \sqrt{2hR}$

(h = Height of the antenna, R = Radius of earth)

(iii) Area covered $A = \pi d^2 = 2\pi hR$

(iv) Population covered = Area $\times$ Population density.

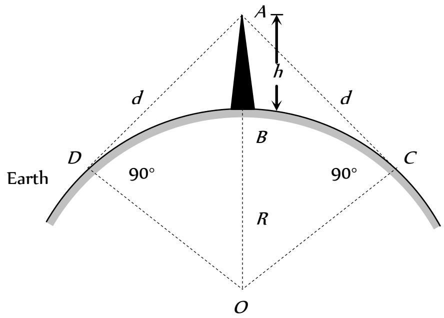


Fig. 30.51

Tips & Tricks

✍ In interference redistribution of energy takes place in the form of maxima and minima.

✍ Average intensity : $I_{av} = \dfrac{I_{\max} + I_{\min}}{2} = I_1 + I_2 = a_1^2 + a_2^2$

✍ Ratio of maximum and minimum intensities :

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}}\right)^2 = \left(\frac{\sqrt{I_1 / I_2} + 1}{\sqrt{I_1 / I_2} - 1}\right)^2$$

$$= \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \left(\frac{a_1 / a_2 + 1}{a_1 / a_2 - 1}\right)^2 \text{ also } \sqrt{\frac{I_1}{I_2}} = \frac{a_1}{a_2} = \left(\frac{\sqrt{\frac{I_{\max}}{I_{\min}}} + 1}{\sqrt{\frac{I_{\max}}{I_{\min}}} - 1}\right)$$

✍ If two waves having equal intensity ($I_1 = I_2 = I$) meets at two locations P and Q with path difference Δ_1 and Δ_2 respectively then the ratio of resultant intensity at point P and Q will be

$$\frac{I_P}{I_Q}=\frac{\cos^2\frac{\phi_1}{2}}{\cos^2\frac{\phi_2}{2}}=\frac{\cos^2\left(\frac{\pi\Delta_1}{\lambda}\right)}{\cos^2\left(\frac{\pi\Delta_2}{\lambda}\right)}$$

✍ The angular thickness of fringe width is defined as $\delta=\frac{\beta}{D}=\frac{\lambda}{d}$, which is independent of the screen distance *D*.

✍ Central maxima means the maxima formed with zero optical path difference. It may be formed anywhere on the screen.

✍ All the wavelengths produce their central maxima at the same position.

✍ The wave with smaller wavelength from its maxima before the wave with longer wavelength.

✍ The first maxima of violet colour is closest and that for the red colour is farthest.

✍ Fringes with blue light are thicker than those for red light.

✍ In an interference pattern, whatever energy disappears at the minimum, appears at the maximum.

✍ In YDSE, the nth maxima always comes before the nth minima.

✍ In YDSE, the ratio $\frac{I_{\max}}{I_{\min}}$ is maximum when both the sources have same intensity.

✍ For two interfering waves if initial phase difference between them is ϕ and phase difference due to path difference between them is ϕ. Then total phase difference will be

$$\phi=\phi_0+\phi'=\phi_0+\frac{2\pi}{\lambda}\Delta .$$

✍ Sometimes maximm number of maximas or minimas are asked in the question which can be obtained on the screen. For this we use the fact that value of sin θ (or cos θ) can't be greater than 1. For example in the first case when the slits are vertical

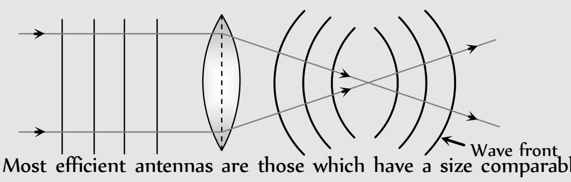
$$\sin\theta=\frac{n\lambda}{d} \qquad \text{(for maximum intensity)}$$

$$\because \ \sin\theta \not> 1 \ \therefore \ \frac{n\lambda}{d}\not> 1 \quad \text{or} \quad n \not> \frac{d}{\lambda}$$

Suppose in some question d/λ comes out say 4.6, then total number of maximuas on the screen will be 9. Corresponding to $n=0,\pm1,\pm2,\pm3$ and ± 4.

✍ ***Shape of wave front***

If rays are parallel, wave front is plane. If rays are converging wave front is spherical of decreasing radius. If rays are diverging wave front is spherical of increasing radius.



✍ Most efficient antennas are those which have a size comparable to the wavelength of the of electromagnetic wave they emit or receive.

✍ A substance (like calcite quartz) which exhibits different properties in different direction is called an anisotopic substance.

Ordinary Thinking

Objective Questions

Wave Nature and Interference of Light

1. By corpuscular theory of light, the phenomenon which can be explained is
   (a) Refraction (b) Interference
   (c) Diffraction (d) Polarisation

2. According to corpuscular theory of light, the different colours of light are due to
   (a) Different electromagnetic waves
   (b) Different force of attraction among the corpuscles
   (c) Different size of the corpuscles
   (d) None of the above

3. Huygen's conception of secondary waves [CPMT 1975]
   (a) Allow us to find the focal length of a thick lens
   (b) Is a geometrical method to find a wavefront
   (c) Is used to determine the velocity of light
   (d) Is used to explain polarisation

4. The idea of the quantum nature of light has emerged in an attempt to explain [CPMT 1990]
   (a) Interference
   (b) Diffraction
   (c) Radiation spectrum of a black body
   (d) Polarisation

5. Two coherent sources of light can be obtained by [MH CET 2001]
   (a) Two different lamps
   (b) Two different lamps but of the same power
   (c) Two different lamps of same power and having the same colour
   (d) None of the above

6. By Huygen's wave theory of light, we cannot explain the phenomenon of [CPMT 1989; AFMC 1993, 99; MP PET 1995, 2003; RPMT 2003; BCECE 2003; Pb PMT 2004]
   (a) Interference (b) Diffraction
   (c) Photoelectric effect (d) Polarisation

7. The phenomenon of interference is shown by [MNR 1994; MP PMT 1997; AIIMS 1999, 2000; JIPMER 2000; UPSEAT 1994, 2000]
   (a) Longitudinal mechanical waves only
   (b) Transverse mechanical waves only
   (c) Electromagnetic waves only
   (d) All the above types of waves

8. Two coherent monochromatic light beams of intensities I and $4I$ are superposed. The maximum and minimum possible intensities in the resulting beam are [IIT-JEE 1988; RPMT 1995; AIIMS 1997; MP PMT 1997; MP PET 1999; BHU 2002; KCET 2000, 05]
   (a) $5I$ and I (b) $5I$ and $3I$
   (c) $9I$ and I (d) $9I$ and $3I$

9. Light appears to travel in straight lines since [RPMT 1997; CPMT 1987, 89, 90, 2001; AIIMS 1998, 2002; KCET 2002; BHU 2002; DCE 2003]
   (a) It is not absorbed by the atmosphere
   (b) It is reflected by the atmosphere
   (c) Its wavelength is very small
   (d) Its velocity is very large

10. The idea of secondary wavelets for the propagation of a wave was first given by [Orissa PMT 2004]
    (a) Newton (b) Huygen
    (c) Maxwell (d) Fresnel

11. By a monochromatic wave, we mean [AFMC 1995]
    (a) A single ray
    (b) A single ray of a single colour
    (c) Wave having a single wavelength
    (d) Many rays of a single colour

12. The similarity between the sound waves and light waves is [KCET 1994]
    (a) Both are electromagnetic waves
    (b) Both are longitudinal waves
    (c) Both have the same speed in a medium
    (d) They can produce interference

13. The ratio of intensities of two waves is 9 : 1. They are producing interference. The ratio of maximum and minimum intensities will be MP PET 1999; AMU (Engg.) 1999; AIIMS 2000]
    (a) 10 : 8 (b) 9 : 1
    (c) 4 : 1 (d) 2 : 1

14. A wave can transmit ...... from one place to another [CPMT 1984]
    (a) Energy (b) Amplitude
    (c) Wavelength (d) Matter

15. If the ratio of intensities of two waves is 1 : 25, then the ratio of their amplitudes will be [CPMT 1984]
    (a) 1 : 25 (b) 5 : 1
    (c) 26 : 24 (d) 1 : 5

16. Two identical light sources S_1 and S_2 emit light of same wavelength λ. These light rays will exhibit interference if [MP PMT 1993]
    (a) Their phase differences remain constant
    (b) Their phases are distributed randomly
    (c) Their light intensities remain constant
    (d) Their light intensities change randomly

17. Wave nature of light follows because [MP PMT 1993]
    (a) Light rays travel in a straight line
    (b) Light exhibits the phenomena of reflection and refraction
    (c) Light exhibits the phenomenon of interference

(d) Light causes the phenomenon of photoelectric effect

18. If L is the coherence length and c the velocity of light, the coherent time is [MP PMT 1996]

(a) cL (b) $\frac{L}{c}$

(c) $\frac{c}{L}$ (d) $\frac{1}{Lc}$

19. If the amplitude ratio of two sources producing interference is 3 : 5, the ratio of intensities at maxima and minima is [MP PMT 1996]

(a) 25 : 16 (b) 5 : 3

(c) 16 : 1 (d) 25 : 9

20. Colours of thin films result from [CPMT 1972, 83, 96; RPMT 1997; DCE 2002; AIIMS 2005]

or

On a rainy day, a small oil film on water show brilliant colours. This is due to [MP PET 2004]

(a) Dispersion of light (b) Interference of light

(c) Absorption of light (d) Scattering of light

21. For constructive interference to take place between two monochromatic light waves of wavelength λ, the path difference should be [MNR 1992; UPSEAT 2001]

(a) $(2n-1)\frac{\lambda}{4}$ (b) $(2n-1)\frac{\lambda}{2}$

(c) $n\lambda$ (d) $(2n+1)\frac{\lambda}{2}$

22. Two sources of waves are called coherent if [NCERT 1984; MNR 1995; RPMT 1996, 97; CPMT 1997; UPSEAT 1995, 2000; Orissa JEE 2002; RPET 2003; MP PMT 1996, 2004]

(a) Both have the same amplitude of vibrations

(b) Both produce waves of the same wavelength

(c) Both produce waves of the same wavelength having constant phase difference

(d) Both produce waves having the same velocity

23. Soap bubble appears coloured due to the phenomenon of [AFMC 1995, 97; RPET 1997; CBSE PMT 1999; Pb PET 2001]

(a) Interference (b) Diffraction

(c) Dispersion (d) Reflection

24. Which of the following statements indicates that light waves are transverse [MP PMT 1995; AFMC 1996]

(a) Light waves can travel in vacuum

(b) Light waves show interference

(c) Light waves can be polarized

(d) Light waves can be diffracted

25. If two light waves having same frequency have intensity ratio 4 : 1 and they interfere, the ratio of maximum to minimum intensity in the pattern will be [BHU 1995; MP PMT 1995; DPMT 1999; CPMT 2003]

(a) 9 : 1 (b) 3 : 1

(c) 25 : 9 (d) 16 : 25

26. Evidence for the wave nature of light cannot be obtained from

(a) Reflection (b) Doppler effect

(c) Interference (d) Diffraction

27. Two light sources are said to be coherent if they are obtained from

(a) Two independent point sources emitting light of the same wavelength

(b) A single point source

(c) A wide source

(d) Two ordinary bulbs emitting light of different wavelengths

28. Wavelength of light of frequency 100Hz [CBSE PMT 1999]

(a) $2\times10^6 m$ (b) $3\times10^6 m$

(c) $4\times10^6 m$ (d) $5\times10^6 m$

29. Two waves having intensity in the ratio 25 : 4 produce interference. The ratio of the maximum to the minimum intensity is

(a) 5 : 2 (b) 7 : 3

(c) 49 : 9 (d) 9 : 49

30. Wavefront means [RPMT 1997, 98]

(a) All particles in it have same phase

(b) All particles have opposite phase of vibrations

(c) Few particles are in same phase, rest are in opposite phase

(d) None of these

31. Wavefront of a wave has direction with wave motion [RPMT 1997]

(a) Parallel (b) Perpendicular

(c) Opposite (d) At an angle of θ

32. Which one of the following phenomena is not explained by Huygen's construction of wavefront [CBSE PMT 1992]

(a) Refraction (b) Reflection

(c) Diffraction (d) Origin of spectra

33. Interference was observed in interference chamber when air was present, now the chamber is evacuated and if the same light is used, a careful observer will see [CBSE PMT 1993; DPMT 2000; BHU 2002]

(a) No interference

(b) Interference with bright bands

(c) Interference with dark bands

(d) Interference in which width of the fringe will be slightly increased

34. The ratio of intensities of two waves are given by 4 : 1. The ratio of the amplitudes of the two waves is [CBSE PMT 1993]

(a) 2 : 1 (b) 1 : 2

(c) 4 : 1 (d) 1 : 4

35. For the sustained interference of light, the necessary condition is that the two sources should [DPMT 1996; RPMT 1998, 2003]

(a) Have constant phase difference

(b) Be narrow

(c) Be close to each other

(d) Of same amplitude

36. If the ratio of amplitude of two waves is 4 : 3, then the ratio of maximum and minimum intensity is [AFMC 1997]

(a) 16 : 18 (b) 18 : 16

(c) 49 : 1 (d) 94 : 1

37. Which of the following is conserved when light waves interfere

(a) Intensity (b) Energy

(c) Amplitude (d) Momentum

38. Intensity of light depends upon [RPMT 1999]

(a) Velocity (b) Wavelength

(c) Amplitude (d) Frequency

39. Ray diverging from a point source from a wave front that is [RPET 2000]

(a) Cylindrical (b) Spherical

(c) Plane (d) Cubical

40. Ratio of amplitude of interfering waves is 3 : 4. Now ratio of their intensities will be [RPET 2000]

(a) $\frac{16}{9}$ (b) 49 : 1

(c) $\frac{9}{16}$ (d) None of these

41. Two coherent sources have intensity in the ratio of $\frac{100}{1}$. Ratio of (intensity) max/(intensity) min is [RPET 2000]

(a) $\frac{1}{100}$ (b) $\frac{1}{10}$

(c) $\frac{10}{1}$ (d) $\frac{3}{2}$

42. If two waves represented by $y_1 = 4\sin\omega t$ and $y_2 = 3\sin\left(\omega t + \frac{\pi}{3}\right)$ interfere at a point, the amplitude of the resulting wave will be about [MP PMT 2000]

(a) 7 (b) 6

(c) 5 (d) 3.5

43. The two waves represented by $y_1 = a\sin(\omega t)$ and $y_2 = b\cos(\omega t)$ have a phase difference of [MP PMT 2000]

(a) 0 (b) $\frac{\pi}{2}$

(c) π (d) $\frac{\pi}{4}$

44. In a wave, the path difference corresponding to a phase difference of ϕ is [MP PET 2000]

(a) $\frac{\pi}{2\lambda}\phi$ (b) $\frac{\pi}{\lambda}\phi$

(c) $\frac{\lambda}{2\pi}\phi$ (d) $\frac{\lambda}{\pi}\phi$

45. Two coherent sources of intensities, I_1 and I_2 produce an interference pattern. The maximum intensity in the interference pattern will be [UPSEAT 2001; MP PET 2001]

(a) $I_1 + I_2$ (b) $I_1^2 + I_2^2$

(c) $(I_1 + I_2)^2$ (d) $(\sqrt{I_1} + \sqrt{I_2})^2$

46. Newton postulated his corpuscular theory on the basis of [UPSEAT 2001; KCET 2001]

(a) Newton's rings

(b) Colours of thin films

(c) Rectilinear propagation of light [MNR 1998]

(d) Dispersion of white light

47. The dual nature of light is exhibited by [KCET 1999; AIIMS 2001; BHU 2001; MH CET 2003; BCECE 2004]

(a) Photoelectric effect

(b) Refraction and interference

(c) Diffraction and reflection

(d) Diffraction and photoelectric effect

48. Two beams of light having intensities I and $4I$ interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\frac{\pi}{2}$ at point A and π at point B. Then the difference between the resultant intensities at A and B is [IIT JEE (Screening) 2001]

(a) $2I$ (b) $4I$

(c) $5I$ (d) $7I$

49. Coherent sources are those sources for which [RPET 2001]

(a) Phase difference remain constant

(b) Frequency remains constant

(c) Both phase difference and frequency remains constant

(d) None of these

50. Wave nature of light is verified by [RPET 2001]

(a) Interference (b) Photoelectric effect

(c) Reflection (d) Refraction

51. Two waves are represented by the equations $y_1 = a\sin\omega t$ and $y_2 = a\cos\omega t$. The first wave [MP PMT 2001]

(a) Leads the second by π

(b) Lags the second by π

(c) Leads the second by $\frac{\pi}{2}$

(d) Lags the second by $\frac{\pi}{2}$

52. Light waves producing interference have their amplitudes in the ratio 3 : 2. The intensity ratio of maximum and minimum of interference fringes is [EAMCET 2001]

(a) 36 : 1 (b) 9 : 4

(c) 25 : 1 (d) 6 : 4

53. Laser beams are used to measure long distance because [DCE 2001]

(a) They are monochromatic

(b) They are highly polarised

(c) They are coherent

(d) They have high degree of parallelism

54. Two coherent sources of different intensities send waves which interfere. The ratio of maximum intensity to the minimum intensity is 25. The intensities of the sources are in the ratio

(a) 25 : 1 (b) 5 : 1

(c) 9 : 4 (d) 25 : 16

55. The frequency of light ray having the wavelength 3000 Å is

[DPMT 2002]

(a) 9×10^{13} *cycles/sec* (b) 10^{15} *cycles/sec*

(c) 90 *cycles/sec* (d) 3000 *cycles/sec*

56. Two waves have their amplitudes in the ratio 1 : 9. The maximum and minimum intensities when they interfere are in the ratio

(a) $\frac{25}{16}$ (b) $\frac{16}{26}$

(c) $\frac{1}{9}$ (d) $\frac{9}{1}$

57. Huygen's principle of secondary wavelets may be used to

[KCET 2002]

(a) Find the velocity of light in vacuum

(b) Explain the particle behaviour of light

(c) Find the new position of the wavefront

(d) Explain photoelectric effect

58. What is the path difference of destructive interference

[AIIMS 2002]

(a) $n\lambda$ (b) $n(\lambda+1)$

(c) $\frac{(n+1)\lambda}{2}$ (d) $\frac{(2n+1)\lambda}{2}$

59. If an interference pattern have maximum and minimum intensities in 36 : 1 ratio then what will be the ratio of amplitudes

(a) 5 : 7 (b) 7 : 4

(c) 4 : 7 (d) 7 : 5

60. Intensities of the two waves of light are I and $4I$. The maximum intensity of the resultant wave after superposition is

(a) 5 I (b) 9 I

(c) 16 I (d) 25 I

61. As a result of interference of two coherent sources of light, energy is **[MP PMT 2002; KCET 2003]**

(a) Increased

(b) Redistributed and the distribution does not vary with time

(c) Decreased

(c) Redistributed and the distribution changes with time

62. To demonstrate the phenomenon of interference, we require two sources which emit radiation **[AIEEE 2003]**

(a) Of the same frequency and having a define phase relationship

(b) Of nearly the same frequency

(c) Of the same frequency

(d) Of different wavelengths

63. When a beam of light is used to determine the position of an object, the maximum accuracy is achieved if the light is

[AIIMS 2003]

(a) Polarised (b) Of longer wavelength

(c) Of shorter wavelength (d) Of high intensity

64. If the distance between a point source and screen is doubled, then intensity of light on the screen will become **[UPSEAT 2002]**

[RPET 1997; RPMT 1999]

(a) Four times (b) Double

(c) Half (d) One-fourth

65. Huygen wave theory allows us to know **[AFMC 2004]**

(a) The wavelength of the wave

(b) The velocity of the wave

(c) The amplitude of the wave

(d) The propagation of wave fronts **[KCET 2002]**

66. The wave theory of light was given by

[J & K CET 2004; KCET 2005]

(a) Maxwell (b) Planck

(c) Huygen (d) Young

67. The phase difference between incident wave and reflected wave is 180° when light ray **[RPMT 1998, 2001]**

(a) Enters into glass from air

(b) Enters into air from glass

(c) Enters into glass from diamond

(d) Enters into water from glass

68. Which of the following phenomena can explain quantum nature of light **[RPMT 2001]**

(a) Photoelectric effect (b) Interference

(c) Diffraction (d) Polarisation

69. Which of the following is not a property of light

[AFMC 2005]

(a) It requires a material medium for propagation **[AFMC 2002]**

(b) It can travel through vacuum

(c) It involves transportation of energy

(d) It has finite speed **[MP PET 2002]**

70. What causes changes in the colours of the soap or oil films for the given beam of light **[AFMC 2005]**

(a) Angle of incidence (b) Angle of reflection

(c) Thickness of film (d) None of these

71. Select the right option in the following **[KCET 2005]**

(a) Christian Huygens a contemporary of Newton established the wave theory of light by assuming that light waves were transverse

(b) Maxwell provided the compelling theoretical evidence that light is transverse wave

(c) Thomas Young experimentally proved the wave behaviour of light and Huygens assumption

(d) All the statements give above, correctly answers the question "what is light"

72. Two waves of intensity I undergo Interference. The maximum intensity obtained is **[BHU 2005]**

(a) $I/2$ (b) I

(c) $2I$ (d) $4I$

Young's Double Slit Experiment

1. Young's experiment establishes that
[CPMT 1972; MP PET 1994, 98; MP PMT 1998]
   (a) Light consists of waves
   (b) Light consists of particles
   (c) Light consists of neither particles nor waves
   (d) Light consists of both particles and waves

2. In the interference pattern, energy is
   (a) Created at the position of maxima
   (b) Destroyed at the position of minima
   (c) Conserved but is redistributed
   (d) None of the above

3. Monochromatic green light of wavelength $5 \times 10^{-7} m$ illuminates a pair of slits 1 *mm* apart. The separation of bright lines on the interference pattern formed on a screen 2 *m* away is
   (a) 0.25 *mm* (b) 0.1 *mm*
   (c) 1.0 *mm* (d) 0.01 *mm*

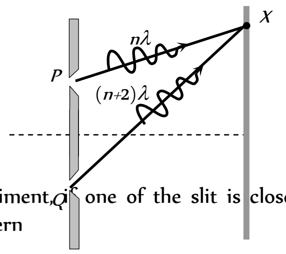
4. In Young's double slit experiment, if the slit widths are in the ratio 1 : 9, then the ratio of the intensity at minima to that at maxima will be [MP PET 1987]
   (a) 1 (b) 1/9
   (c) 1/4 (d) 1/3

5. In Young's double slit interference experiment, the slit separation is made 3 fold. The fringe width becomes
[CPMT 1982, 89]
   (a) 1/3 *times* (b) 1/9 *times*
   (c) 3 *times* (d) 9 *times*

6. In a certain double slit experimental arrangement interference fringes of width 1.0 *mm* each are observed when light of wavelength 5000 Å is used. Keeping the set up unaltered, if the source is replaced by another source of wavelength 6000 Å, the fringe width will be [CPMT 1988]
   (a) 0.5 *mm* (b) 1.0 *mm*
   (c) 1.2 *mm* (d) 1.5 *mm*

7. Two coherent light sources S_1 and S_2 (λ= 6000 Å) are 1*mm* apart from each other. The screen is placed at a distance of 25 *cm* from the sources. The width of the fringes on the screen should be
   (a) 0.015 *cm* (b) 0.025 *cm*
   (c) 0.010 *cm* (d) 0.030 *cm*

8. The figure shows a double slit experiment *P* and *Q* are the slits. The path lengths *PX* and *QX* are $n\lambda$ and $(n+2)\lambda$ respectively, where *n* is a whole number and λ is the wavelength. Taking the central fringe as zero, what is formed at *X*
   (a) First bright
   (b) First dark
   (c) Second bright
   (d) Second dark



9. In Young's double slit experiment, one of the slit is closed fully, then in the interference pattern
   (a) A bright slit will be observed, no interference pattern will exist
   (b) The bright fringes will become more bright
   (c) The bright fringes will become fainter
   (d) None of the above

10. In Young's double slit experiment, a glass plate is placed before a slit which absorbs half the intensity of light. Under this case
   (a) The brightness of fringes decreases
   (b) The fringe width decreases
   (c) No fringes will be observed
   (d) The bright fringes become fainter and the dark fringes have finite light intensity

11. In Young's experiment, the distance between the slits is reduced to half and the distance between the slit and screen is doubled, then the fringe width
[IIT 1981; MP PMT 1994; RPMT 1997; KCET 2000; CPMT 2003; AMU (Engg.) 2000; DPMT 2003; UPSEAT 2000, 04; Kerala PMT 2004]
   (a) Will not change (b) Will become half
   (c) Will be doubled (d) Will become four times

[CPMT 1977; DPMT 1999]

12. The maximum intensity of fringes in Young's experiment is *I*. If one of the slit is closed, then the intensity at that place becomes I_0. Which of the following relation is true ?
[NCERT 1982; MP PMT 1994, 99; BHU 1998; RPMT 1996; RPET 1999; AMU (Engg.) 1999]
   (a) $I = I_0$
   (b) $I = 2I_0$
   (c) $I = 4I_0$
   (d) There is no relation between I and I_0

13. In the Young's double slit experiment, the ratio of intensities of bright and dark fringes is 9. This means that [IIT 1982]
   (a) The intensities of individual sources are 5 and 4 units respectively
   (b) The intensities of individual sources are 4 and 1 units respectively
   (c) The ratio of their amplitudes is 3
   (d) The ratio of their amplitudes is 2

14. An oil flowing on water seems coloured due to interference. For observing this effect, the approximate thickness of the oil film should be
[DPET 1987; JIPMER 1997; RPMT 2002, 04]
   (a) 100 Å (b) 10000 Å
   (c) 1 *mm* (d) 1 *cm*

[CPMT 1990]

15. The Young's experiment is performed with the lights of blue (λ = 4360 Å) and green colour (λ = 5460 Å), If the distance of the 4th fringe from the centre is *x*, then [CPMT 1987]
   (a) *x* (Blue) = *x* (Green) (b) *x* (Blue)> *x* (Green)
   (c) *x* (Blue) < *x* (Green) (d) $\frac{x(Blue)}{x(Green)} = \frac{5460}{4360}$

16. In the Young's double slit experiment, the spacing between two slits is 0.1 *mm*. If the screen is kept at a distance of 1.0 *m* from the slits and the wavelength of light is 5000 Å, then the fringe width is [MP PMT 1993; R
   (a) 1.0 *cm* (b) 1.5 *cm*
   (c) 0.5 *cm* (d) 2.0 *cm*

17. In Young's double slit experiment, if *L* is the distance between the slits and the screen upon which interference pattern is observed, *x* is

the average distance between the adjacent fringes and d being the slit separation. The wavelength of light is given by

(a) $\frac{xd}{L}$ (b) $\frac{xL}{d}$

(c) $\frac{Ld}{x}$ (d) $\frac{1}{Ldx}$

18. In a Young's double slit experiment, the central point on the screen is [MP PMT 1996]

(a) Bright (b) Dark

(c) First bright and then dark (d) First dark and then bright

19. In a Young's double slit experiment, the fringe width is found to be 0.4 *mm*. If the whole apparatus is immersed in water of refractive index 4/3 without disturbing the geometrical arrangement, the new fringe width will be

[CBSE PMT 1990]

(a) 0.30 *mm* (b) 0.40 *mm*

(c) 0.53 *mm* (d) 450 *micron*

20. Young's experiment is performed in air and then performed in water, the fringe width [CPMT 1990; MP PMT 1994; RPMT 1997; Kerala PMT 2004]

(a) Will remain same (b) Will decrease

(c) Will increase (d) Will be infinite

21. In double slits experiment, for light of which colour the fringe width will be minimum [MP PMT 1994]

(a) Violet (b) Red

(c) Green (d) Yellow

22. In Young's experiment, light of wavelength 4000 Å is used to produce bright fringes of width 0.6 *mm*, at a distance of 2 meters. If the whole apparatus is dipped in a liquid of refractive index 1.5, then fringe width will be [MP PMT 1994]

(a) 0.2 *mm* (b) 0.3 *mm*

(c) 0.4 *mm* (d) 1.2 *mm*

23. In Young's double slit experiment, the phase difference between the light waves reaching third bright fringe from the central fringe will be (λ =6000 Å) [MP PMT 1994]

(a) Zero (b) 2π

(c) 4π (d) 6π

24. In Young's double slit experiment, if the widths of the slits are in the ratio 4 : 9, the ratio of the intensity at maxima to the intensity at minima will be [Manipal MEE 1995]

(a) 169 : 25 (b) 81 : 16

(c) 25 : 1 (d) 9 : 4

25. In Young's double slit experiment when wavelength used is 6000 Å and the screen is 40 *cm* from the slits, the fringes are 0.012 *cm* wide. What is the distance between the slits

[MP PMT 1995; Pb PET 2002]

(a) 0.024 *cm* (b) 2.4 *cm*

(c) 0.24 *cm* (d) 0.2 *cm*

26. In two separate set - ups of the Young's double slit experiment, fringes of equal width are observed when lights of wavelengths in the ratio 1 : 2 are used. If the ratio of the slit separation in the two cases is 2 : 1, the ratio of the distances between the plane of the slits and the screen in the two set - ups is [MP PET 1993]

(a) 4 : 1 (b) 1 : 1

(c) 1 : 4 (d) 2 : 1

27. In an interference experiment, the spacing between successive maxima or minima is [MP PET 1996]

(a) $\frac{\lambda d}{D}$ (b) $\frac{\lambda D}{d}$

(c) $\frac{dD}{\lambda}$ (d) $\frac{\lambda d}{4D}$

(Where the symbols have their usual meanings)

28. If yellow light in the Young's double slit experiment is replaced by red light, the fringe width will [MP PMT 1996]

(a) Decrease

(b) Remain unaffected

(c) Increase

(d) First increase and then decrease

29. In Young's double slit experiment, the fringe width is $1\times10^{-4}\,m$ if the distance between the slit and screen is doubled and the distance between the two slit is reduced to half and wavelength is changed from $6.4\times10^{7}\,m$ to $4.0\times10^{-7}\,m$, the value of new fringe width will be

(a) $0.15\times10^{-4}\,m$ (b) $2.0\times10^{-4}\,m$

(c) $1.25\times10^{-4}\,m$ (d) $2.5\times10^{-4}\,m$

30. In Young's experiment, one slit is covered with a blue filter and the other (slit) with a yellow filter. Then the interference pattern

(a) Will be blue (b) Will be yellow

(c) Will be green (d) Will not be formed

31. Two sources give interference pattern which is observed on a screen, D distance apart from the sources. The fringe width is $2w$. If the distance D is now doubled, the fringe width will

(a) Become $w/2$ (b) Remain the same

(c) Become w (d) Become $4w$

32. In double slit experiment, the angular width of the fringes is 0.20° for the sodium light (λ =5890 Å). In order to increase the angular width of the fringes by 10%, the necessary change in the wavelength is [MP PMT 1997]

(a) Increase of 589 Å (b) Decrease of 589 Å

(c) Increase of 6479 Å (d) Zero

33. In a biprism experiment, by using light of wavelength 5000 Å, 5 *mm* wide fringes are obtained on a screen 1.0 *m* away from the coherent sources. The separation between the two coherent sources is

(a) 1.0 *mm* (b) 0.1 *mm*

(c) 0.05 *mm* (d) 0.01 *mm*

34. The slits in a Young's double slit experiment have equal widths and the source is placed symmetrically relative to the slits. The intensity at the central fringes is I. If one of the slits is closed, the intensity at this point will be

[MP PMT 1999; Orissa JEE 2004; Kerala PET 2005]

(a) I (b) $I/4$

(c) $I/2$ (d) $4I$

35. A thin mica sheet of thickness $2\times10^{-6}m$ and refractive index $(\mu=1.5)$ is introduced in the path of the first wave. The wavelength of the wave used is 5000 Å. The central bright maximum will shift [CPMT 1999]

(a) 2 fringes upward (b) 2 fringes downward

(c) 10 fringes upward (d) None of these

36. In a Young's double slit experiment, the fringe width will remain same, if (D = distance between screen and plane of slits, d = separation between two slits and λ = wavelength of light used)

(a) Both λ and D are doubled

(b) Both d and D are doubled

(c) D is doubled but d is halved

(d) λ is doubled but d is halved

37. In Young's double slit experiment, the slits are 0.5 mm apart and interference pattern is observed on a screen placed at a distance of 1.0 m from the plane containing the slits. If wavelength of the incident light is 6000 Å, then the separation between the third bright fringe and the central maxima is

(a) 4.0 mm (b) 3.5 mm

(c) 3.0 mm (d) 2.5 mm

38. In Young's double slit experiment, 62 fringes are seen in visible region for sodium light of wavelength 5893 Å. If violet light of wavelength 4358 Å is used in place of sodium light, then number of fringes seen will be [RPET 1997]

(a) 54 (b) 64

(c) 74 (d) 84

39. In Young's double slit experiment, angular width of fringes is 0.20° for sodium light of wavelength 5890 Å. If complete system is dipped in water, then angular width of fringes becomes

(a) 0.11° (b) 0.15°

(c) 0.22° (d) 0.30°

40. In Young's double slit experiment, the distance between the slits is 1 mm and that between slit and screen is 1 meter and 10th fringe is 5 mm away from the central bright fringe, then wavelength of light used will be [RPMT 1997]

(a) 5000 Å (b) 6000 Å

(c) 7000 Å (d) 8000 Å

41. In Young's double slit experiment, carried out with light of wavelength λ = 5000 Å, the distance between the slits is 0.2 mm and the screen is at 200 cm from the slits. The central maximum is at x =0. The third maximum (taking the central maximum as zeroth maximum) will be at x equal to

[CBSE PMT 1992; MH CET 2002]

(a) 1.67 cm (b) 1.5 cm

(c) 0.5 cm (d) 5.0 cm

42. In a Young's experiment, two coherent sources are placed 0.90 mm apart and the fringes are observed one metre away. If it produces the second dark fringe at a distance of 1 mm from the central fringe, the wavelength of monochromatic light used would be

[CBSE PMT 1992; KCET 2004]

(a) $60\times10^{-4}\,cm$ (b) $10\times10^{-4}\,cm$

(c) $10\times10^{-5}\,cm$ (d) $6\times10^{-5}\,cm$

43. In Young's double slit experiment, the distance between the two slits is 0.1 mm and the wavelength of light used is $4\times10^{-7}\,m$. If the width of the fringe on the screen is 4 mm, the distance between screen and slit is

[Bihar CMEET 1995]

(a) 0.1 mm (b) 1 cm

(c) 0.1 cm (d) 1 m

44. In Young's double slit experiment, the distance between sources is 1 mm and distance between the screen and source is 1 m. If the fringe width of [Bihar MEE 1995] 0.06 cm, then λ =

(a) 6000 Å (b) 4000 Å

(c) 1200 Å (d) 2400 Å

45. In Young's double slit experiment, a mica slit of thickness t and refractive index μ is introduced in the ray from the first source S. By how much distance the fringes pattern will be displaced [RPMT 1996, 97; JIPM

(a) $\frac{d}{D}(\mu-1)\,t$ (b) $\frac{D}{d}(\mu-1)\,t$

[AMU 1995]

(c) $\frac{d}{(\mu-1)D}$ (d) $\frac{D}{d}(\mu-1)$

46. In Young's double slit experiment using sodium light (λ = 5898 Å), 92 fringes are seen. If given colour (λ = 5461 Å) is used, how many fringes will be seen

[CPMT 1989; RPET 1996; JIPMER 2001, 02]

(a) 62 (b) 67

(c) 85 (d) 99

47. If a torch is used in place of monochromatic light in Young's experiment what will happens [RPET 1997]

[MH CET 1999; KCET 1999]

(a) Fringe will appear for a moment then it will disappear

(b) Fringes will occur as from monochromatic light

(c) Only bright fringes will appear

(d) No fringes will appear

48. When a thin metal plate is placed in the path of one of the interfering beams of light [KCET 1999]

(a) Fringe width increases (b) Fringes disappear

(c) Fringes become brighter (d) Fringes becomes blurred

49. In Young's experiment, the distance between slits is 0.28 mm and distance between slits and screen is 1.4 m. Distance between central bright fringe and third bright fringe is 0.9 cm. What is the wavelength of used light [KCET 1999]

(a) 5000 Å (b) 6000 Å

(c) 7000 Å (d) 9000 Å

50. Two parallel slits 0.6 *mm* apart are illuminated by light source of wavelength 6000 Å. The distance between two consecutive dark fringes on a screen 1 *m* away from the slits is

(a) 1 *mm* (b) 0.01 *mm*

(c) 0.1 *m* (d) 10 *m*

51. In young's double slit experiment with a source of light of wavelength 6320Å, the first maxima will occur when

[Roorkee 1999]

(a) Path difference is 9480 Å

(b) Phase difference is 2π radian

(c) Path difference is 6320 Å

(d) Phase difference is π radian

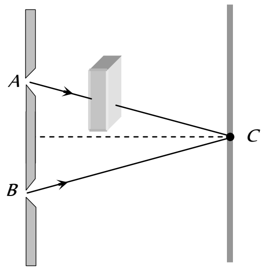
52. If a transparent medium of refractive index $\mu = 1.5$ and thickness $t = 2.5 \times 10^{-5}$ *m* is inserted in front of one of the slits of Young's Double Slit experiment, how much will be the shift in the interference pattern? The distance between the slits is 0.5 *mm* and that between slits and screen is 100 *cm*

(a) 5 *cm* (b) 2.5 *cm*

(c) 0.25 *cm* (d) 0.1 *cm*

53. In Young's experiment, monochromatic light is used to illuminate the two slits *A* and *B*. Interference fringes are observed on a screen placed in front of the slits. Now if a thin glass plate is placed normally in the path of the beam coming from the slit

[UPSEAT 1993, 2000; AIIMS 1999, 2004]



(a) The fringes will disappear

(b) The fringe width will increase

(c) The fringe width will increase

(d) There will be no change in the fringe width but the pattern shifts

54. The fringe width in Young's double slit experiment increases when

(a) Wavelength increases

(b) Distance between the slits increases

(c) Distance between the source and screen decreases

(d) The width of the slits increases

55. In a double slit experiment, instead of taking slits of equal widths, one slit is made twice as wide as the other. Then in the interference pattern [IIT-JEE (Screening) 2000]

(a) The intensities of both the maxima and the minima increase

(b) The intensity of maxima increases and the minima has zero intensity

(c) The intensity of maxima decreases and that of the minima increases

(d) The intensity of maxima decreases and the minima has zero intensity

56. Two slits, 4 *mm* apart, are illuminated by light of wavelength 6000 Å. What will be the fringe width on a screen placed 2*m* from the slits [MP PET 2000]

(a) 0.12 *mm* (b) 0.3 *mm*

(c) 3.0 *mm* (d) 4.0 *mm*

57. In the Young's double slit experiment, for which colour the fringe width is least [UPSEAT 2001, MP PET 2001]

(a) Red (b) Green

(c) Blue (d) Yellow

58. In a Young's double slit experiment, the separation of the two slits is doubled. To keep the same spacing of fringes, the distance *D* of the screen from the sl[JIPMER 1999] made

[MNR 1998; AMU (Engg.) 2001]

(a) $\dfrac{D}{2}$ (b) $\dfrac{D}{\sqrt{2}}$

(c) $2D$ (d) $4D$

59. Young's double slit experiment is performed with light of wavelength 550 *nm*. The separation between the slits is 1.10 *mm* and screen is placed at distance of 1 *m*. What is the distance between the consecutive bright or dark fringes

[Pb. PMT 2000]

(a) 1.5 *mm* (b) 1.0 *mm*

(c) 0.5 *mm* (d) None of these

60. In Young's experiment, the ratio of maximum to minimum intensities of the fringe system is 4 : 1. The amplitudes of the coherent sources are in the ratio [AIIMS 1999]

[RPMT 1996; MP PET 2000; RPET 2001; MP PMT 2001]

(a) 4 : 1 (b) 3 : 1

(c) 2 : 1 (d) 1 : 1

61. An interference pattern was made by using red light. If the red light changes with blue light, the fringes will become

[BHU 2001]

(a) Wider (b) Narrower

(c) Fainter (d) Brighter

62. If a white light is used in Young's double slit experiments then a very large number of coloured fringes can be seen

[KCET 2001]

(a) With first order violet fringes being closer to the central white fringes

(b) First order red fringes being closer to the central white fringes

(c) With a central white fringe [MP PMT 2000]

(d) With a central black fringe

63. In a Young's double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600 *nm* is used. If the wavelength of light is changed to 400 *nm*, number of fringes observed in the same segment of the screen is given by [IIT-JEE (Screening) 2001]

(a) 12 (b) 18

(c) 24 (d) 30

64. In the Young's double slit experiment with sodium light. The slits are 0.589 *m* a part. The angular separation of the third maximum from the central maximum will be (given $\lambda = 589$ *mm*)

(a) $\sin^{-1}(0.33\times 10^{8})$ (b) $\sin^{-1}(0.33\times 10^{-6})$

(c) $\sin^{-1}(3\times 10^{-8})$ (d) $\sin^{-1}(3\times 10^{-6})$

65. In Young's double slit experiment, the distance between the two slits is made half, then the fringe width will become

[RPMT 1999; BHU 2002]

(a) Half (b) Double

(c) One fourth (d) Unchanged

66. In Young's double slit experiment, the central bright fringe can be identified [KCET 2002]

(a) By using white light instead of monochromatic light

(b) As it is narrower than other bright fringes

(c) As it is wider than other bright fringes

(d) As it has a greater intensity than the other bright fringes

67. In Young's double slit experiment, the wavelength of the light used is doubled and distance between two slits is half of initial distance, the resultant fringe width becomes [AIEEE 2002]

(a) 2 *times* (b) 3 *times*
(c) 4 *times* (d) 1/2 *times*

68. In a Young's double slit experiment, the source illuminating the slits is changed from blue to violet. The width of the fringes

(a) Increases (b) Decreases
(c) Becomes unequal (d) Remains constant

69. In Young's double slit experiment, the intensity of light coming from the first slit is double the intensity from the second slit. The ratio of the maximum intensity to the minimum intensity on the interference fringe pattern observed is [KCET 2002]

(a) 34 (b) 40
(c) 25 (d) 38

70. If the sodium light in Young's double slit experiment is replaced by red light, the fringe width will [MP PMT 2002]

(a) Decrease
(b) Increase
(c) Remain unaffected
(d) First increase, then decrease

71. In Young's double slit experiment the wavelength of light was changed from 7000 Å to 3500 Å. While doubling the separation between the slits which of the following is not true for this experiment [Orissa JEE 2002]

(a) The width of the fringes changes
(b) The colour of bright fringes changes
(c) The separation between successive bright fringes changes
(d) The separation between successive dark fringes remains unchanged

72. When a thin transparent plate of thickness t and refractive index μ is placed in the path of one of the two interfering waves of light, then the path difference changes by [MP PMT 2002]

(a) $(\mu+1)t$ (b) $(\mu-1)t$
(c) $\frac{(\mu+1)}{t}$ (d) $\frac{(\mu-1)}{t}$

73. In Young's double-slit experiment, an interference pattern is obtained on a screen by a light of wavelength 6000 Å, coming from the coherent sources S_1 and S_2. At certain point P on the screen third dark fringe is formed. Then the path difference $S_1P - S_2P$ in microns is [EAMCET 2003]

(a) 0.75 (b) 1.5
(c) 3.0 (d) 4.5

74. In a Young's double slit experiment, the slit separation is 1 *mm* and the screen is 1 *m* from the slit. For a monochromatic light of wavelength 500 *nm*, the distance of 3rd minima from the central maxima is [Orissa JEE 2003]

(a) 0.50 *mm* (b) 1.25 *mm*
(c) 1.50 *mm* (d) 1.75 *mm*

75. In Young's double-slit experiment the fringe width is β. If entire arrangement is placed in a liquid of refractive index n, the fringe width becomes [KCET 2003]

(a) $\frac{\beta}{n+1}$ (b) $n\beta$
(c) $\frac{\beta}{n}$ (d) $\frac{\beta}{n-1}$

76. In an interference experiment, third bright fringe is obtained at a point on the screen with a light of 700 *nm*. What should be the wavelength of the light source in order to obtain 5th bright fringe at the same point [KCET 2003] [KCET 2002]

(a) 500 *nm* (b) 630 *nm*
(c) 750 *nm* (d) 420 *nm*

77. If the separation between slits in Young's double slit experiment is reduced to $\frac{1}{3}rd$, the fringe width becomes n times. The value of n is [MP PET 2003]

(a) 3 (b) $\frac{1}{3}$
(c) 9 (d) $\frac{1}{9}$

78. A double slit experiment is performed with light of wavelength 500 *nm*. A thin film of thickness 2 μm and refractive index 1.5 is introduced in the path of the upper beam. The location of the central maximum will [AIIMS 2003]

(a) Remain unshifted
(b) Shift downward by nearly two fringes
(c) Shift upward by nearly two fringes
(d) Shift downward by 10 fringes

79. The two slits at a distance of 1 *mm* are illuminated by the light of wavelength $6.5\times10^{-7}\,m$. The interference fringes are observed on a screen placed at a distance of 1*m*. The distance between third dark fringe and fifth bright fringe will be [NCERT 1982; MP PET 1995; BVP 2003]

(a) 0.65 *mm* (b) 1.63 *mm*
(c) 3.25 *mm* (d) 4.88 *mm*

80. In a Young's double-slit experiment the fringe width is 0.2 *mm*. If the wavelength of light used is increased by 10% and the separation between the slits is also increased by 10%, the fringe width will be

(a) 0.20 *mm* (b) 0.401 *mm*
(c) 0.242 *mm* (d) 0.165 *mm*

81. Two coherent sources of intensity ratio 1 : 4 produce an interference pattern. The fringe visibility will be [J & K CET 2004]

(a) 1 (b) 0.8
(c) 0.4 (d) 0.6

82. In Young's double slit experiment the amplitudes of two sources are 3*a* and a respectively. The ratio of intensities of bright and dark fringes will be [J & K CET 2004]

(a) 3 : 1 (b) 4 : 1
(c) 2 : 1 (d) 9 : 1

83. In Young's double slit experiment, distance between two sources is 0.1 *mm*. The distance of screen from the sources is 20 *cm*. Wavelength of light used is 5460 Å. Then angular position of the first dark fringe is [DCE 2002]

(a) 0.08° (b) 0.16°

(c) 0.20° (d) 0.313°

84. In a Young's double slit experiment, the slit separation is 0.2 *cm,* the distance between the screen and slit is 1*m*. Wavelength of the light used is 5000 Å. The distance between two consecutive dark fringes (in *mm*) is

[DCE 2004]

(a) 0.25 (b) 0.26

(c) 0.27 (d) 0.28

85. A light of wavelength 5890 Å falls normally on a thin air film. The minimum thickness of the film such that the film appears dark in reflected light [Pb. PMT 2003]

(a) $2.945 \times 10^{-7} m$ (b) $3.945 \times 10^{-7} m$

(c) $4.95 \times 10^{-7} m$ (d) $1.945 \times 10^{-7} m$

86. In Young's double slit experiment, a minimum is obtained when the phase difference of super imposing waves is

[MH CET 2004]

(a) Zero (b) $(2n-1)\,\pi$

(c) $n\pi$ (d) $(n+1)\,\pi$

87. In Fresnel's biprism $(\mu = 1.5)$ experiment the distance between source and biprism is 0.3 *m* and that between biprism and screen is 0.7*m* and angle of prism is 1°. The fringe width with light of wavelength 6000 Å will be

[RPMT 2002]

(a) 3 *cm* (b) 0.011 *cm*

(c) 2 *cm* (d) 4 *cm*

88. In Young double slit experiment, when two light waves form third minimum, they have [RPMT 2003]

(a) Phase difference of 3π (b) Phase difference of $\frac{5\pi}{2}$

(c) Path difference of 3λ (d) Path difference of $\frac{5\lambda}{2}$

89. In Fresnel's biprism experiment, on increasing the prism angle, fringe width will [RPMT 2003]

(a) Increase

(b) Decrease

(c) Remain unchanged

(d) Depend on the position of object

90. If prism angle $\alpha = 1°$, $\mu = 1.54$, distance between screen and prism $(b) = 0.7\,m$, distance between prism and source $a = 0.3\,m$, $\lambda = 180\pi\ nm$ then in Fresnal biprism find the value of β (fringe width) [RPMT 2002]

(a) $10^{-4}\,m$ (b) $10^{-3}\,mm$

(c) $10^{-4} \times \pi m$ (d) $\pi \times 10^{-3}\,m$

91. If Fresnel's biprism experiment as held in water inspite of air, then what will be the effect on fringe width [RPMT 1997, 98]

(a) Decrease (b) Increase

(c) No effect (d) None of these

92. What is the effect on Fresnel's biprism experiment when the use of white light is made [RPMT 1998]

(a) Fringe are affected

(b) Diffraction pattern is spread more

(c) Central fringe is white and all are coloured

(d) None of these

93. What happens to the fringe pattern when the Young's double slit experiment is performed in water instead or air then fringe width

(a) Shrinks (b) Disappear

(c) Unchanged (d) Enlarged

94. In Young's doubled slit experiment, the separation between the slit and the screen increases. The fringe width

[BCECE 2005]

(a) Increases (b) Decreases

(c) Remains unchanged (d) None of these

95. In Young's double slit experiment, the aperture screen distance is 2*m*. The fringe width is 1 *mm*. Light of 600 *nm* is used. If a thin plate of glass (μ = 1.5) of thickness 0.06 *mm* is placed over one of the slits, then there will be a lateral displacement of the fringes by

(a) 0 *cm* (b) 5 *cm*

(c) 10 *cm* (d) 15 *cm*

96. In which of the following is the interference due to the division of wave front [UPSEAT 2005]

(a) Young's double slit experiment

(b) Fresnel's biprism experiment

(c) Lloyd's mirror experiment

(d) Demonstration colours of thin film

97. Two slits are separated by a distance of 0.5 *mm* and illuminated with light of $\lambda = 6000$ Å. If the screen is placed 2.5 *m* from the slits. The distance of the third bright image from the centre will be

(a) 1.5 *mm* (b) 3 *mm*

(c) 6 *mm* (d) 9 *mm*

Doppler's Effect of Light

1. The observed wavelength of light coming from a distant galaxy is found to be increased by 0.5% as compared with that coming from a terrestrial source. The galaxy is

[MP PMT 1993, 2003]

(a) Stationary with respect to the earth

(b) Approaching the earth with velocity of light

(c) Receding from the earth with the velocity of light

(d) Receding from the earth with a velocity equal to $1.5 \times 10^{6}\,m/s$

2. A star producing light of wavelength 6000 Å moves away from the earth with a speed of 5 *km/sec*. Due to Doppler effect the shift in wavelength will be $(c = 3 \times 10^{8}\,m/\sec)$

[MP PMT 1990]

(a) 0.1 Å (b) 0.05 Å

(c) 0.2 Å (d) 1 Å

3. If the shift of wavelength of light emitted by a star is towards violet, then this shows that star is

[RPET 1996; RPMT 1999]

(a) Stationary
(b) Moving towards earth
(c) Moving away from earth
(d) Information is incomplete

4. Assuming that universe is expanding, if the spectrum of light coming from a star which is going away from earth is tested, then in the wavelength of light

(a) There will be no change
(b) The spectrum will move to infrared region
(c) The spectrum will seems to shift to ultraviolet side
(d) None of the above

5. Doppler's effect in sound in addition to relative velocity between source and observer, also depends while source and observer or both are moving. Doppler effect in light depend only on the relative velocity of source and observer. The reason of this is

(a) Einstein mass - energy relation
(b) Einstein theory of relativity
(c) Photoelectric effect
(d) None of these

6. A rocket is moving away from the earth at a speed of $6\times10^7 m/s$. The rocket has blue light in it. What will be the wavelength of light recorded by an observer on the earth (wavelength of blue light = 4600 Å)

(a) 4600 Å (b) 5520 Å
(c) 3680 Å (d) 3920 Å

7. A spectral line λ = 5000 Å in the light coming from a distant star is observed as a 5200 Å. What will be recession velocity of the star

(a) $1.15\times10^7 cm/\sec$ (b) $1.15\times10^7 m/\sec$
(c) $1.15\times10^7 km/\sec$ (d) 1.15 *km/sec*

8. The apparent wavelength of the light from a star moving away from the earth is 0.01% more than its real wavelength. Then the velocity of star is [CPMT 1979]

(a) 60 *km/sec* (b) 15 *km/sec*
(c) 150 *km/sec* (d) 30 *km/sec*

9. A star emits light of 5500 Å wavelength. Its appears blue to an observer on the earth, it means [DPMT 2002]

(a) Star is going away from the earth
(b) Star is stationary
(c) Star is coming towards earth
(d) None of the above

10. The velocity of light emitted by a source *S* observed by an observer *O*, who is at rest with respect to *S* is *c*. If the observer moves towards *S* with velocity *v*, the velocity of light as observed will be

(a) *c* + *v* (b) $c-v$
(c) *c* (d) $\sqrt{1-\frac{v^2}{c^2}}$

11. In the context of Doppler effect in light, the term 'red shift' signifies

(a) Decrease in frequency
(b) Increase in frequency
(c) Decrease in intensity
(d) Increase in intensity

12. The sun is rotating about its own axis. The spectral lines emitted from the two ends of its equator, for an observer on the earth, will show [MP PMT 1994]

(a) Shift towards red end
(b) Shift towards violet end
(c) Shift towards red end by one line and towards violet end by other
(d) No shift

13. A star is moving away from the earth with a velocity of 100 *km/s*. If the velocity of light is $3\times10^8 m/s$ then the shift of its spectral line of wavelength 5700 Å due to Doppler's effect will be [MP PET/PMT 1988]

(a) 0.63 Å (b) 1.90 Å
(c) 3.80 Å (d) 5.70 Å

14. If a source of light is moving away from a stationary observer, then the frequency of light wave appears to change because of

(a) Doppler's effect
(b) Interference
(c) Diffraction
(d) None of these

15. A star emitting radiation at a wavelength of 5000 Å is approaching earth with a velocity of $1.5\times10^6 m/s$. The change in wavelength of the radiation as received on the earth, is

(a) 25 Å (b) Zero
(c) 100 Å (d) 2.5 Å

16. A star emitting light of wavelength 5896 Å is moving away from the earth with a speed of 3600 *km / sec*. The wavelength of light observed on earth will

[MP PET 1995, 2002]

(a) Decrease by 5825.25 Å
(b) Increase by 5966.75 Å
(c) Decrease by 70.75 Å
(d) Increase by 70.75 Å

($c=3\times10^8 m/\sec$ is the speed of light)

17. A star moves away from earth at speed 0.8 *c* while emitting light of frequency $6\times10^{14} Hz$. What frequency will be observed on the earth (in units of 10 *Hz*) (*c* = speed of light) [NCERT 1980]

(a) 0.24 (b) 1.2
(c) 30 (d) 3.3

18. A light source approaches the observer with velocity 0.8 *c*. The doppler shift for the light of wavelength 5500 Å is

[MP PET 1996]

(a) 4400 Å (b) 1833 Å

(c) 3167 Å (d) 7333 Å

19. Light coming from a star is observed to have a wavelength of 3737 Å, while its real wavelength is 3700 Å. The speed of the star relative to the earth is [Speed of light $3\times10^8 m/s$] **[MP PET 1997]**

(a) $3\times10^5 m/s$ (b) $3\times10^6 m/s$

(c) $3.7\times10^7 m/s$ (d) $3.7\times10^6 m/s$

20. In the spectrum of light of a luminous heavenly body the wavelength of a spectral line is measured to be 4747 Å while actual wavelength of the line is 4700 Å. The relative velocity of the heavenly body with respect to earth will be (velocity of light is $3\times10^8 m/s$)

(a) $3\times10^5 m/s$ moving towards the earth

(b) $3\times10^5 m/s$ moving away from the earth

(c) $3\times10^6 m/s$ moving towards the earth

(d) $3\times10^6 m/s$ moving away from the earth

21. The wavelength of light observed on the earth, from a moving star is found to decrease by 0.05%. Relative to the earth the star is

(a) Moving away with a velocity of $1.5\times10^5 m/s$

(b) Coming closer with a velocity of $1.5\times10^5 m/s$

(c) Moving away with a velocity of $1.5\times10^4 m/s$

(d) Coming closer with a velocity of $1.5\times10^4 m/s$

22. A star is going away from the earth. An observer on the earth will see the wavelength of light coming from the star **[MP PMT 1999]**

(a) Decreased

(b) Increased

(c) Neither decreased nor increased

(d) Decreased or increased depending upon the velocity of the star

23. A star is moving towards the earth with a speed of 4.5×10^6 *m/s*. If the true wavelength of a certain line in the spectrum received from the star is 5890 Å, its apparent wavelength will be about $[c = 3\times10^8 m/s]$ **[MP PMT 1999]**

(a) 5890 Å (b) 5978 Å

(c) 5802 Å (d) 5896 Å

24. Due to Doppler's effect, the shift in wavelength observed is 0.1 Å for a star producing wavelength 6000 Å. Velocity of recession of the star will be

(a) 2.5 *km/s* (d) 10 *km/s*

(c) 5 *km/s* (d) 20 *km/s*

25. A rocket is going away from the earth at a speed of 10^6 *m/s* If the wavelength of the light wave emitted by it be 5700 Å, what will be its Doppler's shift **[RPMT 1996]**

(a) 200 Å (b) 19 Å

(c) 20 Å (d) 0.2 Å

26. A rocket is going away from the earth at a speed 0.2*c*, where *c* = speed of light. It emits a signal of frequency $4\times10^7 Hz$. What will be the frequency observed by an observer on the earth

(a) $4\times10^6 Hz$ (b) $3.2\times10^7 Hz$

(c) $3\times10^6 Hz$ (d) $5\times10^7 Hz$

27. If a star is moving towards the earth, then the lines are shifted towards **[AIIMS 1997]**

(a) Red (b) Infrared

(c) Blue (d) Green

28. When the wavelength of light coming from a distant star is measured it is found shifted towards red. Then the conclusion is **[MP PMT/PET 1998]**

(a) The star is approaching the observer

(b) The star recedes away from earth

(c) There is gravitational effect on the light

(d) The star remains stationary

29. A heavenly body is receding from earth such that the fractional change in λ is 1, then its velocity is **[DCE 2000]**

(a) C **[MP PMT/PET 1998]** (b) $\frac{3C}{5}$

(c) $\frac{C}{5}$ (d) $\frac{2C}{5}$

30. The 6563 Å line emitted by hydrogen atom in a star is found to be red shifted by 5 Å. The speed with which the star is receding from the earth is **[Pb. PMT 2002]**

(a) 17.29×10^9 *m/s* (b) 4.29×10^7 *m/s*

(c) 3.39×10^5 *m/s* (d) 2.29×10^5 *m/s*

31. Three observers *A*, *B* and *C* measure the speed of light coming from a source to be v_A, v_B and v_C. The observer *A* moves towards the source, the observer *C* moves away from the source with the same speed. The observer *B* stays stationary. the surrounding space is vacuum every where. Then **[KCET 2002]**

(a) $v_A > v_B > v_C$ (b) $v_A < v_B < v_C$

(c) $v_A = v_B = v_C$ (d) $v_A = v_B > v_C$

32. Light from the constellation Virgo is observed to increase in wavelength by 0.4%. With respect to Earth the constellation is

(a) Moving away with velocity 1.2×10^6 *m/s*

(b) Coming closer with velocity 1.2×10^6 *m/s*

(c) Moving away with velocity 4×10^6 *m/s*

(d) Coming closer with velocity 4×10^6 *m/s*

33. It is believed that the universe is expanding and hence the distant stars are receding from us. Light from such a star will show

(a) Shift in frequency towards longer wavelengths

(b) Shift in frequency towards shorter wavelength

(c) No shift in frequency but a decrease in intensity

(d) A shift in frequency sometimes towards longer and sometimes towards shorter wavelengths

Diffraction of Light

1. A slit of width *a* is illuminated by white light. For red light (λ = 6500 Å), the first minima is obtained at $\theta = 30^o$. Then the value of *a* will be [MP PMT 1987; CPMT 2002]

(a) 3250 Å (b) $6.5 \times 10^{-4} mm$

(c) 1.24 *microns* (d) $2.6 \times 10^{-4} cm$

2. The light of wavelength 6328 Å is incident on a slit of width 0.2 *mm* perpendicularly, the angular width of central maxima will be [MP PMT 1987; Pb. PMT 2002]

(a) 0.36^o (b) 0.18^o

(c) 0.72^o (d) 0.09^o

3. The bending of beam of light around corners of obstacles is called [NCERT 1990; AFMC 1995; RPET 1997; RPMT 1997; CPMT 1999; JIPMER 2000]

(a) Reflection (b) Diffraction

(c) Refraction (d) Interference

4. The penetration of light into the region of geometrical shadow is called [CPMT 1999; JIPMER 2000]

(a) Polarisation (b) Interference

(c) Diffraction (d) Refraction

5. A slit of size 0.15 *cm* is placed at 2.1 *m* from a screen. On illuminated it by a light of wavelength 5×10^{-5} *cm*. The width of central maxima will be [RPMT 1999]

(a) 70 *mm* (b) 0.14 *mm*

(c) 1.4 *mm* (d) 0.14 *cm*

6. A diffraction is obtained by using a beam of red light. What will happen if the red light is replaced by the blue light [KCET 2000; BHU 2001]

(a) Bands will narrower and crowd full together

(b) Bands become broader and further apart

(c) No change will take place

(d) Bands disappear

7. What will be the angle of diffracting for the first minimum due to Fraunhoffer diffraction with sources of light of wave length 550 *nm* and slit of width 0.55 *mm*

(a) 0.001 *rad* (b) 0.01 *rad*

(c) 1 *rad* (d) 0.1 *rad*

8. Angular width (β) of central maximum of a diffraction pattern on a single slit does not depend upon [DCE 2000; 01]

(a) Distance between slit and source

(b) Wavelength of light used

(c) Width of the slit

(d) Frequency of light used

9. A single slit of width 0.20 *mm* is illuminated with light of wavelength 500 *nm*. The observing screen is placed 80 *cm* from the slit. The width of the central bright fringe will be [AMU (Med.) 2002]

(a) 1 *mm* (b) 2 *mm*

(c) 4 *mm* (d) 5 *mm*

10. Yellow light is used in single slit diffraction experiment with slit width 0.6 *mm*. If yellow light is replaced by *X*-rays then the pattern will reveal [IIT-JEE (Screening) 1999; MP PMT 2002; KCET 2003]

(a) That the central maxima is narrower

(b) No diffraction pattern

(c) More number of fringes

(d) Less number of fringes

11. Which statement is correct for a zone plate and a lens [RPMT 2002]

(a) Zone plate has multi focii whereas lens has one

(b) Zone plate has one focus whereas lens has multiple focii

(c) Both are correct

(d) Zone plate has one focus whereas a lens has infinite

12. In Fresnel diffraction, if the distance between the disc and the screen is decreased, the intensity of central bright spot will

(a) Increase (b) Decrease

(c) Remain constant (d) None of these

13. A plane wavefront $(\lambda = 6 \times 10^{-7} m)$ falls on a slit $0.4\,mm$ wide. A convex lens of focal length $0.8\,m$ placed behind the slit focusses the light on a screen. What is the linear diameter of second maximum [RPMT 2001]

(a) $6\,mm$ (b) $12\,mm$

(c) $3\,mm$ (d) $9\,mm$

14. A zone plate of focal length $60\,cm$, behaves as a convex lens, If wavelength of incident light is $6000\ Å$, then radius of first half period zone will be [RPMT 2001]

(a) $36 \times 10^{-8} m.$ (b) $6 \times 10^{-8} m.$

(c) $\sqrt{6} \times 10^{-8} m.$ (d) $6 \times 10^{-4} m.$

15. Radius of central zone of circular zone plate is $2.3\,mm$. Wavelength of incident light is $5893\ Å$. Source is at a distance of $6\,m$. Then the distance of first image will be [RPMT 2001]

(a) $9\,m$ (b) $12\,m$

(c) $24\,m$ (d) $36\,m$

16. Red light is generally used to observe diffraction pattern from single slit. If blue light is used instead of red light, then diffraction pattern [Pb. PMT 2001] [RPMT 2001; BCECE 2005; CPMT 2005]

(a) Will be more clear (b) Will contract

(c) Will expanded (d) Will not be visualized

17. In the experiment of diffraction at a single slit, if the slit width is decreased, the width of the central maximum [KCET 2001]

(a) Increases in both Fresnel and Fraunhofer diffraction

(b) Decreases both in Fresnal and Fraunhofer diffraction

(c) Increases in Fresnel diffraction but decreases in Fraunhofer diffraction

(d) Decreases in Fresnel diffraction but increases is Fraunofer diffraction.

18. Conditions of diffraction is [RPET 2001]

(a) $\dfrac{a}{\lambda} = 1$ (b) $\dfrac{a}{\lambda} >> 1$

(c) $\frac{a}{\lambda} << 1$ (d) None of these

19. Light of wavelength $589.3\,nm$ is incident normally on the slit of width $0.1\,mm$. What will be the angular width of the central diffraction maximum at a distance of $1\,m$ from the slit

(a) $0.68°$ (b) $1.02°$

(c) $0.34°$ (d) None of these

20. The phenomenon of diffraction of light was discovered by [KCET 2000]

(a) Hygens (b) Newton

(c) Fresnel (d) Grimaldi

21. The radius r of half period zone is proportional to [RPMT 1998, 2002]

(a) $\sqrt{n}$ (b) $\frac{1}{\sqrt{n}}$

(c) n^2 (d) $\frac{1}{n}$

22. In a diffraction pattern by a wire, on increasing diameter of wire, fringe width [RPMT 1998]

(a) Decreases

(b) Increases

(c) Remains unchanged

(d) Increasing or decreasing will depend on wavelength

23. What will be the angular width of central maxima in Fraunhoffer diffraction when light of wavelength $6000\,Å$ is used and slit width is $12 \times 10^{-5}\,cm$. [RPMT 2004]

(a) 2 *rad* (b) 3 *rad*

(c) 1 *rad* (d) 8 *rad*

24. When a compact disc is illuminated by a source of white light. Coloured 'lanes' are observed. This is due to [DCE 2003; AIIMS 2004]

(a) Dispersion (b) Diffraction

(c) Interference (d) Refraction

25. The diffraction effect can be observed in [J & K CET 2004]

(a) Only sound waves

(b) Only light waves

(c) Only ultrasonic waves

(d) Sound as well as light waves

26. If we observe the single slit Fraunhoffer diffraction with wavelength λ and slit width *e*, the width of the central maxima is 2θ. On decreasing the slit width for the same λ [UPSEAT 2004]

(a) θ increases

(b) θ remains unchanged

(c) θ decreases

(d) θ increases or decreases depending on the intensity of light

27. When light is incident on a diffraction grating the zero order principal maximum will be [KCET 2004]

(a) One of the component colours

(b) Absent

(c) Spectrum of the colours

(d) White

28. A beam of light of wavelength 600 *nm* from a distant source falls on a single slit 1 *mm* wide and the resulting diffraction pattern is observed on a screen 2 *m* away. The distance between the first dark fringes on either side of the central bright fringe is [IIT-JEE 1994; KCET 2004]

(a) 1.2 *mm* (b) 1.2 *cm*

(c) 2.4 *cm* [BHU (Med.) 1999] (d) 2.4 *mm*

29. In order to see diffraction the thickness of the film is [J&K CEE 2001]

(a) 100 Å (b) 10,000 Å

(c) 1 *mm* (d) 1 *cm*

30. Diffraction effects are easier to notice in the case of sound waves than in the case of light waves because [RPET 1978; KCET 1994, 2000]

(a) Sound waves are longitudinal

(b) Sound is perceived by the ear

(c) Sound waves are mechanical waves

(d) Sound waves are of longer wavelength

31. Direction of the first secondary maximum in the Fraunhofer diffraction pattern at a single slit is given by (*a* is the width of the slit) [KCET 1999]

(a) $a\sin\theta = \frac{\lambda}{2}$ (b) $a\cos\theta = \frac{3\lambda}{2}$

(c) $a\sin\theta = \lambda$ (d) $a\sin\theta = \frac{3\lambda}{2}$

32. A parallel monochromatic beam of light is incident normally on a narrow slit. A diffraction pattern is formed on a screen placed perpendicular to the direction of incident beam. At the first maximum of the diffraction pattern the phase difference between the rays coming from the edges of the slit is

(a) 0 (b) $\frac{\pi}{2}$

(c) π (d) 2π

33. Diffraction and interference of light suggest [CPMT 1995; RPMT 1998]

(a) Nature of light is electro-magnetic

(b) Wave nature

(c) Nature is quantum

(d) Nature of light is transverse

34. A light wave is incident normally over a slit of width $24 \times 10^{-5}\,cm$. The angular position of second dark fringe from the central maxima is 30°. What is the wavelength of light

(a) 6000 Å (b) 5000 Å

(c) 3000 Å (d) 1500 Å

35. A parallel beam of monochromatic light of wavelength 5000 Å is incident normally on a single narrow slit of width 0.001 *mm*. The light is focused by a convex lens on a screen placed on the focal plane. The first minimum will be formed for the angle of diffraction equal to [CBSE PMT 1993]

(a) 0° (b) 15°

(c) 30° (d) 60°

36. The condition for observing Fraunhofer diffraction from a single slit is that the light wavefront incident on the slit should be

(a) Spherical (b) Cylindrical

(c) Plane (d) Elliptical

37. To observe diffraction the size of an obstacle **[CPMT 1982]**

(a) Should be of the same order as wavelength

(b) Should be much larger than the wavelength

(c) Have no relation to wavelength

(d) Should be exactly $\frac{\lambda}{2}$

38. In the far field diffraction pattern of a single slit under polychromatic illumination, the first minimum with the wavelength λ_1 is found to be coincident with the third maximum at λ_2. So

(a) $3\lambda_1 = 0.3\lambda_2$ (b) $3\lambda_1 = \lambda_2$

(c) $\lambda_1 = 3.5\lambda_2$ (d) $0.3\lambda_1 = 3\lambda_2$

39. Light of wavelength λ = 5000 $\mathring{A}$ falls normally on a narrow slit. A screen placed at a distance of 1 *m* from the slit and perpendicular to the direction of light. The first minima of the diffraction pattern is situated at 5 *mm* from the centre of central maximum. The width of the slit is

(a) 0.1 *mm* (b) 1.0 *mm*

(c) 0.5 *mm* (d) 0.2 *mm*

40. The width of the *n HPZ* will be

(a) $\sqrt{nb\lambda}$ (b) $\sqrt{b\lambda}\,[\sqrt{n}-\sqrt{n-1}]$

(c) $(\sqrt{n}-\sqrt{n-1})$ (d) $\frac{\sqrt{b\lambda}}{[\sqrt{n}-\sqrt{n-1}]}$

41. A single slit of width *a* is illuminated by violet light of wavelength 400 *nm* and the width of the diffraction pattern is measured as *y*. When half of the slit width is covered and illuminated by yellow light of wavelength 600 *nm*, the width of the diffraction pattern is

(a) The pattern vanishes and the width is zero

(b) $y/3$

(c) $3y$

(d) None of these

Polarization of Light

1. A polariser is used to **[CPMT 1999]**

(a) Reduce intensity of light

(b) Produce polarised light

(c) Increase intensity of light

(d) Produce unpolarised light

2. Light waves can be polarised as they are

[CBSE PMT 1993; KCET 1994; AFMC 1997; J & K CET 2002; CPMT 2005]

(a) Transverse (b) Of high frequency

(c) Longitudinal (d) Reflected

3. Through which character we can distinguish the light waves from sound waves **[CBSE PMT 1990; RPET 2000, 02]**

(a) Interference (b) Refraction

(c) Polarisation **[MP PMT 1987]** (d) Reflection

4. The angle of polarisation for any medium is 60°, what will be critical angle for this **[UPSEAT 1999]**

(a) $\sin^{-1}\sqrt{3}$ (b) $\tan^{-1}\sqrt{3}$

(c) $\cos^{-1}\sqrt{3}$ (d) $\sin^{-1}\frac{1}{\sqrt{3}}$

5. The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refraction index *n*) is

(a) $\sin^{-1}(n)$ (b) $\sin^{-1}\left(\frac{1}{n}\right)$

(c) $\tan^{-1}\left(\frac{1}{n}\right)$ (d) $\tan^{-1}(n)$

6. Which of following can not be polarised **[Kerala PMT 2001]**

(a) Radio waves (b) Ultraviolet rays

(c) Infrared rays (d) Ultrasonic waves

7. A polaroid is placed at 45° to an incoming light of intensity I_0. Now the intensity of light passing through polaroid after polarisation would be **[CPMT 1995]**

(a) I_0 (b) $I_0/2$

(c) $I_0/4$ (d) Zero

8. Plane polarised light is passed through a polaroid. On viewing through the polaroid we find that when the polariod is given one complete rotation about the direction of the light, one of the following is observed **[MNR 1993]**

(a) The intensity of light gradually decreases to zero and remains at zero

(b) The intensity of light gradually increases to a maximum and remains at maximum

(c) There is no change in intensity

(d) The intensity of light is twice maximum and twice zero

9. Out of the following statements which is not correct

[KCET 2005] **[CPMT 1991]**

(a) When unpolarised light passes through a Nicol's prism, the emergent light is elliptically polarised

(b) Nicol's prism works on the principle of double refraction and total internal reflection

(c) Nicol's prism can be used to produce and analyse polarised light

(d) Calcite and Quartz are both doubly refracting crystals

10. A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angle ϕ. If μ represents the refractive index of glass with respect to air, then the angle between reflected and refracted rays is

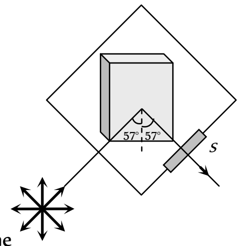
[CPMT 1989]

(a) $90+\phi$ (b) $\sin^{-1}(\mu\cos\phi)$

(c) 90° (d) $90^o - \sin^{-1}(\sin\phi/\mu)$

11. Figure represents a glass plate placed vertically on a horizontal table with a beam of unpolarised light falling on its surface at the polarising angle of 57° with the normal. The electric vector in the reflected light on screen *S* will vibrate with respect to the plane of incidence in a

[CPMT 1988]



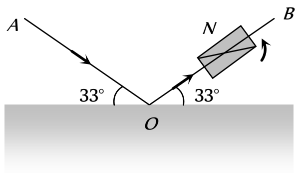
(a) Vertical plane

(b) Horizontal plane

(c) Plane making an angle of 45° with the vertical

(d) Plane making an angle of 57° with the horizontal

12. A beam of light *AO* is incident on a glass slab $(\mu = 1.54)$ in a direction as shown in figure. The reflected ray *OB* is passed through a Nicol prism on viewing through a Nicole prism, we find on rotating the prism that [CPMT 1986]



(a) The intensity is reduced down to zero and remains zero

(b) The intensity reduces down some what and rises again

(c) There is no change in intensity

(d) The intensity gradually reduces to zero and then again increases

13. Polarised glass is used in sun glasses because [CPMT 1981]

(a) It reduces the light intensity to half an account of polarisation

(b) It is fashionable

(c) It has good colour

(d) It is cheaper

14. In the propagation of electromagnetic waves the angle between the direction of propagation and plane of polarisation is

(a) 0° (b) 45°

(c) 90° (d) 180°

15. The transverse nature of light is shown by [CPMT 1972, 74, 78; RPMT 1999; AFMC 2001; AIEEE 2002; MP PET 2004; MP PMT 2000, 04; UPSEAT 2005]

(a) Interference of light (b) Refraction of light

(c) Polarisation of light (d) Dispersion of light

16. A calcite crystal is placed over a dot on a piece of paper and rotated, on seeing through the calcite one will be see

[CPMT 1971]

(a) One dot

(b) Two stationary dots

(c) Two rotating dots

(d) One dot rotating about the other

17. A light has amplitude *A* and angle between analyser and polariser is 60°. Light is reflected by analyser has amplitude

[UPSEAT 2001]

(a) $A\sqrt{2}$ (b) $A/\sqrt{2}$

(c) $\sqrt{3}A/2$ (d) $A/2$

18. When light is incident on a doubly refracting crystal, two refracted rays-ordinary ray (*O*-ray) and extra ordinary ray (*E*-ray) are produced. Then [KCET 2001]

(a) Both *O*-ray and *E*-ray are polarised perpendicular to the plane of incidence

(b) Both *O*-ray and *E*-ray are polarised in the plane of incidence

(c) *E*-ray is polarised perpendicular to the plane of incidence and *O*-ray in the plane of incidence

(d) *E*-ray is polarised in the plane of incidence and *O*-ray perpendicular to the plane of incidence

19. Light passes successively through two polarimeters tubes each of length 0.29*m*. The first tube contains dextro rotatory solution of concentration 60*kgm*⁻³ and specific rotation 0.01*rad m*²*kg*⁻¹. The second tube contains laevo rotatory solution of concentration 30*kg*/*m*³ and specific rotation 0.02 *rad m*²*kg*⁻¹. The net rotation produced is [KCET 2002]

(a) 15° (b) 0°

(c) 20° (d) 10°

20. V_o and V_E represent the velocities, μ_o and μ_E the refractive indices of ordinary and extraordinary rays for a doubly refracting crystal. Then [KCET 2002]

(a) $V_o \geq V_E,\ \mu_o \leq \mu_E$ if the crystal is calcite

(b) $V_o \leq V_E,\ \mu_o \leq \mu_E$ if the crystal is quartz

(c) $V_o \leq V_E,\ \mu_o \geq \mu_E$ if the crystal is calcite

(d) $V_o \geq V_E,\ \mu_o \geq \mu_E$ if the crystal is quartz

21. Polarising angle for water is 53°4'. If light is incident at this angle on the surface of water and reflected, the angle of refraction is

(a) 53°4' (b) 126°56'

(c) 36°56' (d) 30°4'

22. When a plane polarised light is passed through an analyser and analyser is rotated through 90°, the intensity of the emerging light [CPMT 1978]

(a) Varies between a maximum and minimum

(b) Becomes zero

(c) Does not vary

(d) Varies between a maximum and zero

23. Consider the following statements *A* to *B* and identify the correct answer

A. Polarised light can be used to study the helical surface of nucleic acids.

B. Optics axis is a direction and not any particular line in the crystal [EAMCET (Med.) 2003]

(a) A and B are correct

(b) A and B are wrong

(c) A is correct but B is wrong

(d) A is wrong but B is correct

24. Two Nicols are oriented with their principal planes making an angle of 60°. The percentage of incident unpolarized light which passes through the system is

(a) 50% (b) 100%

(c) 12.5% (d) 37.5%

25. Unpolarized light falls on two polarizing sheets placed one on top of the other. What must be the angle between the characteristic directions of the sheets if the intensity of the final transmitted light is one-third the maximum intensity of the first transmitted beam

(a) 75° (b) 55°

(c) 35° (d) 15°

26. Unpolarized light of intensity $32\,Wm^{-2}$ passes through three polarizers such that transmission axes of the first and second polarizer makes and angle 30° with each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of final emerging light will be

(a) $32\ Wm^{-2}$ (b) $3\ Wm^{-2}$

(c) $8\ Wm^{-2}$ (d) $4\ Wm^{-2}$

27. In the visible region of the spectrum the rotation of the place of polarization is given by $\theta = a + \frac{b}{\lambda^2}$. The optical rotation produced by a particular material is found to be 30° *per mm* at $\lambda = 5000\,Å$ and 50° *per mm* at $\lambda = 4000\,Å$. The value of constant *a* will be

(a) $+\frac{50°}{9}$ *per mm* (b) $-\frac{50°}{9}$ *per mm*

(c) $+\frac{9°}{50}$ *per mm* (d) $-\frac{9°}{50}$ *per mm*

28. When an unpolarized light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is **[AIEEE 2005]**

(a) Zero (b) I_0

(c) $\frac{1}{2}I_0$ (c) $\frac{1}{4}I_0$

29. Refractive index of material is equal to tangent of polarising angle. It is called **[AFMC 2005]**

(a) Brewster's law (b) Lambert's law

(c) Malus's law (d) Bragg's law

30. In case of linearly polarized light, the magnitude of the electric field vector: **[AIIMS 2005]**

(a) Does not change with time

(b) Varies periodically with time

(c) Increases and decreases linearly with time

(d) Is parallel to the direction of propagation

31. When unpolarised light beam is incident from air onto glass (n = 1.5) at the polarising angle **[KCET 2005]**

(a) Reflected beam is polarised 100 percent

(b) Reflected and refracted beams are partially polarised

(c) The reason for (a) is that almost all the light is reflected

(d) All of the above

32. An optically active compound **[DCE 2005]**

(a) Rotates the plane polarised light

(b) Changing the direction of polarised light

(c) Do not allow plane polarised light to pass through

(d) None of the above

33. When the angle of incidence on a material is 60°, the reflected light is completely polarized. The velocity of the refracted ray inside the material is (in ms^{-1}) **[Kerala PMT 2005]**

(a) 3×10^8 (b) $\left(\frac{3}{\sqrt{2}}\right) \times 10^8$

(c) $\sqrt{3} \times 10^8$ (d) 0.5×10^8

34. Two polaroids are placed in the path of unpolarized beam of intensity I_0 such that no light is emitted from the second polaroid. If a third polaroid whose polarization axis makes an angle θ with the polarization axis of first polaroid, is placed between these polaroids then the intensity of light emerging from the last polaroid will be **[UPSEAT 2005]**

(a) $\left(\frac{I_0}{8}\right)\sin^2 2\theta$ (b) $\left(\frac{I_0}{4}\right)\sin^2 2\theta$

(c) $\left(\frac{I_0}{2}\right)\cos^4 \theta$ (d) $I_0 \cos^4 \theta$

35. For the study of the helical structure of nucleic acids, the property of electromagnetic radiation generally used is **[EAMCET 2005]**

(a) Reflection (b) Interference

(c) Diffraction (d) Polarization

EM Waves

1. Which of the following statement is wrong **[NCERT 1976]**

(a) Infrared photon has more energy than the photon of visible light

(b) Photographic plates are sensitive to ultraviolet rays

(c) Photographic plates can be made sensitive to infrared rays

(d) Infrared rays are invisible but can cast shadows like visible light rays

2. Pick out the longest wavelength from the following types of radiations **[CBSE PMT 1990]**

(a) Blue light (b) γ-rays

(c) X-rays (d) Red light

3. Wave which cannot travel in vacuum is **[MP PMT 1994]**

(a) X-rays (b) Infrasonic

(c) Ultraviolet (d) Radiowaves

4. Light is an electromagnetic wave. Its speed in vacuum is given by the expression **[CBSE PMT 1993; MP PMT 1994; RPMT 1999; MP PET 2001; Kerala PET 2001; AIIMS 2002]**

(a) $\sqrt{\mu_o \varepsilon_o}$ (b) $\sqrt{\frac{\mu_o}{\varepsilon_o}}$

(c) $\sqrt{\frac{\varepsilon_o}{\mu_o}}$ (d) $\frac{1}{\sqrt{\mu_o \varepsilon_o}}$

5. The range of wavelength of the visible light is **[MP PMT 2000; MP PET 2002]**

(a) 10 Å to 100 Å (b) 4,000 Å to 8,000 Å

(c) 8,000 Å to 10,000 Å (d) 10,000 Å to 15,000 Å

6. Which radiation in sunlight, causes heating effect **[AFMC 2001]**

(a) Ultraviolet (b) Infrared

(c) Visible light (d) All of these

7. Which of the following represents an infrared wavelength

[CPMT 1975; MP PET/PMT 1988]

(a) 10^{-4} *cm* (b) 10^{-5} *cm*

(c) 10^{-6} *cm* (d) 10^{-7} *cm*

8. The wavelength of light visible to eye is of the order of [CPMT 1982, 84]

(a) 10^{-2} *m* (b) 10^{-10} *m*

(c) 1 *m* (d) 6×10^{-7} *m*

9. The speed of electromagnetic wave in vacuum depends upon the source of radiation [Kerala PMT 2004]

(a) Increases as we move from γ-rays to radio waves

(b) Decreases as we move from γ-rays to radio waves

(c) Is same for all of them

(d) None of these

10. Which of the following radiations has the least wavelength [AIEEE 2003]

(a) γ-rays (b) β-rays

(c) α-rays (d) X-rays

11. The maximum distance upto which TV transmission from a TV tower of height h can be received is proportional to [AIIMS 2003]

(a) $h^{1/2}$ (b) h

(c) h (d) h^2

12. Which of the following are not electromagnetic waves [AIEEE 2002; CBSE PMT 2003]

(a) Cosmic rays (b) Gamma rays

(c) β-rays (d) X-rays

13. Ozone is found in [DPMT 2002]

(a) Stratosphere (b) Ionosphere

(c) Mesosphere (d) Troposphere

14. The electromagnetic waves travel with a velocity [J & K CET 2002]

(a) Equal to velocity of sound

(b) Equal to velocity of light

(c) Less than velocity of light

(d) None of these

15. The ozone layer absorbs [Kerala PET 2002]

(a) Infrared radiations (b) Ultraviolet radiations

(c) X-rays (d) γ-rays

16. Electromagnetic radiation of highest frequency is [Kerala PMT 2002]

(a) Infrared radiations (b) Visible radiation

(c) Radio waves (d) γ-rays

17. Which of the following shows green house effect [CBSE PMT 2002]

(a) Ultraviolet rays (b) Infrared rays

(c) X-rays (d) None of these

18. Which of the following waves have the maximum wavelength [AFMC 2002]

(a) X-rays (b) I.R. rays

(c) UV rays (d) Radio waves

19. Electromagnetic waves are transverse in nature is evident by [AIEEE 2002]

(a) Polarization (b) Interference

(c) Reflection (d) Diffraction

20. If $\vec{E}$ and $\vec{B}$ are the electric and magnetic field vectors of E.M. waves then the direction of propagation of E.M. wave is along the direction of [CBSE PMT 1992, 2002; DCE 2002, 05]

(a) $\vec{E}$ (b) $\vec{B}$

(c) $\vec{E}\times\vec{B}$ (d) None of these

21. Biological importance of Ozone layer is [CBSE PMT 2001]

(a) It stops ultraviolet rays

(b) Ozone rays reduce green house effect

(c) Ozone layer reflects radio waves

(d) Ozone layer controls O_2 / H_2 radio in atmosphere

22. What is ozone hole [AFMC 2001]

(a) Hole in the ozone layer

(b) Formation of ozone layer

(c) Thinning of ozone layer in troposphere

(d) Reduction in ozone thickness in stratosphere

23. Which rays are not the portion of electromagnetic spectrum [Haryana CEET 2000]

(a) X-rays (b) Microwaves

(c) α-rays (d) Radio waves

24. Radio wave diffract around building although light waves do not. The reason is that radio waves [AMU 2000]

(a) Travel with speed larger than c

(b) Have much larger wavelength than light

(c) Carry news

(d) Are not electromagnetic waves

25. The frequencies of X-rays, γ-rays and ultraviolet rays are respectively a, b and c. Then [CBSE PMT 2000]

(a) $a < b, b > c$ (b) $a > b, b > c$

(c) $a > b, b < c$ (d) $a < b, b < c$

26. Radio waves and visible light in vacuum have [KCET 2000]

(a) Same velocity but different wavelength

(b) Continuous emission spectrum

(c) Band absorption spectrum

(d) Line emission spectrum

27. Energy stored in electromagnetic oscillations is in the form of [Haryana CEET 2000; AFMC 1994]

(a) Electrical energy (b) Magnetic energy

(c) Both (a) and (b) (d) None of these

28. Heat radiations propagate with the speed of [AMU 2000]

(a) α-rays (b) β-rays

(c) Light waves (d) Sound waves

29. If a source is transmitting electromagnetic wave of frequency 8.2×10^6 *Hz*, then wavelength of the electromagnetic waves transmitted from the source will be [DPMT 1999]

(a) 36.6 *m* (b) 40.5 *m*

(c) 42.3 *m* (d) 50.9 *m*

30. In an apparatus, the electric field was found to oscillate with an amplitude of 18 *V/m*. The magnitude of the oscillating magnetic field will be [Pb. PMT 1999]

(a) $4 \times 10^{-6}\ T$ (b) $6 \times 10^{-8}\ T$

(c) $9 \times 10^{-9}\ T$ (d) $11 \times 10^{-11}\ T$

31. According to Maxwell's hypothesis, a changing electric field gives rise to [AIIMS 1998]

(a) An e.m.f. (b) Electric current

(c) Magnetic field (d) Pressure radiant

32. In an electromagnetic wave, the electric and magnetising fields are $100\ V m^{-1}$ and $0.265\ A m^{-1}$. The maximum energy flow is

(a) $26.5\ W/m^2$ (b) $36.5\ W/m^2$

(c) $46.7\ W/m^2$ (d) $765\ W/m^2$

33. The 21 *cm* radio wave emitted by hydrogen in interstellar space is due to the interaction called the hyperfine interaction is atomic hydrogen. the energy of the emitted wave is nearly

(a) 10^{-17} *Joule* (b) 1 *Joule*

(c) 7×10^{-8} *Joule* (d) 10^{-24} *Joule*

34. TV waves have a wavelength range of 1-10 *meter*. Their frequency range in *MHz* is [KCET 1998]

(a) 30-300 (b) 3-30

(c) 300-3000 (d) 3-3000

35. Maxwell's equations describe the fundamental laws of [CPMT 1996]

(a) Electricity only (b) Magnetism only

(c) Mechanics only (d) Both (a) and (b)

36. The oscillating electric and magnetic vectors of an electromagnetic wave are oriented along [CBSE PMT 1994]

(a) The same direction but differ in phase by 90°

(b) The same direction and are in phase

(c) Mutually perpendicular directions and are in phase

(d) Mutually perpendicular directions and differ in phase by 90°

37. In which one of the following regions of the electromagnetic spectrum will the vibrational motion of molecules give rise to absorption [SCRA 1994]

(a) Ultraviolet (b) Microwaves

(c) Infrared (d) Radio waves

38. An electromagnetic wave travels along *z*-axis. Which of the following pairs of space and time varying fields would generate such a wave

(a) E_x, B_y (b) E_y, B_x

(c) E_z, B_x (d) E_y, B_z

39. Which of the following rays has the maximum frequency [CBSE PMT 1994]

(a) Gamma rays (b) Blue light

(c) Infrared rays (d) Ultraviolet rays

40. A signal emitted by an antenna from a certain point can be received at another point of the surface in the form of [CPMT 1993]

(a) Sky wave (b) Ground wave

(c) Sea wave (d) Both (a) and (b)

41. Approximate height of ozone layer above the ground is [CBSE PMT 1991]

(a) 60 to 70 *km* (b) 59 *km* to 80 *km*

(c) 70 *km* to 100 *km* (d) 100 *km* to 200 *km*

42. The electromagnetic waves do not transport [Pb. PET 1991]

(a) Energy (b) Charge

(c) Momentum [Pb. PMT 1997, 98] (d) Information

43. A plane electromagnetic wave is incident on a material surface. If the wave delivers momentum *p* and energy *E*, then

(a) $p = 0,\ E = 0$ (b) $p \neq 0,\ E \neq 0$

(c) $p \neq 0,\ E = 0$ (d) $p = 0,\ E \neq 0$

44. An electromagnetic wave, going through vacuum is described by $E = E_0 \sin(kx - \omega t)$. Which of the following is independent of wavelength [CBSE PMT 1998]

(a) k (b) ω

(c) k/ω (d) $k\omega$

45. An electromagnetic wave going through vacuum is described by $E = E_0 \sin(kx - \omega t)$; $B = B_0 \sin(kx - \omega t)$. Which of the following equation is true

(a) $E_0 k = B_0 \omega$ (b) $E_0 \omega = B_0 k$

(c) $E_0 B_0 = \omega k$ (d) None of these

46. An *LC* resonant circuit contains a 400 *pF* capacitor and a 100 μH inductor. It is set into oscillation coupled to an antenna. The wavelength of the radiated electromagnetic waves is

(a) 377 *mm* (b) 377 *metre*

(c) 377 *cm* (d) 3.77 *cm*

47. A radio receiver antenna that is 2 *m* long is oriented along the direction of the electromagnetic wave and receives a signal of intensity $5 \times 10^{-16}\ W/m^2$. The maximum instantaneous potential difference across the two ends of the antenna is

(a) 1.23 μV (b) 1.23 *mV*

(c) 1.23 *V* (d) 12.3 *mV*

48. Television signals broadcast from the moon can be received on the earth while TV broadcast from Delhi cannot be received at places about 100 *km* distant from Delhi. This is because [CBSE PMT 1994]

(a) There is no atmosphere around the moon

(b) Of strong gravity effect on TV signals

(c) TV signals travel straight and cannot follow the curvature of the earth

(d) There is atmosphere around the earth

49. A TV tower has a height of 100 *m*. The average population density around the tower is 1000 per *km*. The radius of the earth is 6.4×10^6 *m*. the population covered by the tower is

(a) 2×10^6 (b) 3×10^6

(c) 4×10^6 (d) 6×10^6

50. The wavelength 21 *cm* emitted by atomic hydrogen in interstellar space belongs to

(a) Radio waves (b) Infrared waves

(c) Microwaves (d) γ-rays

51. Which scientist experimentally proved the existence of electromagnetic waves [AFMC 2004]

(a) Sir J.C. Bose (b) Maxwell

(c) Marconi (d) Hertz

52. An electromagnetic wave of frequency $\nu = 3.0\,MHz$ passes from vacuum into a dielectric medium with permitivity $\varepsilon = 4.0$. Then

(a) Wavelength is doubled and the frequency remains unchanged

(b) Wavelength is doubled and frequency becomes half

(c) Wavelength is halved and frequency remains unchanged

(d) Wavelength and frequency both remain unchanged

53. Frequency of a wave is $6\times10^{15}\,Hz$. The wave is [Orissa PMT 2004]

(a) Radiowave (b) Microwave

(c) X-ray (d) None of these

54. The region of the atmosphere above troposphere is known as

(a) Lithosphere (b) Uppersphere

(c) Ionosphere (d) Stratosphere

55. Which of the following electromagnetic waves have minimum frequency [Pb PET 2000]

(a) Microwaves (b) Audible waves

(c) Ultrasonic waves (d) Radiowaves

56. Which one of the following have minimum wavelength [Pb PET 2001]

(a) Ultraviolet rays (b) Cosmic rays

(c) *X*-rays (d) $\gamma-$rays

57. Radiations of intensity $0.5\ W/m^2$ are striking a metal plate. The pressure on the plate is [DCE 2004]

(a) $0.166\times10^{-8}\ N/m^2$ (b) $0.332\times10^{-8}\ N/m^2$

(c) $0.111\times10^{-8}\ N/m^2$ (d) $0.083\times10^{-8}\ N/m^2$

58. Electromagnetic waves travel in a medium which has relative permeability 1.3 and relative permittivity 2.14. Then the speed of the electromagnetic wave in the medium will be

(a) $13.6\times10^6\ m/s$ (b) $1.8\times10^2\ m/s$

(c) $3.6\times10^8\ m/s$ (d) $1.8\times10^8\ m/s$

59. The intensity of gamma radiation from a given source is *I*. On passing through 36 *mm* of lead, it is reduced to $\frac{I}{8}$. The thickness of lead which will reduce the intensity to $\frac{I}{2}$ will be [AIEEE 2005]

(a) 18 *mm* (b) 12 *mm*

(c) 6 *mm* (d) 9 *mm*

60. If λ_v, λ_r and λ_m represent the wavelength of visible light *x*-rays and microwaves respectively, then [CBSE PMT 2005]

(a) $\lambda_m > \lambda_x > \lambda_v$ (b) $\lambda_v > \lambda_m > \lambda_y$

(c) $\lambda_m > \lambda_v > \lambda_v$ (d) $\lambda_v > \lambda_x > \lambda_m$

61. For skywave propagation of a 10 *MHz* signal, what should be the minimum electron density in ionosphere [AFMC 2005]

(a) $\sim 1.2\times10^{12}m^{-3}$ (b) $\sim 10^6 m^{-3}$

(c) $\sim 10^{14}m^{-3}$ (d) $\sim 10^{22}m^{-3}$

62. The pressure exerted by an electromagnetic wave of intensity *I* (*watts/m*) on a nonreflecting surface is [*c* is the velocity of light]

(a) Ic (b) Ic^2

(c) I/c (d) I/c^2

63. Infrared radiation was discovered in 1800 by [AIEEE 2004] [KCET 2005]

(a) William Wollaston (b) William Herschel

(c) Wilhelm Roentgen (d) Thomas Young

64. Which of the following is electromagnetic wave [BCECE 2005]

(a) *X*-rays and light waves

(b) Cosmic rays and sound waves

(c) Beta rays and sound waves

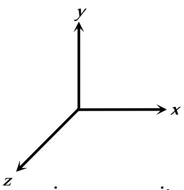
(d) Alpha rays and sound waves

65. Which one of the following is not electromagnetic in nature [Kerala PMT 2005] [BCECE 2004]

(a) *X*-rays (b) Gamma rays

(c) Cathode rays (d) Infrared rays

66. Light wave is travelling along y-direction. If the corresponding $\vec{E}$ vector at any time is along the x-axis, the direction of $\vec{B}$ vector at that time is along [UPSEAT 2005]



(a) *y*-axis

(b) *x*-axis

(c) + *z*-axis

(d) $-$ *z* axis

67. If *c* is the speed of electromagnetic waves in vacuum, its speed in a medium of dielectric constant *K* and relative permeability μ_r is [MH CET 2003]

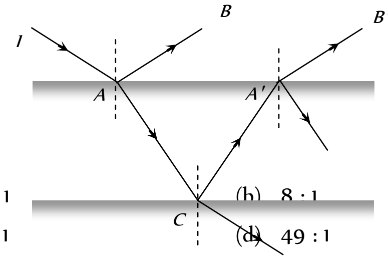
(a) $v = \frac{1}{\sqrt{\mu_r K}}$ (b) $v = c\sqrt{\mu_r K}$

(c) $v = \frac{c}{\sqrt{\mu_r K}}$ (d) $v = \frac{K}{\sqrt{\mu_r C}}$

Critical Thinking

Objective Questions

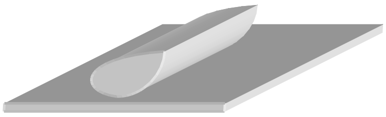
1. A ray of light of intensity I is incident on a parallel glass-slab at a point A as shown in fig. It undergoes partial reflection and refraction. At each reflection 25% of incident energy is reflected. The rays AB and $A'B'$ undergo interference. The ratio I_{max} / I_{min} is



(a) 4 : 1 (b) 8 : 1

(c) 7 : 1 (d) 49 : 1

2. A thin slice is cut out of a glass cylinder along a plane parallel to its axis. The slice is placed on a flat glass plate as shown. The observed interference fringes from this combination shall be



(a) Straight

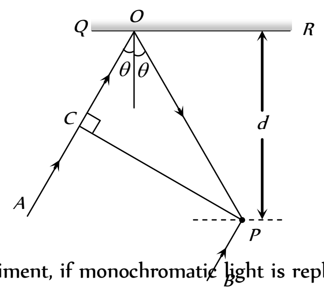
(b) Circular

(c) Equally spaced

(d) Having fringe spacing which increases as we go outwards

3. In the adjacent diagram, CP represents a wavefront and AO & BP, the corresponding two rays. Find the condition on θ for constructive interference at P between the ray BP and reflected ray OP

[IIT-JEE (Screening) 2003]



(a) $\cos\theta = 3\lambda/2d$

(b) $\cos\theta = \lambda/4d$

(c) $\sec\theta - \cos\theta = \lambda/d$

(d) $\sec\theta - \cos\theta = 4\lambda/d$

4. In Young's double slit experiment, if monochromatic light is replaced by white light

[AIIMS 2001; Kerala PET 2000; KCET 2004]

(a) All bright fringes become white

(b) All bright fringes have colours between violet and red

(c) Only the central fringe is white, all other fringes are coloured

(d) No fringes are observed

5. In Young's double slit experiment, if the two slits are illuminated with separate sources, no interference pattern is observed because

(a) There will be no constant phase difference between the two waves

(b) The wavelengths are not equal

(c) The amplitudes are not equal

(d) None of the above

6. In Young's double slit experiment, white light is used. The separation between the slits is b. The screen is at a distance d ($d >> b$) from the slits. Some wavelengths are missing exactly in front of one slit. These wavelengths are

[IIT 1984; AIIMS 1995]

(a) $\lambda = \frac{b^2}{d}$ (b) $\lambda = \frac{2b^2}{d}$

(c) $\lambda = \frac{b^2}{3d}$ (d) $\lambda = \frac{2b^2}{3d}$

7. In a Young's double slit experiment the source S and the two slits A and B are vertical with slit A above slit B. The fringes are observed on a vertical screen K. The optical path length from S to B is increased very slightly (by introducing a transparent material of higher refractive index) and the optical path length from S to A is not changed, as a result the fringe system on K moves

[IIT 1990]

(a) Vertically downwards slightly

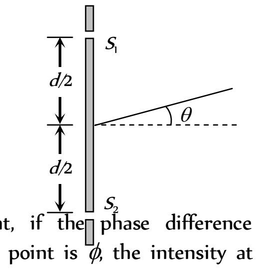
(b) Vertically upwards slightly

(c) Horizontally, slightly to the left

(d) Horizontally, slightly to the right

8. In an interference arrangement similar to Young's double slit experiment, the slits S_1 and S_2 are illuminated with coherent microwave sources each of frequency 10^6 Hz. The sources are synchronized to have zero phase difference. The slits are separated by distance $d = 150$ m. The intensity $I(\theta)$ is measured as a function of θ, where θ is defined as shown. If I_0 is maximum intensity, then $I(\theta)$ for $0 \le \theta \le 90^o$ is given by

[IIT-JEE (Screening) 1999]



(a) $I(\theta) = I_0$ for $\theta = 0^o$

(b) $I(\theta) = I_0/2$ for $\theta = 30^o$

(c) $I(\theta) = I_0/4$ for $\theta = 90^o$

(d) $I(\theta)$ is constant for all values of θ

9. In the Young's double slit experiment, if the phase difference between the two waves interfering at a point is ϕ, the intensity at that point can be expressed by the expression

[MP PET 1998; MP PMT 2003]

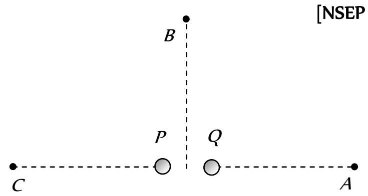
(a) $I = \sqrt{A^2 + B^2\cos^2\phi}$ (b) $I = \frac{A}{B}\cos\phi$

(c) $I = A + B\cos\frac{\phi}{2}$ (d) $I = A + B\cos\phi$

Where A and B depend upon the amplitudes of the two waves.

10. Figure here shows P and Q as two equally intense coherent sources emitting radiations of wavelength 20 m. The separation PQ is 5.0 m and phase of P is ahead of the phase of Q by 90°. A, B and C are three distant points of observation equidistant from the mid-point of PQ. The intensity of radiations at A, B, C will bear the ratio

[NSEP 1994]



(a) 0 : 1 : 4

(b) 4 : 1 : 0

(c) 0 : 1 : 2

(d) 2 : 1 : 0

11. In Young's double slit experiment, the intensity on the screen at a point where path difference is λ is K. What will be the intensity at the point where path difference is $\lambda/4$

[RPET 1996]

(a) $\frac{K}{4}$ (b) $\frac{K}{2}$

(c) K (d) Zero

12. When one of the slits of Young's experiment is covered with a transparent sheet of thickness 4.8 *mm*, the central fringe shifts to a position originally occupied by the 30th bright fringe. What should be the thickness of the sheet if the central fringe has to shift to the position occupied by 20th bright fringe

(a) 3.8 *mm* (b) 1.6 *mm*

(c) 7.6 *mm* (d) 3.2 *mm*

13. In the ideal double-slit experiment, when a glass-plate (refractive index 1.5) of thickness t is introduced in the path of one of the interfering beams (wavelength λ), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass-plate is

(a) 2λ (b) $\frac{2\lambda}{3}$

(c) $\frac{\lambda}{3}$ (d) λ

14. The time period of rotation of the sun is 25 days and its radius is $7\times10^{8}m$. The Doppler shift for the light of wavelength 6000 Å emitted from the surface of the sun will be

(a) 0.04 Å (b) 0.40 Å

(c) 4.00 Å (d) 40.0 Å

15. In hydrogen spectrum the wavelength of H_α line is 656 *nm* whereas in the spectrum of a distant galaxy, H_α line wavelength is 706 *nm*. Estimated speed of the galaxy with respect to earth is [IIT-JEE 1999; UPSEAT 2003]

(a) $2\times10^{8}m/s$ (b) $2\times10^{7}m/s$

(c) $2\times10^{6}m/s$ (d) $2\times10^{5}m/s$

16. A rocket is going towards moon with a speed *v*. The astronaut in the rocket sends signals of frequency ν towards the moon and receives them back on reflection from the moon. What will be the frequency of the signal received by the astronaut (Take *v*<<*c*)

(a) $\frac{c}{c-v}\nu$ (b) $\frac{c}{c-2v}\nu$

(c) $\frac{2v}{c}\nu$ (d) $\frac{2c}{v}\nu$

17. The periodic time of rotation of a certain star is 22 days and its radius is 7 × 10^{8} *metres*. If the wavelength of light emitted by its surface be 4320 Å, the Doppler shift will be (1 day = 86400 *sec*)

(a) 0.033 Å (b) 0.33 Å

(c) 3.3 Å (d) 33 Å

18. In a two slit experiment with monochromatic light fringes are obtained on a screen placed at some distance from the sits. If the screen is moved by $5\times10^{-2}m$ towards the slits, the change in fringe width is $3\times10^{-5}m$. If separation between the slits is $10^{-3}m$, the wavelength of light used is

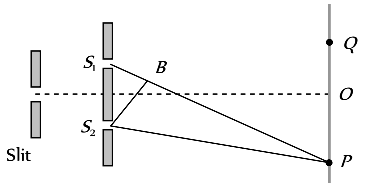
[Roorkee 1992]

(a) 6000 *Å* (b) 5000 *Å*

(c) 3000 *Å* (d) 4500 *Å*

19. In the figure is shown Young's double slit experiment. Q is the position of the first bright fringe on the right side of O. P is the 11th fringe on the other side, as measured from Q. If the wavelength of the light used is $6000\times10^{-10}m$, then S_1B will be equal to

[KCET 2002]



(a) $6\times10^{-6}m$

(b) $6.6\times10^{-6}m$

(c) $3.138\times10^{-7}m$

(d) $3.144\times10^{-7}m$

20. In Young's double slit experiment, the two slits act as coherent sources of equal amplitude A and wavelength λ. In another experiment with the same set up the two slits are of equal amplitude A and wavelength λ but are incoherent. The ratio of the intensity of light at the mid-point of the screen in the first case to that in the second case is [IIT-JEE (Screening) 2002]

[IIT-JEE 1986; RPMT 2002]

(a) 1 : 2 (b) 2 : 1

(c) 4 : 1 (d) 1 : 1

21. Four light waves are represented by [MP PMT 1994]

(i) $y = a_1 \sin\omega t$ (ii) $y = a_2\sin(\omega t+\phi)$

(iii) $y = a_1\sin 2\omega t$ (iv) $y = a_2\sin 2(\omega t+\phi)$

Interference fringes may be observed due to superposition of

(a) (i) and (ii) (b) (i) and (iii)

(c) (ii) and (iv) (d) (iii) and (iv)

22. In Young's double slit experiment the *y*-coordinates of central maxima and 10th maxima are 2 *cm* and 5 *cm* respectively. When the YDSE apparatus is immersed in a liquid of refractive index 1.5 the corresponding *y*-coordinates will be

[RPMT 1996; DPMT 2000]

(a) 2 *cm*, 7.5 *cm*

(b) 3 *cm*, 6 *cm*

(c) 2 *cm*, 4 *cm*

(d) 4/3 *cm*, 10/3 *cm*

23. The maximum intensity in Young's double slit experiment is I_0. Distance between the slits is d = 5 λ, where λ is the wavelength of monochromatic light used in the experiment. What will be the intensity of light in front of one of the slits on a screen at a distance $D = 10\,d$

[MP PET 2001]

(a) $\frac{I_0}{2}$ (b) $\frac{3}{4}I_0$

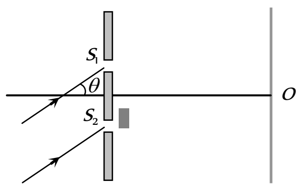
(c) I_0 (d) $\frac{I_0}{4}$

24. A monochromatic beam of light falls on YDSE apparatus at some angle (say θ) as shown in figure. A thin sheet of glass is inserted in front of the lower slit S_2. The central bright fringe (path difference = 0) will be obtained

(a) At O

(b) Above O

(c) Below O



(d) Anywhere depending on angle θ, thickness of plate t and refractive index of glass μ

25. In Young's double slit experiment how many maximas can be obtained on a screen (including the central maximum) on both sides of the central fringe if $\lambda = 2000\,Å$ and $d = 7000\,Å$

(a) 12 (b) 7

(c) 18 (d) 4

26. In a Young's double slit experiment, the slits are 2 *mm* apart and are illuminated with a mixture of two wavelength $\lambda_0 = 750nm$ and $\lambda = 900nm$. The minimum distance from the common central bright fringe on a screen $2m$ from the slits where a bright fringe from one interference pattern coincides with a bright fringe from the other is

(a) 1.5 *mm* (b) 3 *mm*

(c) 4.5 *mm* (d) 6 *mm*

27. A flake of glass (refractive index 1.5) is placed over one of the openings of a double slit apparatus. The interference pattern displaces itself through seven successive maxima towards the side where the flake is placed. if wavelength of the diffracted light is $\lambda = 600nm$, then the thickness of the flake is

(a) 2100 *nm* (b) 4200 *nm*

(c) 8400 *nm* (d) None of these

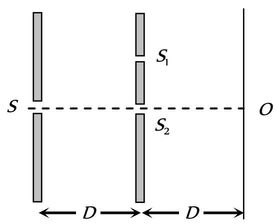
28. Two ideal slits S_1 and S_2 are at a distance d apart, and illuminated by light of wavelength λ passing through an ideal source slit S placed on the line through S_2 as shown. The distance between the planes of slits and the source slit is D. A screen is held at a distance D from the plane of the slits. The minimum value of d for which there is darkness at O is

(a) $\sqrt{\dfrac{3\lambda D}{2}}$

(b) $\sqrt{\lambda D}$

(c) $\sqrt{\dfrac{\lambda D}{2}}$

(d) $\sqrt{3\lambda D}$



29. In a double slit arrangement fringes are produced using light of wavelength 4800 Å. One slit is covered by a thin plate of glass of refractive index 1.4 and the other with another glass plate of same thickness but of refractive index 1.7. By doing so the central bright shifts to original fifth bright fringe from centre. Thickness of glass plate is

(a) 8 μm (b) 6 μm

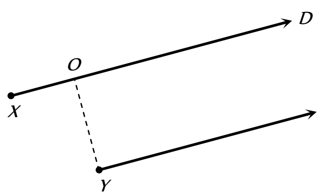
(c) 4 μm (d) 10 μm

30. Two point sources X and Y emit waves of same frequency and speed but Y lags in phase behind X by $2\pi l$ radian. If there is a maximum in direction D the distance XO using n as an integer is given by

(a) $\dfrac{\lambda}{2}(n - l)$

(b) $\lambda(n + l)$

(c) $\dfrac{\lambda}{2}(n + l)$



(d) $\lambda(n - l)$

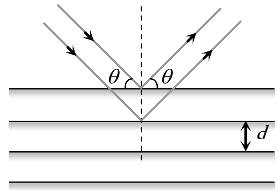
31. A beam with wavelength λ falls on a stack of partially reflecting planes with separation d. The angle θ that the beam should make with the planes so that the beams reflected from successive planes may interfere constructively is (where n =1, 2, ...)

(a) $\sin^{-1}\left(\dfrac{n\lambda}{d}\right)$

(b) $\tan^{-1}\left(\dfrac{n\lambda}{d}\right)$

(c) $\sin^{-1}\left(\dfrac{n\lambda}{2d}\right)$

(d) $\cos^{-1}\left(\dfrac{n\lambda}{2d}\right)$



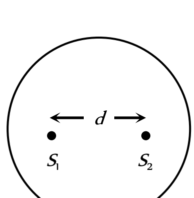
32. Two coherent sources separated by distance d are radiating in phase having wavelength λ. A detector moves in a big circle around the two sources in the plane of the two sources. The angular position of n = 4 interference maxima is given as

(a) $\sin^{-1}\dfrac{n\lambda}{d}$

(b) $\cos^{-1}\dfrac{4\lambda}{d}$

(c) $\tan^{-1}\dfrac{d}{4\lambda}$

(d) $\cos^{-1}\dfrac{\lambda}{4d}$



33. Two coherent sources S_1 and S_2 are separated by a distance four times the wavelength λ of the source. The sources lie along y axis whereas a detector moves along + x axis. Leaving the origin and far off points the number of points where maxima are observed is

(a) 2 (b) 3

(c) 4 (d) 5

34. A circular disc is placed in front of a narrow source. When the point of observation is at a distance of 1 *meter* from the disc, then the disc covers first HPZ. The intensity at this point is I_0. The intensity at a point distance 25 *cm* from the disc will be

(a) $I_1 = 0.531 I_0$ (b) $I_1 = 0.053 I_0$

(c) $I_1 = 53 I_0$ (d) $I_1 = 5.03 I_0$

35. A wavefront presents one, two and three HPZ at points A, B and C respectively. If the ratio of consecutive amplitudes of HPZ is 4 : 3, then the ratio of resultant intensities at these point will be

(a) 169 : 16 : 256 (b) 256 : 16 : 169

(c) 256 : 16 : 196 (d) 256 : 196 : 16

36. A circular disc is placed in front of a narrow source. When the point of observation is 2 *m* from the disc, then it covers first HPZ. The intensity at this point is I. When the point of observation is 25 *cm* from the disc then intensity will be

(a) $\left(\dfrac{R_6}{R_2}\right)^2 I$ (b) $\left(\dfrac{R_7}{R_2}\right)^2 I$

(c) $\left(\dfrac{R_8}{R_2}\right)^2 I$ (d) $\left(\dfrac{R_9}{R_2}\right)^2 I$

37. In a single slit diffraction of light of wavelength λ by a slit of width *e*, the size of the central maximum on a screen at a distance *b* is

(a) $2b\lambda + e$ (b) $\frac{2b\lambda}{e}$

(c) $\frac{2b\lambda}{e} + e$ (d) $\frac{2b\lambda}{e} - e$

38. Angular width of central maxima in the Fraunhoffer diffraction pattern of a slit is measured. The slit is illuminated by light of wavelength 6000 Å. When the slit is illuminated by light of another wavelength, the angular width decreases by 30%. The wavelength of this light will be

(a) 6000 Å (b) 4200 Å

(c) 3000 Å (d) 1800 Å

39. In a single slit diffraction experiment first minimum for red light (660 *nm*) coincides with first maximum of some other wavelength λ'. The value of λ' is

(a) 4400 Å (b) 6600 Å

(c) 2000 Å (d) 3500 Å

40. The ratio of intensities of consecutive maxima in the diffraction pattern due to a single slit is

(a) 1 : 4 : 9 (b) 1 : 2 : 3

(c) $1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2}$ (d) $1 : \frac{1}{\pi^2} : \frac{9}{\pi^2}$

41. Light is incident normally on a diffraction grating through which the first order diffraction is seen at 32°. The second order diffraction will be seen at

(a) 48°

(b) 64°

(c) 80°

(d) There is no second order diffraction in this case

42. White light may be considered to be a mixture of waves with λ ranging between 3900 Å and 7800 Å. An oil film of thickness 10,000 Å is examined normally by reflected light. If μ = 1.4, then the film appears bright for

(a) 4308 Å, 5091 Å, 6222 Å

(b) 4000 Å, 5091 Å, 5600 Å

(c) 4667 Å, 6222 Å, 7000 Å

(d) 4000 Å, 4667 Å, 5600 Å, 7000Å

43. Among the two interfering monochromatic sources *A* and *B*; *A* is ahead of *B* in phase by 66°. If the observation be taken from point *P*, such that *PB* – *PA* = λ/4. Then the phase difference between the waves from *A* and *B* reaching *P* is

(a) 156° (b) 140°

(c) 136° (d) 126°

44. The ratio of the intensity at the centre of a bright fringe to the intensity at a point one-quarter of the distance between two fringe from the centre is

(a) 2 (b) 1/2

(c) 4 (d) 16

45. A parallel plate capacitor of plate separation 2 *mm* is connected in an electric circuit having source voltage 400 *V*. if the plate area is 60 *cm*, then the value of displacement current for 10^{-6} sec will be

(a) 1.062 *amp* (b) 1.062×10^{-2} *amp*

(c) 1.062×10^{-3} *amp* (d) 1.062×10^{-4} *amp*

46. A long straight wire of resistance *R*, radius *a* and length *l* carries a constant current *I*. The Poynting vector for the wire will be

(a) $\frac{IR}{2\pi al}$ (b) $\frac{IR^2}{al}$

(c) $\frac{I^2R}{al}$ (d) $\frac{I^2R}{2\pi al}$

47. In an electromagnetic wave, the amplitude of electric field is 1 *V/m*. the frequency of wave is $5 \times 10^{14}\ Hz$. The wave is propagating along *z*-axis. The average energy density of electric field, in *Joule/m*, will be

(a) 1.1×10^{-11} (b) 2.2×10^{-12}

(c) 3.3×10^{-13} (d) 4.4×10^{-14}

48. A laser beam can be focussed on an area equal to the square of its wavelength *A He-Ne* laser radiates energy at the rate of 1*mW* and its wavelength is 632.8 *nm*. The intensity of focussed beam will be

(a) $1.5 \times 10^{13}\ W/m^2$ (b) $2.5 \times 10^{9}\ W/m^2$

(c) $3.5 \times 10^{17}\ W/m^2$ (d) None of these

49. A lamp emits monochromatic green light uniformly in all directions. The lamp is 3% efficient in converting electrical power to electromagnetic waves and consumes 100 *W* of power. The amplitude of the electric field associated with the electromagnetic radiation at a distance of 10*m* from the lamp will be

(a) 1.34 *V/m* (b) 2.68 *V/m*

(c) 5.36 *V/m* (d) 9.37 *V/m*

50. A point source of electromagnetic radiation has an average power output of 800 *W*. The maximum value of electric field at a distance 4.0 *m* from the source is

(a) 64.7 *V/m* (b) 57.8 *V/m*

(c) 56.72 *V/m* (d) 54.77 *V/m*

51. A wave is propagating in a medium of electric dielectric constant 2 and relative magnetic permeability 50. The wave impedance of such a medium is

(a) 5 Ω (b) 376.6 Ω

(c) 1883 Ω (d) 3776 Ω

52. A plane electromagnetic wave of wave intensity 6 *W/m* strikes a small mirror area 40 *cm*, held perpendicular to the approaching wave. The momentum transferred by the wave to the mirror each second will be

(a) $6.4 \times 10^{-7}\ kg - m/s^2$ (b) $4.8 \times 10^{-8}\ kg - m/s^2$

(c) $3.2 \times 10^{-9}\ kg - m/s^2$ (d) $1.6 \times 10^{-10}\ kg - m/s^2$

53. Specific rotation of sugar solution is 0.01 SI units. 200 kgm^{-3} of impure sugar solution is taken in a polarimeter tube of length 0.25*m* and an optical rotation of 0.4*rad* is observed. The percentage of purity of sugar is the sample is

[KCET 2004]

(a) 80% (b) 89%

(c) 11% (d) 20%

54. A 20 *cm* length of a certain solution causes right-handed rotation of 38°. A 30*cm* length of another solution causes left-handed rotation of 24°. The optical rotation caused by 30*cm* length of a mixture of the above solutions in the volume ratio 1 : 2 is

(a) Left handed rotation of 14°

(b) Right handed rotation of 14°

(c) Left handed rotation of 3°

(d) Right handed rotation of 3°

55. A beam of natural light falls on a system of 6 polaroids, which are arranged in succession such that each polaroid is turned through 30° with respect to the preceding one. The percentage of incident intensity that passes through the system will be

(a) 100% (b) 50%

(c) 30% (d) 12%

56. A beam of plane polarized light falls normally on a polarizer of cross sectional area $3\times10^{-4}m^2$. Flux of energy of incident ray in 10^{-3} *W*. The polarizer rotates with an angular frequency of 31.4 *rad/sec*. The energy of light passing through the polarizer per revolution will be

(a) 10^{-4} *Joule* (b) 10^{-3} *Joule*

(c) 10^{-2} *Joule* (d) 10^{-1} *Joule*

57. In a YDSE bi-chromatic light of wavelengths 400 *nm* and 560 *nm* are used. The distance between the slits is 0.1 *mm* and the distance between the plane of the slits and the screen is 1*m*. The minimum distance between two successive regions of complete darkness is

[IIT JEE (Screening) 2004]

(a) 4 *mm* (b) 5.6 *mm*

(c) 14 *mm* (d) 28 *mm*

58. The maximum number of possible interference maxima for slit-separation equal to twice the wavelength in Young's double-slit experiment is [AIEEE 2004]

(a) Infinite (b) Five

(c) Three (d) Zero

59. The *k* line of singly ionised calcium has a wavelength of 393.3 *nm* as measured on earth. In the spectrum of one of the observed galaxies, this spectral line is located at 401.8 *nm*. The speed with which the galaxy is moving away from us, will be

(a) 6480 *km/s* (b) 3240 *km/s*

(c) 4240 *km/sec* (d) None of these

60. A Young's double slit experiment uses a monochromatic source. The shape of the interference fringes formed on a screen is

(a) Straight line (b) Parabola

(c) Hyperbola (d) Circle

61. If I_0 is the intensity of the principal maximum in the single slit diffraction pattern, then what will be its intensity when the slit width is doubled [AIEEE 2005]

(a) I_0 (b) $\frac{I_0}{2}$

(c) $2I_0$ (d) $4\,I_0$

62. In Young's double slit experiment intensity at a point is (1/4) of the maximum intensity. Angular position of this point is

[IIT-JEE (Screening) 2005]

(a) $\sin^{-1}(\lambda/d)$ (b) $\sin^{-1}(\lambda/2d)$

(c) $\sin^{-1}(\lambda/3d)$ (d) $\sin^{-1}(\lambda/4d)$

63. A beam of electron is used in an *YDSE* experiment. The slit width is d. When the velocity of electron is increased, then

[IIT-JEE (Screening) 2005]

(a) No interference is observed

(b) Fringe width increases [KCET 2001]

(c) Fringe width decreases

(d) Fringe width remains same

[Pb. PET 2003]

[AIEEE 2005]

Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) *If both assertion and reason are true and the reason is the correct explanation of the assertion.*
(b) *If both assertion and reason are true but reason is not the correct explanation of the assertion.*
(c) *If assertion is true but reason is false.*
(d) *If the assertion and reason both are false.*
(e) *If assertion is false but reason is true.*

1. Assertion : When a light wave travels from a rarer to a denser medium, it loses speed. The reduction in speed imply a reduction in energy carried by the light wave.

Reason : The energy of a wave is proportional to velocity of wave.

2. Assertion : A narrow pulse of light is sent through a medium. The pulse will retain its shape as it travels through the medium.

Reason : A narrow pulse is made of harmonic waves with a large range of wavelengths.

3. Assertion : No interference pattern is detected when two coherent sources are infinitely close to each other.

Reason : The fringe width is inversely proportional to the distance between the two slits.

4. Assertion : Newton's rings are formed in the reflected system. When the space between the lens and the glass plate is filled with a liquid of refractive index greater than that of glass, the central spot of the pattern is dark.

Reason : The reflection is Newton's ring cases will be from a denser to a rarer medium and the two interfering rays are reflected under similar conditions. **[AIIMS 1998]**

5. Assertion : The film which appears bright in reflected system will appear dark in the transmitted light and vice-versa.

Reason : The conditions for film to appear bright or dark in reflected light are just reverse to those in the transmitted light.

6. Assertion : For best contrast between maxima and minima in the interference pattern of Young's double slit experiment, the intensity of light emerging out of the two slits should be equal.

Reason : The intensity of interference pattern is proportional to square of amplitude.

7. Assertion : In Young's double slit experiment, the fringes become indistinct if one of the slits is covered with cellophane paper.

Reason : The cellophane paper decrease the wavelength of light.

8. Assertion : The unpolarised light and polarised light can be distinguished from each other by using polaroid.

Reason : A polaroid is capable of producing plane polarised beams of light.

9. Assertion : Nicol prism is used to produce and analyse plane polarised light.

Reason : Nicol prism reduces the intensity of light to zero.

10. Assertion : In everyday life the Doppler's effect is observed readily for sound waves than light waves.

Reason : Velocity of light is greater than that of sound.

[AIIMS 1995]

11. Assertion : In Young's experiment, the fringe width for dark fringes is different from that for white fringes.

Reason : In Young's double slit experiment the fringes are performed with a source of white light, then only black and bright fringes are observed. **[AIIMS 2001]**

12. Assertion : Coloured spectrum is seen when we look through a muslin cloth.

Reason : It is due to the diffraction of white light on passing through fine slits. **[AIIMS 2002]**

13. Assertion : When a tiny circular obstacle is placed in the path of light from some distance, a bright spot is seen at the centre of shadow of the obstacle.

Reason : Destructive interference occurs at the centre of the shadow. **[AIIMS 2002]**

14. Assertion : Thin films such as soap bubble or a thin layer of oil on water show beautiful colours when illuminated by white light.

Reason : It happens due to the interference of light reflected from the upper surface of the thin film. **[AIIMS 2002]**

15. Assertion : Microwave communication is preferred over optical communication.

Reason : Microwaves provide large number of channels and band width compared to optical signals. **[AIIMS 2003]**

16. Assertion : Corpuscular theory fails in explaining the velocities of light in air and water.

Reason : According to corpuscular theory, light should travel faster in denser medium than, in rarer medium. **[AIIMS 1998]**

17. Assertion : Interference pattern is made by using blue light instead of red light, the fringes becomes narrower.

Reason : In Young's double slit experiment, fringe width is given by relation $B = \frac{\lambda D}{d}$.

[AIIMS 1999]

18. Assertion : The cloud in sky generally appear to be whitish.

Reason : Diffraction due to clouds is efficient in equal measure at all wavelengths. **[AIIMS 2005]**

19. Assertion : Television signals are received through sky-wave propagation.

Reason : The ionosphere reflects electromagnetic waves of frequencies greater than a certain critical frequency. **[AIIMS 2005]**

20. Assertion : It is necessary to use satellites for long distance T.V. transmission.

Reason : The television signals are low frequency signals.

21. Assertion : The electrical conductivity of earth's atmosphere decrease with altitude.

Reason : The high energy particles (*i.e.* γ-rays and cosmic rays) coming from outer space and entering our earth's atmosphere causes ionisation of the atoms of the gases present there and the pressure of gases decreases with increase in altitude.

22. Assertion : Only microwaves are used in radar.

Reason : Because microwaves have very small wavelength. [AIIMS 1998]

23. Assertion : In Hertz experiment, the electric vector of radiation produced by the source gap is parallel to the gap.

Reason : Production of sparks between the detector gap is maximum when it is placed perpendicular to the source gap.

24. Assertion : For cooking in a microwave oven, food is always kept in metal containers.

Reason : The energy of microwave is easily transferred to the food in metal container.

25. Assertion : X-ray astronomy is possible only from satellites orbiting the earth.

Reason : Efficiency of X-rays telescope is large as compared to any other telescope.

26. Assertion : Short wave bands are used for transmission of ratio waves to a large distance

Reason : Short waves are reflected by ionosphere

[AIIMS 1994]

27. Assertion : Ultraviolet radiation are of higher frequency waves are dangerous to human being.

Reason : Ultraviolet radiation are absorbed by the atmosphere [AIIMS 1995]

28. Assertion : Environmental damage has increased the amount of ozone in the atmosphere.

Reason : Increase of ozone increases the amount of ultraviolet radiation on earth. [AIIMS 1996]

29. Assertion : Radio waves can be polarised.

Reason : Sound waves in air are longitudinal in nature.

30. Assertion : The earth without atmosphere would be inhospitably cold.

Reason : All heat would escape in the absence of atmosphere. [AIIMS 2002]

Answers

Wave Nature and Interference of Light

1	a	2	c	3	b	4	c	5	d
6	c	7	d	8	c	9	c	10	b
11	a	12	d	13	c	14	a	15	d
16	a	17	c	18	b	19	c	20	b
21	c	22	c	23	a	24	c	25	a
26	a	27	b	28	b	29	c	30	a
31	b	32	d	33	d	34	a	35	a
36	c	37	b	38	c	39	b	40	c
41	d	42	b	43	b	44	c	45	d
46	c	47	d	48	b	49	c	50	a
51	d	52	c	53	d	54	c	55	b
56	a	57	c	58	d	59	d	60	b
61	b	62	a	63	c	64	d	65	d
66	c	67	a	68	a	69	a	70	c
71	b	72	d						

Young's Double Slit Experiment

1	a	2	c	3	c	4	c	5	a
6	c	7	a	8	c	9	a	10	d
11	d	12	c	13	bd	14	b	15	c
16	c	17	a	18	a	19	a	20	b
21	a	22	c	23	d	24	c	25	d
26	a	27	b	28	c	29	d	30	d
31	d	32	a	33	b	34	b	35	a
36	b	37	b	38	d	39	b	40	a
41	b	42	d	43	d	44	a	45	b
46	d	47	d	48	b	49	b	50	a
51	bc	52	b	53	d	54	a	55	a

56	b	57	c	58	c	59	c	60	b
61	b	62	c	63	b	64	d	65	b
66	a	67	c	68	b	69	a	70	b
71	d	72	b	73	b	74	b	75	c
76	d	77	a	78	c	79	b	80	a
81	b	82	b	83	d	84	a	85	a
86	b	87	b	88	d	89	b	90	a
91	a	92	c	93	a	94	a	95	b
96	b	97	d						

Doppler's Effect of Light

1	d	2	a	3	b	4	b	5	b
6	b	7	b	8	d	9	c	10	c
11	a	12	c	13	b	14	a	15	a
16	d	17	b	18	c	19	b	20	d
21	b	22	b	23	c	24	c	25	b
26	b	27	c	28	b	29	a	30	d
31	c	32	a	33	a				

Diffraction of Light

1	c	2	a	3	b	4	c	5	c
6	a	7	a	8	a	9	c	10	b
11	a	12	b	13	a	14	d	15	a
16	b	17	a	18	a	19	a	20	d
21	a	22	a	23	c	24	b	25	d
26	a	27	d	28	d	29	b	30	d
31	d	32	d	33	b	34	a	35	c
36	c	37	a	38	c	39	a	40	b
41	c								

Polarisation of Light

1	b	2	a	3	c	4	d	5	d
6	d	7	b	8	d	9	a	10	c
11	a	12	d	13	a	14	a	15	c
16	d	17	d	18	d	19	b	20	c
21	c	22	d	23	a	24	c	25	b
26	b	27	b	28	c	29	a	30	b
31	a	32	a	33	c	34	a	35	d

EM Waves

1	a	2	d	3	b	4	d	5	b
6	b	7	a	8	d	9	c	10	a
11	a	12	c	13	a	14	b	15	b
16	d	17	b	18	d	19	a	20	c
21	a	22	d	23	c	24	b	25	a
26	a	27	c	28	c	29	a	30	b
31	c	32	a	33	d	34	a	35	d
36	c	37	b	38	a	39	a	40	d
41	a	42	b	43	b	44	c	45	a
46	b	47	a	48	c	49	c	50	a
51	c	52	c	53	d	54	d	55	b
56	b	57	a	58	d	59	b	60	c
61	a	62	c	63	b	64	a	65	c
66	d	67	c						

Critical Thinking Questions

1	d	2	a	3	b	4	c	5	a
6	ac	7	a	8	ab	9	d	10	d
11	b	12	d	13	a	14	a	15	b
16	b	17	a	18	a	19	a	20	b
21	ad	22	c	23	a	24	d	25	b
26	c	27	c	28	c	29	a	30	b
31	c	32	b	33	b	34	a	35	b
36	d	37	c	38	b	39	a	40	c
41	d	42	a	43	a	44	a	45	b
46	d	47	b	48	b	49	a	50	d
51	c	52	d	53	a	54	d	55	d
56	a	57	d	58	b	59	a	60	c
61	d	62	c	63	b				

Assertion and Reason

1	d	2	e	3	b	4	a	5	a
6	b	7	c	8	a	9	c	10	b
11	d	12	a	13	c	14	c	15	a
16	a	17	a	18	c	19	d	20	c
21	e	22	a	23	c	24	d	25	c
26	b	27	b	28	d	29	b	30	a

Answers and Solutions

Wave Nature and Interference of Light

1. (a) Corpuscular theory explains refraction of light.

2. (c) According to Corpuscular theory different colour of light are due to different size of Corpuscules.

3. (b)

4. (c) According to Plank's hypothesis, black bodies emits radiations in the form of photons.

5. (d) The coherent source cannot be obtained from two different light sources.

6. (c) Huygen's wave theory fails to explain the particle nature of light (*i.e.* photoelectric effect)

7. (d) Interference is shown by transverse as well as mechanical waves.

8. (c) $I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2 = (\sqrt{I} + \sqrt{4I})^2 = 9I$

$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2 = (\sqrt{I} - \sqrt{4I})^2 = I$

9. (c)

10. (b) The idea of secondary wavelets is given by Huygen.

11. (c) Monochromatic wave means of single wavelength not the single colour.

12. (d) Sound wave and light waves both shows interference.

13. (c) $\frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{\frac{I_1}{I_2}}+1}{\sqrt{\frac{I_1}{I_2}}-1}\right)^2 = \left(\frac{\sqrt{\frac{9}{1}}+1}{\sqrt{\frac{9}{1}}-1}\right) = \frac{4}{1}$

14. (a) A wave can transmit energy from one place to another.

15. (d) $\frac{I_1}{I_2} = \frac{1}{25}$; $\therefore \frac{a_1^2}{a_2^2} = \frac{1}{25} \Rightarrow \frac{a_1}{a_2} = \frac{1}{5}$

16. (a) For interference phase difference must be constant.

17. (c) Interference is explained by wave nature of light.

18. (b) Coherent time $= \frac{\text{Coherence length}}{\text{Velocity of light}} = \frac{L}{c}$

19. (c) $\frac{a_1}{a_2} = \frac{3}{5}$

$\therefore \frac{I_{max}}{I_{min}} = \frac{(a_1+a_2)^2}{(a_1-a_2)^2} = \frac{(3+5)^2}{(3-5)^2} = \frac{16}{1}$

20. (b) Colour's of thin film are due to interference of light.

21. (c) For constructive interference path difference is even multiple of $\frac{\lambda}{2}$.

22. (c) Two coherent source must have a constant phase difference otherwise they can not produce interference.

23. (a) Phenomenon of interference of light takes place.

24. (c) Transverse waves can be polarised.

25. (a) $\frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{\frac{I_1}{I_2}}+1}{\sqrt{\frac{I_1}{I_2}}-1}\right)^2 = \left(\frac{\sqrt{\frac{4}{1}}+1}{\sqrt{\frac{4}{1}}-1}\right)^2 = \frac{9}{1}$

26. (a) Reflection phenomenon is shown by both particle and wave nature of light.

27. (b) When two sources are obtained from a single source, the wavefront is divided into two parts. These two wavefronts acts as if they emanated from two sources having a fixed phase relationship.

28. (b) $\lambda = \frac{c}{\nu} = \frac{3\times10^8}{100} = 3\times10^6 m$

29. (c) $\frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{\frac{I_1}{I_2}}+1}{\sqrt{\frac{I_1}{I_2}}-1}\right)^2 = \left(\frac{\sqrt{\frac{25}{4}}+1}{\sqrt{\frac{25}{4}}-1}\right)^2 = \frac{49}{9}$

30. (a) Wavefront is the locus of all the particles which vibrates in the same phase.

31. (b) Direction of wave is perpendicular to the wavefront.

32. (d) Origin of spectra is not explained by Huygen's theory.

33. (d) The refractive index of air is slightly more than 1. When chamber is evacuated, refractive index decreases and hence the wavelength increases and fringe width also increases.

34. (a) $I \propto a^2 \Rightarrow \frac{a_1}{a_2} = \left(\frac{4}{1}\right)^{1/2} = \frac{2}{1}$

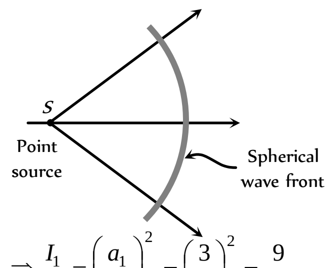
35. (a) The essential condition for sustained interference is constancy of phase difference.

36. (c) $\frac{I_{max}}{I_{min}} = \left(\frac{\frac{a_1}{a_2}+1}{\frac{a_1}{a_2}-1}\right)^2 = \left(\frac{\frac{4}{3}+1}{\frac{4}{3}-1}\right)^2 = \frac{49}{1}$

37. (b) Energy is conserved in the interference of light.

38. (c) $I \propto a^2$

39. (b)



40. (c) $I \propto a^2 \Rightarrow \frac{I_1}{I_2} = \left(\frac{a_1}{a_2}\right)^2 = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$

41. (d) $\frac{I_1}{I_2} = \frac{100}{1}$

Now $\dfrac{I_{max}}{I_{min}} = \left(\dfrac{\sqrt{\dfrac{I_1}{I_2}}+1}{\sqrt{\dfrac{I_1}{I_2}}-1}\right)^2 = \left(\dfrac{\sqrt{100}+1}{\sqrt{100}-1}\right)^2 = \dfrac{121}{81} \approx \dfrac{3}{2}$

42. (b) $\phi = \pi/3, a_1 = 4, a_2 = 3$

So, $A = \sqrt{a_1^2 + a_2^2 + 2a_1.a_2\cos\phi} \Rightarrow A \approx 6$

43. (b) $y_1 = a\sin\omega t$, and $y_2 = b\cos\omega t = b\sin\left(\omega t + \dfrac{\pi}{2}\right)$

So phase difference $\phi = \pi/2$

44. (c) For 2π phase difference $\rightarrow$ Path difference is λ

$\therefore$ For ϕ phase difference $\rightarrow$ Path difference is $\dfrac{\lambda}{2\pi}\times\phi$

45. (d) Resultant intensity $I_R = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\phi$

For maximum I_R, $\phi = 0^o$

$\Rightarrow I_R = I_1 + I_2 + 2\sqrt{I_1I_2} = \left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

46. (c) Newton first law of motion states that every particle travels in a straight line with a constant velocity unless disturbed by an external force. So the corpuscles travels in straight lines.

47. (d) Diffraction shows the wave nature of light and photoelectric effect shows particle nature of light.

48. (b) At point *A*, resultant intensity

$I_A = I_1 + I_2 = 5I$; and at point *B*

$I_B = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\pi = 5I + 4I$

$I_B = 9I$ so $I_B - I_A = 4I.$

49. (c)

50. (a) Photoelectric effect varifies particle nature of light. Reflection and refraction varifies both particle nature and wave nature of light.

51. (d) $y_1 = a\sin\omega t, y_2 = a\cos\omega t = a\sin\left(\omega t + \dfrac{\pi}{2}\right)$

52. (c) $\dfrac{I_{max}}{I_{min}} = \left(\dfrac{\dfrac{a_1}{a_2}+1}{\dfrac{a_1}{a_2}-1}\right)^2 = \dfrac{25}{1}$

53. (d) Laser beams are perfectly parallel. So that they are very narrow and can travel a long distance without spreading. This is the feature of laser while they are monochromatic and coherent these are characteristics only.

54. (c) $\dfrac{I_{max}}{I_{min}} = \left(\dfrac{\sqrt{\dfrac{I_1}{I_2}}+1}{\sqrt{\dfrac{I_1}{I_2}}-1}\right)^2 \Rightarrow \dfrac{I_1}{I_2} = \dfrac{9}{4}$

55. (b) $\nu = \dfrac{c}{\lambda} = \dfrac{3\times10^8}{3000\times10^{-10}} = 10^{15}$ *cycles/sec*

56. (a) $\dfrac{I_{max}}{I_{min}} = \left(\dfrac{\dfrac{a_1}{a_2}+1}{\dfrac{a_1}{a_2}-1}\right)^2 = \left(\dfrac{\dfrac{1}{9}+1}{\dfrac{1}{9}-1}\right)^2 = \left(\dfrac{5}{4}\right)^2 = \dfrac{25}{16}.$

57. (c)

58. (d) For destructive interference path difference is odd multiple of $\dfrac{\lambda}{2}$.

59. (d) $\dfrac{I_{max}}{I_{min}} = \left(\dfrac{\dfrac{a_1}{a_2}+1}{\dfrac{a_1}{a_2}-1}\right)^2 \Rightarrow \dfrac{a_1+a_2}{a_1-a_2} = 6$

$\dfrac{a_2}{a_1} = 7:5$

60. (b) $I_{max} = I_1 + I_2 + 2\sqrt{I_1I_2}$

So, $I_{max} = I + 4I + 2\sqrt{I.4I} = 9I$

61. (b) In interference energy is redistribution.

62. (a) For interference frequency must be same and phase difference must be constant.

63. (c) When a beam of light is used to determine the position of an object, the maximum accuracy is achieved if the light is of shorter wavelength, because

$$\text{Accuracy} \propto \frac{1}{\text{Wavelength}}$$

64. (d) Intensity $\propto \dfrac{1}{(\text{Distance})^2}$

65. (d) Huygen's theory explains propagation of wavefront.

66. (c) Wave theory of light is given by Huygen.

67. (a) When light reflect from denser surface phase change of π occurs.

68. (a) Photoelectric effect explain the quantum nature of light while interference, diffraction and polarization explain the wave nature of light.

69. (a) Light is electromagnetic in nature it does not require any material medium for its propagation.

70. (c) For viewing interference in oil films or soap bubble, thickness of film is of the order of wavelength of light.

71. (b)

72. (d) For maximum intensity $\phi = 0°$

$\therefore\ I = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\phi = I + I + 2\sqrt{II}\cos 0° = 4I$

Young's Double Slit Experiment

1. (a)

2. (c) In interference of light the energy is transferred from the region of destructive interference to the region of constructive interference. The average energy being always equal to the sum of the energies of the interfering waves. Thus the phenomenon of interference is in complete agreement with the law of conservation of energy.

3. (c) $\beta = \frac{\lambda D}{d} = \frac{5\times10^{-7}\times 2}{10^{-3}} m = 10^{-3} m = 1.0\ mm$.

4. (c) Slit width ratio = 1 : 9
Since slit width ratio is the ratio of intensity and intensity $\propto$ (amplitude)
$\therefore I_1 : I_2 = 1 : 9$
$\Rightarrow a_1^2 : a_2^2 = 1:9 \Rightarrow a_1 : a_2 = 1:3$
$I_{\max} = (a_1 + a_2)^2,\ I_{\min} = (a_1 - a_2)^2 \Rightarrow \frac{I_{\min}}{I_{\max}} = \frac{1}{4}$

5. (a) $\beta \propto \frac{1}{d} \Rightarrow$ If d becomes thrice, then β become becomes $\frac{1}{3}$ times.

6. (c) $\frac{\beta_1}{\beta_2} = \frac{\lambda_1}{\lambda_2}$ or $\frac{1.0}{\beta_2} = \frac{5000}{6000}$ or $\beta_2 = \frac{6000}{5000} = 1.2\ mm$.

7. (a) $\beta = \frac{6000\times10^{-10}\times 25\times10^{-2}}{10^{-3}}$
$= 150000\times10^{-9} = 0.15\times10^{-3} m = 0.015\ cm$.

8. (c) For brightness, path difference $= n\lambda = 2\lambda$
So second is bright.

9. (a) If one of slit is closed then interference fringes are not formed on the screen but a fringe pattern is observed due to diffraction from slit.

10. (d)

11. (d) $\beta = \frac{\lambda D}{d} \Rightarrow$ If D becomes twice and d becomes half so β becomes four times.

12. (c) Suppose slit width's are equal, so they produces waves of equal intensity say I'. Resultant intensity at any point $I_R = 4\,I'\cos^2\phi$ where ϕ is the phase difference between the waves at the point of observation.
For maximum intensity $\phi = 0^o \Rightarrow I_{\max} = 4I' = I$ ...(i)
If one of slit is closed, Resultant intensity at the same point will be I' only *i.e.* $I' = I_O$ ...(ii)
Comparing equation (i) and (ii) we get
$I = 4I_O$

13. (b, d) $\frac{I_{\max}}{I_{\min}} = 9 \Rightarrow \left(\frac{a_1+a_2}{a_1-a_2}\right)^2 = 9 \Rightarrow \frac{a_1+a_2}{a_1-a_2} = 3$
$\Rightarrow \frac{a_1}{a_2} = \frac{3+1}{3-1} \Rightarrow \frac{a_1}{a_2} = 2$. Therefore $I_1 : I_2 = 4:1$

14. (b)

15. (c) Distance of n bright fringe $y_n = \frac{n\lambda D}{d}$ *i.e.* $y_n \propto \lambda$
$\therefore \frac{x_{n_1}}{x_{n_2}} = \frac{\lambda_1}{\lambda_2} \Rightarrow \frac{x(\text{Blue})}{x(\text{Green})} = \frac{4360}{5460}$
$\therefore x$ (Green) > x (Blue).

16. (c) $\beta = \frac{\lambda D}{d} = \frac{5000\times10^{-10}\times 1}{0.1\times10^{-3}} m = 5\times10^{-3} m = 0.5\ cm$.

17. (a) We know that fringe width $\beta = \frac{D\lambda}{d}$
$\therefore x = \frac{L\lambda}{d} \Rightarrow \lambda = \frac{xd}{L}$

18. (a) In the normal adjustment of young's, double slit experiment, path difference between the waves at central location is always zero, so maxima is obtained at central position.

19. (a) $\beta = \frac{\lambda D}{d}$; $\therefore B \propto \lambda$
$\frac{\lambda'}{\lambda} = \frac{0.4}{4/3} \Rightarrow \lambda' = 0.3\ mm$.

20. (b) $\beta \propto \lambda,\ \therefore \lambda \propto \frac{1}{\mu}$

21. (a) $\beta \propto \lambda,\ \therefore \lambda_v =$ minimum.

22. (c) $\beta_{\text{medium}} = \frac{\beta_{\text{air}}}{\mu} = \frac{0.6}{1.5} = 0.4\ mm$.

23. (d) $\because n = 3,\ \therefore 2n\pi = 2\times3\pi = 6\pi$

24. (c) Slit width ratio $= 4:9$; hence $I_1 : I_2 = 4:9$
$\therefore \frac{a_1^2}{a_2^2} = \frac{4}{9} \Rightarrow \frac{a_1}{a_2} = \frac{2}{3}$
$\therefore \frac{I_{\max}}{I_{\min}} = \frac{(a_1+a_2)^2}{(a_1-a_2)^2} = \frac{25}{1}$

25. (d) $\beta = \frac{\lambda D}{d} \Rightarrow d = \frac{\lambda D}{\beta} = \frac{6000\times10^{-10}\times(40\times10^{-2})}{0.012\times10^{-2}} = 0.2\,cm$.

26. (a) As $\beta = \frac{D\lambda}{d} \Rightarrow \frac{\beta_1}{\beta_2} = \left(\frac{D_1}{D_2}\right)\left(\frac{\lambda_1}{\lambda_2}\right)\left(\frac{d_2}{d_1}\right)$
$\Rightarrow 1 = \left(\frac{D_1}{D_2}\right)\times\left(\frac{1}{2}\right)\times\left(\frac{1}{2}\right) \Rightarrow \frac{D_1}{D_2} = \frac{4}{1}$

27. (b)

28. (c) Fringe width $(\beta) = \frac{D\lambda}{d} \Rightarrow \beta \propto \lambda$
As $\lambda_{\text{red}} > \lambda_{\text{yellow}}$, hence fringe width will increase.

29. (d) $\beta = \frac{\lambda D}{d} \Rightarrow \frac{\beta_2}{\beta_1} = \frac{\lambda_2 D_2 d_1}{\lambda_1 D_1 d_2} \Rightarrow \beta_2 = 2.5\times10^{-4}\,m$.

30. (d) For interference, λ of both the waves must be same.

31. (d) $\beta \propto D$

32. (a) $\theta = \frac{\lambda}{d}$; θ can be increased by increasing λ, so here λ has to be increased by 10%
i.e., % Increase $= \frac{10}{100}\times 5890 = 589\text{Å}$

33. (b) $d = \frac{D\lambda}{\beta} = \frac{1\times5\times10^{-7}}{5\times10^{-3}} = 10^{-4} m = 0.1\ mm$.

34. (b) If intensity of each wave is I, then initially at central position $I_o = 4I$. when one of the slit is covered then intensity at central position will be I only *i.e.*, $\frac{I_o}{4}$.

35. (a) Shift $= \frac{\beta}{\lambda}(\mu-1)\,t = \frac{\beta}{(5000\times10^{-10})}(1.5)\times2\times10^{-6} = 2\beta$
i.e., 2 fringes upwards.

36. (b) $\beta = \dfrac{\lambda D}{d}$

37. (b) Separation n^{th} bright fringe and central maxima is $x_n = \dfrac{n\lambda D}{d}$

So, $x_3 = \dfrac{3 \times 6000 \times 10^{-10} \times 1}{0.5 \times 10^{-3}} = 3.5\, mm.$

38. (d) $n_1\lambda_1 = n_2\lambda_2 \Rightarrow 62 \times 5893 = n_2 \times 4358 \Rightarrow n_2 = 84.$

39. (b) Angular fringe width $\theta = \dfrac{\lambda}{d} \Rightarrow \theta \propto \lambda$

$\lambda_w = \dfrac{\lambda_a}{\mu_w}$

So $\theta_w = \dfrac{\theta_{\text{air}}}{\mu_w} = \dfrac{0.20}{\frac{4}{3}} = 0.15°$

40. (a) By using $x_n = \dfrac{n\lambda D}{d}$

$\Rightarrow (5 \times 10^{-3}) = \dfrac{10 \times \lambda \times 1}{(1 \times 10^{-3})} \Rightarrow \lambda = 5 \times 10^{-7} m = 5000\, Å$

41. (b) Distance of third maxima from central maxima is $x = \dfrac{3\lambda D}{d} = \dfrac{3 \times 5000 \times 10^{-10} \times (200 \times 10^{-2})}{0.2 \times 10^{-3}} = 1.5\, cm.$

42. (d) Distance of n^{th} dark fringe from central fringe

$x_n = \dfrac{(2n-1)\lambda D}{2d}$

$\therefore\ x_2 = \dfrac{(2 \times 2 - 1)\lambda D}{2d} = \dfrac{3\lambda D}{2d}$

$\Rightarrow 1 \times 10^{-3} = \dfrac{3 \times \lambda \times 1}{2 \times 0.9 \times 10^{-3}} \Rightarrow \lambda = 6 \times 10^{-5}\, cm$

43. (d) $\beta = \dfrac{\lambda D}{d} \Rightarrow (4 \times 10^{-3}) = \dfrac{4 \times 10^{-7} \times D}{0.1 \times 10^{-3}} \Rightarrow D = 1\, m$

44. (a) $\beta = \dfrac{\lambda D}{d} \Rightarrow (0.06 \times 10^{-2}) = \dfrac{\lambda \times 1}{1 \times 10^{-3}} \Rightarrow \lambda = 6000\, Å$

45. (b)

46. (d) $(n_1\lambda_1 = n_2\lambda_2)\ \dfrac{n_1}{n_2} = \dfrac{\lambda_2}{\lambda_1} \Rightarrow \dfrac{n_1}{92} = \dfrac{5898}{5461} \Rightarrow n_1 = 99$

47. (d) If we use torch light in place of monochromatic light then overlapping of fringe pattern take place. Hence no fringe will appear.

48. (b)

49. (b) Position of 3rd bright fringe $x_3 = \dfrac{3D\lambda}{d}$

$\Rightarrow \lambda = \dfrac{x_3 d}{3D} = \dfrac{(0.9 \times 10^{-2}) \times (0.28 \times 10^{-3})}{3 \times 1.4} = 6000 Å$

50. (a) Distance between two consecutive

Dark fringes $= \dfrac{\lambda D}{d} = \dfrac{6000 \times 10^{-10} \times 1}{0.6 \times 10^{-3}}$

$= 1 \times 10^{-3} m = 1\, mm.$

51. (b, c) For maxima, path difference $\Delta = n\lambda$

So for $n = 1$, $\Delta = \lambda = 6320 Å$

52. (b) Shift in the fringe pattern $x = \dfrac{(\mu - 1)t.D}{d}$

$= \dfrac{(1.5 - 1) \times 2.5 \times 10^{-5} \times 100 \times 10^{-2}}{0.5 \times 10^{-3}} = 2.5\, cm.$

53. (d) In the presence of thin glass plate, the fringe pattern shifts, but no change in fringe width.

54. (a) $\beta = \dfrac{\lambda D}{d} \Rightarrow \beta \propto \lambda$

55. (a) In interference between waves of equal amplitudes a, the minimum intensity is zero and the maximum intensity is proportional to $4a$. For waves of unequal amplitudes a and $A (A > a)$, the minimum intensity is non zero and the maximum intensity is proportional to $(a + A)^2$, which is greater than $4a$.

56. (b) $\beta = \dfrac{\lambda D}{d} = \dfrac{6000 \times 10^{-10} \times 2}{4 \times 10^{-3}} = 3 \times 10^{4} m = 0.3\, mm$

57. (c) $\beta \propto \lambda$

58. (c) $\beta = \dfrac{\lambda D}{d}$

59. (c) Distance between consecutive bright fringes or dark fringes $= \beta$

$\beta = \dfrac{\lambda D}{d} = \dfrac{550 \times 10^{-9} \times 1}{1.1 \times 10^{-3}} = 500 \times 60^{-6} = 0.5\, mm$

60. (b) $\dfrac{I_{\max}}{I_{\min}} = \dfrac{\left(\dfrac{a_1}{a_2} + 1\right)^2}{\left(\dfrac{a_1}{a_2} - 1\right)^2} = \dfrac{4}{1} \Rightarrow \dfrac{a_1}{a_2} = \dfrac{3}{1}$

61. (b) $\beta = \dfrac{\lambda D}{d} \Rightarrow \beta \propto \lambda$

62. (c)

63. (b) $n_1\lambda_1 = n_2\lambda_2 \Rightarrow n_2 = n_1 \times \dfrac{\lambda_1}{\lambda_2} = 12 \times \dfrac{600}{400} = 18$

64. (d) Using relation, $d\sin\theta = n\lambda \Rightarrow \sin\theta = \dfrac{n\lambda}{d}$

For $n = 3$, $\sin\theta = \dfrac{3\lambda}{d} = \dfrac{3 \times 589 \times 10^{-9}}{0.589}$

$= 3 \times 10^{-6}$ or $\theta = \sin^{-1}(3 \times 10^{-6})$

65. (b) $\beta \propto \dfrac{1}{d}$

66. (a) When white light is used, central fringe will be white with red edges, and on either side of it, we shall get few coloured bands and then uniform illumination.

67. (c) $\beta \propto \dfrac{\lambda}{d}$

68. (b) $\beta \propto \lambda$

69. (a) $\dfrac{I_{\max}}{I_{\min}} = \dfrac{\left(\sqrt{\dfrac{I_1}{I_2}} + 1\right)^2}{\left(\sqrt{\dfrac{I_1}{I_2}} - 1\right)^2} = \left(\dfrac{\sqrt{2} + 1}{\sqrt{2} - 1}\right)^2 \approx 34;$ (given $I_1 = 2I_2$)

70. (b) $B \propto \lambda$

71. (d) $\beta \propto \frac{\lambda}{d}$

72. (b)

73. (b) For dark fringe at P

$S_1P - S_2P = \Delta = (2n-1)\lambda/2$

Here n =3 and λ = 6000

So, $\Delta = \frac{5\lambda}{2} = 5 \times \frac{6000}{2} = 15000 Å = 1.5\ micron$

74. (b) Distance of n minima from central bright fringe

$x_n = \frac{(2n-1)\lambda D}{2d}$

For n=3 *i.e.* 3rd minima

$x_3 = \frac{(2\times3-1)\times500\times10^{-9}\times1}{2\times1\times10^{-3}}$

$= \frac{5\times500\times10^{-6}}{2} = 1.25\times10^{-3} m = 1.25\ mm.$

75. (c) $\beta = \frac{\lambda D}{d}$ and $\lambda \propto \frac{1}{\mu}$

76. (d) $n_1\lambda_1 = n_2\lambda_2 \Rightarrow 3\times700 = 5\times\lambda_2 \Rightarrow \lambda_2 = 420\ nm$

77. (a) $\beta \propto \frac{\lambda}{d}$ as $d \to \frac{d}{3}$ so $\beta \to 3\beta\ \therefore n = 3$

78. (c) If shift is equal to n fringes width, then

$n = \frac{(\mu-1)t}{\lambda} = \frac{(1.5-1)\times2\times10^{-6}}{500\times10^{-9}} = \frac{1}{500}\times10^3 = 2$

Since a thin film is introduced in upper beam. So shift will be upward.

79. (b) Distance between n^{th} Bright fringe and m^{th} dark fringe $(n > m)$

$\Delta x = \left(n - m + \frac{1}{2}\right)\beta = \left(5 - 3 + \frac{1}{2}\right)\times\frac{6.5\times10^{-7}\times1}{1\times10^{-3}}$

$= 1.63\ mm$

80. (a) $\beta = \frac{\lambda D}{d}$; If λ and d both increase by 10%, there will be no change in fringe width (β).

81. (b) $\frac{I_1}{I_2} = \frac{1}{4} \Rightarrow I_1 = k$ and $I_2 = 4k$

$\therefore$ Fringe visibility $V = \frac{2\sqrt{I_1I_2}}{(I_1+I_2)} = \frac{2\sqrt{k\times4k}}{(k+4k)} = 0.8$

82. (b) $\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1+a_2}{a_1-a_2}\right)^2 = \left(\frac{3a+a}{3a-a}\right)^2 = \frac{4}{1}$

83. (d) Angular position of first dark fringe

$\theta = \frac{\lambda}{d} = \frac{5460\times10^{-10}}{0.1\times10^{-3}}\times\frac{180}{\pi}$ (in degree)

= 0.313°

84. (a) Distance between two consecutive dark fringes $\beta = \frac{\lambda D}{d}$

$= \frac{5000\times10^{-10}\times1}{0.2\times10^{-2}} = 0.25\ mm.$

85. (a) If thin film appears dark

$2\mu t \cos r = n\lambda$ for normal incidence $r = 0^o$

$\Rightarrow 2\ \mu\ t = n\lambda \Rightarrow t = \frac{n\lambda}{2\mu}$

$\Rightarrow t_{\min} = \frac{\lambda}{2\mu} = \frac{5890\times10^{-10}}{2\times1} = 2.945\times10^{-7} m$.

86. (b) In case of destructive interference (minima) phase difference is odd multiple of π .

87. (b) $\beta = \frac{(a+b)\lambda}{2a(\mu-1)\alpha}$

where a = distance between source and biprism = 0.3 m

b = distance between biprism and screen = 0.7 m.

α = Angle of prism = 1°, μ = 1.5, λ = 6000 × 10⁻ m

Hence, $\beta = \frac{(0.3+0.7)\times6\times10^{-7}}{2\times0.3(1.5-1)\times(1^o\times\frac{\pi}{180})}$

= 1.14 × 10⁻ m = 0.0114 cm.

88. (d) For minima, path difference $\Delta = (2n-1)\frac{\lambda}{2}$

For third minima $n = 3 \Rightarrow \Delta = (2\times3-1)\frac{\lambda}{2} = \frac{5\lambda}{2}$

89. (b) Fringe width $(\beta) \propto \frac{1}{\text{prism Angle}(\alpha)}$

90. (a) By using $\beta = \frac{(a+b)\lambda}{2a(\mu-1)\alpha} = \frac{(0.3+0.7)\times180\pi\times10^{-9}}{2\times0.3(1.54-1)\times\left(1\times\frac{\pi}{180}\right)}$

$= 10^{-4} m$

91. (a) $\because\ \beta \propto \lambda \Rightarrow \lambda_w < \lambda_a$ so $\beta_w < \beta_a$

92. (c) With white light, the rays reaching the centre has zero path difference. So we get white fringe at the centre and coloured near the central fringe.

93. (a) $\beta_{water} = \frac{\beta_{air}}{\mu_w}$

94. (a) $\beta = \frac{\lambda D}{d}$

95. (b) Lateral displacement of fringes = $\frac{\beta}{\lambda}(\mu-1)t$

$= \frac{1\times10^{-3}}{600\times10^{-9}}(1.5-1)\times0.06\times10^{-3} = \frac{1}{20} m = 5\ cm.$

96. (b)

97. (d) Distance of the n bright fringe from the centre $x_n = \frac{n\lambda D}{d}$

$\Rightarrow x_3 = \frac{3\times6000\times10^{-10}\times2.5}{0.5\times10^{-3}} = 9\times10^{-3} m = 9\ mm.$

Doppler's Effect of Light

1. (d) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c}$, Now $\Delta\lambda = \frac{0.5}{100}\lambda \Rightarrow \frac{\Delta\lambda}{\lambda} = \frac{0.5}{100}$

$\therefore v = \frac{0.5}{100}\times c = \frac{0.5}{100}\times3\times10^8 = 1.5\times10^6 m/s$

Increase in λ indicates that the star is receding.

2. (a) Doppler's shift is given by

$$\Delta\lambda = \frac{v\lambda}{c} = \frac{5000 \times 6000}{3 \times 10^8} = 0.1 \mathring{A}$$

3. (b) Shifting towards ultraviolet region shows that Apparent wavelength decreased. Therefore the source is moving towards the earth.

4. (b) Due to expansion of universe, the star will go away from the earth thereby increasing the observed wavelength. Therefore the spectrum will shift to the infrared region.

5. (b) With reference to this theory the velocity of the observer is neglected *w.r.t.* the light velocity.

6. (b) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} = \frac{6 \times 10^7}{3 \times 10^8} = 0.2$

$$\Delta\lambda = \lambda' - \lambda = 0.2\lambda \Rightarrow \lambda' = 1.2\lambda = 1.2 \times 4600 = 5520 \mathring{A}$$

7. (b) $\Delta\lambda = 5200 - 5000 = 200 \mathring{A}$

Now $\frac{\Delta\lambda}{\lambda'} = \frac{v}{c} \Rightarrow v = \frac{c\Delta\lambda}{\lambda'} = \frac{3 \times 10^8 \times 200}{5000}$

$$= 1.2 \times 10^7 m/\sec \approx 1.15 \times 10^7 m/\sec$$

8. (d) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow v = \frac{c}{\lambda}\Delta\lambda = \frac{c}{\lambda}(\lambda' - \lambda) = c \times \frac{0.01}{100}$

$$= 3 \times 10^4 m/s = 30\ km/\sec$$

9. (c) Blue radiations have the wavelength around 4600 $\overset{o}{A}$. It shows that apparent wavelength is smaller than the real wavelength. It means that the star is proceeding towards earth.

10. (c)

11. (a)

12. (c)

13. (b) $\Delta\lambda = \lambda\frac{v}{c} = 5700 \times \frac{100 \times 10^3}{3 \times 10^8} = 1.90 \mathring{A}$

14. (a) According to Doppler's effect, wherever there is a relative motion between source and observer, the frequency observed is different from that given out by source.

15. (a) $\Delta\lambda = \lambda.\frac{v}{c} = \frac{1.5 \times 10^6}{3 \times 10^8} \times 5000 = 25 \mathring{A}$

16. (d) $\Delta\lambda = \frac{v}{c}\lambda = \frac{3600 \times 10^3}{3 \times 10^8} \times 5896 = 70.75 \mathring{A}$

So the increased wavelength of light is observed.

17. (b) Observed frequency $\nu' = \nu\left(1 - \frac{v}{c}\right)$

$$\Rightarrow \nu' = 6 \times 10^{14}\left(1 - \frac{0.8\,c}{c}\right) = 1.2 \times 10^{14} Hz$$

18. (c) According to Doppler's principle $\lambda' = \lambda\sqrt{\frac{1 - v/c}{1 + v/c}}$ for $v = c$

$$\lambda' = 5500\sqrt{\frac{(1 - 0.8)}{1 + 0.8}} = 1833.3$$

$\therefore$ Shift = 5500 – 1833.3 = 3167 Å

19. (b) $\Delta\lambda = \lambda\frac{v}{c}$

$$\Rightarrow (3737 - 3700) = 3700 \times \frac{v}{3 \times 10^8} \Rightarrow v = 3 \times 10^6 m/s$$

20. (d) $\Delta\lambda = \frac{v_s}{c}\lambda \Rightarrow v_s = \frac{\Delta\lambda.c}{\lambda} = \frac{47 \times 3 \times 10^8}{4700}$

$= 3 \times 10^6 m/s$ away from earth

21. (b) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow \frac{0.05}{100} = \frac{v}{3 \times 10^8} \Rightarrow v = 1.5 \times 10^5$ *m/s*

(Since wavelength is decreasing, so star coming closer)

22. (b)

23. (c) $\lambda' = \lambda\left(1 - \frac{v}{c}\right) = 5890\left(1 - \frac{4.5 \times 10^6}{3 \times 10^8}\right) \approx$ 5802Å

24. (c) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \therefore v = \frac{\Delta\lambda}{\lambda}c = \frac{0.1}{6000} \times 3 \times 10^5 km/s = 5\ km/s$

25. (b) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow \Delta\lambda = \frac{5700 \times 10^6}{3 \times 10^3} = 19 \mathring{A}$

26. (b) $\nu' = \nu\left(1 - \frac{v}{c}\right) = 4 \times 10^7\left(1 - \frac{0.2c}{c}\right) = 3.2 \times 10^7 Hz$

27. (c) When the source and observer approach each other, apparent frequency increases and hence wavelength decreases.

28. (b)

29. (a) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow 1 = \frac{v}{c} \Rightarrow v = c$

30. (d) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow v = \frac{\Delta\lambda}{\lambda}.c = \frac{5}{6563} \times (3 \times 10^8) = 2.29 \times 10^5$ *m/sec*

31. (c)

32. (a) Using $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow v = \frac{\Delta\lambda}{\lambda}c$

$\Rightarrow v = 0.004 \times 3 \times 10^8 = 1.2 \times 10^6$ *m/sec*

33. (a)

Diffraction of Light

1. (c) For first minima $\theta = \frac{\lambda}{a}$ or $a = \frac{\lambda}{\theta}$

$\therefore a = \frac{6500 \times 10^{-8} \times 6}{\pi}$ (As 30° = $\frac{\pi}{6}$ radian)

$= 1.24 \times 10^{-4} cm = 1.24$ microns

2. (a) The angular half width of the central maxima is given by

$\sin\theta = \frac{\lambda}{a} \Rightarrow \theta = \frac{6328 \times 10^{-10}}{0.2 \times 10^{-3}}$ *rad*

$= \frac{6328 \times 10^{-10} \times 80}{0.2 \times 10^{-3} \times \pi}$ *degree* = 0.18°

Total width of central maxima $= 2\theta = 0.36^o$

3. (b)

4. (c) It is caused due to turning of light around corners.

5. (c) Width of central maxima $= \frac{2\lambda D}{d}$

$$= \frac{2 \times 2.1 \times 5 \times 10^{-7}}{0.15 \times 10^{-2}} = 1.4 \times 10^{-3} m = 1.4\ mm$$

6. (a) Band width $\propto \lambda$,

$\because \lambda_{Blue} < \lambda_{Red}$, hence for blue light the diffraction bands becomes narrower and crowded together.

7. (a) Using $d\sin\theta = n\lambda$, for $n = 1$

$$\sin\theta = \frac{\lambda}{d} = \frac{550\times10^{-9}}{0.55\times10^{-3}} = 10^{-3} = 0.001\,rad$$

8. (a) For single slit diffraction pattern $d\sin\theta = \lambda$ (d = slit width)

Angular width = $2\theta = 2\sin^{-1}\left(\frac{\lambda}{d}\right)$

It is independent of D *i.e.* distance between screen and slit

9. (c) Width of central bright fringe.

$$= \frac{2\lambda D}{d} = \frac{2\times500\times10^{-9}\times80\times10^{-2}}{0.20\times10^{-3}} = 4\times10^{-3}m = 4mm.$$

10. (b) Diffraction is obtained when the slit width is of the order of wavelength of EM waves (or light). Here wavelength of X-rays (1-100 Å) is very-very lesser than slit width (0.6 *mm*). Therefore no diffraction pattern will be observed.

11. (a) Multiple focii of zone plate given by $f_p = \frac{r_n^2}{(2p-1)\lambda}$, where $p = 1, 2, 3......$

12. (b) $A = n\pi d\lambda \Rightarrow nd = \frac{A}{\pi\lambda}$ = constant $\Rightarrow n \propto \frac{1}{d}$ (n = number of blocked HPZ) on decreasing d, n increases, hence intensity decreases.

13. (a) For secondary maxima $d\sin\theta = \frac{5\lambda}{2}$

$$\Rightarrow d\theta = d.\frac{x}{D(\approx f)} = \frac{5\lambda}{2}$$

$$\Rightarrow 2x = \frac{5\lambda f}{d} = \frac{5\times0.8\times10^{-7}}{4\times10^{-4}} = 6\times10^{-3}m = 6mm$$

14. (d) By using $f_p = \frac{r^2}{(2p-1)\lambda}$

For first HPZ $r = \sqrt{f_p\lambda} = \sqrt{0.6\times6000\times10^{-10}}$

$= 6\times10^{-4}m.$

15. (a) $f_1 = \frac{r^2}{\lambda} = \frac{(2.3\times10^{-3})^2}{5893\times10^{-10}} = 9\,m.$

16. (b) $\lambda_{Blue} < \lambda_{Red}$. Therefore fringe pattern will contract because fringe width $\propto \lambda$

17. (a)

18. (a) For diffraction size of the obstacle must be of the order of wavelength of wave *i.e.* $a \approx \lambda$

19. (a) Angular width of central maxima

$$= \frac{2\lambda}{d} = \frac{2\times589.3\times10^{-9}}{0.1\times10^{-3}}rad = 0.0117\times\frac{180}{\pi} = 0.68°$$

20. (d)

21. (a) $r_n = \sqrt{nd\lambda} \Rightarrow r_n \propto \sqrt{n}$

22. (a) $\beta = \frac{\lambda.D}{d}$ where D = distance of screen from wire, d = diameter of wire

23. (c) Angular width $= \frac{2\lambda}{d} = \frac{2\times6000\times10^{-10}}{12\times10^{-5}\times10^{-2}} = 1\,rad.$

24. (b)

25. (d)

26. (a) $2\theta = \frac{2\lambda}{d}$ (where d = slit width)

As d decreases, θ increases.

27. (d)

28. (d) Distance between the first dark fringes on either side of central maxima = width of central maxima $= \frac{2\lambda D}{d} = \frac{2\times600\times10^{-9}\times2}{1\times10^{-3}}$ = 2.4 *mm.*

29. (b) Thickness of the film must be of the order of wavelength of light falling on film (*i.e.* visible light)

30. (d)

31. (d) For n secondary maxima path difference

$$d\sin\theta = (2n+1)\frac{\lambda}{2} \Rightarrow a\sin\theta = \frac{3\lambda}{2}$$

32. (d) The phase difference (ϕ) between the wavelets from the top edge and the bottom edge of the slit is $\phi = \frac{2\pi}{\lambda}(d\sin\theta)$ where d is the slit width. The first minima of the diffraction pattern occurs at $\sin\theta = \frac{\lambda}{d}$ so $\phi = \frac{2\pi}{\lambda}\left(d\times\frac{\lambda}{d}\right) = 2\pi$

33. (b)

34. (a) For second dark fringe $d\sin\theta = 2\lambda$

$$\Rightarrow 24\times10^{-5}\times10^{-2}\times\sin30 = 2\lambda$$

$$\Rightarrow \lambda = 6\times10^{-7}\,m = 6000\,Å$$

35. (c) For the first minima $d\sin\theta = \lambda$

$$\Rightarrow \sin\theta = \frac{\lambda}{d} \Rightarrow \theta = \sin^{-1}\left(\frac{5000\times10^{-10}}{0.001\times10^{-3}}\right) = 30^o$$

36. (c)

37. (a)

38. (c) Position of first minima = position of third maxima *i.e.*, $\frac{1\times\lambda_1 D}{d} = \frac{(2\times3+1)}{2}\frac{\lambda_2 D}{d} \Rightarrow \lambda_1 = 3.5\lambda_2$

39. (a) Position of nth minima $x_n = \frac{n\lambda D}{d}$

$$\Rightarrow 5\times10^{-3} = \frac{1\times5000\times10^{-10}\times1}{d}$$

$$\Rightarrow d = 10^{-4}\ m = 0.1\ mm.$$

40. (b) Width of n HPZ $B_n = r_n - r_{n-1}$

$$r_n = \sqrt{nb\lambda},\ r_{n-1} = \sqrt{(n-1)b\lambda}$$

$$B_n = \sqrt{nb\lambda} - \sqrt{(n-1)b\lambda} = \sqrt{b\lambda}\,[\sqrt{n} - \sqrt{(n-1)}]$$

41. (c) In single slit experiment,

Width of central maxima (y) = $2\lambda D/d$

$$\Rightarrow \frac{y'}{y} = \frac{\lambda'}{d'}\times\frac{d}{\lambda} = \frac{600}{d/2}\times\frac{d}{400} \Rightarrow y' = 3y.$$

Polarisation of Light

1. (b) Polariser produced prolarised light.
2. (a) Only transverse waves can be polarised.
3. (c) Polarisation is not shown by sound waves.
4. (d) By using $\mu = \tan\theta_p \Rightarrow \mu = \tan 60 = \sqrt{3}$,

   also $C = \sin^{-1}\left(\frac{1}{\mu}\right) \Rightarrow C = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$
5. (d) $\mu = \tan\theta_p \Rightarrow \theta_p = \tan^{-1} n$
6. (d) Ultrasonic waves are longitudinal waves.
7. (b) $I = I_0 \cos^2\theta = I_0 \cos^2 45 = \frac{I_0}{2}$
8. (d)
9. (a) It magnitude of light vector varies periodically during it's rotation, the tip of vector traces an ellipse and light is said to be elliptically polarised. This is not in nicol prism.
10. (c) At polarizing angle, the reflected and refracted rays are mutually perpendicular to each other.
11. (a) When unpolarised light is made incident at polarising angle, the reflected light is plane polarised in a direction perpendicular to the plane of incidence.

    Therefore $\vec{E}$ in reflected light will vibrate in vertical plane with respect to plane of incidence.
12. (d) In the arrangement shown, the unpolarised light is incident at polarising angle of 90° – 33° = 57°. The reflected light is thus plane polarised light. When plane polarised light is passed through Nicol prism (a polariser or analyser), the intensity gradually reduces to zero and finally increases.
13. (a)
14. (a) A plane which contains $\vec{E}$ and the propagation direction is called the plane of polarization.
15. (c)
16. (d) Light suffers double refraction through calcite.
17. (d) The amplitude will be $A\cos 60^o = A/2$
18. (d)
19. (b) Rotation produced $\theta = Slc$

    Net rotation produced $\theta_r = \theta_1 - \theta_2 = l(S_1c_1 - S_2c_2)$

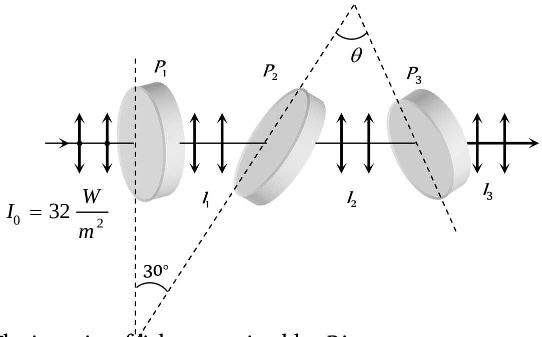
    $= 0.29 \times [0.01 \times 60 - 0.02 \times 30] = 0$
20. (c) In double refraction light rays always splits into two rays (*O*-ray & *E*-ray). *O*-ray has same velocity in all direction but *E*-ray has different velocity in different direction.

    For calcite $\mu_e < \mu_o \Rightarrow v_e > v_o$

    For quartz $\mu_e > \mu_o \Rightarrow v_o > v_e$
21. (c) $\theta_P + r = 90^\circ$ or $r = 90^\circ - \theta_p = 90^\circ - 53^\circ 4' = 36^\circ 56'$.
22. (d)
23. (a)
24. (c) Intensity of polarized light from first polarizer $= \frac{100}{2} = 50$

    $I = 50\cos^2 60^\circ = \frac{50}{4} = 12.5$
25. (b) $I' = \frac{I}{2}\cos^2\theta = \frac{I}{6}$ or $\cos\theta = \frac{1}{\sqrt{3}}$ $\therefore \theta = 55^o$
26. (b) Angle between P_1 and P_2 = 30° (given)

    Angle between P_2 and P_3 = θ = 90° – 30° = 60°



    The intensity of light transmitted by P_1 is

    $I_1 = \frac{I_0}{2} = \frac{32}{2} = 16\,\frac{W}{m^2}$

    According to Malus law the intensity of light transmitted by P_2 is $I_2 = I_1\cos^2 30^\circ = 16\left(\frac{\sqrt{3}}{2}\right)^2 = 12\,\frac{W}{m^2}$

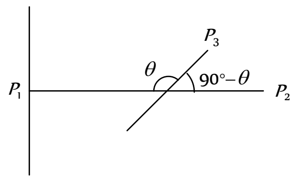
    Similarly intensity of light transmitted by P_3 is

    $I_3 = I_2\cos^2\theta = 12\cos^2 60^\circ = 12\left(\frac{1}{2}\right)^2 = 3\,\frac{W}{m^2}$
27. (b) $\theta = a + \frac{b}{\lambda^2}$

    $30 = a + \frac{b}{(5000)^2}$ and $50 = a + \frac{b}{(4000)^2}$

    Solving for a, we get $a = -\frac{50^\circ}{9}\,per\,mm$
28. (c) If an unpolarised light is converted into plane polarised light by passing through a polaroid, it's intensity becomes half.
29. (a)
30. (b) The magnitude of electric field vector varies periodically with time because it is the form of electromagnetic wave.
31. (a) According to Brewster's law, when a beam of ordinary light (i.e. unpolarised) is reflected from a transparent medium (like glass), the reflected light is completely plane polarised at certain angle of incidence called the angle of polarisation.
32. (a) When the plane-polarised light passes through certain substance, the plane of polarisation of the light is rotated about the direction of propagation of light through a certain angle.
33. (c) From Brewster's law $\mu = \tan i_p \Rightarrow \frac{c}{v} = \tan 60^\circ = \sqrt{3}$

    $\Rightarrow v = \frac{c}{\sqrt{3}} = \frac{3\times 10^8}{\sqrt{3}} = \sqrt{3}\times 10^8$ *m/sec.*
34. (a) No light is emitted from the second polaroid, so P_1 and P_2 are perpendicular to each other



Let the initial intensity of light is I_0 . So Intensity of light after transmission from first polaroid = $\frac{I_0}{2}$.

Intensity of light emitted from P_3 $I_1 = \frac{I_0}{2}\cos^2\theta$

Intensity of light transmitted from last polaroid *i.e.* from $P_2 = I_1\cos^2(90^o - \theta) = \frac{I_0}{2}\cos^2\theta.\sin^2\theta$

$= \frac{I_0}{8}(2\sin\theta\cos\theta)^2 = \frac{I_0}{8}\sin^2 2\theta$.

35. (d)

EM Waves

1. (a)

2. (d) $\lambda_{Red} > \lambda_{Blue} > \lambda_{X-ray} > \lambda_\gamma$

3. (b) Infrasonic waves are mechanical waves.

4. (d) $\mu_0 = 4\pi\times10^{-7}, \varepsilon_0 = 8.85\times10^{-12}\ \frac{N\text{-}m^2}{C^2}$

so $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 3\times10^8\ \frac{meter}{sec}$.

5. (b) Wavelength of visible spectrum is 3900 Å – 7800 Å.

6. (b) Infrared causes heating effect.

7. (a)

8. (d)

9. (c) Speed of EM waves in vacuum = $\frac{1}{\sqrt{\mu_0\epsilon_0}}$ =constant

10. (a) $\lambda_{\gamma-rays} < \lambda_{x-rays} < \lambda_{\alpha-rays} < \lambda_{\beta-rays}$.

11. (a) Distance covered by T.V. signals = $\sqrt{2hR}$

$\Rightarrow$ maximum distance $\propto h$

12. (c) β-rays are beams of fast electrons.

13. (a)

14. (b) Velocity of EM waves = $\frac{1}{\sqrt{\mu_0\epsilon_0}}$ $3\times10^8 m/s$ =velocity of light

15. (b) Ozone layer absorbs most of the *UV* rays emitted by sun.

16. (d) $\nu_{\gamma-rays} > \nu_{\text{visibleradiation}} > \nu_{\text{Infrared}} > \nu_{\text{Radio waves}}$

17. (b) Infrared radiations reflected by low lying clouds and keeps the earth warm.

18. (d) $\lambda_{Radiowaves} > \lambda_{UV\ rays} > \lambda_{J\ Rays} > \lambda_{X-rays}$

19. (a) Polarization is shown by only transverse waves.

20. (c) EM waves travels with perpendicular to *E* and *B*. Which are also perpendicular to each other $\vec{v} = \vec{E}\times\vec{B}$

21. (a)

22. (d) Ozone hole is depletion of ozone layer in stratosphere because of gases like CFC'S etc.

23. (c)

24. (b)

25. (a) $\nu_{\gamma-rays} > \nu_{X-rays} > \nu_{UV-rays}$

26. (a) In vacuum velocity of all EM waves are same but their wavelengths are different.

27. (c)

28. (c)

29. (a) $\lambda = \frac{c}{\nu} = \frac{3\times10^8}{8.2\times10^6} = 36.5\ m$

30. (b) $c = \frac{E}{B} \Rightarrow B = \frac{E}{c} = \frac{18}{3\times10^8} = 6\times10^{-8}T$.

31. (c) According to the Maxwell's EM theory, the EM waves propagation contains electric and magnetic field vibration in mutually perpendicular direction. Thus the changing of electric field give rise to magnetic field.

32. (a) Here $E_0 = 100\ V/m,\ B_0 = 0.265\ A/m.$

$\therefore$ Maximum rate of energy flow $S = E_0\times B_0$

$= 100\times.265 = 26.5\frac{W}{m^2}$

33. (d) $E = \frac{hc}{\lambda} = \frac{6.6\times10^{-34}\times3\times10^8}{21\times10^{-2}} = 0.94\times10^{-24} \approx 10^{-24}\ J$

34. (a) $\nu = \frac{C}{\lambda} \Rightarrow \nu_1 = \frac{3\times10^8}{1} = 3\times10^8\ Hz = 300\ MHz$

and $\nu_2 = \frac{3\times10^8}{10} = 3\times10^7\ Hz = 30 MHz$

35. (d)

36. (c) $\vec{E}$ and $\vec{B}$ are mutually perpendicular to each other and are in phase i.e. they become zero and minimum at the same place and at the same time.

37. (b) Molecular spectra due to vibrational motion lie in the microwave region of EM-spectrum. Due to Kirchhoff's law in spectroscopy the same will be absorbed.

38. (a) E_x and B_y would generate a plane EM wave travelling in *z*-direction. $\vec{E}$, $\vec{B}$ and $\vec{k}$ form a right handed system $\vec{k}$ is along z-axis. As $\hat{i}\times\hat{j} = \hat{k}$

$\Rightarrow E_x\hat{i}\times B_y\hat{j} = C\hat{k}$ *i.e.* *E* is along *x*-axis and *B* is along *y-axis.*

39. (a) $\nu_{\gamma-rays} > \nu_{UV\text{-rays}} > \nu_{\text{Blue light}} > \nu_{\text{Infrared rays}}$

40. (d) Ground wave and sky wave both are amplitude modulated wave and the amplitude modulated signal is transmitted by a transmitting antenna and received by the receiving antenna at a distance place.

41. (a)

42. (b) EM waves transport energy, momentum and information but not charge. EM waves are uncharged

43. (b) EM waves carry momentum and hence can exert pressure on surfaces. They also transfer energy to the surface so $p \neq 0$ and $E \neq 0$.

44. (c) The angular wave number $k = \frac{2\pi}{\lambda}$; where λ is the wave length. The angular frequency is $w = 2\pi\nu$.

The ratio $\frac{k}{\omega} = \frac{2\pi/\lambda}{2\pi\nu} = \frac{1}{\nu\pi} = \frac{1}{c}$ = constant

45. (a) $\frac{E_0}{B_0} = C$. also $k = \frac{2\pi}{\lambda}$ and $\omega = 2\pi\nu$

These relation gives $E_0K = B_0\omega$

46. (b) $\nu = \frac{1}{2\pi\sqrt{LC}}$ and $\lambda = \frac{C}{\nu}$

47. (a) $I = \frac{1}{2}\varepsilon_0 C E_0^2$

$\Rightarrow E_0 = \sqrt{\frac{2I}{\varepsilon_0 C}} = \sqrt{\frac{2\times5\times10^{-16}}{8.85}} = 0.61\times10^{-6}\,\frac{V}{m}$

Also $E_0 = \frac{V_0}{d} \Rightarrow V_0 = E_0 d = 0.61\times10^{-6}\times2 = 1.23\mu V$

48. (c)

49. (c) Population covered = $2\pi hR \times$ Population density

$= 2\pi \times 100 \times 6.4 \times 10^6 \times \frac{1000}{(10^3)^2} = 4 \times 10^6$

50. (a)

51. (c)

52. (c) Refractive index $= \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}}$

Here μ is not specified so we can consider $\mu = \mu_0$

then refractive index $= \sqrt{\frac{\varepsilon}{\varepsilon_0}} = 2$

$\therefore$ Speed and wavelength of wave becomes half and frequency remain unchanged.

53. (d)

54. (d)

55. (b)

56. (b)

57. (a) Intensity or power per unit area of the radiations $P = fv \Rightarrow$

$f = \frac{P}{v} = \frac{0.5}{3\times10^8} = 0.166\times10^{-8}\ N/m^2$

58. (d) $v = \frac{c}{\sqrt{\mu_r\varepsilon_r}} = \frac{3\times10^8}{\sqrt{1.3\times2.14}} = 1.8\times10^8 m/sec$

59. (b) $I' = Ie^{-\mu x} \Rightarrow x = \frac{1}{\mu}\log_e\frac{I}{I'}$ (where I = original intensity, I' = changed intensity)

$36 = \frac{1}{\mu}\log_e\frac{I}{I/8} = \frac{3}{\mu}\log_e 2$ ....(i)

$x = \frac{1}{\mu}\log_e\frac{I}{I/2} = \frac{1}{\mu}\log_e 2$ .....(ii)

From equation (i) and (ii), $x = 12\ mm$.

60. (c) $\lambda_m > \lambda_v > \lambda_x$

61. (a) If maximum electron density of the ionosphere is N_{max} per m^3 then the critical frequency f_c is given by $f_c = 9(N_{\max})^{1/2}$.

$\Rightarrow 1\times10^6 = 9(N)^{1/2} \Rightarrow N = 1.2 \times 10^{12}\ m^{-3}$

62. (c)

63. (b)

64. (a)

65. (c)

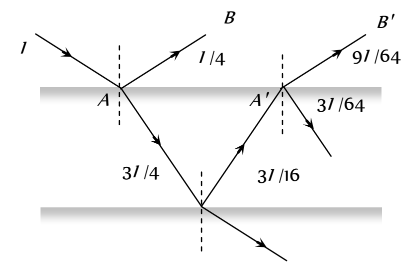
66. (d) Direction of wave propagation is given by $\vec{E}\times\vec{B}$.

67. (c) Speed of light of vacuum $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$ and in another medium $v = \frac{1}{\sqrt{\mu\varepsilon}}$

$\therefore \frac{c}{v} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} = \sqrt{\mu_r K} \Rightarrow v = \frac{c}{\sqrt{\mu_r K}}$

Critical Thinking Questions

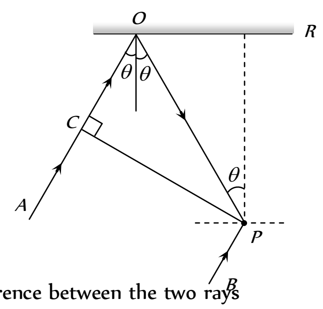
1. (d) From figure $I_1 = \dfrac{I}{4}$ and $I_2 = \dfrac{9I}{64} \Rightarrow \dfrac{I_2}{I_1} = \dfrac{9}{16}$



By using $\dfrac{I_{max}}{I_{min}} = \left(\dfrac{\sqrt{\dfrac{I_2}{I_1}}+1}{\sqrt{\dfrac{I_2}{I_1}}-1}\right) = \left(\dfrac{\sqrt{\dfrac{9}{16}}+1}{\sqrt{\dfrac{9}{16}}-1}\right) = \dfrac{49}{1}$

2. (a) The cylindrical surface touches the glass plate along a line parallel to axis of cylinder. The thickness of wedge shaped film increases on both sides of this line. Locus of equal path difference are the lines running parallel to the axis of the cylinder. Hence straight fringes are obtained.

3. (b) $\because PR = d \Rightarrow PO = d\sec\theta$ and $CO = PO\cos 2\theta = d\sec\theta\cos 2\theta$ is



Path difference between the two rays

$\Delta = CO + PO = (d\sec\theta + d\sec\theta\cos 2\theta)$

Phase difference between the two rays is

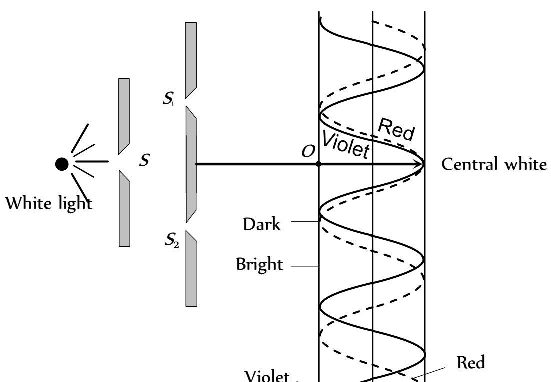
$\phi = \pi$ (One is reflected, while another is direct)

Therefore condition for constructive interference should be $\Delta = \dfrac{\lambda}{2}, \dfrac{3\lambda}{2}$ ......

or $d\sec\theta(1+\cos 2\theta) = \dfrac{\lambda}{2}$

or $\dfrac{d}{\cos\theta}(2\cos^2\theta) = \dfrac{\lambda}{2} \Rightarrow \cos\theta = \dfrac{\lambda}{4d}$

4. (c) In young's double slit experiment, if white light is used in place of monochromatic light, then the central fringe is white and some coloured fringes around the central fringe are formed.



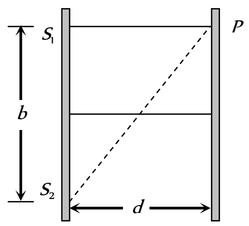
Since $\beta_{red} > \beta_{violet}$ etc., the bright fringe of violet colour forms first and that of the red forms later.

It may be noted that, the inner edge of the dark fringe is red, while the outer edge is violet. Similarly, the inner edge of the bright fringe is violet and the outer edge is red.

5. (a) In conventional light source, light comes from a large number of independent atoms, each atom emitting light for about 10^{-} *sec i.e.* light emitted by an atom is essentially a pulse lasting for only 10^{-} *sec*. Light coming out from two slits will have a fixed phase relationship only for 10^{-} *sec*. Hence any interference pattern formed on the screen would last only for 10^{-} *sec*, and then the pattern will change. The human eye can notice intensity changes which last at least for a tenth of a second and hence we will not be able to see any interference pattern. In stead due to rapid changes in the pattern, we will only observe a uniform intensity over the screen.

6. (a,c) Path difference between the rays reaching infront of slit S_1 is.

$S_1P - S_2P = (b^2+d^2)^{1/2} - d$



For distructive interference at P

$S_1P - S_2P = \dfrac{(2n-1)\lambda}{2}$

i.e., $(b^2+d^2)^{1/2} - d = \dfrac{(2n-1)\lambda}{2}$

$\Rightarrow d\left(1+\dfrac{b^2}{d^2}\right)^{1/2} - d = \dfrac{(2n-1)\lambda}{2}$

$\Rightarrow d\left(1+\dfrac{b^2}{2d^2}+......\right) - d = \dfrac{(2n-1)\lambda}{2}$

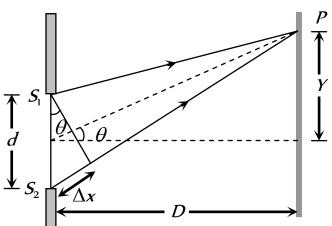
(Binomial Expansion)

$\Rightarrow \dfrac{b}{2d} = \dfrac{(2n-1)\lambda}{2} \Rightarrow \lambda = \dfrac{b^2}{(2n-1)d}$

For $n = 1, 2$.......... .., $\lambda = \dfrac{b^2}{d}, \dfrac{b^2}{3d}$

7. (a)

8. (a,b) For microwave $\lambda = \dfrac{c}{f} = \dfrac{3\times10^8}{10^6} = 300\ m$



As $\Delta x = d\sin\theta$

Phase difference $\phi = \frac{2\pi}{\lambda}$ (Path difference)

$$= \frac{2\pi}{\lambda}(d\sin\theta) = \frac{2\pi}{300}(150\sin\theta) = \pi\sin\theta$$

$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

Here $I_1 = I_2$ and $\phi = \pi\sin\theta$

$$\therefore I_R = 2I_1[1+\cos(\pi\sin\theta)] = 4I_1\cos^2\left(\frac{\pi\sin\theta}{2}\right)$$

I will be maximum when $\cos^2\left(\frac{\pi\sin\theta}{2}\right) = 1$

$$\therefore (I_R)_{\max} = 4I_1 = I_o$$

Hence $I = I_o\cos^2\left(\frac{\pi\sin\theta}{2}\right)$

If $\theta = 0$, then $I = I_o\cos\theta = I_o$

If $\theta = 30°$, then $I = I_o\cos^2(\pi/4) = I_o/2$

If $\theta = 90^o$, then $I = I_o\cos^2(\pi/2) = 0$

9. (d) $I = a_1^2 + a_2^2 + 2a_1a_2\cos\phi$

Put $a_1^2 + a_2^2 = A$ and $a_1a_2 = B$; $\therefore I = A + B\cos\phi$

10. (d) Since P is ahead of Q by $90°$ and path difference between P and Q is $\lambda/4$. Therefore at A, phase difference is zero, so intensity is $4I$. At C it is zero and at B, the phase difference is $90°$, so intensity is $2I$.

11. (b) By using phase difference $\phi = \frac{2\pi}{\lambda}(\Delta)$

For path difference λ, phase difference $\phi_1 = 2\pi$ and for path difference $\lambda/4$, phase difference $\phi_2 = \pi/2$.

Also by using $I = 4I_0\cos^2\frac{\phi}{2} \Rightarrow \frac{I_1}{I_2} = \frac{\cos^2(\phi_1/2)}{\cos^2(\phi_2/2)}$

$$\Rightarrow \frac{K}{I_2} = \frac{\cos^2(2\pi/2)}{\cos^2\left(\frac{\pi/2}{2}\right)} = \frac{1}{1/2} \Rightarrow I_2 = \frac{K}{2}.$$

12. (d) If shift is equivalent to n fringes then

$$n = \frac{(\mu-1)t}{\lambda} \Rightarrow n \propto t \Rightarrow \frac{t_2}{t_1} = \frac{n_2}{n_1} \Rightarrow t_2 = \frac{n_2}{n_1}\times t$$

$$t_2 = \frac{20}{30}\times 4.8 = 3.2\ mm.$$

13. (a) According to given condition

$(\mu-1)t = n\lambda$ for minimum t, $n = 1$

So, $(\mu-1)t_{\min} = \lambda$

$$t_{\min} = \frac{\lambda}{\mu-1} = \frac{\lambda}{1.5-1} = 2\lambda$$

14. (a) $\Delta\lambda = \lambda\frac{v}{c}$ and $v = r\omega$

$$v = 7\times10^8\times\frac{2\pi}{25\times24\times3600},\ c = 3\times10^8\ m/s$$

$$\therefore \Delta\lambda = 0.04Å$$

15. (b) $v = \frac{c\Delta\lambda}{\lambda} = \frac{3\times10^8\times(706-656)}{656} = \frac{1500}{656}\times10^7$

$$= 2\times10^7\ m/s$$

16. (b) In this case, we can assume as if both the source and the observer are moving towards each other with speed v. Hence $\nu' = \frac{c-u_o}{c-u_s}\nu = \frac{c-(-v)}{c-v}\nu = \frac{c+v}{c-v}\nu$

$$= \frac{(c+v)(c-v)}{(c-v)^2}\nu = \frac{c^2-v^2}{c^2+v^2-2vc}\nu$$

Since $v<<c$, therefore $\nu' = \frac{c^2}{c^2-2vc} = \frac{c}{c-2v}\nu$

17. (a) $\Delta\lambda = \lambda.\frac{v}{c}$ where $v = r\omega = r\times\left(\frac{2\pi}{T}\right)$

$$\therefore \Delta\lambda = \frac{4320\times7\times10^8\times2\times3.14}{3\times10^8\times22\times86400} = 0.033Å$$

18. (a) $\beta = \frac{\lambda D}{d} \Rightarrow \beta \propto D$

$$\Rightarrow \frac{\beta_1}{\beta_2} = \frac{D_1}{D_2} \Rightarrow \frac{\beta_1-\beta_2}{\beta_2} = \frac{D_1-D_2}{D_2} \Rightarrow \frac{\Delta\beta}{\Delta D} = \frac{\beta_2}{D_2} = \frac{\lambda_2}{d_2}$$

$$= \lambda_2 = \frac{3\times10^{-5}}{5\times10^{-2}}\times10^{-3} = 6\times10^{-7}m = 6000Å$$

19. (a) P is the position of 11th bright fringe from Q. From central position O, P will be the position of 10th bright fringe.

Path difference between the waves reaching at $P = SB = 10\ \lambda = 10\times6000\times10^{-10} = 6\times10^{-6}m$.

20. (b) Resultant intensity $I = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\phi$

At central position with coherent source (and $I_1 = I_2 = I_0$

$I_{con} = 4I_0$ ... (i)

In case of incoherent at a given point, ϕ varies randomly with time so $(\cos\phi)_{av} = 0$

$\therefore\ I_{Incoh} = I_1 + I_2 = 2I_0$ ... (ii)

Hence $\frac{I_{coh}}{I_{Incoh}} = \frac{2}{1}$.

21. (a, d) These waves are of same frequencies and they are coherent

22. (c) Fringe width $\beta \propto \lambda$. Therefore, λ and hence β decreases 1.5 times when immersed in liquid. The distance between central maxima and 10th maxima is 3 cm in vacuum. When immersed in liquid it will reduce to 2 cm. Position of central maxima will not change while 10th maxima will be obtained at $y = 4cm$.

23. (a) Suppose P is a point infront of one slit at which intensity is to be calculated from figure it is clear that $x = \frac{d}{2}$. Path difference between the waves reaching at P

$$\Delta = \frac{xd}{D} = \frac{\left(\frac{d}{2}\right)d}{10d} = \frac{d}{20} = \frac{5\lambda}{20} = \frac{\lambda}{4}$$

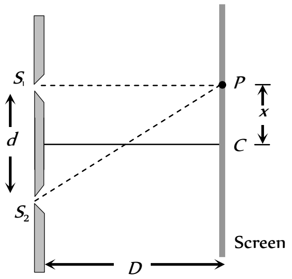
Hence corresponding phase difference

$$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$$

Resultant intensity at P

$$I = I_{\max}\cos^2\frac{\phi}{2}$$

$$= I_0\cos^2\left(\frac{\pi}{4}\right) = \frac{I_0}{2}$$



24. (d) If $d\sin\theta = (\mu - 1)t$, central fringe is obtained at O

If $d\sin\theta > (\mu - 1)t$, central fringe is obtained above O and

If $d\sin\theta < (\mu - 1)t$, central fringe is obtained below O.

25. (b) For maximum intensity on the screen

$$d\sin\theta = n\lambda \Rightarrow \sin\theta = \frac{n\lambda}{d} = \frac{n(2000)}{7000} = \frac{n}{3.5}$$

Since maximum value of $\sin\theta$ is 1

So $n = 0, 1, 2, 3,$ only. Thus only seven maximas can be obtained on both sides of the screen.

26. (c) From the given data, note that the fringe width (β) for $\lambda_1 = 900\,nm$ is greater than fringe width (β) for $\lambda_2 = 750\,nm$. This means that at though the central maxima of the two coincide, but first maximum for $\lambda_1 = 900\,nm$ will be further away from the first maxima for $\lambda_2 = 750\,nm$, and so on. A stage may come when this mismatch equals β, then again maxima of $\lambda_1 = 900\,nm$, will coincide with a maxima of $\lambda_2 = 750\,nm$, let this correspond to n order fringe for λ. Then it will correspond to $(n+1)^{th}$ order fringe for λ.

Therefore $\frac{n\lambda_1 D}{d} = \frac{(n+1)\lambda_2 D}{d}$

$$\Rightarrow n \times 900 \times 10^{-9} = (n+1)750 \times 10^{-9} \Rightarrow n = 5$$

Minimum distance from

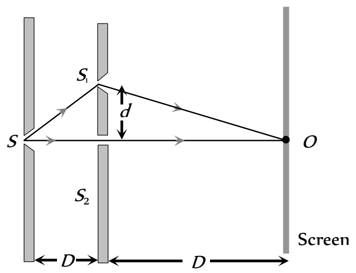
Central maxima $= \frac{n\lambda_1 D}{d} = \frac{5 \times 900 \times 10^{-9} \times 2}{2 \times 10^{-3}}$

$$= 45 \times 10^{-4}\,m = 4.5\,mm$$

27. (c) Shift $= \frac{\beta}{\lambda}(\mu - 1)t$

$$\Rightarrow 7\beta = \frac{\beta}{\lambda}(\mu - 1)t \Rightarrow t = \frac{7\lambda}{(\mu - 1)} = \frac{7 \times 600}{(1.5 - 1)} = 8400\,nm.$$

28. (c) Path difference between the waves reaching at P, $\Delta = \Delta_1 + \Delta_2$



where $\Delta_1 =$ Initial path difference

$\Delta_2 =$ Path difference between the waves after emerging from slits.

$$\Delta_1 = SS_1 - SS_2 = \sqrt{D^2 + d^2} - D$$

and $\Delta_2 = S_1O - S_2O = \sqrt{D^2 + d^2} - D$

$$\therefore\ \Delta = 2\left\{(D^2 + d^2)^{\frac{1}{2}} - D\right\} = 2\left\{(D^2 + \frac{d^2}{2D}) - D\right\}$$

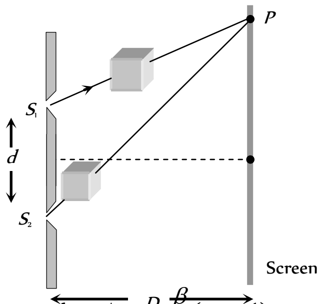
$$= \frac{d^2}{D} \quad \text{(From Binomial expansion)}$$

For obtaining dark at O, Δ must be equals to $(2n-1)\frac{\lambda}{2}$ *i.e.*

$$\frac{d^2}{D} = (2n-1)\frac{\lambda}{2} \Rightarrow d\sqrt{\frac{(2n-1)\lambda D}{2}}$$

For minimum distance $n = 1$ so $d = \sqrt{\frac{\lambda D}{2}}$

29. (a) Shift $\Delta x = \frac{\beta}{\lambda}(\mu - 1)t$



Shift due to one plate $\Delta x_1 = \frac{\beta}{\lambda}(\mu_1 - 1)$

Shift due to another path $\Delta x_2 = \frac{\beta}{\lambda}(\mu_2 - 1)t$

Net shift $\Delta x = \Delta x_2 - \Delta x_1 = \frac{\beta}{\lambda}(\mu_2 - \mu_1)t$ .....(i)

Also it is given that $\Delta x = 5\beta$ .....(ii)

Hence $5\beta = \frac{\beta}{\lambda}(\mu_1 - \mu_2)t$

$$\Rightarrow t = \frac{5\lambda}{(\mu_2 - \mu_1)} = \frac{5 \times 4800 \times 10^{-10}}{(1.7 - 1.4)} = 8 \times 10^{-6}\,m = 8\,\mu m.$$

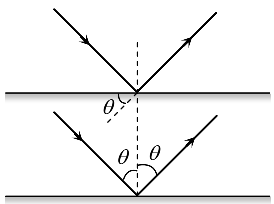
30. (b) For maxima $2\pi n = \frac{2\pi}{\lambda}(XO) - 2\pi l$

or $\frac{2\pi}{\lambda}(XO) = 2\pi(n + l)$ or $(XO) = \lambda(n + l)$

31. (c) Path difference $= 2d\sin\theta$

$\therefore$ For constructive interference

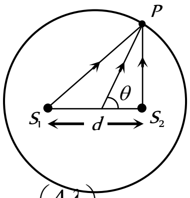
$$2d\sin\theta = n\lambda$$



$$\Rightarrow \theta = \sin^{-1}\left(\frac{n\lambda}{2d}\right)$$

32. (b) Here path difference at a point P on the circle is given by

$\Delta x = d\cos\theta$ ..... (i)



For maxima at P

$\Delta x = n\lambda$ ..... (ii)

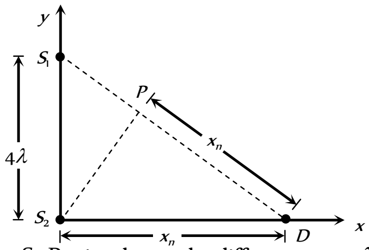
From equation (i) and (ii)

$$n\lambda = d\cos\theta \Rightarrow \theta\cos^{-1}\left(\frac{n\lambda}{d}\right) = \cos^{-1}\left(\frac{4\lambda}{d}\right)$$

33. (b) From ΔS_1S_2D,

$$(S_1D)^2 = (S_1S_2)^2 + (S_2D)^2$$

$$(S_1P + PD)^2 = (S_1S_2)^2 + (S_2D)^2$$



Here S_1P is the path difference $= n\lambda$ for maximum intensity.

$$\therefore (n\lambda + x_n)^2 = (4\lambda)^2 + (x_n)^2$$

or $x_n = \dfrac{16\lambda^2 - n^2\lambda^2}{2n\lambda}$

Then $x_1 = \dfrac{16\lambda^2 - \lambda^2}{2\lambda} = 7.5\,\lambda$

$$x_2 = \frac{16\lambda^2 - 4\lambda^2}{4\lambda} = 3\lambda$$

$$x_3 = \frac{16\lambda^2 - 9\lambda^2}{6\lambda} = \frac{7}{6}\lambda$$

$x_4 = 0$.

$\therefore$ Number of points for maxima becomes 3.

34. (a) $I_0 = R^2 = \dfrac{R_2^2}{4}$

Number of *HPZ* covered by the disc at $b = 25\,cm$

$$n_1b_1 = n_2b_2$$

$$n_2 = \frac{n_1b_1}{b_2} + \frac{1\times 1}{0.25} = 4$$

Hence the intensity at this point is

$$I = R'^2 = \left(\frac{R_5}{2}\right)^2 = \left(\frac{R_5}{R_4}\times\frac{R_4}{R_3}\times\frac{R_3}{R_2}\right)^2\times\left(\frac{R_2}{2}\right)^2$$

or $1 = [0.9]^6$

$$I_1 = 0.531\ I_0$$

Hence the correct answer will be (a).

35. (b) $I_A = R_1^2$

$$I_B = (R_1 - R_2)^2 = R_1^2\left(1 - \frac{R_2}{R_1}\right)^2 = R_1^2\left(1 - \frac{3}{4}\right)^2 = \frac{R_1^2}{16}$$

$$I_C = (R_1 - R_2 + R_3)^2 = R_1^2\left(1 - \frac{R_2}{R_1} + \frac{R_3}{R_1}\right)^2$$

$$= R_1^2\left(1 - \frac{R_2}{R_1} + \frac{R_3}{R_2}\times\frac{R_2}{R_1}\right)^2$$

$$= R_1^2\left(1 - \frac{3}{4} + \frac{3}{4}\times\frac{3}{4}\right)^2 = \left(\frac{13}{16}\right)^2 R_1^2 = \frac{169}{256}R_1^2$$

$$\therefore I_A : I_B : I_C = R_1^2 : \frac{R_1^2}{16} : \frac{169}{256}R_1^2 = 256 : 16 : 169$$

36. (d) $I = \dfrac{R_2^2}{4}$ given $n_1b_1 = n_2b_2 \Rightarrow 1\times 200 = n_2\times 25$

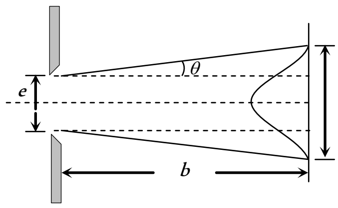
$\therefore n_2 = 8\ HPZ$

$$\therefore I = \left(\frac{R_9}{2}\right)^2$$

$$= \left(\frac{R_9}{R_8}\times\frac{R_8}{R_7}\times\frac{R_7}{R_6}\times\frac{R_6}{R_5}\times\frac{R_5}{R_4}\times\frac{R_4}{R_3}\times\frac{R_3}{R_2}\times\frac{R_2}{R_2}\right)^2$$

$$= \left(\frac{R_9}{R_2}\right)^2 I$$

37. (c) The direction in which the first minima occurs is θ (say). Then $e\sin\theta = \lambda$ or $e\theta = \lambda$ or, $\theta = \dfrac{\lambda}{e}$ ($\because\ \theta = \sin\theta$ when θ small)



Width of the central maximum $= 2b\theta + e = 2b.\dfrac{\lambda}{e} + e$

38. (b) Angular width $\beta = \dfrac{2\lambda}{d} \Rightarrow \beta \propto \lambda$

$$\Rightarrow \frac{\beta_1}{\beta_2} = \frac{\lambda_1}{\lambda_2} \Rightarrow \frac{\beta}{\frac{70}{100}\beta} = \frac{6000}{\lambda_2} \Rightarrow \lambda_2 = 4200\,Å$$

39. (a) In a single slit diffraction experiment, position of minima is given by $d\sin\theta = n\lambda$

So for first minima of red $\sin\theta = 1\times\left(\dfrac{\lambda_R}{d}\right)$

and as first maxima is midway between first and second minima, for wavelength λ', its position will be

$$d\sin\theta' = \frac{\lambda' + 2\lambda'}{2} \Rightarrow \sin\theta' = \frac{3\lambda'}{2d}$$

According to given condition $\sin\theta = \sin\theta'$

$$\Rightarrow \lambda' = \frac{2}{3}\lambda_R \text{ so } \lambda' = \frac{2}{3}\times 6600 = 440\,nm = 4400\,Å$$

40. (c) $I = I_0\left[\frac{\sin\alpha}{\alpha}\right]^2$, where $\alpha = \frac{\phi}{2}$

For n^{th} secondary maxima $d\sin\theta = \left(\frac{2n+1}{2}\right)\lambda$

$$\Rightarrow \alpha = \frac{\phi}{2} = \frac{\pi}{\lambda}[d\sin\theta] = \left(\frac{2n+1}{2}\right)\pi$$

$$\therefore\ I = I_0\left[\frac{\sin\left(\frac{2n+1}{2}\right)\pi}{\left(\frac{2n+1}{2}\right)\pi}\right]^2 = \frac{I_0}{\left\{\frac{(2n+1)}{2}\pi\right\}^2}$$

So $I_0 : I_1 : I_2 = I_0 : \frac{4}{9\pi^2}I_0 : \frac{4}{25\pi^2}I_0$

$$= 1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2}$$

41. (d) For a grating $(e+d)\sin\theta_n = n\lambda$

where $(e+d) =$ grating element

$$\sin\theta_n = \frac{n\lambda}{(e+d)}$$

For $n = 1$, $\sin\theta_1 = \frac{\lambda}{(e+d)} = \sin 32°$

This is more than 0.5. Now $\sin\theta_2$ will be more than 2×0.5, which is not possible.

42. (a) The film appears bright when the path difference $(2\mu t\cos r)$ is equal to odd multiple of $\frac{\lambda}{2}$

i.e. $2\mu t\cos r = (2n-1)\,\lambda/2$ where $n = 1, 2, 3$ .....

$$\therefore\ \lambda = \frac{4\mu t\cos r}{(2n-1)}$$

$$= \frac{4\times 1.4\times 10{,}000\times 10^{-10}\times\cos 0}{(2n-1)} = \frac{56000}{(2n-1)}Å$$

$\therefore\ \lambda = 56000\,Å$ $18666\,Å$, $8000\,Å$, $6222\,Å$, $5091\,Å$, $4308\,Å$, $3733\,Å$.

The wavelength which are not within specified range are to be refracted.

43. (a) Total phase difference

= Initial phase difference + Phase difference due to path

$$= 66° + \frac{360°}{\lambda}\times\Delta x = 66° + \frac{360°}{\lambda}\times\frac{\lambda}{4} = 66° + 90 = 156°$$

44. (a) $I = 4I_0\cos^2\frac{\phi}{2}$

At central position $I_1 = 4I_0$ ......(i)

Since the phase difference between two successive fringes is 2π, the phase difference between two points separated by a distance equal to one quarter of the distance between the two, successive fringes is equal to $\delta = (2\pi)\left(\frac{1}{4}\right) = \frac{\pi}{2}$ radian

$$\Rightarrow I_2 = 4I_0\cos^2\left(\frac{\frac{\pi}{2}}{2}\right) = 2I_0 \quad\text{......(ii)}$$

Using (i) and (ii), $\frac{I_1}{I_2} = \frac{4I_0}{2I_0} = 2$

45. (b) $I_D = \varepsilon_0\frac{d\phi_E}{dt} = \varepsilon_0\frac{EA}{t} = \varepsilon_0\left(\frac{V}{d}\right).\frac{A}{t}$.

$$= \frac{8.85\times 10^{-12}\times 400\times 60\times 10^{-4}}{10^{-3}\times 10^{-6}} = 1.602\times 10^{-2}amp$$

46. (d) Electric field $E = \frac{V}{l} = \frac{iR}{l}$ (R = Resistance of wire)

Magnetic field at the surface of wire $B = \frac{\mu_0 i}{2\pi a}$ (a = radius of wire)

Hence poynting vector, directed radially inward is given by

$$S = \frac{EB}{\mu_0} = \frac{iR}{\mu_0 l}.\frac{\mu_0 i}{2\pi a} = \frac{i^2R}{2\pi al}$$

47. (b) Average energy density of electric field is given by

$$u_e = \frac{1}{2}\varepsilon_0E^2 = \frac{1}{2}\varepsilon_0\left(\frac{E_0}{\sqrt{2}}\right)^2 = \frac{1}{4}\varepsilon_0E_0^2$$

$$= \frac{1}{4}\times 8.85\times 10^{-12}(1)^2 = 2.2\times 10^{-12}\,J/m^3.$$

48. (b) Area through which the energy of beam passes

$$= (6.328\times 10^{-7}) = 4\times 10^{-13}m^2$$

$$\therefore\ I = \frac{P}{A} = \frac{10^{-3}}{4\times 10^{-13}} = 2.5\times 10^9\,W/m^2$$

49. (a) $S_{av} = \frac{1}{2}\varepsilon_0cE_0^2 = \frac{P}{4\pi R^2}$

$$\Rightarrow E_0 = \sqrt{\frac{P}{2\pi R^2\varepsilon_0 C}}$$

$$= \sqrt{\frac{3}{2\times 3.14\times 100\times 8.85\times 10^{-12}\times 3\times 10^8}}$$

= 1.34 V/m

50. (d) Intensity of EM wave is given by

$$I = \frac{P}{4\pi R^2} = v_{av}.c = \frac{1}{2}\varepsilon_0E_0^2\times c$$

$\Rightarrow E_0 = \sqrt{\frac{P}{2\pi R^2 \varepsilon_0 c}}$

$= \sqrt{\frac{800}{2\times 3.14\times (4)^2\times 8.85\times 10^{-12}\times 3\times 10^8}}$

$= 54.77\ \frac{V}{m}$

51. (c) Wave impedance $Z = \sqrt{\frac{\mu_r}{\varepsilon_r}}\times\sqrt{\frac{\mu_0}{\varepsilon_0}}$

$= \sqrt{\frac{50}{2}}\times 376.6 = 1883\,\Omega$

52. (d) Momentum transferred in one second

$p = \frac{2U}{c} = \frac{2S_{av}A}{c} = \frac{2\times 6\times 40\times 10^{-4}}{3\times 10^8}$

$= 1.6\times 10^{-10}$ *kg-m/s*.

53. (a) Specific rotation

$(\alpha) = \frac{\theta}{l\,c} \Rightarrow c = \frac{\theta}{\alpha l} = \frac{0.4}{0.01\times 0.25} = 160\,kg/m^3$

Now percentage purity of sugar solution

$= \frac{160}{200}\times 100 = 80\%$

54. (d) As $\theta \propto l$

Volume ratio 1 : 2 in a tube of length 30 *cm* means 10 *cm* length of first solution and 20 *cm* length of second solution .

Rotation produced by 10 *cm* length of first solution

$\theta_1 = \frac{38°}{20}\times 10 = 19°$

Rotation produced by 20*cm* length of second solution

$\theta_2 = -\frac{24°}{30}\times 20 = -16°$

$\therefore$ Total rotation produced = 19° – 16° = 3°

55. (d) If I is the final intensity and I is the initial intensity then

$I = \frac{I_0}{2}(\cos^2 30°)^5$ or $\frac{I}{I_0} = \frac{1}{2}\times\left(\frac{\sqrt{3}}{2}\right)^{10} = 0.12$

56. (a) Using Matus law, $I = I_0\cos^2\theta$

As here polariser is rotating *i.e.* all the values of θ are possible.

$I_{av} = \frac{1}{2\pi}\int_0^{2\pi} I\,d\theta = \frac{1}{2\pi}\int_0^{2\pi} I_0\cos^2\theta\,d\theta$

On integration we get $I_{av} = \frac{I_0}{2}$

where $I_0 = \frac{\text{Energy}}{\text{Area}\times\text{Time}} = \frac{p}{A} = \frac{10^{-3}}{3\times 10^{-4}} = \frac{10}{3}\frac{Watt}{m^2}$

$\therefore I_{av} = \frac{1}{2}\times\frac{10}{3} = \frac{5}{3}Watt$

and Time period $T = \frac{2\pi}{\omega} = \frac{2\times 3.14}{31.4} = \frac{1}{5}sec$

$\therefore$ Energy of light passing through the polariser per revolution

$= I_{av}\times\text{Area}\times T = \frac{5}{3}\times 3\times 10^{-4}\times\frac{1}{5} = 10^{-4}\ J.$

57. (d) Let *n*th minima of 400 *nm* coincides with *m*th minima of 560 *nm* then

$(2n-1)400 = (2m-1)560 \Rightarrow \frac{2n-1}{2m-1} = \frac{7}{5} = \frac{14}{10} = \frac{21}{15}$

i.e. 4th minima of 400 *nm* coincides with 3rd minima of 560 *nm*.

The location of this minima is

$= \frac{7(1000)(400\times 10^{-6})}{2\times 0.1} = 14\,mm$

Next, 11th minima of 400 *nm* will coincide with 8th minima of 560 *nm*

Location of this minima is

$= \frac{21(1000)(400\times 10^{-6})}{2\times 0.1} = 42\,mm$

$\therefore$ Required distance = 28 *mm*

58. (b) For maxima $\Delta = d\sin\theta = n\lambda$

$\Rightarrow 2\lambda\sin\theta = n\lambda \Rightarrow \sin\theta = \frac{n}{2}$

since value of sin θ can not be greater 1.

$\therefore$ n = 0, 1, 2

Therefore only five maximas can be obtained on both side of the screen.

59. (a) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow \frac{(401.8-393.3)}{393.3} = \frac{v}{3\times 10^8}$

$\Rightarrow v = 6.48\times 10^6\ m/s = 6480\ km/sec.$

60. (c) The interference fringes for two slits are hyperbolic.

61. (d) If you divide the original slit into N strips and represents the light from each strip, when it reaches the screen, by a phasor, then at the central maximum in the diffraction pattern you add N phasors, all in the same direction and each with the same amplitude. The intensity is therefore N. If you double the slit width, you need $2N$ phasors, if they are each to have the amplitude of the each to have the amplitude of the phasors you used for the narrow slit. The intensity at the central maximum is proportional to $(2N)^2$ and is, therefore, four times the intensity for the narrow slit.

62. (c) $I = 4I_0\cos^2(\phi/2) \Rightarrow \phi = 2\pi/3$

$\Rightarrow \Delta x\times(2\pi/\lambda) = 2\pi/3 = \lambda/3$

sin $\theta = \Delta x/d \Rightarrow$ sin $\theta = \lambda/3d$

63. (b) Momentum of the electron will increase. So the wavelength ($\lambda = h/p$)of electrons will decrease and fringe width decreases as $\beta \propto \lambda$.

Assertion and Reason

1. (d) When a light wave travel from a rarer to a denser medium it loses speed, but energy carried by the wave does not depend on its speed. Instead, it depends on the amplitude of wave.

2. (e) A narrow pulse is made of harmonic waves with a large range of wavelength. As speed of propagation is different for different wavelengths, the pulse cannot retain its shape while travelling through the medium.

3. (b) When d is negligibly small, fringe width β which is proportional to $1/d$ may become too large. Even a single fringe may occupy the whole screen. Hence the pattern cannot be detected.

4. (a) The central spot of Newton's rings is dark when the medium between plano convex lens and plane glass is rarer than the medium of lens and glass. The central spot is dark because the phase change of π is introduced between the rays reflected from surfaces of denser to rarer and rarer to denser media.

5. (a) For reflected system of the film, the maxima or constructive interference is $2\mu t\cos r = \frac{(2n-1)\lambda}{2}$ while the maxima for transmitted system of film is given by equation $2\mu t\cos r = n\lambda$

where t is thickness of the film and r is angle of reflection.

From these two equations we can see that condition for maxima in reflected system and transmitted system are just opposite.

6. (b) When intensity of light emerging from two slits is equal, the intensity at minima,

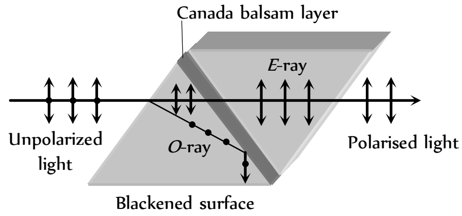
$$I_{\min} = \left(\sqrt{I_a} - \sqrt{I_b}\right)^2 = 0, \text{ or absolute dark.}$$

It provides a better contrast.

7. (c) When one of slits is covered with cellophane paper, the intensity of light emerging from the slit is decreased (because this medium is translucent). Now the two interfering beam have different intensities or amplitudes. Hence intensity at minima will not be zero and fringes will become indistinct.

8. (a) When a polaroid is rotated in the path of unpolarised light, the intensity of light transmitted from polaroid remains undiminished (because unpolarised light contains waves vibrating in all possible planes with equal probability). However, when the polaroid is rotated in path of plane polarised light, its intensity will vary from maximum (when the vibrations of the plane polarised light are parallel to the axis of the polaroid) to minimum (when the direction of the vibrations becomes perpendicular to the axis of the crystal). Thus using polaroid we can easily verify that whether the light is polarised or not.

9. (c) The nicol prism is made of calcite crystal. When light is passed through calcite crystal, it breaks up into two rays (i) the ordinary ray which has its electric vector perpendicular to the principal section of the crystal and (ii) the extra ordinary ray which has its electric vector parallel to the principal section. The nicol prism is made in such a way that it eliminates one of the two rays by total internal reflection, thus produces plane polarised light. It is generally found that the ordinary ray is eliminated and only the extra ordinary ray is transmitted through the prism. The nicol prism consists of two calcite crystal cut at – 68° with its principal axis joined by a glue called Canada balsam.



10. (b) Doppler's effect is observed readily in sound wave due to larger wavelengths. The same is not the case with light due to shorter wavelength in every day life.

11. (d) In Young's experiments fringe width for dark and white fringes are same while in Young's double slit experiment when a white light as a source is used, the central fringe is white around which few coloured fringes are observed on either side.

12. (a) It is quite clear that the coloured spectrum is seen due to diffraction of white light on passing through fine slits made by fine threads in the muslin cloth.

13. (c) As the waves diffracted from the edges of circular obstacle, placed in the path of light interfere constructively at the centre of the shadow resulting in the formation of a bright spot.

14. (c) The beautiful colours are seen on account of interference of light reflected from the upper and the lower surfaces of the thin films.

15. (a) Microwave communication is preferred over optical communication because microwaves provide large number of channels and wider band width compared to optical signals as information carrying capacity is directly proportional to band width. So, wider the band width, greater the information carrying capacity.

16. (a)

17. (a) $\beta = \frac{\lambda D}{d}$

18. (c) The clouds consists of dust particles and water droplets. Their size is very large as compared to the wavelength of the incident light from the sun. So there is very little scattering of light. Hence the light which we receive through the clouds has all the colours of light. As a result of this, we receive almost white light. Therefore, the cloud are generally white.

19. (d) In sky wave propagation, the radio waves having frequency range 2 *MHz* to 30 *MHz* are reflected back by the ionosphere. Radio waves having frequency nearly greater than 30 *MHz* penetrates the inosphere and is not reflected back by the ionosphere. The TV signal having frequency greater than 30 *MHz* therefore cannot be propagated through sky wave propagation.

In case of sky wave propagation, critical frequency is defined as the highest frequency is returned to the earth by the considered layer of the ionosphere after having sent straight to it. Above this frequency, a wave will penetrate the inosphere and is not reflected by it.

20. (c) The television signals being of high frequency are not reflected by the ionosphere. So the T.V. signals are broadcasted by tall antenna to get large coverage, but for transmission over large distance satellites are needed. That is way, satellites are used for long distance T.V. transmission.

21. (e) We know, with increase in altitude, the atmospheric pressure decreases. The high energy particles (*i.e.* γ-rays and cosmic rays) coming from outer space and entering out earth's atmosphere cause ionisation of the atoms of the gases present there. The ionising power of these radiation decreases rapidly as they approach to earth, due to increase in number of collisions with the gas atoms. It is due to this reason the electrical conductivity of earth's atmosphere increase with altitude.

22. (a) In a radar, a beam signal is needed in particular direction which is possible if wavelength of wave is very small. Since the wavelength of microwaves is a few millimeter, hence they are used in radar.

23. (c) Hertz experimentally observed that the production of spark between the detector gap is maximum when it is placed parallel to source gap. This means that the electric vector of radiation produced by the source gap is parallel to the two gaps *i.e.,* in the direction perpendicular to the direction of propagation of the radiation.

24. (d) The atoms of the metallic container are set into forced vibrations by the microwaves. Hence, energy of the microwaves is not efficiently transferred to the metallic container. Hence food in metallic containers cannot be cooked in microwave oven. Normally in microwave oven the energy of waves is transferred to the kinetic energy of the molecules. This raises the temperature of any food.

25. (c) The earth's atmosphere is transparent to visible light and radio waves, but absorbs X-rays. Therefore X-rays telescope cannot be used on earth surface.

26. (b) Short wave (wavelength 30 *km* to 30 *cm*). These waves are used for radio transmission and for general communication purpose to a longer distance from ionosphere.

27. (b) The wavelength of these waves ranges between 4000 Å to 100 Å that is smaller wavelength and higher frequency. They are absorbed by atmosphere and convert oxygen into ozone. They cause skin diseases and they are harmful to eye and cause permanent blindness.

28. (d) Ozone layer in the stratosphere helps in protecting life of organism from ultraviolet radiation on earth. Ozone layer is depleted due to of several factors like use of chlorofluoro carbon (CFC) which is the cause of environmental damages.

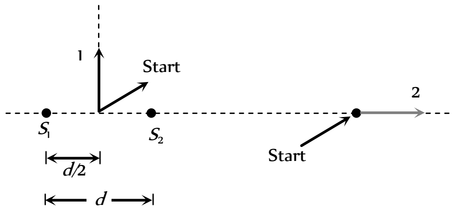
29. (b) Radio waves can be polarised becomes they are transverse in nature. Sound waves in air are longitudinal in nature.

30. (a) In the absence of atmosphere, all the heat will escape from earth's surface which will make earth in hospitably cold.

Wave Optics

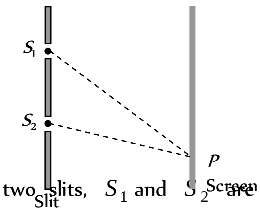
SET Self Evaluation Test - 30

1. Following figure shows sources S_1 and S_2 that emits light of wavelength λ in all directions. The sources are exactly in phase and are separated by a distance equal to 1.5λ. If we start at the indicated start point and travel along path 1 and 2, the interference produce a maxima all along



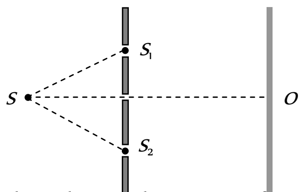
(a) Path 1 (b) Path 2
(c) Any path (d) None of these

2. In a Young's double slit experimental arrangement shown here, if a mica sheet of thickness t and refractive index μ is placed in front of the slit S_1, then the path difference $(S_1P - S_2P)$



(a) Decreases by $(\mu - 1)t$
(b) Increases by $(\mu - 1)t$
(c) Does not change
(d) Increases by μt

3. In the set up shown in Fig the two slits, S_1 and S_2 are not equidistant from the slit S. The central fringe at O is then



(a) Always bright
(b) Always dark
(c) Either dark or bright depending on the position of S
(d) Neither dark nor bright.

4. The intensity ratio of two coherent sources of light is p. They are interfering in some region and produce interference pattern. Then the fringe visibility is

(a) $\frac{1+p}{2\sqrt{p}}$ (b) $\frac{2\sqrt{p}}{1+p}$
(c) $\frac{p}{1+p}$ (d) $\frac{2p}{1+p}$

5. Three waves of equal frequency having amplitudes $10\mu m$, $4\mu m$, $7\mu m$ arrive at a given point with successive phase difference of $\frac{\pi}{2}$, the amplitude of the resulting wave in μm is given by

(a) 4 (b) 5
(c) 6 (d) 7

6. Four different independent waves are represented by

(i) $y_1 = a_1 \sin \omega t$ (ii) $y_2 = a_2 \sin 2\omega t$
(iii) $y_3 = a_3 \cos \omega t$ (iv) $y_4 = a_4 \sin\left(\omega t + \frac{\pi}{3}\right)$

With which two waves interference is possible
(a) In (i) and (iii) (b) In (i) and (iv)
(c) In (iii) and (iv) (d) Insufficient data to predict.

7. A beam of light consisting of two wavelengths 650 nm and 520 nm is used to illuminate the slit of a Young's double slit experiment. Then the order of the bright fringe of the longer wavelength that coincide with a bright fringe of the shorter wavelength at the least distance from the central maximum is
(a) 1 (b) 2
(c) 3 (d) 4

8. Two identical radiators have a separation of $d = \lambda / 4$ where λ is the wavelength of the waves emitted by either source. The initial phase difference between the sources is $\pi / 4$. Then the intensity on the screen at a distant point situated at an angle $\theta = 30°$ from the radiators is (here I_o is intensity at that point due to one radiator alone)
(a) I_o (b) $2I_o$
(c) $3I_o$ (d) $4I_o$

9. In Young's double slit experiment, the $8th$ maximum with wavelength λ_1 is at a distance d_1 from the central maximum and the $6th$ maximum with a wavelength λ_2 is at a distance d_2. Then (d_1 / d_2) is equal to

(a) $\frac{4}{3}\left(\frac{\lambda_2}{\lambda_1}\right)$ (b) $\frac{4}{3}\left(\frac{\lambda_1}{\lambda_2}\right)$
(c) $\frac{3}{4}\left(\frac{\lambda_2}{\lambda_1}\right)$ (d) $\frac{3}{4}\left(\frac{\lambda_1}{\lambda_2}\right)$

10. Light of wavelength $500\,nm$ is used to form interference pattern in Young's double slit experiment. A uniform glass plate of refractive index 1.5 and thickness $0.1\,mm$ is introduced in the path of one of the interfering beams. The number of fringes which will shift the cross wire due to this is
(a) 100 (b) 200
(c) 300 (d) 400

11. The two coherent sources of equal intensity produce maximum intensity of 100 units at a point. If the intensity of one of the sources is reduced by 36% by reducing its width then the intensity of light at the same point will be
(a) 90 (b) 89
(c) 67 (d) 81

12. The path difference between two interfering waves of equal intensities at a point on the screen is $\frac{\lambda}{4}$. The ratio of intensity at this point and that at the central fringe will be
(a) 1 : 1 (b) 1 : 2

(c) 2 : 1 (d) 1 : 4

13. In a Young's double slit experiment, I_o is the intensity at the central maximum and β is the fringe width. The intensity at a point P distant x from the centre will be

(a) $I_o \cos\frac{\pi x}{\beta}$ (b) $4I_o \cos^2\frac{\pi x}{\beta}$

(c) $I_o \cos^2\frac{\pi x}{\beta}$ (d) $\frac{I_o}{4}\cos^2\frac{\pi x}{\beta}$

14. In a Fresnel's diffraction arrangement, the screen is at a distance of 2 meter from a circular aperture. It is found that for light of wavelengths λ_1 and λ_2, the radius of $4th$ zone for λ_1 coincides with the radius of 5 zone for λ_2. Then the ratio $\lambda_1 : \lambda_2$ is

(a) $\sqrt{4/5}$ (b) $\sqrt{5/4}$

(c) $5/4$ (d) $4/5$

15. If n represents the order of a half period zone, the area of this zone is approximately proportional to n^m where m is equal to

(a) Zero (b) Half

(c) One (d) Two

16. A screen is placed $50\,cm$ from a single slit, which is illuminated with $6000\mathring{A}$ light. If distance between the first and third minima in the diffraction pattern is $3\,mm$, the width of the slit is

(a) $0.1\,mm$ (b) $0.2\,mm$

(c) $0.3\,mm$ (d) $0.4\,mm$

17. In Young's double slit experiment, the fringes are displaced by a distance x when a glass plate of one refractive index 1.5 is introduced in the path of one of the beams. When this plate in replaced by another plate of the same thickness, the shift of fringes is $(3/2)x$. The refractive index of the second plate is

(a) 1.75 (b) 1.50

(c) 1.25 (d) 1.00

18. Two waves of equal amplitude and frequency interfere each other. The ratio of intensity when the two waves arrive in phase to that when they arrive $90°$ out of phase is

(a) 1 : 1 (b) $\sqrt{2}$: 1

(c) 2 : 1 (d) 4 : 1

19. In Young's double slit experiment, we get 60 fringes in the field of view of monochromatic light of wavelength $4000\mathring{A}$. If we use monochromatic light of wavelength $6000\mathring{A}$, then the number of fringes obtained in the same field of view is

(a) 60 (b) 90

(c) 40 (d) 1.5

20. A parallel plate capacitor with plate are A and seperation between the plates d, is charged by a constant current i, consider a plane surface of area $A/2$ parallel to the plates and drawn symmetrically between the plates, the displacement current through this area, will be.

(a) i (b) $\frac{i}{2}$

(c) $\frac{i}{4}$ (d) None of these

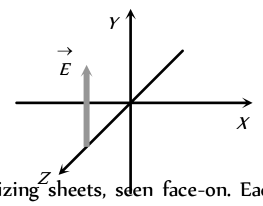
21. The figure here gives the electric field of an EM wave at a certain point and a certain instant. The wave is transporting energy in the negative z direction. What is the direction of the magnetic field of the wave at that point and instant

(a) Towards + X direction

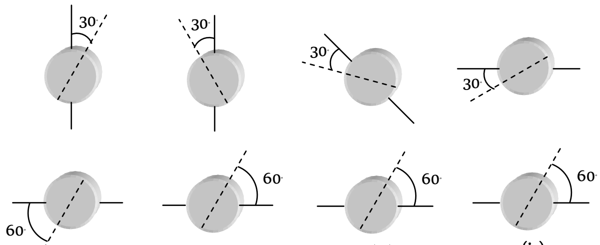
(b) Towards – X direction

(c) Towards + Z direction

(d) Towards – Z direction



22. The figure shows four pairs of polarizing sheets, seen face-on. Each pair is mounted in the path of initially unpolarized light. The polarizing direction of each sheet (indicated by the dashed line) is referenced to either a horizontal x-axis or a vertical y axis. Rank the pair according to the fraction of the initial intensity that they pass, greatest first



(a) (i) > (ii) > (iii) > (iv) (b) (i) > (iv) > (ii) > (iii)

(c) (i) > (iii) > (ii) > (iv) (d) (iv) > (iii) > (ii) > (i)

23. An astronaut floating freely in space decides to use his flash light as a rocket. He shines a 10 *watt* light beam in a fixed direction so that he acquires momentum in the opposite direction. If his mass is 80 *kg*, how long must he need to reach a velocity of 1 *ms*

(a) 9 *sec* (b) 2.4 × 10 *sec*

(c) 2.4 × 10 *sec* (d) 2.4 × 10 *sec*

Answers and Solutions

(SET -30)

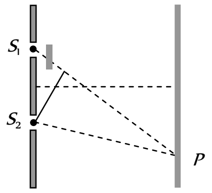
1. (a) At any point along the path 1, path difference between the waves is 0.

Hence maxima is obtained all along the path 1.

At any point along the path 2, path difference is 1.5 λ which is odd multiple of $\frac{\lambda}{2}$, so minima is obtained all along the path 2.

2. (b) Path difference at P $\Delta = (S_1P + (\mu - 1)t) - S_2P$

$= (S_1P - S_2P) + (\mu - 1)t$



3. (c) If path difference $\Delta = (SS_1 + S_1O) - (SS_2 + S_2O) = n\lambda$ n = 0, 1, 2, 3, ... the central fringe at O is a bright fringe and if the path difference $\Delta = \left(n - \frac{1}{2}\right)\lambda$, n = 1, 2, 3, ... the central bright fringe will be a dark fringe.

4. (b) Visibility $V = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{2\sqrt{I_1 I_2}}{(I_1 + I_2)}$

$$= \frac{2\sqrt{I_1 / I_2}}{\left(\frac{I_1}{I_2} + 1\right)} = \frac{2\sqrt{P}}{(P+1)}$$

5. (b) The amplitudes of the waves are

$a_1 = 10\ \mu m$, $a_2 = 4\ \mu m$ and $a_3 = 7\mu m$

and the phase difference between 1st and 2nd wave is $\frac{\pi}{2}$ and that between 2nd and 3rd wave is $\frac{\pi}{2}$. Therefore, phase difference between 1st and 3rd is π. Combining 1st with 3rd, their resultant amplitude is given by

$$A_1^2 = a_1^2 + a_3^2 + 2a_1a_3\cos\phi$$

or $A_1 = \sqrt{10^2 + 7^2 + 2\times 10 \times 7\cos\pi} = \sqrt{100 + 49 - 140}$

$= \sqrt{9} = 3\mu m$ in the direction of first.

Now combining this with 2nd wave we have, the resultant amplitude

$$A^2 = A_1^2 + a_2^2 + 2A_1a_2\cos\frac{\pi}{2}$$

or $A = \sqrt{3^2 + 4^2 + 2\times 3\times 4\cos 90^o} = \sqrt{9+16} = 5\mu m$

6. (d) Since the sources are independent, interference will not occur unless they are coherent (such as laser beams *etc*). So, insufficient data to predict.

7. (d) $n\beta_1 = (n+1)\beta_2$

$$\Rightarrow \frac{n\times 650\times 10^{-19}D}{d} = \frac{(n+1)\times 520\times 10^{-19}\times D}{d}$$

$\Rightarrow n = 4$

8. (b) The intensity at a point on screen is given by

$$I = 4I_0\cos^2(\phi/2)$$

where ϕ is the phase difference. In this problem ϕ arises (i) due to initial phase difference of $\pi/4$ and (ii) due to path difference for the observation point situated at $\theta = 30^o$. Thus

$$\phi = \frac{\pi}{4} + \frac{2\pi}{\lambda}(d\sin\theta) = \frac{\pi}{4} + \frac{2\pi}{\lambda}.\frac{\lambda}{4}(\sin 30°) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$$

Thus $\frac{\phi}{2} = \frac{\pi}{4}$ and $I = 4I_0\cos^2(\pi/4) = 2I_0$

9. (b) Position of n^{th} maxima from central maxima is given by $x_n = \frac{n\lambda D}{d}$

$$\Rightarrow x_n \propto n\lambda \Rightarrow \frac{d_1}{d_2} = \frac{n_1\lambda_1}{n_2\lambda_2} = \frac{8\lambda_1}{6\lambda_2} = \frac{4}{3}\left(\frac{\lambda_1}{\lambda_2}\right)$$

10. (a) The number of fringes shifting is decided by the extra path difference produced by introducing the glass plate. The extra path difference is $(\mu - 1)\ t = n\lambda$

or $(1.5-1)\times 0.1\times 10^{-3} = n\times 500\times 10^{-9}$

$\Rightarrow n = 100$

11. (d) Intensity of each source $= I_0 = \frac{100}{4} = 25\,unit$

If the intensity of one of the source is reduced by 36% then

$I_1 = 25\,unit$ and $I_2 = 25 - 25\times\frac{36}{100} = 16\,(unit)$

Hence resultant intensity at the same point will be

$$I = I_1 + I_2 + 2\sqrt{I_1I_2} = 25 + 16 + 2\sqrt{25\times 16} = 81\,unit$$

12. (b) By using $I = 4I_0\cos^2\left(\frac{\phi}{2}\right) = 4I_0\cos^2\left(\frac{\pi\Delta}{\lambda}\right)$

$$\left\{\because \phi = \frac{2\pi}{\lambda}\Delta\right\}$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{\cos^2\left(\frac{\pi\Delta_1}{\lambda}\right)}{\cos^2\left(\frac{\pi\Delta_2}{\lambda}\right)} = \frac{\cos^2\left(\frac{\pi.\frac{\lambda}{4}}{\lambda}\right)}{\cos^2(0)} = \frac{1}{2}$$

13. (c) Path difference at point $P = \frac{xd}{D}$

Phase difference at point $P = \frac{2\pi}{\lambda}\frac{xd}{D} = \frac{2\pi x}{\beta}$

$I_0 = 4I_1$, intensity at point P

$$I = I_1 + I_1 + 2I_1\cos\frac{2\pi x}{\beta} = 2I_1\left[1 + \cos\frac{2\pi x}{\beta}\right]$$

$$= I_0\cos^2\frac{\pi x}{\beta}$$

14. (c) It is given that $r_4 = \sqrt{4b\lambda_1}$ and $r_5 = \sqrt{5b\lambda_2}$ are equal. Therefore $\sqrt{4b\lambda_1} = \sqrt{5b\lambda_2}$

or $4b\lambda_1 = 5b\lambda_2$

or $\frac{\lambda_1}{\lambda_2} = \frac{5}{4}$.

15. (a) Area of half period zone is independent of order of zone. Therefore, m is equal to zero in n.

16. (b) Position of n^{th} minima $y_n = \frac{n\lambda D}{d}$

$$\Rightarrow (y_3 - y_1) = \frac{\lambda D}{d}(3-1) = \frac{2\lambda D}{d}$$

$$\Rightarrow 3\times 10^{-3} = \frac{2\times 6000\times 10^{-10}\times 0.5}{d}$$

$$\Rightarrow d = 0.2\times 10^{-3} m = 0.2\, mm$$

17. (b) Fringe shift is given by $x = \frac{(\mu - 1)t\,\beta}{\lambda}$

For first plate, $x = \frac{(\mu_1 - 1)t\,\beta}{\lambda}$

For second plate $\frac{3}{2}x = \frac{(\mu_2 - 1)t\beta}{\lambda}$

$$\Rightarrow \left(\frac{\mu_2 - 1}{\mu_1 - 1}\right) = \frac{3}{2} \Rightarrow \left(\frac{\mu_2 - 1}{1.5 - 1}\right) = \frac{3}{2}$$

$$\Rightarrow \mu_2 = 1.75$$

18. (c) Resultant intensity $I = 4I_0\cos^2(\phi/2)$

$$\Rightarrow \frac{I_1}{I_2} = \frac{\cos^2(\phi_1/2)}{\cos^2(\phi_2/2)} = \frac{\cos^2 0}{\cos^2(90/2)} = \frac{2}{1}$$

19. (c) $n_1\lambda_1 = n_2\lambda_2 \Rightarrow 60\times 4000 = n_2\times 6000 \Rightarrow n_2 = 40$

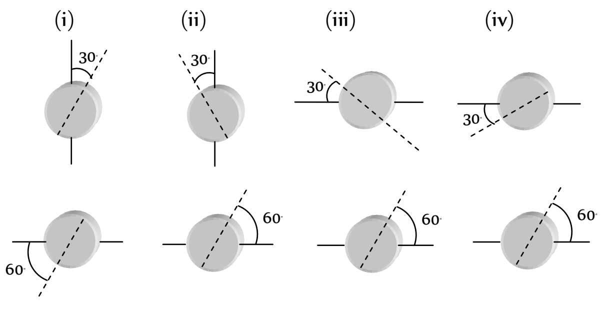
20. (b) Suppose the charge on the capacitor at time t is Q, the electric field between the plates of the capacitor is $E = \frac{Q}{\varepsilon_0 A}$. The flux through the area considered is $\phi_E = \frac{Q}{\varepsilon_0 A}\cdot\frac{A}{2} = \frac{Q}{2\varepsilon_0}$

$\therefore$ The displacement current

$$i_d = \varepsilon_0\frac{d\phi_E}{dt} = \varepsilon_0\left(\frac{1}{2\varepsilon_0}\right)\frac{dQ}{dt} = \frac{i}{2}.$$

21. (a) The direction of EM wave is given by the direction of $\vec{E}\times\vec{B}$.

22. (b) Final intensity of light is given by Brewster's law $I = I_0\cos^2\theta$; where θ = Angle between transmission axes of polariser and analyser.



Hence decreasing order of intensity is (i) > (iv) > (ii) > (iii)

23. (d) Let it take t *sec* for astronaut to acquire a velocity of 1 ms^{-1}. Then energy of photons = 10 t

Momentum $= \frac{10t}{C}$ = 80 × 1

$$t = \frac{80\times 1\times 3\times 10^8}{10} = 2.4\times 10^9\ sec$$

***

Chapter

31

Universe

Universe is the limitless expanse of space around us consisting of solar system. star, galaxies *etc*.

Solar System

Solar system is a family of nine planets, satellites, asteroids, comets, meteors, meteorites and dust particles orbiting around the Sun.

(1) **Planets :** Nine planets revolving around the sun in elliptical orbits. In order of increasing distance from Sun, these are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune and Pluto.

(i) The gravitational pull of the Sun on the planets control their motion.

(ii) There are other heavenly bodies (about 32) which revolve around the planets called satellites (or moons) of the planets.

(iii) A planet does not emit light of its own.

(iv) A planet do not twinkle at night.

(v) The planets are very small in size as compared to stars or Sun.

(vi) The relative positions of planets keep on changing day by day.

(vii) Most of the planets move around the sun from west to east.

(viii) The planets are made of rocks and metals.

(ix) The temperature of planet depends upon its distance from sun.

(2) **Asteroids :** The small pieces of planet revolving around the sun between orbits of Mars and Jupiter are called Asteroids.

(i) Astronomers have identified about 2000 asteroids ranging from the largest 770 *km* diameter to bodies 1.5 *km* in diameter.

Table 31.1 : Some information about planets

Planet	Radius $R \times 10^3$ *km*	Mean distance from sun $\times 10^6$ *km*	Mass as compared to earth	Time of revolution around the sun	Time taken to complete one rotation around its own axis	Number of satellites
Mercury	2.4	57.9	0.055	88 days	59 days	—
Venus	6.1	108.2	0.815	225 days	243 days	—
Earth	6.3	149.6	1	1 Year	23 hrs. 56 min.	1
Mars	3.4	227.9	0.108	1.9 Year	24 hrs. 27 min	2
Jupiter	71.4	778.3	317.9	11.8 Year	9 hrs. 50 min	14
Saturn	60.0	1427	95.2	29.5 Year	10 hrs. 14 min	10 + Ring
Uranus	23.4	2870	14.6	85 Year	10 hrs. 49 min	5 + Ring
Neptune	22.3	4594	17.2	165 Year	15 hrs.	2
Pluto	3.2	5900	0.002	248 Year	6.39 days	—

(ii) The largest asteroids are called Ceres.

(iii) The largest asteroid complete one revolution around the sun in 4.6 years.

(3) **Comets :** These are composed of rock like materials surrounded by large masses of easily vaporisable substances like, ice, water, ammonia and methane.

(i) They revolve around the Sun in highly elliptical orbits.

(ii) Their time period of revolution around the Sun is very large.

(iii) Comets appear to be having a bright head and a long tail while passing close to the Sun and when away from sun generally they show no tail.

(iv) The tail of comet is formed when the comet is passing close to the Sun and the heat of Sun exerts a pressure on the material which gets evaporated due to heat of Sun.

(v) Hally comet was seen in early 1986 and is expected to be seen again in 2062.

(4) **Meteors and meteorites :** Meteors are the smaller pieces of stones and metals which may be produced due to the breaking up of comets while approaching the Sun. When they reach earth's atmosphere due to friction they start buring. They are also called shooting stars.

Sometimes, the large pieces of stones (acting as meteors) do not burn completely and reach the surface of the earth as stony, iron balls resulting in crators on the earth surface. These are called meteorites.

Measurement of Size of Planet

We can measure the size of a planet by measuring the angle subtended by its diameter AB at a point on the earth. This angle is called angular diameter of planet. If d denotes diameter of planet and D its distance from the earth

$$\alpha \simeq \frac{d}{D}$$

or $\quad d \simeq D\alpha$

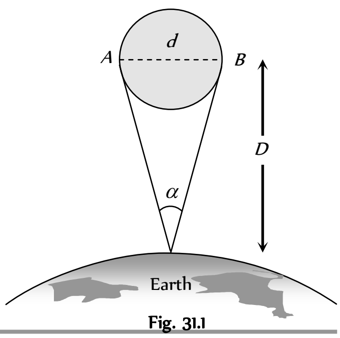


Fig. 31.1

Measurement of Distance of Planet From the Earth

(1) **Parallax method :** The planet O is observed from two points P_1 and P_2 on the surface of the earth. The distance between these two points, $P_1P_2 = b$, is called basis. The angle subtended by planet at these two points is called parallax angle or parallactic angle

From figure $\theta \simeq \dfrac{b}{D}$

or $\quad D \simeq \dfrac{b}{\theta}$

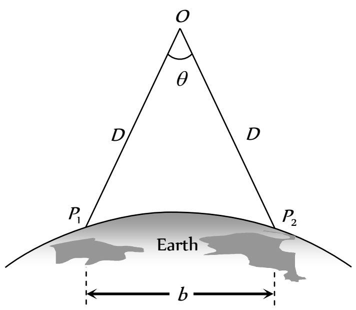


Fig. 31.2

(2) **Copernicus method :** The inferior planets (Mercury and Venus) have nearly circular orbits. Angle between directions of observation from earth to sun and earth to planet is called planet's elongation.

r_{es} = The distance of earth from Sun, r_{ps} = The distance of planet from Sun and r_{pe} = The distance of planet from Earth

The r_{ps} and r_{es} are fixed distances as orbits have been assumed to be circular. During orbital motion of the planet the distance r_{pe} changes. Planet's elongation is maximum when earth and sun subtend an angle 90° at the planet. From figure,

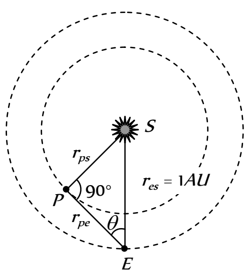


Fig. 31.3

$$\sin\theta = \frac{PS}{ES} = \frac{r_{ps}}{1\,AU}$$

where 1 AU = 1.496 × 10^{11} m

Thus $\sin\theta = r_{ps}$, similarly $\cos\theta = \dfrac{PE}{ES} = \dfrac{r_{pe}}{1\,AU} \Rightarrow r_{pe} = \cos\theta$

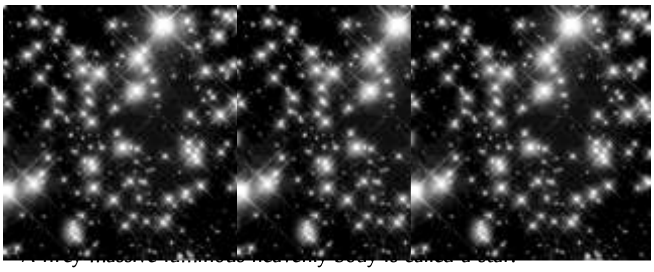
(3) **Kepler's law :** According to Kepler's law the square of time period of planet around sun is proportional to cube of semi-major axis of the orbit of planet around sun *i.e.* $\dfrac{a^3}{T^2} = \text{constant}$, if a_1 and a_2 are semi-major axes of planets 1 and 2 and T_1 and T_2 their respective periods of revolution, then

$$\frac{a_1^3}{T_1^2} = \frac{a_2^3}{T_2^2} \quad \text{or} \quad a_2 = \left(\frac{T_2}{T_1}\right)^{2/3} a_1$$

For circular orbits a_1 and a_2 represent the radii of orbits.

(4) **Spectroscopic method :** In this method, photograph of two different planets P_1 and P_2 are taken on similar photographic plates from one place of the earth. Let I_1 and I_2 be the intensities of the images of these two planets. If R_1 and R_2 be the distances of these planets from the earth then $\dfrac{I_1}{I_2} = \dfrac{R_2^2}{R_1^2}$ ($\because$ intensity at a point is inversely proportional to the square of the distance)

Stars



A very massive luminous heavenly body is called a star.

(1) **Some features**

(i) Stars twinkle at night.

(ii) Stars are countless in number ; about 10^{22} in a universe.

(iii) Stars are very big in size but they appear small because they are very far off.

(iv) The relative positions of stars do not change day by day.

(v) Stars appear to be moving from west to east.

(vi) The temperature of stars is very high.

(vii) The Sun is the nearest star to the earth. Its light reaches the earth in 8.3 minutes.

(viii) After Sun the next nearest star to earth is Alpha centuri. Its distance is 4.3 light year from earth.

(ix) Other bright stars in the sky are known as Spica (Chitra), Arcturus (Swati), Polaris (Dhruva), Sirius (Vydha), Canopus (Agasti) *etc.*

(x) The temperature of a star is estimated from the colour of its light received on earth. The blue coloured star is at higher temperature than red coloured star.

(2) **Constellation :** Many of the stars appear to be bunched together in groups. These groups are called constellations.

(i) The Great-bear (Saptarishi), Taurus (Vrishabha) Aries (Mesha) *etc.* are the other constellations near the north and south celestial poles.

(ii) According to modern astronomy there are 88 constellations in the sky.

(3) **Brightness of star :** Brightness of stars is represented through system of magnitudes. Magnitude of star is the measure of its brightness when observed from earth.

(i) Hipporacus, a Greek astronomer divided the stars (visible with naked eye) into six magnitude classes. Brightness goes on decreasing as the magnitude increases. A first magnitude star is about 100 times as bright as a sixth magnitude star. Decrease in magnitude number by one increases brightness by ratio $100^{1/5} = 2.5119.$ In general

$$\frac{\text{The brightness of star in } n^{\text{th}} \text{ magnitude class}}{\text{The brightness of star in } (n+m)^{\text{th}} \text{ magnitude class}} = (2.512)^m$$

(ii) If two stars have magnitudes m_1 and m_2 ($m_2 > m_1$) and brightness l_1 and l_2 ($l_2 < l_1$), then $\frac{l_1}{l_2} = 100^{(m_2 - m_1)/5}$

Taking logarithm to base 10 of both sides, we get

$$(m_2 - m_1) = -2.5 \log \frac{l_2}{l_1}$$

(iii) For a star of zero magnitude $m_1 = 0$, $l_1 = l_0$, $m_2 = m$ and $l_2 = l$.

$$\Rightarrow m = 2.5 \log \frac{l_0}{l}$$

(iv) The star vega is of zero magnitude and of brightness $l_0 = 2.52 \times 10^{-8} W/m^2$.

(v) A star having negative magnitude is brighter; *e.g.*, a star having magnitude – 5 will be 100 times more bright than a star of zero magnitude.

(4) **Absolute luminosity :** The total energy radiated into space per second from the surface of a star is called absolute luminosity of the star. The absolute luminosity of the Sun is $\approx 3.9 \times 10^{26} J/sec$.

(5) **Birth of a star :** Star dust and gases present in interstellar space come closer together with a gravitational force in the form of a cloud.

(i) When the cloud is quite big, due to compression cloud heats up and starts radiating

(ii) At this temperature, fusion of hydrogen atom into helium atom takes place and a star is said to have come into existence.

(iii) This process result in the release of energy, which keeps the star shining for millions of years.

(6) **Death of a star :** When large number of hydrogen atoms of a star are converted into *He*, the core of star begins to contract and other layers begin to expand. At this stage star appears red, the stage is called Red Giant.

(i) Now a violet explosion occurs in star. This is called nova or super nova explosion.

(ii) Due to explosion, the outer layers are thrown back into interstellar leaving behind the core of the star. This is known as **death of the star**.

The core of the star may further end up into one of the following three dead bodies (stellar dead materials) :

(a) White dwarf (b) Neutron star (c) Black hole

(a) **White dwarf :** When the original mass of the star is less than $2M_s$ (M_s being solar mass), the core of the star tends to die as White dwarf. It was theoretically discovered by S. Chandrasekhar in1930 and is known as **Chandrasekhar limit**. As the core keeps on emitting heat and light for millions of year, it colour changes from white to yellow, then to red finally it becomes black. Now this becomes invisible for ever.

(c) **Neutron star :** When the original mass of the star is lies between $2M_s$ *and* $5\,M_s$ the core of the star tends to finish up as neutron star. In such a case, when super nova explosion occurs, the core of the star is compressed and electrons and protons combine to form neutrons. Due to this reason, this is called as neutron star. It is found to have a radius of about 10 *km*.

(7) **Black hole :** When the original mass of the star is more than $5M_s$, then on supernova explosion, the core continues suffering compression indefinitely due to recoil. This gives rise to a black hole. The mass of the black hole is greater than the mass of the Sun but its size is very small. Due to this fact, the gravitational pull of black hole is very strong. This is the reason that the photon of radiation emitted by it cannot escape from its surface. On the other hand, a photon approaching a black hole is swallowed by it. Hence it is called a black hole.

The black hole is said to have been formed if the star of mass *M* has contracted within a radius *r* which is given by $r \le \frac{2GM}{c^2}$

Sun

The Sun called the centre of the solar system, is a star nearest to the earth.

(1) **Properties of the Sun**

(i) It's average distance from earth is 1.49×10^{8} *km* = 1 *AU*

(ii) It's mass is 1.99×10^{30} *kg*

(iii) It's mean diameter is 1.392×10^{6} *km*

(iv) The density of the Sun varies from 10^{-4} kg/m^3 at the surface to 10^{4} kg/m^3 at the centre. It's mean density is 1410 $kg\ m^{-3}$.

(v) The pressure at the centre of the Sun is about 2×10^{6} N/m^2.

(vi) Light takes 8 minute to reach earth from the Sun.

(vii) 70% of Sun's mass is H_2, 28% *He* and 2% Lithium or Uranium.

(viii) The Sun is also called Yellow Dwarf.

(2) **Structure of Sun :** The Sun structure consist of four parts : Photosphere (*P*), Reversing layer (*R*), Chromosphere (*CH*) and Corona (*C*).

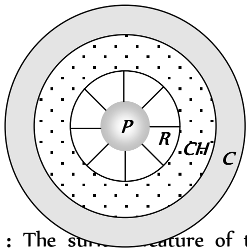


Fig. 36.4

(3) **Solar activity :** The surface feature of the Sun are called Solar activity. This can be classified as follows

(i) **Sun spots :** These are dark spots on the surface of sun associated with strong magnetic fields. The sun spots move across Sun slowly, so there numbers vary over a cycle of 11 year called Sun spot cycle. After every eleven year activity of sun spots tends to be maximum. Movements of sun spots have revealed the time period of rotation of sun on its own axis as about 25 days.

(ii) **Faculae :** These are bright patches near Sun spots.

(iii) **Granules :** Small granules form a covering over photosphere.

(iv) **Flares :** Sudden increase in magnetic activity is called flare. During these flares Sun emits streams of protons, α-particles and electrons.

(v) **Spicules :** Bright spikes emerging from chromosphere are termed spicules. Spicules are source of large number of charged particles into the corona.

(vi) **Prominences :** Surface of photosphere is covered by rising clouds called prominences.

(vii) **Filaments :** These are thin markings on the photosphere.

(4) **Solar constant (*S*) :** Energy falling in one second on the unit area of the earth's surface held normal to Sun's rays is called solar constant. It is given by $S = \dfrac{\sigma T^4 R^2}{r^2} = 1.388 \times 10^3 W/m^2$

where σ = Stefan's constant

= 5.68×10^{-8} S.I. unit

T = Surface temperature of Sun

R = Radius of Sun

r = Radius of Earth's orbit

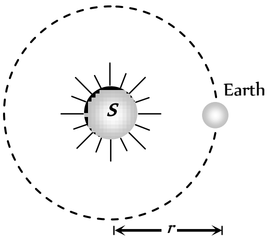


Fig. 31.5

(5) **Solar Luminosity (*L*) :** It is defined as the amount of energy emitted by the Sun per second in all directions.

$$L_s = (4\pi r^2)S = 3.9 \times 10^{26} W$$

(6) **Temperature of Sun (*T*) :** The surface temperature of the Sun is given by $T = \left(\dfrac{r}{R}\right)^{1/2} \left(\dfrac{S}{\sigma}\right)^{1/4}$

(7) **Mass of the Sun (*M*) :** Let M be the mass of sun and m be the mass of a planet moving around it, then as gravitational force of attraction between them supplies the necessary centripetal force

$$F = \frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow \text{Mass of Sun } M = \frac{v^2 r}{G}$$

$$= \frac{r^2\omega^2 r}{G} = \frac{r^3\omega^2}{G} = \frac{r^3\left(\frac{2\pi}{T}\right)^2}{G} = \frac{4\pi^2 r^3}{GT^2}$$

where $G = 6.67 \times 10^{-11}$ $Nm^2\,kg^{-2}$ and r is distance between the sun and planet. T period of revolution of planet around the sun.

If we consider the planet and its satellite, mass of the planet can similarly be found

Stellar Radii, Mass and Spectra

(1) **Stellar radii :** The total energy radiated by the star per second is given by $E = \sigma T^4 \times$ Surface area of the star

$$\Rightarrow E = \sigma T^4 \times 4\pi R^2 \Rightarrow \text{Radius of star } (R) = \left(\frac{E}{4\pi\sigma}\right)^{1/2} T^2$$

Usually, the radius of star is expressed in terms of solar radius $(R_s = 6.95 \times 10^8 m)$. Thus star radius $= \left(\dfrac{E}{4\pi\sigma}\right)^{1/2} \dfrac{T^2}{6.95 \times 10^8}$ solar radius.

$\Rightarrow$ The radii of most of the stars lie in the range 0.02 to 220 solar radii.

(2) **Stellar masses :** Let M_1 and M_2 be the masses of two stars revolving about their common centre of mass in circular orbits of radii r_1 and r_2 respectively such that $r_1 + r_2 = r$. Now

$$M_1 + M_2 = \frac{4\pi^2}{G} \times \frac{r^3}{T^2} \qquad \text{......(i)}$$

where T is common period of revolution.

If a planet of mass M_1 moves round the Sun of mass M_S, then the mass M_1 can be neglected in comparison with M_S because $M_S >> M_1$. Then equation (i) can be written as

$$M_S = \frac{4\pi^2}{G} \times \frac{r^3}{T^2} \qquad \text{.....(ii)}$$

As M_S is constant, it implies that $\dfrac{r^3}{T^2}$ = constant

which is Kepler's third law.

In binary system, r = 1 AU, T = 1 year and $M_1 + M_2 = 1$ solar mass.

Hence equation (i) gives $G = 4\pi^2$

$$\therefore \quad M_1 + M_2 = \frac{r^3}{T^2} \qquad \text{..... (iii)}$$

Equation (iii) can be used to find the masses of two stars in binary system.

(3) **Spectra of stars :** The different stars are of different colours and the spectrum of a star is related to its colour, There are seven classes of stellar spectra denoted by letters *O*, *B*, *A*, *F*, *G*, *K* and *M*. Our sun belongs to *G* class star.

Table 31.2 : Spectrum of stars

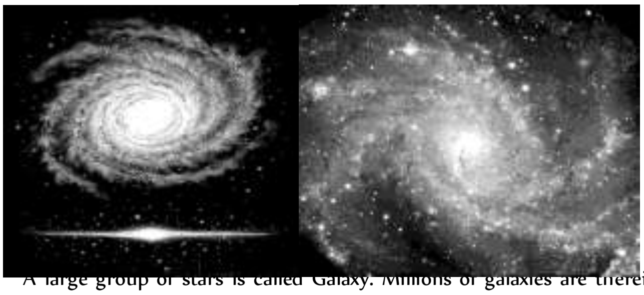
Spectra type	Colour	Surface temp (*K*)	Description of absorption spectra
O	Dark blue	3×10^4 to 4×10^4	Ionized helium lines
B	Blue	$1.s5 \times 10^4$ to 2.3×10^4	Lines of neutral helium
A	White	9.5×10^3 to 1.1×10^4	Lines of H_2
F	Green	6.5×10^3 to 7.5×10^3	Lines of H_2 and ionised metals
G	Yellow	5800	Lines of ionised *Ca*, *Fe*, *C*
K	Orange	4500	Bands due to hydrocarbons
M	Red	3500	Bands of Titanium oxide

These relationship between the colour of a star and its temperature is expressed by Wien's displacement law. According to this law

$$\lambda_m \propto \frac{1}{T} \quad \text{or} \quad \lambda_m T = b \quad \text{or} \quad T = \frac{b}{\lambda_m}$$; where $b = 2.89 \times 10^{-3}\ mK$.

So those stars which appear blue (minimum wavelength) such as class *O* and *B*, are very hot and which appear red (maximum wavelength) such as class *M* are less hot.

Galaxies



A large group of stars is called Galaxy. Millions of galaxies are therein the sky. Each galaxy contains about 10^{11} stars.

The Sun and the planets of the solar system belong to the galaxy, called Milky way (Akash Ganga).

(1) **Types of galaxies :** There are two types of galaxies

(i) Normal galaxies, and (ii) Radio-galaxies.

(i) **Normal galaxies :** Besides milky way, there are billions of other galaxies in the universe. All these galaxies are called normal galaxies. There are three types of normal galaxies. (a) Elliptical galaxies (18%), (ii) Spiral galaxies (80%), and (iii) Irregular galaxies (2%).

(a) **Elliptical galaxies :** The galaxies which look like the flat elliptical discs are called elliptical galaxies. These generally consist of red giants, white dwarfs *etc. i.e.*, those stars which are nearing their ends.

(b) **Spiral galaxies :** The galaxies have lens-shaped central portion surrounded by a flat disc. It has two spiral arms which spiral around the central portion.

Example : Milky way and Andromeda.

(c) **Irregular galaxies :** These have no specific form of their own. Irregular galaxies are youngest normal galaxies and are middle aged and elliptical galaxies are quite old galaxies.

(2) **Radio galaxies :** The galaxies which emit electromagnetic radiations in the radio frequency are called radio galaxies. These have been classified as (i) Ordinary radio galaxies (ii) Quasars.

(i) **Ordinary radio galaxies :** A normal optical galaxy (O) which has two strong radio sources (R_1 and R_2) occurring symmetrically on either side of it, is called an ordinary radio galaxy. It appears like two ears on the two sides of the face of a person. The radio power output lies in the range 10^{30} to 10^{38} *watt*.

(ii) **Quasars :** Quasars are quasi-stellar radio sources. They are star like in structure and they emit powerful radio waves. They have a radio output of 10^{37} to 10^{38} *watt*. Quasars are farthest objects known. They are millions of light years away from Earth. These seem to be lying at the limit of the universe. They are moving away from Earth with a velocity of about 0.9 times the velocity of light. Their size is much smaller. It is of the order of light days. They form very dense galaxies. The density is also very large and their gravitational field is also very high. The cause of tremendous energy of the quasars is unknown. About 150 quasars have been detected so far.

(3) **Milky way (Akash Ganga) :** It is the name of the galaxy to which our earth belongs. The milky way is the glowing belt of the sky formed by the combined light of a very large number of stars. It is called milky way or Akash Ganga because the light from the various stars together gives the impression of a stream of milk flowing across the sky.

Milky way is a spiral galaxy. Its mass is 150 solar masses (*i.e.* 3×10^{41} *kg*).

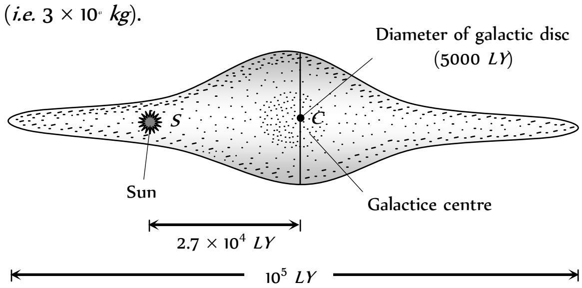


Fig. 31.6

Milky way contains 150 billions sun like stars.

Milky way contains clouds of dust and gases.

Pulsars

As the age of a star increases, its hydrogen content goes on decreasing. Ultimately, the star explodes as a supernova, in the universe. After explosion of a supernova, a variable star is born. It is not an ordinary star. It is the remaining part of a supernova. The variable star is called a pulsar. A pulsar emits electromagnetic waves in pulses and not continuously. The pulses are of very short duration (0.033 *s* to 0.088 *s*). The pulses may lie either in visible region or in radio region. About 50 pulsars have been detected, two in visible region and others in radio region. It is expected that there are about 100 pulsars in the universe.

Evolution of the Universe

Important theories about the origin and evolution of universe are as follows.

(1) **Big Bang theory :** The whole of the matter of the universe was concentrated in a very dense and hot fire ball about 20 billion years ago. An explosion occurred. The matter was broken into pieces in the form of stars and galaxies. The faster moving galaxies have gone farther than the slower ones. A galaxy situated at 20 billion light years is the boundary of the universe.

(2) **Expanding universe theory :** All the galaxies would continue to move away from the Earth and we will have an empty universe because on account of continuous expansion of the universe, more and more galaxies will go beyond the boundary of the universe and will be lost.

The motions of galaxies relative to the earth can be measured by observing the shifts in the wavelengths of their spectra. For distant galaxies these shifts are always toward longer wavelength, so they appear to be receding from us and from each other. Astronomers first assumed that these were Doppler shifts and used a relation between the wavelength λ_0 of light measured now from a source receding at speed v and the wavelength λ_s measured in the rest frame of the source when it was emitted.

$$\lambda_0 = \lambda_S \sqrt{\frac{c+v}{c-v}}$$

For $v << c$, Red shift (or Doppler's shift) $\Delta\lambda = \lambda_S \dfrac{v}{c}$

(3) **Pulsating universe theory :** As the galaxies move away, the expansion of the galaxies would be stopped by the gravitational pull. The galaxies would come so close that again a new explosion would take place. The same sequence will be repeated. Thus, we have alternate expansion and contraction of the universe giving rise to a pulsating universe. This takes place after every 80 billion years.

(4) **Steady state theory :** As the farthest galaxies speed away from each other, new galaxies are born to take their places. The total number of galaxies in the universe remains constant.

It is certain that : (i) The age of the universe is about 20 to 30 billion years. (ii) The most distant galaxy is situated at a distance of two billion light years away from the Earth. (iii) This galaxy is receding away from the Earth with a velocity 0.3 times that of light. (iv) The universe will live for about 100 million years more. Thus, the universe is quite young at present.

Hubble's law

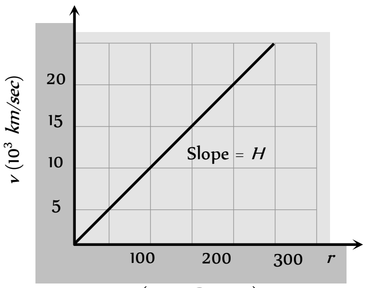
(1) The speed of recession v of a galaxy is proportional to it's distance r from us *i.e.* $v \propto r \Rightarrow v = Hr$ this relation is called Hubble's law.

(2) Here H = An experimental quantity, called Hubble's constant. It's value is 19.3 *mm/sec* for each light year.

(3) Determining H has been a key goal of the Hubble's space telescope.

(4) The quantity $\dfrac{1}{H}$ has the dimensions of time.

(5) This time is called Hubble's times, which is an estimate of the order of magnitude of time that has elapsed since the Big Bange, and thus of the age of universe.



(Mega Parsecs)

Fig. 31.7

Tips & Tricks

✍ The name black hole is given because it's gravity is so high that it prevents even light to radiate into space.

✍ Visible light is restricted from entering a telescope by dust particles in universe. Therefore range of a telescope is limited. Observation made in visible range are referred to as optical astronomy. Whereas observations made in radio range is called Radio-Astronomy.

✍ **Albedo :** The presence of atmosphere, clouds, *etc.* is acknowledged by a parameter known as albedo. It is the ratio of energy reflected by a planet to that incident on it. Clouds being good reflectors of light, they considerably increase the reflecting power of the planet and hence it's albedo is large. Venus has an albedo of 85% (highest).

✍ Mercury, Pluto and Venus do not have any satellites.

✍ On a clear night 5000 stars can be observed with naked eye.

✍ Closet star is alpha centuri (after the Sun) which is 4.3 light years away.

✍ Astronomy is branch of science which deals with the study of universe.

✍ Study of heavenly bodies is based upon visible light (λ ranging from 4000 Å to 8000 Å) and radio waves (λ ranging from 1 *mm* to 20 *m*).

✍ Hipparchus, a Greek astronomer, divided naked eye stars into six magnitude classes, on the basis of their brightness. The brightest stars were placed in the first magnitude class. Faintest visible stars were put in the sixth magnitude class.

✍ A comet does not have any tail when it is far from the Sun.

✍ **Mercury**

(i) Smallest planet (ii) Closest to the Sun

(iii) Fastest (iv) No atmosphere.

✍ Cygnus is a group of five stars. Which forms a cross like a swan.

✍ The clouds of dusty gas are called nebulae.

Ordinary Thinking

Objective Questions

Universe

1. A study of binary stars is most helpful in **[CBSE PMT 1993]**

(a) Finding their distances

(b) Finding their temperature

(c) Finding their masses

(d) Verifying Newton's force law of gravitation

2. A group of bright and faint stars is called **[AFMC 1994]**

(a) Galaxy (b) Comet

(c) Black hole (d) Constellation

3. According to modern astronomers into how many constellations, the whole sky is divided **[BHU 1994]**

(a) 10 (b) 88

(c) 880 (d) 5000

4. Which of the following theories is the most satisfactory about the origin of the universe **[CBSE PMT 1994]**

(a) Big Bang theory (b) Pulsating theory

(c) Steady state theory (d) None of these

5. Which of the planet is brightest **[BHU 1999]**

(a) Mercury (b) Venus

(c) Mars (d) Jupiter

6. A star which appears blue will be **[CPMT 1998]**

(a) As hot as the sun (b) Cooler than the sun

(c) Very cold indeed (d) Much hotter than the sun

7. Hubble showed that the universe as a whole is expanding and the distant stars are receding from us. The spectral line from a star, when compared with the corresponding line from an source will then show **[Haryana CEE 1996]**

(a) A shift in frequency towards the red end

(b) A shift in frequency towards the violet end

(c) No shift in frequency at all

(d) A shift in frequency towards the violet end as well as a decrease in intensity

8. The solar constant on the surface of the earth is *S*. What will be its value on the surface of another planet which is about 5.3 *A.U.* away from sun [AMU 1996, 97]

(a) $\frac{S}{5.3}$ (b) $\frac{S}{(5.3)^2}$

(c) 5.3 *S* (d) $(5.3)^2 S$

9. CO_2 gas is found in which of the following pairs of the planet [AFMC 1994]

(a) Earth and Mercury (b) Mercury and Saturn

(c) Venus and Saturn (d) Venus and Mars

10. The wavelength of maximum energy, released during an atomic explosion, was 2.93×10^{-10} *m*. Given that the Wien's constant is 2.93×10^{-3} *m K*, the maximum temperature attained must be of the order of [Haryana CEE 1996]

(a) 10^{-7} *K* (b) 10^{7} *K*

(c) 10^{-13} *K* (d) 5.86×10^{7} *K*

11. Black hole is a [BHU 1995; MH CET 2003]

(a) Hole in the ozone layer of atmosphere

(b) Hole in earth's centre

(c) Highly dense matter available in the atmosphere

(d) Hole in troposphere

12. A planet of mass M has a satellite of mass m, revolving around the planet in a circular orbit of radius r and time period T. The mass (M) of the planet is [AMU 2000]

(a) $\frac{4\pi^2 r^3}{GT^2}$ (b) $\frac{4\pi^2 r^2}{GT^3}$

(c) $\frac{GT^2}{4\pi r^3}$ (d) $\frac{r^3 G}{4\pi T^2}$

13. The age of universe is believed to be [NTSE 1995]

(a) 1 billion years (b) 10 billion years

(c) 10-20 billion years (d) 1000 billion years

14. A planet which is born sister of earth is [AFMC 2000]

(a) Mercury (b) Venus

(c) Mars (d) Jupiter

15. Source of Sun's energy is [CBSE PMT 1992; KCET 1994; AFMC 1998; BHU 2000; DCE 2001]

(a) Burning of hydrogen

(b) Fission reactions involving hydrogen

(c) Fusion reactions involving hydrogen

(d) Some other source

16. Asteroids are [DPMT 2000]

(a) Small planets

(b) Shooting stars

(c) Found in a belt between Earth ad Venus

(d) None of these

17. Sun radiates continuously and maintains its brightness because [MP PMT 1990; JIPMER 1997]

(a) Helium is converted into iron in its core

(b) Of fusion of hydrogen nuclei into helium

(c) Fusion of helium in hydrogen

(d) Burning of carbon, in its core

18. Venus appears brighter than other stars because [MP PMT 1990]

(a) It is heavier than other planets

(b) Its density is more than other planets

(c) It is nearer to earth in comparison to other planets

(d) Nuclear fusion takes place at its surface

19. There is no atmosphere on moon because [MP PMT 1990]

(a) There is no vegetation

(b) The escape velocity at its surface is very low

(c) Diffusion constant of gases is high

(d) There is vacuum in space

20. Which of the following planets have rings around it [MP PMT 1991]

(a) Uranus (b) Mars

(c) Jupiter (d) Saturn

21. Milky way is [MP PMT 1991; Kerala PMT 2001]

(a) A planet of our system

(b) A sun

(c) One of the solar system

(d) One of the enormous galaxies of universe

22. Hubble's law states that the velocity with which the 'milky way' is moving away from the earth is proportional to [MP PMT 1991; Kerala PMT 2004]

(a) Square of the distance of the milky way from the earth

(b) Distance of milky way from the earth

(c) Mass of the milky way

(d) Product of the mass of the milky way and its distance from the earth

23. The hottest planet of solar system is [CBSE PMT 1992]

(a) Mars (b) Mercury

(c) Venus (d) Pluto

24. Towards the centre of sun [MP PMT 1992]

(a) Density decreases

(b) Pressure decreases

(c) Temperature decreases

(d) Density and pressure increases

25. Period of revolution increases in the order of [MP PMT 1992]

(a) Saturn, Uranus, Venus (b) Mars, Saturn, Pluto

(c) Mercury, Neptune, Mars (d) Mars, Jupiter, Venus

26. The length of Milky way is [MP PMT 1992]

(a) 100,000 light years (b) 10,000 light years

(c) 1000 light years (d) 100 light years

27. Which of the nine planets is nearest to sun [CBSE PMT 1992]

(a) Venus (b) Mercury

(c) Mars (d) Jupiter

28. An extremely hot star would appear to be [AMU 1996, 97]

(a) Red (b) Blue

(c) Yellow (d) Orange

29. The sun emits a light with maximum wavelength 510 mm while another star X emits a light with maximum wavelength of 350 nm. What is the ratio of surface temperature of sun and the star X

(a) 2.1 (b) 0.68

(c) 0.46 (d) 1.45

30. A double star is a system of two stars rotating about their centre of mass only under their mutual gravitational attraction. Let the star have mass m and $2m$ and their separation be l. Their time period of rotation about their centre of mass will be proportional to [JIPMER 2000]

(a) $l^{2/3}$ (b) l

(c) $m^{1/2}$ (d) $m^{-1/2}$

31. Hubble's law is related with [AIIMS 2002; Pb. PET 2002]

(a) Comet (b) Speed of galaxy

(c) Black hole (d) Planetary motion

32. 'Albedo' is [Pb. PET 2001; BHU 2001; Kerala PET 2002; AFMC 2002]

(a) Reflecting power of a heavenly body

(b) Transmitive power of a heavenly body

(c) Absorptive power of a heavenly body

(d) Refracting power of a heavenly body

33. According to the pulsating theory the expansion and contraction of the universe repeats after every [TNPCEE 2002]

(a) 11 years (b) 8 billion years

(c) 8 million years (d) 80 billion years

34. Meteors are [TNPCEE 2002]

(a) Small stars

(b) Burnt pieces of comets that fall on earth

(c) Comets without tails

(d) None of these

35. Which of the following helps us in the determination of the temperature of sun [CBSE PMT 2001]

(a) Kirchhoff's law (b) Maxwell Boltzmann law

(c) Planck's law (d) Stefan's law

36. How does the red shift confirms that the universe is expanding

(a) Due to Wien's law (b) Due to Stefan's law

(c) Due to Kirchhoff's law (d) Due to Doppler's effect

37. Two stars P and Q are observed at night. Star P appears reddish while, star Q is white. From this we conclude [Roorkee 1992]

(a) Temperature of Q is higher than that of P

(b) Temperature of Q is lower than that of P

(c) Star Q is at the same distance at that of star P

(d) Star P is farther than star Q

38. Albedo is maximum for [Pb. PET 2000]

(a) Pluto (b) Venus

(c) Earth (d) Mercury

39. When original mass of star is greater than 5 M (M = mass of the sun). The death of this star will give rise to [Pb. PET 2000]

(a) White dwarf (b) Black hole

(c) Quasars (d) Nebula

40. The tail of the comet is due to [Pb. PET 2002]

(a) Vaporisation of water on the comet

(b) Sublimation of vapour in the comet

(c) Cooling of water in the comet

(d) Vaporisation of heat in the comet

41. In our solar system, there is one sun and [BHU 2004]

(a) Seven planets

(b) Nine planets

(c) Eleven planets

(d) Indefinite number of planets

42. Which one of the following planet has the longest day [AFMC 2003]

(a) Venus (b) Mars

(c) Mercury (d) Earth

43. Which one of the following is known as Saptarishi [AFMC 2003]

(a) Orion (b) Ursa major

(c) Ursa minor (d) Scorpion

44. Smaller pieces of heavy stones and metals which on entering earth's atmosphere burns out are [AFMC 2003]

(a) Comets (b) Meteorites

(c) Asteroids (d) All of these

45. In determining the temperature of a distant star, one makes use of

(a) Kirchhoff's law (b) Stefan's law

(c) Wien's displacement law (d) None of these

46. The motion of planets in the solar system is an example of conservation of [DCE 2001, 03]

(a) Mass (b) Momentum

(c) Angular momentum (d) Kinetic energy

47. Mass of earth has been determined through [Kerala (Engg.) 2002]

(a) Use of Kepler's T/R constancy law

(b) Sampling the density of earth's crust and using R

(c) Cavendish's determination of G and using R and 'g' at the surface

[Pb. PMT 1997; AIIMS 2001]

(d) Use of periods of satellites at different heights above earth's surface

48. The galaxies are moving away from each other. It is explained by

(a) White dwarf star (b) Red shift

(c) Neutron star (d) None of these

49. Speed of recession of galaxy is proportional to it's distance [DCE 1999]

(a) Directly (b) Inversely

(c) Exponentially (d) None of these

50. Great bear is a [DCE 1998]

(a) Star (b) Galaxy

(c) Constellation (d) Planet

51. Surface temperature of the sun is of the order of [DCE 1996]

(a) 5000 K (b) 7000 K

(c) 6000 K (d) 12000 K

52. The colour of a star is an indication of its [BCECE 2005]

(a) Weight (b) Distance

(c) Surface temperature (d) Size

53. Which of the following is coldest planet [BCECE 2005]

(a) Mercury (b) Pluto

(c) Earth (d) Venus

54. According to Hubble's law, the redshift (Z) of a receding galaxy and its distance r from earth are related as [AIIMS 2005]

(a) $Z \propto r$ (b) $Z \propto 1/r$

(c) $Z \propto 1/r^2$ (d) $Z \propto r^{3/2}$

55. The condition for a uniform spherical mass m of radius r to be a black hole is [G = gravitational constant and g = acceleration due to gravity] [AIIMS 2005]

(a) $(2Gm/r)^{1/2} \le c$ (b) $(2Gm/r)^{1/2} = c$

(c) $(2Gm/r)^{1/2} \ge c$ (d) $(gm/r)^{1/2} \ge c$

56. Fraunhofer lines of the solar system is an example of [AIIMS 2001]

(a) Emission spectrum

(b) Emission band spectrum

(c) Continuous emission spectrum

(d) Line absorption spectrum

57. The difference in the lengths of a mean solar day and a sidereal day is about [AIIMS 2003]

(a) 1 min [DCE 2003] (b) 4 min

(c) 15 min (d) 56 min

Critical Thinking

Objective Questions

1. A bright star is indicated to have a brightness magnitude of – 5 compared to a star of brightness zero magnitude. It means that this star compared to the reference star of zero brightness is
   (a) 100 times less bright (b) 5 times more bright
   (c) 5 times less bright (d) 100 times more bright

2. The sun revolves around the galaxy with a speed of 250 *km/sec* and it's radius is 3×10^4 light year. The mass of the milky way is
   (a) 3×10^{40} *kg* (b) 3×10^{41} *kg*
   (c) 5×10^{40} *kg* (d) 6×10^{41} *kg*

3. There are certain types of stars called visible stars which undergo periodic change in their light output. If such a star quadruple it's light output, how much does it's magnitude change
   (a) – 1.25 (b) – 1.5
   (c) – 1.75 (d) – 2

4. A particular emission line, detected in the light from a galaxy, has a wavelength $\lambda' = 1.1\lambda$, where λ is the proper wavelength of the line. The galaxy distance from us
   (a) $1.6 \times 10^9 ly$ (b) $0.97 \times 10^9 ly$
   (c) $2.4 \times 10^9 ly$ (d) $1.62 \times 10^{11} ly$

5. Assuming that the dimmest visible star to the naked eye has a magnitude of about 6. Brightness of planet Venus (magnitude = – 4) *w.r.t.* this star is
   (a) 10,000 times brighter (b) 2000 times brighter
   (c) 15000 times brighter (d) 4000 times brighter

6. A galaxy is observed to be moving with a velocity of 8600 *km-sec*$^{-1}$. If it is at a distance of 430 million light year from us, Hubble constant and corresponding age of the universe are respectively
   (a) $2 \times 10^{-5} \dfrac{kms^{-1}}{ly}, 1.49 \times 10^{10}$ *year*
   (b) $2 \times 10^{-6} \dfrac{kms^{-1}}{ly}, 1.58 \times 10^{3}$ *year*
   (c) $10^{6} \dfrac{kms^{-1}}{ly}, 1.49 \times 10^{10}$ *year*
   (d) None of these

7. Consider a binary star system consisting of two stars of masses M_1 and M_2 separated by a distance of 30 AU with a period of revolution equal to 30 years. If one of the two stars is 5 times farther from the centre of mass than the other. The masses of the two stars in terms of solar masses are
   (a) 5, 15 (b) 25, 5
   (c) 25, 10 (d) 7, 25

8. A planet of mass *m* moves in an ellipse around the sun of mass M_S so that its maximum and minimum distances are r_1 and r_2 respectively. The angular momentum of the planet relative to the centre of the sun is
   (a) $\sqrt{\dfrac{2GM_S r_1}{(r_1 + r_2)}}$ (b) $\sqrt{\dfrac{2GM_S m^2 r_1 r_2}{(r_1 + r_2)}}$
   (c) $\sqrt{\dfrac{GM_S r_1 r_2}{(r_1 + r_2)}}$ (d) $\sqrt{\dfrac{2GM_S}{r_1 r_2 (r_1 + r_2)}}$

9. The percentage of Sun's total energy which reaches the earth's surface is [Kerala PMT 2003]
   (a) 10^{-7} % (b) 10^{-6} %
   (c) 10^{-3} % (d) 10^{-2} %

10. Suppose a planet goes around Sun with a linear speed twice as fast that of earth. What will be it's orbit size as compared to that of earth ? (Radius of earth = *R*) [BHU 1993]
    (a) $R/4$ (b) $R/2$
    (c) R (d) $2R$

Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.
(e) If assertion is false but reason is true.

1. Assertion : The stars twinkle while the planets do not.
   Reason : The stars are much bigger in size than the planets. [AIIMS 2003
2. Assertion : A pulsor is a source of radio waves which change in terms of intensity at regular interval of time
   Reason : A pulsor is a rotating neutron star [AIIMS 1998, 2002]
3. Assertion : The comet do not obey Kepler's laws of planetary motion
   Reason : The comet do not have elliptical orbit [AIIMS 1995]
4. Assertion : A star which appears blue will be much hotter than the sun
   Reason : It is based on Wien's law
5. Assertion : There is no atmosphere on moon
   Reason : Escape velocity at the surface of moon is low.
6. Assertion : Red shift confirms that the universe is expanding
   Reason : Wavelength of red light is maximum in the visible region
7. Assertion : Sun is at the galactic centre C of the milky way
   Reason : All planets of solar system revolve around the sun.
8. Assertion : Moon is seen as it partly reflects the sun light falling on it
   Reason : Moon is a satellite of earth. It does not emit light of its own
9. Assertion : The value of Hubble's constant is 16 km/s

Reason : Hubble's constant means that a galaxy at 1 million light years away is receding at the rate of 16 *km/s*.

Answers

Universe

1	d	2	d	3	b	4	a	5	b
6	d	7	a	8	b	9	d	10	b
11	c	12	a	13	c	14	b	15	c
16	a	17	b	18	c	19	b	20	d
21	d	22	b	23	b	24	d	25	b
26	a	27	b	28	b	29	b	30	d
31	b	32	a	33	d	34	b	35	d
36	d	37	a	38	b	39	b	40	a
41	b	42	a	43	b	44	b	45	c
46	c	47	c	48	b	49	a	50	c
51	c	52	c	53	b	54	a	55	c
56	d	57	b						

Critical Thinking Questions

1	d	2	b	3	b	4	a	5	a
6	a	7	b	8	b	9	a	10	a

Assertion and Reason

1	b	2	b	3	b	4	a	5	a
6	b	7	e	8	a	9	e		

Answers and Solutions

Universe

1. (d) A study of binary star is most helpful in verifying Newton's law of gravitation.

2. (d) A group of bright and faint stars is called a constellation

3. (b) The sky is divided into 88 constellations.

4. (a) Big Bang theory is the most satisfactory theory about the origin of universe.

5. (b) Venus is the brightest planet.

6. (d) A star which appears blue will be much hotter than the sun.

7. (a) When distant stars are receding from us, spectral line from the star, when compared to with the corresponding line from source will show red shift *i.e.* a shift in frequency towards the red end.

8. (b) Solar constant is the energy crossing per unit area per sec at earth's distance, area being normal to the sun's rays. Also energy falling is inversely proportional to the square of distance from the source.

$$\therefore\ S' = \frac{S}{(5.3)^2}$$

9. (d) Venus and Mars have both CO_2 present.

10. (b) $\lambda_m T = b \Rightarrow 2.93\times10^{-10}\times T = 2.93\times10^{-3} \Rightarrow T = 10^7\ K$

11. (c) Black hole is highly dense malter in the atmosphere which has very large value of gravitational pull, so that nothing escapes from it.

12. (a) $F = \frac{GMm}{r^2} = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2$

$$M = \frac{mr^3 4\pi^2}{GmT^2} = \frac{4\pi^2 r^3}{GT^2}$$

13. (c) The age of universe is believed to be 10-20 billion years.

14. (b) Planet Venus is called Earth's sister.

15. (c) Source of Sun's energy is fusion reactions involving hydrogen.

16. (a) Asteroids are a group of rock pieces moving around the Sun in between Mars and Jupiter. They are believed to be the remains of a large planet which exploded due to gravitative attraction of Sun and that planet, may be called small planets.

17. (b) The energy of the sun is due to fusion of hydrogen nuclei into helium.

18. (c) Venus appears brighter as it is nearest to the earth and the light of sun reflected form sun reaches earth with greater intensity.

19. (b)

20. (d) Saturn only has ring around it.

21. (d) Milky way is one of the enormous galaxies of the universe.

22. (b) According to Hubble's law, $v \propto r$.

23. (b) The hottest planet of solar system is one which is nearest to sun and has no atmosphere.

24. (d) As we move towards the centre of the sun, the density and pressure increases.

25. (b) As $T^2 \propto r^3$ and distance of planet from sun in increasing order is for Mars, Saturn and Pluto.

26. (a) Length of milky way is 10^5 light years.

27. (b) Mercury is the nearest planet to sun.

28. (b) According to Wien's law, $\lambda_m \propto \frac{1}{T}$. It means higher the temperature of a star, the lower is the wavelength of maximum intensity radiation emitted from star which tells the colour of star.

29. (b) As $\lambda \propto \frac{1}{T}$; so $\frac{T_1}{T_2} = \frac{\lambda_2}{\lambda_1} = \frac{350}{510} = 0.68$

30. (d) $\frac{Gm\times 2m}{l^2} = m\times\frac{2l}{3}\frac{4\pi^2}{T^2}$ or $T = \left(\frac{4\pi^2 l^3}{3Gm}\right)^{1/2}$ *i.e.* $T \propto m^{-1/2}$

31. (b) Speed of galaxy is proportional to it's distance from us *i.e.* $U \propto r$. This is Hubble's law

32. (a) Reflecting power of a heavenly body is called albedo.

33. (d)

34. (b) Meteors are burnt piece of comet. When they reach earth's atmosphere, they start burning due to friction.

35. (d) According to Stefan's law $E = \sigma T^4$

36. (d) If the light received from galaxies indicates a shift towards the red end of spectrum of light, it means that the galaxies should be receding away (Doppler's effect). Therefore we conclude that the universe is expanding.

37. (a) The star which appears red is at less temperature, than the star which appears white. Therefore, temperature of *Q* is higher than that of *P*.

38. (b) The albedo (reflection power) is maximum for Venus, because it reflects 85% of incident light. It's value of albedo is 0.85.

39. (b) It is well known that if the mass of the star is more than that of mass of Sun, it explodes after it's red giant stage and dies out giving rise to supernova and a black hole.

40. (a) If a comet approaches the sun, the substances like water *etc.* on the comet, get vaporised due to the heat of Sun, and radiation pressure forces of these vapours move away from the Sun. Hence, it forms the tail of the comet.

41. (b)

42. (a) Venus has the longest day.

43. (b) Ursa major is known as saptarishi.

44. (b)

45. (c) The temperature of stars can be determined by Wiens displacement law which is $\lambda_m.T = \text{constant}$.

46. (c) The motion of planets in the solar system is based on the conservation of angular momentum.

47. (c)

48. (b)

49. (a) Hubble's law state that. Speed of recession (v) $\propto$ distance (r).

50. (c) Great bear is a constellation, which is a group of some stars.

51. (c) Surface temperature of Sun is about 6000 *K*.

52. (c) By using $\lambda_m T = \text{constant}$

53. (b) Because pluto is farthest from Sun.

54. (a) Hubble's law is a statement of a direct correlation between the distance (r) to a galaxy and its recessional velocity as determined by the red shift (Z). It is stated as $Z = Hr$.

55. (c) The criterion for a star to be black hole is

$$\frac{GM}{c^2R} \geq \frac{1}{2} \text{ or, } \sqrt{\frac{2GM}{R}} \geq c.$$

56. (d) Fraunhofer lines are produced by the absorption of rays of the Sun in the atmosphere. When white light from photosphere passes through chromosphere, the vapours and gases present in it absorbs certain wavelengths and produces dark lines (Fraunhofer lines).

57. (b) The difference in the length of mean solar day and a sidereal day is about 4 min.

Critical Thinking Questions

1. (d) Given that magnitude for brightest star = – 5
and magnitude of given star = 0
Now $m_2 - m_1 = 0 - (-5) = 5$
The brightness ratio is given by

$$\frac{l_1}{l_2} = 100^{(m_2 - m_1)/s} = 100^{5/5} = 100$$

So bright star is 100 time bright that the dim star.

2. (b) The mass of galaxy is given by $M = \frac{v^2 r}{G}$

where $v = 250\ km/sec = 250 \times 10^3\ \frac{m}{\sec}$

$r = 3 \times 10^4\ ly = 3 \times 10^4 \times 9.46 \times 10^{12}\ km \approx 3 \times 10^{20}\ m$

$$\therefore\ m = \frac{(250 \times 10^3)^2 \times (3 \times 10^{20})}{6.6 \times 10^{-11}} \approx 3 \times 10^{41}\ kg.$$

3. (b) $\frac{l_2}{l_1} = 4 \Rightarrow m_2 - m_1 = -2.5 \log\left(\frac{l_2}{l_1}\right) = -2.5 \log 4$

$= -2.5 \times 0.6021 = -1.5$.

4. (a) From Hubble's law $v = Hr$ where H = Hubble's constant = 19.3 *mm/sec-ly* and r = Distance of Galaxy from us.

According to Doppler's effect speed of Galaxy $v = \frac{c\Delta\lambda}{\lambda}$

$$\Rightarrow r = \frac{c\Delta\lambda}{H\lambda} = \frac{c \times 0.1\lambda}{H\lambda} = \frac{0.1 \times 3 \times 10^8}{19.3 \times 3 \times 10^{-3}} = 1.6 \times 10^9\ ly$$

5. (a) Here, for Venus $m_1 = -4$, for star $m_2 = 6$ using

$$\frac{l_1}{l_2} = 100^{(m_2 - m_1)/5} = 100^{[6-(-4)]/5} = 100^2 = 10{,}000.$$

6. (a) $H = \frac{v}{r} = \frac{8600}{430 \times 10^6} = 2 \times 10^{-5}\ \frac{kms^{-1}}{ly}$

Age of the universe, $t_0 = \frac{1}{H} = \frac{r}{v}$

Taking $r = 430 \times 10^6\ ly = 430 \times 10^6 \times 9.46 \times 10^{12}\ km$

$$\Rightarrow t_0 = \frac{430 \times 10^6 \times 9.46 \times 10^{12}}{8600} \sec$$

$$= \frac{430 \times 10^6 \times 9.46 \times 10^{12}}{8600 \times 3600 \times 24 \times 365} = 1.49 \times 10^{10}\ year$$

7. (b) $M_1 + M_2 = \frac{4\pi^2}{G} \cdot \frac{r^3}{T^2}$

If T is measured in years, r in *A.U.* and masses in Solar masses then $G = 4\pi^2$.

$$\therefore\ M_1 + M_2 = \frac{r^3}{T^2} = \frac{(30)^3}{(30)^2} = 30 \qquad .....(i)$$

Now $r_1 + r_2 = 30 \Rightarrow r_1 + 5r_1 = 60$

$\Rightarrow r_1 = 5$ and $r_2 = 25$

Again $M_1 r_1 = M_2 r_2 \Rightarrow \frac{M_1}{M_2} = 5$ .....(ii)

After solving (i) and (ii) we get $M_1 = 25$ and $M_2 = 5$

8. (b) From conservation of energy

$\frac{1}{2}mv_1^2 - \frac{GM_S m}{r_1} = \frac{1}{2}mv_2^2 - \frac{GM_S m}{r_2}$. Angular momentum is conserved, that is $mv_1 r_1 = mv_2 r_2$

$$\text{or } v_2 = v_1 \frac{r_1}{r_2} \Rightarrow \frac{1}{2}mv_1^2 - \frac{GM_S m}{r_1} = \frac{1}{2}m\left(\frac{v_1 r_1}{r_2}\right)^2 - \frac{GM_S m}{r_2}$$

$$\text{or } v_1 = \sqrt{\frac{2GM_S r_2}{r_1(r_1 + r_2)}} \Rightarrow L = mv_1 r_1 = \sqrt{\frac{2GM_S m^2 r_1 r_2}{r_1 + r_2}}$$

9. (a) If S is the total energy emitted by Sun per second and r is the distance of earth from Sun; then energy reaching earth of radius R per second $= \frac{S}{4\pi r^2} \times 2\pi R^2 = \frac{SR^2}{2r^2}$.

$\therefore$ Percentage of energy reaching earth

$$= \frac{SR^2}{2r^2 S} \times 100 = \frac{(6.4 \times 10^6)^2 \times 100}{2 \times (1.5 \times 10^{11})^2} \simeq 10^{-7}\%$$

10. (a) From Kepler's law $T \propto R^{3/2}$ and also $T = \frac{2\pi R}{v}$

$$\Rightarrow v \propto \frac{1}{R^{1/2}} \Rightarrow \frac{v_1}{v_2} = \left(\frac{R_2}{R_1}\right)^{1/2} \Rightarrow \frac{v_1}{2v_1} = \left(\frac{R_2}{R_1}\right)^{1/2}$$

$$\Rightarrow R_2 = \frac{R_1}{4} = \frac{R}{4}$$

Assertion and Reason

1. (b) Stars twinkles due to variation in density of atmospheric layer. Also stars are much bigger in size than planets but it has nothing to deal with twinkling phenomenon.

2. (b) Pulsar is a source of radio waves which emits pulses of radio waves at short and regular time of intervals.

Pulsar is formed, due to super nova explosion, when super nova explosion occurs, the core of the star is compressed and electrons and protons combine to form a neutron. Due to this region pulsar is called neutron star.

3. (b) Comets do not revolve around the sun in fixed elliptical orbit like other planets and don't obey Kepler's law for planetary motion.

4. (a) According to Wien's law, $\lambda_m T = b =$ constant. As λ_m for the star is blue, which is less than λ_m for sun, which is yellow, therefore temp. T of star will be much higher than the temperature of the sun.

5. (a) At the surface of moon $v_{rms} > v_{escape}$ hence molecules escape out before reaching their *rms* velocity that's why there is no atmosphere present.

6. (b) Red shift means that wavelength of light received from stars is increasing *i.e.,* apparent frequency is decreasing. Therefore, the stars/galaxies must be receding away. Hence the universe is expanding. Reason is also true, but it does not explain the assertion appropriately.

7. (e) The reason is true, but the assertion is false. Infect, distance of sun of our solar system from galactic centre is 3×10^4 light years.

8. (a) Both the assertion and reason are true and reason is a correct explanation of the assertion.

9. (e) The assertion is not true. Infect, the value of Hubble's constant is 16 km per sec per million light years.

Universe

SET Self Evaluation Test -31

1. "The universe is expanding" means
   (a) Size of the hole in Ozon layer is increasing
   (b) Universe is expanding into something
   (c) Infinite universe is becoming more infinite
   (d) None of these

2. The galaxy in which we live is
   (a) Spiral galaxy (b) Radio galaxy
   (c) Irregular galaxy (d) None of these

3. The distance of Venus from the sun is 0.72 AU. the orbital period of the Venus is
   (a) 200 days (b) 320 days
   (c) 225 days (d) 325 days

4. Suppose the sun was located at the position occupied by the nearest star, say, alphacenturi 4 light years away. By what factor the solar radiation received per sec per unit area decrease
   (a) 1.5×10^{-6} (b) 1.5×10^{-8}
   (c) 1.5×10^{-9} (d) 1.5×10^{-11}

5. If a galaxy is at a distance 430 million light years from us, determine Hubble's constant. Its speed being $6.48\times10^{6} ms^{-1}$
   (a) 16 kms^{-1} per million light year
   (b) 15 kms^{-1} per million light year
   (c) 14 kms^{-1} per million light year
   (d) None of these

6. The magnitude of two stars A and B are 2.5 and –5 respectively. The brightness ratio of $\frac{B}{A}$ is
   (a) 7.5 (b) 10^{2}
   (c) 10^{3} (d) $10^{7.5}$

7. A body at 1500 K emits maximum energy at a wavelength 20,000 $\mathring{A}$. If the Sun emits maximum energy at wavelength 5500 $\mathring{A}$, then the temperature of Sun is
   (a) 5454 (b) 4454
   (c) 4550 (d) 5400

8. The hottest type of stars are called
   (a) A type (b) B type
   (c) O type (d) M type

9. Venus appears brighter than other stars because
   (a) It is heavier than other planets
   (b) Its density is more than other planets
   (c) It is nearer to earth in comparison to other planets
   (d) Nuclear fusion takes place at its surface

10. The death of a star results is in a neutron star if the original mass of star in terms of mass of Sun (M) is
    (a) Less than $2M$
    (b) Between $2M$ and $4M$
    (c) Greater than 5 M
    (d) Exactly equal to M

11. The tail of a comet points
    (a) Towards the Sun
    (b) Away from the Sun
    (c) In arbitrary
    (d) Away from the earth

12. The angle of maximum elongation for Venus is 47°. The distance of Venus from earth in A.U. is
    (a) 0.68 A.U. (b) 0.86 A.U.
    (c) 1 A.U. (d) 0.73 A.U.

13. The number of stars in our solar system is
    (a) 9 (b) 5
    (c) 1 (d) More than 9

14. If angular diameter of Sun is about 30′ and it's distance from earth is 1.5×10^{11} m, then solar diameter is
    (a) 1.1×10^{7} m
    (b) 1.5×10^{8} m
    (c) 1.4×10^{9} m
    (d) 1.9×10^{11} m

AS Answers and Solutions

(SET -31)

1. (c)

2. (a) The galaxy in which we live is spiral galaxy. Our galaxy Milky way is a spiral galaxy.

3. (c) $\frac{T_2^2}{T_1^2} = \left(\frac{r_2}{r_1}\right)^3$ or $T_2 = T_1\left(\frac{r_2}{r_1}\right)^{3/2} = 1\left(\frac{0.72}{1}\right)^{3/2}$

= 0.62 year or 225 days.

4. (d) $\frac{E_1}{E_2} = \frac{r_2^2}{r_1^2}$ or $\frac{E_2}{E_1} = \frac{r_1^2}{r^2} \Rightarrow \frac{(1.5\times10^{11})^2}{(4\times9.46\times10^{15})^2} = 1.5\times10^{-11}$

where r_1 = Distance of Sun from earth = 1.5×10^{11} m = 1 AU, r_2 = 4 ly = $4 \times 9.46 \times 10^{15}$ m

5. (b) $H = \frac{v}{r} = \frac{6.48\times10^6}{430} = 15.07\, kms^{-1}$ per million light year

6. (c) $m_B - m_A = -2.5\log_{10}\left(\frac{I_B}{I_A}\right)$

$\Rightarrow -5-(2.5) = -2.5\log_{10}\frac{I_B}{I_A} \Rightarrow \log_{10}\frac{I_B}{I_A} = 3$

$\Rightarrow \frac{I_B}{I_A} = 10^3$.

7. (a) According to Wien's displacement law $\lambda_m T =$ constant

or $\lambda_m T = \lambda_m' T'$

or $T' = \frac{\lambda_m}{\lambda_m'}\times T = \frac{20{,}000Å\times1500K}{5500Å} = 5454K.$

8. (c) O type stars are hottest.

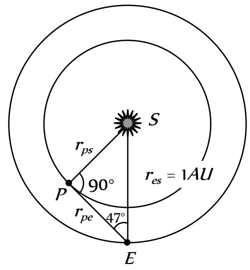
9. (c) Venus appears brighter than other stars because it is nearest to earth than other stars.

10. (b)

11. (b) Tail of comet points away from the sun.

12. (a) The angle formed at earth between earth planet and earth sun direction is called planet's elongation represented by ε, when planet appears farthest from the Sun, the angle subtended by the Sun and earth at the planet is $90°$.

From the geometry of figure



$\frac{r_{PE}}{r_{sE}} = \cos\varepsilon = \cos 47°$

$r_P = r_{sE}\cos 47°$

$= (\cos 47°)\times 1AU = 0.68AU$

Choice (a) is correct

[$\cos 45° = \frac{1}{\sqrt{2}} = 0.707$. As angle increases its cosine decreases $\cos 47°$ can not be 0.86, 0.73 or 1]

13. (c) The number of stars in our solar system in one (our Sun).

14. (c) We know that

$$D = r\theta = 1.5\times10^{11}\times\frac{1}{2}\times\frac{\pi}{180°} = 1.4\times10^9 m$$

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